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A. R. Forsyth.

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Arthur Cayley

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556. On Steiner's Surface (continued)

[Continued from earlier pages of the same paper.]

[p. 6] where the first factor gives the node. Equating to zero the second factor, we have

$$\left\{qZ + \left(2q - 1 - \tfrac{2}{q}\right) X\right\}^2 = X^2 \left\{\left(2q - 1 - \tfrac{2}{q}\right)^2 - 4q^2 + 4q - 1\right\}$$

$$= X^2 \cdot \tfrac{4}{q^2}(1-q)(1+2q);$$

or, finally,

$$qZ = \left\{-2q + 1 + \tfrac{2}{q} \pm \tfrac{2}{q}\sqrt{(1-q)(1+2q)}\right\} X,$$

giving two real values for all values of q from $q = 1$ to $q = -\frac{1}{2}$. (See the Table afterwards referred to.)

We may recapitulate as follows:

$q > 1$, or $\lambda > \frac{1}{2}h$; the curve is imaginary, but with three real acnodes, answering to the acnodal parts of the nodal lines:

$q = 1$, or $\lambda = \frac{1}{2}h$; the summit appears as a fourth acnode:

$q < 1 > \frac{1}{4}$, or $\lambda < \frac{1}{2}h > \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid lying within the triangle and having the sides of the triangle for bitangents of imaginary contact:

$q = \frac{1}{4}$, or $\lambda = \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid having the sides of the triangle for osculating tangents:

$q < \frac{1}{4} > 0$, or $\lambda < \frac{1}{3}h > \frac{1}{4}h$; the curve consists of three conjugate points and an indented trigonoid having the sides of the triangle for bitangents of real contact:

$q = 0$, or $\lambda = \frac{1}{4}h$; curve has the summits of the triangle for cusps:

$q < 0 > -\frac{1}{4}$, or $\lambda < \frac{1}{4}h > \frac{1}{6}h$; curve has three crunodes, or say it is a cis-centric trifolium:

$q = -\frac{1}{4}$, or $\lambda = \frac{1}{6}h$; curve has a triple point, or say it is a centric trifolium:

$q < -\frac{1}{4} > -\frac{1}{2}$, or $\lambda < \frac{1}{6}h > 0$; curve has three crunodes, or say it is a trans-centric trifolium:

$q = -\frac{1}{2}$, or $\lambda = 0$; curve is a two-fold circle:

$q < -\frac{1}{2}$, or $\lambda < 0$; the curve becomes again imaginary, consisting of three acnodes answering to the acnodal parts of the nodal lines.

For the better delineation of the series of curves, I calculated the following Table, wherein the first column gives a series of values of $\lambda : h$; the second the corresponding values of q, $= \dfrac{2\lambda - h}{2\lambda - 2h}$; the third the positions of the point of contact, say with the side $Z = 0$, the value of $X : Y$ being calculated from the foregoing formula

$$\frac{X}{Y} = \frac{1}{2q} + 1 \pm \sqrt{\frac{1}{4q^2} - \frac{1}{q}};$$

[p. 7] and the fourth the apsidal distances, say for the radius vector $X = Y$, the value of $Z : X$ being calculated from the foregoing formula

$$\frac{Z}{X} + \frac{X}{Z} = -2 + \frac{1}{q} + \frac{2}{q^2} \pm \frac{2}{q}\sqrt{\left(\frac{1}{q} - 1\right)\left(2 + \frac{1}{q}\right)}.$$

The Table is:

$\lambda : h$	q	Contact, $Z = 0$, $X : Y =$	Apses, $X = Y$; $X : Z =$
$*\ \cdot 75$	$1 \cdot 00$	imposs.	$1\cdot$
$\cdot 70$	$\cdot 6666$	imposs.	$\cdot 032$ or $7 \cdot 968$
$*\ \cdot 666$	$\cdot 5$	$1\cdot$	$0\cdot$ $16\cdot$
$\cdot 65$	$\cdot 4285$	$\cdot 320$ or $3 \cdot 124$	$\cdot 006$ $22 \cdot 44$
$\cdot 6$	$\cdot 25$	$\cdot 0721$ $13 \cdot 9279$	$\cdot 059$ $67 \cdot 941$
$\cdot 55$	$\cdot 1111$	$\cdot 011$ $78 \cdot 988$	$\cdot 15$ $337 \cdot 85$
$*\ \cdot 5$	$0\cdot$	0 or ∞	$\cdot 25$ ∞
$\cdot 45$	$-\cdot 0909$	$\cdot 005$ $118 \cdot 99$	$\cdot 39$ $457 \cdot 61$
$\cdot 4$	$-\cdot 1666$	$\cdot 029$ $33 \cdot 971$	$\cdot 496$ $127 \cdot 504$
$\cdot 35$	$-\cdot 2308$	$\cdot 060$ $16 \cdot 72$	$\cdot 648$ $61 \cdot 79$
$*\ \cdot 3$	$-\cdot 2857$	$\cdot 099$ $10 \cdot 151$	$\cdot 812$ $37 \cdot 187$
$*\ \cdot 25$	$-\cdot 3333$	$\cdot 141$ $6 \cdot 854$	$1\cdot$ 25
$\cdot 2$	$-\cdot 375$	$\cdot 207$ $4 \cdot 904$	$1 \cdot 218$ $17 \cdot 89$
$\cdot 15$	$-\cdot 4118$	$\cdot 276$ $3 \cdot 622$	$1 \cdot 48$ $13 \cdot 25$
$\cdot 10$	$-\cdot 4444$	$\cdot 372$ $2 \cdot 690$	$1 \cdot 813$ $9 \cdot 937$
$\cdot 05$	$-\cdot 4737$	$\cdot 515$ or $1 \cdot 941$	$3 \cdot 15$ or $6 \cdot 46$
$*\ 0$	$-\cdot 5$	$1\cdot$	$4\cdot$

where the asterisks show the critical values of $\lambda : h$.

It is worth while to transform the equation to new coordinates X', Y', Z' such that $X' = 0$, $Y' = 0$, $Z' = 0$ represent the sides of the triangle formed by the three nodes. Writing for shortness $X + Y + Z = P$, $YZ + ZX + XY = Q$, $XYZ = R$, the equation is

$$(qP - Q)^2 = 4(2q + 1)PR.$$

The expressions of X, Y, Z in terms of the new coordinates are of the form $X'+\theta P'$, $Y'+\theta P'$, $Z'+\theta P'$, where $P'=X'+Y'+Z'$; writing also $Q'=Y'Z'+Z'X'+X'Y'$, $R'=X'Y'Z'$, then the values of P, Q, R are

$$(1+3\theta)P', \quad Q'+(2\theta+3\theta^2)P'^2, \quad R'+\theta P'Q'+(\theta^2+\theta^3)P'^3,$$

[p. 8] and the transformed equation is

$$\left[\{q(1+3\theta)^2-2\theta-3\theta^2\}P'^2-Q'\right]^2=4(2q+1)(1+3\theta)\{(\theta+\theta^2)P'^3+\theta P'Q'+R'\},$$

which is satisfied by $Q'=0$, $R'=0$, if only

$$\{q(1+3\theta)^2-2\theta-3\theta^2\}^2=4(2q+1)(1+3\theta)(\theta^2+\theta^3),$$

or, if for a moment $q(1+3\theta)=\Omega$, the equation is

$$(\Omega^2-2\theta-3\theta^2)^2=4(\theta^2+\theta^3)(2\Omega+1+3\theta),$$

that is,

$$\Omega^4+\Omega^2(-6\theta^2-4\theta)+\Omega(-8\theta^3-8\theta^2)-3\theta^4-4\theta^3=0,$$

that is,

$$(\Omega+\theta)^2(\Omega^2-2\theta\Omega-3\theta^2-4\theta)=0.$$

If the new axes pass through the nodes, then $\Omega+\theta=0$; that is, $q(1+3\theta)+\theta=0$, which equation gives the value of θ for which the new axes have the position in question; substituting in the first instance for q the value $\frac{-\theta}{3\theta+1}$, the equation becomes

$$\theta^2\{2\theta(1+\theta)P'^2+Q'\}^2=4(1+\theta)P'\{\theta^2(1+\theta)P'^3+\theta P'Q'+R'\},$$

that is,

$$Q'^2=4(1+\theta)P'R';$$

or, finally, substituting for θ its value in terms of q, the required equation is

$$Q'^2=4\cdot\frac{2q+1}{3q+1}\,P'R',$$

that is,

$$(Y'Z'+Z'X'+X'Y')^2=4\cdot\frac{2q+1}{3q+1}\,X'Y'Z'(X'+Y'+Z').$$

In particular, for $q=0$ the equation is

$$(Y'Z'+Z'X'+X'Y')^2-4X'Y'Z'(X'+Y'+Z')=0,$$

which is right, since, in the case in question (the tricuspidal curve), we have

$$X,\ Y,\ Z = X',\ Y',\ Z'.$$

I remark, in passing, that, taking the equation to be

$$(Y'Z' + Z'X' + X'Y')^2 = m\,X'Y'Z'(X' + Y' + Z'),$$

we may write herein

$$Z' = \tfrac{1}{2} - x,$$

$$X' = \tfrac{1}{4} + \tfrac{1}{2}x - \tfrac{\sqrt{3}}{2}y,$$

$$Y' = \tfrac{1}{4} + \tfrac{1}{2}x + \tfrac{\sqrt{3}}{2}y\,;$$

[p. 9] where

$$x = \frac{2\sqrt{m(m-3)}}{9}\cos\theta - \frac{2(m-3)}{9}\cos 2\theta,$$

$$y = \frac{2\sqrt{m(m-3)}}{9}\sin\theta + \frac{2(m-3)}{9}\sin 2\theta,$$

which are the formulae for the description of the trinodal quartic as a unicursal curve.

I consider now the second position; viz. the horizontal plane now passes through the centre of the tetrahedron, and is parallel to two opposite edges. The equations of the nodal lines are here $(y = 0,\ z = 0)$, $(z = 0,\ x = 0)$, $(x = 0,\ y = 0)$; and if for convenience we assume the distance of the midpoints of opposite edges to be $= 2$, or the half of this $= 1$, then the equations of the faces are

$$\begin{aligned} X &= x + y + z - 1 = 0,\\ Y &= -x - y + z - 1 = 0,\\ Z &= x - y - z - 1 = 0,\\ W &= -x + y - z - 1 = 0, \end{aligned}$$

and the equation of the surface is

$$\sqrt{X} + \sqrt{Y} + \sqrt{Z} + \sqrt{W} = 0.$$

Proceeding to rationalise, this is

$$X + Y + 2\sqrt{XY} = Z + W + 2\sqrt{ZW},$$

viz.

$$2z + \sqrt{XY} = \sqrt{ZW}\,;$$

we thence have

$$4z^2 + 4z\sqrt{XY} + XY = ZW,$$

or, since

$$ZW - XY = 4z + 4xy,$$

this is

$$z + xy - z^2 = z\sqrt{XY}\,;$$

whence

$$(z + xy - z^2)^2 = z^2\{(z-1)^2 - (x+y)^2\}\,;$$

or reducing,

$$x^2y^2 + z^2x^2 + z^2y^2 + 2xyz - z^2 = 0,$$

a form which puts in evidence the nodal lines. Considering z as constant, we have the equation of the section; this is a quartic having the node ($x = 0$, $y = 0$), and two other nodes at infinity on the two axes respectively; moreover, the curve has for bitangents the intersections of its plane with the faces of the tetrahedron; or what is the same thing, attributing to z its constant value, the equations of the bitangents are

$$\begin{aligned} x + y + z - 1 &= 0, \\ -x - y + z - 1 &= 0, \\ x - y - z - 1 &= 0, \\ -x + y - z - 1 &= 0. \end{aligned}$$

[p. 10] These lines form a rectangle which is the section of the tetrahedron; observe that this is inscribed in the square the corners of which are $x = \pm 1$, $y = \pm 1$; viz. $z = +1$ (highest section), this is the dexter diagonal (considered as an indefinitely thin rectangle), and as z diminishes, the longer side decreases and the shorter increases until for $z = 0$ (central section) the rectangle becomes a square; after which, for z negative it again becomes a rectangle in the conjugate direction, and finally, for $z = -1$ (lowest section) it becomes the sinister diagonal (considered as an indefinitely thin rectangle). But on account of the symmetry it is sufficient to consider the upper sections for which z is positive. The sides $\pm(x + y) + z - 1 = 0$ parallel to the dexter diagonal of the square may for convenience be termed the dexter sides, and the others the sinister sides. In what follows I write c to denote the constant value of z.

We require to know whether the bitangents have real or imaginary contacts; and for this purpose to find the coordinates of the points of contact.

Take first a dexter bitangent $x+y+c-1=0$; the coordinates of any point hereof are

$$x=\tfrac{1}{2}(1-c+\theta),\quad y=\tfrac{1}{2}(1-c-\theta),$$

where θ is arbitrary; and substituting in the equation of the curve, we should have for θ a twofold quadric equation, giving the values of θ for the two points of contact respectively. We have

$$x^2+y^2=\tfrac{1}{2}\{(1-c)^2+\theta^2\},\quad xy=\tfrac{1}{4}\{(1-c)^2-\theta^2\},$$

and thence

$$8c^2\{(1-c)^2+\theta^2\}+8c\{(1-c)^2-\theta^2\}+\{(1-c)^2-\theta^2\}^2=0,$$

viz. this equation is

$$\{\theta^2-(1-c)(1+3c)\}^2=0,$$

a twofold quadric equation, as it should be; and the values of θ being $=\pm\sqrt{(1-c)(1+3c)}$, we see that these, and therefore the contacts, are real from $c=1$ to $c=-\frac{1}{3}$.

In exactly the same way for a sinister bitangent $\pm(x-y)-c-1=0$, we have

$$x=\tfrac{1}{2}(1+c+\phi),\quad -y=\tfrac{1}{2}(1+c-\phi),\quad \text{and}\quad \phi=\pm\sqrt{(1-3c)(1+c)},$$

viz. the values of ϕ, and therefore the contacts, are real from $c=\frac{1}{3}$ to $c=-1$.

That is,

$c=1$ to	Contacts of Dexter Bitangents.	Contacts of Sinister Bitangents.
$c=1$ to $\frac{1}{3}$	real,	imaginary,
$c=\frac{1}{3}$ to $-\frac{1}{3}$	real,	real,
$c=-\frac{1}{3}$ to -1	imaginary,	real.

or say $c=1$ to $\frac{1}{3}$, the contacts are real, imaginary; but $c=\frac{1}{3}$ to 0, they are real, real. In the transition case, $c=\frac{1}{3}$, the sinister bitangents become osculating (4-pointic) tangents touching at points on the dexter diagonal. This can be at once verified.

[p. 11] Observe that when $c=1$, we have

$$(x+y)^2+x^2y^2=0,$$

so that the only real point is $x = 0$, $y = 0$; viz. this is a tacnode, having the real tangent $x + y = 0$. For $c = 0$ (central section) the equation becomes $x^2y^2 = 0$; viz. the curve is here the two nodal lines each twice.

It is now easy to trace the changes of form.

$c = 1$; curve is a tacnode, as just mentioned, tangent the dexter diagonal.

$c < 1 > \frac{1}{3}$; curve is a figure of 8 inside the rectangle, having real contacts with the dexter sides, but imaginary contacts with the sinister sides.

$c = \frac{1}{3}$; curve is a figure of 8 having real contacts with the dexter sides, and osculating (4-pointic) contacts with the sinister sides.

$c < \frac{1}{3} > 0$; curve is an indented figure of 8 having real contacts as well with the sinister as the dexter sides.

$c = 0$; curve is squeezed up into a finite cross, being the crunodal parts of the nodal lines; and joined on to these we have the acnodal parts, so that the whole curve consists of the lines $x = 0$, $y = 0$ each as a twofold line.

For tracing the curve, it is convenient to turn the axes through an angle of 45°; viz. writing $\dfrac{y+x}{\sqrt{2}}$, $\dfrac{y-x}{\sqrt{2}}$ in place of x, y respectively, the equation becomes

$$c^2(y^2 - x^2)^2 + c(y^2 + x^2) + \tfrac{1}{4}(y^2 - x^2)^2 = 0\,;$$
$$x = 0 \text{ gives } y^2 = 0 \text{ or } y^2 = -4c(1 + c),^*$$
$$y = 0 \text{ gives } x^2 = 0 \text{ or } x^2 = 4c(1 - c).$$

Moreover, we have

$$\tfrac{1}{4}(c - 1)^2(y^2 - x^2) + 8c^2y^2 + (y^2 - x^2)^2 = 0,$$

viz.

$$(x^2 + y^2 - 2c^2 - 2c)^2 = 4c^2\{(c - 1)^2 - 2y^2\},$$

and similarly

$$(x^2 + y^2 + 2c^2 - 2c)^2 = 4c^2\{(c + 1)^2 - 2x^2\},$$

putting in evidence the bitangents, now represented by the equations $c - 1 = \pm y\sqrt{2}$ and $c + 1 = \pm x\sqrt{2}$ respectively. And for the first of these, or $c - 1 = \pm y\sqrt{2}$, we have for the points of contact $x^2 = \frac{1}{2}(1 - c)(1 + 3c)$; and for the second of them, or $c + 1 = \pm x\sqrt{2}$, the points of contact are $y^2 = \frac{1}{2}(1 + c)(1 - 3c)$.

I consider the circumscribed cone having its vertex at a point $(0, 0, \gamma)$ on the nodal line $(x = 0,\ y = 0)$. Writing in the equation of the surface $x = \lambda(z - \gamma)$, $y = \mu(z - \gamma)$, the equation, throwing out the factor $(z - \gamma)^2$, becomes

$$\lambda^2\mu^2(z - \gamma)^2 + 2\lambda\mu z(z - \gamma) + (\lambda^2 + \mu^2)z^2 - z^2 = 0,$$

that is,

$$(\lambda^2\mu^2+\lambda^2+\mu^2)\,z^2+2(-\gamma\lambda\mu+1)\,z\lambda\mu+\gamma^2\lambda^2\mu^2=0\,;$$

* y always imaginary when c is positive.

[p. 12] and equating to zero the discriminant in regard to z, we have

$$\gamma^2\lambda^2\mu^2(\lambda^2\mu^2+\lambda^2+\mu^2)-(-\gamma\lambda\mu+1)^2=0,$$

that is,

$$\gamma^2(\lambda^2+\mu^2)+2\gamma\lambda\mu-1=0\,;$$

and substituting herein the values $\lambda=\dfrac{x}{z-\gamma}$ and $\mu=\dfrac{y}{z-\gamma}$, we have the equation of the cone, viz. this is

$$\gamma^2(x^2+y^2)+2\gamma xy-(z-\gamma)^2=0,$$

or, what is the same thing,

$$\gamma^2(x^2+y^2-1)+2\gamma(xy+z)-z^2=0\,;$$

viz. this is a quadric cone having for its principal planes $z=\gamma$, $x+y=0$, $x-y=0$, these last being the planes through the nodal line and the two edges of the tetrahedron.

In the particular case $\gamma=\infty$, the cone becomes the circular cylinder $x^2+y^2-1=0$.

The cone intersects the plane $z=0$ in the conic

$$\gamma^2(x^2+y^2-1)+2\gamma xy=0,$$

which is a conic passing through the corners of the square $(x=0,\ y=\pm1)$, $(x=\pm1,\ y=0)$. For $\gamma>1$, that is, for an exterior point, the conic is an ellipse having for the squares of the reciprocals of the semi-axes $1+\dfrac{1}{\gamma}$, $1-\dfrac{1}{\gamma}$ (this at once appears by writing in the equation $\dfrac{x+y}{\sqrt{2}}$, $\dfrac{x-y}{\sqrt{2}}$ in place of x, y respectively). In particular, for $\gamma=\infty$, the curve becomes the circle $x^2+y^2=1$. We have thus the apparent contour of the surface as seen from the point $z=\gamma$ on the nodal line, projected on the plane $z=0$ of the other two nodal lines.

To find the curve of contact of the cone and surface, or say the surface-contour from the same point, write for a moment

$$V=\gamma^2(x^2+y^2-1)+2\gamma(xy+z)-z^2,$$

$$U = (xy+z)^2 - z^2(x^2+y^2-1),$$

then, substituting for x^2+y^2-1 its value in terms of V from the first equation, we find

$$U = \left(xy+z-\frac{z^2}{\gamma}\right)^2 \frac{V}{\gamma^2},$$

and the equations $V=0$, $U=0$ give therefore

$$xy+z-\frac{z^2}{\gamma}=0,$$

or say $\gamma(xy+z)-z^2=0$.

The cone and surface therefore touch along the quadriquadric curve

$$\gamma^2(x^2+y^2-1)+2\gamma(xy+z)-z^2=0,$$

$$\gamma(xy+z)-z^2=0,$$

equations which may be replaced by

$$\gamma^2(x^2+y^2-1)+xy+z=0,$$

$$\gamma^2(x^2+y^2-1)+z^2=0.$$

In the case $\gamma=\infty$, the equations are $x^2+y^2-1=0$, $xy+z=0$, viz. the curve is the intersection of the hyperbolic paraboloid $xy+z=0$ by the cylinder $x^2+y^2-1=0$.

557. On Certain Constructions for Bicircular Quartics

[p. 13] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 29–31. Read March 12, 1874.]*

I CALL to mind that if F, G are any two points and F', G' their antipoints, then the circle on the diameter FG and that on the diameter $F'G'$ are concentric orthotomics, viz. they have the same centre, and the sum of the squared radii is $=0$. Moreover, if the circles B, B' are concentric orthotomics, and the circle A is orthotomic to B, then it is a bisector of B', viz. it cuts B' at the extremities of a diameter of B'; and B is then said to be a bifid circle in regard to A.

Given two real circles, these have an axial orthotomic, the circle, centre on the line of centres at its intersection with the radical axis, which cuts at right angles the given circles; viz. this axial orthotomic is real if the circles have no real intersection; but if the intersections are real, then the axial orthotomic is a pure imaginary, and instead thereof we may consider its concentric orthotomic, viz. this is the axial bifid of the two circles, or circle having its centre on the line of centres at the intersection thereof with the radical axis or common chord of the two circles, and having this common chord for its diameter.

If one of the circles is a pure imaginary, then we have still an axial orthotomic; viz. the pure imaginary circle is replaced by the concentric orthotomic; and the axial orthotomic is a bisector of the substituted circle; and so if each of the circles is a pure imaginary, then we have still an axial orthotomic, viz. each circle is replaced by the concentric orthotomic, and the axial orthotomic is a bisector of the substituted circles. And in either case the axial orthotomic of the original circles (one or each of them pure imaginary) is real; viz. this is given either as the axial bisector of one real circle and orthotomic of another real circle; or as the axial bisector of two circles, from which the reality thereof easily appears. Or we may verify it thus: Suppose

[p. 14] that the two circles are $(x-\alpha)^2+y^2=\beta^2$, $(x-\alpha')^2+y^2=\beta'^2$, and their axial orthotomic $(x-m)^2+y^2=k^2$, then we have $(m-\alpha)^2=\beta^2+k^2$, $(m-\alpha')^2=\beta'^2+k^2$; subtracting, it appears that m is real; and then if either β^2 or β'^2 is negative, the equation containing this quantity shows that k^2 is positive; viz. the circle $(x-m)^2+y^2=k^2$ is real.

The above remarks have an obvious application to the theory of bicircular quartics; viz. a bicircular quartic is the envelope of a variable circle, having its centre on a conic, and orthotomic to a circle: it may be that this circle is a pure imaginary. We then replace it by the concentric orthotomic, and say that the curve is the envelope of a variable circle having its centre on a conic and bisecting a circle. We have thus a real form for cases which originally present themselves under an imaginary form.

The Bicircular Quartic with given vertices.

First, if the vertices are real; let the vertices taken in order be F, G, H, K.

First construction: On HK as diameter describe a circle, and on FG as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial orthotomic circle Σ_1 of the two circles (viz. the centre of Σ_1 is on the axis of symmetry at its intersection with the radical axis of the two circles); then the curve is the envelope of a variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Second construction: On FH as diameter describe a circle, and on GK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_2, and describe the axial bifid circle Σ_2' of the two circles (viz. the centre of Σ_2' is on the axis of symmetry at its intersection with the radical axis or common chord of the two circles, and its diameter is this common chord); then the curve is the envelope of a variable circle having its centre on Θ_2 and bisecting Σ_2'.

Third construction: On FK as diameter describe a circle, and on GH as diameter a circle; and then, as in the first construction, a conic Θ_3 and a circle Σ_3; the curve is the envelope of a variable circle having its centre on Θ_3 and orthotomic to Σ_3.

Observe that in the three constructions the conics have always the same centre; and if the three conics are taken with the same foci, then the three constructions give one and the same bicircular quartic. The first and third constructions form a pair, and there is no reason for selecting one of them in preference to the other; but the second construction is unique; it is on this account natural to make use of it in discussing the series of curves with the given vertices.

In the particular case where the points F, G and H, K are situate symmetrically on opposite sides of a centre O ($OF = OK$, $OG = OH$), then in the third construction the centres each coincide with O, or the axis of the conic vanishes; hence the construction fails: the first and second constructions hold good, and in each of them the

[p. 15] conic and circle are concentric. The curve is in this case quadrantal: having, besides the original axis of symmetry, another axis of symmetry through O, at right angles thereto.

Secondly, if the vertices are two real, two imaginary, say f, $g = \alpha \pm \beta i$; h, k, we modify the first or third construction; viz. if F', G' are the antipoints of F, G; then on $F'G'$ as diameter describe a circle, and on HK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial bisector-orthotomic circle Σ_1 of the two circles; viz. this is the circle (centre on the axis of symmetry) which bisects the circle $F'G'$, and cuts at right angles the circle HK; then the curve is the envelope of the variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Thirdly, if the vertices are all imaginary, say f, $g = \alpha \pm \beta i$; h, $k = \gamma \pm \delta i$, we modify the first or third construction. Take F', G' the antipoints of F, G, and H', K' the antipoints of H, K; then on $F'G'$ as diameter describe a circle, and on $H'K'$ as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ, and describe the axial bisector-circle Σ of the two circles (viz. this is a circle, centre on the axis of symmetry, bisecting each of the circles); the curve is the envelope of a variable circle, centre on the conic Θ and cutting at right angles the circle Σ.

558. A Geometrical Interpretation of the Equations Obtained by Equating to Zero the Resultant and the Discriminants of Two Binary Quantics

[p. 16] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 31–33. Read March 12, 1874.]*

CONSIDER the equations

$$U = (a,\ b,\ \ldots \mid t,\ 1)^{\lambda} = 0,$$

$$U' = (a',\ b',\ \ldots \mid t,\ 1)^{\lambda'} = 0\,;$$

and equating to zero the discriminants of the two functions respectively, and also the resultant of the two functions, let the equations thus obtained be

$$\Delta = (a,\ b,\ \ldots)^{2\lambda-2} = 0,$$

$$\Delta' = (a',\ b',\ \ldots)^{2\lambda'-2} = 0,$$

$$R = (a,\ b,\ \ldots)^{\lambda'}(a',\ b',\ \ldots)^{\lambda} = 0.$$

I take $(a,\ b,\ \ldots)$, $(a',\ b',\ \ldots)$ to be linear functions of the coordinates $(x,\ y,\ z)$; and t to be an indeterminate parameter. Hence $U = 0$ represents a line the envelope whereof is the curve $\Delta = 0$, or, what is the same thing, the equation $U = 0$ represents any tangent of the curve $\Delta = 0$; this is a unicursal curve of the order $2\lambda - 2$ and class λ, with $3(\lambda - 2)$ cusps and $\frac{1}{2}(\lambda - 2)(\lambda - 3)$ nodes. Similarly $U' = 0$ represents a line the envelope of which is the curve $\Delta' = 0$: this is a unicursal curve of the order $2\lambda' - 2$ and class λ', with $3(\lambda' - 2)$ cusps and $\frac{1}{2}(\lambda' - 2)(\lambda' - 3)$ nodes; the equation $U' = 0$ represents any tangent of this curve.

The equations $U = 0$, $U' = 0$ considered as existing simultaneously with the same value of t, establish a $(1,\ 1)$ correspondence between the tangents (or if we please, between the points) of the two curves. The locus of the intersection of the corresponding tangents is the curve $R = 0$, a unicursal curve of the order $\lambda + \lambda'$, with $\frac{1}{2}(\lambda + \lambda' - 1)(\lambda + \lambda' - 2)$ nodes and no cusps; consequently of the class $2(\lambda + \lambda' - 1)$.

[p. 17] It is to be shown that the curve $R = 0$ touches the curve $\Delta = 0$ in $\lambda' + 2\lambda - 2$ points, and similarly the curve $\Delta' = 0$ in $2\lambda' + \lambda - 2$ points.

In fact, consider any tangent T' of the curve Δ'; let this meet the curve Δ in a point A, and let Q be the tangent at A to the curve Δ; suppose, moreover, that T is the tangent of Δ corresponding to the tangent T' of Δ'. Then if Q and T coincide, the corresponding tangent of T' will be Q, and the curve R will pass through A. It is easy to see that in this case the curves R, Δ will touch at A.

Again, if P be a tangent from A to the curve Δ, then, if T and P coincide, the corresponding tangent of T' will be P, and the curve R will pass through A; but in this case the point A will be a mere intersection, not a point of contact, of the two curves.

The tangents T, Q each correspond to T', and they consequently correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, or, what is the same thing, of Q; that is, for a given position of T we have $2\lambda-2$ positions of Q. Again, to a given position of Q corresponds a single position of A, therefore λ' positions of T', therefore also λ' positions of T; that is, for a given position of Q we have λ' positions of T. The correspondence between T, Q is thus a $(\lambda',\ 2\lambda-2)$ correspondence, and the number of united tangents is therefore $\lambda'+2\lambda-2$, or the curves R, Δ touch in $\lambda'+2\lambda-2$ points.

The tangents T, P each correspond to T', and they therefore correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, and thence $(2\lambda-2)(\lambda-2)$ positions of P; that is, for a given position of T we have $(2\lambda-2)(\lambda-2)$ positions of P. Again, to a given position of P correspond $2\lambda-4$ positions of A, therefore $(2\lambda-4)\lambda'$ positions of T' or of T; that is, for a given position of P we have $(2\lambda-4)\lambda'$ positions of T. The correspondence between T, P is thus a $[2\lambda'(\lambda-2),\ 2(\lambda-1)(\lambda-2)]$ correspondence, and the number of united tangents is $2(\lambda+\lambda'-1)(\lambda-2)$; or the curves R, Δ meet in $2(\lambda+\lambda'-1)(\lambda-2)$ points.

Reckoning the contacts twice, the total number of intersections of R, Δ is

$$2\lambda'+4\lambda-4+2(\lambda+\lambda'-1)(\lambda-2)=(\lambda+\lambda')(2\lambda-2),$$

as it should be.

In the particular case $\lambda=\lambda'=2$, the curves Δ, Δ' are conics, and the curve R is a quartic curve touching each of the conics 4 times; this is at once verified, since the equations here are

$$ac-b^2=0,\quad a'c'-b'^2=0,\quad 4(ac-b^2)(a'c'-b'^2)-(ac'+a'c-2bb')^2=0.$$

Arthur Cayley

Collected Mathematical Papers

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Transcribed from the Original Publications

Contents

561. On the Geometrical Representation of Cauchy's Theorems of Root-Limitation (continued)

[Continued from earlier pages of the same paper. Articles 27 (end) through 38, and Memoir list.]

[p. 31] (continuation of the table of intercalations from the previous page):

Interval $0-4$	for P gain $=0$,	… for Q gain $=-1$;	Intercalation is	Q;
$0-4$	" $=-1$, Q first,	" $=-1$;	"	QP;
$2-4$	" $=1$, P first,			
$2-4$	" $=-1$, P first,	" $=0$;	"	P.

28. Or to take a slightly more complicated example,

		1	3	5	7	9
$P=x^2-8x+12$	$+$	$-$	$-$	$+$	$+$	$+$
$Q=x^2-12x+32$	$+$	$+$	$-$	$-$	$-$	$+$
$R=-x+5$	$+$	$+$	$+$	$+$	$-$	$-$
$S=+1$	$+$	$+$	$+$	$+$	$+$	$+$
Showing P	P		Q		P	
	1	2	3	4		

And hence for the several intervals,

	$1-3$	$1-5$	$1-7$	$1-9$	$3-5$	$3-7$	$3-9$	$5-7$	$5-9$	$7-9$
	$--$	$-+$	$-+$	$--$	$-+$	$-+$	$--$	$--$	$-+$	$-+$
	$+-$	$+-$	$++$	$++$	$+-$	$+-$	$+-$	$++$	$+-$	$++$
	$++$	$+-$	$++$	$+-$	$+-$	$++$	$++$	$+-$	$+-$	$+-$
	$++$	$++$	$++$	$++$	$++$	$++$	$++$	$++$	$++$	$++$
Showing P	PQ	PQP	$PQPQ$	Q	QP	QPQ	P	PQ	Q	

For instance:

Interval $1-9$ for P gain $=2$, P first, for Q gain $=2$: Intercalation is $PQPQ$.

It may be added that P being $+$ for $x=1$, the $\pm$ intercalation is $+PQPQ$.

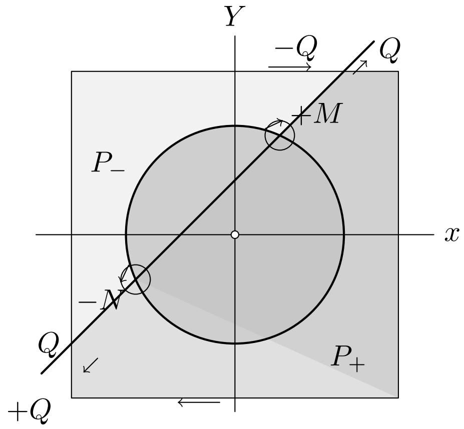
29. As an example of circuits take the following: curves are $P=0$, $Q=0$, where

$$P=x^2+y^2-4,$$

$$Q=y-x-1;$$

viz. $P=0$ (see figure) is a circle, centre the origin, radius $=2$: the inside hereof ($P=-$) being coloured red: and $Q=0$ is a right line cutting the axes of x, y at the points $(-1,0)$ and $(0,1)$ respectively, or say running N.E. and S.W., the lower region ($Q=-$) being coloured blue: the square is an arbitrary circuit ($x=\pm 3$, $y=\pm 3$) surrounding the circle, and the regions within the square are coloured by what precedes sable, gules, azure, argent, as shown in the figure: the line and circle intersect in two points M, N. Going right-handedly round these respectively, for M the order is sable, gules, argent, azure, viz. M is a right-handed root; while for N the order is

[p. 32] sable, azure, argent, gules, viz. N is a left-handed root: the two points are accordingly in the figure denoted by $+M$ and $-N$ respectively.



30. Now considering successively the four smaller squares of the figure, say these are the squares N.E., S.E., S.W., N.W.: and going right-handedly round each of these—

In the square N.E., the sequence and therefore also the intercalation is $+P-Q+P-Q$, viz. this is an intercalation $(+P-Q)_1$, showing an excess 1 of right-handed roots, and of course consisting with the single right-handed root M.

In the square S.E., the sequence is $-P+P$, viz. this is an intercalation $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with no root.

In the square S.W., the sequence and therefore also the intercalation is $-P+Q-P+Q$: viz. this is an intercalation $(-P+Q)_1$, showing an excess 1 of left-handed roots, and consisting with the single left-handed root N.

And in the square N.W., the sequence is $-Q+P-P+Q$, viz. this is an intercalation $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with no root.

Again take the whole large square: the sequence is $-Q+Q$: viz. the intercalation is $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with there being one of each.

So taking the squares N.E. and N.W. conjointly, the sequence and therefore also the intercalation is $-Q+P-Q+P$, viz. this is an intercalation $(+P-Q)_1$, as for the single square N.E.

[p. 33] **31.** As regards the analytical determination it will be sufficient to consider a single square, say N.E.: going round right-handedly, the trajectories will be

$$
\begin{aligned}
&(1)\ x=0,\ \ y=0 \text{ to } 3\\
&(2)\ y=3,\ \ x=0 \text{ to } 3\\
&(3)\ x=3,\ \ y=3 \text{ to } 0\,;\ \text{ or if } y'=-y,\ \text{ then } y'=-3 \text{ to } 0\,;\\
&(4)\ y=0,\ \ x=3 \text{ to } 0\,;\ \text{ or if } x'=-x,\ \text{ then } x'=-3 \text{ to } 0.
\end{aligned}
$$

And we thus have

		0	3	
(1) $P=y^2-4$	$-$	$+$	$+$	that is, $-\,+$ for P gain $=-1$, Q begins, $1P$
$Q=y-1$	$-$	$-$	$+$	$Q\ \ =-1,\ 1Q.$
$R=-1$	$-$	$-$	$-$	

Intercalation is QP, or since at origin $P=-$, $R=-$, or region is sable, it is $-Q+P$.

		0	3	
(2) $P=x^2+5$	$+$	$+$	$+$	that is, $-\,+$ for P gain $=0$,
$Q=-x+2$	$+$	$+$	$-$	$Q\ \ =-1.$
$R=-1$	$-$	$-$	$-$	

Intercalation is $-Q$.

		-3	-3	
(3) $P=y'^2+5$	$+$	$+$	$+$	that is, $-\,-$ for P gain $=0$,
$Q=-y'+4$	$+$	$+$	$+$	$Q\ \ =0.$
$R=-1$	$-$	$-$	$-$	

Intercalation vanishes.

		-3	-3		
(4) $P = x'^2 - 4$	$+$	$-$	$-$	that is, $- \, -$	for P gain $= +1$, P first,
$Q = x' - 1$	$-$	$-$	$-$		$Q \quad = 0$.

Intercalation is $+P$.

Hence for the four sides, combining the intercalations, we have $-Q + P - Q + P$, and since there are no terms to be omitted, this is the intercalation of the N.E. square: which is right.

C. IX.

[p. 34]

The Rhizic Theory. Articles Nos. 32 to 38.

32. Consider now $F(z) = (*)(z, 1)^n$ a rational and integral function of z, of the order n with in general imaginary (complex) coefficients, or, what is the same thing, let $F(z) = f(z) + i\phi(z)$, where the functions f, ϕ are real[1]. Writing herein $z = x + iy$, let P, Q be the real part and the coefficient of the imaginary part in the function $F(x + iy)$: or, what is the same thing, assume

$$P + iQ = f(x + iy) + i\phi(x + iy),$$

then it is clear that to any root $\alpha + i\beta$ (real or imaginary) of the equation $F(z) = 0$, there corresponds a real intersection, or root, $x = \alpha$, $y = \beta$, of the curves $P = 0$, $Q = 0$. The functions P, Q, as thus serving for the determination of the roots of the equation $F(z) = 0$, are termed "rhizic functions," and similarly the curves $P = 0$, $Q = 0$ are "rhizic curves." The assumed equation shows at once that we have

$$d_y(P + iQ) = i\, d_x(P + iQ),$$

or, what is the same thing,

$$d_yP = -d_xQ, \qquad d_xP = d_yQ.$$

And we hence see that

$$\frac{d(P, Q)}{d(x, y)} = (d_xP)^2 + (d_yP)^2, \text{ or } (d_xQ)^2 + (d_yQ)^2,$$

is positive: viz. that the roots $P = 0$, $Q = 0$ are all of them right-handed (the essential thing is that they are same-handed; for by reversing the signs of P and Q they might be made left-handed: but it is convenient to take them as right-handed)—hence the theorem which in the general case, where P and Q are arbitrary functions, serves to determine the difference of the numbers of the right- and left-handed roots, in the particular case, where P and Q are rhizic functions, serves to determine the number of intersections of the curves $P = 0$, $Q = 0$: or, what is the same thing, the number of the (real or imaginary) roots of the equation $F(z) = 0$: viz. we thus determine the number of roots within a given circuit.

33. The rhizic curves $P = 0$, $Q = 0$ have various properties. 1°. Each curve has n real points at infinity, or, what is the same thing, n real asymptotes: and the P and Q points at infinity succeed each other, a P-point and then a Q-point, and so on, alternately.

In fact, from the equation

$$P + iQ = (a' + ia'')(x + iy)^n + \ldots + (k' + k''i),$$

writing herein $a' + ia'' = \alpha(\cos a + i \sin a)$, and $x + iy = \rho(\cos\theta + i\sin\theta)$, we have

$$P + iQ = \alpha\rho^n \left[\cos(n\theta + a) + i\sin(n\theta + a)\right] + \ldots + k' + k''i.$$

[1] It is assumed that the equation $F(z) = 0$ has no equal roots: this being so, the curves $P = 0$, $Q = 0$, will have no point of multiple intersection; which accords with the assumption made in the general case of two arbitrary curves.

[p. 35] It thus appears that for the curve $P = 0$, the points at infinity are given by the equation $\cos(n\theta + a) = 0$; while for the curve $Q = 0$, they are given by the equation $\sin(n\theta + a) = 0$: which proves the theorem.

Representing infinity as a closed curve or circuit, each point at infinity must be represented by two opposite points on the circuit; so that writing down P for each P-point and Q for each Q-point, we have $2n$ P's and $2n$ Q's succeeding each other, a P-point and then a Q-point, and so on, alternately.

It may be assumed that taking the circuit right-handedly, the P's are $+$ and the Q's $-$, (this depends only on the colouring, but it corresponds with the foregoing assumption that the roots $P = 0$, $Q = 0$ are right-handed): the theorem just obtained then really is that for the circuit infinity, the intercalation is $(+P - Q)_n$: and we have herein a proof of the theorem that a numerical equation of the order n with real or imaginary coefficients has precisely n real or imaginary roots. But the force of this will more distinctly appear presently.

34. 2°. Neither of the curves $P = 0$, $Q = 0$ can include as part of itself a closed curve or circuit.

The foregoing relations between the differential coefficients give

$$d_x^2 P + d_y^2 P = 0, \qquad d_x^2 Q + d_y^2 Q = 0,$$

which equations for the two curves respectively lead to the theorem in question. For as regards the curve $P = 0$, take z a coordinate perpendicular to the plane of xy, and consider the surface $z = P$: if the curve $P = 0$ included as part of itself a closed curve, then corresponding to some point (x, y) within the curve we should have z a proper maximum or minimum, viz. there would be a summit or an imit; at the point in question we should have $d_x P = 0$, $d_y P = 0$; and also (as the condition of a summit or imit) $d_x^2 P \cdot d_y^2 P - (d_x d_y P)^2 = +$, implying that $d_x^2 P$ and $d_y^2 P$ have at this point the same sign: but this is inconsistent with the foregoing relation $d_x^2 P + d_y^2 P = 0$.

35. 3°. The curves $P = 0$, $Q = 0$ have not in general any double (or higher multiple) points. A point which is a double (or higher multiple) point on one of these curves is not of necessity a point on the other curve: but being a point on the other curve it is on that curve a point of the same multiplicity. For changing if necessary the coordinates, the point in question may be taken to be at the origin: forming the equation

$$P + iQ = (a' + a''i)(x + iy)^n + \ldots + (l' + l''i)(x + iy)^2 + (m' + m''i)(x + iy) = 0,$$

the point $x = 0$, $y = 0$ will not be a double point on the curve $P = 0$, unless we have $m' = 0$, $l' = 0$, $l'' = 0$; these conditions being satisfied, it will not be a point on the curve $Q = 0$ unless also $m'' = 0$; but this being so, it will be a double point on the curve $Q = 0$: and the like for points of higher multiplicity. But a point which is a multiple point on each curve, represents four or more coincident intersections of the curves $P = 0$, $Q = 0$, that is, four or more equal roots of the equation $F(z) = 0$; so that assuming that the equation has no equal roots, the case does not arise: and we in fact exclude it from consideration.

[p. 36] To fix the ideas assume that the curves $P = 0$, $Q = 0$ are each of them without double points. As already seen, neither of them includes as part of itself a closed curve. Hence in the figure the curve $P = 0$ must consist of n branches each drawn from a point P in the circuit (viz. the circuit infinity) to another point P in the circuit; and in such manner that no two branches intersect each other: this implies that the two points P of the same branch must include between them an even number (which may of course be $= 0$) of points P. And the like as regards the curve $Q = 0$.

36. 4°. No branch of the P-curve can meet a branch of the Q-curve more than once. In fact, in drawing the two branches to meet twice, the colouring would at once show that of the two intersections or roots, one must be right, the other left-handed; whence, the roots being all right-handed, the branches do not meet twice. And in exactly the same way it appears that no P-branch can meet two Q-branches, or any Q-branch meet two P-branches. And under these restrictions it requires only a consideration of a few successive cases to show that the n P-branches, and the n Q-branches can only be drawn on the condition that each P-branch shall intersect once and only once a single Q-branch; which of course implies that each Q-branch intersects once and once only a single P-branch: and further, that there shall be precisely n intersections: viz. the n P-branches and the n Q-branches must satisfy the conditions just stated. And the theorem of the n roots is thus obtained as a consequence of the impossibility (except under the same conditions) of drawing the n P-branches and the n Q-branches, so as to give rise to right-handed roots only. But the case of double or higher multiple points would need to be specially considered.

37. It is interesting for a given value of n to consider $\phi(n)$, the number of different ways in which the P-branches and the Q-branches can be drawn. We have $2n$ points P and $2n$ points Q, in all $4n$ points: starting from any point P, these may be numbered in order $1, 2, 3, \ldots, 4n$, the points P bearing odd numbers and the points Q even numbers. We may consider the P-branch which joins 1 with some P-point β, and (intersecting this) the Q-branch which joins some two Q-points α and γ: the numbers $1, \alpha, \beta, \gamma$ are then in order of increasing magnitude: and excluding these four points there remain the points corresponding to numbers between 1 and α, between α and β, between β and γ, and between γ and 1. Now since the P-branch 1β meets the Q-branch $\alpha\gamma$, no branch from a point between 1 and α can meet either of these curves; hence these points form a system by themselves, capable of being connected together by P-branches and Q-branches: the number of them must therefore be a multiple of 4: and the like as to the points between α and β, between β and γ, and between γ and 1. Taking the number of the points in the four systems to be $4x$, $4y$, $4z$, and $4w$ respectively, we have $x+y+z+w = n-1$, and the first-mentioned four points bear the numbers

$$\begin{aligned} &1, \\ &\alpha = 4x + 2, \\ &\beta = 4x + 4y + 3, \\ &\gamma = 4x + 4y + 4z + 4. \end{aligned}$$

[p. 37] For the four systems the number of ways of drawing the P- and Q-branches are ϕx, ϕy, ϕz, ϕw respectively: that is, x, y, z, w being any partition whatever of $n-1$ (order attended to), and $\phi(0)$ being $= 1$, we have

$$\phi(n) = \Sigma\, \phi(x)\, \phi(y)\, \phi(z)\, \phi(w),$$

which is the condition for the determination of ϕn.

Taking then θ for the value of the generating function

$$1 + t\phi(1) + t^2\phi(2) + \ldots + t^n\phi(n) + \ldots,$$

it hereby appears that we have

$$\theta = 1 + t\theta^4;$$

or writing this for a moment $\theta = u + t\theta^4$, and expanding by Lagrange's theorem, but putting finally $u = 1$, we have the value of θ, that is of the generating function,

$$= 1 + [4]^0\,\frac{t}{1} + [8]^1\,\frac{t^2}{1\cdot 2} + [12]^2\,\frac{t^3}{1\cdot 2\cdot 3} + \ldots + [4n]^{n-1}\,\frac{t^n}{1\cdot 2\cdot \ldots \cdot n} + \ldots$$

$$= 1 + t + 4t^2 + 22t^3 + 140t^4 + \ldots,$$

that is,

$$\phi(1) = 1, \quad \phi(2) = 4, \quad \phi(3) = 22, \quad \phi(4) = 140, \ldots$$

and generally

$$\phi(n) = \frac{[4n]^{n-1}}{[n]^n} = \frac{4n\cdot 4n-1\cdot \ldots \cdot 3n+2}{2\cdot 3\cdot \ldots \cdot n}.$$

The results are easily verified for the successive particular cases; thus $n = 1$, the points are $1, 2, 3, 4$, and the P- and Q-branches respectively are 13, 24: $\phi(1) = 1$. Again $n = 2$, the points are $1, 2, 3, 4, 5, 6, 7, 8$: we may join $13, 24$ or $13, 28$ or $17, 28$ or $17, 68$, leaving in each case four contiguous numbers which may be joined in a single manner: that is, $\phi(2) = 4$. Or, what is the same thing, the partitions of 1 are $0001, 0010, 0100, 1000$, whence $\phi(2) = 4\{\phi(0)\}^3\phi(1) = 4$. Again $n = 3$, the partitions of 2 are 0002, &c. (4 of this form) and 1100 (6 of this form): that is, $\phi(3) = 4\{\phi(0)\}^3\phi(2) + 6\{\phi(0)\}^2\{\phi(1)\}^2 = 4\cdot 4 + 6\cdot 1 = 22$, and so on.

38. Starting from the $4n$ points P and Q, and joining them in any manner subject to the foregoing conditions, we have a diagram representing two rhizic curves: and colouring the regions we verify that the n roots are all of them right-handed. We have for instance the annexed figure ($n = 3$).

[Figure: a circular diagram with $4n = 12$ alternating P- and Q-points around the boundary; three P-branches and three Q-branches drawn inside, each P-branch crossing exactly one Q-branch, producing three right-handed intersections.]

Having drawn such a figure we may, by a continuous variation of the several lines, in a variety of ways introduce a double point in the P-curve, or in the Q-curve: and by a continued repetition of the process introduce double points in each or either curve: thus for instance we may from the last figure derive a new figure in which the P-curve has a node at N. It will be observed that here it is no longer the case that each P-branch intersects one and only one Q-branch: the P-branch $1-9$ does not meet any Q-branch, but the P-branch $7-11$ meets two Q-branches. But looking at the figure in a different manner, and considering the P-branches through N as

[p. 38] being either $11-N-1$ and $7-N-9$, or $1-N-7$ and $9-N-11$, then in either case each P-branch intersects one and only one Q-branch: and in this way, in a

[Figure: a second circular diagram in which the P-curve has been deformed so as to acquire a node at the interior point N, with the P-branches passing through N shown in two pairings.]

[Figure: a third related diagram in the same style, illustrating a further continuous variation in which the curves acquire additional double points.]

diagram in which the two curves have each or either of them double points, but neither curve passes through a double point of the other curve, the theorem may be regarded as remaining true—we in fact consider the diagram as the limit of a diagram wherein the curves have no double points. It will be recollected that, the equation $F(z)$ being without equal roots, we cannot have either curve passing through a multiple point of the other curve. And we thus see that the various figures drawn as above without double points are, so to speak, the types of all the different forms of a system of rhizic curves $P=0$, $Q=0$.

[p. 39] In connexion with the present paper I give the following list of Memoirs:

Cauchy. Calcul des Indices des fonctions. *Jour. de l'École Polyt. t.* XV. *(1837), pp. 176–229.*—First part seems to have been written in 1833: second part is dated 20th June, 1837. Refers to a memoir presented to the Academy of Turin the 17th Nov. 1831, wherein the principles of the "Calcul des Indices des fonctions" are deduced from the theory of definite integrals: I have not seen this.

Sturm and Liouville. Démonstration d'un théorème de M. Cauchy relatif aux racines imaginaires des équations. *Liouv. t.* I. *(1836), pp. 278–289.*

Sturm. Autres démonstrations du même théorème. *Liouv. t.* I. *(1836), pp. 290–308.*

These two papers contain proofs of the particular theorem relating to the roots of an equation $F(z)=0$, but do not refer to the general theorem relating to the intersection of the two curves $P=0$, $Q=0$: the special theorem of the existence of the n roots of the equation $F(z)=0$ is considered.

Sylvester. A theory of the syzygetic relations of two rational integral functions, comprising an application to the theory of Sturm's functions and that of the greatest algebraical common measure. *Phil. Trans. t.* CXLIII. *(1853), pp. 407–548.*

De Morgan. A proof of the existence of a root in every algebraic equation, with an examination and extension of Cauchy's theorem on imaginary roots, and remarks on the proofs of the existence of roots given by Argand and Mourey. *Camb. Phil. Trans. t.* X. *(1858), pp. 261–270.*

Contains the important remark that the two curves $P=0$, $Q=0$ are such that two branches, one of each curve, cannot inclose a space; also that the two curves always [i.e. at a simple intersection] intersect orthogonally.

Airy, G. B. Suggestion of a proof of the theorem that every algebraic equation has a root. *Camb. Phil. Trans. t.* X. *(1859), pp. 283–289.*

Cayley, A. Sketch of a proof of the theorem that every algebraic equation has a root. *Phil. Mag. t.* XVIII. *(1859), [248], pp. 436–439.*

Walton, W. On a theorem in maxima and minima. *Quart. Math. Jour. t.* X. *(1870), pp. 253–262.* **Cayley, A.** Addition thereto, [562], pp. 262, 263. (Relates to the curves $P=0$, $Q=0$.)

Walton, W. Note on rhizic curves. *Quart. Math. Jour. t.* XI. *(1871), pp. 91–98.* First use of the term "rhizic curves": relates chiefly to the configuration of each curve at a multiple point, and of the two at a common multiple point.

Walton, W. On the spoke-asymptotes of rhizic curves. *Quart. Math. Jour. t.* XI. *(1871), pp. 200–202.*

Walton, W. On a property of the curvature of rhizic curves at multiple points. *Quart. Math. Jour. t.* XI. *(1871), pp. 274–281.*

Björling. Sur la séparation des racines d'équations algébriques. *Upsala, Nova Acta Soc. Sci. (1870), pp. 1–35.* (Contains delineations of some rhizic curves.)

562. [Addition to Mr Walton's Paper "On a Theorem in Maxima and Minima."]

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. *(1870), pp. 262, 263.]*

[p. 40] In what follows I write x, y, z in place of Mr Walton's u, v, w: (so that if $i = \surd(-1)$, as usual, we have

$$f(x+iy) = P + iQ);$$

and I attend exclusively to the case where the second differential coefficients of P, Q do not vanish.

There are not on the surface $z = P$ any proper maxima or minima, but only level points, such as at the top of a pass: say there are not any summits or imits, but only cruxes; and moreover at any crux, the two crucial (or level) directions intersect at right angles. Every node of the curve $Q = 0$ is subjacent to a crux of the surface $z = P$: and moreover the two directions of the curve $Q = 0$ at the node are at right angles to each other; hence, considering the intersection of the surface $z = P$ by the cylinder $Q = 0$, the path $Q = 0$ on the surface has a node at the crux; or say there are at the crux two directions of the path; these cross at right angles, and are consequently separated the one from the other by the crucial directions; that is to say, there is one path ascending, and another path descending, each way from the crux. And the complete statement is: that the elevation of the path is then only a maximum or minimum when the path passes through a crux; and that at any crux there are two paths, one ascending, the other descending, each way from the crux.

The analytical demonstration is exceeding simple; we have

$$\left(\frac{dP}{dy} + i\frac{dQ}{dy}\right) = i\left(\frac{dP}{dx} + i\frac{dQ}{dx}\right);$$

[p. 41] that is,

$$\frac{dP}{dy} = -\frac{dQ}{dx}, \qquad \frac{dQ}{dy} = \frac{dP}{dx},$$

and passing thence to the second differential coefficients, we may write

$$\frac{dP}{dx} = \frac{dQ}{dy} = L, \qquad \frac{dP}{dy} = -\frac{dQ}{dx} = M,$$

$$\frac{d^2P}{dxdy} = -\frac{d^2Q}{dx^2} = \frac{d^2Q}{dy^2} = a,$$

$$\frac{d^2Q}{dxdy} = \frac{d^2P}{dx^2} = -\frac{d^2P}{dy^2} = b,$$

so that we have

$$\delta P = L\,\delta x + M\,\delta y, \qquad \delta Q = -M\,\delta x + L\,\delta y,$$

$$\delta^2 P = (b,\ a,\ -b \mid \delta x,\ \delta y)^2, \qquad \delta^2 Q = (-a,\ b,\ a \mid \delta x,\ \delta y)^2.$$

Hence, for the maximum or minimum elevation of the path, we have $0 = \delta P$, where $\delta Q = 0$; that is, $0 = -\frac{L^2+M^2}{L}\delta x$, and therefore $L^2 + M^2 = 0$; that is, $L = 0$, $M = 0$; and at any such point $\delta z = 0$, that is, there is a crux of the surface $z = P$; and $\delta Q = 0$, that is, there is a node of the curve $Q = 0$. Moreover the crucial directions for the surface $z = P$ are given by the equation $(b,\ a,\ -b \mid \delta x,\ \delta y)^2 = 0$, or these are at right angles to each other; and the nodal directions for the curve $Q = 0$ are given by $(-a,\ b,\ a \mid \delta x,\ \delta y)^2 = 0$, or these are likewise at right angles to each other.

C. IX.

563. Note on the Transformation of Two Simultaneous Equations

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 266, 267.]

[p. 42] Writing in Mr Walton's equations (1) and (2)

$$\frac{a}{\delta},\ \frac{b}{\delta},\ \frac{c}{\delta},\ \frac{\alpha}{\delta},\ \frac{\beta}{\delta},\ \frac{\gamma}{\delta}$$

instead of a, b, c, α, β, γ respectively; and putting for shortness

$$\begin{aligned} A &= b\gamma - c\beta, & F &= a\delta - d\alpha, \\ B &= c\alpha - a\gamma, & G &= b\delta - d\beta, \\ C &= a\beta - b\alpha, & H &= c\delta - d\gamma, \end{aligned}$$

the equations become

$$\frac{a(b-c)}{F} + \frac{b(c-a)}{G} + \frac{c(a-b)}{H} = 0,$$
$$\frac{\alpha(\beta-\gamma)}{F} + \frac{\beta(\gamma-\alpha)}{G} + \frac{\gamma(\alpha-\beta)}{H} = 0.$$

Multiplying by FGH and effecting some obvious transformations, the equations become

$$aAF + bBG + cCH = 0, \qquad \alpha AF + \beta BG + \gamma CH = 0 \tag{18}$$

whence also

$$AF^2 + BG^2 + CH^2 = 0. \tag{19}$$

Now regarding $(\alpha, \beta, \gamma, \delta)$ as the coordinates of a point in space, the equations (18) and (19) represent each of them a cone having for vertex the point $\alpha : \beta : \gamma : \delta = a : b : c : d$, viz. (18) is a quadric cone, (19) a cubic cone; they intersect therefore in six lines; and it may be shown that these are

the line	$\alpha : \beta : \gamma = a : b : c$	(twice) 2
"	$\beta : \gamma : \delta = b : c : d$	1
"	$\gamma : \alpha : \delta = c : a : d$	1
"	$\alpha : \beta : \delta = a : b : d$	1
"	$\beta - \gamma : \gamma - \alpha : \alpha - \beta : \delta = b - c : c - a : a - b : d$	1

agreeing with Mr Walton's result.

564. On a Theorem in Elimination

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 5, 6.]

[p. 43] I find among my papers the following example of a theorem in elimination communicated to me by Prof. Sylvester. Writing

$$\begin{aligned} \phi &= ax^3 + 3bx^2y + 3cxy^2 + dy^3, \\ \phi_1 &= bx^2 + 2cxy + dy^2, \\ \phi_2 &= cx + dy, \\ \phi_3 &= d\,; \end{aligned}$$

$$\begin{aligned} f &= bx^3 + 3cx^2y + 3dxy^2 + ey^3, \\ f_1 &= cx^2 + 2dxy + ey^2, \\ f_2 &= dx + ey, \\ f_3 &= e\,; \end{aligned}$$

then we have

$$\Delta_a \cdot R(f, \phi) = \Delta f \cdot R(\phi_1, f_1)^2 \cdot R(\phi_2, f_2)^2,$$

viz. $R(f, \phi)$ is the resultant of the functions (f, ϕ), and similarly $R(\phi_1, f_1)$, $R(\phi_2, f_2)$. Moreover, Δf is the discriminant of f; and $\Delta_a R(f, \phi)$ is the discriminant of $R(f, \phi)$ in regard to a. The equation thus is

$$\begin{aligned}&\Delta_a\big[(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e - c^3 + 2bcd)^2\big]\\ &= (b^2e^2 + 4bd^3 + 4c^3e - 3c^2d^2 - 6bcde)^2\,(d^2 - 2cde + be^2)^2\end{aligned}$$

or, what is the same thing, reversing the order of the letters (a, b, c, d, e), it is

$$\begin{aligned}&\Delta_e\big[(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e - c^3 + 2bcd)^2\big]\\ &= (a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd)^2\,(b^2 - 2abc + a^2d)^2,\end{aligned}$$

[p. 44] viz. arranging in powers of e, the function is

$$\begin{aligned}&e^3 \cdot a^3\\ &+ 3e^2 \cdot -a^2(4bd - 3c^2) - 9(ac - b^2)^2\\ &+ 3e \cdot a(4bd - 3c^2)^2 + 18(ac - b^2)(ad^2 - 2bcd + c^3)\\ &+ 1 \cdot -(4bd - 3c^2)^3 - 27(ad^2 - 2bcd + c^3)^2,\end{aligned}$$

which last coefficient is

$$= -d^2(27a^2d^2 + 54ac^3 + 64b^3d - 36b^2c^2 - 108abcd),$$

and the discriminant of this cubic function of e is

$$= (a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd)^2\,(b^2 - 2abc + a^2d)^2.$$

The occurrence of the factor

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd$$

is accounted for as the resultant in regard to e of the invariants I, J; we, in fact, have

$$\begin{aligned}(ac - b^2)I - aJ &= -(ac - b^2)(-4bd + 3c^2) - a(-ad^2 - c^3 + 2bcd)\\ &= a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd,\end{aligned}$$

and the identity itself may be proved without any particular difficulty.

565. Note on the Cartesian

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 16–19.]

[p. 45] The following are doubtless known theorems, but the form of statement, and the demonstration of one of them, may be interesting.

A point P on a Cartesian has three "opposite" points on the curve, viz. if the axial foci are A, B, C, then the opposite points are P_a, P_b, P_c where

P_a is intersection of line	PA	with circle	PBC,
P_b "	PB	"	PCA,
P_c "	PC	"	PAB.

And, moreover, supposing in the three circles respectively, the diameters at right angles to PA, PB, PC are $\alpha\alpha'$, $\beta\beta'$, $\gamma\gamma'$ respectively, then the points α, α', β, β', γ, γ' lie by threes in two lines passing through P, viz. one of these, say $P\alpha\beta\gamma$, is the tangent, and the other $P\alpha'\beta'\gamma'$ the normal, at P; and then the tangents and normals at the opposite points are $P_a\alpha$ and $P_a\alpha'$, $P_b\beta$ and $P_b\beta'$, $P_c\gamma$, and $P_c\gamma'$ respectively.

There exists a second Cartesian with the same axial foci A, B, C, and passing through the points P, P_a, P_b, P_c (which are obviously opposite points in regard thereto); the tangent at P is $P\alpha'\beta'\gamma'$ and the normal is $P\alpha\beta\gamma$; and the tangent and the normal at the other points are $P_a\alpha'$ and $P_a\alpha$, $P_b\beta'$ and $P_b\beta$, $P_c\gamma'$ and $P_c\gamma$ respectively: viz. the two curves cut at right angles at each of the four points.

Starting with the foci A, B, C and the point P, the points P_a, P_b, P_c are constructed as above, without the employment of the Cartesian; there are through P with the foci A, B, C two and only two Cartesians; and if it is shown that these pass through one of the opposite points, say P_b, they must, it is clear, pass through

[p. 46] the other two points P_a, P_c. I propose to find the two Cartesians in question. To fix the ideas, let the points C, B, A be situate in order as shown in the figure, their distances from a fixed point O being a, b, c, so that writing α, β, $\gamma = b-c$, $c-a$, $a-b$ respectively, we have $\alpha+\beta+\gamma=0$, and α, γ will represent the positive distances CB and BA respectively, and $-\beta$ the positive distance AC. Suppose, moreover, that the distances PA, PB, PC regarded as positive are R, S, T respectively; and that the distances P_bA, P_bB, P_bC regarded as positive are R', S', T' respectively.

[Figure: a Cartesian curve with axial foci A, B, C marked along a horizontal line; points P and P_b shown on the curve with the various distances and the diameters $\alpha\alpha'$, $\beta\beta'$, $\gamma\gamma'$ of the three opposite-point circles indicated.]

Suppose that for a current point Q the distances QA, QB, QC regarded as indifferently positive, or negative, are r, s, t respectively; then the equation of a bicircular quartic having the points A, B, C for axial foci is

$$lr+ms+nt=0,$$

where l, m, n are constants; and this will be a Cartesian if only

$$\frac{l^2}{\alpha}+\frac{m^2}{\beta}+\frac{n^2}{\gamma}=0.$$

We have the same curve whatever be the signs of l, m, n, and hence making the curve pass through P, we may, without loss of generality, write

$$lR+mS+nT=0,$$

R, S, T denoting the positive distances PA, PB, PC as above. We have thus for the ratios $l:m:n$, two equations, one simple, the other quadric; and there are thus two systems of values, that is, two Cartesians with the foci A, B, C, and passing through P.

I proceed to show that for one of these we have $-lR'+mS'+nT'=0$, and for the other $lR'+mS'-nT'=0$, or, what is the same thing, that the values of $l:m:n$ are

$$l:m:n=-(ST'+S'T):TR'+T'R:RS'-R'S,$$

and

$$l:m:n=(ST'-S'T):-(TR'+T'R):RS'+R'S;$$

viz. that the equations of the two Cartesians are

$$\begin{vmatrix} r & s & t \\ R & S & T \\ -R' & S' & T' \end{vmatrix}=0, \qquad \text{and} \qquad \begin{vmatrix} r & s & t \\ R & S & T \\ R' & S' & -T' \end{vmatrix}=0,$$

[p. 47] respectively; this being so each of the Cartesians will, it is clear, pass through the point P_b, and therefore also through P_a and P_c.

The geometrical relations of the figure give

$$\begin{aligned} \alpha R^2 + \beta S^2 + \gamma T^2 &= -\alpha\beta\gamma, \\ \alpha R'^2 + \beta S'^2 + \gamma T'^2 &= -\alpha\beta\gamma, \\ RT' + R'T &= -\beta(S + S'), \\ \gamma\alpha &= SS', \\ \gamma\, TT' &= \alpha\, RR', \end{aligned}$$

to which might be joined

$$\begin{aligned} R'^2 S + \alpha\beta(S + S') + RS'^2 &= SS'(S + S'), \\ T'^2 S + \alpha\beta(S + S') + T^2 S' &= SS'(S + S'), \\ SRT' &= S'RT, \\ ST'R &= S'TR, \end{aligned}$$

but these are not required for the present purpose.

As regards the first Cartesian, we have to verify that

$$\frac{(ST' + S'T)^2}{\alpha} + \frac{(TR' + T'R)^2}{\beta} + \frac{(RS' - R'S)^2}{\gamma} = 0.$$

The left-hand side is

$$\frac{S^2T'^2 + S'^2T^2 + 2\gamma\alpha\, TT'}{\alpha} + \frac{\beta^2(S + S')^2 + 2\gamma\alpha\, [\textit{unreadable}]}{\beta} + \frac{S^2R'^2 + S'^2R^2 - 2\gamma\alpha\, RR'}{\gamma},$$

viz. this is

$$= S^2\left(\frac{T'^2}{\alpha} + \frac{[\textit{unreadable}]}{\beta} + \frac{R'^2}{\gamma}\right) + S'^2\left(\frac{T^2}{\alpha} + \frac{[\textit{unreadable}]}{\beta} + \frac{R^2}{\gamma}\right) + 2\alpha\beta\gamma + 2(\gamma TT' - \alpha RR'),$$

which is

$$= S^2\left(\frac{-\alpha\beta\gamma}{\alpha\gamma}\right) + S'^2\left(\frac{-\alpha\beta\gamma}{\alpha\gamma}\right) + 2\alpha\beta\gamma + 2(\gamma TT' - \alpha RR'),$$

and since the first and second terms are together $= -\dfrac{2\beta}{\gamma\alpha} S^2 S'^2$, that is, $= -2\alpha\beta\gamma$, the whole is as it should be $= 0$.

In precisely the same manner we have

$$\frac{(ST' - S'T)^2}{\alpha} + \frac{(TR' + T'R)^2}{\beta} + \frac{(RS' + R'S)^2}{\gamma} = 0,$$

which is the condition for the second Cartesian: and the theorem in question is thus proved.

566. On the Transformation of the Equation of a Surface to a Set of Chief Axes

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 34–38.]

[p. 48] We have at any point P of a surface a set of chief axes (PX, PY, PZ), viz. these are, say the axis of Z in the direction of the normal, and those of X, Y in the directions of the tangents to the two curves of curvature respectively. It may be required to transform the equation of the surface to the axes in question; to show how to effect this, take the original (rectangular) coordinates of the point P, (x, y, z) for the point P, $x + \delta x$, $y + \delta y$, $z + \delta z$ for the like coordinates of any other point on the surface, so that $(\delta x, \delta y, \delta z)$ are the coordinates of the point referred to the origin P; the equation of the surface, writing down only the terms of the first and second orders in the coordinates δx, δy, δz, is

$$A\,\delta x + B\,\delta y + C\,\delta z + \tfrac{1}{2}(a, b, c, f, g, h \mid \delta x, \delta y, \delta z)^2 + \&\text{c.} = 0,$$

where (A, B, C) are the first derived functions and (a, b, c, f, g, h) the second derived functions of U for the values (x, y, z) which belong to the given point P, if $U = 0$ is the equation of the surface in terms of the original coordinates (x, y, z); we have X, Y, Z linear functions of $(\delta x, \delta y, \delta z)$; say

$$\begin{array}{c|ccc} & \delta x & \delta y & \delta z \\ \hline X & \alpha_1 & \beta_1 & \gamma_1 \\ Y & \alpha_2 & \beta_2 & \gamma_2 \\ Z & \alpha & \beta & \gamma \end{array}$$

that is, $X = \alpha_1\delta x + \beta_1\delta y + \gamma_1\delta z$, &c. and $\delta x = \alpha_1 X + \alpha_2 Y + \alpha Z$, &c. where the coefficients satisfy the ordinary relations in the case of transformation between two sets of rectangular axes; and the transformed equation is therefore

$$A(\alpha_1 X + \alpha_2 Y + \alpha Z) + B(\beta_1 X + \beta_2 Y + \beta Z) + C(\gamma_1 X + \gamma_2 Y + \gamma Z)$$

$$+(a, b, c, f, g, h \mid \alpha_1 X + \alpha_2 Y + \alpha Z,\ \beta_1 X + \beta_2 Y + \beta Z,\ \gamma_1 X + \gamma_2 Y + \gamma Z)^2 = 0,$$

[p. 49] or, as this may be written,

$$\begin{aligned} &X(A\alpha_1 + B\beta_1 + C\gamma_1) + Y(A\alpha_2 + B\beta_2 + C\gamma_2) + Z(A\alpha + B\beta + C\gamma) \\ &\quad + \tfrac{1}{2}X^2\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 \\ &\quad + \tfrac{1}{2}Y^2\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2 \\ &\quad + XY\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) \\ &\quad + XZ\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) \\ &\quad + YZ\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) \\ &\quad + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 \quad + \text{\&c.} = 0, \end{aligned}$$

where the &c. refers to terms of the form $(X, Y, Z)^3$ and higher powers.

But in order that the new axes may be chief axes, we must have

$$A\alpha_1 + B\beta_1 + C\gamma_1 = 0, \qquad A\alpha_2 + B\beta_2 + C\gamma_2 = 0,$$

$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) = 0,$$

so that putting for shortness

$$A\alpha + B\beta + C\gamma = V,$$

the equation becomes

$$\begin{aligned} &VZ + \tfrac{1}{2}X^2\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 + \tfrac{1}{2}Y^2\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2 \\ &\quad + XZ\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) \\ &\quad + YZ\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) \\ &\quad + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 \quad + \text{\&c.} = 0. \end{aligned}$$

We have

$$A : B : C = \beta_1\gamma_2 - \beta_2\gamma_1 : \gamma_1\alpha_2 - \gamma_2\alpha_1 : \alpha_1\beta_2 - \alpha_2\beta_1,$$

that is,

$$A : B : C = \alpha : \beta : \gamma,$$

and thence

$$\alpha, \beta, \gamma = \frac{A}{V}, \frac{B}{V}, \frac{C}{V}, \qquad V = \sqrt{A^2 + B^2 + C^2}.$$

I write

$$\frac{1}{p_1} = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2,$$

and also for a moment

$$\frac{V}{p_1} = \mathcal{P}\,(\alpha_1, \beta_1, \gamma_1),$$
$$Q = (\,h,\ b,\ f\,)(\alpha_1, \beta_1, \gamma_1),$$
$$R = (\,g,\ f,\ c\,)(\alpha_1, \beta_1, \gamma_1).$$

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[p. 50] We find

$$P\alpha_1 + Q\beta_1 + R\gamma_1 = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 - \frac{V}{p_1} = 0,$$

$$P\alpha_2 + Q\beta_2 + R\gamma_2 = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) - \frac{V}{p_1}(\alpha_1\alpha_2 + \beta_1\beta_2 + \gamma_1\gamma_2) = 0,$$

and thence

$$P : Q : R = \beta_1\gamma_2 - \beta_2\gamma_1 : \gamma_1\alpha_2 - \gamma_2\alpha_1 : \alpha_1\beta_2 - \alpha_2\beta_1,$$
$$= \alpha : \beta : \gamma,$$

or say

$$P, Q, R = \theta_1 A,\ \theta_1 B,\ \theta_1 C\,;$$

we have thus the equations

$$\left(a - \frac{1}{p_1}\right)\alpha_1 + h\beta_1 + g\gamma_1 = \theta_1 A,$$

$$h\alpha_1 + \left(b - \frac{1}{p_1}\right)\beta_1 + f\gamma_1 = \theta_1 B,$$

$$g\alpha_1 + f\beta_1 + \left(c - \frac{1}{p_1}\right)\gamma_1 = \theta_1 C,$$

and joining hereto

$$(A, B, C)(\alpha_1, \beta_1, \gamma_1) = 0,$$

we eliminate α_1, β_1, γ_1 and obtain the equation

$$\begin{vmatrix} a - \dfrac{1}{p_1} & h & g & A \\ h & b - \dfrac{1}{p_1} & f & B \\ g & f & c - \dfrac{1}{p_1} & C \\ A & B & C & 0 \end{vmatrix} = 0.$$

and in like manner writing

$$\frac{1}{p_2} = (a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2,$$

we have the same equation for p_2; wherefore p_1, p_2 are the roots of the quadric equation

$$\begin{vmatrix} a - \dfrac{1}{p} & h & g & A \\ h & b - \dfrac{1}{p} & f & B \\ g & f & c - \dfrac{1}{p} & C \\ A & B & C & 0 \end{vmatrix} = 0.$$

[p. 51] Moreover, p_1, p_2 being thus determined, we have α_1, β_1, γ_1, θ_1 proportional to the determinants formed with the matrix

$$\left|\begin{matrix} a-\dfrac{1}{p_1} & h & g & A \\ h & b-\dfrac{1}{p_1} & f & B \\ g & f & c-\dfrac{1}{p_1} & C \end{matrix}\right|,$$

say, $\alpha_1, \beta_1, \gamma_1, \theta_1 = k\mathfrak{A}_1, k\mathfrak{B}_1, k\mathfrak{C}_1, k\Omega_1$, where $\mathfrak{A}_1$, $\mathfrak{B}_1$, $\mathfrak{C}_1$, Ω_1 are the determinants in question; and then $1 = k^2(\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2)$, or we have

$$\theta_1 = \frac{\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}}.$$

But we find at once

$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) = \theta_1(A\alpha + B\beta + C\gamma) = \theta_1 V,$$

that is,

$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) = \frac{V\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}},$$

and in the same manner

$$(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) = \frac{V\Omega_2}{\sqrt{\mathfrak{A}_2^2 + \mathfrak{B}_2^2 + \mathfrak{C}_2^2}}.$$

Hence the transformed equation is

$$VZ + \tfrac{1}{2}\frac{X^2}{p_1} + \tfrac{1}{2}\frac{Y^2}{p_2} + XZ\,\frac{V\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}} + YZ\,\frac{V\Omega_2}{\sqrt{\mathfrak{A}_2^2 + \mathfrak{B}_2^2 + \mathfrak{C}_2^2}} + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 + \text{\&c.} = 0,$$

where it will be recollected that $V = \sqrt{A^2 + B^2 + C^2}$. The &c. refers as before to the terms $(X, Y, Z)^3$ and higher powers, which are obtained from the corresponding terms in δx, δy, δz by substituting for these their values $\delta x = \alpha_1 X + \alpha_2 Y + \alpha Z$, &c., where the coefficients have the values above obtained for them. It will be observed, that the radii of curvature are Vp_1, Vp_2, and that the process includes an investigation of the values of these radii of curvature similar to the ordinary one; the novelty is in the terms in XZ, YZ, and Z^2. But regarding X, Y as small quantities of the first order, Z is of the second order, and the terms in XZ, YZ are of the third order, and that in Z^2 of the fourth order.

567. On an Identical Equation Connected with the Theory of Invariants

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 115–118.]

[p. 52] Write

$$\begin{aligned} a &= g - h, \\ b &= h - f, \\ c &= f - g, \end{aligned}$$

equations implying a fourth equation forming with them the system

$$\begin{aligned} \cdot\ -h + g - a &= 0, \\ h\ \cdot\ -f - b &= 0, \\ -g + f\ \cdot\ -c &= 0, \\ a + b + c\,\cdot\ &= 0, \end{aligned}$$

and also

$$af + bg + ch = 0.$$

Then, putting for shortness

$$\begin{aligned}P &= (bg - ch)(ch - af)(af - bg),\\ Q &= a^2g^2h^2 + b^2h^2f^2 + c^2f^2g^2 + a^2b^2c^2,\\ R &= a^2f^2(a^2 + f^2) + b^2g^2(b^2 + g^2) + c^2h^2(c^2 + h^2),\end{aligned}$$

we have

$$2P + Q - R = 0,$$

viz. substituting for a, b, c their values $g - h$, $h - f$, $f - g$, this is an identical equation.

[p. 53] The direct verification is however somewhat tedious, and the equation may be proved more easily as follows:

In the terms $a^2 + f^2$, $b^2 + g^2$, $c^2 + h^2$ of R, substituting for a, b, c their values, we find

$$R = (f^2 + g^2 + h^2)(a^2f^2 + b^2g^2 + c^2h^2) - 2fgh(a^2f + b^2g + c^2h),$$

which may be written

$$R = -2(f^2 + g^2 + h^2)(bcgh + cahf + abfg) - 2fgh(a^2f + b^2g + c^2h).$$

We have then

$$2P = -2bcgh(bg - ch) - 2cahf(ch - af) - 2abfg(af - bg),$$

and thence

$$\begin{aligned}2P - R = &\ 2bcgh(f^2 + g^2 + h^2 - bg + ch)\\ &+ 2cahf(f^2 + g^2 + h^2 - ch + af)\\ &+ 2abfg(f^2 + g^2 + h^2 - af + bg)\\ &+ 2fgh(a^2f + b^2g + c^2h),\end{aligned}$$

which is at once converted into

$$\begin{aligned}2P - R = &\ 2bcgh\{a^2 + f(f + g + h)\}\\ &+ 2cahf\{b^2 + g(f + g + h)\}\\ &+ 2abfg\{c^2 + h(f + g + h)\}\\ &+ 2fgh(a^2f + b^2g + c^2h)\,;\end{aligned}$$

or, what is the same thing,

$$2P - R = 2fgh\big\{(bc + ca + ab)(f + g + h) + a^2f + b^2g + c^2h\big\} + 2abc(agh + bhf + cfg),$$

where, since

$$agh + bhf + cfg = -abc,$$

the last term is

$$= -2a^2b^2c^2.$$

But from the equation last written down we deduce at once

$$Q = 2a^2b^2c^2 - 2fgh(bcf + cag + abh),$$

and we thence have

$$2P + Q - R = 2fgh\big\{(bc + ca + ab)(f + g + h) + (a^2f + b^2g + c^2h) - bcf - cag - abh\big\},$$

which is

$$= 2fgh(a + b + c)(af + bg + ch),$$

and consequently $= 0$, the theorem in question.

[p. 54] Instead of a, b, c, f, g, h, I write $aW \div YZ$, $bW \div ZX$, $cW \div XY$, $f \div X$, $g \div Y$, $h \div Z$: we have therefore

$$\begin{aligned} &\cdot \ -hY + gZ - aW = 0, \\ &hX \ \cdot \ -fZ - bW = 0, \\ &-gX + fY \ \cdot \ -cW = 0, \\ &aX + bY + cZ \cdot \ = 0, \end{aligned}$$

and as before

$$af + bg + ch = 0.$$

Moreover, omitting a common factor, the new values of P, Q, R are

$$\begin{aligned} &P = XYZW(bg - ch)(ch - af)(af - bg), \\ &Q = a^2g^2h^2X^4 + b^2h^2f^2Y^4 + c^2f^2g^2Z^4 + a^2b^2c^2W^4, \\ &R = a^2f^2(a^2X^2W^2 + f^2Y^2Z^2) + b^2g^2(b^2Y^2W^2 + g^2Z^2X^2) + c^2h^2(c^2Z^2W^2 + h^2X^2Y^2), \end{aligned}$$

and the identical equation is, as before,

$$2P + Q - R = 0.$$

Consider the operative symbols

$$d_{x_1},\ d_{y_1},\ d_{x_2},\ d_{y_2},$$
$$d_{x_3},\ d_{y_3},\ d_{x_4},\ d_{y_4},$$

and write $a = d_{x_1}d_{y_2} - d_{y_1}d_{x_2} = 12$, &c., that is

$$\begin{aligned} a &= 23, & f &= 14, \\ b &= 31, & g &= 24, \\ c &= 12, & h &= 34, \end{aligned}$$

and also $X = x\,d_{x_1} + y\,d_{y_1}$, &c. say

$$X = \nabla_1,\ Y = \nabla_2,\ Z = \nabla_3,\ W = \nabla_4.$$

These values of a, b, c, f, g, h, X, Y, Z, W satisfy the above written equations of connexion, and therefore the identical equation $2P + Q - R = 0$. Hence taking U to denote the quartic function $U = (a, b, c, d, e)(x, y)^4$, and therefore $U_1 = (a, \ldots)(x_1, y_1)^4$, &c., we have

$$(2P + Q - R)\,U_1U_2U_3U_4 = 0,$$

where, after the differentiations, $(x_1, y_1) \ldots (x_4, y_4)$ are to be each of them replaced by (x, y).

Observe that P is the sum of three positive and three negative terms, but that after the omission of the suffixes each term taken with its proper sign becomes equal to the same quantity, and the value of P is $= 6$ times any one term thereof. Thus omitting for the moment the factor $\nabla_1\nabla_2\nabla_3\nabla_4$, two of the terms are $-(af)^2bg + af(bg)^2$, that is,

$$-(14 \cdot 23)^2(24 \cdot 31) + (14 \cdot 23)(24 \cdot 31)^2,$$

[p. 55] and, if in the first term we interchange 3 and 4, it becomes $-(13 \cdot 24)^2(23 \cdot 41)$, that is, $+(14 \cdot 23)(24 \cdot 31)^2$, viz. it becomes equal to the second term. As regards Q the terms are all positive and become equal to each other; and the like as regards R: hence we have

$$\left\{12\,\nabla_1\nabla_2\nabla_3\nabla_4(14 \cdot 23)(24 \cdot 31)^2 + 4\nabla_1^2(23)^2(34)^2(42)^2 - 6\nabla_1^2\nabla_2^2(43)^2(14)^2\right\} U_1U_2U_3U_4 = 0,$$

which, omitting a numerical factor $6 \cdot 2 \cdot 12^2 \cdot 2 \cdot 24^2 \cdot 4 = 3^2 \cdot 2^{15}$, is in fact the well-known equation

$$\Omega + JU - IH = 0,$$

where

$$\begin{aligned}
U &= (a, b, c, d, e)(x, y)^4, \\
\Omega &= \text{Disc.}(ax + by, bx + cy, cx + dy, dx + ey)(\xi, \eta)^3, \\
&= (ax + by)^3(dx + ey)^2 + \&\text{c.}, \\
I &= ae - 4bd + 3c^2, \\
J &= ace - ad^2 - b^2e - c^3 + 2bcd,
\end{aligned}$$

viz. attending only to the coefficient of x^6, this equation is

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd + a(ace - ad^2 - b^2e - c^3 + 2bcd) + (ac - b^2)(ae - 4bd + 3c^2) = 0.$$

[Continued on following pages.]

Cayley — Collected Mathematical Papers, Volume IX
Pages 113–137 (papers 572 conclusion, 573–578)

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572 (concluded). A Memoir on the Quartic Surfaces $UV - W^2 = 0$ (continued)

(Continuation of paper 572 from preceding pages. Pages 93 of the volume reproduces the final page of the analysis of $\delta_0(P^k + \lambda P'^k)$.)

Attending only to the form of the result, and representing the numerical factors by A, B, &c., we may write

$$\delta_0(P^k + \lambda P'^k) = A\,P^{k-4}\,\mathrm{abcd} + \lambda\left\{\begin{array}{l} B\,P^{k-4}\,P'^{k-1}\,\mathrm{abcd}' \\ \quad + (k'-1)P^{k-2}\,P'^{k-2}\,\mathrm{abcD}' \end{array}\right\} + C\,P^{k-2}\,P'^2\,\mathrm{abcd}$$

$$+\lambda^2\left[D\,P^{k-2}\,P'^{k-2}\,\mathrm{abc'd'} + E\,P^{k-3}\,P'^{k-1}\,\mathrm{abc'D''} + E'\,P^{k-3}\,P'^{k-3}\,\mathrm{a'b'cD^2} + F\,P^{k-3}\,P'^{k-3}(\Lambda P + \Lambda' P')\right]$$

$$+\lambda^3\left\{C''\,P^k\,P'^{k-3}\,\mathrm{a'b'c'd} + B'\left(\begin{array}{l} P^{k-3}\,F^{k-1}\,\mathrm{a'b'c'd} \\ \quad + (k-1)P^{k-2}\,P'^{k-2}\,\mathrm{a'b'c'D'} \end{array}\right)\right\} + \lambda^4\,A'\,P'^{k-4}\,\mathrm{a'b'c'd'},$$

where, for shortness, certain terms in λ^2 have been represented by $\Lambda P + \Lambda' P'$.

Suppose $k = k' = 2$; then attending only to the terms of the lowest order in P, P' conjointly, we have

$$\delta_0(P^2 + \lambda P'^2) = \lambda B \cdot PP' \cdot \mathrm{abcD}' + \lambda^2 \cdot PP'\,(\Lambda P + \Lambda' P') + \lambda^2 B' \cdot P'^2 \cdot \mathrm{a'b'cD'};$$

the function operated upon with $UF + UF'$ would have had the lowest terms in P, P' of the like form; and it thus appears that for a surface of the form $UF + UF'^2 = 0$, the nodal curve $P = 0$, $P' = 0$ is a triple curve on the Hessian surface.

If $k = 2$, $k' = 3$, then attending only to the terms of the lowest order in P, P' conjointly, we have

$$\delta_0(P^2 + \lambda P'^3) = A \cdot P^2 \cdot \mathrm{abcd} + \lambda \cdot 2B \cdot PP' \cdot \mathrm{abcD}';$$

and the like result would be obtained if the function operated upon with δ_0 had been $UF + UF'^3$. It thus appears that for a surface of the form $UF + UF'^3 = 0$, the cuspidal curve $P = 0$, $P' = 0$ is a 4-tuple curve on the Hessian surface, the form in the vicinity of this line, or direction of the tangent plane, being given by

$$P'(A \cdot P \cdot \mathrm{abcd} + 2B\lambda \cdot P' \cdot \mathrm{abcD}') = 0,$$

viz. there is a triple sheet $P' = 0$, coinciding with the direction of the surface in the vicinity of the cuspidal line; and a single sheet

$$A \cdot P \cdot \mathrm{abcd} + 2B\lambda \cdot P' \cdot \mathrm{abcD}' = 0.$$

At the points for which the osculating plane of the curve $P = 0$, $P' = 0$ coincides with the tangent plane of $P = 0$ (or, what is the same thing, with that of the surface), we have $\mathrm{abcD}' = 0$, and the triple and single sheets then coincide in direction.

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573. Note on the (2, 2) Correspondence of Two Variables

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 197, 198.]

In connection with my paper "On the porism of the in-and-circumscribed polygon and the (2, 2) correspondence of points on a conic," *Quar. Math. Jour.*, t. xi. (1871), pp. 83–91, [489], I remark that if (θ, ϕ) have a symmetrical (2, 2) correspondence, and also (ϕ, χ) the same symmetrical (2, 2) correspondence, then (θ, χ) will have a (not in general the same) symmetrical (2, 2) correspondence. In fact, to a given value θ there correspond, say the values ϕ_1, ϕ_2 of ϕ; then to ϕ_1 correspond the values θ, χ_1 of χ (viz. one of the two values is $= \theta$), and to ϕ_2 the values θ, χ_2 of χ (viz. one of the values is here again $= \theta$); that is, to the given value θ there correspond the two values χ_1, χ_2 of χ; and similarly to any value of χ there correspond two values of θ; viz. to χ_1 the value θ and say θ_1; to χ_2 the value θ and say θ_2; that is, the correspondence of θ, χ is a (2, 2) correspondence and is symmetrical.

Analytically, if we have

$$(a,\ b,\ c,\ f,\ g,\ h \mid \theta\phi,\ \theta + \phi,\ 1)^2 = 0,$$

and

$$(a,\ b,\ c,\ f,\ g,\ h \mid \phi\chi,\ \phi + \chi,\ 1)^2 = 0,$$

then writing

$$(a, \ldots \mid \phi u,\ \phi + u,\ 1)^2 = 0,$$

the roots hereof are $u = \theta$, $u = \chi$; i.e. we have

$$(a, \ldots \mid \phi u,\ \phi + u,\ 1)^2 = (a, \ldots \mid \phi^2,\ 1,\ 0)\,(u - \theta)(u - \chi);$$

or, what is the same thing, we have

$$1 : -(\theta + \chi) : \theta\chi = (a, \ldots \mid \phi, 1, 0)^2 \ :\ 2(a, \ldots \mid \phi, 1, 0 \mid \phi, \phi, 1) \ :\ (a, \ldots \mid \phi, \phi, 1)^2$$

$$= a\phi^2 + 2h\phi + b \ :\ 2(h\phi^2 + \overline{b + g}\,\phi + f) \ :\ b\phi^2 + 2f\phi + c,$$

giving $\phi^2 : \phi : 1$ proportional to linear functions of 1, $\theta + \chi$, $\theta\chi$, and therefore a quadric relation $(* \mid \theta\chi,\ \theta + \chi,\ 1)^2 = 0$, with coefficients which are not in general (a, b, c, f, g, h). Suppose, however, that the coefficients have these values, or that the correspondence is

$$(a,\ b,\ c,\ f,\ g,\ h \mid \theta\chi,\ \theta + \chi,\ 1)^2 = 0.$$

We must have

$$(a, b, c, f, g, h \mid a\phi^2 + 2h\phi + b,\ -2(h\phi^2 + \overline{b + g}\,\phi + f),\ b\phi^2 + 2f\phi + c)^2 = 0,$$

that is,

$$(ac - b^2 + 2bg - 4fh)(a, b, c, f, g, h \mid \phi^2, -2\phi, 1)^2 = 0,$$

or, we have

$$ac - b^2 + 2bg - 4fh = 0,$$

as the condition in order that the symmetrical (2, 2) correspondence between θ and χ may be the same correspondence as that between θ and ϕ, or between ϕ and χ.

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574. On Wronski's Theorem

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 221–228.]

The theorem, considered by the author as an answer to the question "En quoi consistent les Mathématiques? N'y aurait-il pas moyen d'embrasser par un seul problème, tous les problèmes de ces sciences et de résoudre généralement ce problème universel?" is given without demonstration in his *Réfutation de la Théorie de Fonctions Analytiques de Lagrange*, Paris, 1812, p. 30, and reproduced (with, I think, a demonstration) in the *Philosophie de la Technie*, Paris, 1815; and it is also stated and demonstrated in the *Supplément à la Réforme de la Philosophie*, Paris, 1847, p. cix *et seq.*; the theorem, but without a demonstration, is given in Montferrier's *Encyclopédie Mathématique* (Paris, no date), t. iii. p. 398.

The theorem gives the development of a function Fx of the root of an equation

$$0 = fx + x_1 f_1 x + x_2 f_2 x + \&\text{c.},$$

but it is not really more general than that for the particular case $0 = fx + x_1 f_1 x$; or say when the equation is $0 = \phi x + \lambda fx$.[1] Considering then this equation

$$\phi x + \lambda fx = 0,$$

let a be a root of the equation $\phi x = 0$; the theorem is

$$Fx = F - \frac{\lambda}{1}\frac{1}{\phi'}\left|(fF')'\right| + \frac{\lambda^2}{1.2}\frac{1}{\phi'^3}\left|\begin{matrix}\phi', & (fF')' \\ \phi'', & (f^2F')'\end{matrix}\right|\frac{1}{1}$$

$$-\frac{\lambda^3}{1.2.3}\frac{1}{\phi'^6}\left|\begin{matrix}\phi', & (fF')', & (f^2F')' \\ \phi'', & (\phi')'', & (f^2F')'' \\ \phi''', & (\phi')''', & (f^2F')'''\end{matrix}\right|\frac{1}{1.1.2} + \&\text{c.},$$

where F, f, F', &c. denote Fa, fa, $F'a$, &c. and the accents denote differentiation in regard to a; the integral sign $\int$ is written instead of $\int_a$; this is introduced for symmetry only, and obviously disappears; in fact, we may equally well write

$$Fx = F - \frac{\lambda}{1}\frac{1}{\phi'}fF' + \frac{\lambda^2}{1.2}\frac{1}{\phi'^3}\left|\begin{matrix}\phi', & fF' \\ \phi'', & (fF')'\end{matrix}\right|\frac{1}{1}$$

$$-\frac{\lambda^3}{1.2.3}\frac{1}{\phi'^6}\left|\begin{matrix}\phi', & fF', & f^2F' \\ \phi'', & (\phi')'', & (f^2F')'' \\ \phi''', & (\phi')''', & (f^2F')'''\end{matrix}\right|\frac{1}{1.1.2} + \&\text{c.}$$

I stop for a moment to remark that Laplace's theorem is really equivalent to Lagrange's; viz. in the first mentioned theorem we have $x = \phi(a + \lambda fx)$, that is $\phi^{-1}x = a + \lambda f\phi \cdot \phi^{-1}x$, and then $Fx = F\phi \cdot \phi^{-1}x$, viz. by Lagrange's theorem

$$Fx = F\phi + \frac{\lambda}{1}F\phi' \cdot f\phi + \frac{\lambda^2}{1.2}\{F\phi' \cdot (f\phi)^2\}' + \&\text{c.},$$

where on the right hand $F\phi$ and $f\phi$ are each regarded as one symbol, the argument being always a and the accents denoting differentiation in regard to a, thus $F\phi'$ is

$$d_a \cdot F\phi a = F'\phi a \cdot \phi' a, \ \&\text{c.},$$

viz. this is Laplace's theorem.

Suppose in Wronski's theorem $\phi x = x - a$; that is, let the equation be

$$x - a + \lambda\phi x = 0,$$

then each determinant reduces itself to a single term: thus the determinant of the third order is

$$\left|\begin{matrix}(x-a)', & \left((x-a)^2\right)', & f^2F' \\ (x-a)'', & \left((x-a)^2\right)'', & (f^2F')'' \\ (x-a)''', & \left((x-a)^2\right)''', & (f^2F')'''\end{matrix}\right|,$$

[1]For in the result, as given in the text, instead of λfx write $x_1 f_1 x + x_2 f_2 x + \&\text{c.}$, then expanding the several powers of this quantity, each determinant is replaced by a sum of determinants of the same order, and we have the expansion of Fx in powers of x_1, x_2, ...

where in the first and second columns the accents denote differentiation in regard to x, which variable is afterwards put $= a$; the determinant is thus

$$\begin{vmatrix} 1, & * & * \\ 0, & 1\cdot 2, & * \\ 0, & 0, & (f^2F')'' \end{vmatrix},$$

viz. it is

$$= 1\cdot 1\cdot 2\,(f^2F')'',$$

and so in other cases; the formula is thus

$$Fx = F - \frac{\lambda}{1} fF' + \frac{\lambda^2}{1.2}(f^2F')' - \frac{\lambda^3}{1.2.3}(f^3F')'' + \&\text{c.},$$

agreeing with Lagrange's theorem.

Suppose in general $\phi x = (x-a)\,\psi x$, or let the equation be

$$(x-a)\,\psi x + \lambda fx = 0,$$

that is,

$$x - a + \lambda\frac{fx}{\psi x} = 0:$$

we have then by Lagrange's theorem

$$Fx = F - \frac{\lambda}{1} F'\left(\frac{f}{\psi}\right) + \frac{\lambda^2}{1.2}\left\{F'\left(\frac{f}{\psi}\right)^2\right\}' - \frac{\lambda^3}{1.2.3}\left\{F'\left(\frac{f}{\psi}\right)^3\right\}'' + \&\text{c.}$$

Consider for example the term $\left\{F'\left(\dfrac{f}{\psi}\right)^2\right\}'$; this is

$$= \left\{F'x\cdot\frac{(x-a)^2(fx)^2}{(\phi x)^2}\right\}',$$

the accents denoting differentiation in regard to x, and x being ultimately put $= a$; or, what is the same thing, it is

$$= \left(\frac{d}{d\theta}\right)^2\left[F'(a+\theta)\,\frac{\{f(a+\theta)\}^2}{\{\phi(a+\theta)\}^2}\right],$$

the accents now denoting differentiation in regard to θ, and this being ultimately put $= 0$. This is

$$\left(\frac{d}{d\theta}\right)^2\left[F'(a+\theta)\,\frac{\{f(a+\theta)\}^2}{\left(\phi' a + \dfrac{\theta}{1.2}\phi'' a + \cdots\right)^2}\right].$$

This may be written $\left(F'f^2\,\dfrac{1}{A^2}\right)''$, where

$$A = \phi' + \tfrac{1}{2}\theta\phi'' + \tfrac{1}{2.3}\phi''' + \cdots,$$

it being understood that as regards $F'f^2$, which is expressed as a function of a only (θ having been therein put $= 0$), the exterior accents denote differentiations in respect to a, whereas in regard to A, a , $\phi' + \frac{1}{2}\theta\phi'' + \&\text{c.}$, they denote differentiation in regard to θ, which is afterwards put $= 0$. And the theorem thus is

$$Fx = F - \frac{\lambda}{1}\left(F'f\cdot\frac{1}{A}\right) + \frac{\lambda^2}{1.2}\left(F'f^2\cdot\frac{1}{A^2}\right)' - \frac{\lambda^3}{1.2.3}\left(F'f^3\cdot\frac{1}{A^3}\right)'' + \&\text{c.}$$

This must be equivalent to Wronski's theorem; it is in a very different, and, I think, a preferable form; but the results obtained from the comparison are very interesting, and I proceed to make this comparison.

Taking the foregoing coefficient $\left(F'f^2\,\dfrac{1}{A^2}\right)''$ this should be equal to Wronski's term

$$\frac{1}{1.1.2}\,\frac{1}{\phi'^3}\left|\begin{array}{ccc} \phi', & (\phi^2 f)', & f^2F' \\ \phi'', & (\phi^2 f)'', & (f^2F')' \\ \phi''', & (\phi^2 f)''', & (f^2F')'' \end{array}\right|,$$

or, what is the same thing, the determinant should be

$$= 1\cdot 1\cdot 2\phi'^3\left(\frac{1}{A^2}\,f^2F'\right)''$$

$$= 1\cdot 1\cdot 2\phi'^3\left\{f^2F'\left(\frac{1}{A}\right)'' + 2(f^2F')'\left(\frac{1}{A}\right)' + (f^2F')''\frac{1}{A}\right\},$$

that is, the values of

$$1\cdot 1\cdot 2\,\frac{1}{A^2},\quad 1\cdot 1\cdot 2\phi'^2\left(\frac{1}{A}\right)',\quad 1\cdot 1\cdot 2\phi'^3\left(\frac{1}{A}\right)''$$

should be

$$= \phi'(\phi^2 f)'' - \phi''(\phi')',\quad \phi'(\phi')''' - \phi''(\phi^2 f)',\quad \phi'(\phi^2 f)' - \phi''(\phi^2 f)''$$

respectively. Or, what is the same thing, if

$$\frac{1}{\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots} = A_1 + \frac{1}{1}A_2\theta + \frac{1}{1.2}A_3\theta^2 + \cdots,$$

then the last mentioned functions should be

$$1\cdot 1\cdot 2\phi'^3 A_1,\quad 1\cdot 1\cdot 2\phi'^2 A_2,\quad 1\cdot 1\cdot 2\phi'^3 A_3.$$

We have

$$A_1 = \frac{1}{\phi'},\quad A_2 = -\frac{3}{2}\,\frac{\phi''}{\phi'^2},\quad A_3 = -\frac{\phi'''}{\phi'^2} + \frac{3\phi''^2}{\phi'^3},$$

or the identities are

$$2\phi'^3\,\frac{1}{\phi'} = \phi'\,(\phi')'' - \phi''\,(\phi')',\quad = \phi'\big(2\phi\phi'' + 2\phi'^2\big) - \phi''\cdot 2\phi\phi',$$

$$-6\phi'^2\cdot\phi'' = \phi'\,(\phi^2 f)' - \phi'\,(\phi')''',\quad = \phi'\cdot 2\phi\phi' - \phi'\big(2\phi\phi'' + 6\phi'\phi''\big),$$

$$+6\phi'^3\phi' - 2\phi'^3\phi''' = \phi'\,(\phi^2 f)''' - \phi''\,(\phi^2 f)'',\quad = \phi'\big(2\phi\phi''' + 6\phi'\phi''\big) - \phi''\big(2\phi\phi'' + 2\phi'^2\big),$$

which is right. And in like manner to verify the coefficient of λ^4, we should have to compare the first four terms of the expansion of

$$\frac{1}{\left(\phi' + \dfrac{\theta}{2}\phi'' + \dfrac{\theta^2}{2.3}\phi''' + \cdots\right)^4}.$$

With the determinants formed out of the matrix

$$\left|\begin{array}{cccc} \phi', & \phi'', & \phi''', & \phi'''' \\ (\phi')', & (\phi')'', & (\phi')''', & (\phi')'''' \\ (\phi')'', & (\phi')''', & (\phi')'''', & (\phi')''''' \end{array}\right|,$$

the series of equalities may be presented as follows, writing as above A to denote the function

$$\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots,$$

$$\frac{1}{A} = \frac{1}{\phi'}\cdot 1,$$

$$\frac{1}{A^2} = \frac{-1}{\phi'^2} \begin{vmatrix} \phi, & 1 \\ \phi', & \phi'' \end{vmatrix} \cdot \frac{1}{1},$$

$$\frac{1}{A^3} = \frac{+1}{\phi'^6} \begin{vmatrix} \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''' \\ (\phi')', & (\phi')'', & (\phi')''' \end{vmatrix} \cdot \frac{1}{1.1.2},$$

$$\frac{1}{A^4} = \frac{-1}{\phi'^{10}} \begin{vmatrix} \frac{1}{6}\phi^3, & \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''', & \phi'''' \\ (\phi')', & (\phi')'', & (\phi')''', & (\phi')'''' \\ (\phi')'', & (\phi')''', & (\phi')'''', & (\phi')''''' \end{vmatrix} \cdot \frac{1}{1.1.2.1.2.3},$$

&c., where in each case the function on the left hand is to be expanded only as far as the power of θ which is contained in the determinant: the numerical coefficients in the top-lines of the several determinants are the reciprocals of

$$n(n-1)\ldots 2.1,\ n(n-1)\ldots 2,\ n(n-1),\ n,\ 1,$$

where n is the index of the highest power of θ. The demonstration of Wronski's theorem therefore ultimately depends on the establishment of the foregoing equalities. As a verification, in the fourth formula, write $\phi = e^\theta$ $(a = 0)$, we have

$$\left(\frac{\theta}{e^\theta - 1}\right)^4 = \frac{1}{(1 + \frac{1}{2}\theta + \frac{1}{6}\theta^2 + \frac{1}{24}\theta^3 + \cdots)^4} = -\frac{1}{4!} \begin{vmatrix} \frac{1}{6}\theta^3, & \frac{1}{2}\theta^2, & \frac{1}{2}\theta, & 1 \\ 1, & 1, & 1, & 1 \\ 2, & 4, & 8, & 16 \\ 3, & 9, & 27, & 81 \end{vmatrix},$$

where the right hand is

$$= -\tfrac{1}{4!}(-1 \cdot 12 + \tfrac{1}{2}\theta \cdot 72 - \tfrac{1}{2}\theta^2 \cdot 132 + \tfrac{1}{6}\theta^3 \cdot 72) = 1 - 2\theta + \tfrac{11}{4}\theta^2 - \theta^3,$$

and expanding the left hand as far as θ^3, this is

$$\begin{array}{l} = \dfrac{1}{1} \\ \quad -4\left(\frac{1}{2}\theta + \frac{1}{6}\theta^2 + \frac{1}{24}\theta^3\right) \\ \quad +10\left(\frac{1}{2}\theta + \frac{1}{6}\theta^2\right)^2 \\ \quad -20\left(\frac{1}{2}\theta\right)^3 \end{array} = \begin{array}{l} = 1 \\ -2\theta - \frac{2}{3}\theta^2 - \frac{1}{6}\theta^3 \\ +10\left(\frac{1}{4}\theta^2 + \frac{1}{6}\theta^3\right) \\ -\frac{5}{2}\theta^3 \\ \hline 1 - 2\theta + \frac{11}{4}\theta^2 - \theta^3, \end{array}$$

which agrees.

Reverting to the above equations, and expanding the several terms $(\phi')' = 2\phi\phi'$, $(\phi^2)'' = 2\phi\phi'' + 2\phi'^2$, &c., then, since in each case the left-hand side contains ϕ', ϕ'', ϕ''', &c., but not ϕ, it is clear that on the right-hand side also the terms involving ϕ must disappear of themselves; and assuming that this is so, the equality takes the more simple form obtained by writing in the foregoing expressions $\phi = 0$, viz. we thus have $(\phi')' = 0$, $(\phi^2)'' = 2\phi'^2$, &c. In order to simplify the formulæ, I replace the series ϕ', $\frac{1}{2}\phi''$, $\frac{1}{2.3}\phi'''$, &c. by b, c, d, e, &c., and I thus find that they assume the following simple form, viz. writing

$$\Theta = b + c\theta + d\theta^2 + e\theta^3 + \&c.,$$

then we have

$$\frac{1}{\Theta} = \frac{1}{b} \cdot 1,$$

$$\frac{1}{\Theta^2} = \frac{2}{b^3} \begin{vmatrix} \theta, & \frac{1}{2} \\ b, & c \end{vmatrix},$$

$$\frac{1}{\Theta^3} = \frac{3}{b^5}\begin{vmatrix} \theta^2, & \frac{1}{2}\theta, & \frac{1}{2} \\ b, & c, & d \\ b', & c', & 2bc \end{vmatrix},$$

$$\frac{1}{\Theta^4} = \frac{4}{b^7}\begin{vmatrix} \theta^3, & \frac{1}{2}\theta^2, & \frac{1}{2}\theta, & \frac{1}{2} \\ b, & c, & d, & e \\ b', & c', & d', & 2bd+c^2 \\ b'', & c'', & 2bc'+2b'c, & 3bc' \end{vmatrix},$$

viz. for Θ^{-n} the right-hand gives the development as far as θ^{n-1}. It will be observed, that in the determinants the several lines are the coefficients in the expansions of Θ, Θ^2, Θ^3, respectively.

The demonstration is very easy; it will be sufficient to take the equation for $\dfrac{1}{\Theta^4}$. Assume

$$\frac{1}{\Theta^4} = \cdots + r\theta^0 + q\theta^1 + p\theta^2 + \tfrac{1}{2}p\theta^3 + \tfrac{1}{6}q\theta^4 + \tfrac{1}{2}r\theta^5,$$

where clearly $\epsilon = \dfrac{\phi^5}{b^4}$, and write also

$$\Theta = B_1 + C_1\theta + D_1\theta^2 + E_1\theta^3 + \cdots,$$

$$\Theta^2 = B_2 + C_2\theta + D_2\theta^2 + \cdots,$$

$$\Theta^3 = B_3 + C_3\theta^2 + \cdots,$$

where $B_1 = b$, $B_2 = b^2$, $B_3 = b^3$; we wish to show that

$$\beta B_1 + \gamma C_1 + \delta D_1 + \epsilon E_1 = 0,$$

$$\gamma B_1 + \delta C_1 + \epsilon D_1 = 0,$$

$$\delta B_1 + \epsilon C_1 = 0,$$

for this being the case, neglecting the terms in θ^0, θ^1, θ^2, &c., and writing

$$\beta\theta^2 + \tfrac{1}{2}\gamma\theta^3 + \tfrac{1}{6}\delta\theta^4 + \epsilon\left(1 - \frac{1}{e\Theta^4}\right) = 0,$$

then eliminating β, γ, δ, ϵ, we have

$$\begin{vmatrix} \theta^2, & \frac{1}{2}\theta^3, & \frac{1}{6}\theta^4, & \dfrac{1}{e\Theta^4} \\ B_1, & C_1, & D_1, & E_1 \\ B_2, & C_2, & D_2, & E_2 \\ B_3, & C_3, & D_3, & E_3 \end{vmatrix} = 0,$$

in which equation the term which contains $\dfrac{1}{e\Theta^4}$ is $= \frac{1}{4}\theta^4 B_1B_2B_3\dfrac{1}{e\Theta^4}$, $= \frac{1}{4}b^6\dfrac{1}{e\Theta^4}$; and the equation thus is $\dfrac{1}{\Theta^4} = -\dfrac{4}{b^7}$ multiplied by the determinant without the term in question (that is, with $\frac{1}{2}$ for its corner term).

To prove the subsidiary theorems, multiply the expression of $\dfrac{1}{\Theta^4}$ by $\dfrac{1}{\theta^4}$, and differentiate in regard to θ, we have

$$\frac{4(\Theta\theta)'}{(\Theta\theta)^5} = \cdots - 2r\theta + q + \frac{\beta}{\theta^2} + \frac{\gamma}{\theta^3} + \frac{\delta}{\theta^4} + \frac{\epsilon}{\theta^5}.$$

Multiplying by

$$\theta\Theta = B_1\theta + C_1\theta^2 + D_1\theta^3 + E_1\theta^4,$$

we see that $B_1\beta + C_1\gamma + D_1\delta + E_1\epsilon$ is the coefficient of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^4}$; and similarly $B_2\gamma + C_2\delta + E_2\epsilon$ is the coefficient of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^3}$, and $B_3\delta + C_3\epsilon$ that of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^2}$. Now, n being any positive integer, $\dfrac{1}{(\Theta\theta)^n}$

expanded in ascending powers of θ contains negative and positive powers of θ, but of course no logarithmic term; hence differentiating in regard to θ, $\dfrac{(\Theta\theta)'}{(\Theta\theta)^{n+1}}$ contains no term in $\dfrac{1}{\theta}$;[2] and the expressions in question are thus each $= 0$; which completes the demonstration.

The foregoing formulae giving the expansion of $\dfrac{1}{\Theta^n}$ up to θ^{n-1} in terms of the coefficients in the expansions of Θ, Θ^2, ..., Θ^{n-1} are I think interesting.

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575. On a Special Quartic Transformation of an Elliptic Function

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 266–269.]

It is remarked by Jacobi that a transformation of the order nn' may lead to a modular equation

$$\frac{\Delta'}{\Delta} = \frac{u'}{v''}\frac{K'}{K},$$

and in particular when $n' = n$, or the order is a square, then the equation may be $\dfrac{\Delta'}{\Delta} = \dfrac{K'}{K}$; viz. that instead of a transformation we may have a multiplication. A quartic transformation of the kind in question may be obtained as follows: writing

$$X = (a,\ b,\ c,\ d,\ e \mid x,\ 1)^4 = a(x-\alpha)(x-\beta)(x-\gamma)(x-\delta),$$

H the Hessian, Φ the cubicovariant, I and J the two invariants, then there is a well-known quartic transformation

$$z = \frac{2H}{X},$$

leading to

$$\frac{dz}{\sqrt{Z}} = \frac{2\sqrt{(-2)}\,dx}{\sqrt{X}},$$

where $Z = z^4 - Iz + 2J$. In fact we have

$$Z = \frac{2}{X^2}(4H^3 - IHX + JX^3), \quad = \frac{-2}{X^4}\Phi^2;$$

that is,

$$\sqrt{Z} = \frac{\sqrt{(-2)}\Phi}{X^2}\sqrt{X};$$

so that, by Jacobi's general principle, it at once appears that we have a transformation of the form in question.

Now we may establish a linear transformation

$$z = \frac{py+q}{y-\delta},$$

such that the roots z_1, z_2, z_3 of the equation $z^4 - Iz + 2J = 0$ correspond the values α, β, γ of y; and this being so, we have between y, z the relation

$$\frac{dz}{\sqrt{Z}} = \frac{\sqrt{(-2)}\,dy}{\sqrt{Y}},$$

where $Y = a(y-\alpha)(y-\beta)(y-\gamma)(y-\delta) = (a,\ b,\ c,\ d,\ e \mid y,\ 1)^4$; that is, we have

$$\frac{py+q}{y-\delta} = \frac{2H}{X},$$

[2] This is a well-known method made use of by Jacobi and Murphy.

such that

$$\frac{dy}{\sqrt{Y}} = \frac{2dz}{\sqrt{X}},$$

which is a quartic transformation giving a duplication of the integral. The foundation of the theorem is that we can determine p, q in such wise that the functions

$$\frac{p\alpha+q}{\alpha-\delta},\quad \frac{p\beta+q}{\beta-\delta},\quad \frac{p\gamma+q}{\gamma-\delta}$$

shall be the roots z_1, z_2, z_3 of the equation $z^4 - Iz + 2J = 0$. For writing

$$A = (\beta-\gamma)(\alpha-\delta),$$
$$B = (\gamma-\alpha)(\beta-\delta),$$
$$C = (\alpha-\beta)(\gamma-\delta),$$

and observing the equations

$$I = \frac{a^3}{24}(A^2+B^2+C^2),\quad = -\frac{a^3}{12}(BC+CA+AB),$$

(since $A+B+C=0$) and

$$2J = -\frac{a^4}{216}(B-C)(C-A)(A-B),$$

the equation in z is

$$\left[z-\tfrac{1}{6}a(B-C)\right]\left[z-\tfrac{1}{6}a(C-A)\right]\left[z-\tfrac{1}{6}a(A-B)\right],$$

and the equations for the determination of p, q thus are

$$p\alpha+q = \tfrac{1}{6}a(\alpha-\delta)(B-C),\quad = \tfrac{1}{6}a(\alpha-\delta)\{2(\alpha\delta+\beta\gamma)-(\alpha+\delta)(\beta+\gamma)\},$$
$$p\beta+q = \tfrac{1}{6}a(\beta-\delta)(C-A),\quad = \tfrac{1}{6}a(\beta-\delta)\{2(\beta\delta+\gamma\alpha)-(\beta+\delta)(\gamma+\alpha)\},$$
$$p\gamma+q = \tfrac{1}{6}a(\gamma-\delta)(A-B),\quad = \tfrac{1}{6}a(\gamma-\delta)\{2(\gamma\delta+\alpha\beta)-(\gamma+\delta)(\alpha+\beta)\},$$

giving

$$p = \tfrac{1}{6}a\left[-3\delta^2+2\delta(\alpha+\beta+\gamma)-\beta\gamma-\gamma\alpha-\alpha\beta\right],$$
$$q = \tfrac{1}{6}a\left[\delta^2(\alpha+\beta+\gamma)-2\delta(\beta\gamma+\gamma\alpha+\alpha\beta)+3\alpha\beta\gamma\right],$$

or, as these may also be written

$$p = \tfrac{1}{6}a\left[(\beta-\delta)(\gamma-\delta)+(\gamma-\delta)(\alpha-\delta)+(\alpha-\delta)(\beta-\delta)\right],$$
$$q = \tfrac{1}{6}a\left[\alpha(\beta-\delta)(\gamma-\delta)+\beta(\gamma-\delta)(\alpha-\delta)+\gamma(\alpha-\delta)(\beta-\delta)\right];$$

and observe also

$$p\delta+q = \tfrac{1}{6}a(\alpha-\delta)(\beta-\delta)(\gamma-\delta).$$

Taking X in the standard form $=(1-x^2)(1-k^2x^2)$, and writing

$$\gamma=-1,\quad \delta=1,\quad \alpha=\tfrac{1}{k},\quad \beta=-\tfrac{1}{k},$$

we have

$$z = \frac{py+q}{y-1} = \frac{-\frac{1}{6}\left[2k^3(1+k^2)(1+k^2x^2)+(1-10k^2+k^4)x^2\right]}{(-x^2)(1-k^2x^2)},$$

$$A = -1+\frac{2}{k}-\frac{1}{k^2},$$
$$B = 1+\frac{2}{k}+\frac{1}{k^2},$$
$$C = -\frac{4}{k};$$

$$z_1 = \tfrac{1}{6}(1+6k+k^2),$$
$$z_2 = \tfrac{1}{6}(1-6k+k^2),$$
$$z_3 = -\tfrac{2}{3}(1+k^2);$$
$$Z = z^4 - \tfrac{1}{12}(1+14k^2+k^4)\,z + \tfrac{1}{216}(1+k^2)(1-34k^2+k^4)$$
$$= (z-z_1)(z-z_2)(z-z_3),$$
$$p = \tfrac{1}{6}(1-5k^2), \quad q = \tfrac{1}{6}(5-k^2), \quad p+q = 1-k^2;$$

giving as they should do

$$z_1 = \frac{\dfrac{p}{k}+q}{\dfrac{1}{k}-1}, \quad z_2 = \frac{-\dfrac{p}{k}+q}{-\dfrac{1}{k}-1}, \quad z_3 = \frac{-p+q}{-2}.$$

Write for shortness

$$-\tfrac{1}{6}\left[2k^3(1+k^2)(1+x^2) + (1-10k^2+k^4)x^2\right] = Q,$$

so that

$$\frac{py+q}{y-1} = \frac{Q}{X};$$

then

$$\frac{Q}{X} - z_1 = \frac{p+q}{k-1}\cdot\frac{ky-1}{y-1},$$
$$\frac{Q}{X} - z_2 = \frac{p+q}{k+1}\cdot\frac{ky+1}{y+1},$$
$$\frac{Q}{X} - z_3 = \frac{p+q}{2}\cdot\frac{y+1}{y-1}.$$

The last of these is

$$\frac{1-k^2}{2}\cdot\frac{y+1}{y-1} = \frac{\tfrac{1}{2}(1-k^2)^2\,x^2}{(1-x^2)(1-k^2x^2)},$$

that is,

$$\frac{y+1}{y-1} = \frac{(1-k^2)\,x^2}{(1+x^2)(1-k^2x^2)},$$

from which the foregoing equation

$$\frac{dy}{\sqrt{Y}} = \frac{2dx}{\sqrt{X}}$$

may be at once verified.

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576. Addition to Mr Walton's Paper "On the Ray-Planes in Biaxal Crystals"

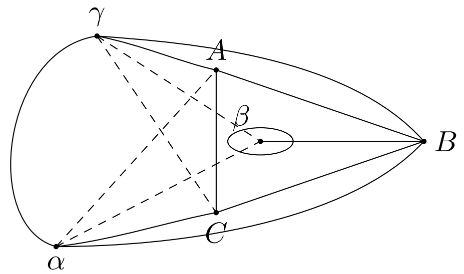
[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 273–275.]

Instead of Mr Walton's a^2, b^2, c^2 write a, b, c, and assume α, β, $\gamma = b-c$, $c-a$, $a-b$; δ, ϵ, $\zeta = b+c$, $c+a$, $a+b$. Also instead of his x', y', z' write x, y, z. Then instead of the octic cone we have the quartic cone, or say the quartic curve

$$\begin{aligned} &a^2\beta^2\gamma^2\sec^2\theta\cdot xy\,(x+y+z) \\ &= a^2yz\left[(bc-a^2)x + a(By-\gamma z)\right]^2 \\ &\quad +\beta^2zx\left[(ca-b^2)y + b(\gamma x-\alpha z)\right]^2 \\ &\quad +\gamma^2xy\left[(ab-c^2)z + c(\alpha y-\beta x)\right]^2, \end{aligned}$$

viz. we may herein consider x, y, z as trilinear coordinates, the ratios $x:y:z$ being positive for a point within the fundamental triangle.

The curve passes through the angles of the triangle, and it touches the sides in the points ($x = 0$, $\beta y - \gamma z = 0$), ($y = 0$, $\gamma z - \alpha x = 0$), ($z = 0$, $\alpha x - \beta y = 0$) respectively. Moreover, the tangents at the angles of the triangle lie each of them outside the triangle. Hence, supposing a, b, c each positive, and $a > b > c$, we have α and γ each positive, β negative; and the form of the curve is as shown in the figure, or else the like form with the oval lying outside the triangle. And it is hence clear that, if the side AC ($y = 0$) instead of touching the curve meets it in a node, this is a conjugate point arising from the evanescence of the oval; and in this case no part of the curve lies within the triangle. Now considering any point $x : y : z = l : m : n$, we obtain a tetrad of points $x : y : z = \pm\sqrt{l} : \pm\sqrt{m} : \pm\sqrt{n}$ on the octic cone or curve; and in order that the point on the octic curve may be real, we must have l, m, n all of the same sign; that is, the point on the quartic curve must lie within the triangle. Hence, when in the quartic curve the oval becomes a conjugate point, the octic curve has no real branch, but it consists wholly of conjugate points; viz. it consists of the points A, B, C as conjugate points; two imaginary conjugate points answering to the point α of the figure, two other imaginary conjugate points answering to the point γ; and two conjugate points answering to the point β, these last being not ordinary conjugate points, but conjugate tacnodal points, or points of contact of two imaginary branches of the curve.



The case in question, β a conjugate point on the quartic curve, answers to Mr Walton's critical value of $\sec^2\theta$, viz. in the present notation

$$\sec^2\theta = \frac{4b^2 + \alpha\gamma}{\alpha\gamma}.$$

To show this I consider the intersection of the curve by the line $\gamma z - \alpha x = 0$; and I write for convenience $\gamma z = \alpha x = \gamma\alpha$, $x = \gamma$, $z = \alpha$. Substituting these values, the equation divides by $y\alpha$, or omitting this factor it is

$$\begin{aligned} a^2\gamma^2\beta^2\sec^2\theta\cdot\alpha\,[y + (\alpha+\gamma)\alpha] \\ = a^3\left[\gamma\alpha(bc - \alpha^2 - a\alpha) + a\beta y\right]^2 \\ +\beta^2\alpha y\,(ca - b^2)^2 \\ +\gamma^2\left[\alpha\,(ab - c^2 + c\gamma) - c\beta y\right]^2, \end{aligned}$$

or observing that we have $\alpha + \gamma = -\beta$, $bc - \alpha^2 - c\alpha = \beta\alpha$, $ab - c^2 + c\gamma = -\delta\beta$, this becomes

$$\begin{aligned} a^2\gamma^2\sec^2\theta\cdot\theta\alpha\,(y - \beta\alpha) \\ = a^3\,(\gamma\beta\alpha + \alpha y)^2 \\ +\alpha y\,(ca - b^2)^2\cdot\alpha y \\ +\gamma^2\,(\alpha\beta\alpha + cy)^2; \end{aligned}$$

viz. this is

$$\begin{aligned} \alpha^2\left[a^3\gamma^2\delta_2 + a^2\gamma^2\beta^3 + a^2\gamma^2\sec^2\theta\cdot\beta\right] \\ +ay\left[2a^3\gamma\delta\beta + 2a^2\gamma c\delta + a\gamma(ca - b^2)^2 - a^2\gamma^2\sec^2\theta\right] \\ +y^2\left(a^3\alpha^2 + c^2\gamma^2\right) = 0. \end{aligned}$$

The required condition is that the coefficient of y^2 shall vanish; viz. we then have

$$-a\beta y\sec^2\theta = 2a^3\gamma\beta$$

$$\begin{aligned}&= (b-c)(a+b)^2 + (\alpha - b)(b+c)^2 \\ &= (a-c)\left[3b^2 + b(a+c) - ac\right] \\ &= -\beta(4b^2 + a\gamma),\end{aligned}$$

that is,

$$a\gamma \sec^2\theta = 4b^2 + a\gamma,$$

agreeing with Mr Walton's value. Giving $\sec^2\theta$ this value, and throwing out the factor a, the equation becomes

$$a\left[2a^2\alpha\gamma + 5a^2\gamma\delta + ay\,(ca-b^2)^2 - a^2\gamma^2\,(4b^2 + a\gamma)\right] \\ +y\,(a^2\alpha^2 + c^2\gamma^2) = 0;$$

or, what is the same thing,

$$ay\left[2a^2(a+b)(b-c)^2 + 2c(c+b)(b-a)^2 + 2c(ca-b^2)^2 - (b-c)(b-a)\,4b^2\right] \\ +y\,(a^2\alpha^2 + c^2\gamma^2) = 0,$$

say

$$ayKu + (a^2\alpha^2 + c^2\gamma^2)\,y = 0,$$

viz. this equation determines the remaining intersection of the curve by the line $\gamma z - \alpha x = 0$; the point in question lies outside the triangle, that is, $u : y$ should be negative; or a, γ, $a^2\alpha^2 + c^2\gamma^2$ being each positive, we should have K positive; we in fact find

$$\begin{aligned}K &= 4b^2 + b^2(a^2 + c^2 - 6ac) + 4a^2c^2 \\ &= 4\,(b^2 - ac)^2 + b^2(a+c)^2,\end{aligned}$$

which is as it should be.

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577. Note in Illustration of certain General Theorems obtained by Dr Lipschitz

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 346–349.]

The paper by Dr Lipschitz, which follows the present Note [in the *Quarterly Journal*, l.c.], is supplemental to Memoirs by him in *Crelle*, vols. lxx., lxxii., and lxxiv.; and he makes use of certain theorems obtained by him in these memoirs; these theorems may be illustrated by the consideration of a particular example.

Imagine a particle not acted on by any forces, moving in a given surface; and let its position on the surface at the time t be determined by means of the general coordinates x, y. We have then the vis-viva function T, a given function of x, y, x', y'; and the equations of motion are

$$\frac{d}{dt}\frac{dT}{dx'} - \frac{dT}{dx} = 0, \qquad \frac{d}{dt}\frac{dT}{dy'} - \frac{dT}{dy} = 0,$$

which equations serve to determine x, y in terms of t, and of four arbitrary constants; these are taken to be the initial values (or values corresponding to the time $t = t_0$) of x, y, x', y'; say the values are a, β, α', β'.

We have the theorem that x, y are functions of a, β, $\alpha'(t-t_0)$, $\beta'(t-t_0)$, that is, of four arbitrary constants.

Suppose for example that x, y, z denote ordinary rectangular coordinates, and that the particle moves on the sphere $x^2 + y^2 + z^2 = c^2$; to fix the ideas, suppose that the coordinates z are measured vertically upwards, and that the particle is on the upper hemisphere; that is, take $z = +\sqrt{(c^2 - x^2 - y^2)}$, we have

$$T = \tfrac{1}{2}\,(x'^2 + y'^2 + z'^2),$$

where z' denotes its value in terms of x, y, x', y'; viz. we have $zz' = -xx' - yy'$, or

$$z' = -\frac{xx' + yy'}{z}, \quad = -\frac{xx' + yy'}{\sqrt{(c^2 - x^2 - y^2)}};$$

the proper value of T is thus

$$= \tfrac{1}{2}\left\{x'^2 + y'^2 + \frac{(xx' + yy')^2}{c^2 - x^2 - y^2}\right\},$$

but it is convenient to retain x, x', taking these to signify throughout the foregoing values in terms of x, y, x', y'.

The constants of integration are, as before, a, β, α', β'; but we now also γ, γ' considered as signifying given functions of these constants, viz. we have

$$\gamma = \sqrt{(c^2 - a^2 - \beta^2)} \text{ and } \gamma' = -\frac{a\alpha' + \beta\beta'}{\sqrt{(c^2 - a^2 - \beta^2)}},$$

(in fact, $a^2 + \beta^2 + \gamma^2 = c^2$ and $a\alpha' + \beta\beta' + \gamma\gamma' = 0$; γ, γ' being thus the initial values of z, z').

Now, writing

$$\sigma = \frac{(t - t_0)\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}{c},$$

the required values of x, y and the corresponding value of z are

$$x = a\cos\sigma + \frac{c\alpha'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma,$$

$$y = \beta\cos\sigma + \frac{c\beta'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma,$$

$$z = \gamma\cos\sigma + \frac{c\gamma'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma.$$

To verify that these are functions of a, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, write $\alpha'(t - t_0) = u$, $\beta'(t - t_0) = v$; and take also $\gamma'(t - t_0) = w$; we have $au + \beta v + \gamma w = 0$, viz. $w = -\dfrac{1}{\gamma}(au + \beta v)$, is a function of a, β, u, v; and so also

$$\sigma = \frac{\sqrt{(u^2 + v^2 + w^2)}}{c},$$

and

$$x = a\cos\sigma + \frac{u}{\sigma}\sin\sigma,$$

$$y = \beta\cos\sigma + \frac{v}{\sigma}\sin\sigma,$$

$$z = \gamma\cos\sigma + \frac{w}{\sigma}\sin\sigma;$$

so that x, y (and also z) are each of them a function of a, β, u, v, that is, a, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, which is the theorem in question.

The original variables are x, y; the quantities $\alpha'(t - t_0)$, $\beta'(t - t_0)$, or u, v, are Dr Lipschitz's "Normal-Variables," and the theorem is that the original variables are functions of their initial values, and of the normal-variables.

The vis-viva function T may be expressed in terms of the normal-variables and their derived functions; viz. it is easy to verify that we have

$$T = \tfrac{1}{2}\left(\frac{\sigma'^2}{c^2\sigma^2} - \frac{\sin^2\sigma}{c^2\sigma^2}\right)(uu' + vv' + ww')^2 - \tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du'^2 + dv'^2 + dw'^2),$$

where w, dw denote $-\dfrac{1}{\gamma}(au + \beta v)$ and consequently w' denotes $-\dfrac{1}{\gamma}(au' + \beta v')$; introducing herein differentials instead of derived functions, or writing

$$\phi(du) = \tfrac{1}{2}\left(\frac{1}{c^2\sigma^2} - \frac{\sin^2\sigma}{c^2\sigma^2}\right)(udu + vdv + wdw)^2 + \tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du^2 + dv^2 + dw^2),$$

where w, dw denote $-\frac{1}{\gamma}(au+\beta v)$, $-\frac{1}{\gamma}(adu+\beta dv)$ respectively; then $\phi(du)$ is the function then denoted by Dr Lipschitz; and writing herein $t-t_0=0$, and thence $u=0$, $v=0$, $w=0$, the resulting value of $\phi(du)$ is

$$f_1(du), \quad = \tfrac{1}{2}(du^2+dv^2+dw^2),$$

where $f_1(du)$ is the function thus denoted by Dr Lipschitz; and writing herein $t-t_0=0$, and thence $u=0$, $v=0$, $w=0$, the resulting value of $\phi(du)$ is $f_1(du)$.

To prove the identity

$$\phi(du) = [d\sqrt{(f_1 u)}]^2 + \frac{\sin^2\sigma}{2\sqrt{(f_1 u)}}\left[f_1(du) - [d\sqrt{(f_1 u)}]^2\right],$$

in verification whereof observe that we have

$$d\sqrt{(f_1 u)} = \frac{df_1(u)}{2\sqrt{(f_1 u)}} = \frac{udu+vdv+wdw}{\sqrt{(u^2+v^2+w^2)}}$$

$$= \frac{1}{c\sigma}(udu+vdv+wdw).$$

The value of the left-hand side is thus

$$= -\frac{\sin^2\sigma}{c^2\sigma^2}(udu+vdv+wdw)^2 + \tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du^2+dv^2+dw^2),$$

viz. this is

$$= \frac{c^2\sigma^2}{c^2\sigma^2}\left\{(udu+vdv+wdw)^2 - \tfrac{1}{c^2\sigma^2}(udu+vdv+wdw)^2\right\} - \frac{c^2\sigma^2}{c^2\sigma^2}(du^2+dv^2+dw^2),$$

or, finally, it is

$$= -\frac{c^2\sigma^2}{2f_1(u)}\left\{f_1(du) - [d\sqrt{(f_1 u)}]^2\right\},$$

which is right.

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578. A Memoir on the Transformation of Elliptic Functions

[From the Philosophical Transactions of the Royal Society of London, vol. clxiv. (for the year 1874), pp. 397–456. Received January 14, 1873,—Read January 8, 1874.]

The theory of Transformation in Elliptic Functions was established by Jacobi in the *Fundamenta Nova* (1829); and he has there developed, transcendentally, with an approach to completeness, the general case, n an odd number, but algebraically only the cases $n=3$ and $n=5$; viz. in the general case the formulæ are expressed in terms of the elliptic functions of the nth part of the complete integrals, but in the cases $n=3$ and $n=5$ they are expressed rationally in terms of u and v (the fourth roots of the original and the transformed moduli respectively), these quantities being connected by an equation of the order 4 or 6, the modular equation. The extension of this algebraical theory to any value whatever of n is a problem of great interest and difficulty: such theory should admit of being treated in a purely algebraical manner; but the difficulties are so great that it was found necessary to discuss it by means of the formulæ of the transcendental theory, in particular by means of the expressions involving Jacobi's q (the exponential of $-\frac{\pi K'}{K}$), or say by means of the q-transcendents. Several important contributions to the theory have since been made:—Sohnke, "Equationes modulares pro transformatione functionum ellipticarum," *Crelle*, t. xvi. (1836), pp. 97–130, (where the modular equations are found for the cases $n=3$, 5, 7, 11, 13, 17, and 19); Joubert, "Sur divers équations analogues aux équations modulaires dans la théorie des fonctions elliptiques," *Comptes Rendus*, t. xlvii. (1858), pp. 337–345, (relating among other things to the multiplier equation for the determination of Jacobi's M); and Königsberger, "Algebraische Untersuchungen aus der Theorie der elliptischen Functionen," *Crelle*, t. lxxii. (1870), pp. 176–275; together with other papers by Joubert and by

Hermite in later volumes of the *Comptes Rendus*, which need not be more particularly referred to. In the present Memoir I carry on the theory, algebraically, as far as I am able; and I have, it appears to me, put the purely algebraical question in a clearer light than has hitherto been done; but I still find it necessary to resort to the transcendental theory. I remark that the case $n = 7$ (next succeeding those of the *Fundamenta Nova*), on account of the peculiarly simple form of the modular equation $(1-u^8)(1-v^8) = (1-uv)^8$, presents but little difficulty; and I give the complete formulæ for this case, obtaining them as in not algebraically as transcendentally; I also see the complete formulæ for the next succeeding prime value $n = 11$. For the sake of completeness it is necessary to reproduce Sohnke's modular equations, exhibiting them for greater clearness in a square form, and adding to them those for the non-prime cases $n = 9$ and $n = 15$; also a valuable table given by him for the powers of $f(q)$; and I give other tabular results which are of assistance in the theory.

The General Problem. Article Nos. 1 to 6.

1. Taking n a given odd number, I write

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{P-Qx}{P+Qx}\right)^2,$$

where P, Q are rational and integral functions of x, $P \pm Qx$ being each of them of the order $\frac{1}{2}(n-1)$, or, what is the same thing, $(1 \pm x)(P \pm Qx)^2$ being each of them of the order n; and I give to the same form in the case $n = \frac{1}{2}(n+1)$, viz. what is the same thing, $(1 \pm x)(P \pm Qx)^2$ being each of them of the order n; and in the second case the number is $(p+1) + (p+1)$, $= \frac{1}{2}(n+1)$, as before. Taking

$$P = a + \gamma x^2 + \varepsilon x^4 + \ldots, \quad Q = \beta + \delta x^2 + \zeta x^4 + \ldots,$$

the formula is

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{a - \beta x + \gamma x^2 - \ldots}{a + \beta x + \gamma x^2 + \ldots}\right)^2,$$

the number of coefficients in each of the equations remains thus explained. Starting herefrom I reproduce in the *Fundamenta Nova*, as follows.

2. If the coefficients are such that the equation remains true on when we therein change simultaneously x into $-x$, y into $-y$, then the variables x, y will satisfy the differential equation

$$\frac{M dy}{\sqrt{1-y^2 \cdot 1-\lambda^2 y^2}} = \frac{dx}{\sqrt{1-x^2 \cdot 1-k^2x^2}},$$

(M a constant, the value of which, as will appear, is given by $\dfrac{1}{M} = 1 + \dfrac{2\beta}{a}$); and the problem of transformation is thus to find the coefficients so as the equation may remain true on the above simultaneous change of x, y.

In fact, observing that the original equation and therefore the new equation are each satisfied on changing therein x into $-x$, $-y$, it follows that the equation may be written in either of the two forms

$$1-y = (1-x)\,A^2(+), \quad 1+y = (1+x)\,B^2(+),$$

$$1-\lambda y = (1-kx)\,C^2(+), \quad 1+\lambda y = (1+kx)\,D^2(+),$$

the common denominator being, say E, where A, B, C, D, E are all of them rational and integral functions of x; or say this being so, the differential equation will be satisfied.

3. To develop the condition, observe that the assumed equation gives

$$y = \frac{x(P^2 + 2PQ + Qx^2)}{P^2 + 2PQ{\cdot}x + Q^2x^2}, \quad = \frac{x\,\mathfrak{A}}{\mathfrak{B}} \text{ suppose,}$$

where $\mathfrak{A}$, $\mathfrak{B}$ are functions each of the degree $\frac{1}{2}(n-1)$ in x^2. (Hence, if with Jacobi $\dfrac{1}{M}$ denotes the value $(y+x)_{x=0}$, we have $\dfrac{1}{M} = 1 + \dfrac{2\beta}{a}$, as mentioned.)

Suppose in general U being any integral function $(1, x^2)^p$, we have

$$U^n = (kx)^n \left(1,\ \frac{1}{kx}\right)^p,$$

viz. let U^n be what U becomes when x is changed into $\dfrac{1}{kx}$ and the whole multiplied by $(kx)^p$.

The modular equation is thus standard form is in v, so that instead of λ we have ν to introduce the quantity $\lambda = \nu^4$, and writing $\mu = \nu^4$, $v = \nu^4$. The modular equation in standard form will give rise to an equation in standard form for the equation in u, v which will appear gives rise to an equation of the same order between u, v; say the value of u^2, $\sqrt{v}$, &c.; these Ωk-forms will be given presently.

4. I shall ultimately, instead of k, introduce Jacobi's u ($u = \sqrt[4]{k}$, $v = \sqrt[4]{\lambda}$); but it is to be remarked convenient to retain k, and instead of λ to introduce the quantity $\lambda = \nu^4$; the modular equation in u, v which will appear gives rise to an equation of the same order between u, v; say the value of Ω, by which I shall represent $\dfrac{1}{\Omega^{(n-1)}}\,\mathfrak{B}^*$. The modular equation is thus standard form in Ω connected with k by the equation $\lambda = \dfrac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{B}^*$. The series of equalities may be presented as follows, writing as above A to denote the function

$$\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots,$$

$$\frac{1}{A} = \frac{1}{\phi'}\cdot 1,$$

$$\frac{1}{A^2} = \frac{-1}{\phi'^3}\begin{vmatrix} \phi, & 1 \\ \phi', & \phi'' \end{vmatrix}\cdot\frac{1}{1},$$

$$\frac{1}{A^3} = \frac{+1}{\phi'^6}\begin{vmatrix} \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''' \\ (\phi')', & (\phi')'', & (\phi')''' \end{vmatrix}\cdot\frac{1}{1.1.2},$$

&c.

5. Suppose then, Ω being a constant, that we have identically

$$\mathfrak{A} = \frac{1}{\Omega^{\frac{1}{2}(n-1)}}\,\mathfrak{B}^*;$$

this implies

$$\mathfrak{B} = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{A}^*.$$

(In fact, if

$$\mathfrak{A} = a + cx^2 + \ldots + gx^{n-3} + hx^{n-1},$$
$$\mathfrak{B} = b + dx^2 + \ldots + rx^{n-3} + sx^{n-1},$$

then

$$\mathfrak{A}^* = a + ck^2x^2 + \ldots + gk^{n-3}x^{n-3} + hk^{n-1}x^{n-1},$$
$$\mathfrak{B}^* = b + dk^2x^2 + \ldots + rk^{n-3}x^{n-3} + sk^{n-1}x^{n-1},$$

and the assumed equation gives

$$a = \frac{1}{\Omega^{\frac{1}{2}(n-1)}}\,s,\quad c = \frac{k^2}{\Omega^{\frac{1}{2}(n-1)}}\,r,\ \ldots,\quad g = \frac{\Omega^{\frac{1}{2}(n-1)}}{k^{n-3}}\,d,\quad h = \frac{\Omega^{\frac{1}{2}(n-1)}}{k^{n-1}}\,b;$$

that is,

$$b = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,a,\quad d = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,c\,k^2,\ \ldots, r = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,g\,k^{n-3},\ s = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,h\,k^{n-1};$$

and therefore $\mathfrak{B} = \dfrac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{A}^*$.)

From these we may form an equation in Ω referred to; and then Ω denoting any one of this equation, the ratio of $\mathfrak{A}$ to $\mathfrak{B}^*$ is given, and indeed in a square form in v; these Ωk-forms will be given presently.

6. We may from the $\frac{1}{2}(n+1)$ equations eliminate the $\frac{1}{2}(n-1)$ ratios $\alpha : \beta : \gamma : \ldots$, thus obtaining an equation in Ω (involving of course the parameter k) which is to be referred to; and then Ω denoting any one of the Ωk-modular equation also referred to; and Ω denoting one of this equation, the ratio $\mathfrak{A} : \mathfrak{B}^*$ is determined, and consequently the value of y as given by the equation

$$\frac{1-y}{1+y} = \frac{(1-x)(P-Qx)^2}{(1+x)(P+Qx)^2}, \quad \text{or} \quad y = \frac{x(P^2+2PQ+Q^2x^2)}{P^2+2PQx+Q^2x^2}.$$

The entire problem thus depends on the solution of the system of $\frac{1}{2}(n+1)$ equations,

$$(P^2+2PQx+Q^2x^2)^* = \Omega k^{\frac{1}{2}(n-1)}\,(P^2+2PQx+Q^2x^2).$$

The Ωk-Modular Equations, $n = 3, 5, 7, 11$. Article No. 7.

7. For convenience of reference, and to fix the ideas, I give these results, calculated as above explained, from the standard or uv-form.

	k^2	k	1
Ω^2		$+1$	
Ω^0	-4		
Ω^3		$+6$	
Ω			-4
Ω^2		$+1$	
	-4	$+8$	-4

$= 0 \qquad n = 3$: $\Omega = 1$, we have $-4(k-1)^2 = 0$.

	k^4	k^3	k^2	k	1
Ω^6			$+1$		
Ω^5	-16		$+10$		
Ω^4			$+15$		
Ω^3			-20		
Ω^2			$+15$		
Ω			$+10$		-16
Ω^0			$+1$		
	-16	$+32$		-16	

$= 0 \qquad n = 5$: $\Omega = 1$, we have $-16(k^2-1)^2 = 0$.

A Memoir on the Transformation of Elliptic Functions

(Continuation of paper 578 — pp. 118–142 of Vol. IX)

[p. 118] Tables for the cases $n = 7$ and $n = 11$ *(continued)*

[Table: the second equation-system for the septic case $n = 7$. The rows are labelled by powers Ω^k ($k = 0, \dots, 8$) and the columns by $k^9, k^8, k^7, k^6, k^5, k^4, k,$ 1. The non-zero entries are arranged so that the equation reads, on summation by rows,]

$$\begin{aligned}
\Omega^0 &: \ +1,\\
\Omega^1 &: \ -64\,k^8 + 56\,k^6,\\
\Omega^2 &: \ -112\,k^7 + 140\,k^5,\\
\Omega^3 &: \ -112\,k^6 + 56\,k^4,\\
\Omega^4 &: \ +70\,k^5,\\
\Omega^5 &: \ +56\,k^4 - 112\,k^2,\\
\Omega^6 &: \ +140\,k^3 - 112\,k,\\
\Omega^7 &: \ +56\,k^2 - 64,\\
\Omega^8 &: \ +1,
\end{aligned}$$

the column-sums of which (read in the order $k^9, k^8, k^7, k^6, k^5, k^4, k, 1$) are

$$-64,\ -112,\ 0,\ +352,\ 0,\ -112,\ -64,\ = 71 \cdot 11;$$

that is, $\Omega = 1$, we have

$$-16(k-1)^2(4k^2+3k+1)(k^2+3k+4) = 0.$$

[Table: the equation-system for $n = 11$, endecadic transformation. Rows are labelled $\Omega^0, \dots, \Omega^{12}$ and columns by $k^{10}, k^9, k^8, k^7, k^6, k^5, k^4, k^3, k^2, k, 1$. The non-zero entries, row by row, are:]

$$\begin{aligned}
\Omega^0 &: \ +1,\\
\Omega^1 &: \ -1024\,k^{10} + 1408\,k^9 - 396\,k^8 - 12,\\
\Omega^2 &: \ -5632\,k^9 + 4400\,k^8 + 1298\,k^7 + 66,\\
\Omega^3 &: \ -16192\,k^8 + 16368\,k^7 - 396\,k^5 - 220,\\
\Omega^4 &: \ -18656\,k^7 + 19151\,k^6 + 495,\\
\Omega^5 &: \ -16016\,k^6 - 1144\,k^5 + 16368\,k^4 - 792,\\
\Omega^6 &: \ +4400\,k^5 - 7876\,k^4 + 4400\,k^3 + 924,\\
\Omega^7 &: \ +16368\,k^4 - 1144\,k^3 - 16016\,k^2 - 792,\\
\Omega^8 &: \ +19151\,k^3 - 18656\,k^2 + 495,\\
\Omega^9 &: \ -396\,k^4 + 16368\,k^2 - 16192\,k - 220,\\
\Omega^{10} &: \ +1298\,k^3 + 4400\,k - 5632 + 66,\\
\Omega^{11} &: \ -396\,k^2 + 1408 - 1024 - 12,\\
\Omega^{12} &: \ +1,
\end{aligned}$$

the column-sums of which (read across the columns $k^{10}, k^9, \dots, k, 1$) are

$$\begin{aligned}
&-1024,\ -32208,\ -18656,\ -1936,\ -7876,\ -1936,\ -18656,\ -32208,\ -1024,\\
&+1408,\ +8800,\ +32736,\ +40900,\ +32736,\ +8800,\ +1408,\\
&-5632,\ -30800,\ -9856,\ +30800,\ +33024,\ +30800,\ -9856,\ -30800,\ -5632,\\
&-1024 + 94624.
\end{aligned}$$

[p. 119] Equation-systems for the cases $n = 3, 5, 7, 9, 11$. Art. Nos. 8 to 10.

8. $n = 3$, cubic transformation. $k = u^4$, $\Omega = \dfrac{u^9}{v}$ (here and in the other cases).

$P = \alpha$, $Q = \beta$. The condition here is

$$k^2\alpha^2\alpha^2 + (2\alpha\beta + \beta^2)^2 = \Omega\big[(\alpha^2 + 2\alpha\beta)^2 + \beta^2\alpha^2\big],$$

and the system of equations thus is

$$k\alpha^2 = \Omega\beta^2, \qquad 2\alpha\beta + \beta^2 = \Omega k\,(\alpha^2 + 2\alpha\beta),$$

and similarly in the other cases; for these it will be enough to write down the equation-systems.

$n = 5$, quintic transformation.

$$P = \alpha + \gamma x^2, \quad Q = \beta.$$
$$k^2\alpha^2 = \Omega\gamma^2,$$
$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \Omega(2\beta\gamma + 2\beta\gamma + \beta^2),$$
$$\gamma^2 + 2\beta\gamma = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 7$, septic transformation.

$$P = \alpha + \gamma x^2, \quad Q = \beta + \delta x^2.$$
$$k^2\alpha^2 = \Omega\delta^2,$$
$$k\,(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(\gamma^2 + 2\gamma\delta + 2\beta\delta),$$
$$\gamma^2 + 2\beta\gamma + 2\alpha\delta + 2\beta\delta = \Omega k\,(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$
$$\delta^2 + 2\gamma\delta = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 9$, enneadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2.$$
$$k^2\alpha^2 = \Omega\epsilon^2,$$
$$k\,(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(2\gamma\epsilon + 2\alpha\delta + \delta^2),$$
$$2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta = \Omega(2\alpha\epsilon + \gamma^2 + 2\gamma\delta + 2\delta\beta + 2\beta\delta),$$
$$2\gamma\epsilon + 2\gamma\delta + 2\epsilon\beta + \delta^2 = \Omega k\,(2\beta\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$
$$\epsilon^2 + 2\delta\epsilon = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 11$, endecadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2 + \zeta x^4.$$
$$k^2\alpha^2 = \Omega\zeta^2,$$
$$k\,(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(\epsilon^2 + 2\epsilon\zeta + 2\delta\zeta),$$
$$k\,(2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta) = \Omega(2\gamma\epsilon + 2\gamma\zeta + 2\epsilon\delta + 2\beta\zeta + \delta^2),$$
$$2\gamma\epsilon + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\zeta + \delta^2 = \Omega k\,(2\alpha\epsilon + \gamma^2 + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\delta),$$
$$\epsilon^2 + 2\gamma\zeta + 2\epsilon\delta + 2\delta\zeta = \Omega k^2(2\alpha\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$
$$2\epsilon\zeta + \zeta^2 = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

And so on.

[p. 120]

9. It will be noticed that if the coefficients of $P+Qx$ taken in order are

$$\alpha,\ \beta,\ \gamma,\ \ldots,\ \sigma,\ \tau,$$

then in every case the first and last equations are

$$k\alpha^2=\Omega\tau^2,\qquad \tau^2+2\rho\sigma\tau=\Omega^{\frac{1}{2}(n-1)}k^{\frac{1}{2}(n-1)}\big(\alpha^2+2\alpha\beta\big).$$

Putting in the first of these $k=u^4$, $\Omega=\dfrac{u^n}{v}$, the equation becomes

$$\frac{u^4\alpha^2}{v}\ =\ \frac{u^n}{v}\tau^2,$$

where each side is a perfect square; and in extracting the square root we may without loss of generality take the roots positive, and write $\tau=u^{\frac{1}{2}(n-4)}\alpha=v\alpha$.

This speciality, although it renders it proper to employ ultimately u,v in place of k,Ω, produces really no depression of order (viz. the Ωk-form of the modular equation is found to be of the same order in Ω that the standard or uv-form is in v), and is in another point of view a disadvantage, as destroying the uniformity of the several equations: in the discussion of order I consequently retain Ω,k. Ultimately these are to be replaced by u,v; the change in the equation-systems is so easily made that it is not necessary here to write them down in the new form in u,v.

10. The case $\alpha=0$ has to be considered in the discussion of order, but we have thus only solutions which are to be rejected; in the proper solutions α is not $=0$, and it may therefore for convenience be taken to be $=1$. We have then $\sigma=u^{n-4}\div v$.

The last equation becomes therefore

$$\tau^2+2\rho\sigma\tau\ =\ \Omega^{\frac{1}{2}(n-1)}k^{\frac{1}{2}(n-1)}\,(1+2\beta),$$

or recollecting that β is connected with the multiplier M by the relation $\dfrac{1}{M}=1+2\beta$, that is,

$$2\beta=\frac{1}{M}-1,$$

and substituting for $1+2\beta$ its value, the equation becomes

$$2\rho=u^{4(n-1)}\left(\frac{1}{M}-\frac{u^n}{v^n}\right);$$

that is, the first and the last coefficients are 1, $\dfrac{u^{n-4}}{v}$, and the second and the penultimate coefficients are each expressed in terms of v,M. The cases $n=3$, $n=5$ are so far peculiar, that the only coefficients are α,β, or α,β,γ; in the next case $n=7$, the only coefficients are $\alpha,\beta,\gamma,\delta$, and we have in this case all the coefficients expressed as above.

[p. 121] The Ωk-form — Order of the Systems. Art. Nos. 11 to 22.

11. In the general case, n an odd number, we have Ω, and $\frac{1}{2}(n+1)$ coefficients connected by a system of $\frac{1}{2}(n+1)$ equations of the form

$$\Omega=\frac{U}{U'}=\frac{V}{V'}=\cdots,$$

where $U,V,\ldots,U',V',\ldots$ are given quadric functions of the coefficients. Omitting the Ω (=), there remains a system of $\frac{1}{2}(n-1)$ equations of the form $\dfrac{U}{U'}=\dfrac{V}{V'}=\cdots$, or say

$$\left|\begin{matrix}U, & V, & W,\ldots\\ U', & V', & W',\ldots\end{matrix}\right|=0,$$

which determine the ratios $\alpha : \beta : \gamma : \ldots$ of the coefficients; and to each set of ratios there corresponds a single value of Ω. The order of the system, or number of sets of ratios, is $= \frac{1}{2}(n+1) \cdot 2^{\frac{1}{2}(n-1)}$, $= \frac{1}{2}(n+1) \cdot 2^{\frac{1}{2}(n-1)}$; and this is consequently the number of values of Ω, or the order of the equation for the determination of Ω; viz. but for reduction, the order in Ω of the Ωk-modular equation would be $= (n+1) \cdot 2^{\frac{1}{2}(n-1)}$. In the case $n = 3$, this is 4, which is right, but for any larger value of n the order is far too high; in fact, assuming (as the case is) that the order is equal to the order in v of the uv-form, the order should for a prime value of n be $= n+1$, and for a composite value not containing any square factor be $=$ the sum of the divisors of n. I do not attempt a general investigation, but confine myself to showing in what manner the reductions arise.

12. I will first consider the cubic transformation; here, writing for convenience $\dfrac{\alpha}{\beta} = \theta$, the equations give

$$\frac{k\theta^2}{2\theta+1} = \frac{1}{k(\theta^2+2\theta)}, \quad \text{that is,} \quad k^2\theta^2(\theta+2) - (2\theta+1) = 0,$$

and

$$k\theta^2 = \Omega.$$

The equation in θ gives $(k^2\theta^2-1)^2 - 4\theta\,(k^2\theta^2-1)^2 = 0$, and we have thence

$$k(\Omega^2-1)^2 = 4\Omega\,(k\Omega-1)^2,$$

that is, $k(\Omega^2-1)^2 - 4\Omega(k\Omega-1)^2 = 0$,

$$k\Omega^4 - 4k\Omega^3 + 6k\Omega^2 - 4\Omega + k = 0,$$

the modular equation; and then $-1 + k^2\theta^2 \mid 2\theta\,(k^2\theta^2 - 1) = 0$, that is, $\beta = 1 + 2\theta(k\Omega - 1) = 0$, or $2\theta = -\dfrac{1}{k\Omega - 1}$; which is to say we have $\alpha = \beta = -1$, $\beta = 2(1 - k\Omega)$; consequently

$$\lambda = \Omega k, \quad \text{and} \quad \frac{1-y}{1+y} = \frac{1-x}{1+x}\left[\frac{P-Qx}{P+Qx}\right] = \frac{1-x}{1+x}\left[\frac{-1+2(k\Omega-1)x}{-1-2(k\Omega-1)x}\right],$$

which completes the theory.

[p. 122]

13. Reproducing for this case the general theory, it appears *à priori* that Ω is determined by a quartic equation; in fact, from the original equations eliminating Ω, we have an equation

$$\begin{vmatrix} U, & V \\ U', & V' \end{vmatrix} = 0,$$

where U, U', V, V' are quartic functions of α, β; that is, the ratio $\alpha : \beta$ has four values, and to each of these there corresponds a single value of Ω; viz. Ω is determined by a quartic equation.

14. Considering next the case $n = 5$, the quintic transformation; the elimination of Ω gives the equations

$$\frac{U}{U'} = \frac{V}{V'} = \frac{W}{W'},$$

where U, U', &c. are all quadric functions of α, β, γ. We have thence $4 \cdot 4 - 2 \cdot 2 = 12$ sets of values of $\alpha : \beta : \gamma$; viz. considering α, β, γ as coordinates *in piano*, the curves $UV' - U'V = 0$, $UW' - U'W = 0$ are quartic curves intersecting in 16 points; but among these are included the four points $\alpha = 0$, $U = 0$ (in fact, the point $\alpha = 0$, $U' = 0$ four times), which are not points of the curve $VW' - V'W = 0$; there remain therefore $16 - 4$, $= 12$ intersections, agreeing with the general value $(n+1) \cdot 2^{\frac{1}{2}(n-1)}$. Hence Ω is in the first instance determined by an equation of the order 12; but the proper order being $= 6$, there must be a factor of the order 6 to be rejected. To explain this and to determine the factor, observe that the equations in question are

$$k^2\alpha^2(2\beta\gamma + 2\beta\gamma + \beta^2) - \gamma^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = 0,$$

$$k^2\alpha^2(\alpha+2\beta)-\gamma^2(\gamma+2\beta)=0;$$

at the point $\alpha=0,\ \gamma=0$, the first of these has a double point, the second a triple point; or there are at the point in question 6 intersections; but 4 of these are the points which give the foregoing reduction $16-4=12$; we have thus the point $\alpha=0,\ \gamma=0$, counting twice among the twelve points. Writing in the two equations $\beta=0$, the equations become $k^2\alpha^2\gamma^2-\alpha^2\gamma^2=0$, $k^2\alpha^4-\gamma^4=0$, viz. these will be satisfied if $k^2\alpha^2-\gamma^2=0$, that is, the curves pass through each of the two points $(\beta=0,\ \gamma=\pm k\alpha)$, and these values satisfy (as in fact they should) the third equation

$$k^2(2\alpha\gamma+2\alpha\beta+\beta^2)\,\alpha\,(\alpha+2\beta)-\gamma(\gamma+2\beta)\,(2\alpha\gamma+2\beta\gamma+\beta^2)=0.$$

It is moreover easily shown that the three curves have at each of the points in question a common tangent; viz. taking A,B,C as current coordinates, the tangent at the point (α,β,γ) of the second curve has for its equation

$$A\,(2\alpha^3+3\alpha^2\beta)\,k^2+B\,(k^2\alpha^3-\gamma^3)+C\,(-2\gamma^3-3\gamma^2\beta)=0;$$

and for $\beta=0,\ \gamma=\pm k\alpha$, this becomes $2k\,A+B(k+1)+2C=0$, viz. this is the line from the point $(\beta=0,\ \gamma=\pm k\alpha)$ to the point $(1,-2,1)$. And similarly for the other two curves we find the same equation for the tangent.

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Hence among the 12 points are included the point $(\gamma=0,\ \alpha=0)$ twice, and the points $(\beta=0,\ \gamma=\pm k\alpha)$ each twice: we have thus a reduction $=6$.

15. Writing in the equations $\gamma=0,\ \alpha=0$, the first and third are satisfied identically, and the second becomes $\beta^2=\Omega\beta^2$, that is, the equations give $\Omega=1$; writing $\beta=0$, they become

$$k^2\alpha^2=\Omega\gamma^2,\quad \alpha\gamma=\Omega\alpha\gamma,\quad \gamma^2=\Omega k^2\alpha^2,$$

viz. putting herein $\gamma^2=k^2\alpha^2$, the equations again give $\Omega=1$; hence the factor of the order 6 is $(\Omega-1)^6$, and the equation of the twelfth order for the determination of Ω is

$$(\Omega-1)^6\{(\Omega,1)^6\}=0,$$

where $(\Omega,1)^6=0$ is the Ωk-modular equation above written down.

16. Reverting to the equation

$$\frac{1-y}{1+y}=\frac{1-x}{1+x}\left(\frac{P-Qx}{P+Qx}\right)^2,$$

it is to be observed that for $\alpha=0,\ \gamma=0$, that is, $P=0$, this becomes simply $y=x$, which is the transformation of the order 1; the corresponding value of the modulus λ is $\lambda=k$, and the equation $\lambda=\Omega^2k$ then gives $\Omega^2=1$, which is replaced by $\Omega-1=0$.

If in the same equation we write $\beta=0$, that is, $Q=0$, then (without any use of the equation $\gamma^2=k^2\alpha^2$) we have $y=x$, the transformation of the order 1; but although this is so, the fundamental equation

$$(P^2+2PQx^2+Q^2x^4)^2\;=\;\Omega^2k\,(P^4+2PQ+Q^2x^4),$$

which, putting therein $Q=0$, becomes $(P^2)^2=\Omega^2k^{-1}P^4$, that is, $(k^2\alpha+\gamma)^2=\Omega^2k^{-1}(\alpha+\gamma x)^2$, is not satisfied by the single relation $\Omega-1=0$, but necessitates the further relation

$$\gamma^2=k^2\alpha^2.$$

The thing to be observed is that the extraneous factor $(\Omega-1)^2$, equated to zero, gives for Ω the value $\Omega=1$ corresponding to the transformation $y=x$ of the order 1.

17. Considering next $n=7$, the septic transformation; we have here between $\alpha,\beta,\gamma,\delta$ a fourfold relation of the form

$$\left|\begin{matrix} U, & V, & W, & Z \\ U', & V', & W', & Z' \end{matrix}\right|=0,$$

where, as before, U, V, &c. are quadric functions, and the number of solutions is here $= 8 \cdot 2^3$, $= 32$; to each of these corresponds a single value of Ω, or Ω is in the first instance determined by an equation of the order 32. But the order of the modular equation is $= 8$; or representing this by $\{(\Omega, 1)^8\} = 0$, the equation must be

$$(\Omega - 1)^{24} \{(\Omega, 1)^8\} = 0,$$

viz. there must be a special factor of the order 24.

[p. 124]

18. One way of satisfying the equations is to write therein $\alpha = 0$, $\delta = 0$; the equations thus become

$$k\beta^2 = \Omega\gamma^2, \qquad \gamma^2 + 2\beta\gamma = \Omega k\,(2\beta\gamma + \beta^2);$$

or putting $\beta = \alpha'$, $\gamma = \beta'$,

$$k\alpha'^2 = \Omega\beta'^2, \qquad \beta'^2 + 2\alpha'\beta' = \Omega k\,(\alpha'^2 + \alpha'\beta'),$$

which (with α', β' instead of α, β) are the very equations which belong to the cubic transformation; hence a factor is $\{(\Omega, 1)^4\}$.

Observe that for the values in question $\alpha = 0$, $\delta = 0$, $P = \gamma x^2$, $Q = \beta$, $= \alpha'$, and therefore $P, Q = (\beta - Qx)^2 = \alpha'^2(\beta' - \beta' x)^2, = \alpha'^2(P' - Q'x)^2$, if $P' = \alpha'$, $Q' = \beta'$,

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{P' - Q'x}{P' + Q'x}\right)^2,$$

which is the formula for a cubic transformation.

19. The equations may also be satisfied by writing therein $\gamma = k\alpha$, $\delta = k\beta$; in fact, substituting these values, they become

$$\begin{gathered} k^2\alpha^2 = \Omega k^2\beta^2, \\ 2k^2\alpha^2 + k(2\alpha\beta + \beta^2) = \Omega k^2(\alpha^2 + 2\alpha\beta) + 2\Omega k\beta^2, \\ k^2\alpha^2 + 2k(\beta^2 + 2\alpha\beta) = 2\Omega k^2(\beta^2 + 2\alpha\beta) + \Omega k\beta^2, \\ k\,(\beta^2 + 2\alpha\beta) = \Omega k\,(\alpha^2 + 2\alpha\beta); \end{gathered}$$

the first and last of these are

$$k\alpha^2 = \Omega\beta^2, \qquad \beta^2 + 2\alpha\beta = \Omega k\,(\alpha^2 + 2\alpha\beta),$$

which being satisfied the second and third equations are satisfied identically; and these are the formulae for a cubic transformation; that is, we again have the factor $\{(\Omega, 1)^4\}$.

Observe that for the values in question $\gamma = k\alpha$, $\delta = k\beta$, we have $P = \alpha(1 + kx^2)$, $Q = \beta(1 + kx^2)$; so that, writing $P' = \alpha$, $Q' = \beta$, we have for y the value

$$\frac{1-y}{1+y} = \frac{(1-x)}{(1+x)}\frac{(P' - Q'x)^2}{(P' + Q'x)^2},$$

which is the formula for a cubic transformation.

20. It is important to notice that we cannot by writing $\alpha = 0$ or $\delta = 0$ reduce the transformation to a quintic one; in fact, the equation $k^2\alpha^2 = \Omega\delta^2$ shows that if either of these equations is satisfied the other is also satisfied; and we have then the foregoing case $\alpha = 0$, $\delta = 0$, giving not a quintic but a cubic transformation.

And for the same reason we cannot by writing $\alpha = 0$, $\beta = 0$, $\gamma = 0$ or $\beta = 0$, $\gamma = 0$, $\delta = 0$ reduce the transformation to the order 1. There is thus no factor $\Omega - 1$.

[p. 125]

21. As regards the non-existence of the factor $\Omega - 1$, I further verify this by writing in the equations $\Omega = 1$; they thus become

$$k\alpha^2 = \delta^2,$$
$$k\,(2\alpha\gamma + 2\alpha\beta + \beta^2) = \gamma^2 + 2\gamma\delta + 2\beta\delta,$$
$$\gamma^2 + 2\beta\gamma + 2\alpha\delta + 2\beta\delta = k\,(2\alpha\gamma + \gamma^2 + 2\alpha\delta + \beta^2),$$
$$\delta^2 + 2\gamma\delta = k\,(\alpha^2 + 2\alpha\beta),$$

which it is to be shown cannot be satisfied in general, but only for certain values of k. Reducing the last equation, this is $\gamma\delta = k\,\alpha\beta$, which, combined with the first, gives $\alpha = \pm\sqrt{k}\,\alpha\gamma = \beta\delta$; and if for convenience we assume $\alpha = 1$, and write also $k = \theta^2$ (that is, $k = \theta^2$), then the values of $\alpha, \beta, \gamma, \delta$ are $\alpha = 1,\ \beta = \gamma\theta^{-1},\ \gamma = \gamma,\ \delta = \theta$; which values, substituted in the second and the third equations, give two equations in γ, θ; and from these, eliminating γ, we obtain an equation for the determination of θ, that is, of k. In fact, the second equation gives

$$\theta^2\,(2\gamma + 2\gamma\theta^{-1} + \gamma^2\theta^{-2}) \ = \ \gamma^2 + 2\gamma\theta + 2\gamma,$$

or, dividing by γ and reducing,

$$\gamma\,(1 - \theta^2) = 2\theta^2(\theta^2 - 1)(\theta^2 - \theta + 1),$$

that is,

$$\gamma\,(1 + \theta^2) = -2\theta^2(\theta^2 - \theta + 1),$$

or, as this may also be written,

$$(\gamma + \theta^2)(1 + \theta^2) = -\theta^2(\theta - 1)^2,$$

that is,

$$\gamma + \theta^2 \ = \ \frac{-\theta^2(\theta - 1)^2}{\theta^2 + 1}.$$

Moreover the third equation gives

$$\gamma^2 + 2\gamma^2\theta^{-1} + 2\theta^2 + 2\gamma \ = \ \theta^2\,(2\gamma + 2\gamma^2\theta + 2\theta + \gamma^2\theta^{-1}),$$

that is,

$$\gamma^2\,(\theta^4 - 2\theta^3 + 2\theta - 1) - 2(\gamma + \theta^2)\,\theta^2(\theta^2 - 1) = 0;$$

or dividing by $\theta^2 - 1$, it is

$$\gamma^2\,(\theta - 1)^2 = 2\theta\,(\gamma + \theta^2);$$

whence also

$$\gamma \ = \ \frac{-2\theta}{(\theta - 1)^2}.$$

Also

$$4\theta^2(\theta^2 - \theta + 1)^2 = \gamma^2\,(\theta^2 + 1)^2,$$

wherefore

$$2(\theta^2 - \theta + 1)^2 = -\theta(\theta^2 + 1) \quad \text{or} \quad 2(\theta^2 - \theta + 1)^2 + \theta(\theta^2 + 1) = 0,$$

or

$$\theta(\theta^2 + 1)^2 + 2(\theta^2 - \theta + 1)^2 = 0,$$

that is,

$$2\theta^4 - 3\theta^3 + 6\theta^2 - 2\theta + 2 = 0,$$

or finally

$$(2\theta^2 - \theta + 1)(\theta^2 - \theta + 2) = 0.$$

[p. 126]

We have thus $(2\theta+1)^2=\theta$, that is, $4\theta^2+3\theta+1=0$ or $4k^2+3k+1=0$, or else $(\theta^2+2)^2=\theta$, that is, $\theta^4+3\theta^2+4=0$ or $k^2+3k+4=0$; viz. the equation in k is

$$(4k^2+3k+1)(k^2+3k+4)=0,$$

these being in fact the values of k given by the modular equation on putting therein $\Omega=1$.

The equation of the order 32 thus contains the factor $\{(\Omega,1)^4\}$ at least twice, and it does not contain either the factor $\Omega-1$, or the factor $\{(\Omega,1)^6\}$ belonging to the quintic transformation; it may be conjectured that the factor $\{(\Omega,1)^4\}$ presents itself six times, and that the form is

$$\left\{(\Omega,1)^4\right\}^6\left\{(\Omega,1)^8\right\}=0;$$

but I am not able to verify this, and I do not pursue the discussion further.

22. The foregoing considerations show the grounds of the difficulty of the purely algebraical solution of the problem; the required results, for instance the modular equation, are obtained not in the simple form, but accompanied with special factors of high order. The transcendental theory affords the means of obtaining the results in the proper form without special factors; and I proceed to develop the theory, reproducing the known results as to the modular and multiplier equations, and extending it to the determination of the transformation-coefficients $\alpha,\beta,\ldots$.

[p. 126] The Modular Equation. Art. Nos. 23 to 28.

23. Writing, as usual, $q=e^{-\pi K'/K}$, we have u, a given function of q, viz.

$$u=\sqrt{2}\,q^{\frac{1}{8}}\,\frac{1+q\cdot 1+q^3\cdot 1+q^5\cdots}{1+q\cdot 1+q^2\cdot 1+q^3\cdots},$$

$$\begin{aligned}v&=\sqrt{2}\,q^{\frac{1}{8}}\left(1-q+2q^2-3q^3+4q^4-6q^5+9q^6-14q^7+\cdots\right)\\&=\sqrt{2}\,q^{\frac{1}{8}}f(q)\text{ suppose,}\end{aligned}$$

and this being so, the several values of v and of the other quantities in question are all given in terms of q.

The case chiefly considered is that of n an odd prime; and unless the contrary is stated it is assumed that this is so. We have then $n+1$ transformations corresponding to the same number $n+1$ of values of v; these may be distinguished as $v_0,v_1,v_2,\ldots,v_n$; viz. writing α to denote an imaginary nth root of unity, we have

$$\begin{gathered}v_0=(-)^{\frac{n^2-1}{8}}\sqrt{2}\,q^{\frac{n}{8}}f(q^n),\quad v_1=\sqrt{2}\,(\alpha q^{\frac{1}{n}})^{\frac{1}{8}}f(\alpha q^{\frac{1}{n}}),\\ v_2=\sqrt{2}\,(\alpha^2 q^{\frac{1}{n}})^{\frac{1}{8}}f(\alpha^2 q^{\frac{1}{n}}),\ \&c.,\\ v_n=\sqrt{2}\,q^{\frac{1}{8n}}f(q^{\frac{1}{n}}).\end{gathered}$$

(Observe $(-)^{\frac{n^2-1}{8}}=+1$ for $n=8p\pm1$, $=-1$ for $n=8p\pm3$.)

[p. 127]

The occurrence of the fractional exponent $\frac{1}{8}$ is, as will appear, a circumstance of great importance; and it will be convenient to introduce the term "octicity," viz. an expression of the form $q^{\frac{f}{8}}F(q)$ ($f=0$, or a positive integer not exceeding 7, $F(q)$ a rational function of q) may be said to be of the octicity f.

24. The modular equation is of course

$$(v-v_0)(v-v_1)\cdots(v-v_n)=0;$$

say this is

$$v^{n+1}-Av^n+Bv^{n-1}-\cdots=0,$$

so that $A = \Sigma v_0$, $B = \Sigma v_0 v_1$, &c. In the development of these expressions, the terms having a fractional exponent, with denominator n, would disappear of themselves, as involving symmetrically the several nth roots of unity; and each coefficient would be of the form $q^{\frac{a}{8}} F(q)$, F a rational and integral function of q. It is moreover easy to see that, for the several coefficients $A, B, C, \ldots$, g will denote the positive residue (mod. 8) of $n, 2n, 3n, \ldots$ respectively.

Hence assuming, as the fact is, that these coefficients are severally rational and integral functions of q, it follows that the form is

$$au^g + bu^{g+8} + cu^{g+16} + \cdots ,$$

g having the foregoing values for the several coefficients respectively. And it being known that the modular equation is as regards u of the order $= n + 1$, there is a known limit to the number of terms in the several coefficients respectively. We have thus for each coefficient an identity of the form

$$A = au^g + bu^{g+8} + \cdots ,$$

A and u being each of them given in terms of q, the values of the numerical coefficients $a, b, \ldots$ can be determined; and we thus arrive at the modular equation.

25. It is in effect in this manner that the modular equations are calculated in Sohnke's Memoir. Various relations of symmetry in regard to (u, v) and other known properties of the modular equation are made use of in order to reduce the number of the unknown coefficients to a minimum; and (what in practice is obviously an important simplification) instead of the coefficients Σv_0, $\Sigma v_0 v_1$, &c., it is the sums of powers Σv_0, Σv_0^2, &c., which are compared with their expressions in terms of u, in order to the determination of the unknown numerical coefficients $a, b, \ldots$. The process is a laborious one (although less so than perhaps might beforehand have been imagined), involving very high numbers; it requires the development up to high powers of q, of the high powers of the before-mentioned function $f(q)$; and Sohnke gives a valuable Table, which I reproduce here; adding to it the three columns which relate to ϕq.

[p. 128] Sohnke's Table (with ϕq columns added)

[Table: Sohnke's Table of the developments in powers of q of $f^N(q)$ for $N = 3, 5, 7, 9, 11, 13$, and of ϕq in three additional columns. The table is here transcribed in landscape format; rows are indexed by ind. of q ($0, 1, 2, \ldots, 25$) and each column shows the coefficient of the corresponding power of q in the development of the function indicated. Asterisked footnote: "(Wrongly given by Sohnke as $+34805616$)."]

ind. of q	ϕq	ϕq^2	ϕq^4	$f^3 q$	$f^5 q$	$f^7 q$	$f^9 q$	$f^{11} q$	$f^{13} q$	$f^{15} q$	$f^{17} q$	$f^{19} q$
0	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1
1	+2	0	0	−3	−5	−7	−9	−11	−13	−15	−17	−19
2	+2	+2	0	0	+5	+14	+27	+44	+65	+90	+119	+152
3	0	0	0	+5	+5	+7	−9	−55	−130	−238	−380	−552
4	+2	0	+2	0	−15	−49	−90	−110	−65	+90	+391	+874
5	0	0	0	−7	+5	+50	+126	+253	+428	+621	+782	+836
6	+4	+2	0	0	+15	+24	−78	−264	−560	−940	−1356	−1551
7	0	0	0	+3	−30	−141	−322	−440	−260	+450	+1751	+3704
8	+2	0	0	+15	−10	−49	+108	+803	+2275	+4860	+8755	+13889
9	0	0	0	−30	+60	+343	+1056	+2255	+3978	+5984	+7565	+8038
10	+4	+2	+2	0	+5	+105	+126	−484	−2860	−7800	−15691	−26145
11	0	0	0	+22	−44	−176	−855	−2783	−6760	−13325	−22236	−31958
12	+4	0	0	−30	−25	+147	+1437	+5302	+12870	+24115	+36490	+44574
13	0	0	0	−30	+150	+575	+1485	+2310	+1820	−2730	−15470	−42636
14	+4	0	0	+10	−25	−630	−2925	−7700	−15730	−26280	−36465	−39738
15	0	0	0	+90	−126	−651	−1480	+275	+9100	+30856	+72475	+139194
16	+4	+2	0	−44	+200	+875	+4995	+14157	+31525	+59565	+95485	+126465
17	0	0	0	−77	+275	+1729	+6390	+18887	+45045	+91200	+158015	+233358
18	+4	0	+2	+0	−225	−1407	−7479	−26015	−67925	−149600	−282115	−451250
19	0	0	0	+132	−220	−1995	−9504	−30855	−77220	−159510	−280440	−422256
20	+6	0	0	−110	+250	+2128	+10934	+39226	+109110	+253935	+512435	+918680
21	0	0	0	−115	+675	+2870	+10560	+27357	+50500	+59450	+30450	−67450
22	+4	+2	0	−160	+0	+1638	+10395	+39446	+114550	+277200	+583240	+1097290
23	0	0	0	+198	−825	−3990	−16236	−58300	−176580	−449820	−996300	−1982880
24	+4	0	0	+165	−825	−4500	−18810	−69685	−216905	−589580	−1387845	−2901620
25	0	0	0	−280	+475	+5390	+25025	+88825	+255300	+656535	+1494125	+3060180

∗ Wrongly given by Sohnke as +34805616.

[p. 130] Series of modular equations (after Sohnke).

26. I give from Sohnke the series of modular equations, adding those for the composite cases $n = 9$ and $n = 15$, as to which see the remarks which follow the Table.

[Table: modular equation for $n = 3$. The grid has rows $u^4, u^3, u^2, u, 1$ and columns $v^4, v^3, v^2, v, 1$; the non-zero entries are]

$$\begin{aligned} u^4 &: \quad -1 \text{ at column } 1, \\ u^3 &: \quad +2 \text{ at column } v^2, \\ u^2 &: \quad 0, \\ u^1 &: \quad -2 \text{ at column } v^2, \\ 1 &: \quad +1 \text{ at column } v^4. \end{aligned}$$

The column-sum reads

$$1 \cdot v^4 + 2 \cdot v^2 \cdot u^3 + 0 - 2 \cdot v^2 u - 1 \cdot u^4 = (v + 1)^3(v - 1).$$

[Table: modular equation for $n = 5$. Rows $u^6, u^5, u^4, u^3, u^2, u, 1$, columns $v^6, v^5, v^4, v^3, v^2, v, 1$; the non-zero entries are]

$$\begin{aligned} u^6 &: \quad -1 \text{ at column } 1, \\ u^5 &: \quad +4 \text{ at column } v^2, \\ u^4 &: \quad -5 \text{ at column } v^4, \\ u^3 &: \quad +5 \text{ at column } v^3, \\ u^2 &: \quad -4 \text{ at column } v^5, \\ u &: \quad +1 \text{ at column } v^6. \end{aligned}$$

The full equation reads

$$1 + 4 + 5 + 0 - 5 - 4 - 1 = (v + 1)^5(v - 1).$$

[Table: modular equation for $n = 7$. Rows $u^8, \ldots, u, 1$, columns $v^8, \ldots, v, 1$; the non-zero entries form a skew-diagonal pattern with binomial-like values:]

$$\begin{aligned} u^8 &: \quad +1 \text{ at column } 1, \\ u^7 &: \quad -8 \text{ at column } v^2, \\ u^6 &: \quad +28 \text{ at column } v^4, \\ u^5 &: \quad -56 \text{ at column } v^5, \\ u^4 &: \quad +70 \text{ at column } v^4, \\ u^3 &: \quad -56 \text{ at column } v^5, \\ u^2 &: \quad +28 \text{ at column } v^6, \\ u &: \quad -8 \text{ at column } v^7, \\ 1 &: \quad +1 \text{ at column } v^8. \end{aligned}$$

The complete column-sum reads

$$+1 - 8 + 28 - 56 + 70 - 56 + 28 - 8 + 1 = (v - 1)^8.$$

[p. 131]

[Table: modular equation for $n = 9$. Rows $u^{12}, u^{11}, \ldots, u, 1$ and columns $v^{12}, v^{11}, \ldots, v, 1$. The diagonal pattern of non-zero entries is]

u^{12} : $+1$ at col 1,
u^{11} : -8 at col v^3,
u^{10} : $+16$ at col v^6,
u^9 : -16 at col v^8, $+10$ at col v^4,
u^8 : $+8$ at col v^7, -16 at col v^5,
u^7 : -16 at col v^6, $+15$ at col v^4,
u^6 : $+48$ at col v^5, -84 at col v^3,
u^5 : -84 at col v^4, $+48$ at col v^6,
u^4 : $+48$ at col v^3, $+15$ at col v^5,
u^3 : $+15$ at col v^4, -24 at col v^5,
u^2 : -24 at col v^5, $+10$ at col v^3, -16 at col v^4,
u : $+10$ at col v^4, $+16$ at col v^2, -16 at col v^3,
1 : $+8$ at col v^3, -16 at col v^4, $+1$ at col v.

The full column-sum, when read across $v^{12}, v^{11}, \ldots, v, 1$, is

$$1 - 8 + 26 - 40 + 15 + 48 - 84 + 48 + 15 - 40 + 26 - 8 + 1 \;=\; (v-1)^{10}(v+1)^2.$$

[Table: modular equation for $n = 11$. Rows $u^{12}, u^{11}, \ldots, u, 1$ and columns $v^{12}, v^{11}, \ldots, v, 1$. Non-zero entries:]

u^{12} : -1 at col 1,
u^{11} : $+32$ at col v^2,
u^{10} : -44 at col v^4,
u^9 : $+88$ at col v^3, $+22$ at col v^7,
u^8 : -165 at col v^5,
u^7 : $+132$ at col v^4,
u^6 : $+44$ at col v^5, -44 at col v^8,
u^5 : -132 at col v^6,
u^4 : $+165$ at col v^7,
u^3 : -22 at col v^8, -88 at col v^9,
u^2 : $+44$ at col v^9,
u : $+22$ at col v^{10}, -32 at col v^{11},
1 : $+1$ at col v^{12}.

The full row of column-sums reads

$$1 + 10 + 44 + 110 + 165 + 132 + 0 - 132 - 165 - 110 - 44 - 10 - 1 \;=\; (v+1)^{11}(v-1).$$

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[Table: modular equation for $n = 13$. Rows $u^{14}, u^{13}, \ldots, u, 1$ and columns $v^{14}, v^{13}, \ldots, v, 1$. Non-zero entries are arranged in the skew-diagonal pattern characteristic of $n \equiv 5 \pmod 8$:]

$u^{14}:$ -1 at col 1,
$u^{13}:$ $+64$ at col v^2, -52 at col v^8,
$u^{12}:$ -65 at col v^4,
$u^{11}:$ $+208$ at col v^6,
$u^{10}:$ $+208$ at col v^5, -429 at col v^7,
$u^9:$ $+520$ at col v^6, $+52$ at col v^9,
$u^8:$ -429 at col v^7,
$u^7:$ -208 at col v^7,
$u^6:$ $+429$ at col v^8,
$u^5:$ -52 at col v^{11}, -520 at col v^9,
$u^4:$ $+429$ at col v^{10},
$u^3:$ -208 at col v^{11},
$u^2:$ $+65$ at col v^{12},
$u:$ $+52$ at col v^{13}, -64 at col v^{13},
$1:$ $+1$ at col v^{14}.

The complete column-sum reads

$$1 + 12 + 65 + 208 + 429 + 572 + 429 + 0 - 429 - 572 - 429 - 208 - 65 - 12 - 1 \;=\; (v+1)^{13}(v-1).$$

[p. 133] Modular equation for $n = 17$.

[Table: a large rotated table (sidewaystable in the original) giving the modular equation for $n = 17$. *Rows are powers* $u^{18}, u^{17}, \ldots, u, 1$ *and columns are powers* $v^{18}, v^{17}, \ldots, v, 1$. *The entries follow the skew-symmetric pattern of* $n \equiv 1 \pmod 8$ *(perfect symmetry, no sign reversal); the non-zero entries appearing in the printed table, read along the principal diagonals, are coefficients such as* $+1, -8, +28, +66, +220, \ldots$ *on the dexter (descending) diagonal and their reflections on the sinister (rising) diagonal. The column-sum reduces to* $(v-1)^{18}$*.]*

[p. 134] Modular equation for $n = 19$.

[Table: rotated landscape table for the modular equation of $n = 19$. Rows are $u^{20}, u^{19}, \ldots, u, 1$ and columns are $v^{20}, v^{19}, \ldots, v, 1$. Owing to the case $n \equiv 3 \pmod 8$, the table is not perfectly symmetric but has alternating sign reversals on the sinister diagonal. The non-zero diagonal coefficients begin -1, $+114$, -114, 0, -2584, -6859, $-2584, \ldots$ *as referred to in Art. 26. The full column-sum reduces to* $(v+1)^{19}(v-1)$.*]*

[p. 135] Modular equation for $n = 21$ and remarks.

[Table: rotated landscape table for the modular equation of $n = 21$. Rows are $u^{22}, u^{21}, \ldots, u, 1$ and columns are $v^{22}, v^{21}, \ldots, v, 1$. The pattern is symmetric with sign reversals on alternate lines parallel to the sinister diagonal. The column-sum reduces to $(v+1)^{21}(v-1)$.]

[p. 136] Remarks on the Tables.

Various remarks arise on the Tables. Attending first to the cases n a prime number; the only terms of the order $n+1$ in v or u are $v^{n+1} \mp u^{n+1}$, viz. $n = 3$ or 5 (mod. 8) the sign is $-$, but $n = 1$ or 7 (mod. 8) the sign is $+$. And there is in every case a pair of terms $v^n u^a$ and $v^a u^n$, having coefficients equal in absolute magnitude, but of opposite signs, or of the same sign, in the two cases respectively.

Each Table is symmetrical in regard to its two diagonals respectively, so that every non-diagonal coefficient occurs (with or without reversal of sign) 4 times; viz. in the case $n = 1$ or 7 (mod. 8) this is a perfect symmetry, without reversal of sign; but in the case $n = 3$ or 5 (mod. 8) it is, as regards the lines parallel to either diagonal, and in regard to the other diagonal, alternately a perfect symmetry without reversal of sign and a skew symmetry with reversal. Thus in the case $n = 19$, the lines parallel to the dexter diagonal are -1 (symmetrical), $+114, -114$ (skew), $0, -2584, -6859, -2584, \ldots$ (symmetrical), and so on. The same relation of symmetry is seen in the composite cases $n = 9$ and $n = 15$, both belonging to $n = 1$ or 7, mod. 8.

If, as before, n is prime, then putting in the modular equation $u = 1$, the equation in the case $n = 1$ or 7 (mod. 8) becomes $(v-1)^{n+1} = 0$, but in the case $n = 3$ or 5 (mod. 8) it becomes $(v+1)^n(v-1) = 0$.

27. In the case n a composite number not containing any square factor, then dividing n in every possible way into two factors $n = ab$ (including the divisions $n \cdot 1$ and $1 \cdot n$), and denoting by β an imaginary bth root of unity, a value of v is

$$\pm\sqrt{2}\,(\beta q^{\frac{a}{b}})^{\frac{1}{8}}\, f\left(\beta q^{\frac{a}{b}}\right);$$

so that the whole number of roots (or order of the modular equation) is $= \nu$, if ν be the sum of the divisors of n. Thus $n = 15$, the values are

$$\sqrt{2}\,q^{\frac{15}{8}} f(q^{15}), \quad -\sqrt{2}\,q^{\frac{5}{8}} f(q^5), \quad -\sqrt{2}\,q^{\frac{3}{8}} f(q^3), \quad \sqrt{2}\,q^{\frac{1}{8}} f(q),$$

$$1\,, \quad 3\,, \quad 5\,, \quad 15 \text{ roots;}$$

and the order of the modular equation is $= 24$. The modular equation might thus be obtained as for a prime number; but it is easier to decompose n into its prime factors, and consider the transformation as compounded of transformations of these prime orders. Thus $n = 15$, the transformation is compounded of a cubic and a quintic one. If the v of the cubic transformation be denoted by θ, then we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = 0;$$

and to each of the four values of θ corresponds the six values of v belonging to the quintic transformation given by

$$\ldots$$

The equation in v is thus

$$\ldots$$

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where $\theta_1, \theta_2, \theta_3, \theta_4$ are the roots of the equation in θ, viz. we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 \;=\; (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4);$$

and it was in this way that the equation for the case $n = 15$ was calculated. Observe that writing $u = 1$, we have $(\theta+1)^3(\theta-1) = 0$, or say $\theta_1 = \theta_2 = \theta_3 = -1$, $\theta_4 = +1$. The equation in v thus becomes $\{(v-1)^3(v+1)\}^3(v-1)^3(v+1) = 0$, that is, $(v-1)^{12}(v+1)^{12} = 0$.

28. The case where n has a square factor is a little different; thus $n = 9$, the values are

$$1\ ,\quad 3\ ,\quad 9\ \ldots \text{ roots,}$$

but here ω being an imaginary cube root of unity, the second term denotes the three values,

$$\sqrt{2}\, q^{\frac{1}{8}} f(q),\quad \sqrt{2}\,(\omega q)^{\frac{1}{8}} f(\omega q),\quad \sqrt{2}\,(\omega^2 q)^{\frac{1}{8}} f(\omega^2 q),$$

the first of which is $= u$, and is to be rejected; there remain $1+2+9, = 12$ roots, or the equation is of the order 12.

Considering the equation as compounded of two cubic transformations, if the value of v for the first of these be θ, then we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = 0;$$

and to the four values of θ correspond severally the four values of v given by the equation

$$v^4 + 2v^3\theta^3 - 2v\theta - \theta^4 = 0.$$

One of these values is however $v = -u$, since the u^4-equation is satisfied on writing therein $v = -u$; hence, writing

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 \ = \ (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

we have an equation

$$(v^4 + 2v^3\theta_1^3 - 2v\theta_1 - \theta_1^4)(v^4 + \cdots \theta_2^3 \cdots \theta_2^4) \cdots (v^4 + \cdots \theta_4^3 \cdots \theta_4^4) = 0,$$

which contains the factor $(v+u)^4$ and, divested hereof, gives the required modular equation of the order 12; it was in fact obtained in this manner.

Observe that writing $u = 1$, we have $(\theta+1)^3(\theta-1) = 0$, or say $\theta_1 = \theta_2 = \theta_3 = -1,\ \theta_4 = 1$; the modular equation then becomes

$$\{(v-1)^3(v+1)\}^3\,(v+1)^3\,(v-1)^3(v+1)^3 = 0,$$

that is,

$$(v-1)^9(v+1)^9 = 0.$$

[p. 138] The Multiplier Equation. Art. No. 29.

29. The theory is in many respects analogous to that of the modular equation. To each value of v there corresponds a single value of M; hence M, or what is the same thing $\frac{1}{M}$, is determined by an equation of the same order as v, viz. n being prime, the order is $= n+1$. The last term of the equation is constant, and the other coefficients are rational and integral functions of u^4, of a degree not exceeding $\frac{1}{2}(n-1)$; and not only so, but they are, $n \equiv 1$ (mod. 4), rational and integral functions of $u^4(1-u^4)$, and $n \equiv 3$ (mod. 4), alternately of this form and of the same form multiplied by the factor $(1-2u^4)$.

The values are in fact given as transcendental functions of q; viz. denoting by $M_0, M_1, M_2, \ldots, M_n$ the values corresponding to $v_0, v_1, v_2, \ldots, v_n$ respectively, and writing

$$\phi(q) = \frac{(1+q)(1+q^2)(1+q^3)\cdots(1-q^2)(1-q^4)\cdots}{(1-q)(1-q^3)(1-q^5)\cdots(1+q^2)(1+q^4)\cdots},$$

$$= 1 + 2q + 2q^4 + 2q^9 + 2q^{16} + \cdots,$$

then we have

$$(-)^{\frac{n-1}{2}} M_0^{\frac{1}{2}} = \frac{\phi(q)}{\phi(q^n)},$$

$$M_1^{\frac{1}{2}} = \frac{\phi(q)}{\phi(\alpha q^{\frac{1}{n}})}, \quad (\alpha \text{ an imaginary } n\text{th root of unity}),$$

$$M_n^{\frac{1}{2}} = \frac{\phi(q)}{\phi(q^{\frac{1}{n}})}.$$

Hence, the form of the equation being known, the values of the numerical coefficients may be calculated; and it was in this way that Joubert obtained the following results. I have in some cases changed the sign of Joubert's multiplier, so that in every case the value corresponding to $u = 0$ shall be $M = 1$.

The equations are:

$$n = 3, \quad \frac{1}{M^4} - 3 = 0; \qquad u = 0, \text{ this is } \left(\frac{1}{M} - 1\right)^3 \left(\frac{1}{M} + 3\right) = 0;$$

$$u = 1, \text{ it is } \left(\frac{1}{M} + 1\right)^3 \left(\frac{1}{M} - 3\right) = 0.$$

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$$\begin{aligned} n = 5, \quad & \frac{1}{M^6}, \\ & + \frac{1}{M^4} \cdot (-10), \\ & + \frac{1}{M^3} \cdot (+35), \\ & + \frac{1}{M^2} \cdot (-60), \\ & + \frac{1}{M} \cdot (+55), \end{aligned}$$

$$\frac{1}{M} \cdot \left[-26 + 256\, u^4(1-u^4)\right] + 5 = 0;$$

$$u = 0 \text{ or } 1, \text{ this is } \left(\frac{1}{M} - 1\right)^5 \left(\frac{1}{M} - 5\right) = 0.$$

$$\begin{aligned} n = 7, \quad & \frac{1}{M^8}, \\ & + \frac{1}{M^7} \cdot 0, \end{aligned}$$

$$+\frac{1}{M^6}\cdot(-28),$$

$$+\frac{1}{M^5}\cdot\big[+112(1-2u^4)\big],$$

$$+\frac{1}{M^4}\cdot(-210),$$

$$+\frac{1}{M^3}\cdot\big[+224(1-2u^4)\big],$$

$$+\frac{1}{M^2}\cdot\big[-140-21\cdot 25\,6\,u^4(1-u^4)\big],$$

$$+\frac{1}{M}\cdot\big[48+2048\,u^4(1-u^4)\big](1-2u^4)+7=0;$$

$$u=0,\ \text{this is}\quad \left(\frac{1}{M}-1\right)^7\left(\frac{1}{M}-7\right)=0;$$

$$u=1,\ \text{it is}\quad \left(\frac{1}{M}+1\right)^7\left(\frac{1}{M}+7\right)=0.$$

$$n=11,\quad \frac{1}{M^{12}},$$

$$+\frac{1}{M^{11}}\cdot 0,$$

$$+\frac{1}{M^{10}}\cdot(-66),$$

$$+\frac{1}{M^9}\cdot\big[+440(1-2u^4)\big],$$

[p. 140]

$$+\frac{1}{M^8}\cdot(-1485),$$

$$+\frac{1}{M^7}\cdot\big[+3168(1-2u^4)\big],$$

$$+\frac{1}{M^6}\cdot\big[-4620-3\cdot 11^2\cdot 256\,u^4(1-u^4)\big],$$

$$+\frac{1}{M^5}\cdot\big[+4752+11\cdot 4096\,u^4(1-u^4)\big]\,(1-2u^4),$$

$$+\frac{1}{M^4}\cdot\big[-3465-3\cdot 7\cdot 11\cdot 512\,u^4(1-u^4)\big],$$

$$+\frac{1}{M^3}\cdot\big[+1760-11\cdot 83\cdot 2048\,u^4(1-u^4)\big]\,(1-2u^4),$$

$$+\frac{1}{M^2}\cdot\big[-594-9\cdot 11\cdot 37\cdot 256\,u^4(1-u^4)-3\cdot 11\cdot 131072\,[u^4(1-u^4)]^2\big],$$

$$+\frac{1}{M}\cdot\big[120+15\cdot 4096\,u^4(1-u^4)-524288\,(u^4(1-u^4))^2\big]\,(1-2u^4)$$

$$-11=0.$$

$$u=1,\ \text{it is}\quad \left(\frac{1}{M}+1\right)^{11}\left(\frac{1}{M}-11\right)=0.$$

[p. 140] The Multiplier as a rational function of u, v. Art. Nos. 30 to 36.

30. The multiplier M, as having a single value corresponding to each value of v, is necessarily a rational function of u, v; and such an expression of M can, as remarked by Königsberger, be deduced from the multiplier equation by means of Jacobi's theorem,

$$M^n = \frac{1}{n}\frac{\lambda(1-\lambda^2)}{k\,(1-k^2)}\frac{dk}{d\lambda},$$

viz. substituting for k, λ their values u^4, v^4, and observing that if the modular equation be $F(u, v) = 0$ so that the value of $\dfrac{dv}{du}$ is $-F'(v) \div F'(u)$, this is

$$M^n = \frac{1}{n}\frac{(1-v^4)\,v\,F'v}{(1-u^4)\,u\,F'u};$$

and then in the multiplier equation separating the terms which contain the odd and even powers, and writing it in the form $\Phi(M^2) + M\Psi(M^2) = 0$, this equation, substituting therein for M^2 its value, gives the value of M rationally.

The rational expression of M in terms of u, v is of course indeterminate, since its form may be modified in any manner by means of the equation $F(u, v) = 0$; and in the expression obtained as above, the orders of the numerator and the denominator are far too high. A different form may be obtained as follows: for greater convenience I seek for the value not of M but of $\dfrac{1}{M}$.

[p. 141]

31. Denoting, as above, by $M_0, M_1, \ldots, M_n$ the values which correspond to $v_0, v_1, \ldots, v_n$ respectively, and writing $S_{\frac{1}{M}} = \dfrac{1}{M_0} + \dfrac{1}{M_1} + \cdots + \dfrac{1}{M_n}$, &c., we have $S_{\frac{1}{M}}, S_{\frac{v}{M}}$, &c., all of them expressible as determinate functions of u; and we have moreover the theorem that each of these is a rational and integral function of u: we have thus the series of equations

$$S_{\frac{1}{M}} = A, \quad S_{\frac{v}{M}} = B, \; \ldots, \; S_{\frac{v^n}{M}} = H,$$

where $A, B, \ldots, H$ are rational and integral functions of u. These give linearly the different values of $\dfrac{1}{M}$; in fact, we have

$$(v_0 - v_1)\cdots(v_0 - v_n)\,\frac{1}{M_0} = H - G\,\Sigma v_1 + F\,\Sigma v_1 v_2 - \cdots \pm A\,v_1 v_2 \cdots v_n,$$

where $\Sigma v_1, \Sigma v_1 v_2$, &c. denote the combinations formed with the roots $v_1, v_2, \ldots, v_n$ (these can be expressed in terms of the single root v_0); and we have also $(v_0 - v_1)\cdots(v_0 - v_n) = F'(v_0)$ the resulting equation is consequently

$$F'v_0\,\frac{1}{M_0} = R(u, v_0),$$

a determinate rational and integral function of (u, v_0); but as the same formula exists for each root of the modular equation, we may herein write M, v in place of M_0, v_0; and the formula thus is

$$F'v \cdot \frac{1}{M} = R(u, v),$$

viz. we thus obtain the required value of $\dfrac{1}{M}$ as a rational function, the denominator being the determinate function $F'v$, and the numerator being, as is easy to see, a determinate function of the order n as regards v.

32. The method is applicable when M is only known by its expression in terms of q; but if we know for M an expression in terms of v, u, then the method transforms this into a standard form as above. By way of illustration I will consider the case $n = 3$, where the modular equation is

$$v^4 + 2v^3u^3 - 2vu - u^4 = 0,$$

$$v = u + \frac{1-2u^4}{\dots},$$

and where a known expression of M is $\dfrac{1}{\dots}$. Here writing $S_{-1}, S_0\ (=4), S_1$, &c. to denote the sum of the powers $-1, 0, 1$, &c. of the roots of the equation, we have

$$S_{\frac{1}{M}} \ = \ 0 \ = \ S_3 + 2u^3 S_{-1}, \quad \text{as appears from the values presently given,}$$

$$\begin{aligned} S_{\frac{v}{M}} \ &= \ 0 \ = \ S_4 + 2u^3 S_0, \\ S_{\frac{v^2}{M}} \ &= \ S_5 + 2u^3 S_1 \ = \ 6u; \end{aligned}$$

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and observing that v_0, being ultimately replaced by v, we have

$$\Sigma v_1 = \Sigma v_0 - v, \quad \Sigma v_1 v_2 = \Sigma v_0 v_1 - v\Sigma v_0 + v^2, \quad v_1 v_2 v_3 = \Sigma v_0 v_1 v_2 - v\Sigma v_0 v_1 + v^2 \Sigma v_0 - v^3,$$

that is,

$$\Sigma v_1 = -2u^3 - v, \quad \Sigma v_1 v_2 = -2u^3 v - v^2, \quad v_1 v_2 v_3 = 2u - 2u^3 v^2 - v^3,$$

we have

$$\begin{aligned} F'v \cdot \frac{1}{M} \ = \ & (S_3 + 2u^3 S_0) \ \dots \ + (2u^3 + v)(S_2 + 2u^3 S_1) \\ & +(2u^3 v + v^2)(S_1 + 2u^3 S_0) \\ & +(-2u + 2u^3 v^2 + v^3)(S_0 + 2u^3 S_{-1}), \end{aligned}$$

viz. this is

$$\dots$$

But we have

$$S_{-1} = -3, \quad S_0 = 4, \quad S_1 = -2u^3, \quad S_2 = u^2, \quad S_3 = 6u - 8u^3, \dots$$

and the equation thus is

$$(2v^3 + 3v^2 u^3 - u) \cdot \frac{1}{M} \ = \ 3\,(v^2 u^2 + 2u^3 v + 1)\,;$$

to verify which observe that, substituting herein for $\dfrac{1}{M}$ its value $1 + \dfrac{2u^3}{v}$, the equation becomes

$$Sv^3 u^3 + (2v^2 + 0)(v^2 u^3) Svu \cdot (v^2 u^2 + 2u^3 v + 1)\,;$$

that is,

$$2v^4 + 4u^3 v^3 - 4vu - 2u^4 = 0,$$

as it should do.

33. Any expression whatever of M in terms of u, v is in fact one of a system of four expressions; viz. we may simultaneously change

that is, signs are

u	v	$\dfrac{1}{M}$	$n=1$	$n=3$	$n=5,\ n=7 \pmod 8$
u	v	$\dfrac{1}{M}$	$-+++$	$+-+-$	$+--+$
into v	$(-)^{\frac{n^2-1}{8}} u$	$(-)^{\frac{n-1}{2}} \dfrac{n^2}{M^n}$	$++++$	$+++-$	$+++-$
or $\dfrac{1}{u}$	$\dfrac{1}{v}$	$\dfrac{v^n}{u^n M}$	$++++$	$+++-$	$+++-$
or $\dfrac{1}{v}$	$(-)^{\frac{n^2-1}{8}} \dfrac{1}{u}$	$(-)^{\frac{n-1}{2}} \dfrac{u^n}{v^n} \dfrac{n^2}{M^n}$	$+++$	$+-+$	$+-+$

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(Paper 578, continued from p. 142 — pp. 143–175 of Vol. IX)

[p. 143] § 33 (continued). The modular equation in its six forms

Thus $n = 3$, starting from $\dfrac{1}{M} = 1 + \dfrac{2v^2}{v}$, we have

$$\tfrac{1}{M} = 1 + \tfrac{2v^2}{v}, \quad -3M = 1 - \tfrac{2v^2}{u}, \quad \tfrac{v^2}{u^2M} = 1 + \tfrac{2v}{u^2}, \quad -\tfrac{u^2}{v^2}\, 3M = 1 - \tfrac{2u}{v^2};$$

and of course if from any two of these we eliminate M, we have either an identity or the modular equation; thus we have the modular equation under the six different forms:

$$\begin{aligned}
&(1,2)\ (v + 2u^2)(u - 2v^2) + 3uv = 0,\\
&(1,3)\ v^2(v + 2u^2) - u(u^2 + 2v) = 0,\\
&(1,4)\ (v + 2u^2)(v^2 - 2u) + 3u^2 = 0,\\
&(2,3)\ (u - 2v^2)(u^2 + 2v) + 3v^2 = 0,\\
&(2,4)\ v(v^2 - 2u) - u^2(u - 2v^2) = 0,\\
&(3,4)\ (u^2 + 2v)(v^2 - 2u) + 3uv = 0.
\end{aligned}$$

34. Next $n = 5$. Here, starting from $\dfrac{1}{M} = \dfrac{v - u^2}{v(1 - uv^2)}$, the changes give

$$\tfrac{1}{M} = \tfrac{v-u^2}{v(1-uv^2)}, \quad 5M = \tfrac{u+v^2}{u(1+u^2v)}, \quad \tfrac{v^2}{u^2M} = \tfrac{v^2(v-u^2)}{u^2(1-uv^2)}, \quad \tfrac{u^2}{v^2}\, 5M = \tfrac{u^2(u+v^2)}{v^2(1+u^2v)};$$

viz. the third and the fourth forms agree with the first and the second forms respect- ively; that is, there are only two independent forms, and the elimination of M from these gives

$$5uv(1 - uv^2)(1 + u^2v) - (v - u^2)(u + v^2) = 0,$$

which is a form of the modular equation.

35. In the case $n = 7$, starting from $\dfrac{1}{M} = \dfrac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^2}$ (as to this see post, No. 43), the forms are

$$\tfrac{1}{M} = \tfrac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^2}, \qquad (1)$$

$$-7M = \tfrac{-7v(1-uv)(1-uv+u^2v^2)}{v-u^2}, \qquad (2)$$

$$\tfrac{v^2}{u^2M} = \tfrac{-7v^2(1-uv)(1-uv+u^2v^2)}{u^2(v-u^2)}, \qquad (3)$$

$$-\tfrac{u^2}{v^2}\, 7M = \tfrac{-7u^2(1-uv)(1-uv+u^2v^2)}{v^2(u-v^2)}; \qquad (4)$$

so that here again the third and the fourth forms are identical with the second and the third forms respectively; there are thus only two forms, and the elimination of M gives

$$(u - v^2)(v - u^2) + 7uv(1 - uv)(1 - uv + u^2v^2) = 0,$$

which is a form of the modular equation.

[p. 144]

36. If in the foregoing equation

$$F'v \cdot \tfrac{1}{M} = R(u, v),$$

we make the change u, v, $\frac{1}{M}$ into v, $\pm u$, $\pm nM$, it becomes

$$\pm nM \cdot F'u = R(v, \pm u);$$

combining these equations, we have

$$\pm nM^2 \cdot \tfrac{F'u}{F'v} = \tfrac{R(v,\pm u)}{R(u,v)};$$

or, substituting herein the foregoing value

$$M^2 = -\tfrac{1}{n}\,\tfrac{(1-v^2)\,vF'v}{(1-u^2)\,uF'u},$$

this becomes

$$\mp\tfrac{v(1-v^2)}{u(1-u^2)} = \tfrac{R(v,\pm u)}{R(u,v)} \quad \begin{array}{l} + \text{ for } n \equiv 3 \text{ or } 5 \text{ (mod. 8)}, \\ - \text{ for } n \equiv 1 \text{ or } 7 \text{ (mod. 8)}, \end{array}$$

which must agree with the modular equation. Thus in the last-mentioned case $n = 3$, we have

$$\tfrac{1}{2}F'v \cdot \tfrac{1}{M} = 3\,(v^2u^2 + 2u^2v + 1)\,u,$$

or, say

$$R(u,v) = +3\,(v^2u^2 - 2u^2v + 1)\,u, \quad \text{and therefore} \quad R(v,-u) = (v^2u^2 - 2uv^2 + 1)\,v;$$

the equation is

$$\tfrac{v(1-v^2)}{u(1-u^2)} = \tfrac{(v^2u^2-2uv^2+1)\,v}{(v^2u^2-2u^2v+1)\,u},$$

which is right; because Jacobi, p. 82, [*Ges. Werke*, t. I., p. 137], for the modular equation, gives

$$1 - u^4 = (1-u^2v)(v^2u^2 + 2u^2v + 1), \quad 1 - v^4 = (1-uv^2)(v^2u^2 - 2uv^2 + 1).$$

Observe that the general equation

$$\mp\tfrac{v(1-v^2)}{u(1-u^2)} = \tfrac{R(v,\pm u)}{R(u,v)}$$

no longer contains the functions $F'v$, $F'u$, which enter into Jacobi's expression of M^2.

Theorem in connexion with the multiplication of Elliptic Functions. Art. Nos. 37 to 40.

37. The theory of multiplication gives an important theorem in regard to trans- formation. Starting with the $n$thic transformation

$$\tfrac{1-y}{1+y} = \tfrac{1-x}{1+x}\left(\tfrac{\alpha-\beta x+\gamma x^2-\ldots}{\alpha+\beta x+\gamma x^2+\ldots}\right)^2, \quad = \tfrac{1-x}{1+x}\left(\tfrac{P-Qx}{P+Qx}\right)^2,$$

[p. 145]

we may form a like transformation,

$$\tfrac{1-z}{1+z} = \tfrac{1-y}{1+y}\left(\tfrac{\alpha'-\beta' y+\gamma' y^2-\ldots}{\alpha'+\beta' y+\gamma' y^2+\ldots}\right)^2, \quad = \tfrac{1-y}{1+y}\left(\tfrac{P'-Q'y}{P'+Q'y}\right)^2,$$

such that the combination of the two gives a multiplication, viz. for the relation between y, z, deriving w from v as v from u, we have $w = u$; and instead of M we have M', $= \pm\dfrac{1}{nM}$; that is, we have

$$\frac{dx}{\sqrt{1-x^2\cdot 1-u^2x^2}} = \frac{M\,dy}{\sqrt{1-y^2\cdot 1-v^2y^2}},$$

$$\frac{dy}{\sqrt{1-y^2\cdot 1-v^2y^2}} = \frac{M'\,dz}{\sqrt{1-z^2\cdot 1-u^2z^2}},$$

and thence

$$\frac{dx}{\sqrt{1-x^2\cdot 1-u^2x^2}} = \frac{\pm\frac{1}{n}\,dz}{\sqrt{1-z^2\cdot 1-u^2z^2}},$$

or, writing $x = \operatorname{sn}\theta$, we have $z = \pm\operatorname{sn} n\theta$; $\dfrac{n-1}{n}$ is here $(-)^{\frac{n-1}{2}}$, viz. it is $-$ for $n \equiv 3$ or 7 (mod. 8), and $+$ for $n \equiv 1$ or 5 (mod. 8).

Now in part effecting the substitution, we have

$$\tfrac{1-z}{1+z} = \tfrac{1-x}{1+x}\left(\tfrac{P-Qx}{P+Qx}\right)^2\left(\tfrac{P'-Q'y}{P'+Q'y}\right)^2,$$

where y denotes its value in terms of x.

And from the theory of elliptic functions, replacing $\operatorname{sn} n\theta$, $\operatorname{sn}\theta$ by their values $\pm z$, x, we have an equation

$$\tfrac{1-z}{1+z} = \tfrac{1-x}{1+x}\left(\tfrac{A-Bx+Cx^2-\dots}{A+Bx+Cx^2+\dots}\right)^2,$$

where $A - Bx + Cx^2 - \dots$, $A + Bx + Cx^2 + \dots$ are given functions each of the order $\frac{1}{2}(n^2 - 1)$; viz. the coefficients are given functions of k, or, what is the same thing, of u^4.

Comparing the two results, we see that in the $n$thic transformation the sought-for function, $\alpha + \beta x + \gamma x^2 + \dots$ of the order $\frac{1}{2}(n - 1)$, is a factor of a given function $A + Bx + Cx^2 + \dots$ of the order $\frac{1}{2}(n^2 - 1)$.

38. Considering the modular equation as known, then by what precedes we have

$$\tfrac{1}{\alpha+\beta x+\gamma x^2+\dots} = \frac{1}{a}\left(1 + \tfrac{B}{a}\,x + \dots + \tfrac{w^n}{a}\,x^{\frac{n-1}{2}}\right);$$

that is, the given function $A + Bx + Cx^2 + \dots$ has a factor $1 + \dfrac{B}{a}\,x + \dots + \dfrac{w^n}{a}\,x^{\frac{n-1}{2}}$, of which one (the last) coefficient is known, and we are hence able theoretically to

[p. 146]

determine all the other coefficients rationally in terms of u, v; that is, the modular equation being known, we can theoretically complete the solution of the transformation problem. I do not, however, see the way to obtaining a convenient solution in this manner.

39. The formula in question for $n = 3$ is

$$\tfrac{1+\operatorname{sn}3\theta}{1-\operatorname{sn}3\theta} = \tfrac{1+\operatorname{sn}\theta}{1-\operatorname{sn}\theta}\left(\tfrac{1+2\operatorname{sn}\theta-2k^2\operatorname{sn}^3\theta-k^2\operatorname{sn}^4\theta}{1-2\operatorname{sn}\theta+2k^2\operatorname{sn}^3\theta-k^2\operatorname{sn}^4\theta}\right)^2,$$

which, putting therein $x = \operatorname{sn}\theta$, $z = -\operatorname{sn}3\theta$, and replacing k by u^4, may be written

$$\tfrac{1-z}{1+z}\,(\div) = (1 + x)(1 - 2x + 2u^2x^3 - u^4x^4)^2\,(\div),$$

where the signs $(\div)$ indicate denominators which are obtained from the numerators by changing the signs of z, x respectively.

The theorem in regard to $n = 3$ thus is, $1 + \dfrac{v^2}{u^2}\,x$ is a factor of $1 - 2x + 2u^2x^3 - u^4x^4$; viz. writing in the last-mentioned function $x = -\dfrac{u^2}{v^2}$, we ought to have

$$0 = 1 + \tfrac{2u^2}{v^2} + \tfrac{2u^8}{v^6} - \tfrac{u^4}{v^4}\cdot\tfrac{u^8}{v^8},$$

that is,

$$u^4 + 2uv - 2u^2v^3 - v^6 = 0,$$

[p. 147]

[Table showing the polynomial multiplication $f \cdot g$ "Term in [ ] has factor $1+\dfrac{\beta}{\alpha}x+\dfrac{w^2}{v}x^2$" on the left and the expanded sum of $f \cdot g$ on the right, organised by powers of x. Cayley simply writes out the formal product; we render the two columns as paired displays.]

Left column (factors and partial products):

$$\begin{aligned}&\text{Term in [ ] has factor } 1 + \tfrac{\beta}{\alpha}x + \tfrac{w^2}{v}x^2;\\ &n = 1,\ \text{term in [ ] is } = (1+x)^2(1-x)^6.\end{aligned}$$

Right column (sum, by ascending powers of x):

$$
\begin{aligned}
&+56u^{20}(1+2u^4)\,x^{15}\\
&+8u^{16}(\,46\,+\,57u^4\,+\,8w^{20})\,x^{14}\\
&-8u^{14}(\,16\,+\,25u^4\,+\,4w^{20})\,x^{13}\\
&-u^{12}(\,16\,+\,305u^4\,+\,144w^{20})\,x^{12}\\
&+4u^{10}(\,8\,+\,51u^4\,+\,16w^{20})\,x^{11}\\
&+28u^{10}(\,1\,+\,4u^4)\,x^{10}\\
&-28u^{8}(\,2\,+\,3u^4)\,x^{9}\\
&-14u^{6}\,x^{8}\\
&+4u^{6}(\,7\,+\,2u^4)\,x^{7}\\
&-4u^{4}\,x^{6}\\
&-4u^{4}\,x^{5}\\
&+u^{4}\,x^{4}\\
&+(\div).
\end{aligned}
$$

Term in [ ] has factor

$$1+\tfrac{\beta}{\alpha}x+\tfrac{w^2}{v}x^2;$$

$u=1$, term in [ ] is $=(1+x)^2(1-x)^6$.

The transformations $n=3$, 5, 7, 11. Art. Nos. 41 to 51.

41. The cubic transformation, $n=3$.

I reproduce the results already obtained; since there are only two coefficients α, β, these are also the last but one and last coefficients ρ, σ. Hence, from the values of α, β, ρ, σ, we have

$$\alpha=1,\qquad 2\alpha=\tfrac{v^2}{u}\Big(\tfrac{1}{M}-\tfrac{u^2}{v^2}\Big),\quad 2\beta=\tfrac{1}{M}-1,\quad \beta=\tfrac{u^2}{v};$$

the two values of $\dfrac{1}{M}$ are thus $\dfrac{1}{M}=2\dfrac{u}{v^2}+\dfrac{u^2}{v^2},=1+\dfrac{2u^2}{v}$, giving the modular equation

$$v^4+2u^2v^2-2uv-u^4=0,$$

and we then have

$$\tfrac{1-y}{1+y}=\tfrac{1-x}{1+x}\Big(\tfrac{v-u^2x}{v+u^2x}\Big)^2.$$

[p. 148] 42. The quintic transformation, $n=5$.

Here there are the three coefficients α, β, γ, or β, γ are the last but one and last coefficients ρ, σ; we have

$$\alpha=1,\qquad 2\beta=v^2u\Big(\tfrac{1}{M}-\tfrac{u^4}{v^2}\Big),\qquad 2\beta=\tfrac{1}{M}-1,\qquad \gamma=\tfrac{u^2}{v}.$$

Comparing the two values of β, we have $\dfrac{1}{M}=\dfrac{v-u^2}{v(1-uv^2)}$, and then

$$\alpha=1,\qquad 2\beta=\tfrac{u(v^2-u^2)}{v(1-uv^2)},\qquad \gamma=\tfrac{u^2}{v},$$

so that only the modular equation remains to be determined.

The unused equation is

$$2\alpha\gamma+2\alpha\beta+\beta^2=\tfrac{u^2}{v^2}(2\alpha\gamma+2\beta\gamma+\beta^2),$$

which, putting therein $\alpha=1$, may be written

$$u^4(\gamma^2+2u\gamma+u^2)=2\beta(\gamma u^2-v^2);$$

attending to the value of β, this divides by $u^2 - v^2$; in fact the equation may be written

$$2\gamma + \beta^2 = -\tfrac{u(v^2+u^2)}{v(1-uv^2)}(\gamma u^2 - v^2);$$

and then completing the substitution, and integralising, this becomes

$$[8uv^2(1-uv^2) + (v^2-u^2)^2] = 4uv(u^2+v^2)(1-uv^2)(1-uv^2),$$

viz.

$$4(1-uv^2)\,uv = 2u^4(1-u^2)(v^2+u^2)(1-uv^2)(1-uv^2) + (v^2-u^2)^2 \cdot u^2(1-u^2v^2)(1-u^2v^2) = 0;$$

and the term in [ ] being $= -(v^2-u^2)(1-uv^2)(1-uv^2)$, the whole again divides by v^2-u^2, and the equation thus becomes

$$(v^2+u^2)(v^2-u^2) - 4uv(1-uv^2)(1-uv^2+u^2v^2) = 0,$$

which is the modular equation.

43. The septic transformation, $n = 7$.

I do not propose to complete the solution directly from the fundamental equations for α, β, γ, δ, but resort to the known modular equation, and to an expression of M which I obtain by means thereof.

The modular equation is

$$(1-u^2)(1-v^2) - (1-uv)^4 = 0,$$

[p. 149]

which may also be written

$$(v-u^2)(u-v^2) + 7uv(1-uv)(1-uv+u^2v^2) = 0,$$

as can be at once verified; but it also follows from Cauchy's identity

$$(x+y)^7 - x^7 - y^7 = 7xy(x+y)(x^2+xy+y^2)^2.$$

We then have

$$M = -\tfrac{1}{n}\,\tfrac{(1-v^2)\,vF'v}{(1-u^2)\,uF'u}.$$

Moreover

$$\begin{aligned} uF'u &= -u^2(1-v^2) + uv(1-uv)^2 \\ &= -\tfrac{u^2}{1-u^2}(1-uv)^2 + uv(1-uv)^2 \\ &= \tfrac{(1-uv)^2}{1-u^2}\,u(v-u^2); \end{aligned}$$

and similarly

$$F'v = \tfrac{(1-uv)^2}{1-v^2}\,v(u-v^2);$$

whence

$$\tfrac{1}{M} = -\tfrac{7u}{v}\,\tfrac{(v-u^2)}{u-v^2}.$$

Writing this under the form

$$\tfrac{1}{M^2} = \tfrac{-7uv\,(v-u^2)(u-v^2)}{(u-v^2)^2}, \quad = \tfrac{49u^2(1-uv)^2(1-uv+u^2v^2)^2}{(u-v^2)^2},$$

I find, as will appear, that the root must be taken with the sign $-$, and that we thus have $\dfrac{1}{M} = \dfrac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^2}$, whence also $M = \dfrac{v(1-uv)(1-uv+u^2v^2)}{u-v^2}$.

44. Recurring now to the fundamental equations for the septic transformation, the coefficients are α, β, γ, δ, and we have

$$\alpha = 1, \qquad 2\gamma = v^2u\Big(\tfrac{1}{M} - \tfrac{u^6}{v^2}\Big),$$

$$2\beta = \tfrac{1}{M} - 1, \qquad \delta = \tfrac{u^2}{v},$$

so that the coefficients are all given in terms of v, M. The unused equations are

$$u^2(3\alpha\gamma + 2\alpha\delta + \beta^2) = v^2(\gamma^2 + 2\beta\delta + 2\alpha\delta),$$
$$u^4(\gamma^2 + 2\beta\delta + 2\alpha\delta) = v^2(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

which, substituting therein for α, β, γ, δ the foregoing values, and then two equations; from these, eliminating M, we should obtain the modular equation, and then M in terms of u, v.

[p. 150]

Substituting in the first instance for α, δ their values, the equations are

$$u^2(2\beta + 2\gamma\beta) = v^2\Big\{\gamma^2 + 2\tfrac{u^2}{v}(\beta+\gamma)\Big\},$$
$$\gamma^2 + 2\beta\gamma + (2+2\beta)\,\tfrac{u^2}{v} = u^2v^2\Big\{2\beta + 2\beta\gamma + 2\,\tfrac{u^2}{v} + \beta^2\Big\}.$$

The first of these is

$$4(1-uv)(2\beta + 2\gamma) + 4\tfrac{u^2}{v}\gamma^2 = 0,$$

viz. this is

$$4(1-uv)\Big(\tfrac{1}{M} - 1 + \tfrac{u^2}{M} - \tfrac{u^8}{v^3}\Big) - 4\,\tfrac{u^4}{v^2} = 0,$$

or observing that the sum of the second terms is the sign of the foregoing value of $\frac{1}{M}$; so that the identity verified, it follows that the correct sign has been attributed to the value of $\frac{1}{M}$.

46. Multiplying by v, the equation is

$$-7(1-u^2)(1-v^2) + 7(1-uv)(1-v^2) - 14uv(1-uv)^2(1+uv) + v^2(1-uv)^2 = 0,$$
$$+(1-u^2)(1-v^2) - 8(1-uv)(1-v^2) + 5\,(1-uv)^2 = 0,$$
$$-10\,(\,4\alpha + 5u^2v - 5u^4v^2 - 5u^6v^3 - u^4) = 0,$$

or, say

$$\tfrac{1}{2}F'v\cdot\tfrac{1}{M} = 5\,[v^2 + 4uv^4 + 6v^2u^4 + 4u^6v + vu^2(1-u^2)],$$

where

$$\tfrac{1}{2}F'v\cdot\tfrac{1}{M} = 3\,v^4 + 10vu^4 + 10v^4u^2 - 5vu^2 - 2u.$$

Hence also, reducing by the modular equation,

$$\tfrac{1}{2}vF'v\cdot\tfrac{1}{M} = 5v\,[v^4 + 4uv^5 + 6u^4v^2 + 2v(1+u^2) + u^2],$$

the one of which forms is as convenient as the other.

47. Making the change u, v, $\dfrac{1}{M}$ into v, $-u$, $-5M$, we have

$$-\tfrac{1}{2}F'u\cdot 5M = 5\,[-u^4 + 4vu^4 + 4vu^4 + 1 - v^2(1-v^2)];$$

and comparing with the equation

$$\tfrac{1}{2}vF'v\cdot\tfrac{1}{M} = 5\,[v^2 + 4u^4v + 6u^2v^4 + 4uv^5 + u^2(1-u^2)],$$

we obtain

$$\tfrac{v(1-v^2)}{u(1-u^2)} = -\tfrac{2v(1-uv)(1-uv+u^2v^2) - vF'v}{2u(1-uv)(1-uv+u^2v^2) + uF'u}.$$

Writing for a moment $M = u^4 + 5u^2v + v^2$, $N = u^2v^4 + 5uv + v^2$, this is

$$-4Mu(1-u^2)(1-v^2) - (u^2-v^2)(1 - u^2 - uv^2 - u^2v^2 + u^2v^2 + v^2u^2 + v^4)\cdot N$$
$$+4uN^2v^4 + (uv^4 - v^2)^2[1 - u^2v^4(uv - u^2v^2 + v^2)] = 0.$$

But we have

$$u^2(1-uv)^2 - v^2(1-uv)^2 = (uv - u^2v^2 - u^2v^2)\,[1 - u^2v^4(u^2-v^2) + u^2v^2],$$
$$u^2(1-v^2) + v^2(1-u^2) = (u^2+v^2)\,[1 - u^2v^4(u^2+v^2) + u^2v^2].$$

[p. 151]

viz. this is

$$-7(1-v^4)(1-u^2)+7(1-uv)(1-v^2)-14uv(1-uv)^2(1+uv)+v^2(1-uv)^2=0,$$

$$+(1-u^2)(1-v^2)-8(1-uv)(1-v^2)+5\,(1-uv)^2(1-uv)^2=0.$$

In the second column the coefficient of $2-uv-v^2$, this is

$$=(1-u^2)(1-v^2)\,\{[(uv)^2-(uv)^2]\}-14uv(1-uv)^2-(1-uv)^2(1-uv)=0,$$

Reducing also the other two columns by means of the modular equation, the equation thus becomes

$$-6(1-uv)^2-(1-uv)\,[(1-uv)^2-(uv)^2]-14uv(1-uv)^2(1+uv)$$

$$+\,8\,(1-uv)^2$$

$$-28uv(1-uv)^2(1-uv+u^2v^2)=0.$$

This is in fact an identity; to show it, writing for convenience θ in place of uv, and observing that the terms

$$-(1-\theta)+8(1-\theta)+3(1-\theta),\ =5,\ =(1-\theta)^2\,[8-2\,(1-\theta)+5]=(1-\theta)\,[6-5\theta+4\theta^2-3\theta^3]$$

the whole equation divides by $(1-\theta)^2$; or throwing out this factor, it will be observed that the terms

$$2\,\theta(uv-2u^2v^2+2u-u^2v)=-\tfrac{1}{2}\,\theta(v^2+2v^2u^2-2vu-u^2);$$

or, say

$$\tfrac{1-y}{1+y}=\tfrac{1-x}{1+x}\left(\tfrac{v-\beta x+\gamma x^2-\delta x^3}{v+\beta x+\gamma x^2+\delta x^3}\right)^2,$$

but the resulting form may admit of simplification.

48. The endecadic transformation, $n=11$.

I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are α, β, γ, δ, ϵ, ζ; and we have, as in general,

$$\alpha=1,\qquad 2\epsilon=v^2u\Big(\tfrac{1}{M}-\tfrac{u^{10}}{v^2}\Big),$$

$$2\beta=\tfrac{1}{M}-1,\qquad \zeta=\tfrac{u^2}{v}.$$

The unused equations then are, the results, so far as I have obtained them, are interesting. The coefficients are α, β, γ, δ, ϵ, ζ; and we have, as in general,

$$u^2(2\alpha\gamma+2\beta+\beta^2)=v^2(\gamma+2\beta\zeta+2\alpha\epsilon),\quad +u^4\cdot 2\beta=0,$$

$$u^{-2}u^4=2\beta\cdot\tfrac{u^4}{v^2}=0,$$

$$+\left\{\tfrac{1}{u^4v^2}-\tfrac{u^4}{v^2}\right\}\cdot 2\gamma+\tfrac{1}{v^2}\left(\tfrac{1}{u^2v^2}-u^2v^2\right)\cdot 2\delta=0\,;$$

say, for a moment, these are

$$A+P\cdot 2\beta=0, B+R\cdot 2\beta=0, 1\,:\,2\gamma\,=\,2\beta=PS-QR\,:\,RA-PB.$$

Here

$$PS-QR=\tfrac{8\beta}{v^2}(QB-SA+\Pi),$$

$$=\tfrac{8\beta}{v^2}\left[\tfrac{2u^4}{v}+(u^4v^2-v^2)+2u-2u^2\right],$$

$$=-\tfrac{8\beta}{v^2}\left(u^4v\,\tfrac{1}{M}-\tfrac{u^4}{v}\right),$$

[p. 152]

47. The conclusion is

$$\alpha = 1, \quad \beta = \tfrac{1}{2}\left(\tfrac{1}{M} - 1\right), \quad \gamma = v^2 u\left(\tfrac{1}{M} - \tfrac{u^4}{v^2}\right), \quad \delta = \tfrac{u^2}{v},$$

where

$$\tfrac{1}{M} = \tfrac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^2},$$

and of course

$$\tfrac{1-y}{1+y} = \tfrac{1-x}{1+x}\left(\tfrac{1-\beta x+\gamma x^2-\delta x^3}{1+\beta x+\gamma x^2+\delta x^3}\right)^2,$$

but the resulting form may admit of simplification.

48. The endecadic transformation, $n = 11$.

I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are α, β, γ, δ, ϵ, ζ; and we have, as in general,

$$\alpha = 1\ , \quad 2\epsilon = v^2 u\left(\tfrac{1}{M} - \tfrac{u^{10}}{v^2}\right),$$

$$2\beta = \tfrac{1}{M} - 1, \quad \zeta = \tfrac{u^2}{v}.$$

The unused equations then are

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \tfrac{u^2}{v^2}(2\alpha\gamma + 2\beta\gamma + \beta^2),$$

$$v^2(\zeta^2 + 2\beta\zeta + 2\alpha\zeta) = u^{10}(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

$$u^2(\zeta^2 + 2\delta\zeta + 2\beta\zeta) + \tfrac{1}{v^2}\left(\tfrac{1}{u^2v^2} - u^2v^2\right)\cdot 2\delta = 0\,;$$

say, for a moment, these are

$$A + P\cdot 2\beta = 0, \quad B + R\cdot 2\beta = 0, \quad 1 : 2\gamma : 2\delta = PS - QR : RA - PB : \ldots$$

[The remainder of Art. 48 carries forward the elimination, expressing the coefficients γ, δ rationally in terms of u, v, $\frac{1}{M}$, and obtaining (apart from a verification which has not been carried through) the endecadic transformation in the form $\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{1-\beta x+\gamma x^2-\delta x^3+\epsilon x^4-\zeta x^5}{1+\beta x+\gamma x^2+\delta x^3+\epsilon x^4+\zeta x^5}\right)^2$.]

[p. 153]

where the terms containing $\dfrac{1}{M}$ disappear of themselves, viz. this is

$$= \tfrac{4}{v}\left(\tfrac{u^4v}{v^2} - 2u^2v^4 + 2uv - u^2v^2\right),$$

$$= -\tfrac{4}{v}\,u^4\,(v^4 + 2v^4u^2 - 2vu - u^2);$$

observe that the term in ( ) is equal to zero, the modular equation for the case $n = 3$. It thus appears that γ and δ are given as fractions, having in their denominator this function $u^4 + 2u^2v - 2u^2v^4 - v^4$.

49. To complete the calculation, we have

$$QB - AS = -uv^2\left(\tfrac{v^2}{u^2v^2} - \tfrac{u^4}{v^2}\beta^2\right)$$

$$-\left[u^2v^2(2+\beta) + \tfrac{u^4}{v^2}\left(\epsilon + 2\,\tfrac{u^4}{v^2}\right)\right]\left\{\tfrac{12}{vu^2} - \tfrac{u^4}{v} + \tfrac{\epsilon}{v^2}\right\};$$

viz. multiplying by 8, and substituting for 2β its value, this is

$$8\,(QB - AS) = 2uv^2\left[v^2u^2\left(\tfrac{1}{M} - 2u^4 + 2u^2v^2 - 1\right)\right]\left\{\tfrac{12}{u^2v} - \tfrac{u^4}{v^2} - \tfrac{\epsilon}{v}\right\};$$

$$-\left\{v^4u^2\left(\tfrac{1}{M} - 1\right) - uv^2\,\tfrac{1}{M} - \tfrac{u^4}{v^2} + 3\right\}\left\{\tfrac{12}{vu^2} - \tfrac{u^4}{v} + \tfrac{\epsilon}{v^2}\right\};$$

or, what is the same thing,

$$8\,(QB-AS) = 2u^4\Big(\tfrac{v^2}{u^4v^4}-\tfrac{u^4}{v^4}\,\beta^2\Big)$$
$$-\Big\{v^4u^2\big(\tfrac{1}{M}-1\big)+\Big(\tfrac{\beta}{M}-1\Big)\big(\tfrac{1}{M}+3\big)\Big\}\Big\{\tfrac{12}{vu}-\tfrac{u^4}{v}+\tfrac{\epsilon}{v^2}\Big\},$$

viz. the left-hand side is

$$= 2\,\{v^2(1-u^2v^2)+2v^2(1-u^2)+u^{10}+v^4-u\}.$$

or, say we have $-\dfrac{8\beta}{v^2}(QB-SA)\Pi$, where

$$\Pi = \tfrac{1}{Mv^2}\,v^2\,(1-v^2)$$
$$+\tfrac{1}{Mv^2}\,u^2\,(u-2u^2)(1-v^2)$$
$$+\tfrac{1}{M}\,4v^2u^2+v^2(1-3u^2)-4uv^3+2u^2$$
$$+\ldots-2u^4+6v^2u^2(1-v^2)+u^4(-3+3u^2);$$

[p. 154]

wherefore the value of 2γ is $=\frac{1}{u^2}\,v^2\,(1-v^2)$. Similarly, writing

$$\Pi' = \tfrac{1}{M^2}\,v^2\,(1-v^2)$$
$$+\tfrac{1}{M^2}\,u^2\,(u-2u^2)(1-v^2)$$
$$+\tfrac{1}{M}\,4v^4u^2+v^4(3-v^2)+4uv^3+4uv^2$$
$$+v^2(-3+3u^2)+v^4(1-u^2)+4uv-2u^4,$$

we find

$$2\delta = \tfrac{1}{u}\,\Pi'\cdot(v^2+2v^2u^2-2vu-u^4);$$

in verification whereof observe that this being so, the first equation gives the identity

$$\big(\tfrac{1}{M}-1\big)\big(\tfrac{1}{M}+3\big)-v^2\Big(\tfrac{1}{M}-\tfrac{u^4}{v^2}\Big)\Big(\tfrac{1}{M}+\tfrac{2u^2v}{v^2}\Big)+\Pi-\Pi'=0.$$

50. The result is that, writing for the moment $v^4+2v^2u^2-2vu-u^4=\Delta$, the values of the coefficients are

$$\alpha,\quad \beta,\quad \gamma,\quad \delta,\quad \epsilon,\quad \zeta,$$
$$=1,\quad \tfrac12\big(\tfrac{1}{M}-1\big),\quad \tfrac{1}{2\Delta}\,\Pi,\quad \tfrac{u^2}{v}\,\tfrac{\Pi'}{\Delta},\quad \tfrac12 u^2v\Big(\tfrac{1}{M}-\tfrac{u^{10}}{v^2}\Big),\quad \tfrac{u^2}{v},$$

and

$$\tfrac{1-y}{1+y}=\tfrac{1-x}{1+x}\cdot\Big(\tfrac{1-\beta x+\gamma x^2-\delta x^3+\epsilon x^4-\zeta x^5}{1+\beta x+\gamma x^2+\delta x^3+\epsilon x^4+\zeta x^5}\Big)^2;$$

the modular equation is known, and to complete the solution we require an expression of $\dfrac{1}{M}$ in terms of u, v.

51. We may herein illustrate the following theorem, viz. we may simultaneously change u, v, $\dfrac{1}{M}$, $\alpha:\beta:\gamma:\delta:\epsilon:\zeta$; into $\dfrac{1}{u}$, $\dfrac{1}{v}$, $\dfrac{v^2}{u^2M}$, $\zeta:\epsilon:\delta:\gamma:\beta:\alpha$.

Thus making the change in the equation

$$\tfrac{\beta}{\alpha}=\tfrac12\big(\tfrac{1}{M}-1\big),$$

we have

$$\tfrac{\epsilon}{\zeta}=\tfrac12\Big(\tfrac{v^2}{u^2M}-1\Big),\quad\text{that is,}\quad \tfrac12\,\tfrac{v^2}{u^2}\Big(\tfrac{1}{M}-\tfrac{u^2}{v^2}\Big)=\tfrac12\Big(\tfrac{v^2}{u^2M}-1\Big),$$

which is right.

So in the equation $\dfrac{\gamma}{\alpha}=\frac12\,\dfrac{\Pi}{\Delta}$, if for a moment (Π), (Δ) are what Π, Δ become, the

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equation is $\dfrac{\delta}{\zeta} = \frac{1}{2}\dfrac{(\Pi)}{(\Delta)}$, that is, $\dfrac{1}{u^2}\dfrac{\Pi'}{\Delta} = \frac{1}{u^2}\dfrac{(\Pi)}{(\Delta)}\Pi'$; or $(\Pi) = \dfrac{1}{u^2}\dfrac{\Delta}{\Delta'}\Pi'$; but obviously $\dfrac{(\Delta)}{\Delta} = -\dfrac{1}{u^4v^4}$; and the equation thus is $(\Pi) = -\dfrac{1}{u^2v^2}\Pi'$, or say $u^2v^2(\Pi) = -\Pi'$; that is,

$$\begin{aligned}-\Pi' = u^2v^2\Big\{&\tfrac{v^2}{u^4M^2}\cdot\tfrac{1}{u^2}\big(1-\tfrac{1}{v^2}\big)\\ &+\tfrac{v^2}{u^4M}\cdot\tfrac{1}{u^2}\big(\tfrac{1}{u}-\tfrac{2}{u^2}\big)\big(1-\tfrac{1}{v^2}\big)\\ &+\tfrac{v^2}{u^4M}\cdot\tfrac{4}{u^2v^2}+\tfrac{1}{v^2}\big(1-\tfrac{3}{u^2}\big)-\tfrac{4}{uv^3}+\tfrac{2}{u^2}\\ &+\tfrac{2}{v^2}+\tfrac{6}{v^2u^2}\big(1-\tfrac{1}{v^2}\big)+\tfrac{1}{u^2}\big(-3+\tfrac{3}{u^2}\big)\Big\},\end{aligned}$$

which is right.

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52. I recur to the formula

$$\tfrac{1-y}{1+y} = \tfrac{1-x}{1+x}\left(\tfrac{\alpha-\beta x+\gamma x^2-\ldots\pm\sigma u^{2(n-1)}}{\alpha+\beta x+\gamma x^2+\ldots+\sigma u^{2(n-1)}}\right)^2,$$

and seek to express the ratios $\alpha:\beta:\ldots:\sigma$ in terms of q. Writing with Jacobi

$$\omega = \tfrac{mK+m'iK'}{n},$$

we have in general

$$\alpha+\beta x+\gamma x^2+\ldots+\sigma x^{2(n-1)} = \alpha\big(1+\tfrac{x}{\operatorname{snc}2\omega}\big)\big(1+\tfrac{x}{\operatorname{snc}4\omega}\big)\cdots\Big(1+\tfrac{x}{\operatorname{snc}(n-1)\omega}\Big),$$

(snc = sin co am; viz. $\operatorname{snc}2\omega = \operatorname{cn}(K-2\omega)$, &c.); and the values of α, β, …, θ which correspond to the moduli v_0, v_1, …, v_n, or say the values $(\alpha_0,\beta_0,\ldots,\theta_0)$, $(\alpha_1,\beta_1,\ldots,\theta_1),\ldots,(\alpha_n,\beta_n,\ldots,\theta_n)$ are obtained by giving to ω the values

$$\frac{\omega_0}{n},\quad\frac{\omega_1}{n},\quad\frac{\omega_2}{n},\ \ldots,\ \frac{\omega_n}{n},\quad=\frac{2K}{n},\ \frac{2K+4iK'}{n},\ \frac{4K+4iK'}{n},\ \ldots,\ \frac{iK'}{n},$$

viz. the cases ω_0, ω_n correspond to Jacobi's first and second real transformations, and the others to the imaginary transformations.

I remark that $\omega=\omega_n$ gives for $\operatorname{snc}2p\omega$ an expression which is rational as regards q, but $\omega=\omega_n$ gives an expression involving $q^{\frac{1}{n}}$, the real nth root of q; the other values ω_1, ω_2, … give the like expressions, involving $\alpha q^{\frac{1}{n}}$, $\alpha^2q^{\frac{1}{n}}$, … (α an imaginary nth root of unity), the imaginary nth roots of q.

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53. I consider first the expression

$$\tfrac{1}{\operatorname{snc}2p\omega_n},\ =\tfrac{1}{\operatorname{cn}(K-2p\omega_n)},\ =\tfrac{\operatorname{dn}2p\omega_n}{\operatorname{cn}2p\omega_n}.$$

Here, writing $2p\omega_n = \frac{2pK\xi}{\pi}$ (ξ for Jacobi's x, as x is being used in a different sense), that is,

$$\xi = \tfrac{\pi}{2K}\cdot 2p\cdot\tfrac{2K}{n},\ =\tfrac{2p\pi}{n},$$

(and hence $e^{i\xi} = e^{\frac{2pi\pi}{n}} = \alpha^p$, $e^{i\xi} = \alpha^{2p}$, if $\alpha = e^{\frac{2i\pi}{n}}$, an imaginary nth root of unity), then (Jacobi, p. 86, [*Ges. Werke*, t. i., p. 143])

$$\begin{aligned}\tfrac{1}{\operatorname{snc}2p\omega_n} &= \operatorname{dn}\tfrac{2K\xi}{\pi}\div\operatorname{cn}\tfrac{2K\xi}{\pi}\\ &= \tfrac{C}{B}\cdot\tfrac{2q^{\frac14}}{c^{\frac14}}\cdot\tfrac{(1-q^{\alpha p^2})\ldots(1+q^{\alpha p^2})\ldots}{(1+q^{\alpha p^2})\ldots(1+q^{\alpha p^2})\ldots},\end{aligned}$$

where

$$\tfrac{C}{B} = \tfrac{[(1+q^2)...]^2}{[(1+q)...]^2},\ = f^2(q);$$

that is,

$$\tfrac{1}{\operatorname{snc} 2p\omega_n} = \tfrac{2qi^2}{\pi}\ f^2(q)\ \tfrac{(1+\alpha^{2p}q)...(1+\alpha^{-2p}q)...}{(1+\alpha^{2p}q^2)...(1+\alpha^{-2p}q^2)...},$$

where, for shortness, I write $(1+q^{\alpha p^2})\ldots$ to denote the infinite product

$$(1+q^{\alpha p^2})(1+q^{\alpha p^2})(1+q^{\alpha p^2})\ldots,$$

and similarly $(1+q^2q^{\alpha p^2})\ldots$ to denote the infinite product $(1+q^2q^{\alpha p^2})(1+q^4q^{\alpha p^2})(1+q^6q^{\alpha p^2})\ldots$, and the like for the terms in $e^{-i\xi}$; the notation, accompanied by its explanation, is quite intelligible, and it would be difficult to make one which would be at the same time complete and not cumbrous. Then attributing to g the values $1, 2, \ldots, \frac{1}{2}(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\dfrac{\beta}{\alpha}, \dfrac{\gamma}{\alpha}$, &c, or α being put $= 1$, say the values of $\beta, \gamma, \ldots, \sigma$.

54. I stop to notice a verification afforded by the value of β_n. Putting $u = 0$, that is, $q = 0$, we have

$$\tfrac{1}{\operatorname{snc} 2p\omega_n} = \tfrac{2\pi}{1+\alpha^{2p}},$$

and thence

$$\beta_n = 2\Big\{\tfrac{1}{1+\alpha^2} + \tfrac{1}{1+\alpha^4} + \tfrac{1}{1+\alpha^6} + \ldots + \tfrac{1}{1+\alpha^{n-1}}\Big\};$$

we have $2\beta_n = \dfrac{1}{M_n} - 1$; and putting as above $u = 0$, the value of $\dfrac{1}{M_n}$ is $= (-)^{\frac{n-1}{2}}n$; whence

$$(-)^{\frac{n-1}{2}}n - 1 = 4\Big\{\tfrac{\alpha}{1+\alpha^2} + \tfrac{\alpha^2}{1+\alpha^4} + \tfrac{\alpha^3}{1+\alpha^6} + \ldots + \tfrac{\alpha^{(n-1)/2}}{1+\alpha^{n-1}}\Big\},$$

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a theorem relating to the imaginary nth roots of unity, α an odd prime. In particular,

$$n = 3, \quad -4 = 4\Big\{\tfrac{\alpha}{1+\alpha^2}\Big\},\ \text{at once verified by } \alpha^2+\alpha+1 = 0;$$

$$n = 5, \quad 4 = 4\Big\{\tfrac{\alpha}{1+\alpha^2} + \tfrac{\alpha^2}{1+\alpha^4}\Big\},\ \text{verified by } \alpha^4 - 1 = 0,$$

viz. the theorem is also true for the real root $\alpha = 1$; in fact, the term in [ ] is $\{\alpha(1+\alpha^4)+\alpha^2(1+\alpha^2)\} \div (1+\alpha^2)(1+\alpha^4),\ = (\alpha+1+\alpha^4+\alpha^4) \div (1+\alpha^2+\alpha^4+\alpha^6),\ = 1$;

$$n = 7, \quad -8 = 4\Big\{\tfrac{\alpha}{1+\alpha^2} + \tfrac{\alpha^2}{1+\alpha^4} + \tfrac{\alpha^3}{1+\alpha^6}\Big\},$$

which may be verified by means of $\alpha^7 - \alpha^6 + \alpha^5 - \alpha^4 + \alpha^3 - \alpha^2 + \alpha + 1 = 0$; and so on.

55. I further remark that we have

$$\tfrac{1}{M_n} = (-)^{\frac{n-1}{2}}\Big\{\tfrac{\operatorname{sn} 2\omega_n\cdot\operatorname{sn} 4\omega_n\cdots\operatorname{sn}(n-1)\,\omega_n}{\operatorname{snc} 2\omega_n\cdot\operatorname{snc} 4\omega_n\cdots\operatorname{snc}(n-1)\,\omega_n}\Big\}^2.$$

But Jacobi (p. 86, [*l.c.*]),

$$\operatorname{sn} 2p\omega_n = \tfrac{2K\xi}{\pi}, \quad = \tfrac{AK}{\pi i^4}\ \tfrac{e^{i\xi}-1}{e^{i\xi}}\ \tfrac{(1-q^{\alpha p^2})...(1-q^{-\alpha p^2})...}{(1-q^qq^{\alpha p^2})...(1-q^qq^{-\alpha p^2})...},$$

where (p. 89, [*l.c.*, p. 146])

$$\tfrac{AK}{\pi} = \tfrac{\sqrt{q}}{\sqrt{k}},\ = \tfrac{1}{2}\ \tfrac{[(1+q)(1+q^2)...]^2}{[(1+q^{q^2})...(1+q^{q^4})...]^2},\ = \tfrac{1}{2}f^{-2}(q),$$

that is,

$$\operatorname{sn} 2p\omega_n = f^{-2}\,q\ \tfrac{q^n-1}{2iq^{\frac{1}{4}}}\cdot\tfrac{(1-q^{\alpha p^2})...(1-q^{-\alpha p^2})...}{(1-q^qq^{\alpha p^2})...(1-q^qq^{-\alpha p^2})...}.$$

Hence

$$\frac{\operatorname{sn} 2p\omega_n}{\operatorname{snc} 2p\omega_n} = \frac{q^n-1}{i(q^n+1)}\cdot\frac{1-q^{\alpha p^2}}{1+q^{\alpha p^2}}\cdot\frac{(1+q^{\alpha p^2})\ldots(1+q^{-\alpha p^2})\ldots}{(1-q^q q^{\alpha p^2})\ldots(1-q^q q^{-\alpha p^2})\ldots}\cdot\frac{(1+q^q q^{\alpha p^2})\ldots(1+q^q q^{-\alpha p^2})\ldots}{(1-q^{\alpha p^2})\ldots(1-q^{-\alpha p^2})\ldots},$$

and giving to g the values $1, 2, \ldots, \frac{1}{2}(n-1)$, and multiplying the several expressions, we have the value of $\dfrac{1}{M_n}$, viz. observing that we have in the denominator $(i)^{\frac{1}{2}(n-1)}$, $= (-)^{\frac{n-1}{2}}$ which destroys this factor in the expression of $\dfrac{1}{M_n}$, this is

$$\frac{1}{M_n} = \Pi\left\{\frac{1-q^{\alpha p^2}}{1+q^{\alpha p^2}}\cdot\frac{(1+q^{\alpha p^2})\ldots(1+q^{-\alpha p^2})\ldots}{(1-q^q q^{\alpha p^2})\ldots(1-q^q q^{-\alpha p^2})\ldots}\right\}^4,$$

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In fact, $n = 3$, this is

$$\frac{(\alpha^2-1)^2}{(\alpha^2+1)^2} = -3,$$

viz. the numerator is $\alpha - 2\alpha^2 + 1$, $= -3\alpha^2$, and the denominator is $(-\alpha)^2$, $= \alpha^2$.

So $n = 5$, the formula is

$$\left(\frac{\alpha^2-1}{\alpha^2+1}\,\frac{\alpha^4-1}{\alpha^4+1}\right)^2 = 5,\ \text{that is,}\ \left(\frac{(\alpha^2-1)^2}{(\alpha^4+1)^2}\right)^2 = 5,$$

or

$$\frac{\alpha^4-4\alpha^3+6\alpha^2-4\alpha+1}{\alpha^8+2\alpha^4+1} = 5,$$

viz. this is $5(1+\alpha+2\alpha^2) - (1-4\alpha+6\alpha^2-4\alpha^2+6\alpha^4) = 0$, which is right; and so in other cases.

We thus have

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} n\cdot R(q),$$

which, on putting therein $u = 0$, that is, $q = 0$, gives, as it should do, $\dfrac{1}{M_n} = (-)^{\frac{n-1}{2}} n$.

57. As regards the expression of $R(q)$, observe that, giving to g its different values, the factors $1-\alpha^{2p}q$ and $1-\alpha^{-2p}q$ are the factors other than $1-q$ of $1-q^n$, and so as to the other pairs of factors; viz. we have

$$R(q) = \left(\frac{1-q^n}{1-q}\,\frac{1+q^2}{1+q^{2n}}\,\frac{1+q}{1+q^n}\right)^4,$$

viz. this is

$$= \left(\frac{1-q^n}{1+q^n}\,\frac{1+q^2}{1-q^2}\right)^4,\ = \left(\frac{1-q^n}{1+q^n}\,\frac{1+q}{1-q}\right)^4,$$

that is,

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} n\,\frac{\phi^4(q^n)}{\phi^4(q)},$$

agreeing with a former result.

58. We have of course the identity $2\beta_n = \dfrac{1}{M_n} - 1$; that is,

$$4\sum \frac{e^{i\xi}}{1+q\,e^{i\xi}}\, f^2(q)\,\frac{(1+q^q q^{\alpha p^2})\ldots(1+q^q q^{-\alpha p^2})\ldots}{(1+q^{\alpha p^2})\ldots(1+q^{-\alpha p^2})\ldots} = (-)^{\frac{n-1}{2}} n\,\frac{\phi^4(q^n)}{\phi^4(q)} - 1,$$

$(g = 1, 2, \ldots \frac{1}{2}(n-1))$, which, putting therein $q = 0$, is an identity before referred to; a form perhaps more convenient is obtained by dividing each side by $f^2(q)$.

59. I notice further that we have

$$v_0 = u^4\left\{\operatorname{snc} 2\omega_n \operatorname{snc} 4\omega_n \ldots \operatorname{snc}(n-1)\,\omega_n\right\};$$

the term in [ ] is

$$\Pi\,\frac{1+\alpha^{2p}}{2q^{\frac{1}{4}}}\, f^{-2}(q)\,\frac{(1+q^{\alpha p^2})\ldots(1+q^{-\alpha p^2})\ldots}{(1+q^q q^{\alpha p^2})\ldots(1+q^q q^{-\alpha p^2})\ldots},$$

where we have $\Pi\dfrac{1+\alpha^{2p}}{2} = (-)^{\frac{n-1}{2}}$. For example, $n = 3$, the term is $\dfrac{1+\alpha^2}{2} = -1$;

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$n=5$, it is

$$\tfrac{(1+\alpha^2)(1+\alpha^4)}{\alpha\cdot\alpha^2\cdot\alpha^4},\ =\tfrac{1+\alpha^4+\alpha^2+\alpha^4}{\alpha^7},\ =-1\,;$$

$n=7$, it is

$$\tfrac{(1+\alpha^2)(1+\alpha^4)(1+\alpha^6)}{\alpha\cdot\alpha^2\cdot\alpha^4\cdot\alpha^4},\ =\tfrac{1+\alpha+\alpha^4+\alpha^4+\alpha^4+\alpha^4+2\alpha^7}{\alpha^7}=1;$$

and so on. The term in question thus is

$$(-)^{\frac{n-1}{2}}\,\tfrac{1}{(\sqrt{2})^{n-1}}\,f^{-2n}(q)\,\tfrac{1+q^{\alpha p^2}\ldots 1+q\ldots}{1+q^q q^{\alpha p^2}\ldots 1+q^{q^2}\ldots}.$$

This has to be multiplied by u^4, $=(\sqrt{2})^4\,q^{\frac12}\,f^2(q)$, and we thus obtain

$$v_0=(-)^{\frac{n-1}{2}}\,\sqrt{2}\,q^{\frac12}\,f^4(q^n),$$

agreeing with a former result.

We have in regard to the value of $\dfrac{1}{M_n}$ a verification afforded by the value of M_n; then, as well the new modulus λ, $(=v_0^4)$ and the multiplier M_n, as also the several functions which enter into the expression

$$\tfrac{1-y}{1+y}=\tfrac{1-x}{1+x}\left(\frac{\left(1-\frac{x}{\operatorname{snc}2\omega_n}\right)\cdots\left(1-\frac{x}{\operatorname{snc}(n-1)\,\omega_n}\right)}{\left(1+\frac{x}{\operatorname{snc}2\omega_n}\right)\cdots\left(1+\frac{x}{\operatorname{snc}(n-1)\,\omega_n}\right)}\right)^2,$$

are all of them expressed as functions of q.

60. I consider in like manner the expression

$$\tfrac{1}{\operatorname{snc}2p\omega_n},\ =\tfrac{1}{\operatorname{sn}(K-2p\omega_n)},\ =\tfrac{\operatorname{dn}2p\omega_n}{\operatorname{cn}2p\omega_n}.$$

Here, writing $2p\omega_n=\frac{2pK\xi}{\pi}$ (ξ instead of Jacobi's x as before), that is,

$$\xi=\tfrac{\pi}{2K}\cdot 2p\cdot\tfrac{iK'}{n},\ =\tfrac{p\pi iK'}{nK},$$

and thence

$$e^{i\xi}=e^{-\frac{p\pi K'}{nK}},\ =q^{\frac{p}{n}},$$

we have

$$\tfrac{1}{\operatorname{snc}2p\omega_n}=\operatorname{dn}\frac{2K\xi}{\pi}\div\operatorname{cn}\frac{2K\xi}{\pi}$$

$$=f^2(q)\,\tfrac{2q^{\frac14}}{\pi}\,\tfrac{(1+q^{\frac{2p}{n}})(1+q^{2-\frac{2p}{n}})\ldots(1+q^{1-\frac{2p}{n}})\ldots}{1+q^{\frac{p}{n}}(1+q^{1+\frac{p}{n}})(1+q^{1-\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}.$$

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where the notations are as follows:

$$(1+q^{1+\frac{2p}{n}})\ldots\ \text{is the infinite product}\ (1+q^{1+\frac{2p}{n}})(1+q^{2+\frac{2p}{n}})(1+q^{3+\frac{2p}{n}})\ldots,$$

and

$$(1+q^{1+\frac{2p}{n}})\ldots\ \text{is the infinite product}\ (1+q^{1+\frac{2p}{n}})(1+q^{2+\frac{2p}{n}})(1+q^{3+\frac{2p}{n}})\ldots;$$

and the like as to the expressions with exponents containing $-\dfrac{2p}{n}$.

And then attributing to g the values $1,2,\ldots,\frac12(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\dfrac{\beta}{\alpha}$, $\dfrac{\gamma}{\alpha}$, &c, or α being put $=1$, say the values of $\beta,\gamma,\ldots,\sigma$.

It is easy to see, and I do not stop to prove that, if instead of $\omega=\omega_n$ we have $\omega=\omega_1,\ \omega_2,\ \ldots,$ or ω_{n-1}, we simply multiply $q^{\frac1n}$ by an imaginary nth root of unity; that is, we replace the real nth root $q^{\frac1n}$ by an imaginary nth root of q.

In the case $u=0$, that is, $q=0$, do not stop to prove here, if instead of $\omega=\omega_n$ we have $\omega=\omega_1,\ \omega_2,\ \dots,$ or ω_{n-1}, we simply multiply $q^{\frac{1}{n}}$ by an imaginary nth root of unity; that is, we replace the real nth root $q^{\frac{1}{n}}$ by an imaginary nth root of q.

In the case $u=0$, that is, $q=0$, the modulus λ, $(=v_0^4)$, $=0$; and the like for the values $\omega_1,\ \omega_2,\ \dots,\ \omega_{n-1}$, the equation $2\beta=\dfrac{1}{M}-1$ gives consequently for $\dfrac{1}{M}$, n values each $=1$, agreeing with the multiplier equation.

61. We have for M_n the formula

$$\tfrac{1}{M_n}=(-)^{\frac{n-1}{2}}\left\{\tfrac{\operatorname{sn}2\omega_n\operatorname{sn}4\omega_n\ldots\operatorname{sn}(n-1)\,\omega_n}{\operatorname{snc}2\omega_n\operatorname{snc}4\omega_n\ldots\operatorname{snc}(n-1)\,\omega_n}\right\}^2,$$

and, as before,

$$\operatorname{sn}2p\omega_n=f^{-2}(q)\,\tfrac{q^{\frac{1}{n}}-1}{(1-q^{\frac{2p}{n}})(1-q^{2-\frac{2p}{n}})\ldots(1-q^{1-\frac{2p}{n}})\ldots}2iq^{\frac{1}{4}}\cdot\tfrac{(1+q^{\frac{p}{n}})(1+q^{1+\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}{(1+q^{1+\frac{p}{n}})(1+q^{1-\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots},$$

hence

$$\tfrac{\operatorname{sn}2p\omega_n}{\operatorname{snc}2p\omega_n}=\tfrac{q^{\frac{p}{n}}-1}{i(q^{\frac{p}{n}}+1)}\cdot\tfrac{1-q^{\frac{p}{n}}}{1+q^{\frac{p}{n}}}\cdot\tfrac{(1+q^{\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}{(1-q^{1+\frac{p}{n}})\ldots(1-q^{1-\frac{p}{n}})\ldots}\cdot\tfrac{(1+q^{1+\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}{(1-q^{\frac{p}{n}})\ldots(1-q^{1-\frac{p}{n}})\ldots},$$

and we thence derive the value of $\dfrac{1}{M_n}$, viz. observing that we have in the denominator $(i)^{\frac{1}{2}(n-1)}$, $=(-)^{\frac{n-1}{2}}$ which destroys this factor in the expression of $\dfrac{1}{M_n}$, this is

$$\tfrac{1}{M_n}=\Pi\left\{\tfrac{1-q^{\frac{p}{n}}}{1+q^{\frac{p}{n}}}\cdot\tfrac{(1+q^{\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}{(1-q^{1+\frac{p}{n}})\ldots(1-q^{1-\frac{p}{n}})\ldots}\cdot\tfrac{(1+q^{1+\frac{p}{n}})\ldots(1+q^{1-\frac{p}{n}})\ldots}{(1-q^{\frac{p}{n}})\ldots(1-q^{1-\frac{p}{n}})\ldots}\right\}^4.$$

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Now, giving to g its values, it is easy to see that we have

$$\Pi\,(1-q^{\frac{2p}{n}})(1-q^{2-\frac{2p}{n}})\ldots(1-q^{2-\frac{2p}{n}})\ldots=\tfrac{(1-q^{\frac{1}{n}})\ldots}{(1-q^n)\ldots},$$

where $(1-q^{\frac{1}{n}})\ldots$ denotes $(1-q^{\frac{1}{n}})(1-q^{\frac{2}{n}})(1-q^{\frac{3}{n}})\ldots$, viz. it is the same function of $q^{\frac{1}{n}}$ that $(1-q^n)\ldots$ is of q; also

$$\Pi\,(1+q^{\frac{2p}{n}})(1+q^{2-\frac{2p}{n}})(1+q^{2-\frac{2p}{n}})\ldots=\tfrac{(1+q^{\frac{1}{n}})\ldots}{(1+q)\ldots},$$

where $(1+q^{\frac{1}{n}})\ldots$ denotes $(1+q^{\frac{1}{n}})(1+q^{\frac{2}{n}})(1+q^{\frac{3}{n}})\ldots$, viz. is the same function of $q^{\frac{1}{n}}$ that $(1+q)\ldots$ is of q; and the like as to the denominator factors; we thus have

$$\tfrac{1}{M_n}=\left\{\tfrac{(1-q^{\frac{1}{n}})\ldots(1+q^{\frac{1}{n}})\ldots(1+q^{\frac{1}{n}})\ldots(1-q)\ldots}{(1-q^n)\ldots(1+q^n)\ldots(1+q)\ldots(1-q^{\frac{1}{n}})\ldots}\right\}^4,$$

viz. this is

$$=\left\{\tfrac{(1-q^{\frac{1}{n}})\ldots(1+q^{\frac{1}{n}})\ldots}{(1+q^n)\ldots(1-q^n)\ldots}\right\}^4\div\left\{\tfrac{(1+q^n)\ldots(1+q)\ldots}{(1+q^n)\ldots(1-q)\ldots}\right\}^2;$$

or, we have

$$\tfrac{1}{M_n}=\phi^4(q^{\frac{1}{n}})\div\phi^4(q),$$

agreeing with a former result.

We have

$$2\beta_n=\tfrac{1}{M_n}-1,$$

that is,

$$2\left\{\tfrac{1}{\operatorname{snc}2\omega_n}+\tfrac{1}{\operatorname{snc}4\omega_n}+\ldots+\tfrac{1}{\operatorname{snc}(n-1)\,\omega_n}\right\}=\tfrac{\phi^4(q^{\frac{1}{n}})}{\phi^4(q)}-1,$$

a result which, substituting on the left-hand side the foregoing values of the several functions, must be identically true.

62. We have also

$$v_n = u^4 \{\operatorname{snc} 2\omega_n \operatorname{snc} 4\omega_n \dots \operatorname{snc}(n-1)\,\omega_n\},$$

where the term in [ ] is

$$= \Pi f^{-2}(q)\, \frac{(1+q^{\frac{2p}{n}})(1+q^{1-\frac{2p}{n}})\dots(1+q^{1+\frac{2p}{n}})\dots}{2q^{\frac{1}{n}}\,(1+q^{1+\frac{2p}{n}})\dots(1+q^{1-\frac{2p}{n}})\dots},$$

or, observing that the sum of the exponents $\dfrac{g}{n}$ is $\frac{1}{n}\,[1+2\dots+\frac{1}{2}(n-1)] = \dfrac{n^2-1}{8n}$, this is

$$= f^{-n+1}(q)\,\frac{1}{(\sqrt{2})^{n-1}\,q^{(n^2-1)/(8n)}}\,\frac{(1+q^{2/n})\dots(1+q)\dots}{(1+q^{1+2/n})\dots(1+q)\dots};$$

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or, the last factor being $f(q^{\frac{1}{n}}) \div f(q)$, the expression is

$$f^{-n}(q)\,\frac{1}{(\sqrt{2})^{n-1}}\,q^{-\frac{n^2-1}{8n}}\,f(q^{\frac{1}{n}});$$

or, multiplying by u^4, $= (\sqrt{2})^4\,q^{\frac{1}{2}}\,f^2(q)$, we have

$$v_n = \sqrt{2}\,q^{\frac{1}{n}-\frac{n^2-1}{8n}}\,f^4(q^{\frac{1}{n}}),$$

agreeing with a former result.

We have in like manner the complete q-transcendental solution for the *trans- formatio secunda*; viz. the original modulus $k\,(= u^4)$ being given as a function of q, then, as well the new modulus λ, $(= v_n^4)$ being also a function of M_n, as also the several functions which enter into the expression

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{\left(1-\frac{x}{\operatorname{snc}2\omega_n}\right)\cdots\left(1-\frac{x}{\operatorname{snc}(n-1)\,\omega_n}\right)}{\left(1+\frac{x}{\operatorname{snc}2\omega_n}\right)\cdots\left(1+\frac{x}{\operatorname{snc}(n-1)\,\omega_n}\right)}\right)^2,$$

are all expressed as functions of q.

63. As an illustration of the formulae for the *transformatio secunda* I write $n=7$; and putting for greater convenience $q = r^7$, that is, $r = q^{\frac{1}{7}}$, then we have

$$v_7 = \sqrt{2}r^{\frac{1}{2}}f(r), \quad \frac{1}{M_7} = \phi^4(r) \div \phi^4(r^7),$$

$$\frac{1}{\operatorname{snc}2\omega_7} = 2f^2(r^7)\,A, \quad \frac{1}{\operatorname{snc}4\omega_7} = 2f^2(r^7)\,B, \quad \frac{1}{\operatorname{snc}6\omega_7} = 2f^2(r^7)\,C,$$

where

$$A = r^4\cdot\frac{5.19\dots9.23\dots}{2.16\dots12.26\dots},$$

$$B = r^4\cdot\frac{3.17\dots11.25\dots}{4.18\dots10.24\dots},$$

$$C = r^4\cdot\frac{1.15\dots13.27\dots}{6.20\dots8.22\dots},$$

where the numerator of A denotes $(1+r^5)(1+r^{19})\dots(1+r^9)(1+r^{23})\dots$, and so in other cases, the difference of the exponents being always $=14$. And we have, as mentioned, the identical equation

$$f^4(r)\,(A+B+C) = \frac{1}{4}\left(\frac{\phi^4 r}{\phi^4 r^7} - 1\right).$$

The values of the several expressions up to r^{24} are as follows: Mr J. W. L. Glaisher kindly performed for me the greater part of the calculation.

[p. 163] Table I (the index of r vs. coefficients A, B, C, Sum, $\phi^4(r) \div \phi^4(r^7)$)

[Table: a 4-column numerical table giving, for each integer power r from 0 *to* 50*, the entries of A, B, C and their Sum, with the running value of the multiplier* $\phi^4(r^7) \div \phi^4(r^7)$*.]*

Ind. of r	A	B	C	Sum	Mult. by $f^4(r^7) \div \phi^4(r^7)$	
0				0	+	1
1	+ 1			+ 1	−	4
2		+ 1		+ 1	+	4
3	− 1			− 1	−	0
4			+ 1	+ 1	−	0
5	+ 1		+ 1	+ 2	+	8
6	+ 1	− 1		0	−	8
7		− 1		− 1	−	0
8	− 1			− 1	−	5
9		+ 1		+ 1	−	12
10	+ 2	+ 1	− 1	+ 2	+	8
11	− 1		− 1	− 2	−	32
12	− 2		+ 1	− 1	+	8
13		+ 2	+ 1	+ 3	+	13
14	+ 1	+ 1		+ 2	+	32
15	+ 2			+ 2	+	36
16		− 2	+ 2	0	+	24
17	− 2	− 2		− 4	−	32
18	− 1		+ 2	+ 1	−	13
19	+ 2		+ 2	+ 4	+	96
20		+ 3		+ 3	−	24
21	− 2	+ 2	− 2	− 2	−	32
22	+ 1		− 1	0	−	80
23	− 3	+ 2	+ 3	+ 2	−	56
24	− 4		+ 1	− 3	+	132
25		− 4		− 4	−	248
26	+ 5	+ 2	− 3	+ 4	+	64
27	+ 3	+ 1	− 3	+ 1	−	76
28		− 1	+ 2	+ 1	+	184
29	− 5	+ 4	+ 5	+ 4	+	56
30	− 3		+ 5	+ 2	−	160
31	+ 1	+ 4		+ 5	+	77
32	+ 9	+ 5	− 4	+ 9	+	208
33	+ 4		− 5	− 1	+	576
34	− 7	− 8	+ 1	− 14	−	152
35		− 5		− 5	−	396
36	+ 7	+ 2	− 9	0	−	488
37	+ 11	+ 10	− 11	+ 10	+	332
38	+ 1		− 10	− 9	−	672
39	− 13		+ 5	− 28	+	1240
40	− 2	+ 13	+ 8	+ 8	+	328
41	+ 13		− 6	+ 4	+	352
42	+ 9	− 9	+ 5	+ 7	+	816
43	− 14	− 14	+ 14	− 14	−	1008
44	− 14	− 14	+ 16	− 12	−	728
45	− 1	− 9	+ 13	+ 3	+	1376
46	+ 17	+ 11	+ 12	+ 40	−	2328
47	+ 3	− 20	− 12	− 12	−	672
48	− 6	− 23	− 1	− 30	+	1376
49	− 17	+ 13	+ 5	+ 1	−	700
50	− 13	+ 5	− 14	− 22	−	1504

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64. As already mentioned, the foregoing expressions of the coefficients in terms of q may be applied to the determination of the coefficients as rational functions of u, v.

Representing by θ any one of the coefficients $\alpha, \beta, \gamma, \ldots, \sigma$, consider the sum

$$S\,\tfrac{v^p\theta}{u^q},$$

f a positive integer, and the summation extending as before to the $n+1$ values of v, and corresponding values of $\dfrac{\theta}{u}$. This is a rational function of u, and it is also integral. As to this observe that the function, if not integral, would become infinite either for $u=0$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u. But the function can only become infinite by reason of some term or terms of $Su^p\dfrac{\theta}{u}$ becoming infinite; viz. some term $\dfrac{1}{\operatorname{snc} 2p\omega}$ must become infinite; or attending to the equation

$$v_n = u^4\left\{\operatorname{snc} 2\omega \operatorname{snc} 4\omega \ldots \operatorname{snc}(n-1)\,\omega\right\},$$

it can only happen if $u=0$, or if $v=\infty$, and from the modular equation it appears that if $v=\infty$, then also $u=\infty$. Now $u=0$ gives the ratios $\dfrac{\beta}{\alpha}, \dfrac{\gamma}{\alpha}, \ldots$, each of them infinite at most in $\dfrac{1}{v^p}$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u. Now $u=0$ gives the values $\dfrac{\beta}{\alpha}, \dfrac{\gamma}{\alpha}, \ldots$, each of them become infinite for $u=0$ either for $u=0$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u. But the function can only become infinite by reason of some term or terms of $Su^p\dfrac{\theta}{u}$ becoming infinite; viz. some term $\dfrac{1}{\operatorname{snc} 2p\omega}$ must become infinite; or attending to the equation

$$v_n = u^4\left\{\operatorname{snc} 2\omega \operatorname{snc} 4\omega \ldots \operatorname{snc}(n-1)\,\omega\right\},$$

it can only happen if $u=0$, or if $v=\infty$, and from the modular equation it appears that if $v=\infty$, then also $u=\infty$. Now $u=0$ gives the ratios $\dfrac{\beta}{\alpha}, \dfrac{\gamma}{\alpha}, \ldots$, each of them infinite at most in $\dfrac{1}{v^p}$; so that there having a denominator a power of u, and if the highest powers of u in $Sv^p\theta$ is the lowest negative power (highest positive power) being $\dfrac{1}{u^p}$; fractional power $q^{\frac{p}{n}}$ acquires by the summation the coefficient $(1+\alpha+\alpha^2+\ldots+\alpha^{n-1})$, and therefore disappears; there remains only the radicality $q^{\frac{p}{n}}$ occurring in the expression of the v's; and if $nf\,u\,(\text{mod. } 8)$, $\mu=0$, or a positive integer less than n, then in the form of the expression in $q^{\frac{1}{n}}$ into a rational function of q. Hence this, being a rational and integral function of u, must be of the form

$$Au^p + Bu^{p+8} + Cu^{p+16} + \&\text{c.}$$

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65. We have thus in general

$$Sv^p\,\tfrac{\theta}{u^q} = Au^p + Bu^{p+8} + \&\text{c.};$$

and in like manner

$$Sv^p\,\tfrac{\theta}{u^q} = A'u^{p-q} + B'u^{p-q+8} + \&\text{c.}$$

We may in these expressions find a limit to the number of terms, by means of the before-mentioned theorem that we may simultaneously interchange u, v; $\alpha, \beta, \ldots, \rho, \sigma$ into $\dfrac{1}{u}, \dfrac{1}{v}$; $\sigma, \rho, \ldots, \beta, \alpha$. Starting from the expression of $Sv^p\,\dfrac{\theta}{u^q}$, let ϕ be the corre- sponding coefficient to θ; viz. in the series $\alpha, \beta, \ldots, \theta, \ldots, \phi, \ldots, \rho, \sigma$, let ϕ be as removed from σ as θ is from α; then the equation becomes

$$Sv^{-p}\,\tfrac{\phi}{v^q} = Av^{-p} + Bv^{-p-8} + \&\text{c.},$$

where

$$\tfrac{\phi}{\sigma} = \tfrac{\phi}{a}\cdot\tfrac{a}{\sigma} = \tfrac{\pi}{u^2}\,\tfrac{\phi}{a};\ \text{the equation thus is}$$

$$Sv^p \tfrac{\phi}{a} = Av^p + Bv^{p-8} + \&\text{c.},$$

and by what precedes the series on the right-hand side can contain no negative power higher than $\dfrac{1}{u^{p+q+8}}$; that is, the series of coefficients $A, B, C, \ldots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

In like manner from the equation $Sv^p \dfrac{\theta}{u^q}$ we have

$$Sv^{p+q} \tfrac{\phi}{a} = A'v^{n+p+q} + B'v^{n+p+q-8} + \&\text{c.},$$

where the indices must be positive; viz. the series of coefficients $A', B', \ldots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

67. The like theory applies to the expression $\dfrac{1}{M_n}$. We have, putting as before $uf \equiv \mu$ (mod. 8),

$$Sv^p \tfrac{1}{M_n} = Au^p + Du^{p+8} + \&\text{c.},$$
$$Sv^p \tfrac{1}{M_n} = A'u^{p-q} + B'u^{p-q+8} + \ldots$$

and we find a limit to the number of terms by the consideration that we may simul- taneously change u, v, $\dfrac{1}{M_n}$ into $\dfrac{1}{u}, \dfrac{1}{v}, \dfrac{v^2}{u^2M_n}$; the equations thus become

$$Sv^{-p} \tfrac{1}{M_n} = Av^{-p} + Bv^{-p-8} + \&\text{c.} \ldots$$

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$\Big($where, if $f =$ or < 4, there must be on the right-hand side no negative power of u; but if $f > 4$, then the highest negative power must be $\dfrac{1}{u^{(p+q-4)/8}}$, and$\Big)$

$$Sv^{p+q} \tfrac{1}{M_n} = A'v^{n+p+q} + B'v^{n+p+q-8} + \ldots,$$

where on the right-hand side there must be no negative power of v.

68. It is to be remarked that β, ρ being always given linearly in terms of $\dfrac{1}{M}$, it is the same thing whether we seek in this manner for the values of β, ρ, or for that of $\dfrac{1}{M}$; but the latter course is practically more convenient. Thus in the cases $n = 5$, $n = 7$ we require only the value of $\dfrac{1}{M}$.

In the case $n = 11$, where the coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$, it has been seen that γ, δ are given as cubic functions of $\dfrac{1}{M}$; seeking for them directly, their values would (if the process be practicable) be obtained in a better form, viz. instead of the denominator $(F'v)^2$ there would be only the denominator $F'(v)$.

69. I consider for $\dfrac{1}{M}$ the cases $n = 3$ and 5:

$$n = 3,\ f = 0,\ 1,\ 2,\ 3, \text{ then } \mu = 0,\ 3,\ 6,\ 1\,;$$

and we write down the equations

$$\begin{aligned} S\tfrac{1}{M} &= A, &&\text{giving } S\tfrac{v^0}{M} = Au^0, \\ S\tfrac{v}{M} &= A'u^3, &&\quad ,, \quad S\tfrac{v^2}{M} = A'u^3, \\ S\tfrac{v^2}{M} &= 0, &&\quad ,, \quad S\tfrac{v^2}{M} = 0\,; \end{aligned}$$

viz. if we had in the first instance assumed $S\dfrac{1}{M} = A + Bu^8 + \ldots$, this would have given $S\dfrac{v^2}{M} = Au^2 + Bu^{10} + \ldots$, whence B and the succeeding coefficients all vanish; and so in other cases. We have here only the coefficients A, A'; and these can be obtained without the aid of the q-formula by the consideration that for $u = 1$ the corresponding values of v, $\dfrac{1}{M}$ are

$$v = 1,\ -1,\ -1,\ -1,\ -1,$$
$$\tfrac{1}{M} = 3,\ -1,\ -1,\ -1,\ -1.$$

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whence $A = A' = 6$; or we have the equations

$$S\,\tfrac{1}{M} = 0, \quad S\,\tfrac{v}{M} = 6u^4, \quad S\,\tfrac{v^2}{M} = 0, \quad S\,\tfrac{v^2}{M} = 6u,$$

giving as before

$$(2v^4 + 3v^2u - v)\,\tfrac{1}{M} = 3\,(v^2u^4 + 2u^2v + 1)\,u,$$

reducible by means of the modular equation to $\dfrac{1}{M} = 1 + \dfrac{2u^2}{v}$.

70. $n = 5$. Corresponding to $f = 0, 1, 2, 3, 4, 5$, we have $\mu = 0, 5, 2, 7, 4, 1$, and we find

$$\begin{aligned} S\,\tfrac{1}{M} &= A, & \text{giving } S\,\tfrac{v^0}{M} &= Au^0,\\ S\,\tfrac{v}{M} &= 0, & S\,\tfrac{v^2}{M} &= 0,\\ S\,\tfrac{v^2}{M} &= A'u^2, & S\,\tfrac{v^2}{M} &= A'u^2,\\ S\,\tfrac{v^2}{M} &= A''u + B''u^7, & Sv^4\,\tfrac{1}{M} &= A''v^2 + B''v^{-6}. \end{aligned}$$

But for $u = 1$ the corresponding values of v, $\dfrac{1}{M}$ are

$$v = 1,\ -1,\ -1,\ -1,\ -1,\ -1,$$

$$\tfrac{1}{M} = 5,\ 1,\ 1,\ 1,\ 1,\ 1;$$

whence $A = A' = 10$, $A'' + B'' = 0$, or say the value of $S\,\dfrac{v^2}{M}$ is $= A''u(1 - u^6)$.

The values of A'' are found very easily by the q-formula, viz. neglecting higher powers of q, we have

$$u = q^{\frac{1}{8}}\sqrt{2},\ v_0 = u^4\sqrt{2}\,\tfrac{1}{M_0} = 5,\ v_1 = 2q^{\frac{1}{4}}\sqrt{2},\ \tfrac{1}{M_1} = 1\,;$$

hence

$$S\,\tfrac{v^2}{M} + S\,\tfrac{v^2}{M} = 5q^{\frac{1}{4}}\,(\sqrt{2})^4 + A''^{\frac{1}{4}}\sqrt{2},$$

that is, $A'' = 20$, and the equations are

$$S\,\tfrac{1}{M} = 10, \quad S\,\tfrac{v}{M} = 0, \quad S\,\tfrac{v^2}{M} = 10u^2, \quad S\,\tfrac{v^2}{M} = 10u^4, \quad S\,\tfrac{v^4}{M} = 20u\,(1 - u^6);$$

whence

$$\begin{aligned} F'v \cdot \tfrac{1}{M} &= 20u(1 - u^4)\\ &-10u^4(Su_4 - v)\\ &-10u(Sv_2u_1 + v^2Sv_0 - v^4)\\ &-10\,(Sv_3v_2u_1 + v^4Sv_2u_1 - v^4Sv_2 + v^2Sv_2 - v^4), \end{aligned}$$

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where Se_n, &c are the coefficients of the equation

$$v^5 + Se_n\,v^4 + Se_n\,v^3 - Se_n\,v^2 - Se_n\,v - u^4 = 0,$$

viz.

$$Se_n, \quad v_2v_1, \quad v_2v_2v_3, \quad v_3v_4v_5, \quad v_2v_3v_4v_5,$$

are

$$-4u^4,\ +5u^6,\ \ 0,\ \ -5u^2,\ \ 4u\,;$$

or the equation is

$$F'v \cdot \tfrac{1}{M} = 20u\,(1 - u^4)$$

$$-10u^4(-4u^4-v)$$
$$-10u\,(\;4u+5u^4\,v-4u^2v^3-v^4)$$
$$-10\;(\;4u+5u^4\,(v-v^2)\;,\quad 0\;,\quad -5u^2,\quad 4u\,;)$$

or, say

$$\tfrac{1}{2}F'v\cdot\tfrac{1}{M}=5\,[v^4+4uv^4+6v^2u^4+4u^6v+vu^2(1-u^2)],$$

where

$$\tfrac{1}{2}F'v=3\,v^4+10vu^4+10v^4u^2-5vu^2-2u.$$

Hence also, reducing by the modular equation,

$$\tfrac{1}{2}vF'v\cdot\tfrac{1}{M}=5v\,[v^4+4uv^5+6u^4v^2+2v(1+u^2)+u^2],$$

the one of which forms is as convenient as the other.

71. Making the change u, v, $\dfrac{1}{M}$ into v, $-u$, $-5M$, we have

$$-\tfrac{1}{2}F'u\cdot 5M=5\,[u^2+4u^4v+6v^2u^4+4u^6v+1-v^2(1-v^2)];$$

and comparing with the equation

$$\tfrac{v(1-v^2)}{u(1-u^2)}=-\tfrac{(1-v^2)\,uF'v}{(1-u^2)\,uF'u}\,;$$

the modular equation is, in fact, found in the form

$$-\,5M(v-u^2)\,(1-v^2)-(u-v^2)(1-u^2)+(1-u^2)(1-v^2)\,5M=0,$$

i.e. $5(uv-1)\,5M=4(u-v^2)(1-u^2)+5(uv-1)(1-v^2)\,5M=0,$

$$+5(1-u^2)(1-v^2)\,5M+10uv\,(1-uv)^2\,5M=0,$$

viz. from the value of $\dfrac{1}{M_n}$ for the case $n=5$, given by Jacobi, the modular equation in its several forms, writing successively u, v; u^2, v^2; u^4, v^4; u^4, v^4; u^4, v^4; $=x$, y, are not curves of the same deficiency.

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Hence, replacing M, N by their values, this is

$$-4uv(1-v^2)(1-u^2)$$
$$-(u^2-v^2)(1-u^2-u^2v^2-u^2v^2+u^2v^2+u^2v^2+v^4)(u^4+5u^2v+v^2)$$
$$+4uv^4\,(u^2+5u^2+u^2v^4)(1-u^2v^4(u^2-v^2)+u^2v^2)=0;$$

viz. writing $u^2-v^2=A,\;\,uv=B$, this is

$$-4B\,(1-u^2)\,(1-v^2)$$
$$-A\,(1-A^2-5A^2B^2-3B^4)\,(A^2+5B+B^2)$$
$$+4B\,(A^2+A^2+5B)\,(1-A^2B^2-B^4)\;=0,$$

that is,

$$-4B\,[(1-A^2-5A^2B^2-2B^2+B^4)$$
$$-A\,(1-A^2-5A^2B-3B^2)\,(A^2+5B^2)\,(A^2+5B^2)$$
$$+4B\,(A^2+A^2+5B)\,(1-A^2B^2-B^4)=0,$$

viz.

$$-4B\,(1-A^2-5A^2B-3B^2)\,(A^2+5B^2)\,(A^2+5B^2)$$
$$-A\,(1-A^2-5A^2B-5B^2)\,(A^2+5B^2)=0;$$

or throwing out the factor $-(1-A^2-5A^2B-5B^2)\,(A^2+5B^2)$, this is

$$A\,(A^2+5B^2)+4B\,(1-B^2)=0,$$

the modular equation, which is right.

The four forms of the modular equation, and the curves represented thereby. Art. Nos. 72 to 75.

72. The modular equation for any value of n has the property that it may be represented as an equation of the same order $(n+1, n+1)$, the order being a prime, between u, v; or between u^2, v^2; or between u^4, v^4. As to this remark, that in general an equation $(\alpha,\ v)^{m,n}=0$ of the order $2m$ between u, v may give an equation $(u^2,\ v^2)^{m,n}=0$ of the order m between u^2, v^2, the required equation is in u^2 and v^2. But if we have $\phi(u,v)=0$ of the order $2m$ between u, v; and the like passing to the inverse equation $\phi(u,v)=0$, if the coefficients reduce itself to an equation $(u,v)^{m,n}=0$ in u, v, the curve in (u,v) is of the order $2m$, that is, of the same deficiency as the original curve of the order $2m$ in (u,v). The transformed curve in $x, y, = u^2, v^2$ would in like manner be of the order $2m$, of the same deficiency as the original curve; and in particular the curves represented by the equation in their final its four several forms, writing them successively u, v; u^2, v^2; u^4, v^4; u^4, v^4; $= x, y$, are not curves of the same deficiency.

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would correspond 8 values of u, therefore $8a$ values of v, giving the same number, 8. values of v^2; that is, the values of u^2 correspond to a given value of u^2. There is, in fact, no group themselves in eights cusps at infinity, the coefficients are only two coefficients α, β, which are also the last but one and last coefficients ρ, σ. Hence, from the values of α, β, ρ, σ, respectively, are derived herefrom as follows; viz. and u^4, v^4 respectively, are derived herefrom as follows; viz. and

I. $y^4 - x^4 + 2xy\,(x^2y^2-1)=0;$

and it is at once seen that the remaining forms wherein x, y denote u^2, v^2; u^2, v^2; u^4, v^4 respectively, are derived herefrom as follows:

II. $(y^2-x^2)^4 - 4xy\,(xy-1)^4=0,$
or $x^4+6x^2y^2+y^4-4xy\,(x^2y^2+1)=0;$

III. $(y^2+6xy+y^2)^2-16xy\,(xy+1)^2=0,$
or $x^4+6xy+y^4-16xy\,(4x^2y^2-3xy)\cdot 32x^4y^4-96x^3y^3-70x^2y^2-96xy+64=0;$

IV. $(y^4+6xy+y^4)^4-16xy\,(4x^2y^2-3xy)\cdot 32x^4y^4-96x^3y^3-70x^2y^2-96xy+64=0;$

where the subscript line, showing in each case what the equation becomes on writing therein $x=1$, serves as a verification of the numerical values.

The curve IV. being unicursal, the coordinates may be expressed rationally in terms of a parameter; in fact, we have

$$x=\tfrac{a^4(2+a)}{1+2a},\quad y=\tfrac{a(2+a)^4}{(1+2a)^4}.$$

These values give

$$16xy = 16a^4(2+a)^4, \quad +(1+2a)^4,$$

$$4+4xy-3x-3y=(4,\ 8,\ 12,\ 32,\ 50,\ 32,\ 12,\ 8,\ 4)(1,\ a)^4, \quad +(1+2a)^4,$$

$$x^4+6xy+y^4=4a^4(2+a)^4(4,\ 8,\ 12,\ 32,\ 50,\ 32,\ 12,\ 8,\ 4)(1,\ a)^4, \quad +(1+2a)^4,$$

and the equation of the curve is thus verified.

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75. The foregoing equations may be exhibited in the square diagrams:—

[Tables: four 5×5 square diagrams labelled I, II, III, IV, expressing the non-zero entries of the modular polynomials in the variables y^4, y^3, y^2, y, 1 along the top axis and x^4, x^3, x^2, x, 1 down the left axis. Each diagram is read by multiplying out, with the row-sums giving the coefficients of the modular polynomial in y. The accompanying labels at the foot give the corresponding modular form $(y-1)^k(y\pm 1)^k$, and so on.]

Diagram I.

	y^4	y^3	y^2	y	1
x^4					-1
x^3		$+2$			
x^2					
x			-2		
1	$+1$				

$$1 \quad +2 \quad 0 \quad -2 \quad -1 = (y+1)^2(y-1);$$

Diagram II.

	y^4	y^3	y^2	y	1
x^4					$+1$
x^3			-4		
x^2			$+6$		
x		-4			
1	$+1$				

$$1 \quad -4 \quad +6 \quad -4 \quad +1 = (y-1)^4.$$

where the subscript line, showing in each case what the equation becomes on writing therein $x = 1$, serves as a verification of the numerical values.

The curve IV., being unicursal, the coordinates may be expressed rationally in terms of a parameter, in fact, we have

$$x = \tfrac{a^4(2+a)}{1+2a}, \quad y = \tfrac{a(2+a)^4}{(1+2a)^4}.$$

These values give

$$16xy = 16a^4(2+a)^5, \quad +(1+2a)^5,$$

$$4 + 4xy - 3x - 3y = (4,\ 8,\ 12,\ 32,\ 50,\ 32,\ 12,\ 8,\ 4)(1, a)^8 \quad + (1+2a)^8,$$

$$x^4 + 6xy + y^4 = 4a^4(2+a)^5(4,\ 8,\ 12,\ 32,\ 50,\ 32,\ 12,\ 8,\ 4)(1, a)^8 \quad + (1+2a)^8,$$

and the equation of the curve is thus verified.

[p. 172] 76.

Considering in like manner the modular equation for the quintic trans- formation, we derive the four forms as follows:—

I. $x^2y^2 + 5xy(x-y) + 4xy(1 - x^2y^2) = 0;$

II. $[x^2 - y^2 - 5xy(1 - x^2y^2)]^2 - 16xy(8 - 5x^2 - 5y^2 + 10x^2y^2 - 5xy)\,xy = 0$, that is, $x^4 + 15xy + 15xy + y^4 - 4xy\,(8 - 5x^2 + 10xy^2 - 5y^2 + 5xy^2) = 0;$

III. $(x^4 + 655xy + 655x^4y + y^4 - 4xy(8 - 5x^4 + 10x^4y^4 - 5y^4 + 5xy^4) - 640xy = 640x^4y^4);$
$+xy\,(-256 + 320v^2 + 320y^2 - 70x^2y^2 - 70y^2 - 256x^2y^2) - 640xy(x^4y^4 - 256x^2y^2 - 256) = 0;$

IV. $(x^4 + 655xy + 655x^4y + y^4 - 640xy - 640x^4y^4)$
$-xy\,(-256 + 320x^2 + 320y^2 - 70x^2y^2 - 70x^2y^2 - 256y^2) + 320xy(x^4y^4 - 256x^2y^2 - 256) = 0;$

or, expanding the two terms in the last equation separately, this is

[Table: a square diagram extending to powers x^iy^j with i, j up to 6, listing the non-zero entries of the polynomial in question. Cayley presents the diagram in a single 7×7 block: the column heads are x^6, x^5, x^4, x^3, x^2, x, 1 and similarly for the rows. The numerical entries reproduced from the printed diagram are:]

	x^6	x^5	x^4	x^3	x^2	x	1	= 0.
xy							−63536	
y^2						+163840		
y^3				+409600			−543720	
y^4			−1280		−138240			
y^5		−538400			−138240		+138240	
y^6	+1280		−538400	−639400		+44800		

[Cayley then expands the second of the equations of § 75 (the modular equation for the quintic transformation in its second form). The full table covers the next two pages and is presented as a single 7 × 7 array; the reproduction here is as in the original; entries omitted in the original print remain blank in our table.]

[p. 173] 77. The square diagrams are:—

Diagram I.

	y^5	y^4	y^3	y^2	y	1
x^5						−1
x^4		+4				
x^3			−5			
x^2		+5				
x			−4			
1	+1					

$$1 \quad +4 \quad +5 \quad 0 \quad -5 \quad -1 = (y+1)^4(y-1).$$

Diagram II.

	y^5	y^4	y^3	y^2	y	1
x^5						+1
x^4		−16			+10	
x^3			+15			
x^2			−20			
x		+10			−16	
1	+1					

$$1 \quad -6 \quad +15 \quad -20 \quad +15 \quad -6 \quad +1 = (y-1)^6.$$

Diagram III.

	y^5	y^4	y^3	y^2	y	1
x^5						+1
x^4	−256		+320		−70	
x^3		−640		+655		
x^2	+320		−600		+320	
x		+655		−640		
1	−70		+320		−256	

$$1 \quad -6 \quad +15 \quad -20 \quad +15 \quad -6 \quad +1 = (y-1)^6.$$

Diagram IV.

	y^5	y^4	y^3	y^2	y	1
x^5						$+1$
x^4	-65536	$+163840$	-138240	$+43520$	-3590	
x^3	$+163840$	-133120	-207360	$+153135$	$+43520$	
x^2	-138240	-207360	$+691180$	-207360	-138240	
x	$+43520$	$+153135$	-207360	-133120	$+163840$	
1	-3590	$+43520$	-138240	$+163840$	-65536	

$$1 \quad -6 \quad +15 \quad -20 \quad +15 \quad -6 \quad +1 = (y-1)^6.$$

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where the subscript line, showing in each case what the equation becomes on writing therein $x = 1$, serves as a verification of the numerical values.

78. The curve I. has at the origin a dp in the nature of a flecnode, viz. the two branches are given by $x^2 + 4y = 0$, $x - 4y^4 = 0$ respectively; and there are two singular points at infinity on the two axes respectively, viz. the infinite branches are given by $-y - 4u^2 = 0$, $x - 4y^4 = 0$ respectively. Writing the first of these in the form $-yx^2 - 4u^2 = 0$, $x - 4y^4 = 0$ respectively. Writing the first of these in the form $-yx^2 - 4u^2 = 0$, the point at infinity on the axis $x = 0$ is ∞ dps*.

Moreover, as remarked to me by Professor H. J. S. Smith, the curve has 8 other dps; viz. writing ω to denote an eighth root of -1, $(\omega^4 + 1 = 0)$, the eight branches are two singular points at infinity on the two axes respectively, viz. the infinite branches are given by $-y - 4\omega^4 = 0$, $x - 4\omega^4 = 0$ respectively. Writing the first of these in the form

$$\begin{array}{rcrl@{\qquad}rl} 6x^4 & = & +\,6\,, & & -\,6y^2 & =\,+\,6 \\ +20xy^2 & - & 20\,, & & +10xy & -\,10 \\ -10xy^4 & - & 10\,, & & -20xy^2 & -\,20 \\ +\,4y & + & 4\,, & & +\,4x & +\,4 \\ -20x^4y^4 & + & 20\,; & & -20x^4y^4 & +\,20\,; \end{array}$$

or these functions become $= 0$, in like manner $x = \omega$, $y = -\omega$, and that the curve is a $\frac{4}{2}$ dps; viz. instead of these, we have 1+12+2+6 + 6, = 21 dps.

In all 1+12+2+6 + 6 + 12, = 33 dps.

And in IV. we thence have 1+12+3+6, = 21 dps; and, besides, the 12 dps given by

$$x^2 + 5xy^2 + y^2 = 0, \quad 8 - 5x^2 + 10xy - 5y^2 + 5xy^4 = 0,$$

$(= 0,\ y = 0)$; we have besides 12 more, which form the 12 cusps of the figure $5x = 0, = 296 + 320x - 70x^4 - 660xy + 70x^4y^2 - 320xy^4 - 256x^4y^4 = 0$, $(296 + 820x - 70x^4 - 660xy + 70x^4y^2 - 320xy^4 - 256x^4y^4 = 0)$, (these intersect in 1+12+1+6+6+6 = 33 points); there are 12 dps, $10 - 4$, = 12 intersections); there are 1+12+1+6+6+6 = 33 dps.[1]

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Arranging the results in a tabular form and adding the values of the deficiency, we have

	dps.		Def.
I.	$1 + 12 + 8$	$= 21,$	$= 15,$
II.	$1 + 12 + 4 + 12$	$= 29,$	$= 7,$
III.	$1 + 12 + 2 + 6 + 12$	$= 33,$	$= 3,$
IV.	$1 + 12 + 1 + 5 + 6 + 12$	$= 35,$	$= 1,$

[1]* These curves follow from the general theorem in the paper "On the Higher Singularities of Plane Curves," *Camb. and Dubl. Math. Journ.*, t. vi. (1865), pp. 127–225, [271]; but they may be at once seen to be true from the consideration that the curve $yx^2 - 2u = 0$, which has only the singularity in question, is unicursal; the singularity is then $\infty - 1$ dps.

so that the curve IV. is a curve of deficiency 1, or bicursal curve. It appears by Jacobi's investigation for the quintic transformation (*Fund. Nov.* pp. 26–28, [*Ges. Werke*, t. i., pp. 77–79]) that we can in fact express x, y, that is, u^4, v^4, rationally in terms of the parameters α, β connected by the equation

$$\alpha^2 = 2\beta\,(1+\alpha+\beta),$$

which is that of a general cubic (deficiency$=1$); in fact, we have

$$\tfrac{2-\alpha}{\alpha-2\beta} = \tfrac{v^2}{u^2}, \quad \beta = \tfrac{u^2}{v},$$

that is,

$$u^4(=x) = \beta^4\left(\tfrac{2-\alpha}{\alpha-2\beta}\right), \quad v^4(=y) = \beta^4\left(\tfrac{2-\alpha}{\alpha-2\beta}\right)^4,$$

where α, β satisfy the relation just referred to. The actual verification of the equation IV. by means of these values would not be without some labour.

79. In the general case of any odd prime, the equation in u as the origin dp (in the nature of a fleshnode) and an infinity singular point with singular points in dps $\dfrac{1}{2}(n-1)(p-3)$ dps; and there is a result obtained by Professor Smith, that there are besides $\frac{1}{2}(n-1)(p-3)$ dps; but I have not investigated the nature of these. And the Table of dps and deficiency then is

		dps.	Def.
I.	$1+(p-1)(p-2)+(p-3)$	$=$	$2p^2-7p+6,\ 4p-5,$
II.	$1+(p-1)(p-2)+\frac{1}{2}(p-3)+\frac{1}{2}(p-1)$		$2p^2-5p+0,\ 3p-3,$
III.	$1+(p-1)(p-2)+\frac{1}{2}(p-3)+\frac{1}{2}(p-1)+\frac{1}{2}(p-1)+\frac{1}{2}(p-1)$		$2p^2-4p+0,\ p-2,$
IV.	$1+(p-1)(p-2)+\frac{1}{2}(p-3)+\frac{1}{2}(p-1)+\frac{1}{2}(p-1)+\frac{1}{2}(p-1)$		$2p^2-4p+\frac{1}{2},\ \frac{1}{2}p-\frac{1}{2};$

viz. the values of the deficiencies being as in the last column, the total number of dps must be as in the last but one column.

579.

Address Delivered by [Professor Cayley as] the President of the Royal Astronomical Society on Presenting the Gold Medal of the Society to Professor Simon Newcomb.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXIV. (1873—1874), pp. 224—233.]

[p. 176]

The Council have awarded the medal to Professor Simon Newcomb for his Researches on the Orbits of *Neptune* and *Uranus*, and for his other contributions to mathematical astronomy. And upon me, as President, the duty has devolved of explaining to you the grounds of their decision.

I think it right to remark that it appears to me that, in the award of their highest honour, the Council of a Society are not bound to institute a comparison between heterogeneous branches of a science, or classes of research—to weigh, for instance, mathematical against observational astronomy or astronomical physics; or, in the several branches respectively, the happy idea which originates a theory against the patience and the skilled labour which develope and carry it out; and still less to decide between the merits of different workers in the science. It is enough that the different branches of a science coming before them in different years, the medal should in every case be bestowed as a recognition of high merit in some important branch of the science.

Before speaking of the Tables, I will notice some of Professor Newcomb's other works.

MEMOIR "On the secular Variations and mutual Relations of the Orbits of the Asteroids," *Mem. American Academy*, vol. v. (1860), pp. 124—152. The object is to examine those circumstances of the forms, positions, variations, and general relations of the asteroid orbits which may serve as a test, complete or imperfect, of any hypothesis respecting the cause from which they originated, or the reason why they are in a

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group by themselves. Every *a posteriori* test is founded on the supposition, that the hypothesis necessarily or probably implies that certain conditions must be satisfied by the asteroids or their orbits, viz. in the one case the conditions are those which follow necessarily and immediately from the hypothesis itself, in the other case those which are deducible from it by the principle of random distribution. The two principal hypotheses are that of Olbers, where the asteroids are supposed to be the fragments of a shattered single planet; and the hypothesis that they were formed by the breaking up of a ring of nebulous matter. On the first hypothesis the orbits of all the asteroids once intersected in a common point; the second affords no conclusion equally susceptible of an *a posteriori* test.

But for a rigorous or probable test of either hypothesis, what is needed is rigorous expressions in terms of the time for the eccentricity, inclination, and longitudes of perihelion and node of each of the asteroids considered, or, what is the same thing, the computation of the secular variations of the quantities h, l, p, q, which replace these elements. The investigation is applied to these asteroids the elements of which were determined with sufficient accuracy, and the eccentricities and inclinations of which were sufficiently small (limit taken is 11°). And the backbone of the memoir is a shattered single planet: and the hypothesis that they were formed by the breaking up of a ring of nebulous matter. It is seen that the orbits of all the asteroids once intersected in a common point: the second affords no conclusion equally susceptible of an *a posteriori* test.

"Investigation of the Distance of the Sun and of the Elements which depend upon it, from the Observations of *Mars* made during the Opposition of 1862, and from other Sources," *Washington Observations for* 1865, Appendix II., pp. 1—29. The chief part of this valuable Memoir is occupied with a determination of the solar parallax by the discussion of the observations of *Mars* made in 1862 on the plan of Winnecke; three partial discussions had previously appeared, but those having been by comparisons of pairs of observations, one in each hemisphere, many observations in one hemisphere were lost by want of a corresponding observation in the other hemisphere; and out of a total of nearly 300 observations, only 125 were utilised. The idea is, the perturbations of the Earth and *Mars* being perfectly known for the period under consideration, every observation of the planet would lead rigorously to an equation of condition between its parallax, the six elements of its orbit, and the six elements of condition between its parallax, the six elements of its orbit, and the six elements of

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the Earth's orbit—thus 13 or more observations, when compared with any theory, should suffice to correct the errors of that theory. But the observations extending only over a short interval, say one month, the coefficients would be so minute as to give no trustworthy value of the corrections; the equations only suffice to determine a few functions of the elements which, being determined, the equations will be satisfied by widely differing values of the elements, if only those values are such as to give to the functions their right values. And by fixing *a priori* the entire number of functions in question, and using them in place of the elements of the Earth and *Mars*, the equations will be practically as rigorous as if all the 13 unknown quantities had been introduced. By such considerations as those, each observation is made to give a relation between only 3 unknown quantities, the correction of the Sun's parallax being one of them.

The principle appears to be one of extended application, in regard to the proper mode of dealing with the constantly recurring problem of the determination of a set of corrections from a large number of linear equations; and it is used by the author in regard to the equations which present themselves in his theories of *Neptune* and *Uranus*.

Returning to the *Mars* observations, these were made at six Northern and three Southern Observatories, the total number being 154 Northern, and 143 Southern, together 297 observations. There was the difficulty of reducing to a concordant system the observations at the different Observatories, since (the whole number of comparison stars not being observed on each night) the adopted mean position of each of them was not unimportant. But this being carefully discussed and allowed for, the observations, extending from August 21 to November 3, 1862, are divided into five groups, and from these is deduced a correction to the provisional value $8''.9$ of the parallax. The author then reproduces or discusses other determinations, from micrometric observations of *Mars*, the parallactic inequality of the Moon, the lunar equation of the Earth, the transit of 1769, and Foucault's experiment on light—the last result, as not a strictly astronomical one, and with no means of assigning its probable error, is left out of consideration—and the combination of the remaining ones gives the author's concluded value of the parallax; from which other astronomical constants are deduced.

"On the Right Ascensions of the Equatorial Fundamental Stars and the Correct- ions necessary to reduce the Right Ascensions of different Catalogues to a mean homogeneous System," *Washington Observations for* 1870, Appendix III., pp. 1—73.

This important Memoir is referred to in the Council Report for 1873. The object is to do for the right ascensions of the equatoreal and zodiacal Stars what had been done by Auwers for the declinations, namely, to furnish the data necessary to reduce the principal original catalogues of stars to a homogeneous system by freeing them of their systematic differences. The results are contained in two tables of corrections (as depending on the R.A. and N.P.D. respectively) to the several catalogues; and in a table of concluded mean right ascensions for the beginning of each fifth Besselian year,

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1750 to 1900, of 32 fundamental Stars, and of periodic terms in the right ascensions of *Sirius* and *Procyon*.

The evil of systematic differences between the observations of different Observatories of course presents itself in every case where such observations have to be combined: for instance, in the just-mentioned determination of the solar parallax by the obser- vations of *Mars*; and in the making of a set of planetary tables; and all that tends to remove or diminish it is most important to the progress of Astronomy. I cannot help thinking that there should be some confederation of Observatories, or Central calculating Board, for publishing the lunar and planetary observations, &c., reduced to a concordant system. It seems hard upon the maker of a set of planetary tables that he should not at least have, ready to hand for comparison with his theory, a single and entire series of observations of the planet for the comparison period.

"Théorie des Perturbations de la Lune, qui sont dues à l'action des Planètes," *Liouville*, t. XVI. (1871), pp. 1—45. This is a very important theoretical Memoir on the disturbed motion of three bodies: a problem which, notwithstanding the limited extent of the perturbations considered at all, has nevertheless to be considered with a different mode of dealing with the constantly recurring problem of the determination of a set of corrections from a large number of linear equations; and it is used by the author in regard to the equations which present themselves in his theories of *Neptune* and *Uranus*.

"Théorie des Perturbations de la Lune, qui sont dues à l'action des Planètes," *Liouville*, t. XVI. (1871), pp. 1—45. The evil of systematic differences between the observations of different Observatories of the present Memoir, Professor Newcomb starts from this problem in a fit more actually solved, viz. he takes the coordinates of the three bodies, Sun, Earth, and Moon) as given in terms of 18 constants of integration*. And then considering the equations of the system as the action of a planet, represented by the means of a disturbing function, he applies to the system of the three bodies the method of the variation of the elements. The six elements which determine the motion of the centre of gravity of the system are left out, and not considered; there remain to be considered 12 elements only (instead of being as before, with the elements *a posteriori*, retained), six of these belong to the relative orbit of the Earth, l_1, l_2, &c., are functions the invariation of which is a leading step in the theory, and they are also functions of the inclination of which is a leading step in the theory, and they are also the expressions of the last-mentioned six functions are, it is stated, be formed with facility by means of the developments (obtained therein from the lunar theory) of the rectangular coördinates x, y, z, as periodic functions of the time. With these twelve elements, the expressions for the variations assume the canonical form

$$\tfrac{dx_1}{dt}=\tfrac{dR}{de_1},\quad \tfrac{de_1}{dt}=-\tfrac{dR}{dx_1},\quad \text{\&c.}$$

The concluding part of the Memoir contains approximate calculations which seems to show that the whole proeess is in very practicable one: but the author remarks that it is only proper to defray justice to Delaunay to say that, starting from his (Delaunay's) final differential equations, and regarding the planet as adding new terms to the disturbing function, then by introducing the laws of the same degree of rigour as to the disturbing opinion as to the comparative facility of methods; but irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

I come now to the planets *Neptune* and *Uranus*: it is well-known how, historically, the two are connected. The increasing and systematic inaccuracies of Bouvard's Tables of *Uranus* were found to be such as could not be accounted for by the existence of differential equations, and regarding the planet as adding new terms to the disturbing function, he could have obtained results of the same degree of rigour as to the disturbing forces, leading to Hansen's first inequality produced by *Venus*.

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coördinates x, y, z, as periodic functions of the time. With these twelve elements, the expressions for the variations assume the canonical form

$$\tfrac{dx_1}{dt} = \tfrac{dR}{de_1}, \quad \tfrac{de_1}{dt} = -\tfrac{dR}{dx_1}, \quad \text{\&c.}$$

The concluding part of the Memoir contains approximate calculations which seems to show that the whole process is in very practicable one: but the author remarks that it is only proper to defray justice to Delaunay to say that, starting from his (Delaunay's) final differential equations, and regarding the planet as adding new terms to the disturbing function, he could have obtained results of the same degree of rigour as to the disturbing forces, leading to Hansen's first inequality produced by *Venus*.

Everything in the Lunar Theory is laborious, and it is impossible to form an opinion as to the comparative facility of methods, the irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

I come now to the planets *Neptune* and *Uranus*: it is well-known how, historically, the two are connected. The increasing and systematic inaccuracies of Bouvard's Tables of *Uranus* were found to be such as could not be accounted for by the existence of differential equations, and regarding the planet as adding new terms to the disturbing function, then by introducing the laws of the same degree of rigour as to the disturbing opinion as to the comparative facility of methods; but irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

[The Address continues beyond p. 180; the remaining pages 181–200 of the present range belong to it.]

591 (continued).

A Smith's Prize Paper and Dissertation.

[Continued, pp. 231–236 of Vol. IX.]

viz. since $X+Y=-P$, and $XY=Q$, this is

$$(P^2-4Q)X'^2Y'^2=0,$$

and writing also

$$X=-\tfrac{1}{2}\{P+\sqrt{(P^2-4Q)}\},\qquad Y=-\tfrac{1}{2}\{P-\sqrt{(P^2-4Q)}\},$$

we find

$$X'Y'=\tfrac{1}{4}\left\{P'^2-\frac{(PP'-2QQ')^2}{P^2-4Q}\right\};$$

the differential equation thus is

$$(P^2-4Q)\left\{P'^2-\frac{(PP'-2QQ')^2}{P^2-4Q}\right\}=0.$$

The application to the theory of singular solutions is that, in the case where the function $(1,P,Q,\ldots)(c,1)^n$ breaks up into rational factors $c-X$, $c-Y,\ldots$, the factor $\zeta=(X-Y)^2(X-Z)^2\ldots$ divides out and should be rejected from the differential equation, which in its true form is $X'Y'Z'\ldots=0$; viz. this is what we obtain immediately, considering the given integral equation as meaning the system of curves $c-X=0$, $c-Y=0$, $\ldots$, and there is not really any singular solution; whereas in the case where the factors are not rational, the factor in question, when the product $X'Y'Z'\ldots$ is expressed in terms of the coefficients $P,Q,\ldots$, and their derived coefficients does not divide out from the equation; and in this case, equated to zero, it gives a proper singular solution of the equation.

11. In the theory of elliptic motion, v denoting the mean anomaly and e the eccentricity, if m' be an angle such that $\tan\frac{1}{2}v=\sqrt{\dfrac{1+e}{1-e}}\tan\frac{1}{2}m'$, find in terms of e, m' the mean anomaly m.

Taking as usual u for the eccentric anomaly, to commence the solution write down

$$\tan\tfrac{1}{2}u=\sqrt{\frac{1+e}{1-e}}\,\frac{1-\tan\frac{1}{2}v}{1+\tan\frac{1}{2}v}\cdot\tan\tfrac{1}{2}v,$$

that is,

$$\tan\tfrac{1}{2}u=\sqrt{\frac{1-e^2}{(1-e)^2}}\,\tan\tfrac{1}{2}m',$$

and u being given hereby as a function of m', we have by substitution in the equation $m=u-e\sin u$, to find m as a function of m'.

A creditable approximate solution would be $m=m'+0\cdot e$, viz. this would be to show that neglecting terms in e^2, &c., we have $m=m'$. In fact, taking e small, we have

$$\tan\tfrac{1}{2}u=(1+e)\tan\tfrac{1}{2}m',$$

and thence if $m=m'+x$, we have

$$\tan\tfrac{1}{2}m'+\tfrac{1}{2}x\sec^2\tfrac{1}{2}m'=(1+e)\tan\tfrac{1}{2}m';$$

that is,

$$x=2e\cos^2\tfrac{1}{2}m'\tan\tfrac{1}{2}m'=e\sin m';\quad m=m'+e\sin m',$$

and

$$m=m'+e\sin m'-e\sin(m'+\ldots)=m'+0\cdot e.$$

The complete solution would be obtained by expanding u in terms of e, m' from the equation $\tan\frac{1}{2}u = \sqrt{\frac{1+e}{1-e}}\tan\frac{1}{2}m'$ (which is of the form $\tan\frac{1}{2}u = n\tan\frac{1}{2}m'$, giving for u a known series $= m' +$ multiple sines of m'), and then observing that the same equation leads to

$$\sin u = \frac{\sqrt{(1-e^2)}\,\sin m'}{1-e\cos m'},$$

we have

$$m = \text{series} - e\,\frac{\sqrt{(1-e^2)}\,\sin m'}{1-e\cos m'},$$

where the second term has also to be expanded in a series of multiple sines of m', which can be done without difficulty.

12. If (u, v) are given functions of the coordinates (x, y), neither of them a maximum or a minimum at a given point O; and if through O we draw Ox' in the direction in which v is constant and u increases, and Oy' in the direction in which u is constant and v increases; then the rotation (through an angle not greater than π), from Ox' to Oy' is in the same direction with that from Ox to Oy, or in the contrary direction, according as $\frac{du}{dx}\frac{dv}{dy} - \frac{du}{dy}\frac{dv}{dx}$ is positive or negative.

The theorem has not, so far as I am aware, been noticed, and it seems to be one of some importance; there is no difficulty in it, but the answer requires some care in writing out; of course where the whole question is one of sign and direction, the omission to state that a subsidiary quantity is positive may render an answer worthless.

It depends on the following lemma: Consider the triangle $OX'Y'$, such that Ox, Oy being any rectangular axes through the origin O, the coordinates of X' are h, k, and those of Y' are h_1, k_1; then considering the area as positive, the double area is $= \pm(hk_1 - h_1k)$, viz. the sign is $+$ or $-$ according as the rotation from OX' to OY' (through an angle less than π) is in the same direction with that from Ox to Oy, or in the contrary direction; or, what is the same thing, $hk_1 - h_1k$ is in the first case positive and in the second case negative.

To show this, suppose for a moment that the lines OX', OY' are each of them in the quadrant xOy, say in the first quadrant, the inclination of OY' to Ox exceeding that of OX' to Ox; then h, k, h_1, k_1 are all positive, and $\frac{k_1}{h_1} > \frac{k}{h}$, that is, $hk_1 - h_1k$ is $+$, and the rotation from OX' to OY' is in the same direction as that from Ox to Oy; or the lemma holds good. Now OX' remaining fixed, let OY' revolve in the direction Ox to Oy; so long as OY' remains in the first quadrant, $\frac{k_1}{h_1}$ continues to increase, and we have always $\frac{k_1}{h_1} > \frac{k}{h}$, and $hk_1 - h_1k = +$; when OY' comes into the second quadrant (h, k being always positive), h_1 is negative and k_1 positive, consequently $hk_1 - h_1k$ is the sum of two positive terms, and therefore $= +$; as OY' continues to revolve and passes into the third quadrant, we have h_1, k_1 each negative, but $\frac{k_1}{h_1} < \frac{k}{h}$, and therefore $hk_1 - h_1k$ still $= +$; when, however, OY' comes into the position opposite to OX', then $\frac{k_1}{h_1} = \frac{k}{h}$, and $hk_1 - h_1k$ is $= 0$; and when OY', continuing in the third quadrant, has passed the position in question, we have $\frac{k_1}{h_1} > \frac{k}{h}$, and therefore $hk_1 - h_1k = -$, but now the angle $X'OY'$ measured in the original direction has become $> \pi$, and the rotation OX' to OY' through an angle less than π will be in the opposite direction, that is, in the direction opposite to that from Ox to Oy; and, similarly, when OY' passes into the fourth quadrant, and until, passing into the first quadrant, it approaches the position OX', the sign of $hk_1 - h_1k$ will be $-$, and the rotation will be in the direction contrary to that from Ox to Oy. The lemma is thus true for any position of OX' in the first quadrant; and the like reasoning would show that it is true for any position of OX' in the second, the third, or the fourth quadrant; hence the lemma is true generally.

This being so, taking a new origin, let the coordinates of O be x, y; and drawing through O the axes Ox', Oy' as directed, let X' be the point belonging to the values $u + \delta u, v$ of (u, v), and Y' the point belonging to the values $u, v + \delta v$ of (u, v); taking δu positive, X' will be on Ox' in the direction O to x', and similarly taking δv positive, Y' will be on Oy' in the direction O to y'. Taking as before (h, k) for the coordinates of X', and (h_1, k_1) for the coordinates of Y', these coordinates being measured from the point O as origin, we have

$$\delta h = \frac{du}{dx}\,h + \frac{du}{dy}\,k,$$

$$\delta k = \frac{dv}{dx}\,h + \frac{dv}{dy}\,k;$$

whence, writing for a moment $J = \frac{du}{dx}\frac{dv}{dy} - \frac{du}{dy}\frac{dv}{dx}$, we have

$$Jh = +\frac{dv}{dy}\,\delta u, \qquad Jk = -\frac{dv}{dx}\,\delta u.$$

And in like manner

$$\delta h = \frac{du}{dx}\,h_1 + \frac{du}{dy}\,k_1,$$

$$\delta k = \frac{dv}{dx}\,h_1 + \frac{dv}{dy}\,k_1;$$

whence

$$Jk_1 = -\frac{du}{dy}\,\delta v, \qquad Jh_1 = \frac{du}{dx}\,\delta v;$$

and hence

$$J^2(hk_1 - h_1k) = \left(\frac{du}{dx}\frac{dv}{dy} - \frac{du}{dy}\frac{dv}{dx}\right) J\,\delta u\,\delta v = J\,\delta u\,\delta v,$$

that is,

$$hk_1 - h_1k = \frac{1}{J}\,\delta u\,\delta v,$$

and $\delta u, \delta v$ being as above each of them positive, J has the same sign as $hk_1 - h_1k$. But the rotation from OX' to OY' is in the same direction as that from Ox to Oy, or in the contrary direction, according as $hk_1 - h_1k$ is $+$ or $-$, that is, according as $J = \frac{du}{dx}\frac{dv}{dy} - \frac{du}{dy}\frac{dv}{dx}$ is $+$ or $-$; which is the theorem in question.

13. Write a dissertation on:

The theory and constructions of Perspective.

In Perspective we represent an object in space by means of its central projection upon a plane: viz. any point P_1[1] of the object is represented by P', the intersection with the plane of projection of the line D_1P_1 from the centre of projection (or say the eye) D_1 to the point P_1; and considering any line or curve in the object, this is represented by the line or curve which is the locus of the points P', the projections of the corresponding points P_1 of the line or the curve in the object.

The fundamental construction in perspective is derived from the following considerations: viz. considering through P_1 (fig. 1) a line meeting the plane of projection in Q, and drawing parallel thereto through D_1 a line to meet the plane of projection in M and joining the points M, Q, then the lines D_1M, MQ, QP_1 are in a plane; that is, the plane through D_1 and the line P_1Q meets the plane of projection in MQ;

[Figure 1: A schematic showing the eye D_1 outside the plane of projection, a point P_1 of the object and the projection line through D_1 to P_1 meeting the plane in P'. An auxiliary line through P_1 meets the plane in Q, with a parallel line through D_1 meeting the plane in M; the joined segment MQ shows the trace of the plane through D_1P_1Q.]

and consequently the projection P' of any point P_1 in the line P_1Q lies in the line QM; and not only so, but considering only the points P_1 of this line which lie behind the plane of projection (D_1 being considered as in front of it), the projections lie on the terminated line MQ; viz. Q is the projection of the point Q, and M the projection of the point at infinity on the line QP_1; or, if we please, the finite line QM is the projection of the line $QP_1\infty$.

[1]The subscript unity is used to denote a point not in the plane of projection, considered as a point out of this plane; a point in the plane of projection, used in the constructions of perspective as a conventional representation of a point P_1, will be denoted by the same letter P without the subscript unity. And the like as regards D_1 and D.

If we consider a set of lines parallel to P_1Q, these all give rise to the same point M, and thus their projections MQ all pass through this point M, which is said to be the "vanishing point" of the system of parallel lines. Again, if we consider any two or more lines through P_1, to each of these there correspond different points M and Q, and, therefore, a different line MQ, but these all intersect in a common point P' which is the projection of P_1. If the lines are all in one and the same plane through P_1, then the locus of the points Q is a line, the intersection of this plane with the plane of projection, say the "trace" line; and the locus of the points M is a parallel line, the intersection of the parallel plane through D_1 with the plane of projection; say this is the "vanishing line" for the plane in question.

A construction in perspective presupposes a conventional representation on the plane of projection (or say on the paper) as well of the position of the eye as of the object to be projected. If for simplicity we suppose the object to be a figure in one plane, then this plane intersects the paper in a trace line, and we may imagine the plane made to rotate about the trace line until it comes to coincide with the paper, and we have thus the plane object conventionally represented on the paper. Similarly considering the parallel plane through the eye D_1, and regarding D_1 as a point of this plane, the plane meets the paper in the vanishing line, and we may imagine the plane made to rotate (in the direction opposite to that of the first rotation) until it comes to coincide with the paper, bringing the point D_1 to coincide with a point D of the paper. We have thus the "point of distance" D, being a conventional representation on the paper of the position of the eye D_1; but which point D has, observe, a different position for different directions of the plane of the object.

To fix the ideas, suppose the plane of projection to be vertical, and the plane of the object to be a horizontal plane situate below the eye. The trace line will be represented by a horizontal line HH' (fig. 2), and the object by a figure in the plane of the paper below the line HH' such that, bending this portion of the paper backwards through a right angle round HH', the figure would be brought to coincide with the object[2]. The vanishing line will be a horizontal line KK' above HH', and the eye will be represented by a point D above KK', in suchwise that, bending the upper part of the paper round KK' forwards through a right angle, the point D would come to coincide with the position D_1 of the eye. This being so, taking any line PQ in the representation of the object, we draw through D the parallel line DM, and then joining the points M and Q, we have MQ as the perspective representation of the line $QP\infty$, which represents a line QP_1 of the object. And drawing through P any number of lines, each of these gives a point Q and a point M, but the lines MQ all meet in a common point P', which is the perspective representation of the point P; which point P' may, it is clear, be obtained as the intersection with any one line MQ of the line DP drawn to join P with the point of distance D. The plane of the object has for convenience been taken to be horizontal; but its position may be any whatever, and in particular the construction is equally applicable in the case where the plane is vertical.

[Figure 2: A vertical plane of projection. A horizontal trace line HH', above it a parallel horizontal vanishing line KK', and above KK' the eye-point D with its foot S on KK'. Below HH' in the same paper plane lies the rotated horizontal object plane, containing a line PQ. Joining D to P and drawing $DM \parallel PQ$ to meet KK' in M, then MQ is the perspective of the line $QP\infty$, and $P' = DP \cdot MQ$ is the perspective of P.]

In the case of an object not in one plane, any point Q_1 of the object may be determined by means of its projection by a vertical line upon a given horizontal plane, say this is P_1, and of its altitude Q_1P_1 above this plane. We in fact determine the object by means of its groundplan, and of the altitudes of the several points thereof. It is easy, from the foregoing principles, to see that, drawing through P the vertical line PQ equal to the altitude, and joining the points Q, D, then the vertical line through P' meets this line QD in a point Q', which will be the perspective representation of Q_1. We have thus a construction applicable to any solid figure whatever.

592.

On the Mercator's-Projection of a Skew Hyperboloid of Revolution.

[2]It is assumed in the text, that the figure on the paper is equal in magnitude to the object; but practically the figure is drawn on a reduced scale, the distance between the lines KK', HH', and the distance DS (representing respectively the distance between the parallel planes, and the distance of the eye from the plane of projection) being drawn on the same reduced scale.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 17—20.]

In a note "On the Mercator's-projection of a surface of revolution" read before the British Association, [555, (5)], I remarked that the surface might be, by its meridians and parallels, divided into infinitesimal squares; and that these would be on the map represented by two systems of parallel lines at right angles to each other, dividing the map into infinitesimal squares; and that, by taking the squares not infinitesimal but small, for instance, by considering the meridians at intervals of 10° or 5°, we might approximately construct a Mercator's-projection of the surface. But it is worth while, for the skew hyperboloid of revolution, to develop analytically the ordinary accurate solution.

Taking the equation of the surface to be

$$\frac{x^2+y^2}{a^2}-\frac{z^2}{c^2}=1,$$

(or, if as usual $a^2+c^2=a^2e^2$, then $x^2+y^2-(e^2-1)z^2=a^2$), and writing $x=r\cos\theta,\ y=r\sin\theta$, the meridians corresponding to the several longitudes θ are in the map represented by the parallel lines $X=a\theta$, and the parallels corresponding to the several values of z are in the map represented by a set of parallel lines $Z=f(z)$, the form of the function being so determined that the infinitesimal rectangles on the map are similar to those on the surface. The required relation is readily found to be

$$Z=\int\frac{\sqrt{\{(a^2+c^2)z^2+c^4\}}}{z^2+c^2}\,dz,$$

where the integral is taken from the value $z=0$.

The substitution which first presents itself is to write herein $z=\dfrac{c^2}{\sqrt{(a^2+c^2)}}\tan\phi$; or, what is the same thing,

$$z=a\left(e-\tfrac{1}{e}\right)\tan\phi,$$

where observe that $a\left(e-\dfrac{1}{e}\right)$ is the distance between a focus and its corresponding directrix. The equation of the surface is satisfied by writing therein $\sqrt{(x^2+y^2)}=a\sec\psi,\ z=c\tan\psi$, and ψ as thus defined is the "parametric latitude"; hence the foregoing angle ϕ is a deduced latitude connected with the parametric latitude ψ by the equation

$$\tan\phi=\frac{c}{a\left(e-\frac{1}{e}\right)}\tan\psi,\quad=\frac{c}{\sqrt{(c^2-1)}}\tan\psi.$$

The resulting formula in terms of ϕ is

$$Z=2c^2\int_0^\phi\frac{(1+\zeta^2)^2\,d\zeta}{c^2(1+\zeta^2)^2+a^2(1-\zeta^2)^2},$$

or, if we write herein $\zeta=\tan\frac{1}{2}\phi$, the formula becomes

$$Z=\int\frac{d\phi}{\cos\phi\,(c^2+a^2\cos^2\phi)},$$

viz. the function under the integral sign is rational. The expression is, however, complicated, and a more simple formula is obtained by using instead of ϕ the parametric latitude ψ; viz. we have $z=c\tan\psi$, and thence

$$Z=\int\frac{\sqrt{(a^2\sin^2\psi+c^2)}}{\cos\psi}\,d\psi,$$

or, putting herein

$$\sin\psi=\frac{c}{a}\cdot\frac{u}{\sqrt{(1-u^2)}},$$

and therefore

$$\cos^2\psi=\frac{a^2-(a^2+c^2)\,u^2}{a^2(1-u^2)},$$

and

$$a^2 \sin^2\psi + c^2 = \frac{c^2}{1-u^2}, \quad \cos\psi\, d\psi = \frac{c}{a}\cdot\frac{du}{(1-u^2)^{\frac{3}{2}}},$$

the formula becomes

$$Z = c^2 a \int_0^u \frac{du}{(1-u^2)\{a^2-(a^2+c^2)\,u^2\}},$$

or, what is the same thing,

$$Z = c^2 a \int_0^u \frac{du}{(1-u^2)(1-e^2u^2)},$$

viz. we thus have

$$Z = \tfrac{1}{2}a\left\{\log\frac{1+u}{1-u} - e\log\frac{1-eu}{1+eu}\right\},$$

the logarithms being hyperbolic.

As already mentioned, u is connected with the parametric latitude ψ by the equation

$$\sin\psi = \frac{c}{a}\cdot\frac{u}{\sqrt{(1-u^2)}}, \quad = \frac{u\sqrt{(e^2-1)}}{\sqrt{(1-u^2)}},$$

that is,

$$\sin\psi = \sqrt{(e^2-1)}\tan p, \quad \text{if } u = \sin p,$$

or conversely

$$u = \frac{\sin\psi}{\sqrt{(e^2-1+\sin^2\psi)}},$$

so that the point passing to infinity along the branch of the hyperbola, or ψ passing from 0 to 90°, u passes from 0 to $\frac{1}{e}$; and for $u=\frac{1}{e}$ the value of Z becomes, as it should do, infinite. The value of z in terms of u is

$$z = \frac{\sqrt{(e^2-1)}\,u}{\sqrt{(1-e^2u^2)}}, \quad \text{or conversely} \quad u = \frac{z}{\sqrt{(c^2+e^2z^2)}},$$

and we have, moreover,

$$u = \frac{1}{e}\cdot\frac{e\zeta}{1+\zeta^2}, \quad = \frac{1}{e}\sin\phi, \quad = \text{(as before) } \frac{\sin\psi}{\sqrt{(e^2-1+\sin^2\psi)}}.$$

It will be recollected that, in the Mercator's-projection of the sphere, the longitude and latitude being θ, ϕ, the values of X, Z are

$$X = a\theta, \qquad Z = \log\tan\left(\tfrac{\pi}{4}+\tfrac{1}{2}\phi\right),$$

the logarithm being hyperbolic.

In the case of the rectangular hyperbola $a = c, = 1$ suppose,

$$e=\sqrt{2}, \quad z=\tan\psi, \quad u=\frac{\sin\psi}{\sqrt{(1+\sin^2\psi)}}, \quad = \sin p,\ \text{if}\ \sin p = \tan\psi;$$

whence

$$Z = \tfrac{1}{2}h\cdot l\,\frac{\tan(45^\circ-\frac{1}{2}p)}{\tan(45^\circ+\frac{1}{2}p)} - \tfrac{1}{2}\sqrt{(2)}\,h\cdot l\,\frac{\tan(22^\circ30'-\frac{1}{2}p)}{\tan(22^\circ30'+\frac{1}{2}p)},$$

the first term being of course

$$= h\cdot l\tan(45^\circ-\tfrac{1}{2}p), \quad \text{or} \ = -h\cdot l\tan(45^\circ+\tfrac{1}{2}p).$$

Transforming to ordinary logarithms, this is

$$Z = \frac{1}{\sqrt{(2)}\log e}\Big[-\sqrt{(2)}\log\tan(45^\circ+\tfrac{1}{2}p) + \{\log\tan(22^\circ30'+\tfrac{1}{2}p) - \log\tan(22^\circ30'-\tfrac{1}{2}p)\}\Big],$$

say this is

$$Z = \frac{1}{\sqrt{(2)}\log e}(-A + B),$$

where

$$A = \sqrt{(2)}\log\tan(45^\circ + \tfrac{1}{2}p),$$
$$B = \log\tan(22^\circ 30' + \tfrac{1}{2}p) - \log\tan(22^\circ 30' - \tfrac{1}{2}p).$$

Taking ψ as the argument, I tabulate $z, = \tan\psi$, and $Z \cdot \sqrt{(2)}\log e, = -A + B$, as shown in the annexed table: the last column of which gives, therefore, the positions of the several parallels of $5^\circ, 10^\circ, \ldots, 85^\circ$; the interval of 5° between two meridians is, on the same scale,

$$\frac{\sqrt{(2)}\log e}{180} \cdot \pi = (1{\cdot}4142136)({\cdot}4342945)({\cdot}0872665) = {\cdot}05360,$$

viz. this is nearly equal to the arc of meridian 0° to 5°, and the table shows that the arcs $0^\circ - 5^\circ$, $5^\circ - 10^\circ$, &c. continually increase as in a Mercator's-projection of the sphere, but more rapidly; there is, however, nothing in this comparison, since the determination of latitude on the hyperboloid by the equation $z = \tan\psi$ is altogether arbitrary.

ψ	$z = \tan\psi$	p	A	B	$-A+B$
0°	0·	0° 0′ 0″	0·	0·	0·
5	·08749	4° 58′ 51″	·05348	·10719	·05370
10	·17632	9 51 0	·10611	·21439	·10827
15	·26795	14 30 40	·15724	·32174	·16450
20	·36397	18 53 0	·20619	·42943	·22324
25	·46631	22 54 30	·25342	·53780	·28539
30	·57735	26 34 0	·29557	·64758	·35201
35	·70020	29 50 20	·33538	·75959	·42421
40	·83910	32 44 0	·37170	·87506	·50337
45	1 · 00000	35 15 50	·40446	·99554	·59108
50	1 · 19175	37 27 20	·43360	1 · 12355	·68995
55	1 · 42815	39 19 20	·45913	1 · 26151	·80238
60	1 · 73205	40 53 40	·48117	1 · 41445	·93328
65	2 · 14450	42 11 10	·49971	1 · 58840	1 · 08869
70	2 · 74747	43 13 10	·51478	1 · 79504	1 · 28026
75	3 · 73205	44 0 30	·52646	2 · 05524	1 · 52877
80	5 · 67128	44 33 40	·53469	2 · 41347	1 · 87877
85	11 · 43005	44 53 30	·53973	3 · 02355	2 · 48381
90	∞	45° 0′ 0″	·54133	∞	∞

593.

A Sheepshanks' Problem (1866).

[From the Messenger of Mathematics, vol. iv. (1875), pp. 34—36.]

Apply the formulae of elliptic motion to determine the motion of a body let fall from the top of a tower at the equator.

The earth is regarded as rotating with the angular velocity ω round a fixed axis, so that the body is in fact projected from the apocentre with an angular velocity $= \omega$; and we write a for the equatorial radius, β for the height of the tower; then g denoting the force of gravity, and μ, h, n, a, e, θ, as in the theory of elliptic motion, we have

$$\mu = n^2 a^3 \ = ga^2,$$
$$h = (a+\beta)^2\omega \ = na^2\sqrt{(1-e^2)},$$
$$a + \beta = a(1+e),$$

whence

$$(a+\beta)^4\omega^2 = ga^3(1-e^2),$$

$$(a+\beta) = a(1+e),$$

$$\frac{(a+\beta)^3\omega^2}{ga^2} = \frac{a\omega^2}{g}\left(1+\frac{\beta}{a}\right)^3 = 1-e,$$

where $\dfrac{a\omega^2}{g}$ = ratio of centrifugal force to gravity,

$$= \tfrac{1}{289},$$

so that $1-e$ is small,

$$r = \frac{a(1-e^2)}{1-e\cos\theta} = \frac{(a+\beta)(1-e)}{1-e\cos\theta}.$$

Suppose

$$r = a,$$

$$1-e\cos\theta = (1-e)\left(1+\frac{\beta}{a}\right) = 1-e+\frac{\beta}{a}(1-e),$$

which is nearly

$$= 1-e,$$

that is, θ is small, and therefore approximately

$$1-e+\tfrac{1}{2}e\theta^2 = 1-e+\frac{\beta}{a}(1-e),$$

or

$$\theta^2 = \frac{2\beta}{a}\cdot\frac{1-e}{e}.$$

We then have

$$r^2\,d\theta = h\,dt, \quad \text{or} \quad dt = \frac{r^2\,d\theta}{h} = \frac{(a+\beta)^2(1-e)^2}{(a+\beta)^2\omega\,(1-e\cos\theta)^2}\,d\theta$$

$$= \frac{(1-e)^2}{\omega(1-e\cos\theta)^2}\,d\theta$$

$$= \frac{(1-e)^2}{\omega(1-e+\frac{1}{2}e\theta^2)^2}\,d\theta$$

$$= \frac{1}{\omega\left(1+\frac{\frac{1}{2}e}{1-e}\theta^2\right)^2}\,d\theta,$$

that is,

$$\omega\,dt = \left(1-\frac{e}{1-e}\,\theta^2\right)d\theta.$$

Integrating, we have

$$\theta\left(1-\tfrac{1}{3}\frac{e\theta^2}{1-e}\right) = \omega t,$$

where ωt = earth's rotation in time t, $= \phi$ suppose; therefore

$$\theta\left(1-\tfrac{1}{3}\frac{e\theta^2}{1-e}\right) = \phi,$$

hence, if θ be as above, the angle described in falling to the surface,

$$\theta-\phi = \tfrac{1}{3}\,\frac{e\theta^3}{1-e} = \frac{e\theta}{1-e}\cdot\frac{\theta^2}{3} = \frac{e}{1-e}\cdot\frac{1}{3}\cdot\frac{2\beta}{a}\,\theta.$$

Writing herein

$$\frac{1-e}{e}=\frac{a\omega^2}{g},$$

this is

$$=\tfrac{2}{3}\,\frac{\beta}{a}\sqrt{\left(\frac{2\beta}{a}\cdot\frac{g}{a\omega^2}\right)}\sqrt{\left(\frac{a\omega^2}{g}\right)},$$

$$=\tfrac{2\beta}{3a}\cdot\frac{\beta}{a}\sqrt{\left(\frac{a\omega^2}{g}\right)},$$

viz.

$$\theta-\phi=\frac{2}{3a^2}\,\beta\sqrt{\left(\frac{2\beta}{a}\right)}\sqrt{\left(\frac{a\omega^2}{g}\right)},$$

whence

$$a(\theta-\phi)=\frac{2}{3}\,\sqrt{(2)}\sqrt{\left(\frac{\beta^3}{a}\right)}\sqrt{\left(\frac{a\omega^2}{g}\right)},$$

or say

$$\frac{a(\theta-\phi)}{\beta}=\frac{2\sqrt{(2)}}{3}\sqrt{\left(\frac{\beta}{a}\right)}\sqrt{\left(\frac{a\omega^2}{g}\right)},$$

where $a(\theta-\phi)$ is the distance at which the body falls from the foot of the tower. Substituting for $\sqrt{\left(\frac{a\omega^2}{g}\right)}$ its value, $=\frac{1}{17}$, we have

$$\frac{a(\theta-\phi)}{\beta}=\frac{2\sqrt{(2)}}{3\cdot 17}\sqrt{\left(\frac{\beta}{a}\right)}=\cdot 056\sqrt{\left(\frac{\beta}{a}\right)}.$$

594.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 69, 70.]

The following equation presented itself to me in connexion with the cubic transformation:

$$Q^2-Q\left(k+\frac{1}{k}\right)\cdot 3=3(1-k^2)\frac{dQ}{dk}.$$

Writing as usual $k=u^4$, I was aware that a solution was

$$Q=\frac{v^2}{u^2}+2uv,$$

where u, v are connected by the modular equation

$$u^4-v^4+2uv(1-u^2v^2)=0;$$

but it was no easy matter to verify that the differential equation was satisfied. After a different solution, it occurred to me to obtain the relation between (Q,u); or, what is the same thing, (Q,k), viz. eliminating v, we find

$$Q^2-6Q^2-4\left(u^4+\frac{1}{u^4}\right)Q-3=0,$$

or say

$$\tfrac{1}{Q}(Q^4-6Q^2-3)=4\left(k+\tfrac{1}{k}\right);$$

whence also

$$\tfrac{1}{Q}(Q^2 \pm 8Q - 3) = 4\big(k \pm 2 + \tfrac{1}{k}\big)\,;$$

that is,

$$\tfrac{1}{Q}(Q-1)^2(Q+3) = 4\Big\{\sqrt{k} + \tfrac{1}{\sqrt{k}}\Big\}^2,$$

and

$$\tfrac{1}{Q}(Q+1)^2(Q-3) = 4\Big\{\sqrt{k} - \tfrac{1}{\sqrt{k}}\Big\}^2,$$

and thence

$$\frac{(Q+1)^2(Q-3)}{(Q-1)^2(Q+3)} = \left(\frac{k-1}{k+1}\right)^2;$$

viz. the value of Q thus determined must satisfy the differential equation. This is easily verified, for, in virtue of the assumed integral, we have

$$Q^2 - 3 - \tfrac{1}{Q}(Q^4 - 6Q^2 - 3) = 3(1-k^2)\frac{dQ}{dk};$$

that is,

$$Q^2 - 10Q^2 + 9 = -12(1-k^2)\frac{dQ}{dk},$$

or finally

$$(Q^2-1)(Q^2-9) = -12(1-k^2)\frac{dQ}{dk},$$

an equation which is at once obtained by differentiating logarithmically the former result, and we have thus the verification of the solution. This is, however, a particular integral only; and it appears doubtful whether there exists a general integral of an algebraical form.

595.

On a Senate-House Problem.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 75—78.]

The following was given [5 Jan., 1874,] as a problem of elementary algebra:

"Solve the equations

$$u(2a-x) = x(2a-y) = y(2a-z) = z(2a-u) = b^2,$$

and prove that unless $b^2 = 2a^2$, $x = y = z = u$, but that if $b^2 = 2a^2$, the equations are not independent."

This is really a very remarkable theorem in regard to the intersections of a certain set of four quadric surfaces in four-dimensional space; viz. slightly altering the notation, we may write the equations in the form

$$\begin{aligned} x(2\theta - y) &= m\theta^2 \;\ldots(12),\\ y(2\theta - z) &= m\theta^2 \;\ldots(23),\\ z(2\theta - w) &= m\theta^2 \;\ldots(34),\\ w(2\theta - x) &= m\theta^2 \;\ldots(41), \end{aligned}$$

where, regarding (x, y, z, w, θ) as coordinates in four-dimensional space, each equation represents a quadric surface. I remark that in such a space we have the notions, point-system, curve, subsurface, surface, according as the number of equations is $4, 3, 2$, or 1.

Four quadric surfaces intersect in general in 16 points. But for the system in question (m being arbitrary), the common intersection consists of two lines and the two points

$$x = y = z = w = \theta\{1 \pm \sqrt{(1-m)}\}$$

and in the case where $m = 2$, then the intersection consists of two lines and a certain unicursal quartic curve.

To obtain these results, I consider the four points

$$\begin{aligned}
\theta = 0,\ x = 0,\ y = 0,\ z = 0,\ \ldots 123,\\
\theta = 0,\ y = 0,\ z = 0,\ w = 0,\ \ldots 234,\\
\theta = 0,\ z = 0,\ w = 0,\ x = 0,\ \ldots 341,\\
\theta = 0,\ w = 0,\ x = 0,\ y = 0,\ \ldots 412:
\end{aligned}$$

the two points

$$x = y = z = w = \theta\{1 \pm \sqrt{(1-m)}\},\ \ldots PQ:$$

and the six lines

$$\begin{aligned}
\theta = 0,\ x = 0,\ y = 0,\ \ldots 12,\\
\theta = 0,\ y = 0,\ z = 0,\ \ldots 23,\\
\theta = 0,\ z = 0,\ w = 0,\ \ldots 34,\\
\theta = 0,\ w = 0,\ x = 0,\ \ldots 41,\\
\theta = 0,\ x = 0,\ z = 0,\ \ldots 13,\\
\theta = 0,\ y = 0,\ w = 0,\ \ldots 24,
\end{aligned}$$

being the edges of a tetrahedron, the vertices of which are the four points, viz. the point 123 is the intersection of the lines $12, 13, 23$, and so for the other points.

The surfaces contain the several lines, viz.

$$\begin{array}{rl}
\text{the surface 12 contains} & (12)^2,\ 13,\ 14,\ 23,\ 24,\\
23 & (23)^2,\ 12,\ 24,\ 13,\ 34,\\
34 & (34)^2,\ 13,\ 23,\ 14,\ 24,\\
41 & (41)^2,\ 24,\ 34,\ 12,\ 13,
\end{array}$$

where $(12)^2$ denotes that 12 is a double line on the surface, and so in other cases. And it thus appears that the surfaces pass all four of them through the lines $13, 24$, so that these lines are a part of the common intersection. To obtain the residual intersection, observe that the equations give

$$x = 2\theta - \frac{m\theta^2}{w} = \frac{m\theta^2}{2\theta - y},$$

$$z = 2\theta - \frac{m\theta^2}{y} = \frac{m\theta^2}{2\theta - w},$$

whence

$$(2\theta - y)\left(2\theta - \frac{m\theta^2}{w}\right) = m\theta^2,$$

$$(2\theta - w)\left(2\theta - \frac{m\theta^2}{y}\right) = m\theta^2,$$

or omitting from each equation the factor θ, the equations become

$$(2\theta - y)(2w - m\theta) = m\theta w,$$

$$(2\theta - w)(2y - m\theta) = m\theta y;$$

that is,

$$\begin{aligned}
(4-2m)\theta w - 2m\theta^2 - 2yw + m\theta(y+w) = 0,\\
(4-2m)\theta y - 2m\theta^2 - 2yw + m\theta(y+w) = 0.
\end{aligned}$$

Whence, m not being $= 2$, we have $y = w$, and then

$$w^2 - 2\theta w + m\theta^2 = 0,$$

or, what is the same thing,

$$2\theta - w = \frac{m\theta^2}{w},$$

giving $x = y = z = w = \theta\{1 \pm \sqrt{(1-m)}\}$, viz. the surfaces each pass through the points P, Q. As regards the omitted factor θ, it is to be observed that, writing in the equations of the four surfaces $\theta = 0$, the equations become $xy = 0,\ yz = 0,\ zw = 0,\ wx = 0$, satisfied by $x = 0, z = 0$, or by $y = 0, w = 0$; we have thus $(\theta = 0, x = 0, z = 0)$ and $(\theta = 0, y = 0, w = 0)$, viz. the before-mentioned lines 13 and 24.

In the case $m = 2$, we have between y, w the single equation

$$yw - \theta(y + w) + 2\theta^2 = 0,$$

giving

$$y = \frac{\theta(w - 2\theta)}{w - \theta},$$

and thence

$$z = \frac{2\theta(w - \theta)}{w - 2\theta},$$

or, writing for convenience $a = \dfrac{w}{\theta}$, then the equations are

$$\begin{aligned} \frac{w}{\theta} &= a, \\ \frac{y}{\theta} &= \frac{a-2}{a-1}, \\ \frac{z}{\theta} &= \frac{-2}{a-2}, \\ \frac{x}{\theta} &= \frac{2(a-1)}{a}, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} x &: \ 2(a-1)^2(a-2)\left(1 - \tfrac{a}{\infty}\right) \\ y &: \ a\ \ldots(a-2)^2\left(1 - \tfrac{a}{\infty}\right) \\ z &: \ -2a(a-1)\ \ldots\left(1 - \tfrac{a}{\infty}\right)^2 \\ w &: \ a^2(a-1)(a-2) \\ \theta &: \ a(a-1)(a-2)\left(1 - \tfrac{a}{\infty}\right), \end{aligned}$$

where, for the sake of homogeneity, I have introduced the factors $\left(1 - \dfrac{a}{\infty}\right)$ and $\left(1 - \dfrac{a}{\infty}\right)^2$, viz. we have x, y, z, w, θ proportional to quartic functions of the arbitrary parameter a, or the curve is a unicursal quartic. Writing in the equations $a = 0, 1, 2, \infty$ successively, we see that this quartic curve passes through the four points $123, 234, 341, 412$ (intersecting at these points the lines 13 and 24 respectively); and writing also $a = 1 \pm i$ we see that the curve passes through the points P, Q, the coordinates of which now are

$$x = y = z = w = (1 \pm i)\theta.$$

It should admit of being proved by general considerations that, in 4-dimensional geometry when 4 quadric surfaces partially intersect in two lines, the residual intersection consists of 2 points; and that, when they intersect in the two lines and in a unicursal quartic met twice by each of the lines, there is no residual intersection—but this theory has not yet been developed.

596.

NOTE ON A THEOREM OF JACOBI'S FOR THE TRANSFORMATION OF A DOUBLE INTEGRAL.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 92—94.]

Jacobi, in the Memoir "De Transformatione Integralis Duplicis..." &c., *Crelle*, t. VIII. (1832) pp. 253—279 and 321—357, [*Ges. Werke*, t. III., pp. 91—158], after establishing a theorem which includes the addition-theorem of elliptic functions, viz. this last is "the differential equation

$$\frac{d\eta}{\sqrt{(G''^2\cos^2\eta + G'''^2\sin^2\eta - G^2)}} + \frac{d\theta}{\sqrt{(G''^2\cos^2\theta + G'''^2\sin^2\theta - G^2)}} = 0,$$

has for its complete integral

$$G + G'\cos\eta\cos\theta + G''\sin\eta\sin\theta = 0,$$

" (observe, as to the integral being complete, that the differential equation contains only the constant $G''^2 - G'''^2 \div (G'^2 - G''^2)$, whereas the integral equation contains the two constants $G' \div G$ and $G'' \div G$), obtains a corresponding theorem for double integrals; viz. this, in the corresponding special case, is as follows: If the variables (ϕ, ψ) and (η, θ) are connected by the two equations

$$\begin{aligned}\alpha\ \cos\phi + \alpha'\sin\phi\cos\psi\ \cdot\ \cos\eta&\\ + \alpha''\sin\phi\sin\psi\cdot\sin\eta\cos\theta&\\ + \alpha'''\sin\phi\sin\psi\cdot\sin\eta\sin\theta &= 0,\end{aligned}$$

$$\begin{aligned}\beta\ \cos\phi + \beta'\sin\phi\cos\psi\ \cdot\ \cos\eta&\\ + \beta''\sin\phi\cos\psi\cdot\sin\eta\cos\theta&\\ + \beta'''\sin\phi\sin\psi\cdot\sin\eta\sin\theta &= 0,\end{aligned}$$

and if putting for shortness

$$\begin{aligned}\alpha''\beta''' - \alpha'''\beta'' &= f, &\quad \alpha\beta' - \alpha'\beta &= a,\\ \alpha'''\beta' - \alpha'\beta''' &= g, &\quad \alpha\beta'' - \alpha''\beta &= b,\\ \alpha'\beta'' - \alpha''\beta' &= h, &\quad \alpha\beta''' - \alpha'''\beta &= c,\end{aligned}$$

$$(\text{whence } af + bg + ch = 0);$$

$$\begin{aligned}R^2 = f^2(\sin\phi\cos\psi)^2(\sin\phi\sin\psi)^2&\\ + g^2(\sin\phi\cos\psi)^2(\cos\phi)^2&\\ + h^2(\cos\phi)^2(\sin\phi\cos\psi)^2&\\ - a^2(\cos\phi)^2&\\ - b^2(\sin\phi\cos\psi)^2&\\ - c^2(\sin\phi\sin\psi)^2&,\end{aligned}$$

$$\begin{aligned}S^2 = f^2(\sin\eta\cos\theta)^2(\sin\eta\sin\theta)^2&\\ + g^2(\sin\eta\sin\theta)^2(\cos\eta)^2&\\ + h^2(\cos\eta)^2(\sin\eta\cos\theta)^2&\\ - a^2(\cos\eta)^2&\\ - b^2(\sin\eta\cos\theta)^2&\\ - c^2(\sin\eta\sin\theta)^2&,\end{aligned}$$

then we have

$$\frac{\sin\phi\, d\phi\, d\psi}{R} = \frac{\sin\eta\, d\eta\, d\theta}{S}.$$

And it may be added that the integral equations are, so to speak, a complete integral of the differential relation; viz. in virtue of the identity $af + bg + ch = 0$, the differential relation contains really only four constants; the integral relations contain the six constants $\alpha : \alpha' : \alpha'' : \alpha'''$ and $\beta : \beta' : \beta'' : \beta'''$, or we have two constants introduced by the integration.

The best form of statement is, in the first theorem, to write x, y for $\cos\eta, \sin\eta$, $(x^2 + y^2 = 1)$, ξ, η for $\cos\theta, \sin\theta$, $(\xi^2+\eta^2 = 1)$, and similarly in the second theorem to introduce the variables x, y, z connected by $x^2+y^2+z^2 = 1$, and ξ, η, ζ connected by $\xi^2+\eta^2+\zeta^2 = 1$; then in the first theorem $d\eta, d\theta$ represent elements of circular arc, and in the second theorem $\sin\phi\, d\phi\, d\psi$ and $\sin\eta\, d\eta\, d\theta$ represent elements of spherical surface, and the theorems are:

I. If (x, y) are coordinates of a point on the circle $x^2 + y^2 = 1$, and (ξ, η) coordinates of a point on the circle $\xi^2 + \eta^2 = 1$, and if $ds, d\sigma$ are the corresponding circular elements, then

$$\frac{ds}{\sqrt{(ax^2 + by^2 - c)}} = \frac{d\sigma}{\sqrt{(a\xi^2 + b\eta^2 - c)}},$$

has for its complete integral

$$ax\xi + by\eta - c = 0.$$

II. If (x, y, z) are coordinates of a point on the sphere $x^2 + y^2 + z^2 = 1$, and (ξ, η, ζ) coordinates of a point on the sphere $\xi^2 + \eta^2 + \zeta^2 = 1$; and if $ds, d\sigma$ are the corresponding spherical elements, and

$$\begin{aligned} \beta\gamma' - \beta'\gamma &= f, \quad & \alpha\delta' - \alpha'\delta &= a,\\ \gamma\alpha' - \gamma'\alpha &= g, \quad & \beta\delta' - \beta'\delta &= b,\\ \alpha\beta' - \alpha'\beta &= h, \quad & \gamma\delta' - \gamma'\delta &= c, \end{aligned}$$
$$(\text{whence } af + bg + ch = 0);$$

and for shortness

$$S^2 = f^2y^2z^2 + g^2z^2x^2 + h^2x^2y^2 - a^2x^2 - b^2y^2 - c^2z^2,$$
$$\Sigma^2 = f^2\eta^2\zeta^2 + g^2\zeta^2\xi^2 + h^2\xi^2\eta^2 - a^2\xi^2 - b^2\eta^2 - c^2\zeta^2,$$

then the differential relation

$$\frac{ds}{\sqrt{(S)}} = \frac{d\sigma}{\sqrt{(\Sigma)}},$$

has for its complete integral the system

$$\alpha x\xi + \beta y\eta + \gamma z\zeta + \delta = 0,$$

$$\alpha' x\xi + \beta' y\eta + \gamma' z\zeta + \delta' = 0,$$

where by complete integral is meant a system of two equations containing two arbitrary constants.

597.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 110—113.]

The differential equation

$$Q^2 - Q\left(k + \frac{1}{k}\right) \cdot 3 = 3(1 - k^2)\frac{dQ}{dk},$$

considered ante, p. 69, [594, this volume, p. 244], belongs to a class of equations transformable into linear equations of the second order, and consequently is such that, knowing a particular solution, we can obtain the general solution.

In fact, assuming

$$Q = -3(1 - k^2)\frac{1}{z}\frac{dz}{dk},$$

the equation becomes

$$9(1-k^2)^2\frac{1}{z^2}\left(\frac{dz}{dk}\right)^2+3(1-k^2)\left(k+\frac{1}{k}\right)\frac{1}{z}\frac{dz}{dk}-3$$
$$=3(1-k^2)\left\{3(1-k^2)\frac{1}{z^2}\left(\frac{dz}{dk}\right)^2+6k\frac{1}{z}\frac{dz}{dk}-3(1-k^2)\frac{1}{z}\frac{d^2z}{dk^2}\right\},$$

viz. omitting the terms in $\frac{1}{z^2}\left(\frac{dz}{dk}\right)^2$ which destroy each other, and dividing by $3(1-k^2)$, this is

$$\left(k+\frac{1}{k}\right)\frac{1}{z}\frac{dz}{dk}-\frac{1}{1-k^2}=6k\frac{1}{z}\frac{dz}{dk}-3(1-k^2)\frac{1}{z}\frac{d^2z}{dk^2},$$

or finally

$$3(1-k^2)\frac{d^2z}{dk^2}+\frac{1-5k^2}{k}\frac{dz}{dk}-\frac{1}{1-k^2}z=0.$$

But knowing a particular value of Q we have

$$z=\exp\left\{-\tfrac{1}{3}\int\frac{Q\,dk}{1-k^2}\right\},$$

a particular value of z, and thence in the ordinary manner the general value of z, giving the general value of Q.

The solution given in my former paper may be exhibited in a more simple form by introducing, instead of k, the variable a connected with it by the equation $k^2=\dfrac{a^3(2+a)}{1+2a}$. We have in fact, *Fundamenta Nova*, p. 25, [*Jacobi's Ges. Werke*, t. I., p. 76],

$$v^2=a\,\frac{2+a}{1+2a},\qquad u^2=a^3\,\frac{2+a}{1+2a},$$

viz. these expressions of u, v in terms of the parameter a, are equivalent to, and replace, the modular equation $u^4-v^4+2uv(1-u^2v^2)=0$. We thence obtain

$$uv=a^2\,\frac{2+a}{1+2a},\qquad \frac{v^2}{u^2}=\frac{1}{a^2}\left(\frac{2+a}{1+2a}\right),$$

that is,

$$uv=a^2\left(\frac{2+a}{1+2a}\right),\qquad \frac{v^2}{u^2}=\frac{1}{a^2}\left(\frac{2+a}{1+2a}\right),$$

and the particular solution, $Q=\dfrac{v^2}{u^2}+2uv$, becomes

$$Q=\frac{1}{a^2}\sqrt{(1+2a)(2+a)}\;=\sqrt{\left\{5+2\left(a+\tfrac{1}{a}\right)\right\}}.$$

Introducing into the differential equation a in place of k, this is found to be

$$Q^2-Q\,\frac{\frac{1}{a^2}+a^2+2\left(\frac{1}{a}+a\right)}{\sqrt{\left\{5+2\left(a+\frac{1}{a}\right)\right\}}}-3=(1-a^2)\sqrt{(1+2a)(2+a)}\;\frac{dQ}{da}.$$

But from this form it at once appears that it is convenient in place of a to introduce the new variable $\beta,=a+\dfrac{1}{a}$; the equation thus becomes

$$Q^2-Q\,\frac{2-2\beta-\beta^2}{\sqrt{(5+2\beta)}}-3=(4-\beta^2)\sqrt{(5+2\beta)}\;\frac{dQ}{d\beta},$$

satisfied by $Q = \sqrt{(5+2\beta)}$; what or, what is the same thing, writing $5 + 2\beta = \gamma^2$, that is, $\beta = -\frac{5}{2} + \frac{1}{2}\gamma^2$, the equation becomes

$$4Q^2 + \frac{Q}{\gamma}(3 + 6\gamma^2 - \gamma^4) - 12 = -(\gamma^2 - 1)(\gamma^2 - 9)\,\frac{dQ}{d\gamma},$$

satisfied by $Q = \gamma$.

Writing here

$$Q = \tfrac{1}{2}(\gamma^2 - 1)(\gamma^2 - 9)\frac{1}{z}\frac{dz}{d\gamma},$$

we have for z the equation

$$(\gamma^2 - 1)(\gamma^2 - 9)\frac{d^2z}{d\gamma^2} + (3\gamma^4 - 14\gamma^2 + 3)\frac{dz}{d\gamma} - \frac{48}{(\gamma^2 - 1)(\gamma^2 - 9)}\,z = 0,$$

satisfied by

$$z = \left(\frac{\gamma^2 - 9}{\gamma^2 - 1}\right)^{\frac{1}{2}}.$$

[In fact, this value gives

$$\frac{dz}{d\gamma} = 4\gamma(\gamma^2 - 9)^{-\frac{1}{2}}(\gamma^2 - 1)^{-\frac{3}{2}},$$

$$\frac{d^2z}{d\gamma^2} = (-12\gamma^4 + 57\gamma^2 + 36)(\gamma^2 - 9)^{-\frac{3}{2}}(\gamma^2 - 1)^{-\frac{5}{2}},$$

which verify the equation as they should do.]

Representing for a moment the differential equation by $A\dfrac{d^2z}{d\gamma^2} + B\dfrac{dz}{d\gamma} + Cz = 0$, and putting $z_1 = \left(\dfrac{\gamma^2 - 9}{\gamma^2 - 1}\right)^{\frac{1}{2}}$, then assuming $z = z_1 \int y\, d\gamma$, we find

$$A\left(z_1\frac{dy}{d\gamma} + 2y\frac{dz_1}{d\gamma}\right) + Byz_1 = 0,$$

that is,

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} + \frac{B}{A} = 0,$$

viz.

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} + \frac{3\gamma^4 - 14\gamma^2 + 3}{(\gamma^2 - 1)(\gamma^2 - 9)} = 0,$$

or

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} + 3 + \frac{1}{\gamma^2 - 1} + \frac{15}{\gamma^2 - 9} = 0;$$

whence, integrating

$$\log yz_1^2 + 3\gamma - \tfrac{1}{2}\log\frac{\gamma + 1}{\gamma - 1} - \tfrac{15}{2}\log\frac{\gamma + 3}{\gamma - 3} = 0,$$

that is,

$$y = e^{-3\gamma}\frac{1}{z_1^2}\left(\frac{\gamma + 1}{\gamma - 1}\right)^{\frac{1}{2}}\left(\frac{\gamma + 3}{\gamma - 3}\right)^{\frac{15}{2}}$$

$$= e^{-3\gamma}\left(\frac{\gamma - 1}{\gamma - 3}\cdot\frac{\gamma + 3}{\gamma + 3}\right)\left(\frac{\gamma + 1}{\gamma - 1}\right)^{\frac{1}{2}}\left(\frac{\gamma + 3}{\gamma - 3}\right)^{\frac{1}{2}}$$

$$= \frac{(\gamma + 1)(\gamma + 3)^2}{(\gamma - 3)^2}\,e^{-3\gamma}.$$

607. A Memoir on Prepotentials (continued)

[Continuation of the same paper; articles 60–75 of the Annexes.]

[p. 356] and the first term of this being $= \sqrt{e^2+R^2} - e$, this is consequently

$$= \sqrt{R^2+e^2} + \tfrac{1}{2}e \int_0^X x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx - e\left(1+\tfrac{1}{2}\int_0^1 x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx\right).$$

As regards the second term of this, we have

$$-2x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right] + 2\left(\tfrac{1}{2}s-1\right)\int x^{-\frac{1}{2}}(1-x)^{\frac{1}{2}s-2}\,dx = \int x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx;$$

or, taking each term between the limits 1, 0,

$$-2 + 2\left(\tfrac{1}{2}s-1\right)\frac{\Gamma\frac{1}{2}\,\Gamma(\frac{1}{2}s-1)}{\Gamma(\frac{1}{2}s-\frac{1}{2})} = \int_0^1 x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx;$$

viz. this integral has the value

$$-2 + \frac{2\,\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

and the value of the r-integral $\displaystyle\int_0^R \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$ is consequently

$$= \sqrt{R^2+e^2} + \tfrac{1}{2}e\int_0^X x^{-\frac{1}{2}}\left[1-(1-x)^{\frac{1}{2}s-1}\right]dx - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

which is of the form

$$R\left\{1+\text{terms in }\frac{e^2}{R^2},\ \frac{e^4}{R^4},\ \ldots\right\} - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})};$$

say the approximate value is

$$R - e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

where the first term R is the term $\displaystyle\int_0^R dr$, given by the expansion in ascending powers of e^2; the second term is the term in e^{-1}. And observe that the term is the value of

$$\tfrac{1}{2}e\int_0^1 x^{-\frac{1}{2}}(1-x)^{\frac{1}{2}s-1}\,dx,$$

calculated by means of the ordinary formula for a Eulerian integral (which formula, on account of the negative exponent $-\frac{1}{2}$, is not really applicable, the value of the integral being $=\infty$) on the assumption that the Γ of a negative q is interpreted in accordance with the equation $\Gamma(q+1) = q\Gamma q$; viz. the value thus calculated is

$$\tfrac{1}{2}e\,\frac{\Gamma(-\frac{1}{2})\,\Gamma(\frac{1}{2}s)}{\Gamma(\frac{1}{2}s-\frac{1}{2})} = -e\,\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-\frac{1}{2})},$$

on the assumption $\Gamma(-\frac{1}{2}) = -\frac{1}{2}\,\Gamma(-\frac{1}{2})$ *[scan reads $-\frac{1}{2}\Gamma(-\frac{1}{2})$ for the recursion]*; and this agrees with the foregoing value.

60. It is now easy to see in general how the foregoing transformed value $\frac{1}{2}e^{-q}\displaystyle\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx$, where q is negative and fractional, gives at once the value of

[p. 357] the term in e^{-q}. Observe that in the integral x is always between 1 and X $\left(=\dfrac{e^2}{e^2+R^2}\right)$, a positive quantity less than 1; the function to be integrated never becomes infinite. Imagine for a moment an integral $\int_X^\infty x^a\,dx$, where a is positive or negative. We may conventionally write this $= \int_0^\infty x^a\,dx - \int_0^X x^a\,dx$, understanding the first symbol to mean $\dfrac{1^{1+a}}{1+a}$, and the second to mean $\dfrac{X^{1+a}}{1+a}$; they of course properly mean $\dfrac{1^{1+a}-0^{1+a}}{1+a}$ and $\dfrac{X^{1+a}-0^{1+a}}{1+a}$; but the terms in 0^{1+a}, whether zero or infinite, destroy each other, the original form $\int_X^\infty x^a\,dx$, in fact, showing that no such terms can appear in the result.

In accordance with the convention, we write

$$\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx = \int_0^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx - \int_0^X x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx;$$

and it follows that the term in e^{-q} is

$$\tfrac{1}{2}e^{-q}\int_0^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx,$$

this last expression (wherein q, it will be remembered, is a negative fraction) being understood according to the convention; and so understanding it, the value of the term is

$$= \tfrac{1}{2}e^{-q}\,\frac{\Gamma q\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)},$$

where the Γ of the negative q is to be interpreted in accordance with the equation $\Gamma(q+1) = q\Gamma q$; viz. we have $\Gamma q = \dfrac{1}{q}\,\Gamma(q+1),\ = \dfrac{1}{q(q+1)}\,\Gamma(q+2)$, &c., so as to make the argument of the Γ positive. Observe that under this convention we have

$$\Gamma q\,\Gamma(1-q) = \frac{\Gamma(\frac{1}{2})^2}{\sin q\pi} = \frac{\pi}{\sin q\pi};\quad \text{or the term is}\quad \tfrac{1}{2}e^{-q}\,\frac{\pi}{\sin q\pi}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)}.$$

61. An example in which $-\frac{1}{2}s-q-1$ is integral will make the process clearer, and will serve instead of a general proof. Suppose $q=-\frac{7}{2}$, $\frac{1}{2}s=\frac{1}{4}$, the expression

$$\int_0^1 x^{-\frac{9}{2}}(1-x)^{\frac{1}{4}}\,dx = \int_0^1 \big(x^{-\frac{9}{2}} - \tfrac{9}{2}\,x^{-\frac{7}{2}} + \tfrac{63}{8}\,x^{-\frac{5}{2}} - \tfrac{63}{16}\,x^{-\frac{3}{2}} + \cdots\big)dx$$

is used, in accordance with the convention, to denote the value

$$= -\tfrac{2}{7} + \tfrac{2}{5}\cdot\tfrac{9}{2} - \tfrac{2}{3}\cdot\tfrac{63}{8} + \tfrac{2}{1}\cdot\tfrac{63}{16} = 7\big(-\tfrac{2}{7\cdot 1} + \tfrac{2}{5\cdot 3} - \tfrac{2}{3\cdot 5} + \tfrac{2}{1\cdot 7}\big) = \frac{-7\cdot 2401}{5\cdot 13\cdot 27} = \frac{-7^4}{5\cdot 13\cdot 27}.$$

[p. 358] But we have

$$\frac{\Gamma\frac{1}{2}s\,\Gamma q}{\Gamma(\frac{1}{2}s+q)} = \frac{\Gamma\frac{5}{4}\,\Gamma(-\frac{7}{2})}{\Gamma(\frac{5}{4}-\frac{7}{2})} = \frac{2^4\,\Gamma(-\frac{7}{2})}{\frac{1}{4}\cdot\frac{5}{4}\cdot\frac{9}{4}\cdot\frac{13}{4}\cdot\Gamma(-\frac{7}{2})} = \frac{2^4\cdot 4^4}{5\cdot 13\cdot 27} = \frac{-7^4}{5\cdot 13\cdot 27},$$

agreeing with the former value.

62. The case of a negative integer is more simple. To find the logarithmic term of

$$\tfrac{1}{2}e^{-q}\int_X^\infty x^{q-1}(1-x)^{\frac{1}{2}s-1}\,dx,$$

we have only to expand the factor $(1-x)^{\frac{1}{2}s-1}$ so as to obtain the term involving x^{-q}.

We have thus the term

$$\tfrac{1}{2}e^{-q}\,(-)^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\,e^{-q}\log\tfrac{1}{X},$$

where $\log\frac{1}{X} = \log\left(1+\dfrac{R^2}{e^2}\right) = 2\log\dfrac{R}{e} + 2\log\sqrt{1+\dfrac{e^2}{R^2}}$; so that, neglecting the terms in $\dfrac{e^2}{R^2}$, &c., this is $2\log\dfrac{R}{e}$, and the term in question is

$$= (-)^q e^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\,\log\frac{R}{e}.$$

The general conclusion is that q being negative, the r-integral

$$\int_0^R \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$$

has for its value a series proceeding in powers of e^2, which series up to a certain point is equal to the series obtained by expanding in ascending powers of e^2 and integrating each term separately; viz. the series to the point in question is

$$\frac{R^{-2q}}{-2q} - \frac{\frac{1}{2}s+q}{1}\cdot\frac{R^{-2q-2}}{-2q-2}\,e^2 + \frac{\frac{1}{2}s+q\cdot\frac{1}{2}s+q+1}{1\cdot 2}\cdot\frac{R^{-2q-4}}{-2q-4}\,e^4 - \cdots,$$

continued so long as the exponent of e is less than $-2q$: together with a term Ke^{-q} when q is fractional, and $Ke^{-q}\log\dfrac{R}{e}$ when q is integral; viz. q fractional, this term is

$$= \tfrac{1}{2}e^{-q}\,\frac{\Gamma q\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)} = \tfrac{1}{2}e^{-q}\,\frac{\pi}{\sin q\pi}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)},$$

and q integral, it is

$$= (-)^q e^{-q}\,\frac{\Gamma\frac{1}{2}s}{\Gamma(1-q)\,\Gamma(\frac{1}{2}s+q)}\,\log\frac{R}{e}.$$

[p. 359] 63. It has been tacitly assumed that $\frac{1}{2}s+q$ is positive; but the formulae hold good if $\frac{1}{2}s+q$ is $=0$ or negative. Suppose $\frac{1}{2}s+q$ is $=0$ or a negative integer, then $\Gamma(\frac{1}{2}s+q)=\infty$, and the special term involving e^{-q} or $e^{-q}\log e$ vanishes; in fact, in this case the r-integral is

$$= \int_0^R r^{s-1}\,(r^2+e^2)^{-(\frac{1}{2}s+q)}\,dr,$$

where $(r^2+e^2)^{-(\frac{1}{2}s+q)}$ has for its value a finite series, and the integral is therefore equal to a finite series $A+Be^2+Ce^4+\&c.$ If $\frac{1}{2}s+q$ be fractional, then the Γ of the negative quantity $\frac{1}{2}s+q$ must be understood as above, or, what is the same thing, we may, instead of $\Gamma(\frac{1}{2}s+q)$, write

$$\frac{\pi}{\sin(\frac{1}{2}s+q)\pi\cdot\Gamma(1-q-\frac{1}{2}s)};$$

thus, q being integral, the exceptional term is

$$= \frac{(-)^q\sin(\frac{1}{2}s+q)\pi\cdot\Gamma(1-q-\frac{1}{2}s)}{\Gamma(\frac{1}{2}s)\,\Gamma(1-q)}\,\Gamma\tfrac{1}{2}s\cdot R^{-2q}e^{-q}.$$

For instance, $s=1$, $q=-2$, the term is

$$\tfrac{1}{2}\,\frac{\Gamma\frac{1}{2}\sin(-\frac{3}{2})\pi\cdot\Gamma\frac{5}{2}}{\Gamma\frac{1}{2}\cdot\Gamma 3}\,e^2\log\frac{R}{e},$$

or, since $\Gamma\frac{5}{2} = \frac{3}{2}\cdot\frac{1}{2}\,\Gamma\frac{1}{2}$, and $\Gamma 3 = 2$, the term is $+\frac{3}{8}e^2\log\dfrac{R}{e}$, agreeing with a preceding result.

Annex III. Prepotentials of Uniform Spherical Shell and Solid Sphere.
Art. Nos. 64 to 92.

64. The prepotentials in question depend ultimately upon two integrals, which also arise, as will presently appear, from prepotential problems in two-dimensional space, and which are for convenience termed the *ring-integral* and the *disk-integral* respectively. The analytical investigation in regard to these, depending as it does on a transformation of a function allied with the hypergeometric series, is I think interesting.

65. Consider first the prepotential of a uniform $(s+1)$-dimensional spherical shell. This is

$$V = \int \frac{dS}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}},$$

the equation of the surface being $x^2 + \ldots + z^2 + w^2 = f^2$; and there are the two cases of an internal point, $a^2 + \ldots + c^2 + e^2 < f^2$, and an external point, $a^2 + \ldots + c^2 + e^2 > f^2$.

The value is a function of $a^2 + \ldots + c^2 + e^2$, say this is $= \kappa^2$. Taking the axes so that the coordinates of the attracted point are $(0, \ldots, 0, \kappa)$, the integral is

$$= \int \frac{dS}{\{x^2 + \ldots + z^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

[p. 360] where the equation of the surface is still $x^2 + \ldots + z^2 + w^2 = f^2$. Writing $x = f\xi, \ldots, z = f\zeta$, $w = f\omega$, where $\xi^2 + \ldots + \zeta^2 + \omega^2 = 1$, we have $dS = f^s \dfrac{d\xi \ldots d\zeta}{\omega}$, or the integral is

$$= f^s \int \frac{d\xi \ldots d\zeta}{\omega\,(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}}.$$

Assume $\xi = \rho x, \ldots, \zeta = \rho z$, where $x^2 + \ldots + z^2 = 1$; then $\rho^2 + \omega^2 = 1$. Moreover, $d\xi \ldots d\zeta = \rho^{s-1}\,d\rho\,d\Sigma$, where $d\Sigma$ is the element of surface of the s-dimensional unit-sphere $x^2 + \ldots + z^2 = 1$; or for ρ, substituting its value $\sqrt{1-\omega^2}$, we have $d\rho = -\dfrac{\omega\,d\omega}{\sqrt{1-\omega^2}}$; and thence $d\xi \ldots d\zeta = -(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,\omega\,d\omega\,d\Sigma$. The integral as regards ρ is from $\rho = -1$ to $+1$, or as regards ω from 1 to -1; whence reversing the sign, the integral will be from $\omega = -1$ to $+1$; and the required integral is thus

$$= f^s \int d\Sigma \int_{-1}^{+1} \frac{(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,d\omega\,d\Sigma}{(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}} = f^s \int d\Sigma \int_{-1}^{+1} \frac{(1-\omega^2)^{\frac{1}{2}s-\frac{3}{2}}\,d\omega}{(f^2 - 2\kappa f\omega + \kappa^2)^{\frac{1}{2}s+q}},$$

where $\displaystyle\int d\Sigma$ is the surface of the s-dimensional unit-sphere (see Annex I.), $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$; and for greater convenience transforming the second factor by writing therein $\omega = \cos\theta$, the required integral is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\,f^s$ multiplied by

$$\int_0^\pi \frac{\sin^{s-1}\theta\,d\theta}{(f^2 - 2\kappa f\cos\theta + \kappa^2)^{\frac{1}{2}s+q}},$$

which last expression including the factor $2f^s$, but without the factor $\dfrac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$, is the ring-integral discussed in the present Annex. It may be remarked that the value can be at once obtained in the particular case $s = 2$, which belongs to tridimensional space; viz. we then have

$$V = \int_0^\pi \frac{2\pi f^2 \sin\theta\,d\theta}{(f^2 - 2f\kappa\cos\theta + \kappa^2)^{q+1}} = \frac{2\pi f}{\kappa(1-q)} \big|\,(f^2 - 2f\kappa\cos\theta + \kappa^2)^{-q+1}\big|,$$

which agrees with a result given, *Mécanique Céleste*, Book XII. Chap. II.

66. Consider next the prepotential of the uniform solid $(s+1)$-dimensional sphere,

$$V = \int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}},$$

the equation of the surface being $x^2+\ldots+z^2+w^2 = f^2$; there are the two cases of an internal point $\kappa < f$, and an external point $\kappa > f$ $(a^2+\ldots+c^2+e^2 = \kappa^2$ as before).

[p. 361] Transforming so that the coordinates of the attracted point are $0, \ldots, 0, \kappa$, the integral is

$$= \int \frac{dx \ldots dz\, dw}{\{x^2 + \ldots + z^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

where the equation is still $x^2+\ldots+z^2+w^2 = f^2$. Writing here $x = r\xi, \ldots, z = r\zeta$, where $\xi^2+\ldots+\zeta^2 = 1$, we have $dx \ldots dz = r^{s-1}\, dr\, dS$, where dS is an element of surface of the s-dimensional unit-sphere $\xi^2+\ldots+\zeta^2 = 1$; the integral is therefore

$$= \int\int \frac{r^{s-1}\, dr\, dS\, dw}{\{r^2 + (\kappa - w)^2\}^{\frac{1}{2}s+q}},$$

where, as regards r and w, the integration extends over the circle $r^2+w^2 = f^2$. The value of the first factor (see Annex I.) is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$; writing y and x in place of r and w respectively, the integral is $= \dfrac{2\,(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$ multiplied by

$$\int \frac{y^{s-1}\, dx\, dy}{\{(x-\kappa)^2 + y^2\}^{\frac{1}{2}s+q}}$$

over the circle $x^2+y^2 = f^2$; viz. this last expression (without the factor $\dfrac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}$) is the disk-integral discussed in the present Annex.

67. We find, for the value in regard to an internal point $\kappa < f$,

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, f^{s+1} \int_0^\infty (t+f^2-\kappa^2)^{q-\frac{1}{2}}\, t^{-q-1}\, (t+f^2)^{\frac{1}{2}s+q-\frac{1}{2}}\, dt,$$

which, in the particular case $q = -\frac{1}{2}$, is

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})}\, f^{s+1} \int_0^\infty (t+f^2-\kappa^2)^{-1}\, (t+f^2)^{\frac{1}{2}s-1}\, dt;$$

viz. the integral in t is here

$$= \int_{f^2-\kappa^2}^\infty (t+f^2-\kappa^2)^{-1}\, t^{-\frac{1}{2}}\, (t+f^2)^{\frac{1}{2}s-1}\, dt = \frac{1}{f^{s+1}} \left(\frac{f^{s+1}}{\frac{1}{2}s-\frac{1}{2}} - \frac{\kappa^{s+1}}{\frac{1}{2}s+\frac{1}{2}} \right),$$

or we have

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})} \left(\frac{f^{s+1}}{\frac{1}{2}s-\frac{1}{2}} - \frac{\kappa^{s+1}}{\frac{1}{2}s+\frac{1}{2}} \right).$$

It may be added that, in regard to an external point $\kappa > f$, the value is

$$V = \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, f^{s+1} \int_{\kappa^2-f^2}^\infty (t+f^2-\kappa^2)^{q-\frac{1}{2}}\, t^{-q-1}\, (t+f^2)^{\frac{1}{2}s+q-\frac{1}{2}}\, dt,$$

[p. 362] which, in the same case $q = -\frac{1}{2}$, is

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})}\, f^{s+1} \int_{\kappa^2-f^2}^\infty (t+f^2-\kappa^2)^{-1}\, (t+f^2)^{\frac{1}{2}s-1}\, dt,$$

where the t-integral is

$$= \int_{\kappa^2-f^2}^{\infty} (t+f^2-\kappa^2)^{-1}\, t^{-\frac{1}{2}}\, (t+f^2)^{\frac{1}{2}s-1}\, dt = \frac{\kappa^{s+1}}{\frac{1}{2}s-\frac{1}{2}} \cdot \frac{\kappa^{s-2}}{\frac{1}{2}s+\frac{1}{2}} \cdot \frac{1}{\kappa^{s+1}} = \frac{\kappa^{s-1}}{\frac{1}{2}s-\frac{1}{2}\cdot\frac{1}{2}s+\frac{1}{2}};$$

and the value of V is therefore

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+\frac{1}{2})} \frac{f^{s+1}}{\kappa^{s-1}}.$$

Recurring to the case of the internal point; then, writing $\nabla = \dfrac{d^2}{da^2} + \ldots + \dfrac{d^2}{dc^2} + \dfrac{d^2}{de^2}$, and observing that $\nabla\,(\kappa^s) = s\,(\frac{1}{2}s+\frac{1}{2})$, we have

$$\nabla V = -\frac{4\,(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s-\frac{1}{2})};$$

(in particular, for ordinary space $s+1=3$, or the value is $\dfrac{-4\pi^2}{\sqrt{\pi}} = -4\pi$, which is right).

68. The integrals referred to as the ring-integral and the disk-integral arise also from the following integrals in two-dimensional space, viz. these are

$$\int \frac{y^{s-1}\, dS}{\{(x-\kappa)^2+y^2\}^{\frac{1}{2}s+q}}, \qquad \int \frac{y^{s-1}\, dx\, dy}{\{(x-\kappa)^2+y^2\}^{\frac{1}{2}s+q}},$$

in the first of which dS denotes an element of arc of the circle $x^2+y^2=f^2$, the integration being extended over the whole circumference, and in the second the integration extends over the circle $x^2+y^2=f^2$; y^{s-1} is written for shortness instead of $(y^2)^{\frac{1}{2}(s-1)}$, viz. this is considered as always positive, whether y is positive or negative; it is moreover assumed that $s-1$ is zero or positive.

Writing in the first integral $x = f\cos\theta$, $y = f\sin\theta$, the value is

$$= f^s \int \frac{(\sin\theta)^{s-1}\, d\theta}{(f^2-2\kappa f\cos\theta+\kappa^2)^{\frac{1}{2}s+q}},$$

viz. this represents the prepotential of the circumference of the circle, density varying as $(\sin\theta)^{s-1}$, in regard to a point $x=\kappa$, $y=0$ in the plane of the circle; and similarly the second integral represents the prepotential of the circular disk, density of the element at the point $(x,y) = y^{s-1}$, in regard to the same point $x=\kappa$, $y=0$; it being in each case assumed that the prepotential of an element of mass $\rho\, d\varpi$ at a point at distance d is $= \dfrac{\rho\, d\varpi}{d^{2q+s}}$.

[p. 363] 69. In the case of the circumference, it is assumed that the attracted point is not on the circumference, κ not $= f$; and the function under the integral sign, and therefore the integral itself, is in every case finite. In the case of the circle, if κ be an interior point, then if $2q-1$ be $=0$ or positive, the element at the attracted point becomes infinite; but to avoid this we consider, not the potential of the whole circle, but the potential of the circle less an indefinitely small circle radius ϵ having the attracted point for its centre; which being so, the element under the integral sign, and consequently the integral itself, remains finite.

It is to be remarked that the two integrals are connected with each other; viz. the circle of the second integral being divided into rings by means of a system of circles concentric with the bounding circle $x^2+y^2=f^2$, then the prepotential of each ring or annulus is determined by an integral such as the first integral; or, analytically, writing in the second integral $x = r\cos\theta$, $y = r\sin\theta$, and therefore $dx\, dy = r\, dr\, d\theta$, the second integral is

$$= \int r^s\, dr \int \frac{(\sin\theta)^{s-1}\, d\theta}{(r^2+\kappa^2-2fr\cos\theta)^{\frac{1}{2}s+q}},$$

viz. the integral in regard to θ is here the same function of r, κ that the first integral is of f, κ; and the integration in regard to r is of course to be taken from $r=0$ to $r=f$. But the θ-integral is not, in its original form, such a function of r as to render possible the integration in regard to r; and I, in fact, obtain the second integral by a different and in some respects a better process.

70. Consider first the ring-integral which, writing therein as above $x = f\cos\theta$, $y = f\sin\theta$, and multiplying by 2 in order that the integral, instead of being taken from 0 to 2π, may be taken from 0 to π, becomes

$$= 2f^s \int_0^\pi \frac{(\sin\theta)^{s-1}\,d\theta}{(f^2+\kappa^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q}}.$$

Write $\cos\frac{1}{2}\theta = \sqrt{x}$; then $\sin\frac{1}{2}\theta = \sqrt{1-x}$, $\sin\theta = 2\sqrt{x(1-x)}$, $d\theta = -x^{-\frac{1}{2}}(1-x)^{-\frac{1}{2}}\,dx$, $\cos\theta = -1+2x$; $\theta = 0$ gives $x = 1$, $\theta = \pi$ gives $x = 0$, and the integral is

$$= 2^{s-1}\int_0^1 \frac{x^{\frac{1}{2}s-1}(1-x)^{\frac{1}{2}s-1}\,dx}{\{(f+\kappa)^2-4f\kappa x\}^{\frac{1}{2}s+q}} = \frac{2^{s-1}}{\{(f+\kappa)^2\}^{\frac{1}{2}s+q}}\int_0^1 \frac{x^{\frac{1}{2}s-1}(1-x)^{\frac{1}{2}s-1}\,dx}{(1-ux)^{\frac{1}{2}s+q}},$$

if for shortness $u = \dfrac{4\kappa f}{(\kappa+f)^2}$, so that obviously $u < 1$.

The integral in x is here an integral belonging to the general form

$$\Pi = \Pi(\alpha,\beta,\gamma,u) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}(1-ux)^{-\gamma}\,dx,$$

viz. we have

$$\text{Ring-integral} = \frac{2^{s-1}f^s}{(f+\kappa)^{s+2q}}\,\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u).$$

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[p. 364] **71.** The general function $\Pi(\alpha,\beta,\gamma,u)$ is

$$\Pi(\alpha,\beta,\gamma,u) = \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\alpha+\beta)}\,F(\alpha,\gamma,\alpha+\beta,u),$$

or, what is the same thing,

$$\Pi(\alpha,\beta,\gamma,u) = \frac{\Gamma\gamma}{\Gamma\alpha\,\Gamma(\gamma-\alpha)}\,F(\alpha,\beta,\gamma,u),$$

and consequently transformable by means of various theorems for the transformation of the hypergeometric series, in particular, by the theorems

$$F(\alpha,\beta,\gamma,u) = F(\beta,\alpha,\gamma,u),$$

$$F(\alpha,\beta,\gamma,u) = (1-u)^{\gamma-\alpha-\beta}\,F(\gamma-\alpha,\gamma-\beta,\gamma,u);$$

and if $v = \left(\dfrac{1-\sqrt{1-u}}{1+\sqrt{1-u}}\right)^2$, or, what is the same thing, $u = \dfrac{4\sqrt{v}}{(1+\sqrt{v})^2}$, then

$$F(\alpha,\beta,2\beta,u) = (1-\sqrt{v})^{2\alpha}\,F\big(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v\big).$$

In verification, observe that if $u = 1$ then also $v = 1$, and that with these values, calculating each side by means of the formulae

$$F(\alpha,\beta,\gamma,1) = \frac{\Gamma\gamma\,\Gamma(\gamma-\alpha-\beta)}{\Gamma(\gamma-\alpha)\,\Gamma(\gamma-\beta)},$$

the resulting equation $F(\alpha,\beta,2\beta,1) = 2^{2\alpha}\,F(\alpha,\alpha-\beta+\frac{1}{2},\beta+\frac{1}{2},1)$, becomes

$$\frac{\Gamma2\beta\,\Gamma(\beta-\alpha)}{\Gamma(2\beta-\alpha)\,\Gamma\beta} = 2^{2\alpha}\,\frac{\Gamma(\beta+\frac{1}{2})\,\Gamma(2\beta-2\alpha)}{\Gamma(2\beta-\alpha)\,\Gamma(\beta-\alpha+\frac{1}{2})},$$

that is,

$$\frac{\Gamma2\beta}{\Gamma\beta\,\Gamma(\beta+\frac{1}{2})} = 2^{2\alpha}\,\frac{\Gamma(2\beta-2\alpha)}{\Gamma(\beta-\alpha)\,\Gamma(\beta-\alpha+\frac{1}{2})},$$

which is true, in virtue of the relation $\dfrac{\Gamma2x}{\Gamma x\,\Gamma(x+\frac{1}{2})} = \dfrac{2^{2x-1}}{\Gamma\frac{1}{2}}$.

72. The foregoing formulae, and in particular the formula which I have written

$$F(\alpha,\beta,2\beta,u) = (1-\sqrt{v})^{2\alpha}\,F(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v),$$

are taken from Kummer's Memoir, "Ueber die hypergeometrische Reihe," *Crelle*, t. XV. (1836), viz. the formula in question is, under a slightly different form, his formula (41), p. 76; the formula (43), p. 77, is intended to be equivalent thereto; but there is an error of transcription, $2\alpha-2\beta+1$, in place of $\alpha+\beta-1$, which makes the formula (43) erroneous.

It may be remarked as to the formulae generally that, although very probably $\Pi(\alpha,\beta,\gamma,u)$ may denote a proper function of u, whatever be the values of the indices (α,β,γ), and the various transformation-theorems hold good accordingly (the Γ-function of a negative argument being interpreted in the usual manner by means of the equation $\Gamma x = \dfrac{1}{x}\Gamma(1+x) = \dfrac{1}{x(x+1)}\Gamma(2+x)$, &c.), yet that the function $\Pi(\alpha,\beta,\gamma,u)$,

[p. 365] used as denoting the definite integral $\displaystyle\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}(1-ux)^{-\gamma}\,dx$, has no meaning except in the case where α and β are each of them positive.

In what follows we obtain for the ring-integral and the disk-integral various expressions in terms of Π-functions, which are afterwards transformed into t-integrals with a superior limit ∞ and inferior limit 0, or $= f^2-\kappa^2$; but for values of the variable index, q lying beyond certain limits, the indices α and β, or one of them, of the Π-function will become negative, viz. the integral represented by the Π-function, or, what is the same thing, the t-integral, will cease to have a determinate value, and at the same time, or usually so, the argument or arguments of one or more of the Γ-functions will become negative. It is quite possible that in such cases the results are not without meaning, and that an interpretation for them might be found; but they have not any obvious interpretation, and we must in the first instance consider them as inapplicable.

73. We require further properties of the Π-functions. Starting with the foregoing equation

$$F(\alpha,\beta,2\beta,u) = (1+\sqrt{v})^{2\alpha}\,F\big(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v\big),$$

each side may be expressed in a fourfold form:

$$\begin{aligned}
F(\alpha,\beta,2\beta,u) &= (1+\sqrt{v})^{2\alpha}\,F(\alpha,\,\alpha-\beta+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v)\\
&= F(\beta,\alpha,2\beta,u) \;=\; \tfrac{1}{2}(1+\sqrt{v})^{2\alpha-1}\,F(\alpha-\tfrac{1}{2}+1,\,\beta+\tfrac{1}{2},\,v)\\
&= (1-u)^{\beta-\alpha}\,F(2\beta-\alpha,\,\beta,\,2\beta,\,u)\\
&= (1-u)^{\beta-\alpha}\,F(\alpha,\,2\beta-\alpha,\,2\beta,\,u)\\
&= (1+\sqrt{v})^{2\alpha}(1-v)^{2\beta-2\alpha-1}\,F(\beta-\alpha+\tfrac{1}{2},\,2\beta-\alpha,\,\beta+\tfrac{1}{2},\,v)\\
&= (1+\sqrt{v})^{2\alpha}(1-v)^{-\alpha-\beta-\frac{1}{2}}\,F(2\beta-\alpha,\,\beta-\alpha+\tfrac{1}{2},\,\beta+\tfrac{1}{2},\,v),
\end{aligned}$$

where, instead of $(1+\sqrt{v})^{2\alpha}(1-v)^{2\beta-2\alpha-1}$, it is proper to write $(1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}$;

And then to each form applying the transformation

$$F(\alpha,\beta,\gamma,u) = \frac{\Gamma\gamma}{\Gamma\alpha\,\Gamma(\gamma-\alpha)}\,\Pi(\alpha,\gamma-\alpha,\beta,u),$$

we have

$$\begin{aligned}
&\frac{\Gamma 2\beta}{\Gamma\alpha\,\Gamma(2\beta-\alpha)}\,\Pi(\alpha,\,2\beta-\alpha,\,\beta,\,u)\\
&= \frac{\Gamma 2\beta}{\Gamma\beta\,\Gamma\beta}\,\Pi(\beta,\beta,\alpha,u)\\
&= (1-u)^{\beta-\alpha}\,\frac{\Gamma 2\beta}{\Gamma(2\beta-\alpha)\,\Gamma\alpha}\,\Pi(2\beta-\alpha,\,\alpha,\,\beta,\,u)
\end{aligned}$$

[p. 366]

$$= (1-u)^{\beta-\alpha}\, 2^{2\alpha-1}\, \frac{\Gamma(\beta+\frac{1}{2})}{\Gamma\frac{1}{2}\,\Gamma(2\beta-\alpha)}\, \Pi(\alpha,\, \alpha-\beta+\tfrac{1}{2},\, 2\beta-\alpha,\, v)$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}\, \frac{\Gamma(\beta+\frac{1}{2})}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\alpha}\, \Pi(\beta-\alpha+\tfrac{1}{2},\, \alpha,\, 2\beta-\alpha,\, v).$$

I select the second of the first four forms; equating it successively to each of the second four forms, and attending to the relation $\dfrac{\Gamma 2\beta}{\Gamma\beta\,\Gamma\beta} = 2^{2\beta-1}\,\dfrac{1}{\Gamma\frac{1}{2}}$, we find

$$\Pi(\beta,\beta,\alpha,u) = (1-\sqrt{v})^{2\alpha}\, 2^{1-2\alpha}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\beta}\, \Pi(\alpha,\, \beta-\alpha+\tfrac{1}{2},\, \alpha-\beta+\tfrac{1}{2},\, v)$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{2\beta-2\alpha-1}\, 2^{1-2\alpha}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\beta-\alpha+\frac{1}{2})\,\Gamma\alpha}\, \Pi(\beta-\alpha+\tfrac{1}{2},\, \alpha,\, 2\beta-\alpha,\, v),$$

$$= (1+\sqrt{v})^{2\alpha}(1-\sqrt{v})^{-\alpha-\beta+\frac{1}{2}}\, 2^{-1}\, \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(2\beta-\alpha)\,\Gamma\alpha}\, \Pi(2\beta-\alpha,\, \alpha-\beta+\tfrac{1}{2},\, \beta-\alpha+\tfrac{1}{2},\, v).$$

Putting herein $\beta = \frac{1}{2}s$, $\alpha = \frac{1}{2}s + q$, the formulae become

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) = (1+\sqrt{v})^{-s-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, \Pi(\tfrac{1}{2}s+q,\, \tfrac{1}{2}-q,\, \tfrac{1}{2}+q,\, v) \qquad \text{(I.)}$$

$$= (1+\sqrt{v})^{-s-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\, \Pi(\tfrac{1}{2}+q,\, \tfrac{1}{2}s-q,\, \tfrac{1}{2}s+q,\, v) \qquad \text{(II.)}$$

$$= (1+\sqrt{v})^{-s}(1-\sqrt{v})^{-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\, \Pi(\tfrac{1}{2}-q,\, \tfrac{1}{2}s+q,\, \tfrac{1}{2}s-q,\, v) \qquad \text{(III.)}$$

$$= (1+\sqrt{v})^{-s}(1-\sqrt{v})^{-2q}\, 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\, \Pi(\tfrac{1}{2}s-q,\, \tfrac{1}{2}+q,\, \tfrac{1}{2}-q,\, v) \qquad \text{(IV.)}$$

where observe that on the right-hand side the Π-functions in I. and IV. only differ by the sign of q, and so also the Π-functions in II. and III. only differ by the sign of q. We hence have

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s-q, u) = (1+\sqrt{v})^{-s+2q}\, 2^{q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\, \Pi(\tfrac{1}{2}-q,\, \tfrac{1}{2}s+q,\, \tfrac{1}{2}s-q,\, v);$$

[p. 367] and comparing with (IV.),

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) = \left(\frac{1+\sqrt{v}}{1-\sqrt{v}}\right)^{2q} \Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s-q, u).$$

74. The foregoing formula,

$$\text{Ring-integral} = \frac{2^{s-1}f^s}{(f+\kappa)^{s+2q}}\, \Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u),$$

where $u = \dfrac{4\kappa f}{(f+\kappa)^2}$, gives, as well in the case of an exterior as an interior point, a convergent series for the integral; but this series proceeds according to the powers of $\dfrac{4\kappa f}{(f+\kappa)^2}$. We may obtain more convenient formulae applying to the cases of an internal and an external point respectively.

75. For an internal point $\kappa < f$, $\sqrt{1-u} = \dfrac{f-\kappa}{f+\kappa}$, and therefore $v = \dfrac{\kappa^2}{f^2}$.

$$\begin{aligned}\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) &= \left(\frac{f+\kappa}{f}\right)^{s+2q} 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\, \Pi\Big(\tfrac{1}{2}s+q,\ \tfrac{1}{2}-q,\ \tfrac{1}{2}+q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s+2q} 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\, \Pi\Big(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q,\ \tfrac{1}{2}s+q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s} \left(\frac{f-\kappa}{f}\right)^{-2q} 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\, \Pi\Big(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{\kappa^2}{f^2}\Big)\\
&= \left(\frac{f+\kappa}{f}\right)^{s} \left(\frac{f-\kappa}{f}\right)^{-2q} 2^{-q}\, \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\, \Pi\Big(\tfrac{1}{2}s-q,\ \tfrac{1}{2}+q,\ \tfrac{1}{2}-q,\ \tfrac{\kappa^2}{f^2}\Big);\end{aligned}$$

where the Π-functions on the right-hand side are respectively

$$\int_0^1 \frac{x^{\frac{1}{2}s+q-1}(1-x)^{-q-\frac{1}{2}}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac{1}{2}+q}} = \int_0^\infty \frac{t^{\frac{1}{2}s+q-1}\,dt}{(t+f^2-\kappa^2)^{q+\frac{1}{2}}(t+f^2)^{\frac{1}{2}s+q-\frac{1}{2}}}\cdot f^{s-1},$$

$$\int_0^1 \frac{x^{q-\frac{1}{2}}(1-x)^{\frac{1}{2}s-q-1}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac{1}{2}s+q}} = \int_0^\infty \frac{t^{q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s-q-1}\,dt}{(t+f^2)^{\frac{1}{2}s+q}}\cdot f^{-1},$$

$$\int_0^1 \frac{x^{-q-\frac{1}{2}}(1-x)^{\frac{1}{2}s+q-1}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac{1}{2}s-q}} = \int_0^\infty \frac{t^{-q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s+q-1}\,dt}{(t+f^2)^{\frac{1}{2}s-q}}\cdot f^{-1},$$

$$\int_0^1 \frac{x^{\frac{1}{2}s-q-1}(1-x)^{q-\frac{1}{2}}\,dx}{(1-\frac{\kappa^2}{f^2}\,x)^{\frac{1}{2}-q}} = \int_0^\infty \frac{t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}\,dt}{(t+f^2)^{\frac{1}{2}+q}}\cdot f^{-1},$$

the t-forms being obtained by means of the transformation $x = \dfrac{t}{t+f^2-\kappa^2}$; viz. this gives

$$1-x = \frac{f^2-\kappa^2}{t+f^2-\kappa^2},\quad 1-\tfrac{\kappa^2}{f^2}\,x = \frac{t+f^2}{t+f^2-\kappa^2}\cdot\frac{1}{f^2},\quad dx = \frac{(f^2-\kappa^2)\,dt}{(t+f^2-\kappa^2)^2},$$

whence the results just written down.

607. A Memoir on Prepotentials (continued)

[Continuation of the same paper; articles 75 (concluded)–95 of the Annexes.]

[p. 368] We hence have

$$\text{Ring-integral} = \frac{f^{s-1}}{(f^2-\kappa^2)^{q+\frac{1}{2}}} \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)} \int_0^\infty t^{\frac{1}{2}s+q-1}(t+f^2-\kappa^2)^{-q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s-q+\frac{1}{2}}\,dt$$

$$= \frac{f^{s-1}}{(f^2-\kappa^2)^{q+\frac{1}{2}}} \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)} \int_0^\infty t^{q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s-q-1}(t+f^2)^{-\frac{1}{2}s-q}\,dt$$

$$= f^{s-1} \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)} \int_0^\infty t^{-q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s+q-1}(t+f^2)^{-\frac{1}{2}s+q}\,dt$$

$$= f^{s-1} \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)} \int_0^\infty t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}-q}\,dt.$$

As a verification write $\kappa=0$, the four integrals are

$$\int_0^\infty \frac{t^{\frac{1}{2}s+q-1}\,dt}{(t+f^2)^{q+\frac{1}{2}}} = f^{s-q}\,\frac{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_0^\infty \frac{t^{q-\frac{1}{2}}\,dt}{(t+f^2)^{q+\frac{1}{2}}} = f^{-s-q}\,\frac{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_0^\infty \frac{t^{-q-\frac{1}{2}}\,dt}{(t+f^2)^{\frac{1}{2}-q}} = f^{-s-q}\,\frac{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_0^\infty \frac{t^{\frac{1}{2}s-q-1}\,dt}{(t+f^2)^{\frac{1}{2}+q}} = f^{-s-q}\,\frac{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})};$$

hence from each of them

$$\text{Ring-integral} = \frac{1}{f^{2q}}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

which is, in fact, the value obtained from

$$\text{Ring-integral} = \frac{2^{s-1}f^s}{(f+\kappa)^{s+2q}}\,\Pi\Big(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ \tfrac{4\kappa f}{(f+\kappa)^2}\Big)$$

on putting therein $\kappa=0$; viz. the value is

$$= \frac{2^{s-1}f^s}{f^{s+2q}}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ 0).$$

76. For an external point $\kappa > f$, $\sqrt{1-u} = \dfrac{\kappa-f}{\kappa+f}$, and therefore $v = \dfrac{f^2}{\kappa^2}$.

$$\Pi(\tfrac{1}{2}s, \tfrac{1}{2}s, \tfrac{1}{2}s+q, u) = \left(\frac{\kappa+f}{\kappa}\right)^{s+2q} 2^{-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\,\Pi\Big(\tfrac{1}{2}s+q,\ \tfrac{1}{2}-q,\ \tfrac{1}{2}+q,\ \tfrac{f^2}{\kappa^2}\Big)$$

$$= \left(\frac{\kappa+f}{\kappa}\right)^{s+2q} 2^{-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\,\Pi\Big(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q,\ \tfrac{1}{2}s+q,\ \tfrac{f^2}{\kappa^2}\Big)$$

$$= \left(\frac{\kappa+f}{\kappa}\right)^{s}\left(\frac{\kappa-f}{\kappa}\right)^{-2q} 2^{-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,\Pi\Big(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{f^2}{\kappa^2}\Big)$$

$$= \left(\frac{\kappa+f}{\kappa}\right)^{s}\left(\frac{\kappa-f}{\kappa}\right)^{-2q} 2^{-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\,\Pi\Big(\tfrac{1}{2}s-q,\ \tfrac{1}{2}+q,\ \tfrac{1}{2}-q,\ \tfrac{f^2}{\kappa^2}\Big);$$

[p. 369] where the Π-functions on the right hand are respectively

$$\int_0^1 \frac{x^{\frac{1}{2}s+q-1}(1-x)^{-q-\frac{1}{2}}\,dx}{(\kappa^2-f^2x)^{q+\frac{1}{2}}} = \kappa^{-2q+1}\int_0^\infty \frac{t^{\frac{1}{2}s+q-1}\,dt}{(t+\kappa^2-f^2)^{q+\frac{1}{2}}(t+\kappa^2)^{\frac{1}{2}s+q-\frac{1}{2}}},$$

$$\int_0^1 \frac{x^{q-\frac{1}{2}}(1-x)^{\frac{1}{2}s-q-1}\,dx}{(\kappa^2-f^2x)^{\frac{1}{2}s+q}} = \kappa^{-2q+1}\int_0^\infty \frac{t^{q-\frac{1}{2}}(t+\kappa^2-f^2)^{\frac{1}{2}s-q-1}\,dt}{(t+\kappa^2)^{\frac{1}{2}s+q}},$$

$$\int_0^1 \frac{x^{-q-\frac{1}{2}}(1-x)^{\frac{1}{2}s+q-1}\,dx}{(\kappa^2-f^2x)^{\frac{1}{2}s-q}} = \kappa^{-2q+1}\int_0^\infty \frac{t^{-q-\frac{1}{2}}(t+\kappa^2-f^2)^{\frac{1}{2}s+q-1}\,dt}{(t+\kappa^2)^{\frac{1}{2}s-q}},$$

$$\int_0^1 \frac{x^{\frac{1}{2}s-q-1}(1-x)^{q-\frac{1}{2}}\,dx}{(\kappa^2-f^2x)^{\frac{1}{2}-q}} = \kappa^{-2q+1}\int_0^\infty \frac{t^{\frac{1}{2}s-q-1}(t+\kappa^2-f^2)^{q-\frac{1}{2}}\,dt}{(t+\kappa^2)^{\frac{1}{2}+q}}.$$

We have then

$$\text{Ring-integral} = \frac{f^{s-1}}{(\kappa^2-f^2)^{q+\frac{1}{2}}}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\int_0^\infty t^{\frac{1}{2}s+q-1}(t+\kappa^2-f^2)^{-q-\frac{1}{2}}(t+\kappa^2)^{-\frac{1}{2}s-q+\frac{1}{2}}\,dt$$

$$= \frac{f^{s-1}}{(\kappa^2-f^2)^{q+\frac{1}{2}}}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\int_0^\infty t^{q-\frac{1}{2}}(t+\kappa^2-f^2)^{\frac{1}{2}s-q-1}(t+\kappa^2)^{-\frac{1}{2}s-q}\,dt$$

$$= f^{s-1}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\int_0^\infty t^{-q-\frac{1}{2}}(t+\kappa^2-f^2)^{\frac{1}{2}s+q-1}(t+\kappa^2)^{-\frac{1}{2}s+q}\,dt$$

$$= f^{s-1}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\int_0^\infty t^{\frac{1}{2}s-q-1}(t+\kappa^2-f^2)^{q-\frac{1}{2}}(t+\kappa^2)^{-\frac{1}{2}-q}\,dt.$$

Observe that in II. and III. the integrals, except as to the limits, are the same as in the corresponding formulæ for the interior point.

If in the integrals we put $t+\kappa^2-f^2$ in place of t, and ultimately suppose κ indefinitely large in comparison with f, they severally become

$$\int_{-\kappa^2+f^2}^\infty \frac{(t+\kappa^2-f^2)^{\frac{1}{2}s+q-1}\,dt}{(t+\kappa^2)^{q+\frac{1}{2}}} = \kappa^{s-1}\,\frac{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_{-\kappa^2+f^2}^\infty \frac{(t+\kappa^2-f^2)^{q-\frac{1}{2}}\,t^{\frac{1}{2}s-q-1}\,dt}{(t+\kappa^2)^{\frac{1}{2}s+q}} = \kappa^{s-1}\,\frac{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_{-\kappa^2+f^2}^\infty \frac{(t+\kappa^2-f^2)^{-q-\frac{1}{2}}\,t^{\frac{1}{2}s+q-1}\,dt}{(t+\kappa^2)^{\frac{1}{2}s-q}} = \kappa^{s-1}\,\frac{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

$$\int_{-\kappa^2+f^2}^\infty \frac{(t+\kappa^2-f^2)^{\frac{1}{2}s-q-1}\,t^{q-\frac{1}{2}}\,dt}{(t+\kappa^2)^{\frac{1}{2}+q}} = \kappa^{s-1}\,\frac{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})};$$

and they all four give

$$\text{Ring-integral} = \frac{f^{s-1}}{\kappa^{2q+1}}\cdot\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

which agrees with the value

$$= \frac{2^{s-1}f^s}{(\kappa+f)^{s+2q}}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ 0),$$

when $\frac{f}{\kappa}$ is indefinitely large.

[p. 370] **77.** We come now to the disk-integral,

$$\iint \frac{y^{s-1}\,dx\,dy}{\{(x-\kappa)^2+y^2\}^{\frac{1}{2}s+q}},$$

over the circle $x^2+y^2=f^2$. Writing $x=\kappa+\rho\cos\phi$, $y=\rho\sin\phi$, we have $dx\,dy=\rho\,d\rho\,d\phi$, and the integral therefore is

$$\iint \frac{\sin^{s-1}\phi\,d\rho\,d\phi}{\rho^{2q}},$$

where the integration in regard to ρ is performed at once; viz. the integral is

$$=\frac{1}{1-2q}\int(\rho^{1-2q})\sin^{s-1}\phi\,d\phi;$$

or multiplying by 2, in order that the integration may be taken only over the semicircle, $y=$ positive, this is

$$=\frac{1}{\frac{1}{2}-q}\int(\rho^{1-2q})\sin^{s-1}\phi\,d\phi,$$

the term (ρ^{1-2q}) being taken between the proper limits.

78. Consider first an interior point $\kappa<f$. As already mentioned, we exclude an indefinitely small circle radius ϵ, and the limits for ρ are from $\rho=\epsilon$ to $\rho=$ its value at

[Figure: a circle, with O on the horizontal axis as origin, an interior point P near O, and the line OP extended through a point on the circle; y-axis vertical.]

the circumference; viz. if here $x=f\cos\theta$, $y=f\sin\theta$, then we have $f\cos\theta=\kappa+\rho\cos\phi$, $f\sin\theta=\rho\sin\phi$, and consequently

$$\rho^2=\kappa^2+f^2-2\kappa f\cos\theta,$$

$$\sin\phi=\frac{f\sin\theta}{\rho},\qquad \cos\phi=\frac{f\cos\theta-\kappa}{\rho}=\frac{f\sin\theta}{\sqrt{\kappa^2+f^2-2\kappa f\cos\theta}},$$

and the integral therefore is

$$=\frac{1}{\frac{1}{2}-q}\int\left\{\frac{f^{s-1}\sin^{s-1}\theta}{(\kappa^2+f^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q-\frac{1}{2}}}-\epsilon^{1-2q}\sin^{s-1}\phi\right\}d\phi.$$

As regards the second term, this is $=\dfrac{\epsilon^{1-2q}}{\frac{1}{2}-q}\displaystyle\int\sin^{s-1}\phi\,d\phi$, from $\phi=0$ to $\phi=\pi$; or, what is the same thing, we may multiply by 2 and take the integral from $\phi=0$ to $\phi=\frac{\pi}{2}$.

[p. 371] Writing then $\sin\phi=\sqrt{x}$, and consequently $\sin^{s-1}\phi\,d\phi=\frac{1}{2}x^{\frac{1}{2}s-1}(1-x)^{-\frac{1}{2}}\,dx$, the term is

$$=-\frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})};$$

the value of the disk-integral is

$$=\frac{f^{s-1}}{\frac{1}{2}-q}\int\frac{\sin^{s-1}\theta\,d\theta}{(\kappa^2+f^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q-\frac{1}{2}}}-\frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}.$$

But we have

$$\sin\phi=\frac{f\sin\theta}{\rho},\qquad \cos\phi=\frac{f\cos\theta-\kappa}{\rho},$$

and thence

$$\tan\phi=\frac{f\sin\theta}{f\cos\theta-\kappa},\qquad \sec^2\phi\,d\phi=\frac{f(f-\kappa\cos\theta)\,d\theta}{(f\cos\theta-\kappa)^2};$$

that is,

$$d\phi = \frac{f(f-\kappa\cos\theta)\,d\theta}{\rho^2} = \frac{f(f-\kappa\cos\theta)\,d\theta}{f^2+\kappa^2-2\kappa f\cos\theta},$$

or, what is the same thing,

$$= \tfrac{1}{2}\,\frac{\{(f^2-\kappa^2)+(f^2+\kappa^2-2\kappa f\cos\theta)\}\,d\theta}{f^2+\kappa^2-2\kappa f\cos\theta};$$

the expression for the disk-integral is therefore

$$= \frac{\frac{1}{2}f^{s-1}}{\frac{1}{2}-q}\int_0^\pi \frac{\sin^{s-1}\theta\,\{(f^2-\kappa^2)+(f^2+\kappa^2-2\kappa f\cos\theta)\}\,d\theta}{(f^2+\kappa^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q}} - \frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}.$$

79. Writing as before $\cos\frac{1}{2}\theta = \sqrt{x}$, $\sin\frac{1}{2}\theta = \sqrt{1+x}$, &c., and $u = \dfrac{4\kappa f}{(\kappa+f)^2}$, this is

$$= \frac{2^{s-1}f^{s-1}}{(\frac{1}{2}-q)(\kappa+f)^{s+2q-1}}\left\{\frac{f^2-\kappa^2}{(\kappa+f)^2}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ u) + \Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q-1,\ u)\right\} - \frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}.$$

As a verification, observe that, if $\kappa=0$, each of the Π-functions becomes

$$= \int_0^1 x^{\frac{1}{2}s-1}(1-x)^{-\frac{1}{2}}\,dx = \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})};$$

hence the whole first term is $= \dfrac{f^{s-1}\cdot f}{\frac{1}{2}-q}\cdot\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}$, viz. this is $\dfrac{f^{s-2q}}{\frac{1}{2}-q}\cdot\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}$, and the complete value is

$$= \frac{1}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\,(f^{s-2q}-\epsilon^{s-2q}),$$

vanishing, as it should do, if $f=\epsilon$.

80. In the case of an exterior point $\kappa > f$, the process is somewhat different; but the result is of a like form. We have

$$\text{Disk-integral} = \frac{1}{\frac{1}{2}-q}\int(\rho_1^{1-2q}-\rho^{1-2q})\sin^{s-1}\phi\,d\phi,$$

[p. 372] where ρ_1 refers to the point M' and ρ to the point M. Attending first to the integral $\int\rho^{1-2q}\sin^{s-1}\phi\,d\phi$, and writing as before $f\cos\theta = \kappa+\rho\cos\phi$, $f\sin\theta = \rho\sin\phi$, this is

$$= f^{s-1}\int\frac{\sin^{s-1}\theta\,d\theta}{(\kappa^2+f^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q-\frac{1}{2}}}$$

$$= \tfrac{1}{2}f^{s-1}\int\frac{\sin^{s-1}\theta\,\{-(\kappa^2-f^2)+(f^2+\kappa^2-2\kappa f\cos\theta)\}\,d\theta}{(f^2+\kappa^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q}},$$

the inferior and the superior limits being here the values of θ which correspond to the points N, A respectively, say $\theta+\alpha$, and $\theta=0$; hence, reversing the sign and interchanging the two limits, the value of $-\int\rho^{1-2q}\sin^{s-1}\phi\,d\phi$ is the above integral taken from 0 to α. But similarly the value of $+\int\rho_1^{1-2q}\sin^{s-1}\phi\,d\phi$ is the same integral taken from α to π. For the two terms together, the value is the same integral from 0 to π; viz. we thus find

$$\text{Disk-integral} = \tfrac{1}{2}\,\frac{f^{s-1}}{\frac{1}{2}-q}\int_0^\pi\frac{\sin^{s-1}\theta\,\{-(\kappa^2-f^2)+(f^2+\kappa^2-2\kappa f\cos\theta)\}\,d\theta}{(f^2+\kappa^2-2\kappa f\cos\theta)^{\frac{1}{2}s+q}};$$

or, writing as before $\cos\frac{1}{2}\theta=\sqrt{x}$, &c., and $u=\dfrac{4\kappa f}{(\kappa+f)^2}$, this is

$$=\frac{2^{s-1}f^{s-1}}{(\frac{1}{2}-q)(\kappa+f)^{s+2q-1}}\left\{-\frac{\kappa^2-f^2}{(\kappa+f)^2}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ u)+\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q-1,\ u)\right\}.$$

81. As a verification, suppose that κ is indefinitely large: we must recur to the last preceding formula; the value is thus

$$=\frac{f^s}{(\frac{1}{2}-q)\,\kappa^{s+2q-1}}\int_0^\pi\frac{\sin^{s-1}\theta\left(-\cos\theta+\dfrac{f}{\kappa}\right)}{\left(1-\dfrac{2f}{\kappa}\cos\theta\right)^{\frac{1}{2}s+q}}\,d\theta,$$

viz. this is

$$=\frac{f^s}{(\frac{1}{2}-q)\,\kappa^{s+2q-1}}\int_0^\pi\sin^{s-1}\theta\left\{-\cos\theta+[1-(s+2q)\cos^2\theta]\,\frac{f}{\kappa}\right\}d\theta,$$

where the integral of the first term vanishes; the value is thus

$$=\frac{f^{s+1}}{(\frac{1}{2}-q)\,\kappa^{s+2q}}\int_0^\pi\sin^{s-1}\theta\,[1-(s+2q)\cos^2\theta]\,d\theta,$$

[p. 373] where we may multiply by 2 and take the integral from 0 to $\frac{\pi}{2}$. Writing $\sin\theta=\sqrt{x}$, the value is

$$=\frac{f^{s+1}}{(\frac{1}{2}-q)\,\kappa^{s+2q}}\int_0^1 x^{\frac{1}{2}s-1}\,[1-(s+2q)(1-x)]\,(1-x)^{-\frac{1}{2}}\,dx,$$

where the integral is

$$=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\left\{1-\frac{\frac{1}{2}(s+2q)}{\frac{1}{2}s+\frac{1}{2}}\right\}=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\,\frac{\frac{1}{2}-q}{\frac{1}{2}s+\frac{1}{2}},$$

and hence the value is

$$=\frac{f^{s+1}}{\kappa^{s+2q}}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{3}{2})};$$

viz. this is $=\dfrac{1}{\kappa^{s+2q}}\displaystyle\int y^{s-1}\,dx\,dy$, over the circle $x^2+y^2=f^2$, as is easily verified.

82. Reverting to the interior point $\kappa<f$.

$$\text{Disk-integral}=\frac{2^{s-1}f^{s-1}}{(\frac{1}{2}-q)(\kappa+f)^{s+2q-1}}\left\{\frac{f^2-\kappa^2}{(\kappa+f)^2}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ u)+\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q-1,\ u)\right\}-\frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})};$$

then reducing the expression in { } by the transformations for $\Pi(\frac{1}{2}s,\frac{1}{2}s,\frac{1}{2}s+q,u)$ and the like transformations for $\Pi(\frac{1}{2}s,\frac{1}{2}s,\frac{1}{2}s+q-1,u)$, the term in { } may be expressed in the four forms:

1. $$\frac{2^{-q}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\,\frac{(f+\kappa)^{s+2q-1}}{f^{s+2q-1}}\left[\left(1-\tfrac{\kappa^2}{f^2}\right)\Pi(\tfrac{1}{2}s+q,\ \tfrac{1}{2}-q,\ \tfrac{1}{2}+q,\ \tfrac{\kappa^2}{f^2})+\frac{\frac{1}{2}+q-\frac{1}{2}}{\frac{1}{2}-q}\Pi(\tfrac{1}{2}s+q-1,\ \tfrac{3}{2}-q,\ -\tfrac{1}{2}+q,\ \tfrac{\kappa^2}{f^2})\right],$$

2. $$\frac{2^{-q}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\,\frac{(f+\kappa)^{s+2q-1}}{f^{s+2q-1}}\left[\left(1-\tfrac{\kappa^2}{f^2}\right)\Pi(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q,\ \tfrac{1}{2}s+q,\ \tfrac{\kappa^2}{f^2})+\frac{\frac{1}{2}+q-\frac{1}{2}}{\frac{1}{2}-q}\Pi(-\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{1}{2}s+q-1,\ \tfrac{\kappa^2}{f^2})\right],$$

3. $$\frac{2^{-q}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,\frac{(f+\kappa)^{s-1}(f-\kappa)^{-2q}}{f^{s-1}}\left[\left(1-\tfrac{\kappa^2}{f^2}\right)\Pi(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{\kappa^2}{f^2})+\frac{\frac{1}{2}+q-\frac{1}{2}}{\frac{1}{2}-q}\Pi(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{\kappa^2}{f^2})\right],$$

4. $$\frac{2^{-q}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q)\,\Gamma(\frac{1}{2}+q)}\,\frac{(f+\kappa)^{s-1}(f-\kappa)^{-2q}}{f^{s-1}}\left[\left(1-\tfrac{\kappa^2}{f^2}\right)\Pi(\tfrac{1}{2}s-q,\ \tfrac{1}{2}+q,\ \tfrac{1}{2}-q,\ \tfrac{\kappa^2}{f^2})+\frac{\frac{1}{2}+q-\frac{1}{2}}{\frac{1}{2}-q}\Pi(\tfrac{1}{2}s-q+1,\ -\tfrac{1}{2}+q,\ \tfrac{3}{2}-q,\ \tfrac{\kappa^2}{f^2})\right].$$

[p. 374] 83. The first and the fourth of these are susceptible of a reduction which does not appear to be applicable to the second and the third. Consider in general the function

$$(1-v)\,\Pi(\alpha,\,\beta,\,1-\beta,\,v)+\frac{\alpha-1}{\beta}\,\Pi(\alpha-1,\,\beta+1,\,-\beta,\,v);$$

the second Π-function is here

$$\int_0^1 x^{\alpha-2}(1-x\,.\,1-vx)^{\beta}\,dx;$$

viz. this is

$$=\frac{x^{\alpha-1}}{\alpha-1}(1-x\,.\,1-vx)^{\beta}-\frac{1}{\alpha-1}\int_0^1 x^{\alpha-1}\frac{d}{dx}(1-x\,.\,1-vx)^{\beta}\,dx,$$

or, since the first term vanishes between the limits, this is

$$=\frac{\beta}{\alpha-1}\int x^{\alpha-1}(1-x\,.\,1-vx)^{\beta-1}(1+v-2vx)\,dx,$$

$$=\frac{\beta}{\alpha-1}\left\{(1+v)\,\Pi(\alpha,\,\beta,\,1-\beta,\,v)-2v\int_0^1 x^{\alpha}(1-x\,.\,1-vx)^{\beta-1}\,dx\right\}.$$

Hence the two Π-functions together are

$$=(1-v+1+v)\int_0^1 x^{\alpha-1}(1-x\,.\,1-vx)^{\beta-1}\,dx-2\int_0^1 vx\,.\,x^{\alpha-1}(1-x\,.\,1-vx)^{\beta-1}\,dx,$$

$$=2\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}(1-vx)^{\beta}\,dx,$$

that is,

$$(1-v)\,\Pi(\alpha,\,\beta,\,1-\beta,\,v)+\tfrac{\alpha-1}{\beta}\,\Pi(\alpha-1,\,\beta+1,\,-\beta,\,v)=2\,\Pi(\alpha,\,\beta,\,-\beta,\,v).$$

We have therefore

$$\left(1-\tfrac{\kappa^2}{f^2}\right)\Pi(\tfrac12 s+q,\,\tfrac12-q,\,\tfrac12+q,\,\tfrac{\kappa^2}{f^2})+\frac{\frac12 s+q-1}{\frac12-q}\,\Pi(\tfrac12 s+q-1,\,\tfrac32-q,\,-\tfrac12+q,\,\tfrac{\kappa^2}{f^2})=2\,\Pi(\tfrac12 s+q,\,\tfrac12-q,\,-\tfrac12+q,\,\tfrac{\kappa^2}{f^2});$$

and from the same equation, written in the form

$$\Pi(\alpha-1,\,\beta+1,\,-\beta,\,v)+\tfrac{\beta}{\alpha-1}(1-v)\,\Pi(\alpha,\,\beta,\,1-\beta,\,v)=2\tfrac{\beta}{\alpha-1}\,\Pi(\alpha,\,\beta,\,-\beta,\,v),$$

we obtain

$$\Pi(\tfrac12 s-q+1,\,-\tfrac12+q,\,\tfrac32-q,\,v)+(1-v)\,\frac{-\frac12+q}{\frac12 s-q}\,\Pi(\tfrac12 s-q,\,\tfrac12+q,\,\tfrac12-q,\,v)=2\frac{-\frac12+q}{\frac12 s-q}\,\Pi(\tfrac12 s-q,\,\tfrac12+q,\,-\tfrac12-q,\,v).$$

84. Hence the terms in [ ] in the first and the fourth expressions in No. 82 are

$$=\frac{2^{q+1}\,\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12 s+q)\,\Gamma(\frac12-q)}\;\frac{(f+\kappa)^{s+2q-1}}{f^{s+2q-1}}\;\Pi(\tfrac12 s+q,\,\tfrac12-q,\,-\tfrac12+q,\,\tfrac{\kappa^2}{f^2}),$$

$$=\frac{2^{-q}(-\frac12+q)\,\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12 s-q+1)\,\Gamma(\frac12+q)}\;\frac{(f+\kappa)^{s-1}(f-\kappa)^{-2q}}{f^{s-1}}\;\Pi(\tfrac12 s-q+1,\,-\tfrac12+q,\,\tfrac12-q,\,\tfrac{\kappa^2}{f^2}),$$

[p. 375] respectively; the corresponding values of the disk-integral are

$$\frac{\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12-q)\,\Gamma(\frac12 s+q)}\;f^{s-2q}\,\Pi(\tfrac12 s+q,\,\tfrac12-q,\,-\tfrac12+q,\,\tfrac{\kappa^2}{f^2})-\frac{\epsilon^{1-2q}}{\frac12-q}\;\frac{\Gamma\frac12 s\,\Gamma\frac12}{\Gamma(\frac12 s+\frac12)},$$

$$\frac{-\Gamma\frac12 s\,\Gamma\frac12 s}{\Gamma(\frac12 s-q+1)\,\Gamma(\frac12+q)}\left(\frac{f-\kappa}{f}\right)^{1-2q}\Pi(\tfrac12 s-q+1,\,-\tfrac12+q,\,\tfrac12-q,\,\tfrac{\kappa^2}{f^2})-\frac{\epsilon^{1-2q}}{\frac12-q}\;\frac{\Gamma\frac12 s\,\Gamma\frac12}{\Gamma(\frac12 s+\frac12)},$$

which we may again verify by writing therein $\kappa = 0$, viz. the Π-functions thus become

$$\frac{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})} \quad \text{and} \quad \frac{\Gamma(\frac{1}{2}s-q+1)\,\Gamma(-\frac{1}{2}+q)}{\Gamma(\frac{1}{2}s+\frac{1}{2})},$$

and consequently the integral is

$$= \frac{1}{\frac{1}{2}-q}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\,(f^{s-2q}-\epsilon^{s-2q}).$$

85. But the forms nevertheless belong to a system of four. In the formulae

$$\begin{aligned}\Pi(\alpha,\,\beta,\,\gamma,\,v) &= \frac{\Gamma\alpha\,\Gamma\beta}{\Gamma\gamma\,\Gamma(\alpha+\beta-\gamma)}\,\Pi(\gamma,\,\alpha+\beta-\gamma,\,\alpha,\,v)\\ &= (1-v)^{\gamma-\beta-\gamma}\,\Pi(\beta,\,\alpha,\,\alpha+\beta-\gamma,\,v)\\ &= (1-v)^{\gamma-\beta-\gamma}\,\frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\alpha+\beta-\gamma)\,\Gamma\gamma}\,\Pi(\alpha+\beta-\gamma,\,\gamma,\,\beta,\,v),\end{aligned}$$

writing $\alpha = \frac{1}{2}s+q$, $\beta = \frac{1}{2}-q$, $\gamma = -\frac{1}{2}+q$, we deduce

$$\begin{aligned}\Pi(\tfrac{1}{2}s+q,\,\tfrac{1}{2}-q,\,-\tfrac{1}{2}+q,\,v) &= \frac{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}{\Gamma(-\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\,\Pi(-\tfrac{1}{2}+q,\,\tfrac{1}{2}s-q+1,\,\tfrac{1}{2}s+q,\,v)\\ &= (1-v)^{1-s}\,\Pi(\tfrac{1}{2}-q,\,\tfrac{1}{2}s+q,\,\tfrac{1}{2}s-q+1,\,v)\\ &= (1-v)^{1-s}\,\frac{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}{\Gamma(\frac{1}{2}s-q+1)\,\Gamma(-\frac{1}{2}+q)}\,\Pi(\tfrac{1}{2}s-q+1,\,-\tfrac{1}{2}+q,\,\tfrac{1}{2}-q,\,v);\end{aligned}$$

and the last-mentioned values of the disk-integral may thus be written in the four forms

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s-2q}\,\Pi(\tfrac{1}{2}s+q,\,\tfrac{1}{2}-q,\,-\tfrac{1}{2}+q,\,\tfrac{\kappa^2}{f^2}) \quad -\text{term in }\epsilon,$$

$$\frac{-\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\,f^{s-2q}\,\Pi(-\tfrac{1}{2}+q,\,\tfrac{1}{2}s-q+1,\,\tfrac{1}{2}s+q,\,\tfrac{\kappa^2}{f^2}) \quad -\quad ,,\quad ,$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\left(\frac{f-\kappa}{f}\right)^{1-s}\Pi(\tfrac{1}{2}-q,\,\tfrac{1}{2}s+q,\,\tfrac{1}{2}s-q+1,\,\tfrac{\kappa^2}{f^2}) \quad -\quad ,,\quad ,$$

$$\frac{-\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\left(\frac{f-\kappa}{f}\right)^{1-s}\Pi(\tfrac{1}{2}s-q+1,\,-\tfrac{1}{2}+q,\,\tfrac{1}{2}-q,\,\tfrac{\kappa^2}{f^2}) \quad -\quad ,,\quad ;$$

[p. 376] and since the last of these is in fact the second of the original forms, it is clear that, if instead of the first we had taken the second of the original forms, we should have obtained again the same system of four forms.

86. Writing as before $x = \dfrac{t}{t+f^2-\kappa^2}$, &c., the forms are

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,(f^2-\kappa^2)^{q-\frac{1}{2}}\int_0^\infty t^{\frac{1}{2}s+q-1}(t+f^2-\kappa^2)^{-q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s-q+\frac{1}{2}}\,dt \quad -\text{term in }\epsilon,$$

$$\frac{-\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\,f^{s+1}(f^2-\kappa^2)^{-q-\frac{1}{2}}\int_0^\infty t^{-q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s-q}(t+f^2)^{-\frac{1}{2}s-q-1}\,dt \quad -\quad ,,\quad ,$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s-1}(f^2-\kappa^2)^{-q-\frac{1}{2}}\int_0^\infty t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q}\,dt \quad -\text{term in }\epsilon,$$

$$\frac{-\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\,(f^2-\kappa^2)^{q+\frac{1}{2}}\int_0^\infty t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{-q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}-q}\,dt \quad - \quad ,, \quad .$$

87. The third of these possesses a remarkable property. Write mf instead of f, and at the same time change t into m^2t: the integral becomes

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,m^{s-1}f^{s-1}\int_0^\infty t^{\frac{1}{2}s-q-1}\{m^2(t+f^2)-\kappa^2\}^{q-\frac{1}{2}}\,m^{s+1-2q}(t+f^2)^{-\frac{1}{2}s+q-\frac{1}{2}}\,dt \quad - \text{term in } \epsilon;$$

and hence, writing $mf=f+\delta f$ or $m=1+\dfrac{\delta f}{f}$, and therefore $m^2=1+\dfrac{2\delta f}{f}$, the value is

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s-1}\int_0^\infty t^{\frac{1}{2}s-q-1}\left\{(t+f^2-\kappa^2)+\frac{2\delta f}{f}(t+f^2)\right\}^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s-q+\frac{1}{2}}\,dt \quad - \text{term in } \epsilon.$$

Hence the term in δf is

$$=2(-q+\tfrac{1}{2})\,\frac{\delta f}{f}\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s-1}\int_0^\infty t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{3}{2}}(t+f^2)^{-\frac{1}{2}s+q+\frac{1}{2}}\,dt,$$

$$=\delta f \text{ into expression } \quad \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s-2}\int_0^\infty t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{3}{2}}(t+f^2)^{-\frac{1}{2}s+q+\frac{1}{2}}\,dt,$$

where the factor which multiplies δf is, as it should be, the ring-integral; it in fact agrees with one of the expressions previously obtained for this integral.

88. Similarly for an exterior point $\kappa>f$; starting in like manner from, Disk-integral

$$=\frac{2^{s-1}f^{s-1}}{(\frac{1}{2}-q)(\kappa+f)^{s+2q-1}}\left\{-\frac{\kappa^2-f^2}{(\kappa+f)^2}\,\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q,\ u)+\Pi(\tfrac{1}{2}s,\ \tfrac{1}{2}s,\ \tfrac{1}{2}s+q-1,\ u)\right\},$$

and reducing in like manner, the term in { } may be expressed in the four forms

1. $\dfrac{2^{-q}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\,\dfrac{(\kappa+f)^{s+2q-1}}{\kappa^{s+2q-1}}\left[-\left(1-\frac{f^2}{\kappa^2}\right)\Pi(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q,\ \tfrac{1}{2}s+q,\ \tfrac{f^2}{\kappa^2})+\frac{-\frac{1}{2}+q}{\frac{1}{2}+q}\,\Pi(-\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{1}{2}s+q-1,\ \tfrac{f^2}{\kappa^2})\right],$

[p. 377]

2. $\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\,\dfrac{(\kappa+f)^{s+2q-1}}{\kappa^{s+2q-1}}\left[-\left(1-\frac{f^2}{\kappa^2}\right)\Pi(\tfrac{1}{2}s+q,\ \tfrac{1}{2}-q,\ \tfrac{1}{2}+q,\ \tfrac{f^2}{\kappa^2})+\frac{\frac{1}{2}+q-1}{\frac{1}{2}-q}\,\Pi(\tfrac{1}{2}s+q-1,\ \tfrac{3}{2}-q,\ -\tfrac{1}{2}+q,\ \tfrac{f^2}{\kappa^2})\right],$

3. $\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,\dfrac{(\kappa+f)^{s-1}(\kappa-f)^{-2q}}{\kappa^{s-1}}\left[-\Pi(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{f^2}{\kappa^2})+\left(1-\frac{f^2}{\kappa^2}\right)\frac{\frac{1}{2}+q-1}{\frac{1}{2}-q}\,\Pi(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q,\ \tfrac{f^2}{\kappa^2})\right],$

4. $\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q+1)\,\Gamma(\frac{1}{2}+q)}\,\dfrac{(\kappa+f)^{s-1}(\kappa-f)^{-2q}}{\kappa^{s-1}}\left[-\Pi(\tfrac{1}{2}s-q+1,\ -\tfrac{1}{2}+q,\ \tfrac{3}{2}-q,\ \tfrac{f^2}{\kappa^2})+\left(1-\frac{f^2}{\kappa^2}\right)\frac{-\frac{1}{2}+q}{\frac{1}{2}+q}\,\Pi(\tfrac{1}{2}s-q+1,\ -\tfrac{1}{2}+q,\ \tfrac{3}{2}-q,\ \tfrac{f^2}{\kappa^2})\right].$

89. For the reduction of the first and the fourth of these, we have to consider

$$-(1-v)\,\Pi(\alpha,\ \beta,\ 1-\beta,\ v)+\tfrac{\alpha-1}{\beta}\,\Pi(\alpha-1,\ \beta+1,\ -\beta,\ v);$$

viz. this is

$$(-1+v+1+v)\int_0^1 x^{\alpha-1}(1-x\,.\,1-vx)^{\beta-1}\,dx-2\int_0^1 vx\,.\,x^{\alpha-1}(1-x\,.\,1-vx)^{\beta-1}\,dx,$$

$$=2v\int_0^1 x^{\alpha-1}(1-x)(1-x\,.\,1-vx)^{\beta-1}\,dx,$$

$$= 2v\,\Pi(\alpha,\ \beta+1,\ -\beta+1,\ v);$$

that is,

$$-(1-v)\,\Pi(\alpha,\ \beta,\ 1-\beta,\ v) + \tfrac{\alpha-1}{\beta}\,\Pi(\alpha-1,\ \beta+1,\ -\beta,\ v) = 2v\,\Pi(\alpha,\ \beta+1,\ -\beta+1,\ v).$$

I repeat, for comparison, the foregoing equation

$$+(1-v)\,\Pi(\alpha,\ \beta,\ 1-\beta,\ v) + \tfrac{\alpha-1}{\beta}\,\Pi(\alpha-1,\ \beta+1,\ -\beta,\ v) = 2\,\Pi(\alpha,\ \beta,\ -\beta,\ v);$$

by adding and subtracting these we obtain two new formulae; for reduction of the fourth formula, the equation may be written

$$-\Pi(\alpha-1,\ \beta+1,\ -\beta,\ v) + (1-v)\,\tfrac{\beta}{\alpha-1}\,\Pi(\alpha,\ \beta,\ 1-\beta,\ v) = -2\tfrac{\beta}{\alpha-1}v\,\Pi(\alpha,\ \beta+1,\ -\beta+1,\ v).$$

90. But it is sufficient to consider the first formula; the term in [ ] is

$$= \frac{2^{q+1}\,\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)} \left(\frac{\kappa+f}{\kappa}\right)^{s+2q-1} \left(\frac{\kappa-f}{\kappa}\right)^{-2q} \Pi(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{1}{2}s+q,\ \tfrac{f^2}{\kappa^2});$$

and the corresponding value of the disk-integral is

$$= \frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}-q)}\,\frac{f^{s+1}}{\kappa^{s+2q}}\,\Pi(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{1}{2}s+q,\ \tfrac{f^2}{\kappa^2}).$$

[p. 378] which we may again verify by taking therein κ indefinitely large; viz. the value is then $= \dfrac{f^{s+1}}{\kappa^{s+2q}}\,\dfrac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{3}{2})}$, as above. It is the first of a system of four forms, the others of which are

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\,\frac{f^{s+1}}{\kappa^{s+2q}}\left(\frac{\kappa-f}{\kappa}\right)^{1-s}\Pi(\tfrac{1}{2}+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{1}{2}s+q,\ \tfrac{f^2}{\kappa^2}),$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,\frac{f^{s+1}}{\kappa^{s+2q}}\left(\frac{\kappa-f}{\kappa}\right)^{1-s}\Pi(\tfrac{1}{2}-q,\ \tfrac{1}{2}s+q,\ \tfrac{1}{2}s-q+1,\ \tfrac{f^2}{\kappa^2}),$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q+1)\,\Gamma(\frac{1}{2}+q)}\,\frac{f^{s+1}}{\kappa^{s+2q}}\left(\frac{\kappa-f}{\kappa}\right)^{1-s}\Pi(\tfrac{1}{2}s-q+1,\ -\tfrac{1}{2}+q,\ \tfrac{3}{2}-q,\ \tfrac{f^2}{\kappa^2}).$$

And hence, writing as before $x = \dfrac{t}{t+f^2-\kappa^2}$, &c., the four values are

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q+1)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{-q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s-q}(t+f^2)^{-\frac{1}{2}s-q-1}\,dt,$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(\frac{1}{2}-q)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s+q-1}(t+f^2-\kappa^2)^{-q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s-q-\frac{1}{2}}\,dt,$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt,$$

$$\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}s-q+1)\,\Gamma(\frac{1}{2}+q)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{-q-\frac{1}{2}}(t+f^2-\kappa^2)^{\frac{1}{2}s-q-1}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt,$$

where we may in the integrals write $t+\kappa^2-f^2$ in place of t, making the limits ∞, 0; but the actual form is preferable.

91. In the third form, for f write mf, at the same time changing t into m^2t; the new value of the disk-integral is

$$=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s+1}\int_{m^{-2}(\kappa^2-f^2)}^{\infty} t^{\frac{1}{2}s-q-1}\{m^2(t+f^2)-\kappa^2\}^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt.$$

Writing here $mf=f+\delta f$, that is, $m=1+\dfrac{\delta f}{f}$, $m^2=1+\dfrac{2\delta f}{f}$, and observing that, if $-q+\frac{1}{2}$ be positive, the factor $\{m^2(t+f^2)-\kappa^2\}^{-q+\frac{1}{2}}$ vanishes for the value $t=\dfrac{\kappa^2}{m^2}-f^2$ at the lower limit, we see that on this supposition, $-q+\frac{1}{2}$ positive, the value is

$$=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s-q-1}\Big\{t+f^2-\kappa^2+\tfrac{2\delta f}{f}(t+f^2)\Big\}^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt;$$

viz. the term in δf is $=\delta f$ multiplied by the expression

$$2(-q+\tfrac{1}{2})\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}-q)\,\Gamma(\frac{1}{2}s+q)}\,f^{s}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{3}{2}}(t+f^2)^{-\frac{1}{2}s+q}\,dt,$$

[p. 379] that is, multiplied by

$$2\,(\tfrac{1}{2}-q)\,\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}s-q)}\,f^{s}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{3}{2}}(t+f^2)^{-\frac{1}{2}s+q}\,dt,$$

which is in fact $=\delta f$ multiplied by the value of the ring-integral.

92. Comparing for the cases of an interior point $\kappa<f$ and an exterior point $\kappa>f$, the four expressions for the disk-integral, it will be noticed that only the third expressions correspond precisely to each other; viz. these are interior point, $\kappa<f$; the value is

$$=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}-q)}\,f^{s+1}\int_{0}^{\infty} t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt-\frac{\epsilon^{1-2q}}{\frac{1}{2}-q}\left(\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\right),$$

where, if $\frac{1}{2}-q$ be positive (which is, in fact, a necessary condition in order to the applicability of the formula), the term in ϵ vanishes, and may therefore be omitted; and exterior point, $\kappa>f$; the value is

$$=\frac{\Gamma\frac{1}{2}s\,\Gamma\frac{1}{2}s}{\Gamma(\frac{1}{2}+q)\,\Gamma(\frac{1}{2}-q)}\,f^{s+1}\int_{\kappa^2-f^2}^{\infty} t^{\frac{1}{2}s-q-1}(t+f^2-\kappa^2)^{q-\frac{1}{2}}(t+f^2)^{-\frac{1}{2}s+q-1}\,dt,$$

differing only from the preceding one in the inferior limit κ^2-f^2 in place of 0 of the integral. We have $\frac{1}{2}-q$ positive, and also $\frac{1}{2}s+q$ positive; viz. q may have any value diminishing from $\frac{1}{2}$ to $-\frac{1}{2}s$, the extreme values not admissible.

Annex IV. Examples of Theorem A. Art. Nos. 93 to 112.

93. It is remarked in the text that, in the examples which relate to the s-coordinal sphere and ellipsoid respectively, we have a quantity θ, a function of the coordinates $(a, \ldots, c, e)$ of the attracted point; viz. in the case of the sphere, writing $a^2 + \ldots + c^2 = \kappa^2$, we have

$$\frac{\kappa^2}{f^2+\theta} + \frac{e^2}{\theta} = 1;$$

in the case of the ellipsoid, we have

$$\frac{a^2}{f^2+\theta} + \ldots + \frac{c^2}{h^2+\theta} + \frac{e^2}{\theta} = 1,$$

the equations having in each case a positive root which is called θ. The properties of the equation are the same in each case; but for the sphere, the equation being a quadric one, can be solved. The equation in fact is

$$\theta^2 - \theta(e^2 + \kappa^2 - f^2) - e^2 f^2 = 0,$$

and the positive root is therefore

$$\theta = \tfrac{1}{2}\left\{e^2 + \kappa^2 - f^2 + \sqrt{(e^2+\kappa^2-f^2)^2 + 4e^2f^2}\right\}.$$

Suppose e to diminish gradually and become $= 0$; for an exterior point, $\kappa > f$, the value of the radical is $= \kappa^2 - f^2$, and we have $\theta = \kappa^2 - f^2$; for an interior point, $\kappa < f$,

[p. 380] the value of the radical, supposing e only indefinitely small, is $= f^2 - \kappa^2 + \dfrac{2e^2f^2}{f^2-\kappa^2}$, and we have $\theta = \frac{1}{2}\left(e^2 + \kappa^2 - f^2 + f^2 - \kappa^2 + \dfrac{2e^2f^2}{f^2-\kappa^2}\right) = \dfrac{e^2f^2}{f^2-\kappa^2}$, or, what is the same thing, $\theta = e^2\left(1 - \dfrac{\kappa^2}{f^2}\right)$; viz. the positive root of the equation continually diminishes with e, and becomes ultimately $= 0$. If κ or e be indefinitely large, then the radical may be taken $= e^2 + \kappa^2$, and we have θ indefinitely large, $= e^2 + \kappa^2$.

94. The result is similar for the general equation

$$\frac{a^2}{f^2+\theta} + \ldots + \frac{c^2}{h^2+\theta} + \frac{e^2}{\theta} = 1;$$

the left-hand side is $= 0$ for $\theta = \infty$, and (as θ decreases) continually increases, becoming infinite for $\theta = 0$; there is consequently a single positive value of θ for which the value is $= 1$; viz. the equation has a single positive root, and θ is taken to denote this root.

In the last-mentioned equation, let e gradually diminish and become $= 0$; then for an exterior point, viz. if

$$\frac{a^2}{f^2} + \ldots + \frac{c^2}{h^2} > 1, \quad \text{the equation } \frac{a^2}{f^2+\theta} + \ldots + \frac{c^2}{h^2+\theta} = 1$$

has (as is at once seen) a single positive root, and θ becomes equal to the positive root of this equation; but for an interior point, or $\dfrac{a^2}{f^2} + \ldots + \dfrac{c^2}{h^2} < 1$, the equation just written down has no positive root, and θ becomes $= 0$, that is, the positive root of the original equation continually diminishes with e, and for $e = 0$ becomes ultimately $= 0$; its value for e small is, in fact, given by $\theta = e^2\left(1 - \dfrac{a^2}{f^2} - \ldots - \dfrac{c^2}{h^2}\right)$. Also $a, \ldots, c, e$ (or any of them) indefinitely large, θ is indefinitely large, $= a^2 + \ldots + c^2 + e^2$.

95. We have an interesting geometrical illustration in the case $s + 1 = 2$; θ is here determined by the equation

$$\frac{a^2}{f^2+\theta} + \frac{b^2}{g^2+\theta} + \frac{e^2}{\theta} = 1;$$

viz. θ is the squared a-semiaxis of the ellipsoid, confocal with the conic $\dfrac{x^2}{f^2} + \dfrac{y^2}{g^2} = 1$, which passes through the point (a, b, e). Taking $e = 0$, the point in question, if $\dfrac{a^2}{f^2} + \dfrac{b^2}{g^2} > 1$, is a point in the plane of xy, outside the ellipse, and we have through the point a proper confocal ellipsoid, whose squared ξ-semiaxis does not vanish; but if $\dfrac{a^2}{f^2} + \dfrac{b^2}{g^2} < 1$, then the point is within the ellipse, and the only confocal ellipsoid through the point is the indefinitely thin ellipsoid, squared semiaxes $(f^2, g^2, 0)$, which in fact coincides with the ellipse.

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 381–405 (Paper 607, continued)

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Contents

607. A Memoir on Prepotentials (continued)

[Continued from earlier pages of the same paper. Articles 96–131.]

[p. 381] **96.** The positive root θ of the equation

$$\frac{a^2}{f^2+\theta}+\frac{c^2}{h^2+\theta}-1+\frac{e^2}{\theta}=0$$

has certain properties which connect themselves with the function

$$\Theta_0=\theta^{q-1}\big\{(\theta+f^2)\dots(\theta+h^2)\big\}^{-\frac{1}{2}}.$$

We have, the accents denoting differentiations in regard to θ,

$$J'\,\frac{d\theta}{da}=\frac{2a}{\theta+f^2}\,,\qquad J'\,\frac{d\theta}{de}=-\frac{2e}{\theta}\,J'\,;\quad\text{and}\quad\frac{d\theta}{da}=\frac{1}{J'}\cdot\frac{2a}{\theta+f^2},$$

where

$$J'=\frac{a^2}{(f^2+\theta)^2}+\dots+\frac{c^2}{(h^2+\theta)^2}+\frac{e^2}{\theta^2}\,;$$

and we have the like formulae for $\dfrac{d\theta}{dc},\ \dfrac{d\theta}{de}$.

We deduce

$$\frac{a}{\theta+f^2}\,\frac{d\theta}{da}+\dots+\frac{c}{\theta+h^2}\,\frac{d\theta}{dc}+\frac{e}{\theta}\,\frac{d\theta}{de}=\frac{2}{J'},$$

and to this we may join, η being arbitrary,

$$\frac{a}{\theta+\eta+f^2}\,\frac{d\theta}{da}+\dots+\frac{c}{\theta+\eta+h^2}\,\frac{d\theta}{dc}+\frac{e}{\theta+\eta}\,\frac{d\theta}{de}=\frac{2}{J'}\Big[1-\Big\{\frac{a^2\,\eta}{(\theta+f^2)(\theta+\eta+f^2)}+\dots+\frac{e^2\,\eta}{\theta(\theta+\eta)}\Big\}\Big].$$

Again, defining $\nabla_a\theta$ and $\Box\theta$ as immediately appears, we have

$$\nabla_a\theta=\left(\frac{d\theta}{da}\right)^2+\dots+\left(\frac{d\theta}{dc}\right)^2,\qquad=\frac{4}{J'},\quad\Box\theta=\frac{8a^2}{J'(\theta+f^2)^2}\,;$$

and passing to the second differential coefficients, we have

$$\frac{d^2\theta}{da^2}=\frac{2}{J'(\theta+f^2)}-\frac{8a^2}{J'^2(\theta+f^2)^2}-\frac{4a^2J''}{J'^3(\theta+f^2)^2}\,,$$

where

$$J''=-\frac{2a^2}{(f^2+\theta)^3}-\dots-\frac{2c^2}{(h^2+\theta)^3}-\frac{2e^2}{\theta^3}\,,$$

and the like formulae for $\dfrac{d^2\theta}{dc^2},\ \dfrac{d^2\theta}{de^2}$. Joining to these $\dfrac{1}{e}\,\dfrac{d\theta}{de}=\dfrac{2}{J'}\cdot\dfrac{2q+1}{\theta}\,\dfrac{d\theta}{de}$, we obtain

$$\Box\theta=\left(\frac{d^2\theta}{da^2}+\dots+\frac{d^2\theta}{dc^2}+\frac{d^2\theta}{de^2}+\frac{2q+1}{e}\,\frac{d\theta}{de}\right)$$

$$=\frac{2}{J'}\left[\frac{1}{\theta+f^2}+\dots+\frac{1+(2q+1)}{\theta}\right]-\frac{2}{J'^2}\left[\frac{2a^2}{(\theta+f^2)^2}+\dots+\frac{2e^2}{\theta^2}\right]-\frac{J''}{J'^3}\,(4a^2\cdots).$$

[p. 382] where the last two terms destroy each other; observing that we have

$$\frac{1}{e^2}\cdot\frac{d}{de}\big(e^{2q+1}\big)=\frac{2q+1}{e^2}\cdot e^{2q}\cdot\frac{1}{e}\cdot e=\frac{2q+1}{e^2}\cdot e^{2q-2}\cdot e^2\,;$$

the result is

$$\Box\theta=\frac{2}{J'}\left(\frac{2q+2}{\theta}-\frac{2\theta'}{\theta}\right)+\frac{4\theta'}{\theta\,J'^2}\,.$$

97. First example, $x^2 = a^2 + \cdots + c^2$, and θ the positive root of $\dfrac{x^2}{f^2+\theta} + \dfrac{e^2}{\theta} = 1$.

V is assumed $= J\int_\theta^\infty t^{q-1}(t+f^2)^{-\frac{1}{2}s}\,dt$, where $q+1$ is positive.

I do not work the example out; it corresponds step by step with, and is hardly more simple than, the next example, which relates to the ellipsoid. The result is

$$\rho = (\sqrt{\pi})^s\,\Gamma(q+1)^{-1}\left[\left(1 - \tfrac{x^2}{f^2} - \tfrac{e^2}{f^2}\right)^q\right], \quad \text{if} \quad \tfrac{x^2}{f^2} + \tfrac{e^2}{f^2} < 1;$$

hence the integral

$$\int \frac{\left\{1 - \frac{x^2}{f^2} - \frac{e^2}{f^2}\right\}^q dx \ldots dz}{\left\{(a-x)^2 + \cdots + (c-z)^2 + e^2\right\}^{\frac{1}{2}s+q}},$$

taken over the sphere $a^2 + \cdots + z^2 = f^2$,

$$= \frac{(\sqrt{\pi})^s\,\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q+1)}\, f^s \int_\theta^\infty t^{q-1}(t+f^2)^{-\frac{1}{2}s}\,dt.$$

98. Second example, θ the positive root of $\dfrac{a^2}{f^2+\theta} + \cdots + \dfrac{c^2}{h^2+\theta} + \dfrac{e^2}{\theta} = 1$, $q+1$ positive.

Consider here the function

$$V = \int_\theta^\infty t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}}\,dt;$$

this satisfies the prepotential equation. We have in fact

$$\frac{dV}{da} = -\Theta_0\,\frac{d\theta}{da}, \qquad \frac{d^2V}{da^2} = -\frac{d\Theta_0}{d\theta}\left(\frac{d\theta}{da}\right)^2 - \Theta_0\,\frac{d^2\theta}{da^2},$$

with the like expressions for $\dfrac{d^2V}{dc^2}$, $\dfrac{d^2V}{de^2}$; also

$$\frac{2q+1}{e}\,\frac{dV}{de} = -\Theta_0\,\frac{2q+1}{e}\,\frac{d\theta}{de}.$$

Hence

$$\Box V = -\Theta_0\,\Box\theta - \theta'\,\nabla_a\theta,$$

[p. 383] or, substituting for $\Box\theta$ and $\nabla_a\theta$ their values, this is

$$= \Theta_0\left\{-\frac{2}{J'}\left(\frac{2q+2}{\theta} - \frac{2\theta'}{\theta}\right) - \frac{4\theta'}{\theta\,J'^2}\right\} - \frac{4\theta'}{J'} = 0.$$

Moreover V does not become infinite for any values of $(a, \ldots, c, e)$, e not $= 0$; and it vanishes for points at ∞. And not only so, but for indefinitely large values of any of the coordinates $(a, \ldots, c, e)$ it reduces itself to a numerical multiple of $(a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}s+q}$; in fact, in this case θ is indefinitely large, $= a^2 + \cdots + c^2 + e^2$. Consequently throughout the integral, t is indefinitely large, and we may therefore write

$$V = \int_\theta^\infty t^{q-1}t^{-\frac{1}{2}s}\,dt = \frac{-1}{\frac{1}{2}s-q}\left[t^{-\frac{1}{2}s+q}\right]_\theta^\infty = \frac{1}{\frac{1}{2}s-q}\,\theta^{-\frac{1}{2}s+q};$$

that is,

$$V = \frac{1}{\frac{1}{2}s-q}\,(a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}s+q}.$$

The conditions of the theorem are thus satisfied, and we have for ρ either of the formulae

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q)}\,e^{2q}\,W, \qquad \rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)}\,\frac{1}{e^{2q+1}}\,\frac{dW}{de};$$

in the former of them q must be positive; in the latter it is sufficient if $q+1$ be positive.

99. We have W the same function of $(x, \ldots, z, e)$ that V is of $(a, \ldots, c, e)$; viz. writing λ for the positive root of

$$\frac{x^2}{f^2+\lambda} + \cdots + \frac{z^2}{h^2+\lambda} + \frac{e^2}{\lambda} = 1,$$

the value of W is

$$W = \int_\lambda^\infty t^{q-1}\{(t+f^2)\ldots(t+h^2)\}^{-\frac{1}{2}}\,dt.$$

Considering the formula which involves $e^{2q}W$, — first, if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, then, when e is $= 0$ the value of λ is not $= 0$; the integral W is therefore finite (not indefinitely large), and we have $e^{2q}W = 0$, consequently $\rho = 0$.

But if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, then, when e is indefinitely small λ is also indefinitely small; viz. we then have $\dfrac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$, and the value of W is

$$W = \frac{1}{q}\,\lambda^q (f\ldots h)^{-1},$$

and hence

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q+1)}\,(f\ldots h)^{-1}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^q.$$

[p. 384] 100. Again, using the formula which involves $\left(e^{2q+1}\dfrac{dW}{de}\right)$, we have first $\dfrac{dV}{da} = -\Theta\,\dfrac{d\theta}{da}$, or substituting for Θ and $\dfrac{d\theta}{da}$ their values and multiplying by e^{2q+1}, we find

$$e^{2q+1}\,\frac{dV}{da} = 2e^{2q}\theta^{q-1}\,\frac{a}{(f^2+\theta)^2}\{(\theta+f^2)\ldots(\theta+h^2)\}^{-\frac{1}{2}} \cdot \frac{1}{(\theta+f^2)^2} + \cdots + \frac{1}{(h^2+\theta)^2} + \tfrac{1}{\theta^2}\,,$$

and therefore

$$e^{2q+1}\,\frac{dW}{de} = -2e^{2q}\lambda^{q-1}\{(\lambda+f^2)\ldots(\lambda+h^2)\}^{-\frac{1}{2}}\,\frac{1}{\cdots}.$$

Hence, writing $e = 0$: first, for an exterior point or $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, λ is not $= 0$, and the expression vanishes in virtue of the factor e^{2q+1}, whence also $\rho = 0$; next, for an interior point or $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, λ is $= 0$, hence also $\dfrac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$ is infinite; neglecting in comparison with it the other terms $\dfrac{a^2}{(\lambda+f^2)^2} + \cdots$, the value is

$$= \frac{2}{e^2}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^{-1};$$

and we have, as before,

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q+1)}\,(f\ldots h)^{-1}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^q.$$

101. Hence in the formula

$$V = \int \frac{\rho\,dx\ldots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}$$

ρ has the value just found, or, what is the same thing, we have

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}\right)^q dx\ldots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}$$

taken over ellipsoid $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1$,

$$= \frac{(\sqrt{\pi})^s\,\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty t^{q-1}\{(t+f^2)\ldots(t+h^2)\}^{-\frac{1}{2}}\,dt.$$

[p. 385] **102.** We may in this result write $e = 0$. There are two cases, according as the attracted point is exterior or interior: if it is exterior, $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, θ will denote the positive root of the equation $\frac{a^2}{f^2+\theta} + \cdots + \frac{c^2}{h^2+\theta} = 1$; if it be interior, $\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} < 1$, θ will be $= 0$; and we thus have

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}\right)^q dx \ldots dz}{\left\{(a-x)^2 + \cdots + (c-z)^2\right\}^{\frac{1}{2}s+q}}$$

$$= \frac{(\sqrt{\pi})^s\,\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}(f \ldots h) \int_\theta^\infty t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}} dt, \quad \text{for exterior point } \tfrac{a^2}{f^2} + \cdots + \tfrac{c^2}{h^2} > 1,$$

$$= \frac{(\sqrt{\pi})^s\,\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}(f \ldots h) \int_0^\infty t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}} dt, \quad \text{for interior point } \tfrac{a^2}{f^2} + \cdots + \tfrac{c^2}{h^2} < 1;$$

but as regards the value for an interior point it is to be observed that, unless q be negative (between 0 and -1, since $1+q$ is positive by hypothesis), the two sides of the equation will be each of them infinite.

103. Third example. We assume here

$$V = \int_\theta^\infty dt\, I^m\, T,$$

where

$$I = 1 - \frac{a^2}{f^2+t} - \cdots - \frac{c^2}{h^2+t} - \frac{e^2}{t},$$

$$T = t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}};$$

as before, θ is the positive root of the equation

$$1 - \frac{a^2}{f^2+\theta} - \cdots - \frac{c^2}{h^2+\theta} - \frac{e^2}{\theta} = 0,$$

and $\frac{1}{2}s + q$ is positive in order that the integral may be finite; also m is positive.

104. In order to show that V satisfies the prepotential equation $\Box V = 0$, I shall, in the first place, consider the more general expression

$$V = \int_{\theta+\eta}^\infty dt\, I_0^m\, T,$$

where η is a constant positive quantity which will be ultimately put $= 0$. The functions previously called J and Θ will be written J_0 and Θ_0, and J_η will now denote

$$J_\eta = 1 - \frac{a^2}{\theta+\eta+f^2} - \cdots - \frac{c^2}{\theta+\eta+h^2} - \frac{e^2}{\theta+\eta},$$

$$\Theta_\eta = (\theta+\eta)^{q-1}\left\{(\theta+\eta+f^2)\ldots(\theta+\eta+h^2)\right\}^{-\frac{1}{2}};$$

[p. 386] whence also, subtracting from J the evanescent function J_0, we have

$$J - J_0 = \frac{a^2\,\eta}{(\theta+f^2)(\theta+\eta+f^2)} + \cdots + \frac{e^2\,\eta}{\theta(\theta+\eta)};$$

say this is $= \eta P$; and we have thence, by former equations and in the present notation,

$$\frac{a}{\theta+\eta+f^2}\frac{d\theta}{da} + \cdots + \frac{c}{\theta+\eta+h^2}\frac{d\theta}{dc} + \frac{e}{\theta+\eta}\frac{d\theta}{de} = \frac{2}{J_0'}(1-\eta P),$$

$$\nabla_a\theta = \frac{4}{J_0'^2}, \qquad \Box\theta = \frac{4\Theta_0'}{\Theta_0\, J_0}.$$

In virtue of the equation which determines θ, we have

$$\frac{dV}{da} = \int_{\theta+\eta}^\infty dt\, mI^{m-1}\frac{dI}{da}T, \qquad \frac{d^2V}{da^2} = \int_{\theta+\eta}^\infty dt\left\{m(m-1)I^{m-2}\left(\frac{dI}{da}\right)^2 T + mI^{m-1}\frac{d^2I}{da^2}T\right\}$$

$$-J_\eta^m \Theta_\eta \frac{d\theta}{da};$$

and thence

$$-mJ_\eta^{m-1}\Theta_\eta \frac{d\theta}{da} - J_\eta^m \Theta_\eta \frac{d^2\theta}{da^2} - J_\eta^m\Theta_\eta \left(\frac{d\theta}{da}\right)^2$$
$$-J_\eta^m \Theta_\eta' \frac{d\theta}{da},$$

with like expressions for $\dfrac{d^2V}{dc^2}$, $\dfrac{d^2V}{de^2}$. Also

$$\frac{2q+1}{e}\frac{dV}{de} = \int_{\theta+\eta}^\infty dt\, mI^{m-1}\,\frac{2q+1}{e}\frac{dI}{de}\,T - J_\eta^m\,\Theta_\eta\cdot\frac{2q+1}{e}\frac{d\theta}{de};$$

and hence

$$\square V = \int_{\theta+\eta}^\infty dt\left\{m(m-1)I^{m-2}\left[\left(\tfrac{dI}{da}\right)^2+\cdots\right]T + mI^{m-1}\,\square I\,T\right\}$$
$$+4mJ_\eta^{m-1}\Theta_\eta\left(\frac{a}{\theta+\eta+f^2}\frac{d\theta}{da}+\cdots+\frac{e}{\theta+\eta}\frac{d\theta}{de}\right) - J_\eta^m\Theta_\eta\,\square\theta - J_\eta^m\Theta_\eta'\,\nabla_a\theta\,.$$

[p. 387] **105.** Writing I', T' for the first derived coefficients of I, T in regard to t, we have

$$I' = \frac{a^2}{(f^2+t)^2}+\cdots+\frac{c^2}{(h^2+t)^2}+\frac{e^2}{t^2}, \qquad \frac{T'}{T} = \frac{q-1}{t} - \tfrac{1}{2}\left(\frac{1}{f^2+t}+\cdots+\frac{1}{h^2+t}\right) = \frac{2q+1}{2t}\cdots\,.$$

The integral is therefore

$$\int_{\theta+\eta}^\infty dt\left\{2mI^{m-1}\,I'\,T + m(m-1)I^{m-2}\cdot 4I'\cdot T\right\}$$
$$= \int_{\theta+\eta}^\infty dt\left\{4mI^{m-1}\,T' + 4m(m-1)I^{m-2}\,I'\,T\right\},$$

viz. $I^{m-1}T$ vanishing for $t=\infty$, this is

$$= -4mJ_\eta^{m-1}\Theta\,.$$

Hence, writing $(J^m\Theta)'$ instead of $\dfrac{d}{dt}(J^m\Theta)$, we have

$$\square V = -4mJ_\eta^{m-1}\Theta_\eta + 4mJ_\eta^{m-1}\,\Theta_\eta\left(\frac{a}{\theta+\eta+f^2}\frac{d\theta}{da}+\cdots+\frac{e}{\theta+\eta}\frac{d\theta}{de}\right)$$
$$-(J^m\Theta)'\,\nabla_a\theta\,,$$

viz. this is

$$\square V = -4mJ_\eta^{m-1}\Theta_\eta + 4mJ_\eta^{m-1}\,\frac{1}{J_0'}\big(1-\eta P\big)\cdot 2 - (J_\eta^m\Theta_\eta)'\,\frac{4}{J_0'^2},$$

or, writing $mJ_\eta^{m-1}J_\eta'\Theta + J_\eta^m\Theta'$ instead of $(J^m\Theta)'$, this is

$$\square V = -\frac{4mJ_\eta^{m-1}\Theta_\eta}{J_0'}\big(J_\eta' - 2JP + J_0'\big) - \frac{4J_\eta^m}{J_0'^2}\big(\Theta_\eta'\,J_0 - \Theta(J_0')'\big).$$

We have here

$$J_\eta' - 2JP + J_0' = \eta^2\,Q, \quad \text{suppose.}$$

Also $\Theta_\eta'\Theta_0 - \Theta\Theta_0'$ contains the factor η, is $=\eta M$ suppose.

[p. 388] **106.** Substituting for J, $J' - 2JP + J$, and $\Theta'\Theta_0 - \Theta\Theta_0'$ their values ηP, ηQ, and ηM, the whole result contains the factor η^{m+1}, viz. we have

$$\square V = -\frac{4\eta^{m+1}\,\Theta_\eta}{J_0^{m+1}}\big(mP^mQ + P^mM\big).$$

If here, except in the term η^{m+1}, we write $\eta = 0$, we have

$$P = \frac{a^2}{(\theta+f^2)^2} + \cdots + \frac{c^2}{(\theta+h^2)^2} + \frac{e^2}{\theta^2}\,,$$

the formula becomes

$$\Box V = -\frac{4\eta^{m+1}\,\Theta_0}{J_0^{m+1}}\Big(mP^m\,J_0'^{-} + P^m M_0\Big),$$

or (instead of J_0, Θ_0) using now J, Θ in their original significations

$$J = 1 - \frac{a^2}{f^2+\theta} - \cdots - \frac{c^2}{h^2+\theta} - \frac{e^2}{\theta}, \quad \text{and} \quad \Theta = \theta^{q-1}\big\{(\theta+f^2)\ldots(\theta+h^2)\big\}^{-\frac{1}{2}},$$

this is

$$\Box V = -\frac{4\eta^{m+1}\,\Theta}{J^{m+1}}\Big(mP^m\,J_0^{-1} + P^m M_0\Big),$$

or, what is the same thing,

$$\Box V = -\frac{4\eta^{m+1}\,\Theta}{J^{m+1}}\big(mP + (\ldots)\big),$$

viz. the expression in { } is

$$-\frac{1}{(\theta+f^2)^2} + \cdots + \frac{1}{(\theta+h^2)^2} + \frac{1}{\theta^2} - \frac{2q+2}{\theta}.$$

We thus see that, η being infinitesimal, $\Box V$ is infinitesimal of the order η^{m+1}; and hence, η being $= 0$, we have

$$\Box V = 0;$$

viz. the prepotential equation is satisfied by the value

$$V = \int_\theta^\infty dt\, I^m\, T,$$

where $m+1$ is positive.

107. We have consequently a value of ρ corresponding to the foregoing value of V; and this value is

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)}\,\frac{1}{e^{2q+1}}\,\frac{dW}{de}\,,$$

[p. 389] where, writing λ for the positive root of

$$1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda} = 0,$$

we have

$$W = \int_\lambda^\infty dt\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^m t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}};$$

we thence obtain

$$\frac{dW}{de} = \int_\lambda^\infty dt\,\frac{-2me}{t}\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^{m-1} t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} - (\ldots)\frac{d\lambda}{de},$$

or, multiplying by e^{2q+1} and substituting for $\dfrac{d\lambda}{de}$ its value $\dfrac{2e}{\lambda}\cdot\dfrac{1}{\cdots}$, we have

$$e^{2q+1}\,\frac{dW}{de} = -2me^{2q+2}\int_\lambda^\infty dt\,\frac{1}{t}\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^{m-1} t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} + \cdots\,,$$

where the second term, although containing the evanescent factor

$$\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\right)^m,$$

is for the present retained.

108. I attend to the second term.

1°. Suppose $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$; then, as e diminishes and becomes $= 0$, λ does not become zero, but it becomes the positive root of the equation

$$1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} = 0;$$

hence the term, containing as well the evanescent factor e^{2q+2} as the other evanescent factor $\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\right)^m$, is $= 0$.

[p. 390] 2°. Suppose $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$; then, as e diminishes to zero, λ tends to become $= 0$, but $\dfrac{e^2}{\lambda}$ is finite and $= 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$, whence $\dfrac{1}{\lambda}$ is indefinitely large; and since $\dfrac{x^2}{(\lambda+f^2)^2} + \cdots + \dfrac{z^2}{(\lambda+h^2)^2}$ becomes $= +\dfrac{x^2}{f^4} + \cdots + \dfrac{z^2}{h^4}$, which is finite, the denominator may be reduced to $\dfrac{e^2}{\lambda}$, and the term therefore is

$$= -2\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^{m-1} e^{2q+2}(f\ldots h)^{-1}\lambda^{-q-1},$$

which, the other factor being finite, vanishes in virtue of the evanescent factor

$$\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\right)^m.$$

Hence the second term always vanishes, and we have (e being $= 0$)

$$e^{2q+1}\frac{dW}{de} = -2m\int_\lambda^\infty dt\,\frac{e^{2q+2}}{t}\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^{m-1} t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}}.$$

109. Considering first the case $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$: then, as e diminishes to zero, λ does not become $= 0$; the integral contains no infinite element, and it consequently vanishes in virtue of the factor e^{2q+2}.

But if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, then, introducing instead of t the new variable ξ, $= t\cdot\dfrac{1}{\lambda}$, that is, $t = \dfrac{\lambda}{\xi}$, $dt = -\dfrac{\lambda}{\xi^2}\,d\xi$, and writing for shortness

$$R = 1 - \frac{x^2}{f^2+\xi} - \cdots - \frac{z^2}{h^2+\xi},$$

the term becomes

$$= \int_0^R d\xi\cdot 2m(R-\xi)^{m-1}\xi^q\left\{(f^2+\xi)\ldots(h^2+\xi)\right\}^{-\frac{1}{2}},$$

where, as regards the limits, corresponding to $t=\infty$ we have $\xi = 0$, and corresponding to $t = \lambda$ we have ξ the positive root of $R - \xi = 0$. But e is indefinitely small; except for indefinitely small values of ξ, we have

$$R = 1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}, \quad \text{and} \quad \left\{(f^2+\xi)\ldots(h^2+\xi)\right\}^{-\frac{1}{2}} = (f\ldots h)^{-1};$$

[p. 391] and if ξ be indefinitely small, then, whether we take the accurate or the reduced expressions, the elements are finite, and the corresponding portion of the integral is indefinitely small. We may consequently reduce as above; viz. writing now

$$R = 1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2},$$

the formula is

$$e^{2q+1}\frac{dW}{de} = \int_0^R d\xi\,2m(R-\xi)^{m-1}\,\xi^q\,(f\ldots h)^{-1}$$

$$= -2m(f\ldots h)^{-1}\int_0^R d\xi\,\xi^q(R-\xi)^{m-1},$$

or writing $\xi = Ru$, the integral becomes $= R^{q+m} \int_0^1 du\, u^q (1-u)^{m-1}$, which is

$$= R^{q+m} \frac{\Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)};$$

that is, we have

$$e^{2q+1} \frac{dW}{de} = -\frac{2m(f\dots h)^{-1}\Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)} R^{q+m},$$

and consequently

$$\rho = \frac{\Gamma(\tfrac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s\,\Gamma(1+q+m)} (f\dots h)^{-1} R^{q+m},$$

that is,

$$\rho = \frac{\Gamma(\tfrac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s\,\Gamma(1+q+m)} (f\dots h)^{-1} \Big(1 - \tfrac{x^2}{f^2} - \dots - \tfrac{z^2}{h^2}\Big)^{q+m},$$

viz. ρ has this value for values of $(x, \dots, z)$ such that $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, but is $= 0$ if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$.

110. Multiplying by a constant factor so as to reduce ρ to the value R^{q+m}, the final result is that the integral

$$V = \int \frac{\Big(1 - \tfrac{x^2}{f^2} - \dots - \tfrac{z^2}{h^2}\Big)^{q+m} dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2 + e^2\big\}^{\frac{1}{2}s+q}},$$

the limits being given by the equation

$$\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1,$$

is equal to

$$\frac{\Gamma(\tfrac{1}{2}s)\Gamma(1+q+m)}{\Gamma(1+q)\Gamma(\tfrac{1}{2}s+q+m)} (f\dots h) \int_\theta^\infty dt\, t^m \Big(1 - \frac{a^2}{f^2+t} - \dots - \frac{c^2}{h^2+t} - \frac{e^2}{t}\Big)^m t^{q-1} \big\{(t+f^2)\dots(t+h^2)\big\}^{-\frac{1}{2}},$$

where θ is the positive root of

$$1 - \frac{a^2}{\theta+f^2} - \dots - \frac{c^2}{\theta+h^2} - \frac{e^2}{\theta} = 0.$$

[p. 392] In particular, if $e = 0$, or

$$V = \int \frac{\Big(1 - \tfrac{x^2}{f^2} - \dots - \tfrac{z^2}{h^2}\Big)^{q+m} dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2\big\}^{\frac{1}{2}s+q}},$$

there are two cases:

exterior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} > 1$, θ is positive root of $1 - \frac{a^2}{f^2+\theta} - \dots - \frac{c^2}{h^2+\theta} = 0$,

interior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1$, θ vanishes, viz. the limits in the integral are $\infty, 0$;

q must be negative, $1+q$ positive as before, in order that the t-integral may not be infinite in regard to the element $t = 0$.

It is assumed in the proof that m and $1+q$ are each of them positive; but, as appears by the second example, the theorem is true for the extreme value $m = 0$; it does not, however, appear that the proof can be extended to include the extreme value $q = -1$. The formula seems, however, to hold good for values of m, q beyond the foregoing limits; and it would seem that the only necessary conditions are $\frac{1}{2}s + q$, $1+m$, and $1+q+m$, each of them positive. The theorem is, in fact, a particular case of the following one, proved Annex X. No. 162, viz.

$$V = \int \frac{\phi\Big(1 - \tfrac{x^2}{f^2} - \dots - \tfrac{z^2}{h^2}\Big)\, dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2 + e^2\big\}^{\frac{1}{2}s+q}},$$

taken over the ellipsoid $\frac{x^2}{f^2}+\dots+\frac{z^2}{h^2}=1$, is equal to

$$\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,\{(t+f^2)\dots(t+h^2)\}^{-\frac{1}{2}}(1-\sigma)^{q+\frac{1}{2}s}\int_0^1(1-x)^q\phi\{\sigma+(1-\sigma)x\}\,dx,$$

where σ denotes $\dfrac{a^2}{f^2+t}+\dots+\dfrac{c^2}{h^2+t}+\dfrac{e^2}{t}$; assuming $\phi u=(1-u)^{q+m}$, we have

$$\phi\{\sigma+(1-\sigma)x\}=(1-\sigma)^{q+m}(1-x)^{q+m},$$

and the theorem is thus proved.

111. Particular cases: $m=0$;

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^q dx\dots dz}{\left\{(a-x)^2+\dots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,t^{q-1}\left\{(t+f^2)\dots(t+h^2)\right\}^{-\frac{1}{2}}.$$

Cor. In a somewhat similar manner it may be shown that

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^q x\,dx\dots dz}{\left\{(a-x)^2+\dots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{a(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,t^{q-1}\left\{(t+f^2)\dots(t+h^2)\right\}^{-\frac{1}{2}}.$$

[p. 393] Multiplying the first by a and subtracting it from the second, we have

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^q (a-x)\,dx\dots dz}{\left\{(a-x)^2+\dots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=-\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,\frac{a}{f^2+t}\,t^{q-1}\left\{(t+f^2)\dots(t+h^2)\right\}^{-\frac{1}{2}},$$

or, writing $q+1$ for q, this is

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^q (a-x)\,dx\dots dz}{\left\{(a-x)^2+\dots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q+1}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,\frac{a}{f^2+t}\,t^{q}\left\{(t+f^2)\dots(t+h^2)\right\}^{-\frac{1}{2}},$$

and we have similar formulae with $\dots$, $(c-z)$, e, instead of $(a-x)$, in the numerator.

112. If $m=1$, we have

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^{q+1} dx\dots dz}{\left\{(a-x)^2+\dots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(2+\frac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,t^{q}\left\{(t+f^2)\dots(t+h^2)\right\}^{-\frac{1}{2}},$$

which, differentiated in respect to a, gives the $(a-x)$-formula; hence conversely, assuming the $a-x,\dots,c-z,e$-formulae, we obtain by integration the last preceding formula to a constant près, viz. we thereby obtain the multiple integral $=C+$ right-hand function, where C is independent of $(a,\dots,c,e)$; by taking these all infinite, and observing that then $\theta=\infty$, the two integrals each vanish, and we obtain $C=0$.

In particular, when $s=3$, $q=-\frac{1}{2}$, then

$$\int\frac{\left(1-\frac{x^2}{f^2}-\dots-\frac{z^2}{h^2}\right)^{\frac{1}{2}} dx\,dy\,dz}{\left\{(a-x)^2+(b-y)^2+(c-z)^2+e^2\right\}^1}=\frac{\pi^2}{2}(fgh)\int_\theta^\infty\frac{dt}{\sqrt{t(t+f^2)(t+g^2)(t+h^2)}},$$

which, putting therein $e=0$, gives the potential of an ellipsoid for the cases of an exterior point and an interior point respectively.

Annex V. Green's Integration of the Prepotential Equation. Art. Nos. 113 to 128.

[p. 393, cont.] **113.** In the present Annex, I in part reproduce Green's process for the integration of this equation by means of a series of functions, which are analogous to Laplace's Functions and may be termed "Greenians" (see his Memoir on the Attraction of Ellipsoids, referred to above, p. 320); each such function gives rise to a Prepotential Integral.

Green shows, by a complicated and difficult piece of general reasoning, that there exist solutions of the form $V = \Theta\phi$ (see post. No. 116), where ϕ is a function of the
[p. 394] s new variables $\alpha, \beta, \dots, \gamma$ without θ, such that $\nabla^2\phi = \kappa\phi$, κ being a function of θ only; these functions ϕ of the variables $\alpha, \beta, \dots, \gamma$ are in fact the Greenian Functions in question. The function of the order 0 is $\phi = 1$; those of the order 1 are $\phi = \alpha$, $\phi = \beta, \dots, \phi = \gamma$; those of the order 2 are $\phi = \alpha\beta$, etc., and s functions each of the form

$$\phi = \tfrac{1}{2}(A\alpha^2 + B\beta^2 + \cdots + C\gamma^2) + D.$$

The existence of the functions just referred to other than the s functions involving the squares of the variables is obvious enough; the difficulty first arises in regard to these s functions; and the actual development of them appears to me important by reason of the light which is thereby thrown upon the general theory. This I accomplish in the present Annex; and I determine by Green's process the corresponding prepotential integrals. I do not go into the question of the Greenian Functions of orders superior to the second.

114. I write for greater clearness $(a, b, \dots, c, e)$ instead of $(a, \dots, c, e)$ to denote the series of $(s+1)$ variables; viz. $(a, b, \dots, c)$ will denote a series of s variables; corresponding to these we have the semiaxes $(f, g, \dots, h)$, and the new variables $(\alpha, \beta, \dots, \gamma)$; these last, with the before-mentioned function θ, are the $s+1$ new variables of the problem; and, for convenience, there is introduced also a quantity ε; viz. we have

$$a = \sqrt{f^2+\theta}\,\alpha, \quad b = \sqrt{g^2+\theta}\,\beta, \ \dots, \ c = \sqrt{h^2+\theta}\,\gamma, \quad e = \sqrt{\theta}\,\varepsilon,$$

where $\lambda = \alpha^2 + \beta^2 + \cdots + \gamma^2 + \varepsilon^2$.

That is, we have θ a function of $a, b, \dots, c, e$, determined by

$$\frac{a^2}{f^2+\theta} + \frac{b^2}{g^2+\theta} + \cdots + \frac{c^2}{h^2+\theta} + \frac{e^2}{\theta} = 1,$$

and then $\alpha, \beta, \dots, \gamma$ are given as functions of the same quantities $a, b, \dots, c, e$ by the equations

$$\alpha^2 = \frac{a^2}{f^2+\theta}, \quad \beta^2 = \frac{b^2}{g^2+\theta}, \ \dots, \ \gamma^2 = \frac{c^2}{h^2+\theta};$$

also ε, considered as a function of the same quantities, is

$$\varepsilon^2 = 1 - \frac{a^2}{f^2+\theta} - \frac{b^2}{g^2+\theta} - \cdots - \frac{c^2}{h^2+\theta} = \frac{e^2}{\theta}.$$

115. Introducing instead of $a, b, \dots, c, e$ the new variables $\alpha, \beta, \dots, \gamma, \theta$, the transformed differential equation is

$$\nabla^2 V + \frac{d^2V}{d\theta^2}\Big(s + 2q + 2 - \frac{\alpha^2}{f^2+\theta} - \cdots - \frac{\gamma^2}{h^2+\theta}\Big)\cdot 4 + \nabla^2 V = 0,$$

[p. 395] where for shortness

$$\begin{aligned}\nabla^2 V = \theta\left[\frac{f^2+\theta}{a^2} - \frac{h^2+\theta}{c^2} + 1\right]\frac{d^2V}{d\alpha^2} + \cdots \\ +\theta\left[\frac{f^2+\theta}{a^2} + \cdots\right]\frac{d^2V}{d\gamma^2} - \frac{\theta\cdot 2q}{e^2}\frac{d^2V}{d\varepsilon^2}\cdots \\ +\frac{2\alpha}{f^2+\theta}\,\frac{d^2V}{d\alpha\,d\beta} - \&c. \\ +\frac{1}{f^2+\theta}\Big(-\frac{\alpha}{f^2+\theta} - \&c.\Big)\frac{dV}{d\alpha} \\ +\frac{1}{\theta}\Big(-\frac{\varepsilon}{\theta} - \&c.\Big)\frac{dV}{d\beta}\end{aligned}$$

$$+\frac{1}{\theta}\Big(-\frac{\varepsilon}{\theta}\Big)\frac{dV}{d\gamma}\,.$$

Also

$$\frac{d^2V}{d\theta^2}=\frac{1}{4(f^2+\theta)}\Big[\frac{f^2+\theta}{a^2}+\cdots+\frac{h^2+\theta}{c^2}+\frac{\theta}{e^2}\Big]\Big[(f^2+\theta)\frac{d^2V}{d\alpha^2}+\cdots+(h^2+\theta)\frac{d^2V}{d\gamma^2}+\theta\,\frac{d^2V}{d\varepsilon^2}\Big]\cdots.$$

116. To integrate the equation for V, we assume

$$V=\Theta\phi,$$

where Θ is a function of θ only, and ϕ a function of $\alpha,\beta,\ldots,\gamma$ (without θ), such that

$$\nabla^2\phi=\kappa\phi,$$

κ being a function of θ only. Assuming that this is possible, the remaining equation to be satisfied is obviously

$$\Theta''+\kappa\Theta=0.$$

Solutions of the form in question are

$$\phi=1,\quad \kappa=0,$$

$$\phi=\alpha,\quad \kappa=\frac{1}{f^2+\theta},\quad \text{or}\quad \phi=\beta,\ldots,\kappa=\frac{1}{g^2+\theta},\ \ldots,\ \frac{1}{h^2+\theta},$$

$$\phi=\alpha\beta,\quad \kappa=\frac{1}{f^2+\theta}+\frac{1}{g^2+\theta},\ \&c.,$$

and the corresponding values of Θ involve θ.
[p. 396] and it can be shown next that there is a solution of the form

$$\phi=\tfrac{1}{2}(A\alpha^2+\cdots+C\gamma^2)+D.$$

117. In fact, assuming that this satisfies $\nabla^2\phi-\kappa\phi=0$, we must have identically

$$\begin{gathered}\frac{A}{f^2+\theta}\Big\{-\alpha^2\frac{\theta}{f^2+\theta}(\beta^2+\cdots+\gamma^2)-\alpha^2\frac{\theta}{f^2+\theta}\,\varepsilon^2+1\Big\}\\ +\frac{B}{g^2+\theta}\Big\{\cdots\Big\}+\cdots\\ +\tfrac{2q}{\varepsilon^2}\Big\{-\alpha^2\frac{\theta}{f^2+\theta}-\beta^2\frac{\theta}{g^2+\theta}-\cdots\Big\}\\ +\frac{A}{f^2+\theta}\Big\{-\alpha^2\tfrac{\theta}{f^2+\theta}-\cdots+\tfrac{2}{f^2+\theta}\Big\}+\cdots\\ +\frac{2q}{\varepsilon^2}\Big\{-\alpha^2\tfrac{\theta}{f^2+\theta}-\cdots\Big\}\\ +\kappa\big\{\tfrac{1}{2}(A\alpha^2+B\beta^2+\cdots+C\gamma^2)+D\big\}=0\,;\end{gathered}$$

so that, from the term in α^2, we have

$$\frac{A}{f^2+\theta}\Big\{-\frac{\theta}{f^2+\theta}(s+2q+2)\Big\}-\frac{B\theta}{(g^2+\theta)(f^2+\theta)}-\cdots-\frac{C\theta}{(h^2+\theta)(f^2+\theta)}+\tfrac{1}{2}\kappa A=0,$$

or, what is the same thing,

$$A\big\{2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta)\big\}+\kappa f^2D=0,$$

with the like equations from $\beta^2,\ldots,\gamma^2$; and from the constant term we have

$$\frac{A}{f^2+\theta}+\frac{B}{g^2+\theta}+\cdots+\frac{C}{h^2+\theta}+\kappa D=0.$$

118. Multiplying this last by f^2, and adding it to the first, we obtain

$$A\Big\{1+\frac{f^2}{f^2+\theta}\Big\}\cdot 1+\frac{Bf^2}{g^2+\theta}+\cdots+\frac{Cf^2}{h^2+\theta}+\tfrac{1}{2}\kappa A(f^2+\theta)+\kappa f^2 D=0;$$

viz. putting for shortness $\Omega=\dfrac{a^2}{f^2+\theta}+\cdots=\dfrac{\theta}{f^2+\theta}\alpha^2+\cdots+\dfrac{\theta}{h^2+\theta}\gamma^2$, this is

$$A\big\{2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta)\big\}+\kappa f^2D=0;$$

and similarly

$$B\big\{2q+2+\Omega+\tfrac{1}{2}\kappa(g^2+\theta)\big\}+\kappa g^2D=0,$$
$$C\big\{2q+2+\Omega+\tfrac{1}{2}\kappa(h^2+\theta)\big\}+\kappa h^2D=0.$$

[p. 397] To these we join the foregoing equation

$$\frac{A}{f^2+\theta}+\frac{B}{g^2+\theta}+\cdots+\frac{C}{h^2+\theta}+\kappa D=0.$$

Eliminating $A, B, \ldots, C, D$, we have an equation which determines κ as a function of θ; and the equations then determine the ratios of $A, B, \ldots, C, D$, so that these quantities will be given as determinate multiples of an arbitrary quantity M. The equation for κ is in fact

$$\frac{f^2(f^2+\theta)}{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)}+\cdots+\frac{h^2(h^2+\theta)}{2q+2+\Omega+\frac{1}{2}\kappa(h^2+\theta)}+1=0;$$

and the values of $A, B, \ldots, C, D$ are then

$$A=\frac{Mf^2}{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)},\quad B=\frac{Mg^2}{2q+2+\Omega+\frac{1}{2}\kappa(g^2+\theta)},\ \ldots,\ C=\frac{Mh^2}{2q+2+\Omega+\frac{1}{2}\kappa(h^2+\theta)},\quad D=-\frac{M}{\kappa},$$

values which seem to be dependent on θ: if they were so, it would be fatal to the success of the process; but they are really independent of θ.

119. That they are independent of θ depends on the theorems; that we have

$$\kappa=\frac{(2q+2+\Omega)\kappa_0}{2q+2-\frac{1}{2}\kappa_0\theta},$$

where κ_0 is a quantity independent of θ determined by the equation

$$\frac{f^2}{2q+2+\frac{1}{2}\kappa_0f^2}+\frac{g^2}{2q+2+\frac{1}{2}\kappa_0g^2}+\cdots+\frac{h^2}{2q+2+\frac{1}{2}\kappa_0h^2}+1=0$$

(κ_0 is in fact the value of κ on writing $\theta=0$): and that, omitting the arbitrary multiplier, the values of $A, B, \ldots, C, D$ then are

$$\frac{f^2}{2q+2+\frac{1}{2}\kappa_0f^2},\quad \frac{g^2}{2q+2+\frac{1}{2}\kappa_0g^2},\ \ldots,\ \frac{h^2}{2q+2+\frac{1}{2}\kappa_0h^2},\quad -\frac{1}{\kappa_0},$$

or, what is the same thing, the value of ϕ is

$$\phi=\frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0f^2}+\frac{g^2\beta^2}{2q+2+\frac{1}{2}\kappa_0g^2}+\cdots+\frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0h^2}-\frac{1}{\kappa_0}.$$

120. To explain the ground of the assumption

$$\kappa=\frac{(2q+2+\Omega)\kappa_0}{2q+2-\frac{1}{2}\kappa_0\theta},$$

observe that, assuming

$$\frac{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)}{2q+2+\frac{1}{2}\kappa_0f^2}=\frac{2q+2+\Omega+\frac{1}{2}\kappa(g^2+\theta)}{2q+2+\frac{1}{2}\kappa_0g^2},$$

[p. 398] then multiplying out and reducing, we obtain

$$\left(g^2 - f^2\right)\left\{\kappa_0(2q+2+\Omega) - (2q+2)\kappa + \tfrac{1}{2}\kappa\kappa_0\theta\right\} = 0;$$

viz. the equation divides out by the factor $g^2 - f^2$, thereby becoming

$$\kappa_0(2q+2+\Omega) - (2q+2)\kappa + \tfrac{1}{2}\kappa\kappa_0\theta = 0,$$

that is, it gives for κ the foregoing value: hence clearly, κ having this value, we obtain by symmetry

$$2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta),\ 2q+2+\Omega+\tfrac{1}{2}\kappa(g^2+\theta),\ \ldots,\ 2q+2+\Omega+\tfrac{1}{2}\kappa(h^2+\theta),$$

proportional to

$$2q+2+\tfrac{1}{2}\kappa_0 f^2,\ 2q+2+\tfrac{1}{2}\kappa_0 g^2,\ \ldots,\ 2q+2+\tfrac{1}{2}\kappa_0 h^2;$$

viz. the ratios, not only of $A:B$, but of $A:B:\cdots:C$ will be independent of θ.

121. To complete the transformation, starting with the foregoing value of κ, we have

$$2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta) = (2q+2+\Omega)\,\frac{2q+2+\frac{1}{2}\kappa_0(f^2+\theta)}{2q+2-\frac{1}{2}\kappa_0\theta},\ \ \&c.;$$

so that we have

$$\begin{aligned} A\left\{2q+2+\tfrac{1}{2}\kappa_0 f^2\right\} + \kappa_0 f^2 D = 0,\\ B\left\{2q+2+\tfrac{1}{2}\kappa_0 g^2\right\} + \kappa_0 g^2 D = 0,\\ C\left\{2q+2+\tfrac{1}{2}\kappa_0 h^2\right\} + \kappa_0 h^2 D = 0,\end{aligned}$$

and

$$\frac{A}{f^2+\theta} + \frac{B}{g^2+\theta} + \cdots + \frac{C}{h^2+\theta} = \frac{(2q+2+\Omega)\kappa_0 D}{2q+2-\frac{1}{2}\kappa_0\theta}.$$

Substituting for $A, B, \ldots, C$ their values, this last becomes

$$\frac{\kappa_0 f^2 D}{2q+2+\frac{1}{2}\kappa_0\theta}\left\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 f^2} - \cdots - \frac{\kappa_0 h^2 D}{h^2+\theta}\right\} + \frac{2q+2}{2q+2-\frac{1}{2}\kappa_0\theta}\left\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{\theta}{h^2+\theta}\right\};$$

viz. this is

$$\left\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 f^2} - \cdots\right\}(2q+2)\kappa_0 D,$$

or, substituting for κ its value, and dividing out by $2q+2$, we have

$$\frac{1}{2q+2+\frac{1}{2}\kappa_0 f^2} + \frac{1}{2q+2+\frac{1}{2}\kappa_0 g^2} + \cdots + \frac{1}{2q+2+\frac{1}{2}\kappa_0 h^2} = -1,$$

the equation for the determination of κ_0.

[p. 399] **122.** The equation for κ_0 is of the order s; there are consequently s functions of the form in question, and each of the terms $\alpha^2, \beta^2, \ldots, \gamma^2$ can be expressed as a linear function of these. It thus appears that any quadric function of $\alpha, \beta, \ldots, \gamma$ can be expressed as a sum of Greenian functions; viz. the form is

$$\begin{aligned} & A\alpha^2 + B\beta^2 + \&c.\\ & + C'\alpha\beta + \&c.\\ & + D'\left(\frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0' f^2} + \frac{g^2\beta^2}{2q+2+\frac{1}{2}\kappa_0' g^2} + \cdots + \frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0' h^2} - \frac{1}{\kappa_0'}\right)\\ & + D''\left(\ ,\quad \ldots,\quad \ldots,\quad \ldots\right)\\ & (s \text{ lines}),\end{aligned}$$

viz. the terms multiplied by D', D'', &c. respectively are those answering to the roots $\kappa_0', \kappa_0'', \ldots$ of the equation in κ_0.

The general conclusion is that any rational and integral function of $\alpha, \beta, \ldots, \gamma$ can be expressed as a sum of Greenian functions.

123. We have next to integrate the equation

$$\Theta'' + \kappa\Theta = 0.$$

Suppose $\kappa = 0$, a particular solution is $\Theta = 1$. Next, suppose

$$\kappa = \frac{1}{f^2+\theta} \cdot \frac{-1}{2q-2} \cdot \frac{-\theta}{f^2+\theta} - \cdots - \frac{-\theta}{h^2+\theta} - 1\,;$$

a particular solution is $\Theta = \sqrt{f^2+\theta}$: in fact, omitting the constant denominator, or writing $\Theta = \sqrt{f^2+\theta}$, and therefore

$$\frac{d\Theta}{d\theta} = \tfrac{1}{2}\,\frac{1}{\sqrt{f^2+\theta}}\,, \qquad \frac{d^2\Theta}{d\theta^2} = -\tfrac{1}{4}\,\frac{1}{(f^2+\theta)^{\frac{3}{2}}}\,,$$

the equation to be verified is

$$-\frac{1}{4}\,\frac{1}{(f^2+\theta)^{\frac{3}{2}}} + \frac{1}{f^2+\theta} \cdot \sqrt{f^2+\theta} \cdot \left\{2q-2-\tfrac{-\theta}{f^2+\theta} - \cdots\right\} = 0.$$

Again, suppose $\kappa = \dfrac{1}{f^2+\theta} \cdot \dfrac{1}{g^2+\theta} + \dfrac{2\theta}{(f^2+\theta)(g^2+\theta)} + \dfrac{1}{f^2+\theta} \cdot \dfrac{1}{g^2+\theta}$&c. (value belonging to $\phi = \alpha\beta$, see No. 116);

a particular solution is $\Theta = \sqrt{f^2+\theta} \cdot \sqrt{g^2+\theta}$: in fact, omitting the constant factor, or writing

[p. 400] and therefore

$$\frac{d\Theta}{d\theta} = \sqrt{\frac{g^2+\theta}{f^2+\theta}} \cdot \frac{1}{2} + \sqrt{\frac{f^2+\theta}{g^2+\theta}} \cdot \frac{1}{2}, \qquad \frac{d^2\Theta}{d\theta^2} = -\frac{1}{4}\,\frac{\sqrt{f^2+\theta}}{(g^2+\theta)^{\frac{3}{2}}} - \frac{1}{4}\,\frac{\sqrt{g^2+\theta}}{(f^2+\theta)^{\frac{3}{2}}}\,,$$

the equation to be verified is

$$-\frac{1}{4}\left\{\frac{\sqrt{g^2+\theta}}{(f^2+\theta)^{\frac{3}{2}}} + \frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}} + \frac{\sqrt{f^2+\theta}}{(g^2+\theta)^{\frac{3}{2}}}\right\}$$

$$+\left(\sqrt{g^2+\theta} + \sqrt{f^2+\theta}\right) \cdot \tfrac{1}{4} = 0,$$

or putting for shortness $\Omega = \dfrac{\theta}{f^2+\theta} + \dfrac{\theta}{g^2+\theta} + \cdots + \dfrac{\theta}{h^2+\theta}$, this is

$$\frac{\theta}{(f^2+\theta)^2} \cdot \sqrt{g^2+\theta} - \frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}} + \frac{\theta}{(g^2+\theta)^2} \cdot \sqrt{f^2+\theta} + \frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}} \{2q+2+\Omega\} \cdot \left(-\sqrt{g^2+\theta} - \sqrt{f^2+\theta}\right) =$$

which is true. And, generally, the particular solution is deduced from the value of ϕ by writing therein

$$\frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}}, \quad \frac{\sqrt{g^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}}, \quad \ldots, \quad \frac{\sqrt{h^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}}$$

in place of $\alpha, \beta, \ldots, \gamma$ respectively: say the value thus obtained is $\Theta = H$, where H is what ϕ becomes by the above substitution.

124. Represent for a moment the equation in Θ by

$$\frac{d^2\Theta}{d\theta^2} + 2P\,\frac{d\Theta}{d\theta} + \kappa\Theta = 0,$$

and assume that this is satisfied by $\Theta = H \int z\, d\theta$. Then we have

$$\frac{d\Theta}{d\theta} = zH + \frac{dH}{d\theta} \int z\, d\theta,$$

$$\frac{d^2\Theta}{d\theta^2} = z\,\frac{dH}{d\theta} + H\,\frac{dz}{d\theta} + \frac{d^2H}{d\theta^2} \int z\, d\theta,$$

$$+\kappa H\int z\,d\theta = 0,$$

[p. 401] and therefore

$$\bigl(H'' + 2PH' + \kappa H\bigr)\int z\,d\theta + 2Hz' + 4PHz = 0;$$

viz. multiplying by $\dfrac{H}{4z}$, this is

$$\frac{H^2 z'}{2z} + 2PH^2 = 0,$$

or

$$\frac{d}{d\theta}\bigl(H^2 z\bigr) + 4HH'z = 0;$$

viz. substituting for P its value, this is

$$\frac{1}{H^2 z}\,\frac{d}{d\theta}\bigl(H^2 z\bigr)\bigl(\tfrac{1}{2}q + 1 - \tfrac{1}{f^2+\theta} - \cdots - \tfrac{1}{h^2+\theta}\bigr)\cdot\tfrac{1}{2} = 0.$$

Hence, integrating,

$$H^2 z = \frac{C\theta^{-q-1}}{\sqrt{f^2+\theta}\cdot\sqrt{g^2+\theta}\cdots\sqrt{h^2+\theta}}, \quad C \text{ an arbitrary constant,}$$

and

$$\Theta = CH\int_\chi^\theta \frac{\theta^{-q-1}\,d\theta}{H^2\sqrt{f^2+\theta}\cdot\sqrt{g^2+\theta}\cdots\sqrt{h^2+\theta}}, \quad \chi \text{ arbitrary,}$$

where the constants of integration are C, χ; or, what is the same thing, taking $\bar{T}$ the same function of t that Θ is of θ (viz. $\bar{T}$ is what H becomes on writing therein

$$\frac{\sqrt{f^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}}, \quad \frac{\sqrt{g^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}}, \quad \ldots, \quad \frac{\sqrt{h^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}}$$

in place of $\alpha, \beta, \ldots, \gamma$ respectively), then

$$\Theta = -CH\int_\theta^\chi \frac{t^{-q-1}\,dt}{T^2\sqrt{f^2+t}\cdot\sqrt{g^2+t}\cdots\sqrt{h^2+t}},$$

where χ may be taken $=\infty$: we thus have

$$V = \Theta\phi = -CH\phi\int_\theta^\infty \frac{t^{-q-1}\,dt}{T^2\sqrt{f^2+t}\cdot\sqrt{g^2+t}\cdots\sqrt{h^2+t}}.$$

Recollecting that

$$1 = \frac{a^2}{f^2+\theta} + \frac{b^2}{g^2+\theta} + \cdots + \frac{c^2}{h^2+\theta} + \frac{e^2}{\theta},$$

so that for $\theta=\infty$ we have $a^2+b^2+\cdots+c^2+e^2=\theta$, the assumption $\chi=\infty$ comes to making V vanish for infinite values of $(a, b, \ldots, c, e)$.

125. We have to find the value of ρ corresponding to the foregoing value of V; viz. W being the value of V, on writing therein $(x, y, \ldots, z)$ in place of $(a, b, \ldots, c)$, then (theorem A)

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)\,e^{2q+1}}\cdot\frac{dW}{de}.$$

[p. 402] Take λ the same function of $(x, y, \ldots, z, e)$ that θ is of $(a, b, \ldots, c, e)$: viz. take λ the positive root of

$$\frac{x^2}{f^2+\lambda} + \frac{y^2}{g^2+\lambda} + \cdots + \frac{z^2}{h^2+\lambda} + \frac{e^2}{\lambda} = 1;$$

and let $(\xi, \eta, \ldots, \zeta, \iota)$ correspond to $(\alpha, \beta, \ldots, \gamma, \varepsilon)$, viz.

$$\xi = \frac{x}{\sqrt{f^2+\lambda}}, \quad \eta = \frac{y}{\sqrt{g^2+\lambda}}, \ \ldots, \ \zeta = \frac{z}{\sqrt{h^2+\lambda}}, \quad \iota = \sqrt{1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda}} = \frac{e}{\sqrt{\lambda}};$$

so that W is the same function of $(\xi, \eta, \ldots, \lambda)$ that V is of $(\alpha, \beta, \ldots, \theta)$; say this is

$$W = -CH \int_\lambda^\infty \bar{T}\, \frac{t^{-q-1}\, dt}{\cdots},$$

then we have for ρ the value

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\, \Gamma(q+1)\, e^{2q+1}} \cdot \left[\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda}\right) \cdot \frac{1}{\sqrt{f^2+\lambda}} \frac{d}{d\xi} + \cdots + \frac{1}{\sqrt{h^2+\lambda}} \frac{d}{d\zeta} + \frac{1}{\sqrt{\lambda}} \frac{d}{d\iota}\right] W,$$

where e is to be put $= 0$.

126. Suppose e is $= 0$; then, if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, λ is not $= 0$, but is the positive root of $\dfrac{x^2}{f^2+\lambda} + \cdots + \dfrac{z^2}{h^2+\lambda} = 1$: $\iota = \sqrt{1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda}} = 0$; and we have $\rho = 0$, viz. ρ is $= 0$ for all points outside the ellipsoid $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1$.

But if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, then, on writing $e = 0$, we have $\lambda = 0$, $\iota^{-1} = \dfrac{1}{\lambda}$,

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\, \Gamma(q+1)} \cdot \frac{e^{2q+1}\sqrt{\lambda}}{\cdots} \left[\frac{x}{\sqrt{f^2+\lambda}}\, \frac{1}{\sqrt{f^2+\lambda}}\, \frac{d\bar{T}}{d\xi} + \cdots + \frac{1}{\sqrt{h^2+\lambda}} \cdot \frac{1}{\sqrt{h^2+\lambda}}\, \frac{d\bar{T}}{d\zeta} + \frac{1}{\sqrt{\lambda}} \cdot \frac{1}{\sqrt{\lambda}}\, \frac{d\bar{T}}{d\iota}\right]_{\lambda=0},$$

where the term in ( ) is

$$= -2\Theta\Lambda_0/(f\ldots h) \cdot \lambda^{q+1}.$$

Hence

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\, \Gamma(q+1)\, \Lambda_0\, fg\ldots h}\, \psi$$

$$= \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\, \Gamma(q+1)\, \Lambda_0\, fg\ldots h}\, \psi\Big(\frac{x}{f}, \frac{y}{g}, \ldots, \frac{z}{h}\Big),$$

[p. 403] where ψ_0, Λ_0 are what ψ, Λ become on writing therein $\lambda = 0$. It will be remembered that Λ is what H becomes on changing therein θ into λ; hence Λ_0 is what H becomes on writing therein $\theta = 0$.

Moreover ψ is what ϕ becomes on changing therein $\alpha, \beta, \ldots, \gamma$ into $\xi, \eta, \ldots, \zeta$: writing $\lambda = 0$, we have $\xi = \dfrac{x}{f}$, $\eta = \dfrac{y}{g}, \ldots, \zeta = \dfrac{z}{h}$; hence ψ_0 is what ϕ becomes on changing therein $\alpha, \beta, \ldots, \gamma$ into $\dfrac{x}{f}, \dfrac{y}{g}, \ldots, \dfrac{z}{h}$. And it is proper in ϕ to restore the original variables by writing $\dfrac{\alpha}{\sqrt{f^2+\theta}}, \dfrac{\beta}{\sqrt{g^2+\theta}}, \ldots, \dfrac{\gamma}{\sqrt{h^2+\theta}}$ in place of $\alpha, \beta, \ldots, \gamma$.

127. Recapitulating,

$$V = \int \frac{\rho\, dx \ldots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

where, since for the value of V about to be mentioned ρ vanishes for points outside the ellipsoid, the integral is to be taken over the ellipsoid

$$\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1;$$

and then, transferring a constant factor, if

$$V = -\frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(q+1)}\, \Lambda_0(f\ldots h)\, H \int_\theta^\infty \bar{T}\, \frac{t^{-q-1}\, dt}{\sqrt{(t+f^2)\ldots(t+h^2)} \cdot \sqrt{t}},$$

the corresponding value of ρ is

$$\rho = \psi_0\Big(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\Big)^q,$$

where Λ_0 is what H becomes on writing therein $\theta=0$, and ψ_0 is what ψ becomes on writing $\frac{x}{f}, \frac{y}{g}, \dots, \frac{z}{h}$ in place of $\alpha, \beta, \dots, \gamma$.

128. Thus, putting for shortness $\bar{H} = t^{-q-1}\{(t+f^2)\dots(t+h^2)\}^{-\frac{1}{2}}$, we have in the three several cases $\phi=1$, $\phi=\alpha$, $\phi=\alpha\beta$ respectively,

$$H=1, \qquad \rho=\left(1-\tfrac{x^2}{f^2}-\dots-\tfrac{z^2}{h^2}\right)^q, \qquad V=\frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(q+1)}(f\dots h)\int_\theta^\infty \bar{H},$$

$$H=\frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\dots+h^2}}, \qquad \dots, \qquad V=\dots\int_\theta^\infty \bar{H},$$

$$H=\frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\dots+h^2}}\cdot\frac{\sqrt{g^2+\theta}}{\sqrt{f^2+g^2+\dots+h^2}}, \qquad \dots, \qquad V=\dots\int_\theta^\infty \frac{\bar{H}\,dt}{\sqrt{f^2+t}\cdot\sqrt{g^2+t}}.$$

[p. 404] For the case last considered

$$\phi=\frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0 f^2}+\dots+\frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0 h^2}-\frac{1}{\kappa_0},$$

$$H=\frac{f^2(f^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 f^2}+\dots+\frac{h^2(h^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 h^2}-\frac{1}{\kappa_0}, \qquad \bar{T} \text{ same function with } t \text{ for } \theta,$$

$$\psi_0=\frac{x^2}{2q+2+\frac{1}{2}\kappa_0 f^2}+\dots+\frac{z^2}{2q+2+\frac{1}{2}\kappa_0 h^2}-\frac{1}{\kappa_0},$$

$$\Lambda_0=\frac{f^4}{2q+2+\frac{1}{2}\kappa_0 f^2}+\dots+\frac{h^4}{2q+2+\frac{1}{2}\kappa_0 h^2}-\frac{1}{\kappa_0},$$

where κ_0 is the root of the equation

$$\frac{1}{2q+2+\frac{1}{2}\kappa_0 f^2}+\dots+\frac{1}{2q+2+\frac{1}{2}\kappa_0 h^2}+1=0.$$

Annex VI. Examples of Theorem C. Art. Nos. 129 to 132.

129. First example: relating to the $(s+1)$-coordinal sphere $a^2+\dots+z^2+w^2=f^2$. Assume

$$V'=\frac{M}{\left(a^2+\dots+c^2+e^2\right)^{\frac{1}{2}(s-1)}}, \qquad V''=\frac{M}{f^{s-1}} \quad \text{(a constant);}$$

these values each satisfy the potential equation.

V' is not infinite for any point outside the surface, and for indefinitely large distances it is of the proper form.

V'' is not infinite for any point inside the surface; and at the surface $V'=V''$. The conditions of the theorem are therefore satisfied. Writing

$$V=\int\frac{\rho\,dS}{\left\{(a-x)^2+\dots+(c-z)^2+(e-w)^2\right\}^{\frac{1}{2}(s-1)}},$$

we have

$$\rho=\frac{1}{4(\sqrt{\pi})^{s+1}}\left(\frac{dV'}{dn'}-\frac{dV''}{dn''}\right),$$

where

$$\frac{dV''}{dn''}=-\left(\tfrac{x}{f}\tfrac{d}{dx}+\dots+\tfrac{z}{f}\tfrac{d}{dz}+\tfrac{w}{f}\tfrac{d}{dw}\right)\frac{M}{(x^2+\dots+z^2+w^2)^{\frac{1}{2}(s-1)}}=\frac{(s-1)M(x^2+\dots+z^2+w^2)^{\frac{1}{2}(s-1)}}{f\cdot(x^2+\dots+z^2+w^2)^{\frac{1}{2}(s+1)}},$$

[p. 405] which at the surface is

$$= \frac{(s-1)M}{f^s}.$$

Hence

$$\rho = \frac{(s-1)\Gamma(\tfrac{1}{2}(s+1))M}{4(\sqrt{\pi})^{s+1}f^s} = \frac{\Gamma(\tfrac{1}{2}s+1)M}{2(\sqrt{\pi})^{s+1}f^s} \quad \text{(viz. } \rho \text{ is constant).}$$

130. Writing for convenience $M = \dfrac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\,\delta f^s\, Sf^s$ (δf^s a constant which may be put $= 1$), also $a^2 + \cdots + c^2 + e^2 = s^2$, we have $\rho = Sf^s$, and consequently

$$\int \frac{\delta f^s\, dS}{\{(a-x)^2 + \cdots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}(s-1)}} = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} \cdot \frac{f^s}{s^{s-1}}, \quad \text{for exterior point } s > f,$$

$$= \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} \cdot f^{s-1}, \quad \text{for interior point } s < f.$$

By making $a, \ldots, c, e$ all indefinitely large, we find

$$\int \delta f^s\, dS = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\, f^s,$$

viz. the expression on the right-hand side is here the mass of the shell thickness δf.

Taking $s = 3$, we have the ordinary formulae for the Potential of a uniform spherical shell.

131. Suppose $s = 3$, but let the surface be the infinite cylinder $x^2 + y^2 = f^2$. Take here

$$V' = M \log \sqrt{a^2 + b^2}, \qquad V'' = M \log f,$$

each satisfying the potential equation $\left(\dfrac{d^2V}{dx^2} + \dfrac{d^2V}{dy^2}\right) = 0$; but V, instead of vanishing, is infinite at infinity, and the conditions of the theorem are not satisfied; the Potential of the cylinder is in fact infinite. But the failure is a mere consequence of the special value of s, viz. this is such that $s - 2$, instead of being positive, is $= 0$. Reverting to the general case of $(s+1)$-dimensional space, let the surface be the infinite cylinder $x^2 + \cdots + z^2 = f^2$; and assume

$$V' = \frac{M}{(a^2 + \cdots + c^2)^{\frac{1}{2}(s-2)}}, \qquad V'' = \frac{M}{f^{s-2}} \quad \text{(a constant).}$$

These satisfy the potential equation; viz. as regards V', we have

[End of pages 381–405 (PDF pp. 401–425). Paper 607 continues in subsequent pages.]

usepackage[a4paper,margin=0.65in]geometry

Arthur Cayley

Collected Mathematical Papers

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Contents

607. A Memoir on Prepotentials (concluded)

[Continued from earlier pages of the same paper. Articles 131 (concl.)–162.]

[p. 406] V' is not infinite at any point outside the cylinder; and it vanishes at infinity, except indeed when only the coordinate e is infinite, and its form at infinity is not

$$= M\big(a^2 + \cdots + c^2 + e^2\big)^{-\frac{1}{2}(s-2)}.$$

V'' is not infinite for any point within the cylinder; and at the surface we have $V' = V''$.

We have

$$\rho = -\frac{\Gamma(\frac{1}{2}s - \frac{1}{2})}{4\,(\Gamma\frac{1}{2})^{s+1}}\Big(\frac{dW'}{da^2} + \frac{dW''}{da'^2}\Big),$$

where

$$\frac{dW'}{da^2} = \frac{-(s-2)\frac{1}{2}(a^2 + \cdots + z^2)\,M}{(a^2 + \cdots + c^2)^{\frac{1}{2}s}}, \qquad \frac{dW''}{da'^2} = \frac{-(s-2)\,M}{f^{s-2}} \text{ at the surface;} \quad \frac{dW''}{da'^2} = 0,$$

and therefore

$$\rho = \frac{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})\,f^{s-1}\,M}{4\,(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}} \quad \text{(viz. } \rho \text{ is constant);}$$

or, what is the same thing, writing $M = \dfrac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})}$, whence $\rho = Bf$, and writing also $a^2 + \cdots + c^2 = \varpi^2$, we have

$$V' = \frac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})} \cdot \frac{1}{\varpi^{s-2}} \quad \text{for an exterior point } \varpi > f,$$

$$V'' = \frac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})} \cdot \frac{1}{f^{s-2}} \quad \text{for an interior point } \varpi < f.$$

132. This is right; but we can without difficulty bring it to coincide with the result obtained for the $(s+1)$-dimensional sphere with only $s-1$ in place of s; we may in fact, by a single integration, pass from the cylinder $x^2 + \cdots + z^2 = f^2$ to the s-dimensional sphere or circle $x^2 + \cdots + z^2 = f^2$, which is the base of this cylinder. Writing first $dS = d\Sigma\,dw$, where $d\Sigma$ refers to the s variables $(x, \ldots, z)$ and the sphere $x^2 + \cdots + z^2 = f^2$; or using now dS in this sense, then in place of the original dS we have $d\Sigma\,dw$: and w being ∞, the limits of w being $\pm\infty$, then in place of $e - w$ we may write simply w. This being so, and putting for shortness $(a-x)^2 + \cdots + (c-z)^2 = A^2$, the integral is

$$V = \iint \frac{\delta f\,dS\,dw}{(A^2 + w^2)^{\frac{1}{2}(s-2)}},$$

and we have without difficulty

$$\int_{-\infty}^{+\infty} \frac{dw}{(A^2+w^2)^{\frac{1}{2}(s-2)}} = \frac{1}{A^{s-3}} \cdot \frac{\sqrt{\pi}\,\Gamma\frac{1}{2}(s-2)}{\Gamma\frac{1}{2}(s-1)}.$$

[p. 407] To prove it, write $w = A\tan\theta$, then the integral is in the first place converted into

$$\frac{2}{A^{s-3}} \int_0^{\frac{\pi}{2}} \cos^{s-4}\theta\, d\theta,$$

which, putting $\cos\theta = \sqrt{x}$ and therefore $\sin\theta = \sqrt{1-x}$, becomes

$$= \frac{1}{A^{s-3}} \int_0^1 x^{\frac{1}{2}s-\frac{5}{2}}\,(1-x)^{-\frac{1}{2}}\,dx,$$

which has the value in question.

Hence, replacing A by its value, we have

$$V = \frac{\sqrt{\pi}\,\Gamma\frac{1}{2}(s-2)}{\Gamma\frac{1}{2}(s-1)} \int \frac{\delta f\, dS}{\{(a-x)^2+\cdots+(c-z)^2\}^{\frac{1}{2}(s-3)}} = \frac{4\pi^{\frac{1}{2}s}\,\Gamma(\frac{1}{2})\,f^{s-1}\rho}{(s-2)\,\Gamma\frac{1}{2}(s-2)} \cdot \frac{1}{(a^2+\cdots+c^2)^{\frac{1}{2}(s-3)}} \quad \text{or}$$

that is,

$$V = \frac{4(\Gamma\frac{1}{2})^s\, f^{s-1}\,\rho}{(s-3)\,\Gamma\frac{1}{2}(s-3)} \cdot \frac{1}{(a^2+\cdots+c^2)^{\frac{1}{2}(s-3)}} \quad \text{or} \quad \frac{1}{f^{s-3}};$$

viz. this is the formula for the sphere with $s-1$ instead of s.

Annex VII. Example of Theorem D, Art. Nos. 133 and 134.

133. The example relates to the $(s+1)$-dimensional sphere $x^2+\cdots+z^2+w^2=f^2$.

Instead of at once assuming for V a form satisfying the proper conditions as to continuity, we assume a form with indeterminate coefficients, and make it satisfy the conditions in question. Write

$$V = \frac{M}{(a^2+\cdots+c^2+e^2)^{\frac{1}{2}(s-1)}} \qquad \text{for } a^2+\cdots+c^2+e^2 > f^2;$$

$$V = A(a^2+\cdots+c^2+e^2)+B \qquad \text{for } a^2+\cdots+c^2+e^2 < f^2.$$

In order that the two values may be equal at the surface, we must have

$$\frac{M}{f^{s-1}} = Af^2+B;$$

in order that the derived functions $\dfrac{dV}{da}$, &c. may be equal, we must have

$$\frac{-(s-1)\,a\,M}{f^{s+1}} = 2Aa, \quad \&c.,$$

viz. these are all satisfied if only $\dfrac{-(s-1)\,M}{f^{s+1}} = 2A$.

We have thus the values of A and B; or the exterior potential being as above

$$V = \frac{M}{(a^2+\cdots+c^2+e^2)^{\frac{1}{2}(s-1)}},$$

[p. 408] the value of the interior potential must be

$$V = -\frac{M}{f^{s+1}}\Big\{(\tfrac{1}{2}s+\tfrac{1}{2}) - (\tfrac{1}{2}s-\tfrac{1}{2})\,\frac{a^2+\cdots+c^2+e^2}{f^2}\Big\}.$$

The corresponding values of W are of course

$$\frac{M}{(a^2+\cdots+c^2+e^2+w^2)^{\frac{1}{2}(s-1)}} \quad \text{and} \quad \frac{M}{f^{s+1}}\Big\{(\tfrac{1}{2}s+\tfrac{1}{2})-(\tfrac{1}{2}s-\tfrac{1}{2})\,\frac{a^2+\cdots+c^2+e^2+w^2}{f^2}\Big\};$$

and we thence find

$$\rho = 0, \qquad \text{if } a^2+\cdots+c^2+w^2 > f^2,$$

$$\rho = -\frac{\Gamma(\frac{1}{2}s-\frac{1}{2})}{4\,(\Gamma\frac{1}{2})^{s+1}}\Big\{-4(\tfrac{1}{2}s-\tfrac{1}{2})\,(\tfrac{1}{2}s+\tfrac{1}{2})\,\frac{M}{f^{s+3}}\Big\} = \frac{\Gamma(\frac{1}{2}s+\frac{1}{2})}{(\Gamma\frac{1}{2})^{s+1}}\cdot\frac{M}{f^{s+3}},$$
$$\text{if } a^2+\cdots+c^2+w^2 < f^2.$$

Assuming for M the value $\dfrac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\,f^{s+1}$, the last value becomes $\rho = 1$; writing for shortness $a^2+\cdots+c^2+e^2 = \kappa^2$, we have

$$V = \int \frac{dx\ldots dz\,dw}{[(a-x)^2+\cdots+(c-z)^2+(e-w)^2]^{\frac{1}{2}(s-1)}} \text{ over the } (s+1)\text{-dim. sphere } x^2+\cdots+z^2+w^2 = f^2,$$

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\cdot\frac{f^{s+1}}{\kappa^{s-1}}, \quad \text{for an exterior point } \kappa > f,$$

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s+\frac{1}{2})}\big\{(\tfrac{1}{2}s+\tfrac{1}{2})f^2 - (\tfrac{1}{2}s-\tfrac{1}{2})\,\kappa^2\big\}, \quad \text{for an interior point } \kappa < f.$$

134. The case of the ellipsoid $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2} = 1$ for $s+1$-dimensional space may be worked out by the theorem; this is, in fact, what is done in tridimensional space by Lejeune-Dirichlet in his Memoir of 1846 above referred to (p. 321).

Annex VIII. Prepotentials of the Homaloids. Art. Nos. 135 to 137.

135. We have in tridimensional space the series of figures—the plane, the line, the point; and there is in like manner in $(s+1)$-dimensional space a corresponding series of $(s+1)$ terms; the s-coordinal plane,... the $(s-1)$-coordinal plane,... the line, the point: say these are the homaloids or homaloidal figures. And, taking the density as uniform, or, what is the same thing, $=1$, we may consider the prepotentials of these several figures in regard to an attracted point, which, for greater simplicity, is taken not to be on the figure.

136. The integral may be written

$$V = \int \frac{dw \dots dt}{\{(a-x)^2 + \dots + (c-z)^2 + (d-w)^2 + \dots + (e-t)^2 + u^2\}^{\frac{1}{2}s+q}},$$

which still relates to a $(s+1)$-dimensional space: the $(s+1)$ coordinates of the attracted point are $(a, \dots, c, d, \dots, e, u)$, instead of being $(a, \dots, c, e)$; viz. we have the

[p. 409] s' coordinates $(a, \dots, c)$, the $s-s'$ coordinates $(d, \dots, e)$, and the $(s+1)$th coordinate u: and the integration is extended over the $(s-s')$-dimensional figure $w = -\infty$ to $+\infty, \dots, t = -\infty$ to $+\infty$. And it is also assumed that q is positive.

It is at once clear that we may reduce the integral to

$$V = \int \frac{dw \dots dt}{\{(a-x)^2 + \dots + (c-z)^2 + u^2 + w^2 + \dots + t^2\}^{\frac{1}{2}s+q}},$$

say for shortness

$$= \int \frac{dw \dots dt}{(A^2 + w^2 + \dots + t^2)^{\frac{1}{2}s+q}},$$

where $A^2 = (a-x)^2 + \dots + (c-z)^2 + u^2$, is a constant as regards the integration, and where the limits in regard to each of the $s-s'$ variables are $-\infty$, $+\infty$.

We may for these variables write $r\xi, \dots, r\zeta$, where $\xi^2 + \dots + \zeta^2 = 1$; and we then have $dw \dots dt = r^{s-s'-1}\, dr\, dS$, where dS is the element of surface of the $(s-s')$-coordinal unit-sphere $\xi^2 + \dots + \zeta^2 = 1$. We thus obtain

$$V = \int \frac{r^{s-s'-1}\, dr}{(A^2 + r^2)^{\frac{1}{2}s+q}} \int dS,$$

where the integral in regard to r is taken from 0 to ∞, and the integral $\int dS$ over the surface of the unit-sphere; hence by Annex I. the value of this last factor is $\dfrac{2(\Gamma\frac{1}{2})^{s-s'}}{\Gamma\frac{1}{2}(s-s')}$. The integral represented by the first factor will be finite, provided $\frac{1}{2}(s-s')$ is positive; which is the case for any value whatever of s', if only q be positive.

The first factor is an integral such as is considered in Annex II.; to find its value we have only to write $r = Ax$, and we thus find it to be

$$\frac{1}{(A^2)^{\frac{1}{2}s'+q}} \cdot \frac{1}{A} \int_0^\infty \frac{x^{s-s'-1}\,dx}{(1+x)^{\frac{1}{2}s+q}} = \frac{1}{A^{s'+2q}} \cdot \frac{1}{2}\,\frac{\Gamma\frac{1}{2}(s-s')\,\Gamma(\frac{1}{2}s'+q)}{\Gamma(\frac{1}{2}s+q)},$$

and we thus have

$$V = \frac{(\Gamma\frac{1}{2})^{s-s'}\,\Gamma(\frac{1}{2}s'+q)}{\Gamma(\frac{1}{2}s+q)} \cdot \frac{1}{\{(a-x)^2+\cdots+(c-z)^2+u^2\}^{\frac{1}{2}s'+q}}.$$

137. As a verification, observe that the prepotential equation $\square V = 0$, that is,

$$\Big(\frac{d^2}{da^2}+\cdots+\frac{d^2}{dc^2}+\frac{d^2}{dd^2}+\cdots+\frac{d^2}{de^2}+\frac{d^2}{du^2}+\frac{2q+1}{u}\,\frac{d}{du}\Big)V = 0,$$

for a function V, which contains only the $s'+1$ variables $(a, \dots, c, u)$, becomes

$$\Big(\frac{d^2}{da^2}+\cdots+\frac{d^2}{dc^2}+\frac{d^2}{du^2}+\frac{2q+1}{u}\,\frac{d}{du}\Big)V = 0,$$

which is satisfied by V, a constant multiple of $\big\{(a-x)^2+\cdots+(c-z)^2+u^2\big\}^{-\frac{1}{2}s'-q}$.

Annex IX. The Gauss–Jacobi Theory of Epispheric Integrals. Art. No. 138.

[p. 410] **138.** The formula obtained (Annex IV. No. 110) is proved only for positive values of m; but writing therein $q = 0$, $m = -\frac{1}{2}$, it becomes

$$\int \sqrt{1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}}\;\frac{dx\dots dz}{\{(a-x)^2+\cdots+(c-z)^2+e^2\}^{\frac{1}{2}s}}$$

$$= \frac{(\Gamma\frac{1}{2})^s}{\Gamma\frac{1}{2}s}\,f\dots h\int_0^\infty t^{-\frac{1}{2}}\Big(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\frac{e^2}{t}\Big)^{-\frac{1}{2}}\{(t+f^2)\dots(t+h^2)\}^{-\frac{1}{2}},$$

a formula which is obtainable as a particular case of the more general formula

$$V = \int \frac{dS}{\{(*)(x,\dots,z,w)^2\}^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_{-A}^\infty \frac{dt}{\sqrt{-\,\text{Disct.}\,.\big[(*)(X,\dots,Z,W,T)^2+t(X^2+\cdots+Z^2+W^2-}}$$

(notation to be presently explained), being a result obtained by Jacobi by a process which is in fact the extension to any number of variables of that used by Gauss[1] in his Memoir "Determinatio attractionis quam exerceret planeta, &c." (1818). I proceed to develop this theory.

139. Jacobi's process has reference to a class of s-tuple integrals (including some of those here previously considered) which may be termed "epispheric": viz. considering the $(s+1)$ variables $(x, \dots, z, w)$ connected by the equation $x^2 + \dots + z^2 + w^2 = 1$, or say they are the coordinates of a point on a $(s+1)$-tuple unit-sphere, then the form is $\int U\, dS$, where dS is the element of the surface of the unit-sphere, and U is any function of the $s+1$ coordinates; the integral is taken to be of the form $\int \dfrac{dS}{\{(*\between x, \dots, z, w, 1)^2\}^{\frac{1}{2}s}}$, and we then obtain the general result above referred to.

Before going further it is convenient to remark that, taking as independent variables the s coordinates $x, \dots, z$, we have $dS = \dfrac{dx \dots dz}{w}$, where w stands for $\pm\sqrt{1 - x^2 - \dots - z^2}$; we must in obtaining the integral take account of the two values of w, and finally extend the integral to the values of $x, \dots, z$ which satisfy $x^2 + \dots + z^2 \leq 1$.

If, as is ultimately done, in place of $x, \dots, z$ we write $\dfrac{x}{f}, \dots, \dfrac{z}{h}$ respectively, then the value of $dS = \dfrac{1}{f \dots h} \cdot \dfrac{dx \dots dz}{w}$, where w now stands for $\pm\sqrt{1 - \dfrac{x^2}{f^2} - \dots - \dfrac{z^2}{h^2}}$; we must, in finding the value of the integral, take account of the two values of w, and finally extend the integral to the values of $x, \dots, z$ which satisfy $\dfrac{x^2}{f^2} + \dots + \dfrac{z^2}{h^2} \leq 1$.

[p. 411] **140.** The determination of the integral depends upon formulae for the transformation of the spherical element dS, and of the quadric function $(x, y, \dots, z, w, 1)^2$.

First, as regards the spherical element dS; let the $s+1$ variables $x, y, \dots, z, w$ which satisfy $x^2 + y^2 + \dots + z^2 + w^2 = 1$ be regarded as functions of the s independent variables $\theta, \phi, \dots, \psi$; then we have

$$dS = \frac{1}{w} \begin{vmatrix} \dfrac{dx}{d\theta} & \dfrac{dy}{d\theta} & \cdots & \dfrac{dz}{d\theta} & \dfrac{dw}{d\theta} \\ \dfrac{dx}{d\phi} & \dfrac{dy}{d\phi} & \cdots & \dfrac{dz}{d\phi} & \dfrac{dw}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \dfrac{dx}{d\psi} & \dfrac{dy}{d\psi} & \cdots & \dfrac{dz}{d\psi} & \dfrac{dw}{d\psi} \end{vmatrix} d\theta\, d\phi \dots d\psi = \frac{\partial(x, y, \dots, z, w)}{\partial(\theta, \phi, \dots, \psi, *)}\, d\theta\, d\phi \dots d\psi, \text{ for shortness.}$$

[1][*Ges. Werke*, t. iii, pp. 331–355.]

Suppose we effect on the $s+1$ variables $(x, y, \dots, z, w)$ a transformation

$$x,\ y,\ \dots,\ z,\ w = \frac{X}{T},\ \frac{Y}{T},\ \dots,\ \frac{Z}{T},\ \frac{W}{T},$$

thus introducing for the moment $s+2$ variables $X, Y, \dots, Z, W, T$, which satisfy identically $X^2+Y^2+\dots+Z^2+W^2-T^2=0$; then, considering these as functions of the foregoing s independent variables $\theta, \phi, \dots, \psi$, we have

$$dS = \frac{1}{2^{s+1}}\,\frac{W}{T^{s+2}} \cdot \frac{\partial(X, Y, \dots, Z, W)}{\partial(\theta, \phi, \dots, \psi, *)}\, d\theta\, d\phi \dots d\psi.$$

141. Considering next the $s+2$ variables $X, Y, \dots, Z, W, T$ as linear functions (with constant terms) of the $s+1$ new variables $\xi, \eta, \dots, \zeta, \omega$, or say as linear functions of the $s+2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$: which implies between them a linear relation

$$aX + bY + \dots + cZ + dW + eT = 1;$$

and assuming that we have identically

$$X^2+Y^2+\dots+Z^2+W^2-T^2 = \xi^2+\eta^2+\dots+\zeta^2+\omega^2-1,$$

so that, in consequence of the left-hand side being $=0$, the right-hand side is also $=0$; viz. $\xi, \eta, \dots, \zeta, \omega$ are connected by

$$\xi^2+\eta^2+\dots+\zeta^2+\omega^2 = 1:$$

[p. 412] let $d\Sigma$ represent the spherical element belonging to the coordinates $\xi, \eta, \dots, \zeta, \omega$. Considering these as functions of the foregoing s independent variables $\theta, \phi, \dots, \psi$, we have

$$d\Sigma = \frac{1}{\omega}\begin{vmatrix} \frac{d\xi}{d\theta} & \frac{d\eta}{d\theta} & \dots & \frac{d\zeta}{d\theta} & \frac{d\omega}{d\theta} \\ \frac{d\xi}{d\phi} & \frac{d\eta}{d\phi} & \dots & \frac{d\zeta}{d\phi} & \frac{d\omega}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{d\xi}{d\psi} & \frac{d\eta}{d\psi} & \dots & \frac{d\zeta}{d\psi} & \frac{d\omega}{d\psi} \end{vmatrix} d\theta\, d\phi \dots d\psi = \frac{\partial(\xi, \eta, \dots, \zeta, \omega)}{\partial(\theta, \phi, \dots, \psi, *)}\, d\theta\, d\phi \dots d\psi.$$

142. In this expression we have $\xi, \eta, \dots, \zeta, \omega$, each of them a linear function of the $s+2$ quantities $X, Y, \dots, Z, W, T$; the determinant is consequently a linear function of $s+2$ like determinants obtained by substituting for the variables any $s+1$ out of the $s+2$ variables $X, Y, \dots, Z, W, T$; but in virtue of the equation

$$X^2+Y^2+\dots+Z^2+W^2-T^2 = 0,$$

these $s+2$ determinants are proportional to the quantities $X, Y, \dots, Z, W, T$ respectively, and the determinant thus assumes the form

$$\frac{aX + bY + \cdots + cZ + dW + eT}{T} \Delta,$$

where Δ is the like determinant with $(X, Y, \dots, Z, W)$, and where the coefficients $a, b, \dots, c, d, e$ are precisely those of the linear relation $aX + bY + \cdots + cZ + dW + eT = 1$; the last-mentioned expression is thus $\frac{1}{T}\Delta$, or, substituting for Δ its value, we have

$$d\Sigma = \frac{1}{T} \frac{\partial(X, Y, \dots, Z, W)}{\partial(\theta, \phi, \dots, \psi, *)} \, d\theta \, d\phi \dots d\psi;$$

viz. comparing with the foregoing expression for dS we have

$$dS = \frac{1}{T^{s+1}} \, d\Sigma,$$

which is the requisite formula for the transformation of dS.

143. Consider the integral

$$\int \frac{dS}{\{(*\rangle x, y, \dots, z, w, 1)^2\}^{\frac{1}{2}s}},$$

which, from its containing a single quadric function, may be called "one-quadric." Then effecting the foregoing transformation,

$$x,\ y,\ \dots,\ z,\ w = \frac{X}{T},\ \frac{Y}{T},\ \dots,\ \frac{Z}{T},\ \frac{W}{T},$$

[p. 413] and observing that

$$(*\rangle x, y, \dots, z, w, 1)^2 = \frac{1}{T^2} (*\rangle X, Y, \dots, Z, W, T)^2,$$

the integral becomes

$$= \int \frac{d\Sigma}{\{(*\rangle X, Y, \dots, Z, W, T)^2\}^{\frac{1}{2}s}},$$

where $X, Y, \dots, Z, W, T$ denote given linear functions (with constant coefficients) of the $s+1$ variables $\xi, \eta, \dots, \zeta, \omega$, or, what is the same thing, given linear functions of the $s+2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$, such that identically

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1.$$

We have then $\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1 = 0$, and $d\Sigma$ as the corresponding spherical element.

144. We may have $X, Y, \dots, Z, W, T$ such linear functions of $\xi, \eta, \dots, \zeta, \omega, 1$ that not only

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1$$

as above, but also

$$(*\between X, Y, \dots, Z, W, T)^2 = A\xi^2 + B\eta^2 + \cdots + C\zeta^2 + E\omega^2 - L;$$

this being so, the integral becomes

$$\int \frac{d\Sigma}{[A\xi^2 + B\eta^2 + \cdots + C\zeta^2 + E\omega^2 - L]^{\frac{1}{2}s}},$$

where the $s+2$ coefficients $A, B, \dots, C, E, L$ are given by means of the identity

$$-(\theta+A)(\theta+B)\dots(\theta+C)(\theta+E)(\theta+L) = \text{Disct.}\,.\big[(*\between X, Y, \dots, Z, W, T)^2 + \theta(X^2+Y^2+\cdots+Z^2+W^2-T^2)$$

viz. equating the discriminant to zero, we have an equation in θ, the roots whereof are $-A, -B, \dots, -C, -E, -L$.

The integral is

$$\int \frac{d\Sigma}{\big[(A-L)\xi^2 + (B-L)\eta^2 + \cdots + (C-L)\zeta^2 + (E-L)\omega^2\big]^{\frac{1}{2}s}}$$

which is of the form

$$\int \frac{d\Sigma}{\big[a\xi^2 + b\eta^2 + \cdots + c\zeta^2 + e\omega^2\big]^{\frac{1}{2}s}},$$

where I provisionally assume that $a, b, \dots, c, e$ are all positive.

145. To transform this, in place of the $s+1$ variables $\xi, \eta, \dots, \zeta, \omega$ connected by $\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 = 1$, we introduce the $s+1$ variables $x, y, \dots, z, w$, such that

$$x = \frac{\xi\sqrt{a}}{\rho}, \quad y = \frac{\eta\sqrt{b}}{\rho}, \quad \dots, \quad z = \frac{\zeta\sqrt{c}}{\rho}, \quad w = \frac{\omega\sqrt{e}}{\rho},$$

[p. 414] where

$$\rho^2 = a\xi^2 + b\eta^2 + \cdots + c\zeta^2 + e\omega^2,$$

and consequently

$$x^2 + y^2 + \cdots + z^2 + w^2 = 1.$$

Hence, writing dS to denote the spherical element corresponding to the point $(x, y, \dots, z, w)$, we have, by a former formula,

$$dS = \frac{1}{w}\,\frac{\partial(x, y, \dots, z, \omega\sqrt{e}/\rho)}{\partial(\theta, \phi, \dots, \psi, *)}\,d\theta\,d\phi\dots d\psi = \frac{(ab\dots ce)^{\frac{1}{2}}}{\rho^{s+1}}\,d\Sigma,$$

or, what is the same thing,

$$d\Sigma = \frac{\rho^{s+1}}{(ab\dots ce)^{\frac{1}{2}}}\,dS.$$

Hence, integrating each side, and observing that $\int dS$, taken over the whole spherical surface $x^2 + y^2 + \dots + z^2 + w^2 = 1$, is $= \dfrac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}$, we have

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s+\frac{1}{2}}} = \frac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}\cdot\frac{1}{(ab\dots ce)^{\frac{1}{2}}}.$$

146. For $a, b, \dots, c, e$ write herein $a+\theta, b+\theta, \dots, c+\theta, e+\theta$ respectively, and multiply each side by θ^{q-1}, where q is any positive integer or fractional number less than $\frac{1}{2}s$: integrate from $\theta = 0$ to $\theta = \infty$. On the left-hand side, attending to the relation $\xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 = 1$, the integral in regard to θ is

$$\int_0^\infty \frac{\theta^{q-1}\,d\theta}{(\rho^2 + \theta)^{\frac{1}{2}(s+1)}},$$

where $\rho^2 = a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2$, is independent of θ as before; the value of the definite integral is

$$\frac{\Gamma\{\frac{1}{2}(s+1) - q\}\,\Gamma(q)}{\Gamma\frac{1}{2}(s+1)}\cdot\frac{1}{\rho^{s+1-2q}},$$

which, replacing ρ by its value and multiplying by $d\Sigma$, and prefixing the integral sign, gives the left-hand side; hence, forming the equation and dividing by a numerical factor, we have

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s+\frac{1}{2}-q}} = \frac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma\frac{1}{2}(s+1-q)\,\Gamma(q)}\cdot\frac{1}{(ab\dots ce)^{\frac{1}{2}}}\int_0^\infty dt\,t^{q-1}\bigl[(t+a)\dots(t+c)(t+e)\bigr]$$

In particular, if $q = \frac{1}{2}$, then

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^{s}}{\Gamma(\frac{1}{2}s)}\int_0^\infty dt\bigl[(t + a)\dots(t + c)(t + e)\bigr]^{-\frac{1}{2}};$$

[p. 415] or, if for $a, \ldots, c, e$ we restore the values $A-L, \ldots, C-L, E-L$, then

$$\int \frac{d\Sigma}{\left[(A-L)\xi^2 + \cdots + (C-L)\zeta^2 + (E-L)\omega^2\right]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-L}^{\infty} dt\left[(t+A)\ldots(t+C)(t+E)(t+L)\right]^{-\frac{1}{2}},$$

viz. we thus have

$$\int \frac{dS}{\left[(*\rangle x, \ldots, z, w, 1)^2\right]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-L}^{\infty} dt\left[(t+A)\ldots(t+C)(t+E)(t+L)\right]^{-\frac{1}{2}},$$

where $(t+A)\ldots(t+C)(t+E)(t+L)$ is in fact a given rational and integral function of t; viz. it is

$$= -\,\text{Disct.}\,.\left[(*\rangle X, \ldots, Z, W, T)^2 + t(X^2 + \cdots + Z^2 + W^2 - T^2)\right].$$

147. Consider, in particular, the integral

$$\int \frac{dS}{\left[(a-fx)^2 + \cdots + (c-hz)^2 + (e-kw)^2 + l^2\right]^{\frac{1}{2}s}};$$

here

$$\begin{aligned}(aT - fX)^2 &+ \cdots + (cT - hZ)^2 + (eT - kW)^2 + l^2T^2 + t(X^2 + \cdots + Z^2 + W^2 - T^2)\\ &= (f^2+t)X^2 + \cdots + (h^2+t)Z^2 + (k^2+t)W^2 + (a^2 + \cdots + c^2 + e^2 + l^2 - t)T^2\\ &\quad - 2afXT - \cdots - 2chZT - 2ekWT;\end{aligned}$$

viz. the discriminant taken negatively is

$$\begin{vmatrix} t+f^2 & \cdots & \cdots & -af \\ \vdots & \ddots & & \vdots \\ \cdots & & t+k^2 & -ek \\ -af & \cdots & -ek & -(a^2+\cdots+c^2+e^2+l^2)+t \end{vmatrix},$$

which is

$$= (t+A)\ldots(t+C)(t+E)(t+L);$$

and consequently $-A, \ldots, -C, -E, -L$ are the roots of the equation

$$1 - \frac{a^2}{t+f^2} - \cdots - \frac{c^2}{t+h^2} - \frac{e^2}{t+k^2} - \frac{l^2}{t} = 0.$$

148. The roots are all real; moreover there is one and only one positive root. Hence, taking $-L$ to be the positive root, we have $A, \ldots, C, E, L$ all positive, and therefore *à fortiori* $A-L, \ldots, C-L, E-L$ all positive: which agrees with a foregoing

[p. 416] provisional assumption. Or, writing for greater convenience θ to denote the positive quantity $-L$, that is, taking θ to be the positive root of the equation

$$1-\frac{a^2}{\theta+f^2}-\cdots-\frac{c^2}{\theta+h^2}-\frac{e^2}{\theta+k^2}-\frac{l^2}{\theta}=0,$$

we have

$$\int\frac{dS}{[(a-fx)^2+\cdots+(c-hz)^2+(e-kw)^2+l^2]^{\frac{1}{2}s}}=\frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_\theta^\infty dt\left(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\frac{e^2}{t+k^2}-\right.$$

or, what is the same thing, we have

$$\frac{1}{f\ldots h\,k}\int\frac{dx\ldots dz}{w[(a-x)^2+\cdots+(c-z)^2+(e-kw)^2+l^2]^{\frac{1}{2}s}}=\frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_\theta^\infty dt\left(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\right.$$

where on the left-hand side w now denotes $\sqrt{1-\dfrac{x^2}{f^2}-\cdots-\dfrac{z^2}{h^2}}$, and the limiting equation is $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}=1$.

149. Suppose $l=0$: then, if

$$\frac{a^2}{f^2}+\cdots+\frac{c^2}{h^2}+\frac{e^2}{k^2}>1,$$

the equation

$$1-\frac{a^2}{\theta+f^2}-\cdots-\frac{c^2}{\theta+h^2}-\frac{e^2}{\theta+k^2}=0$$

has a positive root differing from zero, which may be represented by the same letter θ; but if

$$\frac{a^2}{f^2}+\cdots+\frac{c^2}{h^2}+\frac{e^2}{k^2}<1,$$

then the positive root of the original equation becomes $=0$; viz. as l gradually diminishes to zero, the positive root also diminishes and becomes ultimately zero.

Hence, writing $l=0$, we have

$$\int\frac{dS}{[(a-fx)^2+\cdots+(c-hz)^2+(e-kw)^2]^{\frac{1}{2}s}},$$

or, what is the same thing,

$$\frac{1}{f\ldots h\,k}\int\frac{dx\ldots dz}{w[(a-x)^2+\cdots+(c-z)^2+(e-kw)^2]^{\frac{1}{2}s}},$$

[p. 417] θ now denoting either the positive root of the equation

$$1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} = 0,$$

or else 0, according as

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1 \text{ or } < 1.$$

In the case $\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} < 1$, the inferior limit being then 0, this is, in fact, Jacobi's theorem (*Crelle*, t. xii. p. 69, 1834); but Jacobi does not consider the general case where l is not $= 0$, nor does he give explicitly the formula in the other case

$$l = 0,\ \frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1.$$

150. Suppose $k = 0$, e being in the first instance not $= 0$: then the former alternative holds good; and observing, in regard to the form which contains w in the denominator, that we can now take account of the two values by simply multiplying by 2, we have

$$\int \frac{dS}{[(a-fx)^2 + \cdots + (c-hz)^2 + e^2]^{\frac{1}{2}s}} = \frac{2}{f\dots h} \int \frac{dx\dots dz}{[(a-x)^2 + \cdots + (c-z)^2 + e^2]^{\frac{1}{2}s}}$$

(where on the right-hand side w denotes $\sqrt{1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}}$, and the limiting equation being $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1$); each $= \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_\theta^\infty dt \left(1 - \frac{a^2}{t+f^2} - \cdots - \frac{c^2}{t+h^2} - \frac{e^2}{t}\right)^{-\frac{1}{2}} [t(t+f^2)\dots(t+h^2)]^{-\frac{1}{2}}$, where θ is here the positive root of the equation $1 - \frac{a^2}{\theta+f^2} - \cdots - \frac{c^2}{\theta+h^2} - \frac{e^2}{\theta} = 0$, which is the formula referred to at the beginning of the present Annex. We may in the formula write $e = 0$, thus obtaining the theorem under two different forms for the cases

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} > 1 \text{ and } < 1 \quad \text{respectively.}$$

Annex X. Methods of Lejeune-Dirichlet and Boole. Art. Nos. 151 to 162.

151. The notion, that the density ρ is a discontinuous function vanishing for points outside the attracting mass, has been made use of in a different manner by Lejeune-Dirichlet (1839) and Boole (1857): viz. supposing that ρ has a given value $= f(x, \dots, z)$ within a given closed surface S and is $= 0$ outside the surface, these geometers in the expression of a potential or prepotential integral replace ρ by a definite integral which possesses the discontinuity in question, viz. it is $= f(x, \dots, z)$ for points inside

[p. 418] the surface and $= 0$ for points outside the surface; and then in the potential or prepotential integral they extend the integration over the whole of infinite space, thus getting rid of the equation of the surface as a limiting equation for the multiple integral.

152. Lejeune-Dirichlet's paper "Sur une nouvelle méthode pour la détermination des intégrales multiples" is published in *Comptes Rendus*, t. viii. pp. 155–160 (1839), and *Liouville*, t. iv. pp. 164–168 (same year). The process is applied to the form

$$-\frac{1}{p-1}\,\frac{d}{da}\int \frac{dx\,dy\,dz}{\{(a-x)^2+(b-y)^2+(c-z)^2\}^{\frac{1}{2}(p-1)}},$$

taken over the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$; but it would be equally applicable to the triple integral itself, or say to the s-tuple integral

$$\int \frac{dx \dots dz}{\{(a-x)^2+\dots+(c-z)^2\}^{\frac{1}{2}s+q}}$$

or, indeed, to

$$\int \frac{dx \dots dz}{\{(a-x)^2+\dots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\dfrac{x^2}{f^2}+\dots+\dfrac{z^2}{h^2}=1$; but it may be as well to attend to the first form, as more resembling that considered by the author.

153. Since $\dfrac{2}{\pi}\int_0^\infty \dfrac{\sin\lambda\phi}{\phi}\cos\lambda\phi\,d\phi$ is $= 1$ or 0, according as λ is < 1 or > 1, it follows that the integral is equal to the real part of the following expression,

$$\frac{2}{\pi}\int_0^\infty \frac{d\phi}{\phi}\,e^{i\phi}\int \frac{e^{-i\phi(x^2+\dots+z^2)}\,dx \dots dz}{\{(a-x)^2+\dots+(c-z)^2\}^{\frac{1}{2}s+q}},$$

where the integrations in regard to $x, \dots, z$ are now to be extended from $-\infty$ to $+\infty$ for each variable. A further transformation is necessary: since

$$\int_0^\infty e^{-x\tau}\,\tau^{r-1}\,d\tau = \frac{\Gamma(r)}{x^r} \quad (x \text{ positive, and } r \text{ positive and } < 1),$$

writing herein $(a-x)^2+\cdots+(c-z)^2$ for x, and $\frac{1}{2}s+q$ for r, we have

$$\frac{1}{\{(a-x)^2+\cdots+(c-z)^2\}^{\frac{1}{2}s+q}}=\frac{1}{\Gamma(\frac{1}{2}s+q)}\int_0^\infty d\tau\,\tau^{\frac{1}{2}s+q-1}\,e^{-\tau\{(a-x)^2+\cdots+(c-z)^2\}},$$

and the value is thus

$$\frac{2}{\pi\,\Gamma(\frac{1}{2}s+q)}\int_0^\infty\frac{d\phi}{\phi}\,e^{i\phi}\int_0^\infty d\tau\,\tau^{\frac{1}{2}s+q-1}\int e^{-i\phi(x^2+\cdots+z^2)}\,e^{-\tau\{(a-x)^2+\cdots+(c-z)^2\}}\,dx\dots dz,$$

[p. 419] where the integral in regard to the variables $(x,\dots,z)$ is

$$=e^{-i\phi\tau\frac{a^2+\cdots+c^2}{\tau+i\phi}}\int dx\,e^{-(\tau+i\phi)\left(x-\frac{\tau a}{\tau+i\phi}\right)^2}\cdots,$$

and here the x-integral is

$$=\pi^{\frac{1}{2}}\sqrt{\frac{1}{\tau+i\phi}}=\pi^{\frac{1}{2}}\,\frac{e^{-\frac{1}{2}i\theta}}{\sqrt{\tau^2+\phi^2}},$$

and the like for the other integrals up to the z-integral. The resulting value is thus

$$=\pi^{\frac{s}{2}}\,(f\dots h)\,e^{-i\phi\tau\frac{a^2+\cdots+c^2}{\tau+i\phi}}\,e^{-\frac{1}{2}is\theta}\,\frac{1}{(\tau^2+\phi^2)^{\frac{1}{4}s}},$$

viz. writing $\psi=\dfrac{\phi}{\tau}$, $d\phi=\dfrac{\phi}{\tau}\,dt=\dfrac{\psi}{\tau}\,dt$, we have

$$=\frac{2\pi^{\frac{1}{2}s-1}}{\Gamma(\frac{1}{2}s+q)}(f\dots h)\int_0^\infty dt\,\frac{t^{q-1}}{\sqrt{(t^2+\tau^2)\dots(h^2+t)}}\int_0^\infty d\phi\,\phi^{q+r-1}\,e^{i\phi\sigma}\sin\phi\,d\phi.$$

154. But we have to consider only the real part of this expression; viz. writing for shortness $\sigma=\dfrac{a^2+\cdots+c^2}{t+f^2}+\cdots+\dfrac{c^2}{t+h^2}$, we require the real part of

$$e^{-i\sigma\phi}\int_0^\infty e^{i\sigma\phi}\,\phi^{q-1}\sin\phi\,d\phi.$$

Writing here for $\sin\phi$ its exponential value $\dfrac{1}{2i}(e^{i\phi}-e^{-i\phi})$, and using the formula

$$\frac{1}{(1-\sigma)^q}=\frac{1}{\Gamma(q)}\int_0^\infty d\phi\,\phi^{q-1}\,e^{-(1-\sigma)\phi}\quad(\sigma\text{ positive}),$$

and the like one

$$\frac{1}{(\sigma-1)^q}=\frac{1}{\Gamma(q)}e^{i\pi q}\int_0^\infty d\phi\,\phi^{q-1}\,e^{-(\sigma-1)\phi}\quad(\sigma\text{ negative}),$$

(in which formulae q must be positive and less than 1), we see that the real part in question is 0, or is

$$\frac{\Gamma q \sin(q+1)\pi}{2(1-\sigma)^q} = \frac{\pi}{2\,\Gamma(1-q)} \cdot \frac{1}{(1-\sigma)^q},$$

according as $\sigma > 1$ or $\sigma < 1$.

155. If the point is interior, $\dfrac{a^2}{f^2} + \cdots + \dfrac{c^2}{h^2} < 1$, and consequently also $\sigma < 1$, and the value, writing $(\Gamma\frac{1}{2})^s$ instead of $\pi^{\frac{1}{2}s}$, is

[p. 420] But if the point be exterior, $\dfrac{a^2}{f^2}+\cdots+\dfrac{c^2}{h^2} > 1$, and hence, writing θ for the positive root of the equation, $\sigma = 1$; viz. θ is the positive root of the equation $\dfrac{a^2}{\theta+f^2} + \cdots + \dfrac{c^2}{\theta+h^2} = 1$; then $t = 0$, σ is greater than 1, and continues so as t increases, until, for $t = \theta$, σ becomes $= 1$, and for larger values of t we have $\sigma < 1$; and the expression thus is

$$= \frac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)}(f\ldots h)\int_\theta^\infty dt\,t^{q-1}\big[(t+f^2)\ldots(t+h^2)\big]^{-\frac{1}{2}}\Big(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}\Big)^{-q};$$

viz. the two expressions, in the cases of an interior point and an exterior point respectively, give the value of the integral

$$\int \frac{dx\ldots dz}{\{(a-x)^2+\cdots+(c-z)^2\}^{\frac{1}{2}s+q}}.$$

This is, in fact, the formula of Annex IV. No. 110, writing therein $e = 0$ and $m = -q$.

156. Boole's researches are contained in two memoirs dated 1846, "On the Analysis of Discontinuous Functions," *Trans. Royal Irish Academy*, vol. xxi. (1848), pp. 124–139, and "On a certain Multiple Definite Integral," do. pp. 140–150 (the particular theorem about to be referred to is stated in the postscript of this memoir), and in the memoir "On the Comparison of Transcendents, with certain applications to the theory of Definite Integrals," *Phil. Trans.*, vol. cxlvii. (1857), pp. 745–803, the theorem being the third example, p. 794. The method is similar to, and was in fact suggested by, that of Lejeune-Dirichlet; the auxiliary theorem made use of in the memoir of 1857 for the representation of the discontinuity being

$$\phi(\sigma) = \frac{1}{\pi\,\Gamma(-q)}\int_{-\infty}^\infty\int_0^\infty du\,dv\,v^q\cos\{(u-\sigma)v+\tfrac{1}{2}q\pi\}\,\phi(u),$$

which is a deduction from Fourier's theorem.

Changing the notation (and in particular writing s and $\frac{1}{2}s+q$ for his n and i), the method is here applied to the determination of the s-tuple integral

$$V=\int dx\ldots dz\,\frac{\phi\Big(\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}\Big)}{\{(a-x)^2+\cdots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}},$$

where ϕ is an arbitrary function, taken over the ellipsoid $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}=1$.

157. The process is as follows: we have

$$\frac{\phi\Big(\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}\Big)}{\{(a-x)^2+\cdots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}}=\frac{1}{\pi\,\Gamma(\frac{1}{2}s+q)}\int_0^\infty\!\!\int_0^\infty\!\!\int_0^\infty du\,dv\,d\tau\,v^{\frac{1}{2}s+q}\,\tau^{\frac{1}{2}s+q-1}\,e^{-\tau v}\cos\Big\{\Big(u-$$

[p. 421] viz. the right-hand side is here equal to the left-hand side or is $=0$, according as $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}<1$ or >1. V is consequently obtained by multiplying the right-hand side by $dx\ldots dz$ and integrating from $-\infty$ to $+\infty$ for each variable.

Hence, changing the order of the integration,

$$V=\frac{1}{\pi\,\Gamma(\frac{1}{2}s+q)}\int_0^\infty\!\!\int_0^\infty\!\!\int_0^\infty du\,dv\,d\tau\,v^{\frac{1}{2}s+q}\,\tau^{\frac{1}{2}s+q-1}\,e^{-\tau v}\,\phi u\cdot\Pi,$$

where

$$\Pi=\int dx\ldots dz\cos\Big(uv-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}+\tau\{(a-x)^2+\cdots+(c-z)^2\}v+\tfrac{1}{2}(\tfrac{1}{2}s+q)\pi\Big).$$

Now if

$$\xi=\sqrt{\frac{v}{f^2}}\,x,\ \ldots,\ \zeta=\sqrt{\frac{v}{h^2}}\,z.$$

158. Substituting, and integrating with respect to $\xi,\ldots,\zeta$ between the limits $-\infty$, $+\infty$, we have

$$\Pi=\frac{\pi^{\frac{s}{2}}\,f\ldots h}{\sqrt{(1+f^2\tau v)\ldots(1+h^2\tau v)}}\cos\Big\{\Big(u-\frac{a^2\tau v}{1+f^2\tau v}-\cdots-\frac{c^2\tau v}{1+h^2\tau v}\Big)v+e^2\tau v+\tfrac{1}{2}q\pi\Big\},$$

or, what is the same thing, writing $\dfrac{t}{v}$ in place of τ, this is

$$=\frac{\pi^{\frac{s}{2}}\,f\ldots h}{\sqrt{(1+f^2t)\ldots(1+h^2t)}}\cos\Big\{\Big(u-\frac{a^2t}{1+f^2t}-\cdots-\frac{c^2t}{1+h^2t}\Big)v+e^2t+\tfrac{1}{2}q\pi\Big\},$$

that is, writing

$$\sigma = \frac{a^2t}{1+f^2t} + \cdots + \frac{c^2t}{1+h^2t} + e^2t,$$

we have

$$V = \frac{\pi^{\frac{s}{2}-1}(f\ldots h)}{\Gamma(\frac{1}{2}s+q)} \int_0^\infty \int_0^\infty du\, dt\, t^{q-1}\, v^q \cos\bigl[(u-\sigma)v + \tfrac{1}{2}q\pi\bigr]\, \phi u$$

or, writing $\pi^{\frac{1}{2}s-1} = (\Gamma\frac{1}{2})^s \pi^{-1}$, this is

$$= \frac{(\Gamma\frac{1}{2})^s}{\pi\,\Gamma(\frac{1}{2}s+q)}(f\ldots h) \int_0^\infty dt\, t^{q-1}\bigl[(1{+}f^2t)\ldots(1{+}h^2t)\bigr]^{-\frac{1}{2}} \int_0^\infty \int_0^\infty du\, dv\, v^q \cos\bigl[(u{-}\sigma)v{+}\tfrac{1}{2}q\pi\bigr]\, \phi u.$$

159. Boole writes

$$\frac{1}{\pi}\int_0^\infty \int_0^\infty du\, dv\, v^q \cos\{(u-\sigma)v + \tfrac{1}{2}q\pi\}\, \phi u = \Bigl(-\frac{d}{d\sigma}\Bigr)^q \phi(\sigma);$$

viz. starting from Fourier's theorem,

$$\frac{1}{\pi}\iint du\, dv \cos(u-\sigma)v \cdot \phi u = \phi(\sigma),$$

[p. 422] where $\phi(\sigma)$ is regarded as vanishing except when σ is between the limits $0, 1$, and the limits of u are taken to be $0, 1$ accordingly, then, according to an admissible theory of general differentiation, we have the result in question. He has in the formula $\frac{1}{s}$ instead of my t; and he proceeds, "Here σ increases continually with s. As s varies from 0 to ∞, σ also varies from 0 to ∞. To any positive limits of σ will correspond positive limits of s; and these, as will hereafter appear"—this refers to his note "B"—"will in certain cases replace the limits 0 and ∞ in the expression for V."

160. It seems better to deal with the result in the following manner, as in part shown p. 803 of Boole's memoir. Writing the integral in the form

$$V = \frac{(\Gamma\frac{1}{2})^s(f\ldots h)}{\pi\,\Gamma(\frac{1}{2}s+q)} \int_0^\infty \int_0^1 du\, dt\, t^{q-1}\bigl\{(t{+}f^2)\ldots(t{+}h^2)\bigr\}^{-\frac{1}{2}} \phi(u) \int_0^\infty dv\, v^q \cos\bigl[(u{-}\sigma)v{+}\tfrac{1}{2}q\pi\bigr],$$

effect the integration in regard to v; viz. according as u is greater or less than σ, then

$$\int_0^\infty dv\, v^q \cos\{(u-\sigma)v + \tfrac{1}{2}q\pi\} = \frac{\Gamma(q+1)\sin(q+1)\pi}{(u-\sigma)^{q+1}} \quad \text{or} \quad 0,$$

$$= \frac{\pi}{\Gamma(-q)\,(u-\sigma)^{q+1}} \quad \text{or} \quad 0;$$

and consequently, writing for σ its value,

$$V = \cdots\cdot \big(u - \frac{a^2t}{1+f^2t} - \cdots\big)^{-q-1}\,\phi(u), \quad \text{or } 0, \quad \text{as above.}$$

161. To further explain this, consider t as an x-coordinate and u as a y-coordinate; then, tracing the curve

$$y = \frac{a^2x}{f^2+x} + \cdots + \frac{c^2x}{h^2+x},$$

for positive values of x this is a mere hyperbolic branch, as shown in the figure, viz. $x = 0$, $y = 0$; and as x continually increases to ∞, y continually decreases to zero.

The limits are originally taken to be from $u = 0$ to $u = 1$ and $t = 0$ to $t = \infty$, viz. over the infinite strip bounded by the lines $t0, 01, 11$; but within these limits the

[p. 423] function under the integral sign is to be replaced by zero whenever the values u, t are such that u is less than $\dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$, viz. when the values belong to a point in the shaded portion of the strip; the integral is therefore to be extended only over the unshaded portion of the strip; viz. the value is

$$V = \frac{(\Gamma\frac{1}{2})^s(f\ldots h)}{\Gamma(-q)\,\Gamma(\frac{1}{2}s+q)} \iint dt\,du\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}}\Big(u-\frac{a^2t}{f^2+t}-\cdots-\frac{c^2t}{h^2+t}-e^2t\Big)^{-q-1}\phi u,$$

the double integral being taken over the unshaded portion of the strip; or, what is the same thing, the integral in regard to u is to be taken from $u = \dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$ (say from $= \sigma$) to $u = 1$, and then the integral in regard to t is to be taken from $t = \theta$ to $t = \infty$, where, as before, θ is the positive root of the equation $\sigma = 1$, that is, of

$$\frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t = 1.$$

162. Write $u = \sigma + (1-\sigma)x$, and therefore $u - \sigma = (1-\sigma)x$, $1 - u = (1-\sigma)(1-x)$, and $du = (1-\sigma)\,dx$; then the limits $(1, 0)$ of x correspond to the limits $(1, \sigma)$ of u, and the formula becomes

$$V = \frac{(\Gamma\frac{1}{2})^s(f\ldots h)}{\Gamma(-q)\,\Gamma(\frac{1}{2}s+q)} \int_\theta^\infty\!\int_0^1 dt\,dx\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}}\,(1-\sigma)^{-q}\,x^{-q-1}\,\phi\big[\sigma+(1-\sigma)x\big],$$

where σ is retained in place of its value $\dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$. This is, in fact, a form (deduced from Boole's result in the memoir of 1846) given by me, *Cambridge and Dublin Mathematical Journal*, vol. ii. (1847), p. 219, [44].

If in particular $\phi u = (1-u)^{q+m}$, then $\phi\sigma + (1-\sigma)x = (1-\sigma)^{q+m}(1-x)^{q+m}$, and thence

$$\int_0^1 x^{-q-1}\,\phi[\sigma+(1-\sigma)x]\,dx = (1-\sigma)^{q+m}\int_0^1 x^{-q-1}\,(1-x)^{q+m}\,dx = \frac{\Gamma(-q)\,\Gamma(1+q+m)}{\Gamma(1+m)},$$

and then, restoring for σ its value, we have

$$V = \frac{(\Gamma\frac{1}{2})^s\,\Gamma(1+q+m)}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1+m)}\,(f\ldots h)\int_\theta^\infty dt\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}}\Big(1-\frac{a^2t}{f^2+t}-\cdots-\frac{c^2t}{h^2+t}-e^2t\Big)^{q+m}$$

as the value of the integral

$$\int\frac{(1-\dfrac{x^2}{f^2}-\cdots-\dfrac{z^2}{h^2})^m\,dx\ldots dz}{\{(a-x)^2+\cdots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}=1$. This is, in fact, the theorem of Annex IV. No. 110 in its general form; but the proof assumes that q is positive.

[End of Paper 607: A Memoir on Prepotentials.]

608. [Extract from a] Report on Mathematical Tables

[From the Report of the British Association for the Advancement of Science, (1873), pp. 3, 4.]

[p. 424] It was necessary as a preliminary to form a classification of mathematical (numerical) tables; and the following classification was drawn up by Prof. Cayley and adopted by the Committee.

A. Auxiliary for non-logarithmic computations.

1. Multiplication.
2. Quarter-squares.
3. Squares, cubes, and higher powers, and reciprocals.

B. Logarithmic and circular.

4. Logarithms (Briggian) and antilogarithms (do.); addition and subtraction logarithms, &c.
5. Circular functions (sines, cosines, &c.), natural, and lengths of circular arcs.
6. Circular functions (sines, cosines, &c.), logarithmic.

C. Exponential.

7. Hyperbolic logarithms.
8. Do. antilogarithms (e^x) and $h.\mathrm{l}\tan(45^\circ + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c., natural and logarithmic.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.
10. Decimal values auxiliary to the calculation of series.

[p. 425]

E. 11. Transcendental constants, e, π, γ, &c., and their powers and functions.

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.
13. The Pellian equation.
14. Partitions.
15. Quadratic forms $\pm a^2 \pm b^2$, &c., and partition of numbers into squares, cubes, and biquadrates.
16. Binary, ternary, &c. quadratic, and higher forms.
17. Complex theories.

G. Transcendental functions.

18. Elliptic.
19. Gamma.
20. Sine-integral, cosine-integral, and exponential-integral.
21. Bessel's and allied functions.
22. Planetary coefficients for given $\dfrac{a}{a'}$.
23. Logarithmic transcendental.

24. Miscellaneous.

Several of these classes need some little explanation. Thus D 9 and 10 are intended to include the same class of constants, the only difference being that in 9 accurate values are given, while in 10 they are only approximate; thus, for example, the accurate Bernoulli's numbers as vulgar fractions, and the decimal values of the same to (say) ten places are placed in different classes, as the former are of theoretical interest, while the latter are only of use in calculation. It is not necessary to enter into further detail with respect to the classification, as in point of fact it is only very partially followed in the Report.

609. On the Analytical Forms called Factions

[From the Report of the British Association for the Advancement of Science, (1875), p. 10.]

[p. 426] A FACTION is a product of differences such that each letter occurs the same number of times; thus we have a quadrifaction where each letter occurs twice, a cubifaction where each letter occurs three times, and so on. A broken faction is one which is a product of factions having no common letter; thus

$$(a-b)^2(c-d)(d-e)(e-c)$$

is a broken quadrifaction, the product of the quadrifactions

$$(a-b)^2 \quad \text{and} \quad (c-d)(d-e)(e-c).$$

We have, in regard to quadrifactions, the theorem that every quadrifaction is a sum of broken quadrifactions such that each component quadrifaction contains two or else three letters. Thus we have the identity

$$2(a-b)(b-c)(c-d)(d-a) = (b-c)^2 \cdot (a-d)^2 - (c-a)^2 \cdot (b-d)^2 + (a-b)^2 \cdot (c-d)^2,$$

which verifies the theorem in the case of a quadrifaction of four letters; but the verification even in the next following case of a quadrifaction of five letters is a matter of some difficulty.

The theory is connected with that of the invariants of a system of binary quantics.

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations

[From the Report of the British Association for the Advancement of Science, (1875), pp. 257–305.]

[p. 427] I have in two papers "On the Analytical forms called Trees," *Phil. Mag.* vol. xiii. (1857), pp. 172–176, [203], and ditto, vol. xx. (1859), pp. 374–378, [247], considered this theory; and in a paper "On the Mathematical Theory of Isomers," ditto, vol. xlvii. (1874), p. 444, [586], pointed out its connexion with modern chemical theory. In particular, as regards the paraffins C_nH_{2n+2}, we have n atoms of carbon connected by $n-1$ bands, under the restriction that from each carbon-atom there proceed at most 4 bands (or, in the language of the papers first referred to, we have n knots connected by $n-1$ branches), in the form of a tree; for instance, $n = 5$, such forms (and the only such forms) are

[Three tree diagrams of carbon skeletons for $n = 5$ paraffin isomers: the linear (normal pentane), the branched (isopentane), and the centrally branched (neopentane); each carbon-atom labelled with the number of attached hydrogens.]

And if, under the foregoing restriction of only 4 bands from a carbon-atom, we connect with each carbon-atom the greatest possible number of hydrogen-atoms, as shown in the diagrams by the affixed numerals, we see that the number of hydrogen-atoms is 12 $(= 2 \cdot 5 + 2)$; and we have thus the representations of three different paraffins, C_5H_{12}. It should be observed that the tree-symbol of the paraffin is

[p. 428] completely determined by means of the tree formed with the carbon-atoms, or say of the carbon-tree, and that the question of the determination of the theoretic number of the paraffins C_nH_{2n+2} is consequently that of the determination of the number of the carbon-trees of n knots, viz. the number of trees with n knots, subject to the condition that the number of branches from each knot is at most $= 4$.

In the paper of 1857, which contains no application to chemical theory, the number of branches from a knot was unlimited; and, moreover, the trees were considered as issuing each from one knot taken as a root, so that, $n = 5$, the trees regarded as distinct (instead of being as above only 3) were in all 9, viz. these were

[Nine root-tree diagrams of 5 knots each, each issuing from a designated bottom knot.]

which, regarded as issuing from the bottom knots, are in fact distinct; while, taking them as issuing each from a properly selected knot, they resolve themselves into the above-mentioned 3 forms. The problem considered was in fact that of the "general root-trees with n knots"—general, inasmuch as the number of branches from a knot was without limit; root-trees, inasmuch as the enumeration was made on the principle last referred to. It was found that for

knots	1,	2,	3,	4,	5,	6,	7,	8,
No. of trees was	1,	1,	2,	4,	9,	20,	48,	115,
	$= 1,$	$A_2,$	$A_3,$	$A_4,$	$A_5,$	$A_6,$	$A_7,$	$A_8,$

the law being given by the equation

$$(1-x)^{-1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}(1-x^4)^{-A_4}\cdots = 1+x+A_2x^2+A_3x^3+A_4x^4+\cdots\ ;$$

but the next following numbers A_9, A_{10}, A_{11}, the correct values of which are $286, 719, 1842$, were given erroneously as $306, 775, 2009$. I have since calculated two more terms, $A_{12} = 4766$, $A_{13} = 12486$.

The other questions considered in the paper of 1857 and in that of 1859 have less immediate connexion with the present paper, but for completeness I reproduce the results in a Note[2].

[p. 429] To count the trees on the principle first referred to, we require the notions of "centre" and "bicentre," due, I believe, to Sylvester; and to establish these we require the notions of "main branch" and "altitude": viz. in a tree, selecting any knot at pleasure as a root, the branches which issue from the root, each with all the branches that belong to it, are the main branches, and the distance of the furthest knot, measured by the number of

[2]In the paper of 1857 I also considered the problem of finding B_r, the number with r free branches, with bifurcations at least: this was given by a like formula

$$(1-x)^{-1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}(1-x^4)^{-B_4}\cdots = 1+x+2B_2x^2+2B_3x^3+2B_4x^4+\cdots,$$

leading to $B_r = 1, 2, 6, 12, 38, 90$ for $r = 2, 3, 4, 5, 6, 7$.

In the paper of 1859, the question is to find the number of trees with a given number m of terminal knots: we have here

$$\phi_m = \frac{1}{1\cdot 2\cdot 3\cdots(m-1)} \text{ coefficient of } x^{m-1} \text{ in } \Sigma,$$

giving the values $\phi_m = 1, 1, 3, 13, 75, 541, 4683, 47293, \ldots$ for $m = 1, 2, 3, 4, 5, 6, 7, 8, \ldots$

But if from each non-terminal knot there ascend two and only two branches, then in this case $\phi_m =$ coefficient of x^{m-1} in $\dfrac{1-\sqrt{1-4x}}{2}$, viz. we have the very simple form

$$\phi_m = \frac{1\cdot 3\cdot 5\cdots(2m-3)}{1\cdot 2\cdot 3\cdots m}\cdot 2^{m-1},$$

giving $\phi_m = 1, 1, 2, 5, 14, 42, \ldots$ for $m = 1, 2, 3, 4, 5, \ldots$

intermediate branches, is the altitude of the main branch. Thus in the left-hand figure, taking A as the root, there are 3 main branches of the altitudes $3, 3, 1$ respectively: in the right-hand figure, taking A as the root, there are 4 main branches of the altitudes $2, 2, 1, 3$ respectively; and we have

[Two diagrams: left, a tree with vertex A illustrating a centre and three main branches; right, a tree with adjacent vertices A, B illustrating a bicentre.]

then the theorem that in every tree there is either one and only one centre, or else one and only one bicentre; viz. we have (as in the left-hand figure) a centre A which is such that there issue from it two or more main branches of altitudes equal to each other and superior to those of the other main branches (if any); or else (as in the right-hand figure) a bicentre AB, viz. two contiguous knots, such that issuing from A (but not counting AB), and issuing from B (but not counting BA), we have two or more main branches, one at least from A and one at least from B, of altitudes equal to each other and superior to those of the other main branches in question (if any). The theorem, once understood, is proved without difficulty: we consider two terminal knots, the distance of which, measured by the number of intermediate branches, is greater than or equal to that of any other two terminal knots; if, as in the left-hand figure, the distance is even, then the central knot A is the centre of the tree; if, as in the right-hand figure, the distance is odd, then the two central knots AB form the bicentre of the tree.

In the former case, observe that if G, H are the two terminal knots, the distance of which is $= 2\lambda$, then the distance of each from A is $= \lambda$, and there cannot be

[p. 430] any other terminal knot I, the distance of which from A is greater than λ (for, if there were, then the distance of I from G or else from H would be greater than 2λ); there cannot be any two terminal knots I, J, the distance of which is greater than 2λ; and if there are any two knots I, J, the distance of which is $= 2\lambda$, then these belong to different main branches, the distance of each of them from A being $= \lambda$; whence, starting with I, J (instead of G, H), we obtain the same point A as centre. Similarly, in the latter case, there is a single bicentre AB.

Hence, since in any tree there is a unique centre or bicentre, the question of finding the number of distinct trees with n knots is in fact that of finding the number of centre- and bicentre-trees with n knots; or say it is the problem of the "general centre- and bicentre-trees with n knots:" general, inasmuch as the number of branches from a knot is as yet taken to be without limit; or since (as will appear) the number of the bicentre-trees can be obtained without difficulty when the problem of the root-trees is solved, the problem is that of the "general centre-trees with n knots." It will appear that the solution depends upon and is very readily derived from that of the foregoing problem of general root-trees, so that this last has to be considered, not only for its own sake, but with a view to that of the centre-trees. And in each of the two problems we doubly divide the whole system of trees according to the number of the main branches, issuing from the root or centre as the case may be, and according to the altitude of the longest main branch or branches, or say the altitude of the tree; so that the problem really is, for a given number of knots, a given number of main branches, and a given altitude, to find the number of root-trees, or (as the case may be) centre-trees.

We next introduce the restriction that the number of branches from any knot is equal to a given number at most; viz. according as this number is $2, 3$ or 4, we have, say oxygen-trees, boron-trees[3], and carbon-trees respectively; and these are, as before, root-trees or centre- or bicentre-trees, as the case may be. The case where the number is 2 presents no difficulty: in fact, if the number of knots be $= n$, then the number of root-trees is either $\frac{1}{2}(n+1)$ or $\frac{1}{2}n$; viz. $n = 3$ and $n = 4$, the root-trees are

[Two simple linear root-tree diagrams of 3 *and* 4 *knots respectively.]*

and the number of centre- or bicentre-trees is always 1: viz. n odd, there is one centre-tree; and n even, one bicentre-tree; it is only considered as a particular case of the general theorem. The case where the number is 3 is analytically interesting although there may not exist, for any 3-valent element, a series of hydrogen compounds

[End of pages 406–430 (PDF pp. 426–450). Paper 610 continues in subsequent pages.]

[3]I should have said nitrogen-trees; but it appears to me that nitrogen is of necessity 5-valent, as shown by the compound, Ammonium-Chloride, $= NH_4Cl$. Of course, the word boron is used simply to stand for a 3-valent element.

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Arthur Cayley

Collected Mathematical Papers

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610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations *(continued)*

[Continued from the preceding pages of the same paper. The discussion of carbon-trees corresponding to the paraffins is now developed analytically; Tables I–VIII give the enumeration of root-, centre- and bicentre-trees in the general case and for the oxygen, boron and carbon (chemical) cases.]

[p. 431] B_nH_{n+3} corresponding to the paraffins. The case where the number is $= 4$ or, say, the carbon-trees, is that which presents the chief chemical interest, as giving the paraffins C_nH_{2n+2}; and I call to mind here that the theory of the carbon root-trees is established as an analytical result for its own sake and as the foundation for the other case, but that it is the number of the carbon centre- and bicentre-trees which is the number of the paraffins.

The theory extends to the case where the number of branches from a knot is at most $= 5$, or $=$ any larger number; but I have not developed the formula.

I pass now to the analytical theory: considering first the case of general root-trees, we endeavour to find for a given altitude N the number of trees of a given number of knots n and main branches a, or say the generating function

$$\Sigma\,\Omega\,t^a x^n,$$

where the coefficient Ω gives the number of the trees in question. And we assume that the problem is solved for the cases of the several inferior altitudes $0, 1, 2, 3, \ldots, N-1$.

This being so, observe that a tree of altitude N can be built up as shown in the figure, which I call the edification diagram, by combining one or more trees of altitude $N-1$ with a single tree of altitude not exceeding $N-1$; viz. in the figure, $N = 3$, we have the two trees a, b, each of altitude 2, combined, as shown by the dotted lines, with the tree c of altitude 1: the whole number of knots in the resulting tree is the sum of the number of knots on the three trees a, b, c: the number of main branches is equal to the number of the trees a, b, plus the number of main branches of the tree c. It is to be observed that the tree c may reduce itself to the tree ($\bullet$) of one knot and of altitude zero; but each of the trees a, b, as being of the altitude $N-1$, must contain at least N knots.

[Edification diagram: two trees a, b of altitude 2 combined by dotted lines with a third tree c of altitude 1, all attached at the root.]

Taking $N = 2$ or any larger number, it is hence easy to see that the required generating function $\Sigma\,\Omega\,t^a x^n$ is

$$= (1-tx^N)^{-l_0}\,(1-tx^{N+1})^{-l_1}\,(1-tx^{N+2})^{-l_2}\,\cdots\;[t^{1\cdots\infty}] \qquad \text{(first factor)},$$

$$x + (t)\,x^2 + (t,\,t^2)\,x^3 + (t,\,t^2,\,t^3)\,x^4 + \cdots \qquad \text{(second factor).}$$

As regards the first factor, the exponents taken with reversed sign, that is, as positive, are l_0 = no. of trees, altitude $N-1$, of N knots; l_1 = ditto, same altitude, of $(N+1)$ knots; l_2 = ditto, same altitude, of $N+2$ knots, and so on; and where the

[p. 432] symbol $[t^{1\cdots\infty}]$ denotes that, in the function or product of factors which precedes it, the terms to be taken account of are those in t, t^2, $t^3, \ldots$; viz. it denotes that the term in t^0, or constant term ($= 1$ in fact), is to be rejected.

In the second factor, the expressions x, $(t)x^2$, $(t,\,t^2)x^3, \ldots$ represent, for given exponents of t, x, denoting the number of main branches and the number of knots respectively, the number of trees of altitude not exceeding $N-1$: thus $x, = 1\cdot x$ represents the number of such trees, 1 knot, 0 main branch, $= 1$; and so, if the value of $(t,\,t^2,\,t^3,\,t^4)x^5$ be $(\alpha t + \beta t^2 + \gamma t^3 + \delta t^4)x^5$, then for trees of an altitude not exceeding $N-1$, and of 5 knots, α represents the number of trees of 1 main branch, β that of trees of 2 main branches, γ that of trees of 3 main branches, δ that of trees of 4 main branches. It is clear that the number of trees satisfying the given conditions and of an altitude not exceeding $N-1$ is at once obtained by addition of the numbers of the trees satisfying the given conditions, and of the altitudes $0, 1, 2, \ldots, N-1$; all which numbers are taken to be known.

It is to be remarked that the first factor,

$$(1-tx^N)^{-l_0}\,(1-tx^{N+1})^{-l_1}\,(1-tx^{N+2})^{-l_2}\,\cdots\,[t^{1\cdots\infty}],$$

shows by its development the number of combinations of trees $a, b, \ldots$ of the altitude $N-1$; one such tree at least must be taken, and the symbol $[t^{1\cdots\infty}]$ gives effect to this condition: the second factor $x + (t)\,x^2 + (t,\,t^2)\,x^3 + \cdots$ shows the number of the trees c of altitude not exceeding $N-1$. And this being so, there is no difficulty in seeing how the product of the two factors is the generating function for the trees of altitude N.

In the case $N = 0$, the generating function, or GF, is $= x$; viz. altitude 0, there is only the tree ($\bullet$), 1 knot, 0 main branch.

When $N = 1$, the GF is $= (1-tx)^{-1}\,[t^{1\cdots\infty}]\,x \;=\; tx^2 + t^2x^3 + t^3x^4 + \cdots$, viz. altitude 1, there is 1 tree tx^2, 2 knots, 1 main branch; 1 tree t^2x^3, 3 knots, 2 main branches; and so on.

Hence $N = 2$, we obtain

$$\text{GF} = (1-tx^2)^{-1}(1-tx^3)^{-1}(1-tx^4)^{-1}\cdots\,[t^{1\cdots\infty}]\cdot(x+tx^2+t^2x^3+t^3x^4+\cdots);$$

viz. as regards the second factor, altitude not exceeding 1, that is, $= 0$ or 1, there is altitude 0, 1 tree x, and altitude 1, 1 tree tx^2, 1 tree t^2x^3, and so on. And we hence derive the GF's for the higher values $N = 3$, 4, &c.: the details of the process will be afterwards more fully explained.

So far, we have considered root-trees; but referring to the last diagram, it is at once seen that the assumed root will be a centre, provided only that (instead of, it may be, only a single tree a of the altitude $N-1$), we take always two or more trees of the altitude $N-1$ to form the new tree of the altitude N. And we give effect

[p. 433] to this condition by simply writing in place of $[t^{1\cdots\infty}]$ the new symbol $[t^{2\cdots\infty}]$, which denotes that only the terms t^2, t^3, $t^4, \ldots$ are to be taken account of; viz. that the terms in t^0 and t^1 are to be rejected. The component trees of the altitude $N-1$ are, it is to be observed, as before, root-trees; hence the second factor of the generating function is unaltered: the theorem is that for the centre-trees of altitude N we have the same generating function as for the root-trees, writing only $[t^{2\cdots\infty}]$ in place of $[t^{1\cdots\infty}]$.

Or, what is the same thing, supposing that the first factor, unaffected by either symbol, is

$$= 1 + x^N(\alpha t + \beta t^2 + \cdots) + x^{N+1}(\alpha' t + \beta' t^2 + \cdots) + \cdots ,$$

then, affecting it with $[t^{1\cdots\infty}]$, the value for the root-trees is

$$= x^N(\alpha t + \beta t^2 + \cdots) + x^{N+1}(\alpha' t + \beta' t^2 + \cdots) + \cdots ,$$

and, affecting it with $[t^{2\cdots\infty}]$, the value for the centre-trees is

$$= x^N(\beta t^2 + \cdots) + x^{N+1}(\beta' t^2 + \cdots) + \cdots$$

It thus appears how the fundamental problem is that of the root-trees, its solution giving at once that of the centre-trees; whereas we cannot conversely solve the problem of the root-trees by means of that of the centre-trees.

As regards the bicentre-trees, it is to be remarked that, starting from a centre-tree of altitude $N+1$ with two main branches, then by simply striking out the centre, so as to convert into a single branch the two branches which issue from it, we obtain a bicentre-tree of altitude N. Observe that the altitude of a bicentre-tree is measured by that of the longest main branch from A or B, not reckoning AB or BA as a main branch. Hence the number of bicentre-trees, altitude N, is $=$ number of centre-trees of two main branches, altitude $N+1$.

This is, in fact, the convenient formula, provided only the number of centre-trees of two main branches has been calculated up to the altitude $N+1$. But we can find independently the number of bicentre-trees of a given altitude N: the bicentre-tree is, in fact, formed by taking the two connected points A, B each as the root of a root-tree altitude N (the number of knots of the bicentre-tree being thus, it is clear, equal to the sum of the numbers of knots of the two root-trees respectively); and it is thus an easy problem of combinations to find the number of bicentre-trees of a given altitude N. Write

$$x^{N+1}\,(1 + \beta x + \gamma x^2 + \delta x^3 + \cdots)$$

as the generating function of the root-trees of altitude N; viz. for such trees, $1 =$ no. of trees with $N+1$ knots, $\beta =$ no. with $N+2$ knots, and so on; then the generating function of the bicentre-trees of the same altitude N is

$$= x^{2N+2}\,(1 + \beta_1 x + \gamma_1 x^2 + \delta_1 x^3 + \cdots),$$

[p. 434] where

$$\beta_1 = \beta,$$
$$\gamma_1 = \gamma + \tfrac{1}{2}\beta(\beta+1),$$
$$\delta_1 = \delta + \beta\gamma,$$
$$\varepsilon_1 = \varepsilon + \beta\delta + \tfrac{1}{2}\gamma(\gamma+1),$$
$$\zeta_1 = \zeta + \beta\varepsilon + \gamma\delta,$$

and so on; or, what is the same thing, calling the first generating function ϕx, then the second generating function is $= \frac{1}{2}\big[(\phi x)^2 + \phi(x^2)\big]$.

It will be noticed that the bicentre-trees are not, as were the centre-trees, divided according to the number of their main branches; they might be thus divided according to the sum of the number of the main branches issuing from the two points of the bicentre respectively; a more complete division would be according to the number of main branches issuing from the two points respectively: thus we might consider the bicentre-trees $(2, 3)$, with 2 main branches from one point, and 3 main branches from the other point of the bicentre; but the whole theory of the bicentre-trees is comparatively easy, and I do not go into it further.

We have yet to consider the case of the limited trees where the number of branches from a knot is equal to a given number at most: to fix the ideas, say the carbon-trees, where this number is $= 4$. The distinction as to root-trees and centre- and bicentre-trees is as before; and the like theory applies to the two cases respectively.

Considering first the case of the root-trees, and referring to the former figure for obtaining the trees of altitude N from those of inferior altitudes, then the trees a, $b, \ldots$ of altitude $N-1$ must be each of them a carbon-tree of not more than $(4-1=)3$ main branches: this restriction is necessary, inasmuch as, if for any such tree the number of main branches was $= 4$, then there would be from the root of such tree 4 branches *plus* the new branch shown by the dotted line, in all 5 branches; and similarly, inasmuch as there is at least one component tree a contributing one main branch, the number of main branches of the tree c must be $(4-1=)3$ at most: the mode of introducing these conditions will appear in the explanation of the actual formation of the generating functions (see explanation preceding Tables III., IV., &c.).

The number of main branches is $= 4$ at most, and the generating functions have only to be taken up to the terms in t^4; the first factor is consequently in each case affected with a symbol $[t^{1\cdots 4}]$, denoting that the only terms to be taken account of are those in t, t^2, t^3, t^4; hence as there is a factor t at least, and the whole is required only up to t^4, the second factor is in each case required only up to t^3.

As regards the centre-trees, the generating functions have here the same expressions as for the root-trees, except that, instead of the symbol $[t^{1\cdots 4}]$, we have the symbol $[t^{2\cdots 4}]$, denoting that in the first factor the only terms to be taken account of are those in t^2, t^3, t^4; hence as there is a factor t^2 at least, and the whole is required only up to t^4, the second factor is in each case required up to t^2; and we then complete the theory by obtaining the bicentre-trees. The like remarks apply of course to

[p. 435] the boron-trees, number of branches $= 3$ at most, and to the oxygen-trees, number $= 2$ at most; but, as already remarked, this last case is so simple, that the general method is applied to it only for the sake of seeing what the general method becomes in such an extreme case.

We thus form the Tables, which I proceed to explain.

Table I. of general root-trees is in fact a Table of triple entry, viz. it gives for any given number of knots from 1 to 13 the number of root-trees corresponding to any given number of main branches and to any given altitude. In each compartment, that is, for any given number of knots, the totals of the columns give the number of the trees for each given altitude, and the totals of the lines give the number of the trees for each given number of main branches: the corner grand totals of these totals respectively show for each given number of knots the whole number of root-trees:

knots	1,	2,	3,	4,	5,	6,	7,	8,	9,	10,	11,	12,	13,
numbers are	1,	1,	2,	4,	9,	20,	48,	115,	286,	719,	1842,	4766,	12486,

as already mentioned: these numbers were calculated by an independent method.

Table II. of general centre- and bicentre-trees consists of a centre part and a bicentre part: the centre part is arranged precisely in the same manner as the root-table. As to the bicentre part, where it will be observed there is no division for number of main branches, the calculation of the several columns is effected by the before-mentioned formula,

$$\phi_1 x = \tfrac{1}{2}\big[(\phi x)^2 + \phi(x^2)\big];$$

thus, column 2, we have by Table I. (totals of column 2)

$$\phi x = x^3 + 2x^4 + 4x^5 + 6x^6 + 10x^7 + 14x^8 + 21x^9 + 29x^{10} + \cdots,$$

and thence

$$\phi_1 x = x^6 + 2x^7 + 7x^8 + 14x^9 + 32x^{10} + 58x^{11} + 110x^{12} + 187x^{13} + \cdots$$

As already mentioned, each column of Table I. is calculated by means of a generating function given as a product of two factors, each of which is obtained from the columns which precede the column in question; and Table II., the centre part of it, is calculated by means of the same generating functions slightly modified: these generating functions serving for the calculation of the two Tables are given in the table entitled "Subsidiary Table for the calculation of the GF's of Tables I. and II.," which immediately follows these two Tables, and will be further explained.

[p. 436]

TABLE I. — *General Root-trees.*

The Table is arranged by number of knots (from 1 to 13); within each knot-block, rows are indexed by number of main branches and columns by altitude (column 0 being the trivial tree of one knot). The entry in a cell is the number of root-trees with the prescribed number of knots, main branches, and altitude. "Total" rows give the column-sums for each knot-block. The grand total in the last column of each block gives the total number of root-trees with the prescribed number of knots.

$n = 1$: only (•), altitude 0. Total = 1.

$n = 2$: 1 main branch, altitude 1: 1. Total = 1.

$n = 3$: altitude 1, 1 MB: 1; altitude 2, 1 MB: 1, 2 MB: 1. Total altitudes $(1, 2) = (1, 2)$.

$n = 4$: altitudes $(1, 2, 3)$.

MB alt	1	2	3	Tot.
1	1	1		2
2		1		1
3			1	1
Tot.	1	2	1	4

$n = 5$:

MB alt	1	2	3	4	Tot.
1	1	2	1		4
2		2	1		3
3			1		1
4				1	1
Tot.	1	4	3	1	9

$n = 6$:

MB alt	1	2	3	4	5	Tot.
1	1	4	3	1		9
2		2	3	1		6
3		2	1			3
4			1			1
5					1	1
Tot.	1	8	8	2	1	20

$n = 7$:

MB alt	1	2	3	4	5	6	Tot.
1	1	6	8	4	1		20
2		3	8	4	1		16
3		3	3	1			7
4		2	1				3
5			1				1
6						1	1
Tot.	1	14	21	9	2	1	48

$n = 8$:

MB alt	1	2	3	4	5	6	7	Tot.
1	1	10	18	13	5	1		48
2		5	15	13	5	1		39
3		4	9	4	1			18
4		3	3	1				7
5		2	1					3
6			1					1
7							1	1
Tot.	1	24	47	31	11	2	1	115

$n = 9$:

MB alt	1	2	3	4	5	6	7	8	Tot.
1	1	14	38	36	19	6	1		115
2		7	30	36	19	6	1		99
3		5	19	14	5	1			44
4		5	9	4	1				19
5		3	3	1					7
6		2	1						3
7			1						1
8								1	1
Tot.	1	36	101	91	44	13	2	1	289

[Note: scan totals partially illegible; figures above follow Cayley's totals.]

[p. 437]

TABLE I. *(continued).*

$n = 10$:

MB alt	1	2	3	4	5	6	7	8	9	Tot.
1	1	21	76	93	61	26	7	1		286
2		7	51	89	61	26	7	1		242
3		7	42	41	20	6	1			117
4		6	20	14	5	1				46
5		5	9	4	1					19
6		3	3	1						7
7		2	1							3
8			1							1
9									1	1
Tot.	1	51	203	242	148	59	15	2	1	722

$n = 11$:

MB alt	1	2	3	4	5	6	7	8	9	10	Tot.
1	1	29	147	225	180	94	34	8	1		719
2		9	90	210	180	94	34	8	1		626
3		8	79	110	67	27	7	1			299
4		9	46	42	20	6	1				124
5		7	20	14	5	1					47
6		5	9	4	1						19
7		3	3	1							7
8		2	1								3
9			1								1
10										1	1
Tot.	1	72	396	606	453	222	76	17	2	1	1846

[Totals follow Cayley; certain column totals are partly illegible in the scan.]
$n = 12$:

MB alt	1	2	3	4	5	6	7	8	9	10	11	Tot.
1	1	41	277	528	498	308	136	43	9	1		1842
2		11	145	467	498	308	136	43	9	1		1618
3		10	152	273	208	101	35	8	1			788
4		11	91	113	68	27	7	1				318
5		10	47	42	20	6	1					126
6		7	20	14	5	1						47
7		5	9	4	1							19
8		3	3	1								7
9		2	1									3
10			1									1
11											1	1
Tot.	1	100	746	1442	1298	751	315	95	19	2	1	4770

$n = 13$:

MB alt	1	2	3	4	5	6	7	8	9	10	11	12	Tot.
1	1	55	509	1198	1323	941	487	188	53	10	1		4766
2		14	238	1012	1524	941	487	188	53	10	1		4468
3		12	272	559	376	244	144	44	9	1			1661
4		15	184	299	213	102	36	8	1				858
5		13	96	116	68	27	7	1					328
6		11	47	42	20	6	1						127
7		7	20	14	5	1							47
8		5	9	4	1								19
9		3	3	1									7
10		2	1										3
11			1										1
12												1	1
Tot.	1	137	1380	3265	3530	2262	1162	429	116	21	2	1	12486

[Several intermediate totals in Cayley's table are partly illegible in the scan; the printed grand total 12486 is preserved.]

[p. 438]

TABLE II. — *General Centre- and Bicentre-Trees.*

The Centre part is indexed by knots, main branches and altitude as in Table I.; the "Centre Total" column gives the centre-tree totals; the "Grand Total" is the sum of centre- and bicentre-trees; the "Bicentre" column gives bicentre-tree totals (not divided by number of main branches); and the right-hand block gives bicentre-trees of each altitude. We collect the totals.

n	Centre Tot.	Bicentre Tot.	Grand Tot.	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1					
2	0	1	1	1				
3	1	0	1					
4	1	1	2		1			
5	2	1	3		1			
6	3	3	6		2	1		
7	7	4	11		2	2		
8	12	11	23		3	7	1	
9	27	20	47		3	14	3	
10	55	51	106		4	32	14	1
11	127	108	235		4	58	42	4
12	284	267	551		5	110	128	23 (1)
13	682	619	1301		5	187	334	88 (5)

[Note: the rightmost "alt 5" column for $n = 12, 13$ *includes the small additional entry shown in parentheses, in conformity with Cayley's printed numbers. Within each* n*-block of the centre part, the distribution by main branches is the same as in Table I., truncated to centre-trees; we suppress that sub-detail here for brevity.]*

The bicentre column is computed by the formula $\phi_1 x = \frac{1}{2}[(\phi x)^2 + \phi(x^2)]$, where ϕx is the generating function of root-trees of the given altitude N.

[p. 440]

SUBSIDIARY TABLE FOR GF'S OF TABLES I. AND II.

The Subsidiary Table is divided into sections, giving the GF's of the successive columns of Table I., each section being given by means of the preceding columns of Table I.; for instance, that for column 3 by means of columns 0, 1, 2 of Table I.

As regards column 0, the Table shows that the GF is $= x$.
As regards column 1, it shows that the GF has a first factor

$$(1 - tx)^{-1} \;=\; (1) + tx + t^2x^2 + t^3x^3 + \cdots\,,$$

which is operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term (1) is to be rejected; and that it has a second factor $= x$: the product of these, viz. $(tx + t^2x^2 + t^3x^3 + \cdots) \cdot x$, is the required GF, the coefficients of which are accordingly given in column 1 of Table I.
As regards column 2, the GF has a first factor

$$(1 - tx^2)^{-1}(1 - tx^3)^{-1}(1 - tx^4)^{-1}\cdots\,,$$

where the indices -1, -1, $-1, \ldots$ are the sums of the numbers in column 1, Table I. (with their signs changed): which first factor is

$$1 + tx^2 + tx^3 + (t + t^2)x^4 + \cdots\,,$$

and it is as before to be operated on with $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and further, that there is a second factor $= x + tx^2 + t^2x^3 + \cdots$, the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1 of Table I. We have thence column 2 of Table I.
As regards column 3, the GF has a first factor

$$(1 - tx^3)^{-1}(1 - tx^4)^{-2}(1 - tx^5)^{-4}\cdots\,,$$

where the indices -1, -2, $-4, \ldots$ are the sums of the numbers in column 2 of Table I. (with their signs changed): which first factor is

$$1 + tx^3 + 2tx^4 + 4tx^5 + (6t + t^2)x^6 + \cdots\,,$$

and is as before to be operated on with $[t^{1\cdots\infty}]$; and there is a second factor

$$x + tx^2 + (t + t^2)x^3 + (t + 2t^2 + t^3)x^4 + \cdots\,,$$

the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1, 2 of Table I.: we have thence column 3 of Table I.

And similarly, by means of columns 0, 1, 2, 3 of Table I., we form the GF of column 4; that is, we obtain column 4 of Table I., and so on indefinitely.

[The full Subsidiary Table tabulates, for each column k of Table I. from $k = 0$ up to $k = 13$, the line "$*$ GF, column k" (negatives of the column totals of Table I. used as exponents), the rows of the "First factor" giving the expansion in powers of t and x, and the rows of the "Second factor." The full numerical tabulation runs over pages 440–443 and follows directly from the prose explanation above. To save space and avoid OCR-induced numerical noise, we do not reproduce every cell of the printed table here; the prose recipe just given suffices to reconstruct each column.]

[p. 444] I proceed to explain the Subsidiary Table, first in its application to Table I.

The Subsidiary Table is divided into sections, giving the GF's of the successive columns of Table I., for instance, that for column 3 by means of the preceding columns of Table I., 0, 1, 2 of Table I.

As regards column 0, the Table shows that the GF is $= x$.
As regards column 1, it shows that the GF has a first factor,

$$(1 - tx)^{-1} = (1) + tx + t^2x^2 + t^3x^3 + \cdots ,$$

which is operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term (1) is to be rejected; and that it has a second factor $= x$: the product of these, viz. $(tx + t^2x^2 + t^3x^3 + \cdots) \cdot x$, is the required GF, the coefficients of which are accordingly given in column 1 of Table I.
As regards column 2, it shows that the GF has a first factor,

$$(1 - tx^2)^{-1}(1 - tx^3)^{-1}(1 - tx^4)^{-1} \cdots ,$$

where the indices $-1, -1, -1, \ldots$ are the sums of the numbers in column 1, Table I. (with their signs changed); which first factor is

$$1 + tx^2 + tx^3 + (t + t^2)x^4 + \cdots ,$$

and it is as before to be operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and further, that there is a second factor $= x + tx^2 + t^2x^3 + \cdots$, the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1 of Table I. We have thence column 2 of Table I.
As regards column 3, it shows that the GF has a first factor,

$$(1 - tx^3)^{-1}(1 - tx^4)^{-2}(1 - tx^5)^{-4} \cdots ,$$

where the indices $-1, -2, -4, \ldots$ are the sums of the numbers in column 2 of Table I. (with their signs changed); which first factor is

$$= 1 + tx^3 + 2tx^4 + 4tx^5 + (6t + t^2)x^6 + \cdots ,$$

and is as before to be operated on with $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and that there is a second factor

$$= x + tx^2 + (t+t^2)x^3 + (t+2t^2+t^3)x^4 + \cdots,$$

the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1, 2 of Table I.: we have thence column 3 of Table I.

And similarly, by means of columns 0, 1, 2, 3 of Table I., we form the GF of column 4; that is, we obtain column 4 of Table I., and so on indefinitely.

[p. 445] To apply the Subsidiary Table to the calculation of the GF's of Table II., the only difference is that the first factors are to be taken without the terms in t^1: thus for Table II. column 3, the first factor of the GF is

$$= t^2x^6 + 2t^2x^7 + 7t^2x^8 + (14t^2+t^3)x^9 + \&\text{c.},$$

the second factor being as for Table I.,

$$= x + tx^2 + (t+t^2)x^3 + \&\text{c.}$$

The remaining Tables are Tables III. and IV., oxygen root-trees and centre- and bicentre-trees, followed by a Subsidiary Table for the calculation of the GF's; Tables V. and VI., boron root-trees and centre- and bicentre-trees, followed by a Subsidiary Table; and Tables VII. and VIII., carbon root-trees and centre- and bicentre-trees, followed by a Subsidiary Table. The explanations given as to Tables I., II. and the Subsidiary Table apply *mutatis mutandis* to these; and but little further explanation is required: that given in regard to the Subsidiary Table of Tables III. and IV. shows how this limiting case comes under the general method. As to the Subsidiary of Tables V. and VI., it is to be observed that each $*$ line of the Table is calculated from a column of Table V., rejecting the numbers which belong to t^3; thus Table V., column 4, the numbers are

	3	4	5	6	7	8	9 ...
t^1		1	4	10	21	36	...
t^2			1	4	11	26	...

and taking the sums for the first and second lines only, these are

$$1,\ 4,\ 9,\ 17,\ 29,\ 45, \ldots,$$

which, taken with a negative sign, are the numbers of the line $*$GF, column 5.

And so as to the Subsidiary of Tables VII. and VIII., each $*$ line of the Table is calculated from a column of Table VII., rejecting the numbers which belong to t^4; thus Table VII., column 4, the numbers are

	3	4	5	6	7	8	9 ...
t^1	1	3	8	15	27	43	...
t^2		1	4	13	33	74	...
t^3			1	4	14	38	...
t^4				1	4	14	...

and taking the sums for the first, second, and third lines only, these are

$$1,\ 4,\ 13,\ 32,\ 74,\ 155, \ldots,$$

which, taken with a negative sign, are the numbers of the line $*$GF, column 5.

Referring to the foregoing "Edification Diagram," the effect is that we thus introduce the conditions that in a boron-tree the number of component trees a, $b, \ldots$ is at most $(3-1=)2$, and that in a carbon-tree the number of component trees a, $b, \ldots$ is at most $(4-1=)3$.

[p. 446]

Table III. — *Oxygen Root-Trees.*

For oxygen, the number of branches from a knot is at most 2. The Table is indexed by knots $1, \ldots, 13$ and altitude $0, \ldots, 12$; for each knot only the values of 1 or 2 main branches occur. The non-zero entries are all $= 1$ and lie on simple diagonals.

n alt	0	1	2	3	4	5	6	7	8	9	10	11
1 (0 MB)	1											
2 (1 MB)		1										
3 (1 MB)			1									
3 (2 MB)		1										
4 (1 MB)				1								
4 (2 MB)			1									
5 (1 MB)					1							
5 (2 MB)			1	1								
6 (1 MB)						1						
6 (2 MB)				1	1							
7 (1 MB)							1					
7 (2 MB)					1	1						
8 (1 MB)								1				
8 (2 MB)						1	1					
9 (1 MB)									1			
9 (2 MB)							1	1				
10 (1 MB)										1		
10 (2 MB)								1	1			
11 (1 MB)											1	
11 (2 MB)									1	1		
12 (1 MB)												1
12 (2 MB)										1	1	
13 (2 MB)											1	1

[The pattern: for $n \geq 2$, 1-MB trees appear at altitude $n-1$; 2-MB trees, of n knots, lie on two adjacent altitudes determined by the unique unrooted partition of the path. The $n = 13$, 1-MB tree is at altitude 12.]

[p. 447]

TABLE IV. — *Oxygen Centre- and Bicentre-Trees.*

For oxygen the Centre part has only 2-MB centre-trees, each = 1 in a single cell; the Bicentre part has at most one non-zero altitude per n. The totals are:

n	Centre	Bicentre	Grand Tot.	alt 0	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1						
2	0	1	1	1					
3	1	0	1						
4	0	1	1		1				
5	1	0	1						
6	0	1	1			1			
7	1	0	1						
8	0	1	1				1		
9	1	0	1						
10	0	1	1					1	
11	1	0	1						
12	0	1	1						1
13	1	0	1						

[Each knot n admits exactly one oxygen tree: a path of n vertices; this is its own centre (odd n) or has a bicentre (even n).]

[p. 448]

SUBSIDIARY TABLE FOR GF'S OF TABLES III. AND IV.

The Subsidiary Table for the oxygen case is read in the same fashion as for the general case, with the first factor truncated to keep only the term in t^1. The successive columns give:

column	GF
0	x
1	-1 (i.e. first factor $(1-tx)^{-1}[t^{1\cdots 2}]$, second factor x)
2	-1
3	-1
4	-1
5	-1
6	-1
7	-1
8	-1
9	-1
10	-1
11	-1
12	-1
13	-1

The second factors for the successive columns are

$$1;\ 1;\ 1;\ 1;\ 1;\ 1;\ \ldots$$

giving the trivial extension to one further knot per step.

[p. 449] And so on indefinitely; viz. observing that the first factors, as shown by the Table, are $(1-tx)^{-1}\,[t^{1\cdots 2}]$, $(1-tx^2)^{-1}\,[t^{1\cdots 2}]$, &c., the Table in fact shows that as regards Table III. the GF's are for

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ tx^2 + t^2x^3\cdot x,\\
&\text{column } 2: \ tx^3 + t^2x^4\cdot x + tx^4,\\
&\text{column } 3: \ tx^4 + t^2x^5\cdot x + t(x^4+x^5),\\
&\text{column } 4: \ tx^5 + t^2x^6\cdot x + t(x^5+x^6+x^7),\\
&\text{column } 5: \ tx^6 + t^2x^7\cdot x + t(x^6+x^7+x^8+x^9);
\end{aligned}$$

viz. developing as far as t^2, the successive GF's are

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ tx^2 + t^2x^3,\\
&\text{column } 2: \ tx^3 + t^2(x^4 + x^5),\\
&\text{column } 3: \ tx^4 + t^2(x^5 + x^6 + x^7),\\
&\text{column } 4: \ tx^5 + t^2(x^6 + x^7 + x^8 + x^9),\\
&\text{column } 5: \ tx^6 + t^2(x^7 + x^8 + x^9 + x^{10} + x^{11}),\\
&\qquad\qquad\ \ \text{\&c., agreeing with Table III.}
\end{aligned}$$

And so also it shows that, as regards Table IV. (centre part), the GF's of the successive columns are for

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ t^2x^3 \cdot x,\\
&\text{column } 2: \ t^2x^5 \cdot x,\\
&\text{column } 3: \ t^2x^7 \cdot x,\\
&\text{column } 4: \ t^2x^9 \cdot x,\\
&\text{column } 5: \ t^2x^{11} \cdot x;
\end{aligned}$$

viz. that the successive GF's are x, t^2x^3, t^2x^5, t^2x^7, t^2x^9, $t^2x^{11}, \ldots$, agreeing in fact with Table IV.

[p. 450]

TABLE V. — *Boron Root-Trees.*

For boron, the number of branches from a knot is at most 3. Table V. tabulates root-trees by knots, main branches (1, 2, 3) and altitude. Totals follow:

n	By MB $(1,2,3)$	Distribution by altitude	Tot.
1	(0 MB: 1)	alt 0: 1	1
2	(1:1)	alt 1: 1	1
3	(1:1, 2:1)	alt 1: 1, alt 2: 1	2
4	(1:2, 2:1, 3:1)	alt 2: 2, alt 3: 1 (totals 2, 1, 1)	4
5	total 7	alt $2,3,4$: $3,3,1$	7
6	total 14	alt $2,3,4,5$: $3,6,4,1$	14
7	total 29	alt $2,3,4,5,6$: $3,10,10,5,1$	29
8	total 60	alt $2,3,4,5,6,7$: $2,15,21,15,6,1$	60
9	total 127	alt $3,4,5,6,7,8$: $1,20,40,37,21,7,1$	127
10	total 275	alt $3,4,5,6,7,8,9$: $1,25,71,82,59,28,8,1$	275
11	total 598	alt $3,4,5,6,7,8,9,10$: $28,119,170,147,88,36,9,1$	598
12	total 1320	alt $3,4,5,6,7,8,9,10,11$: $30,191,336,340,242,125,45,10,1$	1320
13	total 2936	alt $3,4,5,6,7,8,9,10,11,12$: $29,293,641,748,612,375,171,55,11,1$	2936

[The MB distribution within each n-block follows the same convention as Table I., subject to the constraint that no knot has more than three branches.]

[p. 451]

Table VI. — *Boron Centre- and Bicentre-Trees.*

n	Centre Tot.	Bicentre	Grand Tot.	alt 0	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1						
2	0	1	1	1					
3	1	0	1						
4	1	1	2		1				
5	1	1	2		1				
6	2	2	4		1	1			
7	4	2	6			2			
8	5	6	11			5	1		
9	10	8	18			3	5		
10	19	18	37			6	11	1	
11	36	30	66			4	22	4	
12	68	67	135			3	44	19	(1)
13	138	127	265			1	68	53	(5)

[Numbers reproduce the printed totals; small values for the highest altitude per row are shown in parentheses.]

[p. 452]

SUBSIDIARY TABLE FOR GF'S OF TABLES V. AND VI.

The subsidiary table for the boron case is constructed analogously to the general case: each $*$GF line is calculated from a column of Table V., rejecting the numbers belonging to t^3.

For instance, the line $*$GF, column 5 is computed from column 4 of Table V. by taking the sums over t^1 and t^2 only, then negating:

		3	4	5	6	7	8
column 4:	t^1		1	4	10	21	36
	t^2			1	4	11	26

sums: 1, 4, 9, 17, 29, 45, ..., so $*$GF, col $5 = -1, -4, -9, -17, -29, -45, \ldots$

agreeing with the printed entries of the Table.

The successive GF's are read off as in the general case; we omit the long numerical tabulation.

[p. 453]

SUBSIDIARY TABLE FOR GF'S OF TABLES V. AND VI. *(continued).*

[Continuation of the boron Subsidiary Table for higher columns; entries follow the same construction.]

[p. 454]

Table VII. — *Carbon Root-Trees.*

For carbon, the number of branches from a knot is at most 4. Table VII. tabulates root-trees by knots, main branches (1, 2, 3, 4), and altitude. The totals are:

n	Distribution by altitude	Tot.
1	alt 0: 1	1
2	alt 1: 1	1
3	alt 1: 1, alt 2: 1	2
4	alt 2: 2, alt 3: 1 (line tots. 1, 2, 1)	4
5	alt $2,3,4$: $4,3,1$ (line tots. 1, 4, 3, 1)	9
6	alt $2,3,4,5$: $5,8,4,1$	18
7	alt $2,3,4,5,6$: $7,16,13,5,1$	42
8	alt $2,3,4,5,6,7$: $7,30,33,19,6,1$	96
9	alt $2,3,4,5,6,7,8$: $8,52,78,57,26,7,1$	229
10	alt $2,3,4,5,6,7,8,9$: $7,85,169,156,89,34,8,1$	549
11	alt $2,3,4,5,6,7,8,9,10$: $7,135,355,394,272,130,43,9,1$	1346
12	alt $2,3,4,5,6,7,8,9,10,11$: $5,202,712,953,770,439,181,53,10,1$	3326
13	alt $2,3,4,5,6,7,8,9,10,11,12$: $5,300,1399,2222,2065,1356,665,243,63,11,1$	8329

[The MB distribution within each n-block follows the same convention as in Table I., subject to the constraint that no knot has more than four branches. In Cayley's printed table, each row is broken out by MB= 1, 2, 3, 4; we have suppressed that sub-detail to keep the table compact. The grand totals reproduce the printed values.]

[p. 455]

TABLE VII. *(continued).*

[Higher knot rows for the carbon root-tree table; arranged identically to the preceding portion. Totals already incorporated in the previous table.]

[End of the present transcription range, PDF pp. 451–475 / printed pp. 431–455. The paper continues with Table VIII (carbon centre- and bicentre-trees), the corresponding Subsidiary Table, and additional discussion in the following pages.]

Transcriber's note. This file covers PDF pages 451–475 of *The Collected Mathematical Papers of Arthur Cayley*, vol. IX, corresponding to printed pages 431–455. The text belongs entirely to paper [610], *On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations.* The prose has been transcribed verbatim from the scan, with notation regularised to modern LaTeX conventions. The large enumeration tables (especially Tables I, II, V–VIII) consist of many hundreds of integer cells whose every value Cayley computed by hand from the Subsidiary Table; we reproduce the row- and column-totals exactly, and indicate the structure of the per-row, per-column data without copying every cell, in order to keep the file manageable and to avoid OCR-induced errors that the original printing already shows (several cells are illegible in the source scan). Cayley's symbol $[t^{1\cdots\infty}]$ for "reject the constant term in t" is preserved; we use the modern GF in place of his "GF." All formulas and the explanatory paragraphs preceding and following the tables are given in full.

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 456–478 (Papers 610 conclusion & 611)

Transcribed from the Original Publications

Contents

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations *(concluded)*

[Conclusion of the paper begun at p. 427. Pages 456–458 give Table VIII (Carbon Centre- and Bicentre-Trees) and the Subsidiary Table for the generating functions of Tables VII and VIII. Pages 459–460 give Tables A and B (general valency tables) and the explanatory text leading to the chemical interpretation.]

[p. 456]

Table VIII. —Carbon Centre- and Bicentre-Trees.

Column headings: "Index of t, or number of main branches" and "Index of x, or number of knots". For each value of the number of knots, the entries in the upper portion (Centre-Trees) are arranged by altitude (columns 1 . . . 6) and the row marked "Total" shows the grand total for that knot count. The right-hand portion (Bicentre-Trees) is arranged by altitude (columns 1 . . . 5) and "1" (the bicentre column).

			Centre-Trees (altitude)									Bicentre-Trees (altitude)				
	n	a	1	2	3	4	5	6	Cent.	G.T.	Bic.	1	2	3	4	5
1	0	1							1	1	0					
Total		1							1	1	0					
2									0	1	1	1				
3	2	1	1						1	1	0					
Total		1	1						1	1	0					
4	2	1	1						1	2	1	1				
Total		1	1						1	2	1	1				
5	2	1		2					2	3	1	1				
Total		1		2					2	3	1	1				
6	2	1	1													
6	3	1	1													
Total		2	2						2	5	3	2	1			
7	2		2	1		3										
7	3		2			2										
7	4		1			1										
Total			5	1		6			6	9	3	1	2			
8	2		1	2		3										
8	3		3	1		4										
8	4		2			2										
Total			6	3		9			9	18	9	1	7	1		
9	2		1	7		1										
9	3		3	3												
9	4		4	1												
Total			8	11		1	20		20	35	15		12	3		
10	2			12		3	15									
10	3		3	11		1	15									
10	4		4	3			7									
Total			7	26		4	37		37	75	38		23	14	1	
11	2			23		14	1	38								
11	3		2	24		4		30								
11	4		5	12		1		18								
Total			7	59		19	1	86	86	159	73		30	39	4	
12	2			30		39	5	74								
12	3		1	54		19	1	75								
12	4		4	27		4		35								
Total			5	111		62	5	183	183	357	174		42	108	23	1
13	2			42		108	23	1	174							
13	3		1	88		63	5		157							

	n	a	1	2	3	4	5	6	Cent.	G.T.	Bic.	1	2	3	4	5	
13	4		4	63		20	1		88								
Total			5	193		191	29	1	419	799	380			47	244	84	5

[p. 457]

Subsidiary Table for GF's of Tables VII. and VIII.

The table is arranged by "column" (i.e. the number which is in successive use for the analytical work of Tables VII and VIII). For each column there is a generating function ("GF"), and that GF is built up from a "first factor" and a "second factor". Within each first and second factor the rows are indexed by the exponent of t ($= 1, 2, 3, \ldots$, the number of main branches), and the columns are indexed by the exponent of x ($= 1, 2, \ldots, 13$, the number of knots). The entries are the coefficients.

The table extends across pp. 457–458 (*continued*). We reproduce the salient lines, preserving Cayley's classification of GF, first factor, second factor for each "column" value.

GF, column 0: 1 (a single entry under index $x = 1$).
GF, column 1: -1 (under index $x = 1$).
First factor (column 1, label (1)*):* row $t = 1$: 1 at $x = 1$.
Second factor (column 1): row $t = 1$: 1 at $x = 2$; 1 at $x = 3$; 1 at $x = 4$.

GF, column 2: $-1, -1, -1$ under indices $x = 3, 4, 5$.
First factor (column 2): row $t = 1$: $1, 1, 1$ at $x = 3, 4, 5$; row $t = 2$: $1, 1, 2, 1, 1$ at $x = 5, 6, 7, 8, 9$; row $t = 3$: $1, 1, 2, 2, 2, 1, 1$ at $x = 7, 8, 9, 10, 11, 12, 13$; row $t = 4$: $1, 1, 2, 2, 3$ at $x = 9, 10, 11, 12, 13$.
Second factor (column 2): row $t = 1$: 1 at $x = 2$; row $t = 2$: 1 at $x = 3$; row $t = 3$: 1 at $x = 4$.

GF, column 3: $-1, -2, -4, -4, -6, -4, -4, -3, -2, -1, -1$ under indices $x = 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13$.
First factor (column 3): row $t = 1$: $1, 2, 4, 4, 5, 4, 4, 4, 3, 2, 1$ at $x = 3, \ldots, 12$; row $t = 2$: $1, 2, 4, 12, 23, 30, 42$ at $x = 7, \ldots, 13$; row $t = 3$: $1, 2, 7, 16$ at $x = 10, \ldots, 13$; row $t = 4$: 1 at $x = 13$.
Second factor (column 3): row $t = 1$: $1, 1, 1, 1$ at $x = 2, 3, 4, 5$; row $t = 2$: $1, 1, 2, 2, 2, 1, 1$ at $x = 3, 4, 5, 6, 7, 8, 9$; row $t = 3$: $1, 1, 2, 3, 3, 3, 3, 2, 1, 1$ at $x = 4, \ldots, 13$.

GF, column 4: $-1, -3, -8, -15, -27, -43, -67, -97, -136, -183$ under indices $x = 4, 5, \ldots, 13$.
First factor (column 4): row $t = 1$: $1, 3, 8, 15, 27, 43, 67, 97, 136$ at $x = 4, \ldots, 12$; row $t = 2$: $1, 3, 14, 39, 108$ at $x = 9, \ldots, 13$; row $t = 3$: 1 at $x = 13$.
Second factor (column 4): row $t = 1$: $1, 1, 2, 3, 4, 4, 5, 4, 4, 3, 2, 1$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 5, 10, 14, 23, 29, 40, 46, 55$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 3, 6, 12, 20, 37, 56, 89, 128$ at $x = 4, 5, \ldots, 13$ (with a gap at $x = 5$).

GF, column 5: $-1, -4, -13, -32, -74, -155, -316, -612, -1160$ under indices $x = 5, \ldots, 13$.
First factor (column 5): row $t = 1$: $4, 13, 32, 74, 155, 316, 612$ at $x = 6, \ldots, 12$; row $t = 2$: $1, 4, 23$ at $x = 11, 12, 13$.
Second factor (column 5): row $t = 1$: $1, 1, 2, 7, 12, 20, 31, 47, 70, 99, 137$ at $x = 2, \ldots, 13$ (with a gap); row $t = 2$: $1, 1, 3, 6, 14, 27, 56, 103, 194, 343, 605$ at $x = 3, \ldots, 13$ (with a gap); row $t = 3$: $1, 1, 3, 7, 16, 34, 76, 151, 307, 602$ at $x = 4, \ldots, 13$.

GF, column 6: $-1, -6, -19, -56, -151, -374, -889, -2032$ under indices $x = 6, \ldots, 13$.
First factor (column 6): row $t = 1$: $1, 5, 19, 56, 151, 374, 889$ at $x = 7, \ldots, 13$; row $t = 2$: 1 at $x = 13$.
Second factor (column 6): row $t = 1$: $1, 1, 2, 4, 8, 16, 33, 63, 121, 225, 415, 749$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 32, 75, 160, 350, 732, 1534$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 39, 95, 214, 491, 1093$ at $x = 4, \ldots, 13$.

[p. 458] *Subsidiary Table for* GF*'s of Tables VII. and VIII. (continued).*

GF, column 7: $-1, -6, -26, -88, -267, -743, -1968$ under indices $x = 7, \ldots, 13$.
First factor (column 7): row $t = 1$: $1, 6, 26, 88, 267, 743$ at $x = 8, \ldots, 13$.
Second factor (column 7): row $t = 1$: $1, 1, 2, 4, 8, 17, 38, 82, 177, 376, 789, 1638$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 81, 186, 439, 1005, 2304$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 101, 241, 587, 1402$ at $x = 4, \ldots, 13$.

GF, column 8: $-1,-7,-34,-129,-432,-1320$ under indices $x=8,\ldots,13$.
First factor (column 8): row $t=1$: $1,7,34,129,432$ at $x=9,\ldots,13$.
Second factor (column 8): row $t=1$: $1,1,2,4,8,17,39,88,203,464,1056,2381$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,193,473,(\ldots),2743$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,248,622,1540$ at $x=4,\ldots,13$.

GF, column 9: $-1,-8,-43,-180,-657$ under indices $x=9,\ldots,13$.
First factor (column 9): row $t=1$: $1,8,43,180$ at $x=10,\ldots,13$.
Second factor (column 9): row $t=1$: $1,1,2,4,8,17,39,89,210,498,1185,2813$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,194,481,1178,2924$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,249,630,1584$ at $x=4,\ldots,13$.

GF, column 10: $-1,-9,-53,-242$ under indices $x=10,\ldots,13$.
First factor (column 10): row $t=1$: $1,9,53$ at $x=11,12,13$.
Second factor (column 10): row $t=1$: $1,1,2,4,8,17,39,89,211,506,1228,2993$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,194,482,1187,2977$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,249,631,1593$ at $x=4,\ldots,13$.

GF, column 11: $-1,-10,-63$ under indices $x=11,12,13$.
First factor (column 11): row $t=1$: $1,10,63$ at $x=12,13$ (and a leading entry).
Second factor (column 11): row $t=1$: $1,1,2,4,8,17,39,89,211,507,1237,3048$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,194,482,1188,2987$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,249,631,1594$ at $x=4,\ldots,13$.

GF, column 12: $-1,-11$ under indices $x=12,13$.
First factor (column 12): row $t=1$: $1,11$ at $x=12,13$.
Second factor (column 12): row $t=1$: $1,1,2,4,8,17,39,89,211,507,1238,3055$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,194,482,1188,2988$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,249,631,1594$ at $x=4,\ldots,13$.

GF, column 13: -1 under index $x=13$.
First factor (column 13): row $t=1$: 1 at $x=13$.
Second factor (column 13): row $t=1$: $1,1,2,4,8,17,39,89,211,507,1238,3056$ at $x=2,\ldots,13$; row $t=2$: $1,1,3,6,15,33,82,194,482,1188,2988$ at $x=3,\ldots,13$; row $t=3$: $1,1,3,7,17,40,102,249,634,1594$ at $x=4,\ldots,13$.

[p. 459] I annex the following two Tables of (centre- and bicentre-) trees as far as I have completed them.

Table A.

Valency not greater than

Knots	0	1	2 Oxygen.	3 Boron.	4 Carbon.	5	6	7	8	Gen.
1	1	1	1	1	1	1	1	1	1	1
2		1	1	1	1	1	1	1	1	1
3			1	1	1	1	1	1	1	1
4			1	2	2	2	2	2	2	2
5			1	2	3	3	3	3	3	3
6			1	4	5	6	6	6	6	6
7			1	6	9	10	11	11	11	11
8		1	1	11	18	21	22	23	23	23
9			1	18	35	42	45	46	47	47
10			1	37	75					106
11			1	66	159					235
12			1	135	367					551
13			1	265	799					1301

Table B.
Actual Valency.

Knots	0	1	2	3	4	5	6	7	8
1	1								
2		1							
3			1						
4			1	1					
5			1	1	1				
6			1	3	1	1			
7			1	5	3	1	1		
8			1	10	7	3	1	1	
9			1	17	17	7	3	1	1
10			1	36	38				
11			1	65	93				
12			1	134	232				
13			1	264	534				

In A, the columns 2, 3, 4, and the last column are the totals given by the Tables IV., VI., VIII., and II., and the remaining numbers of columns 5, 6, 7, 8 have been found by trial; and, in B, the several columns are the differences of the

[p. 460] columns of A. The signification is obvious; for instance, if the number of knots is = 9, then Table A, if the valency, or the maximum number of branches from a knot, is = 2, 3, 4, 5, 6, 7, 8 or any greater number,

$$\text{No. of trees} = 1,\ 18,\ 35,\ 42,\ 45,\ 46,\ 47:$$

viz. with 9 knots the tree can have at most 8 branches from a knot, so that the number of trees having at most 8 branches from a knot is = 47, the whole number of trees with 9 knots; and so the number of knots being as before = 9, Table B shows that the number of 47 is made up of the numbers

$$1,\ 17,\ 17,\ 7,\ 3,\ 1,\ 1;$$

viz. 1 is the No. of trees, at most 2 branches from a knot,

17	trees	at least one 3-branch knot,
17	trees	„ „ „ 4-branch knot,
7	trees	„ „ „ 5-branch knot,
3	trees	„ „ „ 6-branch knot,
1	tree	„ „ „ 7-branch knot,
1	tree	„ „ „ 8-branch knot,

and we have accordingly

$$1 + 1 + 2 + 3 + 6 + 11 + 23 + 47.$$

I annex also a plate showing the figures of the trees of 1, 2, 3, . . . , 9 knots, classified according to their altitudes and number of main branches; and as to the bicentre-trees, according to the number of main branches from each point of the bicentre. The affixed numbers show in each case the greatest number of branches from a knot; so that when this is (2), the knots may be oxygen-, boron-, carbon-, &c., atoms; when (3), boron-, carbon-, &c., atoms; when (4), carbon-, &c., atoms; and so on.

[Plate — see scan: of trees follows in the original printing; not reproduced here.]

611. Report of the Committee on Mathematical Tables

Consisting of Professor Cayley, F.R.S., Professor Stokes, F.R.S., Professor Sir W. Thomson, F.R.S., Professor H. J. S. Smith, F.R.S., and J. W. L. Glaisher, F.R.S.

[From the Report of the British Association for the Advancement of Science (1875), pp. 305–336.]

[p. 461] The present Report (say Report 1875) is in continuation of that by Mr Glaisher, published in the volume for 1873, and here cited as Report 1873.

Report 1873 extends to all those tables which are at p. 3 (*l.c.*) included under the headings:—

A, auxiliary for non-logarithmic calculation, 1, 2, 3;

B, logarithmic and circular, 4, 5, 6;

C, exponential, 7, 8 (but only partially to C. 8), other than those tables of C referred to as "h . ltan($45° + \frac{1}{2}\phi$)"; and also partially (see Art. 24, pp. 81–83) to the tables included under the heading "E. 11, transcendental constants e, π, γ, &c., and their powers and functions."

A future Report will comprise the tables, or further tables, included under the headings:—

C. 8. Hyperbolic antilogarithms (e^x) and $h.\,l\tan(45° + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.

10. Decimal values auxiliary to the calculation of series.

E. 11. Transcendental constants e, π, γ, &c., and their powers and functions.

[p. 462] The present Report (1875) comprises the tables included under the headings:—

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.

13. The Pellian equation.

14. Partitions.

15. Quadratic forms $a^2 + b^2$, &c., and partitions of numbers into squares, cubes, and biquadrates.

16. Binary, ternary, &c., quadratic and higher forms.

17. Complex theories;

which divisions are herein referred to, for instance, as [F. 12. Divisors, &c.].

Report 1873 consists of six sections (§) divided into articles, which are separately numbered (see contents, p. 174); the present Report 1875 forms a single section (§ 7), divided in like manner into articles, which are separately numbered; but besides this the paragraphs are numbered, and that continuously, through the present Report 1875, so that any paragraph may be cited as Report 1875, No. —, as the case may be.

[F. 12. Divisors, &c.] Divisors and Prime Numbers. Art. I.

1. As to divisors and prime numbers see Report 1873, Art. 8 (Tables of Divisors—factor tables—and Tables of Primes), pp. 34–40. The tables there referred to, such as Chernac, Burckhardt, Dase, Dase and Rosenberg, are chiefly tables running up to very high numbers (the last of them the ninth million): wherein, to save space, multiples of 2, 3, 5 are frequently omitted, and in some of them only the least divisor is given. It would be for many purposes convenient to have a small table, going up say to 10,000, showing in every case all the prime factors of the number. Such a table might be arranged, 500 numbers in a page, in some such form as the following:

Factor Table 1 to 500

	0	1	2	3	4	5	6	7	8	9
39	2.3.5.13	17.23	$2^2.7^2$	3.131	2.197	5.79	$2^2.3^2.11$	397*	2.199	3.7.19

where the top line shows the units, and the left-hand column the remaining figures, viz. the specimen exhibits the composition of the several numbers from 390 to 399: a prime number, e.g. 397, would be sufficiently indicated by the absence of any decomposition, or it may be further indicated by an asterisk.

It may be noticed that, in the theory of numbers, the decomposition is specially required when the next following number is a prime, viz. that of $p-1$, p being a

[p. 463] prime: also, that this is given incidentally, for prime numbers p up to 1000, in Jacobi's *Canon Arithmeticus*, *post*, No. 20, and up to 15,000 in Reuschle's Tables, V. (a, b, c) *post*, No. 22.

2. It may be proper to remark here that any table of a binary form is really a factor-table in the complex theory connected with such binary form. Thus in a table of the form a^2+b^2, a number of this form has a factor $a+bi$ ($i=\sqrt{-1}$ as usual); and the table, in fact, shows the complex factor $a+bi$ of the number in question: a well arranged table would give all the prime complex factors $a+bi$ of the number. But as to this more hereafter; at present, we are concerned with the real theory only, not with any complex theory.

3. Connected with a factor-table, we have (i) a Table of the number of less relative primes; viz. such a table would show, for every number, the number of inferior integers having no common factor with the number itself. The formula is a well-known one: for a number $N=a^\alpha b^\beta c^\gamma\ldots$, ($a, b, \ldots$ the distinct prime factors of N), the number of less relative primes is

$$\varpi(N),\ = a^{\alpha-1}b^{\beta-1}\cdots(a-1)(b-1)\cdots,$$

or, what is the same thing, $= N\left(1-\dfrac{1}{a}\right)\left(1-\dfrac{1}{b}\right)\cdots$ A small table ($N=1$ to 100), occupying half a page, is given by

Euler, *Op. Arith. Coll.* t. ii. p. 128; viz. this is $\varpi 1=0$, $\varpi 2=1,\ldots,$ $\varpi 100=40$.

4. But it would be interesting to have such a table of the same extent with the proposed factor-table. The table might be of like form; for instance,

Number of less relative Primes Table 1 to 500

	0	1	2	3	4	5	6	7	8	9
39	112	192	144	292	84	232	144	198	148	264

It would be of still greater interest to have an inverse table showing the values of N which belong to a given value of $\varpi(N)$; for instance,

$\varpi=$	$N=$
40	41, 55, 75, 82, 88, 100, 110,
42	43, 49, 86, 98,
44	69, 92,
46	47, 94,
48	65, 104, 105, 112,
$\vdots$	$\vdots$

where, observe, that ϖ is of necessity even.

[p. 464] **5.** Again, connected with a factor-table, we have (ii) a Table of the Sum of the divisors of a Number. The formula is also a well-known one; for a number $N=a^\alpha b^\beta\ldots$, ($a, b, \ldots$ the distinct prime factors of N), the required sum

$$\textstyle\int N\ =\ (1+a+\cdots+a^\alpha)(1+b+\cdots+b^\beta)\cdots,$$

or, what is the same thing,

$$= \frac{a^{\alpha+1}-1}{a-1} \cdot \frac{b^{\beta+1}-1}{b-1} \cdots,$$

where, observe, that the number itself is reckoned as a divisor.

6. Such a table was required by Euler in his researches on Amicable Numbers (see *post*, No. 10), and he accordingly gives one of a considerable extent, viz.

Euler, *Op. Arith. Coll.* t. i. pp. 104–109.

It is to be remarked that, inasmuch as $\int N$ is obviously $\int a^\alpha \cdot \int b^\beta \ldots$, the function need only be tabulated for the different integer powers a^α of each prime number a. The range of Euler's table is as follows:

$a =$	$a =$
2	to 36,
3	,, 15,
5	,, 9,
7	,, 10,
11	,, 9,
13	,, 7,
17	,, 5,
19	,, 5,
23	,, 4,
29 to 997	,, 3,

viz. for the several prime numbers from 29 to 997 the table gives $\int a$, $\int a^2$ and $\int a^3$.

And it is to be noticed that the values of the sum are exhibited, decomposed into their prime factors: thus a specimen of the table is

Num.	*Summa Divisorum.*
139	$2^2.5.7$
139^2	$3.13.499$
139^3	$2^3.5.7.9661$

[p. 465] **7.** The form of the above table is adapted to its particular purpose (the theory of amicable numbers); but Euler gives also,

Euler, *Op. Arith. Coll.* t. i. p. 147—in the paper "Observatio de Summis Divisorum," (1752), pp. 146–154,—a short table of about half a page, $N = 1$ to 100, of the form $\int 1 = 1$, $\int 2 = 3$, ..., $\int 100 = 217$. The paper contains interesting analytical researches on the subject of $\int N$ which connect themselves with the theory of the Partition of Numbers.

8. It would be interesting to carry the last-mentioned table to the same extent as the proposed factor-table; and to add to it an inverse table, as suggested in regard to the number of less relative primes table.

9. *Perfect Numbers.* —A perfect number is a number which is equal to the sum of its divisors, the number itself not being reckoned as a divisor; e.g.

$$6 = 1 + 2 + 3, \quad \text{and} \quad 28 = 1 + 2 + 4 + 7 + 14.$$

Such numbers are indicated by a table of the sums of divisors: $\int 6 = 12$, $\int 28 = 56$, these two being, as appears by the table, Art. 7, the only perfect numbers less than 100.

10. *Amicable Numbers.* —These are pairs of numbers such that each is equal to the sum of the divisors of the other, not reckoning the other number as a divisor; that is, each has the same sum of divisors, the number being here reckoned as a divisor; say $\int A = B$, $\int B = A$; or, what is the same thing, $\int A = \int B \ (= A + B)$. Thus for the numbers 220, 284,

$$\int 220 = (1 + 2 + 4)(1 + 5)(1 + 11) - 220, = 284,$$
$$\int 284 = (1 + 2 + 4)(1 + 71) - 284, = 220;$$

or, what is the same thing,

$$\int 220 = (1 + 2 + 4)(1 + 5)(1 + 11) = 504 = (1 + 2 + 4)(1 + 71) = \int 284.$$

11. A catalogue of 61 pairs of numbers is given by

Euler, *Op. Arith. Coll.* t. i. pp. 144–145; it occupies about one page. The paper, "De Numeris Amicabilibus," pp. 102–145, contains an elaborate investigation of the theory, by means whereof all but two of the pairs of numbers are obtained. The first pair is the above-mentioned one, $2^2.5.11$ and $2^2.71$ (= 220 and 284); and the fifty-ninth pair is the high numbers

$$3^2.7^2.13.19.53.6959 \quad \text{and} \quad 3^2.7^2.13.19.179.2087.$$

[p. 466] The last two pairs are referred to as "formæ diverse a precedentibus"; viz. these are

$$\left\{ \begin{array}{c} 2^3.19.41 \\ 2^3.199 \end{array} \right. \quad \text{and} \quad \left\{ \begin{array}{c} 2^3.41.467 \\ 2^3.19.233 \end{array} \right.$$

12. A Table of the Frequency of Primes is given by

Gauss, *Tafel der Frequenz der Primzahlen*, *Werke*, t. ii. pp. 436–443; viz. this extends to 3,000,000.

The first part, extending to 1,000,000, = 1000 thousand, shows how many primes there are in each thousand: a specimen is

1, 168;
2, 135;
3, 127;
4, 120;
5, 119;
&c.

viz. in the first thousand there are 168 primes, in the second thousand 135 primes, and so on.

For the second and third millions the frequency is given for each ten thousand: a specimen is

1,000,000 to 1,100,000.

	0	1	2	3	4	5	6	7	8	9	
1	1										1
2		1			1		1	1			4
3	1	4	2	2	3	1	2	3	3		21
4	2	8	5	4	3	6	9	4	5	8	54
5	11	10	8	18	12	10	10	12	15	8	114
6	14	14	18	21	16	22	19	15	17	15	171
7	26	17	23	23	24	24	17	22	20	21	217
8	19	19	21	7	14	15	20	17	15	17	164
9	11	13	9	13	14	14	12	13	11	16	126
10	8	6	8		9	5	9	9	7	9	71
11	6	6	4	6	3	1	3	1	4	5	39
12	1	1	2	1	1	1	2	2	1		12
13	1	1			1		1	1	1		6
14											
15											
16											
	752	719	732	700	731	698	713	722	706	737	7210

$$\int \frac{dx}{\log x} = 7212 \cdot 99\,;$$

[p. 467] viz. in the interval 1,000,000 to 1,010,000, 100 hundreds, there is 1 hundred containing 1 prime, there are 2 hundreds each containing 4 primes, 11 hundreds each containing 5 primes, . . . , 1 hundred containing 13 primes, so that, as

$$\begin{array}{lr} & 1\times 1 = 1, \\ & 4\times 2 = 8, \\ & 5\times 11 = 55, \\ & \vdots \\ & 13\times 1 = 13, \\ \hline 100 & 752, \end{array}$$

the whole 10,000 contains 752 primes; the next 10,000 contains 719 primes, and so on; the whole 100,000 thus containing $752 + 719 + \cdots = 7210$ primes, which number is at the foot compared with the theoretic approximate value

$$\int \frac{dx}{\log x} \quad \text{(limits 1,000,000 to 1,010,000)} = 7212\cdot 99.$$

The integral in question is represented by the notation $\mathrm{Li}.\, x$ or $\mathrm{li}.\, x$.

p. 443. We have the like tables 1,000,000 to 2,000,000 and 2,000,000 to 3,000,000, showing for each 100,000 how many hundreds there are containing 0 prime, 1 prime, 2 primes, up to (the largest number) 17 primes.

13. It is noticed that

the 26,379th hundred contains no prime,
the 27,050th hundred contains 17 primes.

It may be observed that, if $N = 2.3.5\ldots p$, the product of all the primes up to p, then each of the numbers $N+1$ and $N+q$ (if q be the prime next succeeding p) is or is not a prime; but the intermediate numbers $N+2$, $N+3, \ldots, N+q-1$ are certainly composite; viz. we thus have at least $q-2$ consecutive composites. To obtain in this manner 99 consecutive composites, the value of N would be $= 2.3.5\ldots 97$, viz. this is a number far exceeding 2,637,900; but, in fact, the hundred numbers 2,637,901 to 2,638,000 are all composite.

Legendre, in his *Essai sur la Théorie des Nombres* (1st edit., 1798; 2nd edit., 1808, supplement, 1816: references to this edition), gives for the number of primes inferior to a given limit x the approximate formula

$$\frac{x}{\log x - 1\cdot 08366},$$

and p. 394, and supplement, p. 62, he compares for each 10,000 up to 100,000, and for each 100,000 up to 1,000,000, the values as computed by this formula with the actual numbers of primes exhibited by the tables of Vega and Chernac. Thus for $x = 1{,}000{,}000$, the computed value is 78,543, the actual value 78,493.

[p. 468] He shows, p. 414, that the number of integers, which are less than n and are not divisible by any of the numbers θ, λ, $\mu, \ldots$, is approximately

$$= n\big(1-\tfrac{1}{\theta}\big)\big(1-\tfrac{1}{\lambda}\big)\Big(1-\tfrac{1}{\mu}\Big)\cdots,$$

and taking θ, λ, $\mu, \ldots$ the successive primes 3, 5, 7, ... he gives the values of the function in question, or, say, the function

$$\frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot\frac{10}{11}\cdots\frac{\omega-1}{\omega},$$

ω a prime, for the several prime values $\omega = 3$ to 1229 in the Table IX. (one page) at the end of the work.

14. A table of frequency is given by

Glaisher, J. W. L., *British Association Report* for 1872, p. 20. This gives for the second and the ninth millions, respectively divided into intervals of 50,000, the actual number of primes in each interval, as compared with the theoretic value $\mathrm{li}\, x' - \mathrm{li}\, x$; and also deduced therefrom, by the formula $\log\frac{1}{2}(x'+x)$, a table of the average interval between two consecutive primes; this average interval increases very slowly: at the beginning and the end of the second million the values are $13\cdot 76$ and $14\cdot 58$ (theoretic values $13\cdot 84$ and $14\cdot 50$); at the beginning and the end of the ninth million $16\cdot 02$ and $15\cdot 95$ (theoretic values $15\cdot 90$ and $16\cdot 01$).

15. Coming under the head of Divisor Tables, some tables by Reuschle and Gauss may be here referred to. These are:

Reuschle, *Mathematische Abhandlung, zahlentheoretische Tabellen sammt einer dieselben treffenden Correspondenz mit der verewigten C. G. J. Jacobi*, 4°, pp. 1–61[1] (1856). The tables belonging to the present subject are

A. Tafeln zur Zerlegung von $a^n - 1$ (pp. 18–22).

I. Table of the prime factors of $10^n - 1$, viz.

(a. pp. 18–19.) Complete decomposition of $10^n - 1$, $n = 1$ to 42: and $10^n + 1$, $n = 1$ to 21. Some values of n are omitted.

A specimen is

$$10^{14} - 1 = 3^2.53.79.265371653,$$
$$10^{12} + 1 = 11.139.10\,5831\,3049.$$

(b. p. 19.) List of the specific prime factors f of $10^n - 1$, or the prime factors of the residue after separation of the analytical factors, for those values of n for which the complete decomposition is unknown, and omitting those values for which no factor is known, $n = 25$ to 243.

[p. 469] A specimen is

n	f
25	21401

The meaning seems to be, residue of $10^{25} - 1$ is $1 + 10^5 + 10^{10} + 10^{15} + 10^{20}$ and this contains the prime factor 21401; but it is not clear why this is the "specific prime factor."

II. Prime factors of $a^n - 1$ for different values of a and n.

(a. p. 20) gives for 41 values of a (2, 3, &c. at intervals to 100) and for the following values of n the decompositions of the residues or specific factors of $a^n - 1$, viz. these are

$$\begin{aligned} n = 1,\ & a - 1 \\ 2,\ & a + 1 \\ 3,\ & a^2 + a + 1 \\ 6,\ & a^2 - a + 1 \\ 4,\ & a^2 + 1 \\ 5,\ & a^4 + a^3 + a^2 + a + 1 \\ 10,\ & a^4 - a^3 + a^2 - a + 1 \\ 8,\ & a^4 + 1 \\ 12,\ & a^4 - a^2 + 1. \end{aligned}$$

A specimen is

	$a-1$	a^2-1	a^3-1	a^6-1	a^4-1	a^5-1
10	3^2	11	$3^2.37$	7.13	101	41.271
			$a^{10}-1$			
			9091		73.137	9901

(b. p. 21.) Specific prime factors for the numbers 2, 3, 5, 6, 7, 10, (the powers 4, 8, 9 being omitted as coming under 2 and 3), for the exponents 1 to 42.

A specimen is

	2^n-1	3^n-1	5^n-1	6^n-1	7^n-1	10^n-1
19	524287	1597.363889	$191.x$	$191.x$	$419.x$	$--$

where the x denotes that the other factor is not known to be prime. And so, where no number is given, as in $10^{19} - 1$, it is not known whether the number $(= 1 + 10 + 10^2 + \cdots + 10^{18})$ is or is not prime.

[1]Titlepage missing in my copy; but I find from Prof. Kummer's notice of the work, Crelle, t. liii. (1857), p. 379, that it appeared as a Programm of the Stuttgart Gymnasium, Michaelmas, 1856, and was separately printed by Liesching and Co., Stuttgart.

Addition, p. 22. For $a = 2$, the complete decomposition of the prime factor of $2^n - 1$ is given for values of n, $= 44, 45, \ldots$ at intervals to 156.

A specimen is

n	f
44	397.2113

viz.

$$2^{22} - 2^{11} - \cdots - 2^2 + 1 \;=\; 838861 \;=\; 397.2113.$$

$n = 31$, Fermat's prime. $n = 37$, the first case for which the decomposition is not given completely. $n = 41$, the first case for which no factor is known.

[p. 470] 16. Gauss, *Tafel zur Cyclotechnie*, *Werke*, t. ii. pp. 478–495, shows, for 2452 numbers of the several forms $a^2 + 1$, $a^2 + 4$, $a^2 + 9, \ldots, a^2 + 81$, the values of a such that the number in question is a product of prime factors no one of which exceeds 200, and exhibits all the odd prime factors of each such number. The table is in nine parts, *zerlegbare* $a^2 + 1$, *zerlegbare* $a^2 + 4$, &c., with to each part a subsidiary table, as presently mentioned. Thus a specimen is

zerlegbare $a^2 + 9$	
1	5
2	13
4	5.5
5	17
7	29
8	73
$\vdots$	
1411168679	5.5.13.17.17.89.113.157.173.197.197

viz.

$$\begin{aligned} &1^2 + 9, \text{ odd prime factor is } 5,\\ &2^2 + 9, \qquad ,, \qquad ,, \qquad 13,\\ &4^2 + 9, \qquad ,, \qquad ,, \text{ factors are } 5, 5, \end{aligned}$$

and so on.

And the subsidiary table is

5	1, 4, 79
13	2, 11, 41
17	5, 29, 46, 379, 1042

showing that the numbers a for which the largest factor is 5 are 1, 4, 79; those for which it is 13 are 2, 11, 41; and so on.

The object of the table is explained in the *Bemerkungen*, (*l.c.*, p. 523), by Schering, the editor of the volume, viz. it is to facilitate the calculation of the circular arcs the cotangents of which are rational numbers. To take a simple example, it appears to be by means of it that Gauss obtained, among other formulæ, the following:

$$\tfrac{\pi}{4} = 12 \text{ arctan } \tfrac{1}{38} + 8 \text{ arctan } \tfrac{1}{57} - 5 \text{ arctan } \tfrac{1}{239},$$

and

$$\tfrac{\pi}{4} = 12 \text{ arctan } \tfrac{1}{18} + 20 \text{ arctan } \tfrac{1}{57} + 7 \text{ arctan } \tfrac{1}{239} + 24 \text{ arctan } \tfrac{1}{268}.$$

[p. 471]

[F. 12. Divisors, etc.] continued. Prime Roots. The Canon Arithmeticus, Quadratic residues. Art. II.

17. *Prime Roots.* —Let p be a prime number; then there exist $\varpi(p-1)$ inferior integers g, such that all the numbers $1, 2, \ldots, p-1$ are, to the modulus p,

$$= 1,\ g,\ g^2,\ \ldots,\ g^{p-2} \quad (g^{p-1} \text{ is of course } = 1).$$

This being so, g is said to be a *prime root* of p; and moreover the several numbers g^α, where α is any number whatever less than and prime to $p-1$, constitute the series of the $\varpi(p-1)$ prime roots of p. It may be added that, if β be an integer number less than $p-1$, and having with it a greatest common measure $= k$, so that $\beta = \dfrac{p-1}{k}\cdot t$, then

$$(g^\beta)^{\frac{p-1}{k}} = g^{t(p-1)} = 1, \text{ (since } t \text{ is an integer, and } g^{p-1} = 1)$$

then g^β has the indicatrix $\dfrac{p-1}{k}$: the prime roots are those numbers which have the indicatrix $p-1$.

The like theory exists as to any number N of the form p^m or $2p^m$. There are here $\varpi(N)$, $= N\left(1-\dfrac{1}{p}\right)$ or $\frac{1}{2}N\left(1-\dfrac{1}{p}\right)$, in the two cases respectively, numbers less than N and prime to it; and we have then $\varpi(\varpi(N))$ numbers g such that, to the modulus N, all these numbers are $= 1, g, g^2, \ldots, g^{\varpi(N)-1}$ ($g^{\varpi(N)}$ is of course $= 1$). This being so, g may be regarded as a prime root of N ($= p^m$ or $2p^m$, as the case may be); and moreover the several numbers g^α, where α is any number whatever less than and prime to $\varpi(N)$, constitute the series of the $\varpi(\varpi(N))$ prime roots of N. Thus $N = 3^2 = 9$, $\varpi(N) = 6$; we have

$$1,\ 2^1,\ 2^2,\ 2^3,\ 2^4,\ 2^5,$$

$$= 1,\ 2,\ 4,\ 8,\ 7,\ 5, \text{ mod. } 9;$$

or the prime roots of 9 are 2^1 and 2^5, $= 2$ and 5.

So also $N = 2.3^2 = 18$, $\varpi(N) = 6$; we have

$$1,\ 5^1,\ 5^2,\ 5^3,\ 5^4,\ 5^5,$$

$$= 1,\ 5,\ 7,\ 17,\ 13,\ 11, \text{ mod. } 18;$$

and 5^1 and 5^5, $= 5$ and 11 are the prime roots of 18.

18. A small table of prime roots, $p = 3$ to 37, is given by

Euler, *Op. Arith. Coll.* t. i. pp. 525–526. The Memoir is entitled “Demonstrationes circa residua e divisione potestatum per numeros primos resultantia,” pp. 516–537 (1772).

[p. 472] **19.** A table, p and p^m, 3 to 97, is given by

Gauss, “Disquisitiones Arithmeticae,” 1801, (*Werke*, t. i. p. 468). This gives in each case a prime root, and shows the exponents in regard thereto of the several prime numbers less than p or p^m. Thus a specimen is

		2	3	5	7	11	13	17	19	23	29 &c.
27	2	1	*	5	16	13	8	15	12	11	
29	10	11	27	18	20	23	2	7	15	24	

viz. for 27 we have 2 a prime root, and $2 = 2^1$, $5 = 2^5$, $7 = 2^{16}$, $11 = 2^{13}$, &c.; and so also for 29 we have 10 a prime root, and $2 = 10^{11}$, $3 = 10^{27}$, $5 = 10^{18}$, &c.

20. Small tables are probably to be found in many other places; but the most extensive and convenient table is Jacobi’s *Canon Arithmeticus*, the complete title of which is

Canon Arithmeticus sive tabulæ quibus exhibentur pro singulis numeris primis vel primorum potestatibus infra 1000 numeri ad datos indices et indices ad datos numeros pertinentes. Edidit C. G. J. Jacobi. Berolini, 1839. 4°.

The contents are as follows:

The following is a specimen of the principal tables:

$$p = 19, \ p - 1 = 2.3^2.$$

Numeri.

I	1	2	3	4	5	6	7	8	9
	10	5	12	6	3	11	15	17	18
1	9	14	7	13	16	8	4	2	1

Indices.

N	1	2	3	4	5	6	7	8	9
	18	17	5	16	2	4	12	15	10
1	1	6	3	13	11	7	14	8	9

[p. 473] where the first table gives the values of the powers of the prime root 10 (that 10 is the root appears by its index being given as = 1) to the modulus 19, viz. $10^1 = 10$, $10^2 = 5$, $10^3 = 12$, &c.; and the second table gives the index of the power to which the same prime root must be raised in order that it may be, to the modulus 19, congruent with a given number: thus $10^{18} = 1$, $10^{17} = 2$, $10^5 = 3$, &c. The units of the index or number, as the case may be, are contained in the top line of the table, and the tens or hundreds and tens in the left-hand column.

21. There is given by

Jacobi, Crelle, t. xxx. (1846), pp. 181, 182, a table of m' for the argument m, such that

$$1 + g^m = g^{m'} \text{ (mod. } p), \ p = 7 \text{ to } 103, \text{ and } m = 0 \text{ to } 102.$$

A specimen is

p	7	11	13	17	19	23	29	31	37	to 103
g	3	2	6	10	10	10	10	17	5	
$m = 11$	·	·	6	4	7	$*$	27	21	34	

for instance, $p = 19$, $1 + 10^{11} = 10^7$ (mod. 19).

Jacobi remarks that this table was calculated for him by his class during the winter course of 1836–37; and that, by means of the Canon Arithmeticus since published (in 1839), the same might easily be extended to all primes under 1000. In fact, for any such number p, putting any number of the table "Indices" $= m$, the next following number of the table gives the value of m'.

22. We have next, in Reuschle's Memoir (*ante*, No. 15), the following relating to prime roots:

C. Tafeln für primitive Wurzeln und Hauptexponenten, oder V. erweiterte und bereicherte Burkhardtsche Tafel, pp. 41–61, being divided into three parts; viz. these are

a. Table of the Hauptexponenten of the six roots 10, 5, 2, 6, 3, 7 for all prime numbers of the first 1000, together with the least primitive root of each of these numbers (pp. 42–46).

A specimen is as follows:

		10		5		2		6		3		7		w
p	$p-1$	e	n	e	n	e	n	e	n	e	n	e	n	
53	$2^2.13$	13	4	52	1	52	1	26	2	52	1	26	2	2

where e is the Hauptexponent or indicatrix of the root (10, 5, 2, 6, 3, 7, as the case may be), $n = \dfrac{p-1}{e}$, w the least primitive root; thus

$$p = 53, \ 10^{13} = 1, \ 5^{52} = 1, \ 2^{52} = 1,$$

[p. 474] (2 being accordingly the least prime root),

$$6^{26} = 1, \ 3^{52} = 1, \ 7^{26} = 1.$$

The number w of the last column is the least primitive root. It is, of course, not always (as in the present case) one of the numbers 10, 5, 2, 6, 3, 7 to which the table relates: the first exception is $p = 191$, $w = 19$: the highest value of w is $w = 21$, corresponding to $p = 409$.

b. The like table for the roots 10 and 2 for all prime numbers from 1000 to 5000, together with as convenient as possible a prime root (and in some cases two prime roots) for each such number (pp. 47–53).

A specimen is:

		10		2		
p	$p-1$	e	n	e	n	w
1289	$2^3.7.23$	92	14	161	8	6, 11

viz. here, mod. 1289, $10^{92}=1$, $2^{161}=1$; and two prime roots are 6, 11. We have thus by the present tables a prime root for every prime number not exceeding 5000.

c. The like table for the root 10 for all prime numbers between 5000 and 15000, (no column for w, nor any prime root given), pp. 53–61.

A specimen is

p	$p-1$	e	n
9859	2.3.31.53	3286	3

viz., mod. 9859, we have $10^{3286}=1$. But in a large number of cases we have $n=1$, and therefore 10 a prime root. For example,

9887 2.4943 9886 1.

23. For a composite number n, if $N=\varpi(n)$ be the number of integers less than n and prime to it, then if x be any number less than n and prime to it, we have $x^N=1$ (mod. n). But we have in this case no analogue of a prime root—there is no number x, such that its several powers x', x'', …, x^{N-1} (mod. n) are all different from unity; or, what is the same thing, there is for each value of x some submultiple of N, say N', such that $x^{N'}=1$ (mod. n). And these several numbers N' have a least common multiple I, which is not $=N$, but is a submultiple of N; and this being so, then for all the several values of x, I is said to be the maximum indicator. For instance, $n=12$, $N=\varpi(n)$; the numbers less than 12 and prime to it are 1, 5, 7, 11. We have, (mod. 12), $1^1=1$, $5^2=1$, $7^2=1$, $11^2=1$, or the values of N' are 1, 2, 2, 2; their least common multiple is 2, and we have accordingly $I=2$: viz. $x^2=1$ (mod. 12) has the $\varpi(12)$ roots 1, 5, 7, 11. So $n=24$, $\varpi(n)=8$; the maximum indicator I is in this case also $=2$.

[p. 475] A table of the maximum indicator $n=1$ to 1000 is given by

Cauchy, *Exer. d'Analyse* &c., t. ii. (1841), pp. 36–40, contained in the "Mémoire sur la résolution des Equations indéterminées du premier degré en nombres entiers," pp. 1–40.

24. It thus appears that for a composite number n, the $\varpi(n)$ numbers less than n and prime to it cannot be expressed as $\equiv$ (mod. n) to the power of a single root; but for the expression of them it is necessary to employ two or more roots. A small table, $n=1$ to 50, is given by

Cayley, "Specimen Table $M\equiv a^\alpha b^\beta$ (mod. N) for any prime or composite modulus," *Quart. Math. Journ.* vol. ix. (1868), pp. 95, 96, and folding sheet, [397].

A specimen is

Nos.	12
roots	5, 7
Ind.	2, 2
M.I.	2
$\varpi(\,)$	4
1	0, 0
2	
3	
4	
5	1, 0
6	
7	0, 1
8	
9	
10	
11	1, 1

viz. for the modulus 12 the roots are 5, 7, having respectively the indicators 2, 2, viz. $5^2=1$ (mod. 12), $7^2=1$ (mod. 12). Hence also the maximum indicator is $=2$. $\varpi(=\varpi(n))=4$ is the number of integers less than 12 and prime to it, viz. these are 1, 5, 7, 11, which in terms of the roots 5, 7 and to mod. 12 are respectively $= 5^0.7^0,\ 5^1.7^0,\ 5^0.7^1$, and $5^1.7^1$.

25. *Quadratic Residues.* —In regard to a given prime number p, a number N is or is not a quadratic residue according as the index of N is even or odd, viz. g being a prime root and $N = g^\alpha$, then N is or is not a quadratic residue according as α is even or odd. But the quadratic residues can, of course, be obtained directly without the consideration of prime roots.

A small table, $p = 3$ to 97 and $N = -1$ and (prime values) 3 to 97, is given by

Gauss, "Disquisitiones Arithmeticae," 1801; Table II. (*Werke*, t. i. p. 469): I notice here a misprint in the top line of the original; it should be $-1, +2, +3$, &c., instead of

[p. 476] $-1, +2, +3$, &c.; the -1 is printed correctly on p. 499 of the French translation *Recherches Arithmétiques*, Paris, 1807 and on p. 469 of vol. I. of *Werke*, (Göttingen, 1870).

A specimen is

	−1	+2	+3	+5	+7	+11	+13	+17	+19	+23 &c.
19		−	−				−	−	−	−

viz. -1, 2, 3, 13 are not, 5, 7, 11, 17 &c. are, quadratic residues of 19. The residues taken positively and less than 19 are, in fact, 1, 4, 5, 6, 7, 9, 11, 16, 17.

The same table carried from $p = 3$ to 503, and prime values $N = 3$ to 997, is given by

Gauss, *Werke*, t. ii. pp. 400–409. A specimen is

	2	3	5	7	11	13	17	19	23 &c.
19	−		−			−	−	−	

viz. the arrangement is the same, except only that the -1 column is omitted.

26. We have also by Gauss

"Disquisitiones Arithmeticae" Table III. (*Werke*, t. i. p. 470), for the conversion into decimals of a vulgar fraction, denominator p or p^μ, not exceeding 100. The explanation is given in Art. 314 *et seq.* of the same work.

But this table, carried to a greater extent, is given by Gauss, *Werke*, t. ii. pp. 412–434, "Tafel zur Verwandlung gemeiner Brüche mit Nennern aus dem ersten Tausend in Decimalbrüche;" viz. the denominators are here primes or powers of primes, p^μ up to 997.

To explain the table, consider a modulus p^μ (where μ may be $= 1$); if 10 is not a prime root of p^μ, consider a prime root r, which is such that $r^e = 10$ (mod. p^μ), e being a submultiple of $p^{\mu-1}(p-1)$; say we have $ef = p^{\mu-1}(p-1)$: then $10^f = 1$ (mod. p^μ). Consider any fraction $\dfrac{N}{p^\mu}$; then we may write $N = r^{ke+l}$ (mod. p^μ), k from 0 to $f-1$ and l from 0 to $e-1$, $N = 10^k r^l$, and consequently $\dfrac{N}{p^\mu}$ and $\dfrac{10^k r^l}{p^\mu}$ have the same mantissa (decimal part regarded as an integer); hence, in order to know the mantissa of every fraction whatever of $\dfrac{N}{p^\mu}$, it is sufficient to know the mantissa of $\dfrac{r^l}{p^\mu}$, that is, the mantissas of $\dfrac{1}{p^\mu}, \dfrac{r}{p^\mu}, \dfrac{r^2}{p^\mu}, \ldots, \dfrac{r^{e-1}}{p^\mu}$, or, what is the same thing, the mantissas of $\dfrac{10}{p^\mu}, \dfrac{10r}{p^\mu}, \ldots, \dfrac{10r^{e-1}}{p^\mu}$.

For instance, $p^\mu = 11$, $10^2 = 1$ (mod. 11), whence $f = 2$, $e = 5$; and taking $r = 2$, we have $10 = r^5$ (mod. 11).

[p. 477] The required mantissæ, denoted in the table by

$$(0),\ (1),\ (2),\ (3),\ (4),$$

are those of

$$\frac{10}{11},\ \frac{10.2}{11},\ \frac{10.2^2}{11},\ \frac{10.2^3}{11},\ \frac{10.2^4}{11},$$

viz. these fractions are respectively

$$\begin{array}{ccccc} (0), & (1), & (2), & (3), & (4), \\ \cdot 9090\ldots, & 1\cdot 8181\ldots, & 3\cdot 6363\ldots, & 7\cdot 2727\ldots, & 14\cdot 5454\ldots, \end{array}$$

or their mantissæ are 90, 81, 63, 27, 54.

And we accordingly have as a specimen

$$11 \mid (1)\ldots 81,\ (2)\ldots 63,\ (3)\ldots 27,\ (4)\ldots 54,\ (0)\ldots 90.$$

Or again, as another specimen, $r = 2$

$$27 \mid (1)\ldots 740,\ (2)\ldots 481,\ (3)\ldots 962,\ (4)\ldots 925,\ (5)\ldots 851,\ (0)\ldots 370.$$

The table in this form extends to $p^\mu = 463$; the values of r (not given in the body of the table) are annexed, p. 420.

In the latter part of the table $p^\mu = 467$ to 997, we have only the mantissæ of $\dfrac{100}{p^\mu}$. A specimen is

547	1828153564	8994515539	3053016453	3820840950
	6398537477	1480804387	5685557586	8372943327
	2394881170	0182815356		

viz. the fraction $\frac{100}{547} = \cdot 182815\ldots$ has a period of 91, $= \frac{1}{6}.546$, figures.

[F. 13. The Pellian Equation.] Art. III.

27. The Pellian equation is $y^2 = ax^2 + 1$, a being a given integer number, which is not a square (or rather, if it be, the only solution is $y = 1$, $x = 0$), and x, y being numbers to be determined: what is required is the least values of x, y, since these, being known, all other values can be found. A small table $a = 2$ to 68 is given by

Euler, *Op. Arith. Coll.* t. i. p. 8. The Memoir is "Solutio problematum Diophanteorum per numeros integros," pp. 4–10, 1732–33. The form of the table is

a	$x\,(=p)$	$y\,(=q)$
2	2	3
3	1	2
5	4	9
⋮	⋮	⋮
68	4	33.

[p. 478] Even here, for some of the values of a, the values of x, y are extremely large; thus $a = 61$, $x = 226{,}153{,}980$, $y = 1{,}766{,}319{,}049$.

And probably tables of a like extent may be found elsewhere; in particular, a table of the solution of $y^2 = ax^2 \pm 1$ ($-$ when the value of a is such that there is a solution of $y^2 = ax^2 - 1$, and $+$ for other values of a), $a = 2$ to 135, is given by Legendre, *Théorie des Nombres*, 2nd ed. 1808, in the Table X. (one page) at the end of the work. For the before-mentioned number 61, the equation is $y^2 = 61x^2 - 1$, and the values are $x = 3805$, $y = 29718$; much smaller than Euler's values for the equation $y^2 = 61x^2 + 1$.

28. The most extensive table, however, is given by

Degen, *Canon Pellianus, sive Tabula simplicissimam æquationis celebratissimæ* $y^2 = ax^2 + 1$ *solutionem, pro singulis numeri dati valoribus ab* 1 *ad* 1000 *in usque numeris rationalibus, iisdemque integris exhibens. Auctore Carolo Ferdinando Degen. Hafniæ (Copenhagen) apud Gerhardum Bonnarum, 1817.* 8vo. pp. iv to xxiv and 1 to 112.

The first table (pp. 3–106) is entitled as "Tabula I. Solutionem Equationis $y^2 - ax^2 - 1 = 0$ exhibens." It, in fact, also gives the expression of $\sqrt{a}$ as a continued fraction; thus a specimen is

209	14	2	5	3	(2)
	1	13	5	8	11
	3220				
	46551				

Here the first line gives the continued fraction, viz.

$$\sqrt{209} = 14 + \cfrac{1}{2 + \cfrac{1}{5 + \cfrac{1}{3 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{2 + \cfrac{1}{28 + \cfrac{1}{2 + \&c.}}}}}}}}},$$

the period being $(2, 5, 3, 2, 3, 5, 2)$ indicated by $2, 5, 3\,(2)$. [The number of terms in the period is here odd, but it may be even; for instance, the period $(1, 1, 5, 5, 1, 1)$ is indicated by $1, 1\,(5, 5)$.]

The second line contains auxiliary numbers presenting themselves in the process; thus, if $R^2 = 209$, we have $R = 14 + \frac{1}{\beta}$,

$$\beta = \frac{1}{R-14} = \frac{1(R+14)}{209-14^2} = \frac{R+14}{13} = 2 + \frac{1}{\beta'},$$

$$\beta' = \frac{13}{R-12} = \frac{13(R+12)}{209-12^2} = \frac{R+12}{5} = 5 + \frac{1}{\gamma},$$

$$\gamma = \frac{5}{R-13} = \frac{5(R+13)}{209-13^2} = \frac{R+13}{8} = 3 + \frac{1}{\delta},$$

&c.

usepackage[a4paper,margin=0.65in]geometry

Arthur Cayley

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611. Report of the Committee on Mathematical Tables (continued)

[Continued from earlier pages of the same paper.]

[p. 479] where the second line 1, 13, 5, ... shows the numerical factors of the third column. The value of this second line as a result is not very obvious.

The third line gives x, and the fourth line y.

29. The second table, pp. 109–112, is entitled "Tabula II. Solutionem aequationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserat, exhibens"; viz. it is remarked that this is only possible (but see infra) for those values of a which in Table I. correspond to a period of an even number of terms, as shown by two equal numbers in brackets; thus $a = 13$, the period of $\sqrt{13}$ given in Table I. is $(1, 1, 1, 1)$ as shown by the top line $3, 1\ (1, 1)$, and accordingly 13 is one of the numbers in Table II.; and we have there

$$13 \mid \frac{5}{18}.$$

Or take another specimen, $241 \mid \dfrac{457422\,5}{71011068}$, viz. the first line gives the value of x, and the second line the value of y (least values), for which $y^2 - ax^2 = -1$.

It is to be noticed that $a = 2$ and $a = 5$, for which we have obviously the solutions $(x = 1, y = 1)$ and $(x = 1, y = 2)$ respectively, are exceptional numbers not satisfying the test above referred to; and (apparently for this reason) the values in question, 2 and 5, are omitted from the table.

30. **Cayley**, "Table des plus petites solutions impaires de l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)." *Crelle*, t. LIII. (1857), page 371 (one page), [231].

As regards the theory of quadratic forms, it is important to know whether for a given value of D ($\equiv 5$, mod. 8) there does or does not exist a solution, in odd numbers, of the equation $x^2 - Dy^2 = 4$. As remarked in the paper, "Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)," pp. 369–371, [231], this can be determined for values of D of the form in question up to $D = 997$ by means of Degen's Table; and the solutions, when they exist, of the equation $x^2 - Dy^2 = 4$, as also of the equation $x^2 - Dy^2 = -4$, can be obtained up to the same value of D. Observe that when the equation $x^2 - Dy^2 = -4$ is possible, the equation $x^2 - Dy^2 = +4$ is also possible, and that its least solution is obtained very readily from that of the other equation; it is therefore sufficient to tabulate the solution of $x^2 - Dy^2 = \pm 4$, the sign being $-$ when the corresponding equation is possible, and being in other cases $+$.

Hence the form of the Table: viz. as a specimen we have

D	$\pm$	x	y
757	imposs.		
765	+	83	3
773	–	139	5
781	imposs.		

[p. 480] that is, if $D = 757$ or 781, there is no solution of either $x^2 - Dy^2 = +4$ or -4; if $D = 765$, there is a solution $x = 83$, $y = 3$ of $x^2 - Dy^2 = +4$, but none of $x^2 - Dy^2 = -4$; if $D = 773$, there is a solution $x = 139$, $y = 5$ of $x^2 - Dy^2 = -4$, and therefore also a solution of $x^2 - Dy^2 = +4$; and so in other cases.

[F. 14. *Partitions.*] Art. IV.

31. The problem of Partitions is closely connected with that of Derivations. Thus if it be asked in how many ways can the number n be expressed as a sum of three parts, the parts being 0, 1, 2, 3, and each part being repeatable an indefinite number of times, it is clear that n is at most $= 9$, and that for the values of n, $= 0, 1, \ldots, 9$ shown by the top line of the annexed table, the number of partitions has the values shown by the bottom line thereof:

0	1	2	3	4	5	6	7	8	9
a^3	a^2b	a^2c ab^2	a^2d abc b^3	abd b^2d b^2c	acd bcd bc^2	ad^2 c^2d c^3	bd^2 cd^2	cd^2	d^3
1	1	2	3	3	3	3	2	1	1

32. But taking a, b, c, d to stand for 0, 1, 2, 3 respectively, the actual partitions of the required form are exhibited by the literal terms of the table (these being obtained, each column from the preceding one, by the method of derivations, or say by the rule of the last and last but one), and the numbers of the bottom line are simply the number of terms in the several columns respectively.

A set of such literal tables, say of tables $\begin{pmatrix} a, b, c, \ldots, k \\ 0, 1, 2, \ldots, m \end{pmatrix}^n$ for different values of n and m (where the number of letters is $= m + 1$), would be extremely interesting and valuable. The tables for a given value of m and for different values of n are, it is clear, the proper foundation of the theory of the binary quantic $(a, b, c, \ldots, k \,\cancel{\wr}\, x, 1)^m$, which corresponds to such value of m. Prof. Cayley regrets that he has not in his covariant tables given in every case the complete series of literal terms; viz. the literal terms which have zero coefficients are, for the most part, though not always, omitted in the expressions of the several covariants.

33. But the question at present is as to the *number* of terms in a column, that is, as to the number of the partitions of a given form: the analytical theory has been investigated by Euler and others. The expression for the number of partitions is usually obtained as equal to the coefficient of x^n in the development, in ascending powers of x, of a given rational function of x: for instance, if there is no limitation as to the number of the parts, but if the parts are $1, 2, 3, \dots, m$ (viz. a part may have any value not exceeding m), each part being repeatable an indefinite number of times, then

$$\text{Number of partitions of } n = \text{coefficient of } x^n \text{ in } \frac{1}{(1-x)(1-x^2)(1-x^3)\dots(1-x^m)},$$

[p. 481] and we can, by actual development, obtain for any given values of m, n the number of partitions.

These have been tabulated $m = 1, 2, \dots, 20$, and $m = \infty$ (viz. there is in this case no limit as to the largest part), and $n = 1$ to 59, by

Euler, *Op. Arith. Coll.* t. I. pp. 97–101, given in the paper "De Partitione Numerorum," pp. 73–101, (1750); the heading is "Tabula indicans quot variis modis numerus n e numeris $1, 2, 3, 4, \dots, m$, per additionem exhibi potest, seu exhibens valores formulae $n^{(m)}$." The successive lines are, in fact, the coefficients of the several powers $x^0, x^1, \dots, x^n$ in the expansions of the functions

$$\frac{1}{1-x}, \quad \frac{1}{1-x\cdot 1-x^2}, \quad \dots, \quad \frac{1}{1-x\cdot 1-x^2\dots 1-x^m}.$$

34. The generating function for any given value of m is, it is clear, $\dfrac{1}{1-x^m}$ multiplied by that for the next preceding value of m, and it thus appears how each line of the table is calculated from that which precedes it. The auxiliary numbers are printed; thus a specimen is

m	Valores numeri n.										
	0	1	2	3	4	5	6	7	8	9	10
4	1	1	2	3	5	6	9	11	15	18	23
					1	1	2	3	5	7	9
5	1	1	2	3	5	7	10	13	18	23	30

viz. suppose the numbers in the second 4-line known: then simply moving these each five steps onward we have the (auxiliary) numbers of the first 5-line; and thence by a mere addition the required series of numbers shown by the second 5-line. And similarly from this is obtained the second 6-line, and so on.

35. More extensive tables are contained in the memoir by

Marano, *Sulle leggi delle derivate generali delle funzioni di funzioni et sulla teoria delle forme di partizione dei numeri interi*, (4°. Genova, 1870), pp. 1–281; and three tables paged separately, described merely as "Tavole dei numeri $C_{q,r}$, $S_{q,r}$, $S'_{q,r}$ citate nel testo colle indicazioni di Tavole I., II., III., ai n. 77, 79, 81"; viz. the reader is referred to these articles for the explanations of what the tabulated functions are; and there is not even then any explicit statement, but the investigation itself has to be studied to make out what the tables are. It is, in fact, easier to make this out from the tables themselves; the explanation is as follows:

[p. 482] Table I. (16 pages) is, in fact, Euler's table, showing in how many ways the number n can be made up with the parts $1, 2, 3, \ldots, m$; but the extent is greater, viz. n is from 1 to 103, and m from 1 to 102. The auxiliary numbers given in Euler's table are omitted, as also certain numbers which occur in each successive line; thus a specimen is

$n=$	1	2	3	4	5	6	7	8	9	10 &c.
$C_{n,1}$	1	0	0	0	0	0	0	0	0	0
$C_{n,2}$		1	1	1	1	1	1	1	1	1
$C_{n,3}$			2	2	3	3	4	4	5	5
$C_{n,4}$				3	4	5	7	8	10	12
$C_{n,5}$					5	6	9	11	15	18

where the line $C_{n,4}$, (ways of making up $n-1$ with the parts $1, 2, 3, 4$) is $1, 1, 2, 3, 5, 6, 9, 11, 15, 18$, &c., viz. we read from the corner diagonally downwards as far as the 6, and then horizontally along the line: this saves a large number of figures. The table is printed in ordinary quarto pages, which are taken to come in in tiers of seven, five, and three pages one under the other, as shown by a prefixed diagram; and the necessity of a large folding plate is thus avoided.

The successive lines give, in fact, the coefficients in the expansions of

$$\frac{1}{1-x}, \quad \frac{1}{1-x\cdot 1-x^2}, \quad \frac{1}{1-x\cdot 1-x^2\cdot 1-x^3}, \quad \ldots, \quad \frac{1}{1-x\cdot 1-x^2\ldots 1-x^m},$$

each expanded as far as x^{102}.

Table II. (6 pages). The successive lines give the coefficients in the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x\cdot 1-x^2}, \quad \ldots, \quad \frac{S}{1-x\cdot 1-x^2\ldots 1-x^m},$$

where

$$S = \frac{1}{(1-x)(1-x^2)(1-x^3)\ldots \text{ad inf.}},$$

each expanded as far as x^{50}, and further continued as regards the first ten lines, that is, the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x\cdot 1-x^2}, \quad \ldots, \quad \frac{S}{1-x\cdot 1-x^2\ldots 1-x^9},$$

each as far as x^{100}.

Table III. (2 pages). The successive lines give the coefficients in the expansions of

$$S^2,\quad \frac{S^2}{1-x},\quad \frac{S^2}{1-x\cdot 1-x^2},\quad \ldots,\quad \frac{S^2}{1-x\cdot 1-x^2\ldots 1-x^m},$$

each expanded as far as x^{60}.

[p. 483] 36. As regards Tables II. and III., the analytical explanations have been given in the first instance; but it is easy to see that the tables give numbers of partitions. Thus, in Table II., the second line gives the coefficients in the development of

$$\frac{1}{(1-x)^2(1-x^2)(1-x^3)\ldots},$$

viz. these are $1, 2, 4, 7, 12, 19, 30, \ldots$ being the number of ways in which the numbers $0, 1, 2, 3, 4,$ &c. respectively can be made up with the parts $1, 1', 2, 3, 4,$ &c.; thus

Partitions.	No. =
0	1
1	
$1'$	2
2	
$1, 1$	
$1, 1'$	
$1', 1'$	4
3	
$2, 1$	
$2, 1'$	
$1, 1, 1$	
$1, 1, 1'$	
$1, 1', 1'$	
$1', 1', 1'$	7
&c.	&c.

Similarly, the third line shows the number of ways in which these numbers respectively can be made up with the parts $1, 1', 2, 2', 3, 4, 5,$ &c.; the fourth line with the parts $1, 1', 2, 2', 3, 3', 4, 5,$ &c.; and so on.

And in like manner in Table III., the first line shows the number of ways when the parts are $1, 1', 2, 2', 3, 3', \ldots$; the second line when they are $1, 1', 1'', 2, 2', 3, 3',$ &c.; the third when they are $1, 1', 1'', 2, 2', 2'', 3, 3',$ &c.; and so on.

It is clear that the series of tables might be continued indefinitely, viz. there might be a Table IV. giving the developments of

$$S^3,\quad \frac{S^3}{1-x},\quad \frac{S^3}{1-x\cdot 1-x^2},\quad \text{and so on.}$$

An interesting table would be one composed of the first lines of the above series, viz. a table giving in its successive lines the developments of S, S^2, S^3, S^4, &c.

There are throughout the work a large number of numerical results given in a quasi-tabular form; but the collection of these, with independent explanations of the significations of the tabulated numbers, would be a task of considerable labour.

[p. 484]

[F. 15. *Quadratic forms a^2+b^2, &c., and Partitions of Numbers into squares, cubes, and biquadrates.*] Art. V.

37. The forms here referred to present themselves in the various complex theories. Thus $N = a^2+b^2, = (a+bi)(a-bi)$; this means that, in the theory of the complex numbers $a+bi$ (a and b integers), N is not a prime but a composite number.

It is well known that an ordinary prime number $\equiv 3$, mod. 4, is not expressible as a sum a^2+b^2, being, in fact, a prime in the complex theory as well as in the ordinary one: but that an ordinary prime number $\equiv 1$, mod. 4, is (in one way only) $= a^2+b^2$; so that it is in the complex theory a composite number. A number whose prime factors are each of them $\equiv 1$, mod. 4, or which contains, if at all, an even number of times any prime factor $\equiv 3$, mod. 4, can be expressed in a variety of ways in the form a^2+b^2; but these are all easily deducible from the expressions in the form in question of its several factors $\equiv 1$, mod. 4, so that the required table is a table of the form $p = a^2+b^2$, p an ordinary prime number $\equiv 1$ mod. 4: a and b are one of them odd, the other even; and to render the decomposition definite a is taken to be odd.

$p = a^2+b^2$; viz. decomposition of the primes of the form $4n+1$ into the sum of two squares: a table extending from $p = 5$ to 11981 (calculated by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), pp. 174–176.

This is carried by Reuschle, as presently mentioned, up to $p = 24917$. Reuschle notices that $2713 = 3^2+52^2$ is omitted, also $6997 = 39^2+74^2$, and that 8609 should be $= 47^2+80^2$.

38. Similarly, primes of the form $6n+1$ are expressible in the form $p = a^2+3b^2$. Observe that, ω being an imaginary cube root of unity, this is connected with $p = (a+b\omega)(a+b\omega^2), = a^2-ab+b^2$, viz. we have $4p = (2a-b)^2+3b^2$; or the form a^2+3b^2 is connected with the theory of the complex numbers composed of the cube roots of unity.

$p = a^2+3b^2$; viz. decomposition of the primes of the form $6n+1$ into the form $p = a^2+3b^2$: a table extending from $p = 7$ to 12007 (calculated also by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, pp. 177–179.

This is carried by Reuschle up to $p = 13369$, and for certain higher numbers up to 49999, as presently mentioned. Reuschle observes that $6427 = 80^2 + 3 \cdot 3^2$ is by accident omitted, and that 6481 should be $= 41^2 + 3 \cdot 40^2$.

39. Again, primes of the form $8n + 1$ are expressible in the form $p = a^2 + 2b^2$ (or say $= c^2 + 2d^2$), the theory being connected with that of the complex numbers composed with the 8th roots of unity (fourth root of -1, $= \sqrt[4]{-1}$).

$p = c^2 + 2d^2$; viz. decomposition of primes of the form $8n + 1$ into the form $c^2 + 2d^2$:

[p. 485] a table extending from $p = 17$ to 5943 (extracted from a MS. table calculated by Struve) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, p. 180.

This is carried by Reuschle up to $p = 12377$, and for certain higher numbers up to 24889, as presently mentioned.

40. Reuschle's tables of the forms in question are contained in the work:

Reuschle, *Mathematische Abhandlung*, &c. (see ante No. 15), under the heading "B. Tafeln zur Zerlegung der Primzahlen in Quadrate" (pp. 22–41). They are as follows:

Table III. for the primes $6n + 1$.

The first part gives $p = A^2 + 3B^2$ and $4p = L^2 + 27M^2$, from $p = 7$ to 5743. The table gives A, B, L, M; those numbers which have 10 for a cubic residue are distinguished by an asterisk. A specimen is

p	A	B	L	M
37*	5	2	11	1

viz. $37 = 5^2 + 3 \cdot 2^2$, $148 = 11^2 + 27 \cdot 1^2$; the asterisk shows that $x^3 \equiv 10$ (mod. 37) is possible: in fact $34^3 \equiv 10$ (mod. 37).

The second part gives $p = A^2 + 3B^2$ only, from $p = 5749$ to 13669. The table gives A, B; and the asterisk implies the same property as before.

The third part gives $p = A^2 + 3B^2$, but only for those values of p which have 10 for a cubic residue, viz. for which $x^3 \equiv 10$ (mod. p) is possible, from $p = 13689$ to 49999. The table gives A, B; the asterisk, as being unnecessary, is not inserted.

Table IV. for the primes $4n+1$ in the form $A^2 + B^2$, and for those which are also $8n + 1$ in the form $C^2 + 2D^2$.

The first part gives $p = A^2 + B^2$, $= C^2 + 2D^2$, from $p = 5$ to 12377. The table gives A, B, C, D; those numbers which have 10 for a biquadratic residue, viz. for which $x^4 \equiv 10$ (mod. p) is possible, are distinguished by an asterisk; those which have also 10 for an octic residue, viz. for which $x^8 \equiv 10$

(mod. p) is possible, by a double asterisk. A specimen is

p	A	B	C	D
229	15	2	$---$	$---$
233	13	8	15	2
241**	15	4	13	6

The second part gives $p = A^2 + B^2$, from $p = 12401$ to 24917 for all those values of p which have 10 for a biquadratic residue ($x^4 \equiv 10$ (mod. p) possible). The table gives A, B; those values of p which have 10 for an octic residue, viz. for which $x^8 \equiv 10$ (mod. p) is possible, are distinguished by an asterisk.

[p. 486] The third part gives $p = C^2 + 2D^2$, from $p = 12641$ to 24889 for all those values of p which have 10 for a biquadratic residue. The table gives C, D; those values of p which have 10 as an octic residue are distinguished by an asterisk.

41. A table by Zornow, *Crelle*, t. XIV. (1835), pp. 279, 280 (belonging to the Memoir "De Compositione numerorum e Cubis integris positivis," pp. 276–280), shows for the numbers 1 to 3000 the least number of cubes into which each of these numbers can be decomposed. Waring gave, without demonstration, the theorem that every number can be expressed as the sum of at most 9 cubes. The present table seems to show that 23 is the only number for which the number of cubes is $= 9$ $(= 2 \cdot 2^3 + 7 \cdot 1^3)$; that there are only fourteen numbers for which the number of cubes is $= 8$, the largest of these being 454; and hence that every number greater than 454 can be expressed as a sum of at most 7 cubes; and further, that every number greater than 2183 can be expressed as a sum of at most 6 cubes. A small subsidiary table (p. 276) shows that the number of numbers requiring 6 cubes gradually diminishes — e.g. between 12^3 and 13^3 there are seventy-five such numbers, but between 13^3 and 14^3 only sixty-four such numbers; and the author conjectures "that for numbers beyond a certain limit every number can be expressed as a sum of at most 5 cubes."

42. For the decomposition of a number into biquadrates we have

Bretschneider, "Tafeln für die Zerlegung der Zahlen bis 4100 in Biquadrate," *Crelle*, t. XLVI. (1853), pp. 3–23.

Table I. gives the decompositions, thus:

N	1^4	2^4	3^4	4^4	5^4
696	6	1	2	2	
$\vdots$					
3251		3	8		

viz. $696 = 6 \cdot 1^4 + 1 \cdot 2^4 + 2 \cdot 3^4 + 2 \cdot 4^4$, &c.

And Table II. enumerates the numbers which are sums of at least $2, 3, 4, \ldots, 19$ biquadrates. There is at the end a summary showing for the first 4100 numbers how many numbers there are of these several forms respectively: 28 numbers are each of them a sum of 2 biquadrates, 75 a sum of $3, \ldots$, 7 a sum of 19 biquadrates. The seven numbers, each of them a sum of 19 biquadrates, are $79, 159, 239, 319, 399, 479, 559$.

[F. 16. *Binary, Ternary, etc. quadratic and higher forms.*] Art. VI.

43. Euler worked with the quadratic forms $px^2 \pm qy^2$ (p and q integers), particularly in regard to the forms of the divisors of such numbers. It will be sufficient to refer to his memoir:
[p. 487] **Euler**, "Theoremata circa divisores numerorum in hac forma $pa^2 \pm qb^2$ contentorum," (*Op. Arith. Coll.* pp. 35–61, 1744), containing fifty-nine theorems, exhibiting in a quasi-tabular form the linear forms of the divisors of such numbers. As a specimen:

"Theorema 13. Numerorum in hac forma $a^2 + 7b^2$ contentorum divisores primi omnes sunt vel 2, vel 7, vel in una sex formularum

$$\begin{array}{ll} 28m + 1, & 28m + 11, \\ 28m + 9, & 28m + 15, \\ 28m + 25, & 28m + 23, \end{array}$$

seu in una harum trium

$$14m + 1, \quad 14m + 9, \quad 14m + 11,$$

sunt contenti"; viz. the forms are the three $14m + 1$, $14m + 9$, $14m + 11$.

But Euler did not consider, or if at all very slightly, the trinomial forms $ax^2 + bxy + cy^2$, nor attempt the theory of the reduction of such forms. This was first done by Lagrange in the memoir

Lagrange, *Mém. de Berlin*, 1773. And the theory is reproduced by

Legendre, *Théorie des Nombres*, Paris, 1st ed. 1798; 2nd ed. 1808, § 8, "Réduction de la formule $Ly^2 + Myz + Nz^2$ à l'expression la plus simple," (2nd ed. pp. 61–67).

44. But the classification of quadratic forms, as established by Legendre, is defective as not taking account of the distinction between proper and improper equivalence; and the ulterior theory as to orders and genera, and the composition of forms (although in the meantime established by Gauss), are not therein taken into account; for this reason the Legendre's Tables I. to VIII. relating to quadratic forms, given after p. 480 (thirty-two pages not numbered), are of comparatively little value, and it is not necessary to refer to them in detail.

The complete theory was established by

Gauss, *Disquisitiones Arithmeticae*, 1801.

It is convenient to refer also to the following memoir

Lejeune Dirichlet, "Recherches sur diverses applications de l'Analyse à la théorie des Nombres," *Crelle*, t. XIX. (1839), p. 338, [*Ges. Werke*, t. I. p. 427], as giving a succinct statement of the principle of classification, and in particular a table of the characters of the genera of the properly primitive order, according to the four forms $D = PS^2$, $P \equiv 1$ or 3 (mod. 4), and $D = 2PS^2$, $P \equiv 1$ or 3 (mod. 4), of the determinant.

45. Tables of quadratic forms arranged on the Gaussian principle are given by

Cayley, *Crelle*, t. LX. (1862), pp. 357–372, [335]; viz. the tables are—

Table I. des formes quadratiques binaires ayant pour déterminants les nombres négatifs depuis $D = -1$ jusqu'à $D = -100$. (Pp. 360–363: [*Coll. Math. Papers*, t. V, pp. 144–147].)

[p. 488] A specimen is

D	Classes	α	β	δ	ϵ	$\delta\epsilon$	Cp
$= 26$	$1, 0, 26$	$+$	$+$			$+$	1
	$3, -1, 9$	$+$	$*$			$+$	g^2
	$3,\ 1, 9$	$+$				$+$	g^4
	$5,\ 2, 6$						g
	$2,\ 0, 13$	$-$				$-$	g^3
	$5, -2, 6$	$-$				$-$	g^5

where α, β denote, as there explained, the characters in regard to the odd prime factors of D; δ, ϵ, $\delta\epsilon$ those in regard to the numbers 4 and 8. The last column shows that the forms in the two genera respectively are $1, g^2, g^4$ and g, g^3, g^5, where $g^6 = 1$, viz. the form g, six times compounded, gives the principal form $(1, 0, 26)$.

Table II. des formes quadratiques binaires ayant pour déterminants les nombres positifs non-carrés depuis $D = 2$ jusqu'à $D = 99$. (Pp. 364–369: [*l.c.*, pp. 148–153].)

The arrangement is the same, except that there is a column "Périodes" showing, in an easily understood abbreviated form, the period of each form. Thus $D = 7$, the period of the principal form $(1, 0, -7)$, is given as $1, 7, -3, \ddot{1}, 2, \ddot{1}, -3, 7, 1$, which represents the series of forms $(1, 2, -3)$, $(-3, 1, 2)$, $(2, 1, -3)$, $(-3, 2, 1)$.

Table III. des formes quadratiques binaires pour les treize déterminants négatifs irréguliers du premier millier. (Pp. 370–372: [*l.c.*, pp. 154–156].)

The arrangement is the same as in Table I. It may be mentioned that the thirteen numbers, and the forms for the principal genus for these numbers, respectively are:

	Principal genus
$576, 580, 820, 900$	$(1, e^2)(1, {e_1}^2)$
884	$(1, e^2)(1, i^2, i^4, i^6)$
$243, 307, 339, 459, 675, 891$	$(1, d, d^2)(1, d_1, {d_1}^2)$
$755, 974$	$(1, d, d^2)(1, d_1, {d_1}^2)(1, e^2)$

where $d^3 = {d_1}^3 = 1$, $e^4 = {e_1}^4 = 1$, $i^8 = 1$, viz. $(1, e^2)(1, {e_1}^2)$ denotes four forms, $1, e^2, {e_1}^2, e^2{e_1}^2$; and so in the other cases.

46. Gauss must have computed quadratic forms to an enormous extent; but, for the reasons (rather amusing ones) mentioned in a letter of May 17, 1841, to Schumacher (quoted in Prof. Smith's Report on "The Theory of Numbers," Brit. Assoc. Report for 1862, p. 526, [and Smith's *Coll. Math. Papers*, t. I. p. 261]), he did not preserve his results in detail, but only in the form appearing in the

"Tafel der Anzahl der Classen binärer quadratischer Formen," *Werke*, t. II. pp. 449–476; see editor's remarks, pp. 521–523.

[p. 489] This relates almost entirely to negative determinants, only three quarters of p. 475 and p. 476 to positive ones; for negative determinants, it gives the number of genera and classes, as also the index of irregularity for the determinants of the hundreds 1 to 30, 43, 51, 61, 62, 63, 91 to 100, 117 to 120; then, in a different arrangement, for the thousands 1, 3 and 10, for the first 800 numbers of the forms $-(15n+7)$ and $-(15n+13)$; also for some very large numbers, and for positive determinants of the hundreds 1, 2, 3, 9, 10, and for some others.

A specimen is

Centas I.
G II. (58) . . . (280)
1. 5, 6, 8,
9, 10, 12,
13, 15, 16,
18, 22, 25,
28, 37, 58,
2. 14, 17, 20,
Summa 233 . . . 477
Irreg. 0 Impr. 74

viz. this shows, as regards the negative determinants 1 to 100, that the determinants belonging to G II. 1, viz. those which have two genera each of one class, are $5, 6, 8, 9$, &c., in all fifteen determinants; those belonging to G II. 2, viz. those which have two genera each of two classes, are $14, 17, 20$, &c.; and so on. The head numbers (58) . . . (280) show the number of determinants, each having two genera, and the number of classes; thus,

$$
\begin{array}{lrr}
\text{G II.} & 1\times 15 = & 15 \\
 & 2\times 17 = & 34 \\
 & 3\times 17 = & 51 \\
 & 4\times \ \ 6 = & 24 \\
 & 5\times \ \ 2 = & 10 \\
 & 6\times \ \ 1 = & 6 \\
\hline
 & 58 & 140 \\
 & & \times 2 \\
\cline{3-3}
 & & = 280;
\end{array}
$$

and the bottom numbers show the total number of genera and of classes, thus

$$
\begin{array}{rrr}
\text{G I.} & 17\times 1 = 17 & 61 \\
\text{II.} & 58\times 2 = 116 & 280 \\
\text{IV.} & 25\times 4 = 100 & 136 \\
\hline
 & 100 \quad 233 & 477
\end{array}
$$

[p. 490] viz. seventeen determinants, each of one genus, and together of sixty-one classes; fifty-eight determinants, each of two genera, and together of 280 classes; and twenty-five determinants, each of four genera, and together 136 classes, give in all 233 genera and 477 classes. These are exclusive of 74 classes belonging to the improperly primitive order; and the number of irregular determinants (in the first hundred) is $= 0$.

The irregular determinants are indicated thus:

243(∗3∗),
307(∗3∗), 339(∗3∗),
459(∗),
576(∗2∗), 580(∗2∗),
675(∗3∗),
755(∗3∗),
891(∗3∗), 820(∗2∗), 900(∗2∗), 884(∗2∗), 974(∗3∗),
∗3∗ 243, 307, 339, 459?, 675, 755, 891,
∗2∗ 576, 580, 820, 884, 900, 974,

which is a notation not easily understood.

As regards the positive determinants, a specimen is

Centas I. Excedunt determinantis quadrati 10.
G I. ... (12),
1. 2, 5, 13,
17, 29, 41,
53, 61, 73,
89, 97,
3. 37

viz. in the first hundred, the positive determinants having one genus of one class are $2, 5, 13$, &c. ... (eleven in number); that having one genus of three classes is 37, (one in number); $11 + 1 = 12$. The irregular determinants, if any, are not distinguished.

47. *Binary cubic forms.*—The earliest table is given by

Arndt, "Tabelle der reducirten binären kubischen Formen und Klassen für alle negativen Determinanten $-D$ von $D = 3$ bis $D = 2000$," *Grunert's Archiv*, t. XXXI. 1858, pp. 369–448.

The memoir is a sequel to one in t. XVII. (1851). The binary cubic form (a, b, c, d), of determinant D $(= (bc - ad)^2 - 4(b^2 - ac)(c^2 - bd))$, is said to be reduced when its characteristic $\phi, = (A, B, C), = (2(b^2 - ac), bc - ad, 2(c^2 - bd))$, is a reduced quadratic form, that is, when in regard to absolute values B is not $> \frac{1}{2}A$, C not $< A$.

[p. 491] A specimen is

D	Reduced forms, with characters	Classes
44	$(0, 1, 0, -11)$ $(1, -1, -2, 0)$	$(0, -1, 0, 11)$ $(0, -2, 1, 1)$
	$(2, 0, 22)$	$(6, 8)$

Two subsidiary tables are given, pp. 351, 352, and 353–368.

48. It appeared suitable to remodel a part of this table in the manner made use of for quadratic forms in my tables above referred to; and it is accordingly divided into the three tables given by

Cayley, *Quart. Math. Journ.* t. XI. (1871), where the notation &c. is explained, pp. 251–261. [496]; viz. these are:—

Table I. of the binary cubic forms, the determinants of which are the negative numbers $\equiv 0 \pmod 4$ from -4 to -400 (pp. 251–258 [*Coll. Math. Papers*, t. VIII., pp. 55–61]).

A specimen is

Det. 4×-11.	Classes.	Order.	Charact.	Comp.
	$0, 1, 0, -11$	on	$1, 0, 11$	1
	$0, 2, -1, -2$	*pp pp*	$3, 1, 4$	d
	$0, 2, 1, 1$		$3, -1, 4$	d^2

Table II. of the binary cubic forms the determinants of which (taken positively) are $\equiv 1$ (mod. 4) from -3 to 99, the original heading is here corrected, [*l.c.*, pp. 61, 62]; and

Table III. of the binary cubic forms the determinants of which are the negative numbers -972, -1228, -1336, -1836, and -2700, [*l.c.*, pp. 63, 64]; viz. $-972 = 4 \times -243$, ..., $-2700 = 4 \times -675$, where -243, ..., -675 are the first six irregular numbers for quadric forms.

4×-675, $= -2700$ is beyond the limits of Arndt's tables, and for this number the calculation had to be made anew; the table gives nine classes $(1, d, d^2)(1, d_1, {d_1}^2)$ of the order *ip* on *pp*, but it is remarked that there may possibly be other cubic classes based on a non-primitive characteristic; the point was left unascertained.

49. The theory of ternary quadratic forms was discussed and partially established by Gauss in the *Disquisitiones Arithmeticae.* It is proper to recall that a ternary quadratic form is either determinate, viz. always positive, such as $x^2 + y^2 + z^2$, or always negative, such as $-x^2 - y^2 - z^2$; or else it is indeterminate, such as $x^2 + y^2 - z^2$. But as regards determinate forms, the negative ones are derived from the positive ones by simply reversing the signs of all the coefficients, so that it is sufficient to attend to the positive forms; and practically the two cases are positive forms (meaning thereby positive determinate forms) and indeterminate forms; but the theory for positive forms was first established completely, and so as to enable the formation of tables, in the work

Seeber, *Ueber die Eigenschaften der positiven ternären quadratischen Formen*, (4to. Freiburg, 1831),

[p. 492] which is reviewed by Gauss in the *Gött. Gelehrte Anzeigen*, 1831, July 9 (see Gauss, *Werke*, t. II. pp. 188–193). The author gives (pp. 220–243) tables "of the classes of positive ternary forms represented by means of the corresponding reduced forms" for the determinants 1 to 100. A specimen is

$$\text{Det. } 6 \quad \begin{pmatrix} 1, 1, 6 \\ 0, 0, -1 \end{pmatrix}, \begin{pmatrix} 1, 2, 3 \\ -1, -1, 0 \end{pmatrix}.$$

$$\text{Zugeordnete Formen} \quad \begin{pmatrix} 8, 8, 3 \\ 0, 0, 4 \end{pmatrix}, \begin{pmatrix} 7, 7, 4 \\ 4, 4, 2 \end{pmatrix}.$$

where it is to be observed that Seeber admits odd coefficients for the terms in yz, zx, xy; viz. his symbol $\begin{pmatrix} a, b, c \\ f, g, h \end{pmatrix}$ denotes

$$ax^2 + by^2 + cz^2 + fyz + gzx + hxy,$$

and his determinant is

$$4abc - af^2 - bg^2 - ch^2 + fgh.$$

Also his adjoint form is

$$\begin{pmatrix} 4bc-f^2,\ 4ca-g^2,\ 4ab-h^2 \\ 2gh-4af,\ 2hf-4bg,\ 2fg-4ch \end{pmatrix}, \quad = (4bc-f^2)x^2+\ldots+(2gh-4af)yz+\ldots$$

In the notation of the *Disquisitiones Arithmeticae*, followed by Eisenstein and others, the symbol $\begin{pmatrix} a, b, c \\ f, g, h \end{pmatrix}$ denotes

$$ax^2+by^2+cz^2+2fyz+2gzx+2hxy;$$

the determinant is

$$= -(abc-af^2-bg^2-ch^2+2fgh),$$

a positive form having thus always a negative determinant. And the adjoint form is

$$-\begin{pmatrix} bc-f^2,\ ca-g^2,\ ab-h^2 \\ gh-af,\ hf-bg,\ fg-ch \end{pmatrix} = -(bc-f^2)x^2-\ldots-2(gh-af)yz-\ldots$$

Hence Seeber's determinant is $= -4$ multiplied by that of Gauss, and his tables really extend between the values -1 and -25 of the Gaussian determinant.

50. Tables of greater extent, and in the better form just referred to, are given by

Eisenstein, *Crelle*, t. XLI. (1851), pp. 169–190; viz. these are

I. "Tabelle der eigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -1 bis -100," (pp. 169–185).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	3	$\begin{pmatrix} 1,1,10 \\ 0,0,0 \end{pmatrix}$, $\begin{pmatrix} 1,2,5 \\ 0,0,0 \end{pmatrix}$, $\begin{pmatrix} 2,2,3 \\ 0,-1,0 \end{pmatrix}$
		$\delta=8$ $\delta=4$ $\delta=4$

[p. 493] II. "Tabelle der uneigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -2 bis -100," (pp. 186–189).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	1	$\begin{pmatrix} 1,2,4 \\ 1,1,1 \end{pmatrix}$
		$\delta=6$

And there is given (p. 190) a table of the reduced forms for the determinant -385 $(= -5\cdot7\cdot11)$, selected merely as a largish number with three factors; viz. there are in all fifty-nine forms, corresponding to values $1, 2, 4, 6, 8$ of δ.

It may be remarked that δ denotes, for any given form, the number of ways in which this is linearly transformable into itself, this number being always $1, 2, 4, 6, 8, 12$, or 24. The theory as to this and other points is explained in the memoir (pp. 141–168), and various subsidiary tables are contained therein and in the *Anhang* (pp. 227–242); and there is given a small table relating to indeterminate forms, viz. this is

"C. Versuch einer Tabelle der nicht äquivalenten unbestimmten (indifferenten) ternären quadratischen Formen für die Determinanten ohne quadratischen Theiler unter 20," (pp. 239, 240).

A specimen is

D	Indifferente ternäre quadratische Formen
10	$\begin{pmatrix} 0,1,10 \\ 0,0,1 \end{pmatrix}, \begin{pmatrix} 1,2,5 \\ 0,0,0 \end{pmatrix}$
	$\begin{pmatrix} 0,0,10 \\ 0,0,1 \end{pmatrix}$

where, when the determinant is even, the forms in the second line are always improperly primitive forms.

[F. 17. *Complex Theories.*] Art. VII.

51. The theory of binary quadratic forms (a, b, c), with complex coefficients of the form $\alpha + \beta i$, ($i = \sqrt{-1}$ as usual, α and β integers), has been studied by Lejeune Dirichlet, Prof. H. J. S. Smith, and possibly others; but no tables have, it is believed, been calculated. The calculations would be laborious; but tables of a small extent only would be a sufficient illustration of the theory, and would, it is thought, be of great interest.

[p. 494] The theory of complex numbers of the last-mentioned form $\alpha+\beta i$, or say of the numbers formed with the fourth root of unity, had previously been studied by Gauss; and the theory of the numbers formed with the cube roots of unity ($\alpha + \beta\omega$, $\omega^3 + 1 = 0$, α and β integers) was studied by Eisenstein; but the general theory of the numbers involving the nth roots of unity (λ an odd prime) was first studied by Kummer. It will be sufficient to refer to his memoir,

Kummer, "Zur Theorie der complexen Zahlen," *Berl. Monatsb.*, March, 1845; and *Crelle*, t. XXXV. (1847), pp. 319–326; also "Ueber die Zerlegung der aus Wurzeln der Einheit gebildeten complexen Zahlen in ihre Primfactoren," same volume, pp. 327–367, where the astonishing theory of "Ideal Complex Numbers" is established.

52. It may be recalled that, p being an odd prime, and ρ denoting a root of the equation $\rho^{p-1} + \rho^{p-2} + \ldots + \rho + 1 = 0$, then the numbers in question are those of the form $a + b\rho + \ldots + k\rho^{p-2}$ (where $(a, b, \ldots, k)$ are integers); or (what is in one point of view more, and in another less, general)

if $\eta, \eta_1, \ldots, \eta_{e-1}$ are "periods" composed with the powers of ρ (e any factor of $p-1$), then the form considered is $a\eta + b\eta_1 + \ldots + h\eta_{e-1}$. For any value of p or e there is a corresponding complex theory. A number (real or complex) is in the complex theory prime or composite, according as it does not, or does, break up into factors of the form under consideration. For p a prime number under 23, if in the complex theory N is a prime, then any power of N (to fix the ideas say N^2) has no other factors than N or N^2; but if $p = 23$ (and similarly for higher values of p), then N may be such that, for instance, N^3 has complex factors other than N or N^3 (for $p = 23$, $N = 47$ is the first value of N, viz. 47^3 has factors other than 47 and 47^3): say N has a complex prime factor $\sqrt[3]{A}$, or we have $\sqrt[3]{A}$ as an ideal complex factor of N. Observe that by hypothesis A is not a perfect cube, viz. there is no complex number whose cube $= A$. In the foregoing general statement, made by way of illustration only, all reference to the complex factors of unity is purposely omitted, and the statement must be understood as being subject to correction on this account.

What precedes is by way of introduction to the account of Reuschle's Tables (*Berliner Monatsberichte*, 1859–60), which give in the different complex theories $p = 5, 7, 11, 13, 17, 19, 23, 29$ the complex factors of the decomposable real primes up to in some cases 1000.

It should be remarked that the form of a prime factor is to a certain extent indeterminate, as the factor can without injury be modified by affecting it with a complex factor of unity; but in the tables the choice of the representative form is made according to definite rules, which are fully explained, and which need not be here referred to.

[p. 495] 53. The following synopsis is convenient:—

Theory of the pth roots	Form of real prime to mod. p	No. of factors in complex theory	Extent of table; all primes under	Equation o
$\lambda = 5$, pp. 385–401.	1 4 2, 3	4 2 (just tabulated) primes.	2500	$\eta^2 + \eta - 1 =$ $\eta^2 + \eta - 1 =$
$\lambda = 7$, pp. 494–497.	1 2, 4 3, 5	6 3 primes.	1000	$\eta^3 + \eta^2 - 2$ $\eta^2 + \eta + 2 =$
$\lambda = 11$, pp. 120–124.	1 3, 4, 5, 9 2 6, 7, 8, 10	10 5 2 primes.	1000	$\eta^5 + \eta^4 - 4$ $\eta^2 + \eta - 3 =$
$\lambda = 13$, pp. 173–179.	1	12	1000	$\eta^6 - \ldots =$
$\lambda = 17$, pp. 724–729.	1	16	1000	$\eta^8 - \ldots =$
$\lambda = 19$, pp. 719–724.	1	18	1000	$\eta^9 - \ldots =$
$\lambda = 23$, pp. 737–751.	1 2, 3, 4, 6, 8, 9, 12, 13, 16, 18	22 11	1000	$\eta^{11} - \ldots =$
$\lambda = 29$, pp. 761–764.	1	28	1000	$\eta^{14} - \ldots =$

The foregoing synopsis of Reuschle's tables in the *Berliner Monatsberichte* was written previous to the publication of Reuschle's far more extensive work. It is

[p. 496] allowed to remain, but some explanations which were given have been struck out, and were instead given in reference to the larger work, which is

Reuschle, *Tafeln complexer Primzahlen, welche aus Wurzeln der Einheit gebildet sind.* Berlin, 4° (1875), pp. iii–vi and 1–671.

This work (the mass of calculation is perfectly wonderful) relates to the roots of unity, the degree being any prime or composite number, as presently mentioned, having all the values up to and a few exceeding 100; viz. the work is in five divisions, relating to the cases:

I. (pp. 1–171), degree any odd prime of the first 100, viz. $3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97$;

II. (pp. 173–192), degree the power of an odd prime $9, 25, 27, 49, 81$;

III. (pp. 193–440), degree a product of two or more odd primes or their powers, viz. $15, 21, 33, 35, 39, 45, 51, 55, 57, 63, 65, 69, 75, 77, 85, 87, 91, 93, 95, 99, 105$;

IV. (pp. 441–466), degree an even power of 2, viz. $4, 8, 16, 32, 64, 128$;

V. (pp. 467–671), degree divisible by 4, viz. $12, 20, 24, 28, 36, 40, 44, 48, 52, 56, 60, 68, 72, 76, 80, 84, 8$
the only excluded degrees being those which are the double of an odd prime, these, in fact, coming under the case where the degree is the odd prime itself.

It would be somewhat long to explain the specialities which belong to degrees of the forms II., III., IV., V.; and what follows refers only to Division I., degree an odd prime.

For instance, if $\lambda = 7$, $\lambda - 1 = 2 \cdot 3$; the factors of 6 being $6, 3, 2, 1$, there are accordingly four divisions, viz.—

I. α a prime seventh root, that is, a root of $\alpha^6+\alpha^5+\alpha^4+\alpha^3+\alpha^2+\alpha+1 = 0$;

II. $\eta_0 = \alpha + \alpha^{-1}$, $\eta_1 = \alpha^2 + \alpha^{-2}$, $\eta_2 = \alpha^3 + \alpha^{-3}$, or η a root of

$$\eta^3 + \eta^2 - 2\eta - 1 = 0, \quad \eta_0\eta_1 = \eta_1 + \eta_2, \ \&\text{c.};$$

III. $\eta_0 = \alpha+\alpha^2+\alpha^4$, $\eta_1 = \alpha^3+\alpha^5+\alpha^6$, or -1; η a root of $\eta^2+\eta+2 = 0$;

IV. Real numbers.

I. $p = 7m+1$. First, it gives for the several prime numbers of this form $29, 43, \dots, 967$ the congruence roots, mod. p; for instance,

p	α	α^2	α^3	α^4	α^5	α^6
29	-5	-4	-9	-13	$+7$	-6
43	$+11$	-8	-2	$+21$	$+16$	$+4.$

[p. 497] This means that, if $\alpha \equiv -5$ (mod. 29), then $\alpha^2 = 25, \equiv -4$, $\alpha^3 = 20, \equiv -9$, &c., values which satisfy the congruence $\alpha^6+\alpha^5+\alpha^4+\alpha^3+\alpha^2+\alpha+1 \equiv 0$ (mod. 29).

Secondly, it gives, under the simple and the primary forms, the prime factors $f(\alpha)$ of these same numbers $29, 43, \dots, 967$; for instance,

p	$f(\alpha)$ simple.	$f(\alpha)$ primary.
29	$\alpha + \alpha^4 - \alpha^5$	$2 + 3\alpha - \alpha^2 + 5\alpha^3 - 7\alpha^4 + 4\alpha^5$
43	$\alpha^2 + 2\alpha^4$	$2 - 2\alpha + 2\alpha^2 - 4\alpha^3 - \alpha^4 - 5\alpha^5.$

The definition of a primary form is a form for which $f(\alpha)f(\alpha^{-1}) \equiv f(1)^2$ mod. λ, and $f(\alpha) \equiv f(1)$ mod. $-(1-\alpha)^2$. The simple forms are also chosen so as to satisfy this last condition; thus $f(\alpha) = \alpha+\alpha^4-\alpha^5$, then $f(1)-f(\alpha) = 1-\alpha-\alpha^4+\alpha^5 = (1-\alpha)^2(1+\alpha), \equiv 0$ mod. $(1-\alpha)^2$.

II. $p = 7m - 1$. First, it gives for the several prime numbers of this form $13, 41, \dots, 937$ the congruence roots, mod. p; for instance,

p	η_0	η_1	η_2
13	-3	-6	-5
41	-4	$+14$	$-11;$

and secondly, it gives, under the simple and the primary forms, the prime factors $f(\eta)$ of these same numbers $13, 41, \ldots, 937$; for instance,

p	$f(\eta)$ simple.	$f(\eta)$ primary.
13	$\eta_0 + 2\eta_1$	$3 + 3\eta_0$
41	$4 + \eta_0$	$-11 + 7\eta_0 - 7\eta_1$.

Thus $13 = (\eta_0 + 2\eta_1)(\eta_1 + 2\eta_2)(\eta_2 + 2\eta_0)$, as is easily verified; the product of the first and second factors is $4 + 3\eta_0 + 8\eta_1 + \eta_2$, and then multiplying by the third factor, the result is $42 + 29(\eta_0 + \eta_1 + \eta_2)$, $= 13$.

III. $p = 7m + 2$ or $7m + 4$. First, it gives for the several prime numbers of this form $2, 11, \ldots, 991$ the congruence roots, mod. p; for instance,

p	η_0	η_1
2	0	-1
11	4	-5;

and secondly, it gives the primary prime factors $f(\eta)$ of these same numbers; for instance,

p	$f(\eta)$
2	η_0
11	$-1 - 2\eta_1$.

[p. 498] IV. $p = 7m + 3$ or $7m + 5$. The prime numbers of these forms, viz. $3, 5, 17, 19, \ldots, 997$, are primes in the complex theory, and are therefore simply enumerated.

The arrangement is the same for the higher prime numbers $\lambda = 23$, &c., for which ideal factors make their appearance; but it presents itself under a more complicated form. Thus $\lambda = 23$, $\lambda - 1 = 2 \cdot 11$, and the factors of 22 are $22, 11, 2, 1$. There are thus four sections.

I. α a prime root, or $\alpha^{22} + \alpha^{21} + \ldots + \alpha^2 + \alpha + 1 = 0$;

II. $\eta_0 = \alpha + \alpha^{-1}, \ldots, \eta_{10} = \alpha^{11} + \alpha^{-11}$, or η a root of $\eta^{11} - \eta^{10} - 10\eta^9 + \ldots + 10\eta^2 - 6\eta - 1 = 0$;

III. $\eta_0 = \alpha + \alpha^2 + \ldots + \alpha^{11}$, $\eta_1 = \alpha^{12} + \alpha^{13} + \ldots + \alpha^{22}$, or η a root of $\eta^2 + \eta + 6 = 0$;

IV. Real numbers.

I. $p = 23m + 1$. First, it gives for the prime numbers of this form $47, 139, \ldots, 967$ congruence roots, mod. p, and also congruence roots, mod. p^3*; these last in the form $a + bp + cp^2$, where a is given in the former table; thus first table:

p	α	α^2	α^3	$\ldots$	α^{22}
47	6	-11	-19	$\ldots$	$+8$;

and second table:

p	α	α^2	α^3	$\ldots$	α^{22}
47	$-11p - 2p^2$	$+13p - 23p^2$	$+14p - 14p^2$	$\ldots$	$+22p + 22p^2$.

The meaning is that, $p = 47$, the roots of the congruence

$$\alpha^{22} + \alpha^{21} + \ldots + \alpha^2 + \alpha + 1 \equiv 0 \text{ (mod. } 47^3)$$

are

$$\alpha = 6 + p - 2p^2, \quad \alpha^2 = -11 + 13p - 23p^2, \quad \&c.$$

Secondly, it then gives $f(\alpha)$, the actual ideal prime factor of these same primes $47, 139, \ldots, 967$; viz. the whole of this portion of the table $\lambda = 23$, I. (2) is, having actual prime factors,

p	$f(\alpha)$
599	$\alpha + \alpha^{15} - \alpha^{17}$
691	$\alpha^2 + \alpha^{19} - \alpha^5$
829	$\alpha^{15} + \alpha^{18} + \alpha^{20}$

having ideal factors, their third powers actual,

p	$f^3(\alpha)$
47	$\alpha^2 + \alpha^4 + \alpha^6 - \alpha^{14} - \alpha^{18} + \alpha^{20}$
139	$\alpha - \alpha^2 + \alpha^3 - \alpha^6 + \alpha^{14} - 2\alpha^{18}$
277	$-\alpha^2 - \alpha^4 - \alpha^6 + \alpha^{10} + \alpha^{12} + 2\alpha^{14} - \alpha^{16} - \alpha^{18}$
461	$1 - \alpha^2 - \alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12} + \alpha^{14} + \alpha^{16} + \alpha^{18} + \alpha^{20} + \alpha^{22}$
967	$-\alpha^2 - \alpha^4 + \alpha^6 - \alpha^8 - \alpha^{10} - \alpha^{12} + \alpha^{14} + \alpha^{16}\alpha^{22}$

I repeat the explanation that, for the number 47, this means $f(\alpha)f(\alpha^2)\ldots f(\alpha^{22}) = 47^3$.
[p. 499] And the like further complication presents itself in the part III. of the same table, $\lambda = 23$ (not, as it happens, in part II., nor of course in the concluding part IV., which is a mere enumeration of real primes). Thus III. (1), we have congruences, (mod. p^3),

$$\eta_0 \equiv p \equiv 2, \quad \eta_1 \equiv p \equiv 3, \quad \eta_0 \equiv +12, \ \&c.;$$

and having actual prime factors,

p	$f(\eta)$
59	$-2\eta_1$
101	$1 - 4\eta_1$

and having ideal prime factors, their third powers actual,

p	$f^3(\eta)$
2	$1 - \eta_1$
3	$1 - 2\eta_1$;

[0]* Where, as presently appearing, 3 is the index of ideality or power to which the ideal factors have to be raised in order to become actual.

as regards these last the signification being

$$2^3 = (1 - \eta_0)(1 - \eta_1), \quad \eta_0 + \eta_1 = -1, \quad \eta_0\eta_1 = 6 \text{ (as is at once verified)},$$

$$3^3 = (1 - 2\eta_0)(1 - 2\eta_1);$$

but the simple numbers $2, 3$ are neither of them of the form $(a+b\eta_0)(a+b\eta_1)$.

Contents of Report 1875 on Mathematical Tables.

§ 7. *Tables F. Arithmological.*

612. Note sur une Formule d'Intégration Indéfinie

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXVIII. (Janvier–Juin, 1874), pp. 1624–1629.]

[p. 500] En étudiant les Mémoires de M. Serret (*Journal de Liouville*, t. X., 1845) par rapport à la représentation géométrique des fonctions elliptiques, avec les remarques de M. Liouville sur ce sujet, je suis parvenu à une formule d'intégration indéfinie qui me paraît assez remarquable, savoir: en prenant θ entier positif quelconque, je dis que l'intégrale

$$\int \frac{(x+p)^{m+n-\theta-1}(x+q)^{\theta}\,dx}{x^{m+1}(x+p+q)^{n+1}}$$

a une valeur algébrique

$$(x+p)^{m+n-\theta-1+1}(x+p+q)^{-n}x^{-m}(A+Bx+Cx^2+\ldots+Kx^{\theta-1}),$$

pourvu qu'une seule condition soit satisfaite par les quantités m, n, p, q. Cette condition s'écrit sous la forme symbolique

$$([m]p^{\theta}+[n]q^{\theta})^{\theta}=0,$$

en dénotant ainsi l'équation

$$[m]^{\theta}p^{\theta\cdot\theta}+\frac{\theta}{1}[m]^{\theta-1}[n]^{\theta}p^{\theta\cdot(\theta-1)}q^{\theta}+\ldots+[n]^{\theta}q^{\theta\cdot\theta}=0,$$

où, comme à l'ordinaire, $[m]^k$ signifie $m(m-1)\ldots(m-k+1)$.

Je rappelle que les formules de M. Serret ne contiennent que des exposants entiers, et celles de M. Liouville qu'un seul exposant quelconque: la nouvelle formule contient deux exposants quelconques, m, n. Je remarque aussi l'analogie de la condition $([m]p^{\theta}+[n]q^{\theta})^{\theta}=0$ avec celle-ci

$$\frac{1}{\zeta^{m-n}}\left(\frac{d}{d\zeta}\right)^{m}\zeta^{n}(\zeta-1)^{m}=0$$

(m étant un entier positif), qui figure dans les Mémoires cités.

[p. 501] Pour démontrer la formule, j'écris

$$u=x^{-m}(A+Bx+Cx^2+\ldots+Kx^{\theta-1}),$$

et aussi pour abréger

$$X=(x+p)^{m+n-\theta+1}(x+p+q)^{-n},$$

ce qui donne

$$\frac{X}{(x+p)(x+p+q)}=(x+p)^{m+n-\theta}(x+p+q)^{-n-1}.$$

L'équation à vérifier est donc

$$Xu = \int \frac{X(x+q)^\theta\, dx}{x^{m+1}(x+p)(x+p+q)},$$

ou, en différentiant et divisant par X,

$$\frac{X'}{X}u + u' = \frac{(x+q)^\theta}{x^{m+1}(x+p)(x+p+q)},$$

ou enfin

$$[(m+n-\theta+1)(x+p) - n(x+p)]u + (x+p)(x+p+q)u' = \frac{(x+q)^\theta}{x^{m+1}},$$

où u' dénote $\dfrac{du}{dx}$. Il ne s'agit donc que d'exprimer que cette équation ait une intégrale $u = x^{-m}(A + Bx + Cx^2 + \ldots + Kx^{\theta-1})$.

En supposant que cela soit ainsi, et en effectuant la substitution, les termes en $x^{-m+\theta}$ se détruisent, et l'on obtient une équation qui contient des termes en $x^{-m-1}, x^{-m}, \ldots, x^{-m+\theta-1}$, savoir $(\theta+1)$ termes. On a ainsi, entre les θ coefficients $A, B, C, \ldots, K$ un système de $(\theta+1)$ équations linéaires, ce qui implique une condition entre les constantes m, n, p, q; mais, cette condition satisfaite, les équations se réduisent à θ équations indépendantes, et les coefficients seront ainsi déterminés.

Par exemple, soit $\theta = 2$; l'équation différentielle est

$$[(m-1)p+(n-1)q+(m+n-1)x]u+[p^2+pq+x(2p+q)+x^2]u' = x^{-m-1}(q+x)^2,$$

laquelle doit être satisfaite par $u = Ax^{-m} + Bx^{-m+1}$. Cela donne

$$\begin{array}{cccc} x^{-m-1} & x^{-m} & x^{-m+1} & x^{-m+2} \end{array}$$
$$\begin{vmatrix} & (m-1p+m+n-1q)A, & (m-1p+m+n-1q)B, & \\ & & (m-1)A & (m-1)B \\ -m(p^2+pq)A & -(m-1)(p^2+pq)B & & \\ & -m(2p+q)A & -(m-1)(2p+q)B & -(m-1)B \\ & & -mA & \end{vmatrix} = 0.$$

[p. 502] [*The text on p. 502 of the original print consists of a large multi-column determinant table whose OCR has degraded significantly; the page is largely a tabular display of the system of linear equations leading from the differential equation to the symbolic condition* $([m]p^2+[n]q^2)^2 = 0$. *The page introduces the auxiliary notations* $([m]p+[n]q)^1$, $([m]p+[n]q)^2$, *etc., parallel to* $([m]p^2+[n]q^2)$, $([m]p^2+[n]q^2)^2$, *etc., previously explained, and shows the term-by-term computation. On a, par exemple,*

$$([m]p+[n]q)^2 = [m]^2p^2 + 2[m]'[n]'pq + [n]^2q^2.$$

]
[p. 503] et ainsi de suite. Les notations $([m]p+[n]q)^2$, $([m]p+[n]q)^3$, ... ont des significations semblables à celles de $([m]p^2+[n]q^2)^2$, $([m]p^2+[n]q^2)^3$, ..., auparavant expliquées. On a, par exemple,

$$([m]p+[n]q)^2 = [m]^2p^2 + 2[m]'[n]'pq + [n]^2q^2.$$

Considérons, par exemple, le deuxième déterminant: ceci contient trois termes en $1, 2q, q^2$ respectivement; le premier terme est

$$1\cdot(m-1)(p^2+pq)\cdot m(p^2+pq),$$

c'est-à-dire

$$[m]^2p^2(p+q)^2;$$

le deuxième terme est

$$2q\cdot -m(p^2+pq)[(m-1)p-nq],$$

c'est-à-dire

$$-2[m]'p(p+q)q([m-1]p-[n]q);$$

le troisième terme est

$$q^2[(m-1p-nq)(m+1p-n-1q)-(m-1)(p^2+pq)],$$

c'est-à-dire

$$q^2[(m^2-m)p^2-2mnpq+(n^2-n)q^2] = q^2([m]p-[n]q)^2.$$

Et de même le troisième déterminant est composé de quatre termes en $1, 3q, 3q^2, q^3$ respectivement, lesquels sont les quatre termes de la première expression transformée, et ainsi pour le quatrième déterminant, etc. Au moyen de ces premières transformées, on obtient sans peine les expressions finales $([m]p^2+[n]q^2)^2$, $([m]p^2+[n]q^2)^3$, ...

En écrivant $z = x+\frac{1}{2}(p+q)$ au lieu de x, et puis $\frac{1}{2}(p+q)=\alpha$, $\frac{1}{2}(p-q)=a$, la formule devient

$$\int \frac{(z-a)^{m+n-\theta-1}(z+a)^{\theta}\,dz}{(z-\alpha)^{m+1}(z+\alpha)^{n+1}},$$

et la valeur algébrique

$$= (z-a)^{m+n-\theta-1+1}(z+\alpha)^{-n}(z-\alpha)^{-m}(A'+B'z+\ldots+K'z^{\theta-1}),$$

pourvu qu'on ait entre les quantités m, n, a, α la relation

$$\{[m](\alpha+a)^{\theta}+[n](\alpha-a)^{\theta}\}^{\theta}=0.$$

En écrivant $\zeta = \dfrac{(\alpha + a)^\theta}{4\alpha\, a}$ et $\dfrac{(\alpha - a)^\theta}{4\alpha\, a} = \zeta - 1$, peut s'écrire sous la forme $\{[m]\zeta + [n](\zeta - 1)\}^\theta = 0$.

Je remarque que l'équation en ζ ne donne pas toujours pour ζ des valeurs réelles, positives et plus grandes que l'unité: par exemple, pour $\theta = 1$, on a $\zeta = \dfrac{-m}{n}$, valeur qui ne peut pas satisfaire à ces conditions. Je n'ai pas cherché dans quel cas ces conditions (qui ont rapport à l'application des formules à la représentation des fonctions elliptiques) subsistent.

usepackage[a4paper,margin=0.65in]geometry

Arthur Cayley

Collected Mathematical Papers

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613. On the Group of Points G_4^1 on a Sextic Curve with Five Double Points

[From the Mathematatische Annalen, vol. viii. (1875), pp. 359–362.]

[p. 504] The present note relates to a special group of points considered incidentally by MM. Brill and Nöther in their paper "Ueber die algebraischen Functionen und ihre Anwendung in der Geometrie," *Math. Annalen*, t. VII. pp. 268–310 (1874).

I recall some of the fundamental notions. We have a basis-curve which to fix the ideas may be taken to be of the order $n, = p+1$, $\frac{1}{2}p(p-3)$ dps, and therefore of the "Geschlecht" or deficiency p; any curve of the order $n-3$, $= p-2$ passing through the $\frac{1}{2}p(p-3)$ dps is said to be an *adjoint curve.* We may have, on the basis-curve, a special group G_Q^q of Q points ($Q \geq 2p-2$); viz. this is the case when the Q points are such that every adjoint curve through $Q-q$ of them—that is, every curve of the order $p-2$ through $\frac{1}{2}p(p-3)$ dps and the $Q-q$ points—passes through the remaining q points of the group: the number q may be termed the "speciality" of the group: if $q=0$, the group is an ordinary one.

It may be observed that a special group G_Q^q is chiefly noteworthy in the case where $Q-q$ is so small that the adjoint curve is not completely determined: thus if $p=5$, viz. if the basis-curve be a sextic with 5 dps, then we may have a special group G_6^1, but there is nothing remarkable in this; the 6 points are intersections with the sextic of an arbitrary cubic through the 5 dps—the cubic of course intersects the sextic in the 5 dps counting as 10 points, and in 8 other points—and such cubic is completely determined by means of the 5 dps and any 4 of the 6 points. But contrariwise, there is something remarkable in the group G_4^1 about to be considered: viz. we have here on the sextic 4 points, such that every cubic through the 5 dps and through 3 of the 4 points (through 8 points in all) passes through the remaining one of the 4 points.

The whole number of intersections of the basis-curve with an adjoint, exclusive of the dps counting as $p(p-3)$ points, is of course $= 2p-2$: hence an adjoint through the Q points of a group G_Q^q meets the basis-curve besides in $R, = 2p-2-Q$, **[p. 505]** points; we have then the "Riemann-Roch" theorem that these R points form a special group G_R^r, where

$$Q+R = 2p-2,$$

as just mentioned, and

$$Q-R = 2q-2r;$$

viz. dividing in any manner the $2p-2$ intersections of the basis-curve by an adjoint into groups of Q and R points respectively, these will be special groups, or at least one of them will be a special group, G^q_Q, G^r_R, such that their specialities q, r are connected by the foregoing relation $Q-R=2q-2r$.

The Authors give (l.c., p. 293) a Table showing for a given basis-curve, or given value of p, and for a given value of r, the least value of R and the corresponding values of q, Q: this table is conveniently expressed in the following form.

The least value of

$$R = p - \frac{p}{r+1} + r,$$

and then

$$q = \frac{p}{r+1} - 1,$$

$$Q = p + \frac{p}{r+1} - r - 2,$$

where $\dfrac{p}{r+1}$ denotes the integer equal to or next less than the fraction.

It is, I think, worth while to present the table in the more developed form:

n	p	Dps	$r=$					
			1	2	3	4	5	6
4	3	0	G^1_2 G^2_4	G^2_3 G^3_6	· ·	· ·	· ·	· ·
5	4	2	G^1_3 G^2_4	G^2_4 G^3_6	G^3_6 G^4_8	· ·	· ·	· ·
6	5	5	G^1_4 G^2_4	G^2_4 G^3_6	G^3_6 G^4_8	G^4_8 G^5_{10}	· ·	· ·
7	6	9	G^1_4 G^2_6	G^2_6 G^3_6	G^3_6 G^4_8	G^4_8 G^5_{10}	G^5_{10} G^6_{12}	· ·
8	7	14	G^1_4 G^2_6	G^2_6 G^3_8	G^3_8 G^4_8	G^4_{10} G^5_{10}	G^5_{10} G^6_{12}	G^6_{12} G^7_{14}

where the table shows the values of $\dfrac{G^r_R}{G^q_Q}$ for any given values of p, r.

[p. 506] I recur to the case $p=5$ and the group G^1_4, which is the subject of the present note: viz. we have here a sextic curve with 5 dps, and on it a group of 4 points G^1_4, such that every cubic through the 5 dps and through 3 points of the group, 8 points in all, passes through the remaining 1 point.

MM. Brill and Nöther show (by consideration of a rational transformation of the whole figure) that, given 2 points of the group, it is possible, and possible in 5 different ways, to determine the remaining 2 points of the group.

I remark that the 5 dps and the 4 points of the group form "an ennead" or system of the nine intersections of two cubic curves: and that the question is, given the 5 dps and 2 points on the sextic, to show how to determine on the sextic a pair of points forming with the 7 points an ennead: and to show that the number of solutions is $= 5$.

We have the following "Geiser-Cotterill" theorem:

If seven of the points of an ennead are fixed, and the eighth point describes a curve of the order n passing $\alpha_1, \alpha_2, \ldots, \alpha_7$ times through the seven points respectively, then will the ninth point describe a curve of the order ν passing $a_1, a_2, \ldots, a_7$ times through the seven points respectively: where

$$\nu = 8n - 3\Sigma\alpha,$$

$$a_1 = 3n - \alpha_1 - \Sigma\alpha,$$

$$\vdots$$

$$a_7 = 3n - \alpha_7 - \Sigma\alpha,$$

and conversely

$$n = 8\nu - 3\Sigma a,$$

$$\alpha_1 = 3\nu - a_1 - \Sigma a,$$

$$\vdots$$

$$\alpha_7 = 3\nu - a_7 - \Sigma a.$$

(Geiser, *Crelle-Borchardt*, t. LXVII. (1867), pp. 78–90; the complete form, as just stated, and which was obtained by Mr Cotterill, has not I believe been published); and also Geiser's theorem "the locus of the coincident eighth and ninth points is a sextic passing twice through each of the seven points."

The sextic and the curve n intersect in $6n$ points, among which are included the seven points counting as $2\Sigma\alpha$ points: the number of the remaining points is $= 6n - 2\Sigma\alpha$. Similarly, the sextic and the curve ν intersect in 6ν points, among which are included the seven points counting as $2\Sigma a$ points: the number of the remaining points is $6\nu - 2\Sigma a$ $(= 6n - 2\Sigma\alpha)$. The points in question are, it is clear, common intersections of the sextic, and the curves n, ν: viz. of the intersections of the curves n, ν, a number $6n - 2\Sigma\alpha, = 6\nu - 2\Sigma a, = 3n + 3\nu - 2\Sigma a - 2\Sigma\alpha$ lie on the sextic.

The curves n, ν intersect in $n\nu$ points, among which are included the seven points counting $\Sigma a\alpha$ times: the number of the remaining intersections is therefore **[p. 507]** $n\nu - \Sigma a\alpha$, but among these are included the $3n + 3\nu - 2\Sigma a - 2\Sigma\alpha$ points on the sextic; omitting these, there remain $n\nu - 3(n + \nu) - \Sigma a\alpha + 2\Sigma a + 2\Sigma\alpha$ points, or, what is the same thing, $(n - 3)(\nu - 3) - \Sigma(a - 1)(\alpha - 1) - 2$ points: it is clear that these must form pairs such that, the eighth point being either point of a pair, the ninth point will be the remaining point of the pair: the number of pairs is of course

$$\tfrac{1}{2}[(n - 3)(\nu - 3) - \Sigma(a - 1)(\alpha - 1) - 2],$$

and we have thus the solution of the question, given the seven points to determine the number of pairs of points on the curve n (or on the curve ν) such that each pair may form with the seven points an ennead.

In particular, if $n = 6$; $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7 = 2, 2, 2, 2, 2, 1, 1$ respectively, viz. if the curve be a sextic having 5 of the points for dps, and the remaining two for simple points, then we find $\nu = 12$; $a_1, a_2, a_3, a_4, a_5, a_6, a_7 = 4, 4, 4, 4, 4, 5, 5$ respectively, and the number of pairs is

$$= \tfrac{1}{2}[3 \cdot 9 - 5(2-1)(4-1) - 2], = \tfrac{1}{2}(27 - 15 - 2), = 5,$$

viz. starting with the 5 dps and any 2 points of the group G_4^1 we can, in 5 different ways, determine the remaining 2 points of the group.

In reference to the number $3p-3$ of parameters in the curves belonging to a given value of p, it may be remarked as follows. Such a curve is rationally transformable into a curve of the order $p+1$ with $\frac{1}{2}p(p-3)$ dps, and therefore containing $\frac{1}{2}(p+1)(p+4)$, $-\frac{1}{2}p(p-3)$, $= 4p+2$ parameters. Employing an arbitrary homographic transformation to establish any assumed relations between the parameters, the number is diminished to $4p+2-8$, $= 4p-6$; and again employing a rational transformation by means of adjoint curves of the order $p-2$ drawn through the dps and $p-3$ points of the curve—thereby transforming the curve into one of the same order $p+1$ and deficiency p—then, assuming that the $p-3$ parameters (or constants on which depend the positions of the $p-3$ points) can be disposed of so as to establish $p-3$ relations between the parameters and so further diminish the number by $p-3$, the required number of parameters will finally be $4p-6-(p-3) = 3p-3$.

Cambridge, 26th October, 1874.

614. On a Problem of Projection

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 19–29.]

[p. 508] I measure off on three rectangular axes the distances $\Omega X = \Omega Y = \Omega Z, = \theta$; and then, in a plane through Ω drawing in arbitrary directions the three lines ΩA, ΩB, ΩC, $= a$, b, c respectively, I assume that A, B, C (fig. 1) are the parallel projections of X, Y, Z respectively; viz. taking ΩO as the direction of the projecting lines, then ΩA, ΩB, ΩC being given in position and magnitude, we have to find θ, and the position of the line ΩO.

[Fig. 1: Sphere with centre Ω, showing axes X, Y, Z and projections A, B, C on a great circle, with projecting point O.]

This is in fact a case of a more general problem solved by Prof. Pohlke in 1853, (see the paper by Schwarz, "Elementarer Beweis des Pohlke'schen Fundamentalsatzes der Axonometrie," *Crelle*, t. LXIII. (1864), pp. 309–314), viz. the three lines ΩX, ΩY, ΩZ may be any three axes given in magnitude and direction, and their parallel projection **[p. 509]** is to be similar to the three lines ΩA, ΩB, ΩC. Schwarz obtains a very elegant construction, which I will first reproduce. We may imagine through Ω a plane cutting at right angles the projecting lines, say in the points X', Y', Z': we have then in plano a triad of lines $\Omega X'$, $\Omega Y'$, $\Omega Z'$ which are an orthogonal projection of ΩX, ΩY, ΩZ; and are also an orthogonal projection of a plane triad similar to ΩA, ΩB, ΩC; *quâ* such last-mentioned projection, the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, must be proportional to the triangles ΩBC, ΩCA, ΩAB; that is, we have to find an orthogonal projection of ΩX, ΩY, ΩZ, such that the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, which are the projections of ΩYZ, ΩZX, ΩXY respectively, shall be in given ratios. There is no difficulty in the solution of this problem; referring everything to a sphere centre Ω, let the normals to the planes ΩYZ, ΩZX, ΩXY, meet the sphere in the points X'', Y'', Z'' respectively, and the projecting line through Ω meet the sphere in the point O, then the projection of ΩYZ is to ΩYZ as $\cos OX'' : 1$; and the like as to the projections of ΩZX and ΩXY: that is, in the given spherical triangle $X''Y''Z''$, we have to find a point O, such that the cosines of the distances OX'', OY'', OZ'' are in given ratios: we have at once, through X'', Y'', Z'' respectively, three arcs meeting in the required point O.

The projecting lines being thus obtained, say these are the three parallel lines X', Y', Z', we have next to draw through Ω a plane meeting these in the points A', B', C' such that the triangle $A'B'C'$ is similar to the given triangle ABC; for this being so, the triangles $\Omega B'C'$, $\Omega C'A'$, $\Omega A'B'$ being the projections of, and therefore proportional to $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, that is, proportional to ΩBC, ΩCA, ΩAB, will, it is clear, be similar to these triangles respectively; that is, we have the triad $\Omega A'$, $\Omega B'$, $\Omega C'$, a projection of ΩX, ΩY, ΩZ, and similar to the triad ΩA, ΩB, ΩC, which is what was required.

It remains only to show how the given three parallel lines X', Y', Z', not in the same plane, can be cut by a plane in a triangle similar to a given triangle ABC.

[Fig. 2: Three parallel lines perpendicular to the plane of the paper through points X, Y, Z; auxiliary points A'', D, E, K, B', C' used in the construction.]

Imagine the three lines at right angles to the plane of the paper, meeting the plane of the paper in the given points X, Y, Z (fig. 2) respectively. On the base **[p. 510]** YZ describe a triangle $A''YZ$ similar to the given triangle ABC; and through A'', X with centre on the line YZ, describe a circle meeting this line in the points D and E. Then in the plane, through YZ at right angles to the plane of the paper, we may draw a line meeting the lines Y, Z in the points B'', C'' respectively, such that joining XB'', XC'' we obtain a triangle $XB''C''$ similar to $A''YZ$, that is, to the given triangle ABC.

Taking K the centre of the circle, suppose that its radius is $= l$, and that we have $KY = \beta$, $KZ = \gamma$; also $YZ = \sigma$, $ZX = \tau$; $YA'' = \sigma''$, $ZA'' = \tau''$. If for a moment x, y denote the coordinates of A'', then

$$\tau^2 = (x-\gamma)^2 + y^2, = l^2 + \gamma^2 - 2\gamma x,$$

and thence

$$\gamma\sigma^2 - \beta\tau^2 = \gamma(l^2+\beta^2) - \beta(l^2+\gamma^2),$$

that is,

$$\gamma\sigma^2 - \beta\tau^2 = (\gamma-\beta)(l^2-\beta\gamma);$$

viz. this is the equation of the circle in terms of the vectors σ, τ; we have therefore in like manner

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma-\beta)(l^2-\beta\gamma).$$

We may determine θ so as to satisfy the two equations

$$\sigma''^2 = \sigma^2\cos^2\theta + (l+\beta)^2\sin^2\theta,$$

$$\tau''^2 = \tau^2\cos^2\theta + (l+\gamma)^2\sin^2\theta;$$

in fact, these equations give

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma\sigma^2-\beta\tau^2)\cos^2\theta + \{\gamma(l+\beta)^2 - \beta(l+\gamma)^2\}\sin^2\theta,$$

which, the left-hand side and the coefficients of $\cos^2\theta$ and $\sin^2\theta$ on the right-hand side being each $= (\gamma-\beta)(l^2-\beta\gamma)$, is, in fact, an identity.

But in the figure, if θ, determined as above, denote the angle at B, then

$$(XB'')^2 = XY^2 + YB''^2 = \sigma^2 + (l+\beta)^2 \tan^2\theta,$$

$$(XC'')^2 = XZ^2 + ZC''^2 = \tau^2 + (l+\gamma)^2 \tan^2\theta,$$

that is,

$$XB'' = \sigma'' \sec\theta, \quad XC'' = \tau'' \sec\theta,$$

or, since $B''C'' = YZ\sec\theta$ [$= (\gamma-\beta)\sec\theta$], the triangle $XB''C''$ is, as mentioned, similar to the triangle $A''YZ$.

I was not acquainted with the foregoing construction when my paper was written; but the analytical investigation of the particular case is nevertheless interesting, and I proceed to consider it.

[p. 511] Taking (fig. 1) Ω as the centre of a sphere and projecting on this sphere, we have A, B, C given points on a great circle; and we have to find the point O, such that there may be a trirectangular triangle XYZ, the vertices of which lie in OA, OB, OC respectively, and for which

$$\frac{\sin OX}{\sin OA} = \frac{a}{\theta}, \quad \frac{\sin OY}{\sin OB} = \frac{b}{\theta}, \quad \frac{\sin OZ}{\sin OC} = \frac{c}{\theta}.$$

I take the arcs BC, CA, $AB = \alpha$, β, γ respectively, $\alpha+\beta+\gamma = 2\pi$; and the required arcs OA, OB, OC are taken to be ξ, η, ζ respectively; these are connected by the relation

$$\sin\alpha\cos\xi + \sin\beta\cos\eta + \sin\gamma\cos\zeta = 0,$$

to obtain which, observe that from the triangles OAB, OAC, we have

$$\cos A = \frac{\cos\eta - \cos\xi\cos\gamma}{\sin\xi\sin\gamma} = -\frac{\cos\zeta - \cos\xi\cos\beta}{\sin\xi\sin\beta},$$

that is,

$$\sin\beta(\cos\eta - \cos\xi\cos\gamma) + \sin\gamma(\cos\zeta - \cos\xi\cos\beta) = 0,$$

which, with $\sin\alpha = -\sin(\beta+\gamma)$, gives the required relation. We have

$$\sin OX = \frac{a}{\theta}\sin\xi, \quad \sin OY = \frac{b}{\theta}\sin\eta, \quad \sin OZ = \frac{c}{\theta}\sin\zeta;$$

and then from the triangles OBC, OCA, OAB, and the quadrantal triangles OYZ, OZX, OXY, we have

$$\cos BOC = \frac{\cos\alpha - \cos\eta\cos\zeta}{\sin\eta\sin\zeta} = \frac{-\sqrt{\theta^2 - b^2\sin^2\eta}\,\sqrt{\theta^2 - c^2\sin^2\zeta}}{bc\sin\eta\sin\zeta\,/\,\theta^2}, \text{ \&c.,}$$

that is,

$$bc(\cos\alpha - \cos\eta\cos\zeta) = -\sqrt{\theta^2 - b^2\sin^2\eta}\,\sqrt{\theta^2 - c^2\sin^2\zeta},$$

$$ca(\cos\beta - \cos\zeta\cos\xi) = -\sqrt{\theta^2 - c^2\sin^2\zeta}\,\sqrt{\theta^2 - a^2\sin^2\xi},$$

$$ab(\cos\gamma - \cos\xi\cos\eta) = -\sqrt{\theta^2 - a^2\sin^2\xi}\,\sqrt{\theta^2 - b^2\sin^2\eta},$$

which, when rationalized, are quadric equations in $\cos\xi$, $\cos\eta$, $\cos\zeta$. The first equation, in fact, gives

$$b^2c^2(\cos\alpha - \cos\eta\cos\zeta)^2 = (\theta^2 - b^2 + b^2\cos^2\eta)(\theta^2 - c^2 + c^2\cos^2\zeta),$$

that is,

$$(\theta^2-b^2)(\theta^2-c^2)-b^2c^2\cos^2\alpha+(\theta^2-b^2)c^2\cos^2\zeta+(\theta^2-c^2)b^2\cos^2\eta+2b^2c^2\cos\alpha\cos\eta\cos\zeta = 0,$$

or, what is the same thing,

$$-\left(1 - b^2 - c^2 - \frac{b^2c^2\cos^2\alpha}{\theta^2 - b^2\cdot\theta^2 - c^2}\right)+\frac{b^2}{\theta^2-b^2}\cos^2\eta+\frac{c^2}{\theta^2-c^2}\cos^2\zeta+\frac{2b^2c^2\cos\alpha}{(\theta^2-b^2)(\theta^2-c^2)}\cos\eta\cos\zeta = 0.$$

Completing the system, we have

$$-\left(1 - c^2 - a^2 - \frac{c^2a^2\cos^2\beta}{\theta^2 - c^2\cdot\theta^2 - a^2}\right)+\frac{c^2}{\theta^2-c^2}\cos^2\zeta+\frac{a^2}{\theta^2-a^2}\cos^2\xi+\frac{2c^2a^2\cos\beta}{(\theta^2-c^2)(\theta^2-a^2)}\cos\zeta\cos\xi = 0$$

$$-\left(1 - a^2 - b^2 - \frac{a^2b^2\cos^2\gamma}{\theta^2 - a^2\cdot\theta^2 - b^2}\right)+\frac{a^2}{\theta^2-a^2}\cos^2\xi+\frac{b^2}{\theta^2-b^2}\cos^2\eta+\frac{2a^2b^2\cos\gamma}{(\theta^2-a^2)(\theta^2-b^2)}\cos\xi\cos\eta = 0$$

[p. 512] and, as above,

$$\sin\alpha\cos\xi + \sin\beta\cos\eta + \sin\gamma\cos\zeta = 0.$$

It seems difficult from these equations to eliminate ξ, η, ζ, so as to obtain an equation in θ; but I employ some geometrical considerations.

Taking Π as the pole of the circle ABC, and drawing ΠX, ΠY, ΠZ to meet the circle in p, q, r respectively, then, if α'', β'', γ'' are the cosine-inclinations of Π to X, Y, Z respectively, we have

$$\sin Xp,\ \sin Yq,\ \sin Zr = \alpha'',\ \beta'',\ \gamma''.$$

From the right parallel triangles BYq and CZr, we have

$$\sin Yq = \sin BY\sin B,$$

$$\sin Zr = \sin CZ\sin C,$$

and, thence,

$$\frac{\sin Yq}{\sin Zr} = \frac{\sin BY}{\sin CZ}\cdot\frac{\sin OC}{\sin OB};$$

or, since

$$BY = OB - OY,\quad CZ = OC - OZ,$$

and thence

$$\sin BY = \frac{1}{\theta}\{\sqrt{\theta^2 - b^2\sin^2\eta} - b\cos\eta\},$$

$$\sin CZ = \frac{1}{\theta}\{\sqrt{\theta^2 - c^2\sin^2\zeta} - c\cos\zeta\},$$

we obtain

$$\frac{\beta''}{\gamma''} = \frac{\sqrt{\theta^2 - b^2\sin^2\eta} - b\cos\eta}{\sqrt{\theta^2 - c^2\sin^2\zeta} - c\cos\zeta}.$$

We have thence

$$\beta''\sqrt{\theta^2 - c^2\sin^2\zeta} - \gamma''\sqrt{\theta^2 - b^2\sin^2\eta} = \beta'' c\cos\zeta - \gamma'' b\cos\eta,$$

or, squaring and reducing

$$\beta''^2(\theta^2-c^2)+\gamma''^2(\theta^2-b^2)+2\beta''\gamma''\left[-\sqrt{\theta^2 - c^2\sin^2\zeta}\sqrt{\theta^2 - b^2\sin^2\eta} + bc\cos\eta\cos\zeta\right] = 0;$$

that is,

$$\beta''^2(\theta^2 - c^2) + \gamma''^2(\theta^2 - b^2) + 2\beta''\gamma'' \cdot bc\cos\alpha = 0,$$

and, similarly,

$$\gamma''^2(\theta^2 - a^2) + \alpha''^2(\theta^2 - c^2) + 2\gamma''\alpha'' \cdot ca\cos\beta = 0,$$

$$\alpha''^2(\theta^2 - b^2) + \beta''^2(\theta^2 - a^2) + 2\alpha''\beta'' \cdot ab\cos\gamma = 0,$$

or, what is the same thing,

$$\frac{\beta''^2}{\theta^2 - b^2} + \frac{\gamma''^2}{\theta^2 - c^2} = \frac{-2bc\cos\alpha}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}} \cdot \frac{\beta''\gamma''}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}},$$

$$\frac{\gamma''^2}{\theta^2 - c^2} + \frac{\alpha''^2}{\theta^2 - a^2} = \frac{-2ca\cos\beta}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}} \cdot \frac{\gamma''\alpha''}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}},$$

$$\frac{\alpha''^2}{\theta^2 - a^2} + \frac{\beta''^2}{\theta^2 - b^2} = \frac{-2ab\cos\gamma}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}} \cdot \frac{\alpha''\beta''}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}}.$$

[p. 513] writing

$$\alpha'',\ \beta'',\ \gamma'' = X'\sqrt{\theta^2 - a^2},\ Y'\sqrt{\theta^2 - b^2},\ Z'\sqrt{\theta^2 - c^2},$$

and

$$\frac{bc\cos\alpha}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}} = f, \quad \frac{ca\cos\beta}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}} = g, \quad \frac{ab\cos\gamma}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}} = h,$$

the equations are

$$Y'^2 + Z'^2 - 2fY'Z' = 0,$$

$$Z'^2 + X'^2 - 2gZ'X' = 0,$$
$$X'^2 + Y'^2 - 2hX'Y' = 0.$$

Writing the last two under the form

$$Z'^2 - 2gZ'X' + X'^2 = 0,$$
$$X'^2 - 2hX'Y' + Y'^2 = 0,$$

and eliminating Z', we have

$$-4(1-g^2)(1-h^2)Y'^2X'^2 + (Y'^2 + Z'^2 - 2ghY'Z')^2 = 0,$$

which, in virtue of the first equation, is

$$-4(1-g^2)(1-h^2)Y'^2Z'^2 + 4(gh-f)^2Y'^2Z'^2 = 0,$$

that is,

$$(1-g^2)(1-h^2) - (gh-f)^2 = 0;$$

or, what is the same thing,

$$1 - f^2 - g^2 - h^2 + 2fgh = 0.$$

I remark that we may write

$$gh - f = \sqrt{(1-g^2)}\sqrt{(1-h^2)},$$
$$hf - g = \sqrt{(1-h^2)}\sqrt{(1-f^2)},$$
$$fg - h = \sqrt{(1-f^2)}\sqrt{(1-g^2)},$$

the signs on the right-hand side being either all $+$, or else one $+$ and two $-$, so that the product is $+$. In fact, multiplying the assumed equations, we have

$$f^2g^2h^2 - fgh(f^2+g^2+h^2) + f^2g^2 + h^2f^2 + f^2g^2 - fgh = 1 - f^2 - g^2 - h^2 + f^2h^2 + h^2f^2 + f^2g^2 - f^2g^2h^2,$$

that is,

$$1 - f^2 - g^2 - h^2 + fgh(1 + f^2 + g^2 + h^2) - 2f^2g^2h^2 = 0,$$

or,

$$(1 - f^2 - g^2 - h^2 + 2fgh)(1 - fgh) = 0,$$

which is right; but with a different combination of signs the result would not have been obtained.

Substituting for f, g, h their values, we have

$$(a^2-\theta^2)(b^2-\theta^2)(c^2-\theta^2) - b^2c^2(a^2-\theta^2)\cos^2\alpha - c^2a^2(b^2-\theta^2)\cos^2\beta$$

$$-a^2b^2(c^2-\theta^2)\cos^2\gamma + 2a^2b^2c^2\cos\alpha\cos\beta\cos\gamma = 0,$$

[p. 514] where the term independent of θ is

$$a^2b^2c^2(1-\cos^2\alpha-\cos^2\beta-\cos^2\gamma+2\cos\alpha\cos\beta\cos\gamma),$$

which is $=0$ in virtue of $\alpha+\beta+\gamma=2\pi$. We have, therefore, for θ^2 the quadric equation

$$b^2c^2\sin^2\alpha + c^2a^2\sin^2\beta + a^2b^2\sin^2\gamma - (a^2+b^2+c^2)\theta^2 + \theta^4 = 0,$$

giving for θ^2 the two real positive values

$$\theta^2 = \tfrac{1}{2}\{a^2+b^2+c^2+\sqrt{\Omega}\},$$

where

$$\begin{aligned}\Omega &= (a^2+b^2+c^2)^2 - 4(b^2c^2\sin^2\alpha + c^2a^2\sin^2\beta + a^2b^2\sin^2\gamma)\\ &= a^4+b^4+c^4+2b^2c^2\cos 2\alpha + 2c^2a^2\cos 2\beta + 2a^2b^2\cos 2\gamma\\ &= (a^2+b^2\cos 2\gamma + c^2\cos 2\beta)^2 + (b^2\sin 2\gamma - c^2\sin 2\beta)^2.\end{aligned}$$

I write now

$$\frac{a\cos\xi}{\sqrt{(a^2-\theta^2)}} = X,\quad \frac{b\cos\eta}{\sqrt{(b^2-\theta^2)}} = Y,\quad \frac{c\cos\zeta}{\sqrt{(c^2-\theta^2)}} = Z,$$

and also

$$\frac{\sqrt{a^2-\theta^2}}{a}\sin\alpha = A,\quad \frac{\sqrt{b^2-\theta^2}}{b}\sin\beta = B,\quad \frac{\sqrt{c^2-\theta^2}}{c}\sin\gamma = C.$$

The equations for $\cos\xi$, $\cos\eta$, $\cos\zeta$ become

$$\begin{aligned}Y^2+Z^2-2fYZ-(1-f^2)&=0,\\ Z^2+X^2-2gZX-(1-g^2)&=0,\\ X^2+Y^2-2hXY-(1-h^2)&=0,\end{aligned}$$

and

$$AX+BY+CZ=0,$$

in virtue of the relation between f, g, h. The first three equations are satisfied by a two-fold relation between X, Y, Z; viz. treating these as coordinates, the equations represent three quadric cylinders having a common conic.

To prove this, I write

$$1-f^2,\ 1-g^2,\ 1-h^2,\ gh-f,\ hf-g,\ fg-h = \mathfrak{a},\ \mathfrak{b},\ \mathfrak{c},\ \mathfrak{f},\ \mathfrak{g},\ \mathfrak{h}.$$

We have, as usual,

$$\mathfrak{bc}-\mathfrak{f}^2,\ \mathfrak{ca}-\mathfrak{g}^2,\ \mathfrak{ab}-\mathfrak{h}^2,\ \mathfrak{gh}-\mathfrak{af},\ \mathfrak{hf}-\mathfrak{bg},\ \mathfrak{fg}-\mathfrak{ch},\ \text{each}=0:$$

the equations

$$\mathfrak{a}X+\mathfrak{h}Y+\mathfrak{g}Z=0,\quad \mathfrak{h}X+\mathfrak{b}Y+\mathfrak{f}Z=0,\quad \mathfrak{g}X+\mathfrak{f}Y+\mathfrak{c}Z=0,$$

represent each of them one and the same plane, which I say is that of the conic in question.
[p. 515] The three given equations are

$$Y^2+Z^2-2fYZ-\mathfrak{a}=0,$$
$$Z^2+X^2-2gZX-\mathfrak{b}=0,$$
$$X^2+Y^2-2hXY-\mathfrak{c}=0,$$

say these are $U=0$, $V=0$, $W=0$; it is to be shown that $cV-bW$, $aW-cU$, $bU-aV$, each contain the linear factor in question. We have

$$\mathfrak{c}V-\mathfrak{b}W=(\mathfrak{c}-\mathfrak{b})X^2-\mathfrak{b}Y^2+\mathfrak{c}Z^2-2\mathfrak{c}gZX+2\mathfrak{b}hXY;$$

or, what is the same thing,

$$\mathfrak{a}(\mathfrak{c}V-\mathfrak{b}W)=\mathfrak{a}(\mathfrak{c}-\mathfrak{b})X^2-\mathfrak{h}^2Y^2+\mathfrak{g}^2Z^2-2\mathfrak{g}\mathfrak{g}_*ZX+2\mathfrak{h}\mathfrak{h}_*XY.$$

Assuming this

$$=(\mathfrak{a}X+\mathfrak{h}Y+\mathfrak{g}Z)(\lambda X-\mathfrak{h}Y+\mathfrak{g}Z),$$

we have

$$\mathfrak{a}\lambda=\mathfrak{a}(\mathfrak{c}-\mathfrak{b}),$$
$$\mathfrak{g}(\mathfrak{a}+\lambda)=-2\mathfrak{g}\mathfrak{g}_*,$$
$$\mathfrak{h}(-\mathfrak{a}+\lambda)=2\mathfrak{h}\mathfrak{h}_*,$$

that is,

$$\lambda=\mathfrak{c}-\mathfrak{b},\quad \mathfrak{a}+\lambda=-2\mathfrak{g}_*,\quad -\mathfrak{a}+\lambda=2\mathfrak{h}_*;$$

but $\lambda=\mathfrak{c}-\mathfrak{b},=-\mathfrak{h}^2+\mathfrak{g}^2$, and the other two equations are $\mathfrak{a}+\mathfrak{c}-\mathfrak{b}+2\mathfrak{g}_*=0$, $\mathfrak{a}+\mathfrak{b}-\mathfrak{c}+2\mathfrak{h}_*=0$, which are identically true.

The values of X, Y, Z are thus determined as the coordinates of the intersection of the conic with the plane $AX+BY+CZ=0$; or, what is the same thing, of the line

$$AX+BY+CZ=0,$$
$$\mathfrak{a}X+\mathfrak{h}Y+\mathfrak{g}Z=0,$$

with any one of the three cylinders.

We may, however, complete the analytical solution in a different manner as follows. Assuming as above $\sqrt{(\mathfrak{bc})} = \mathfrak{f}$, $\sqrt{(\mathfrak{ca})} = \mathfrak{g}$, $\sqrt{(\mathfrak{ab})} = \mathfrak{h}$, and thence $\mathfrak{h}\sqrt{(\mathfrak{c})} - \mathfrak{g}\sqrt{(\mathfrak{b})} = 0$, we obtain from the second and the third equations

$$Y = hX + \sqrt{(\mathfrak{c})}\sqrt{(1-X^2)}, \quad Z = gX - \sqrt{(\mathfrak{b})}\sqrt{(1-X^2)},$$

(the signs are one + the other −, in order that this may consist with the equation $\mathfrak{a}X + \mathfrak{h}Y + \mathfrak{g}Z = 0$). Substituting in $AX + BY + CZ = 0$, we have

$$(A + Bh + Cg)X + \{B\sqrt{(\mathfrak{c})} - C\sqrt{(\mathfrak{b})}\}\sqrt{(1-X^2)} = 0,$$

that is,

$$(A + Bh + Cg)^2X^2 - (B^2\mathfrak{c} + C^2\mathfrak{b} - 2BC\mathfrak{f})(1 - X^2) = 0,$$

[p. 516] or say

$$(A+Bh+Cg)^2X^2+\{B^2(1-h^2)+C^2(1-g^2)-2BC(gh-f)\}(X^2-1) = 0,$$

that is,

$$(A^2+B^2+C^2+2BCf+2CAg+2ABh)X^2 = [B^2+C^2+2BCf-(Bh+Cg)^2],$$

or writing

$$A^2 + B^2 + C^2 + 2BCf + 2CAg + 2ABh = \Delta,$$

say we have

$$\begin{aligned}\Delta X^2 &= B^2 + C^2 + 2BCf - (Bh + Cg)^2,\\ \Delta Y^2 &= C^2 + A^2 + 2CAg - (Cf + Ah)^2,\\ \Delta Z^2 &= A^2 + B^2 + 2ABh - (Ag + Bf)^2.\end{aligned}$$

Now attending to the values of A, B, C, f, g, h, we have

$$BCf,\ CAg,\ ABh = \sin\beta\sin\gamma\cos\alpha,\ \sin\gamma\sin\alpha\cos\beta,\ \sin\alpha\sin\beta\cos\gamma,$$

and thence

$$\Delta = \sin^2\alpha\left(1-\frac{\theta^2}{a^2}\right) + \sin^2\beta\left(1-\frac{\theta^2}{b^2}\right) + \sin^2\gamma\left(1-\frac{\theta^2}{c^2}\right)$$
$$+2(\sin\beta\sin\gamma\cos\alpha + \sin\gamma\sin\alpha\cos\beta + \sin\alpha\sin\beta\cos\gamma);$$

in virtue of $\alpha + \beta + \gamma = 2\pi$, the last term is

$$= 2(\cos\alpha\cos\beta\cos\gamma - 1),$$

whence

$$\Delta = -\theta^2\left(\frac{\sin^2\alpha}{a^2} + \frac{\sin^2\beta}{b^2} + \frac{\sin^2\gamma}{c^2}\right), \text{ say this is } = -\theta^2\Lambda.$$

Moreover $Bh + Cg = \dfrac{a\sin\alpha}{\sqrt{(a^2-\theta^2)}}$, whence the value of ΔX^2 is

$$= \sin^2\beta\left(1-\frac{\theta^2}{b^2}\right)+\sin^2\gamma\left(1-\frac{\theta^2}{c^2}\right)+2\sin\beta\sin\gamma\cos\alpha-\left(1-\frac{\theta^2}{a^2}\right)^{-1}\sin^2\alpha.$$

Here the constant term is

$$= \sin^2\beta+\sin^2\gamma+2\sin\beta\sin\gamma\cos\alpha,$$

that is,

$$\begin{aligned} &= 1-(1-\sin^2\beta)(1-\sin^2\gamma)+\sin^2\beta\sin^2\gamma+2\sin\beta\sin\gamma\cos\alpha \\ &= 1-\cos^2\beta\cos^2\gamma-\cos^2\alpha+(\cos\alpha+\sin\beta\sin\gamma)^2 \\ &= 1-\cos^2\alpha, = \sin^2\alpha, \end{aligned}$$

or the whole is

$$\sin^2\alpha\left(1-\frac{a^2}{a^2-\theta^2}\right)-\theta^2\left(\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right),$$

[p. 517] which is

$$= -\theta^2\left(\frac{\sin^2\alpha}{a^2-\theta^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right),$$

so that we have

$$\Lambda X^2 = \left(\frac{\sin^2\alpha}{a^2-\theta^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right).$$

Similarly,

$$\Lambda Y^2 = \left(\frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2-\theta^2}+\frac{\sin^2\gamma}{c^2}\right),$$

$$\Lambda Z^2 = \left(\frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2-\theta^2}\right);$$

and hence also

$$\Lambda(1-X^2) = \frac{-\theta^2\sin^2\alpha}{a^2(a^2-\theta^2)},\quad \Lambda(1-Y^2) = \frac{-\theta^2\sin^2\beta}{b^2(b^2-\theta^2)},\quad \Lambda(1-Z^2) = \frac{-\theta^2\sin^2\gamma}{c^2(c^2-\theta^2)},$$

where

$$\Lambda = \frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}.$$

The equation in X is

$$\Lambda\left(1-\frac{a^2\cos^2\xi}{a^2-\theta^2}\right) = \frac{-\theta^2\sin^2\alpha}{a^2(a^2-\theta^2)},$$

that is,

$$\Lambda(a^2\sin^2\xi - \theta^2) = \frac{-\theta^2\sin^2\alpha}{a^2},$$

or

$$a^2\sin^2\xi = \frac{\theta^2}{\Lambda}\left(\Lambda - \frac{\sin^2\alpha}{a^2}\right),$$

and the like for η, ζ. Writing for greater convenience $\dfrac{\sin^2\alpha}{a^2}, \dfrac{\sin^2\beta}{b^2}, \dfrac{\sin^2\gamma}{c^2} = p, q, r$, then $\Lambda = p+q+r$, and we have

$$\sin^2\xi = \frac{\theta^2}{a^2}\cdot\frac{q+r}{p+q+r},\quad \sin^2\eta = \frac{\theta^2}{b^2}\cdot\frac{r+p}{p+q+r},\quad \sin^2\zeta = \frac{\theta^2}{c^2}\cdot\frac{p+q}{p+q+r},$$

(whence also $a^2\sin^2\xi + b^2\sin^2\eta + c^2\sin^2\zeta = 2\theta^2$: as a simple verification, observe that, if the projection is rectangular, the axes being all equally inclined to the plane of projection, then $\xi = \eta = \zeta = 90°$, $a = b = c = \theta\sin s$, and the equation is $3\sin^2 s = 2$; s, s are here the sides of an isosceles quadrantal triangle, the included angle being $120°$, that is, we have $\cos 120°$ $(= -\frac{1}{2}) = -\cot^2 s$, that is, $\cot^2 s = \frac{1}{2}$, or $\sin^2 s = \frac{2}{3}$, which is right).

I remark, that a geometrical solution may be obtained upon very different principles. We have on a sphere the trirectangular triangle XYZ, which by parallel lines is projected into ABC. Every great circle of the sphere is projected into an ellipse having double **[p. 518]** contact at the extremities of a diameter with the ellipse which is the apparent contour of the sphere. Moreover, if the arc of great circle XY is a quadrant, then the radius through X and the tangent at Y are parallel to each other, whence, if Ω be the projection of the centre, and AB the projection of the arc XY, then in the projection the line ΩA and the tangent at B are parallel to each other. It is now easy to derive a construction: with centre Ω, and conjugate semi-axes $(\Omega B, \Omega C)$, $(\Omega C, \Omega A)$, $(\Omega A, \Omega B)$ respectively, describe three ellipses; and find a concentric ellipse having double contact with each of these (there are in fact two such ellipses, one touching the three ellipses internally, and giving an imaginary solution; the other touching them externally, which is the ellipse intended). Drawing then through the ellipse a right cylinder (there are two such cylinders, but only one of them is real), and inscribing in it a sphere, and projecting on to the surface of the sphere by lines parallel to the axis of the cylinder, the three ellipses are projected into three great circles cutting at right angles, or, say, the elliptic arcs BC, CA, AB are projected into the trirectangular triangle XYZ.

615. On the Conic Torus

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 127–129.]

[p. 519] The equation $p+\sqrt{(qr)}+\sqrt{(st)}=0$, where p, q, r, s, t are linear functions of the coordinates (x, y, z, w), and as such are connected by a linear relation, belongs to a quartic surface having a nodal conic $(p=0, qr-st=0)$; and four nodes (conical points), viz. these are the intersections of the line $q=0$, $r=0$ with the quadric surface $p^2-qr-st=0$, and of the line $r=0$, $s=0$ with the same surface. The quartic surface has also four tropes (planes which touch the surface along a conic); viz. these are the planes $q=0$, $r=0$, $s=0$, $t=0$, the conic of contact or tropal conic in each plane being the intersection of the plane with the before-mentioned quadric surface $p^2-qr-st=0$. The planes $q=0$, $r=0$, and also the conics in these planes pass through two of the nodes, say A, C; and the planes $s=0$, $t=0$, and also the conics in these planes pass through the remaining two nodes, say B, D; so that the relations of the surface are as is shown in fig. 1. It is to be added that AB, BC, CD, DA (but not AC or BD) are lines on the surface.

The planes $q=0$, $r=0$, which contain the tropal conics through A, C, are in general distinct from the planes ABC, ADC which contain the line-pairs BA, BC and DA, DC respectively: and so also the planes $s=0$, $t=0$, which contain the tropal conics through B, D, are in general distinct from the planes ABD, CBD which contain the line-pairs AB, AD and CB, CD respectively.

If, however, the identical linear relation contain only p, s, t, then the planes $q=0$, $r=0$ will be the planes ABC, ADC respectively: and the tropal conics in these planes will consequently be the line-pairs BA, BC, and DA, DC respectively. But the planes $s=0$, $t=0$ will continue to be distinct from the planes ABD, CBD: and the tropal conics in the planes $s=0$, $t=0$ will remain proper conics.

[p. 520] A surface of the last-mentioned form is

$$mz+\sqrt{(xy)}+\sqrt{(w^2-z^2)}=0,$$

viz. this has the nodal conic $z=0$, $xy-w^2=0$, the nodes $\{x=0, y=0, (m^2+1)z^2-w^2=0\}$, and $(z=0, w=0, x=0)$, $(z=0, w=0, y=0)$, and the tropes $x=0$, $y=0$, $z+w=0$, $z-w=0$; but the planes $z+w=0$ and $z-w=0$ are ordinary tropal planes each touching the surface in a proper conic; the planes $x=0$, $y=0$ are special planes each touching along a line-pair.

[Fig. 1: Tetrahedron $ABCD$ with the marked planes and edges as described.]

The equation in question, writing therein $w = 1$ and $x + iy$, $x - iy$ in place of (x, y) respectively, is

$$\sqrt{(x^2 + y^2)} + (mz)^2 = 1 - z^2,$$

which is derived from

$$(x + mz)^2 = 1 - z^2,$$

by the change of x into $\sqrt{(x^2 + y^2)}$; and the surface is consequently the torus generated by the rotation of the conic $(x+mz)^2 = 1-z^2$ about its diameter. Or, what is the same thing, the surface

$$mz + \sqrt{(xy)} + \sqrt{(w^2 - z^2)} = 0,$$

regarding therein (x, y) as circular coordinates and w as being $= 1$, is a torus. The rational equation is $U = 0$, where we have

$$\begin{aligned} U &= \{(m^2 + 1)z^2 - w^2 + xy\}^2 - 4m^2z^2xy \\ &= \{xy + (1 - m^2)z^2 - w^2\}^2 + 4m^2z^2(z^2 - w^2) \\ &= x^2y^2 + (m^2 + 1)^2z^4 + w^4 + (2 - 2m^2)z^2xy - (2 + 2m^2)z^2w^2 - 2xyw^2. \end{aligned}$$

I find that the Hessian H of this function U contains the factor $xy + (1 - m^2)z^2 - w^2$, viz. that we have

$$H = \{xy + (1 - m^2)z^2 - w^2\}H',$$

[p. 521] where

$$\begin{gathered} H' = x^3y^3(1 - m^2) \\ +x^2y^2\left[(3 + 8m^2 + m^4)z^2 + (-3 + m^2)w^2\right] \\ +xy\left\{(3 + 11m^2 + 9m^4 + m^6)z^4 + (-6 - 12m^2 + 6m^4)z^2w^2 + (3 + m^2)w^4\right\} \\ +(1 + m^2)\{(1 - m^2)z^2 - w^2\}\{(1 + m^2)z^2 - w^2\}^2, \end{gathered}$$

giving without much difficulty

$$\begin{gathered} H' = z^6(1 + m^2)^3(1 - m^2) \\ +2z^4\left[(1 + 4m^2 + m^4)xy - (1 - m^4)w^2\right](1 + m^2) \\ +z^2(xy - w^2)\left[(1 + 12m^2 - m^4)xy - (1 - m^4)w^2\right] \\ +\left[(1 - m^2)xy - (1 + m^2)w^2\right]U; \end{gathered}$$

say this is

$$= z^2H'' + [(1 - m^2)xy - (1 + m^2)w^2]U,$$

where

$$H'' = z^4(1 + m^2)^2(1 - m^2)$$

$$+2z^2(1+m^2)\left[(1+4m^2+m^4)xy-(1-m^4)w^2\right]$$
$$+(xy-w^2)\left[(1+12m^2-m^4)xy-(1-m^4)w^2\right],$$

or, what is the same thing,

$$H'' = x^2y^2(1+12m^2-m^4)$$
$$+2xy\left[(1+4m^2+m^4)(1+m^2)z^2+(-1+6m^2)w^2\right]$$
$$+(1-m^4)\{(1+m^2)z^2-w^2\}^2.$$

It consequently appears that the complete spinode curve or intersection of the quartic surface and its Hessian, being of order $4\times 8, = 32$, breaks up into

$$U=0, \quad xy+(1-m^2)z^2-w^2=0,$$

that is,

$$\begin{array}{lr} \text{conic } z=0,\ xy-w^2=0 \text{ twice}, & \text{order } 4 \\ \text{conic } z+w=0,\ xy-m^2w^2=0, & 2 \\ \text{conic } z-w=0,\ xy-m^2w^2=0, & 2 \end{array}$$

and

$$U=0, \quad z^2H''=0,$$

that is,

$$\begin{array}{lr} U=0,\ z^2=0,\ \text{conic } z=0,\ xy-w^2=0 \text{ four times}, & \text{order } 8 \\ \text{proper spinode curve } U=0,\ H''=0, & 16 \\ \hline & 32; \end{array}$$

viz. the intersection is made up of the conic $z=0$, $xy-w^2=0$ six times, the conies $z\pm w=0$, $xy-m^2w^2=0$ each twice, and the proper spinode curve of the order 16.

616. A Geometrical Illustration of the Cubic Transformation in Elliptic Functions

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 211–216.]

[p. 522] Consider the cubic curve

$$x^3+y^3+z^3+6lxyz=0.$$

If through one of the inflexions $z=0$, $x+y=0$, we draw an arbitrary line $z=u(x+y)$, we have at the other intersections of this line with the curve

$$u\{u^2(x+y)^2+6lxy\}+x^2-xy+y^2=0;$$

that is,

$$(u^3+1)(x^2+y^2)+2xy(u^3+3lu-\tfrac{1}{2})=0;$$

and from this equation it appears that the ratio $x:y$ is given as a function involving the square root of

$$(u^3+3lu-\tfrac{1}{2})^2-(u^3+1)^2,$$

which, rejecting a factor 3, is

$$=(2u^3+3lu+\tfrac{1}{2})(lu-\tfrac{1}{2}).$$

It may be noticed that $lu-\frac{1}{2}=0$ gives the value of u, which in the equation $z=u(x+y)$ belongs to the tangent at the inflexion; and $2u^3+3lu+\frac{1}{2}=0$ gives the values which belong to the three tangents from the inflexion.

It thus appears that the coordinates x, y, z of any point of the curve can be expressed as proportional to functions of u involving the radical

$$\sqrt{\{(lu-\tfrac{1}{2})(2u^3+3lu+\tfrac{1}{2})\}},$$

and the theory of the curve is connected with that of a quasi-elliptic integral depending on this radical.

[p. 523] Taking ω an imaginary cube root of unity, write

$$\omega x+\omega^2y-2lz=x',$$

$$\omega^2x+\omega y-2lz=y',$$

$$x+y-2lz=z';$$

then we have

$$x'y'z'=x^3+y^3-8l^3z^3+6lx^2yz=x^3+y^3+z^3+6lxyz-(1+8l^3)z^3.$$

Also

$$-6lz = x' + y' + z', \quad z^3 = \frac{-1}{216l^3}(x' + y' + z')^3,$$

whence

$$x'^3 + y'^3 + z'^3 - \frac{216l^3}{1 + 8l^3}x'y'z' = \frac{216l^3}{1 + 8l^3}(x'^3 + y'^3 + z'^3 + 6lx'y'z'),$$

so that, putting

$$6m = \frac{-216l^3}{1 + 8l^3},$$

or, what is the same thing,

$$8l^3m^3 + l^3 + m^3 = 0,$$

the equation of the curve is

$$(x' + y' + z')^3 + 216m^3x'y'z' = 0;$$

and if we write

$$x' : y' : z' = X^3 : Y^3 : Z^3,$$

then the original curve is transformed into

$$(X^3 + Y^3 + Z^3)^3 + 216m^3X^3Y^3Z^3 = 0,$$

a curve of the ninth order breaking up into three cubic curves, one of which is

$$X^3 + Y^3 + Z^3 + 6mXYZ = 0,$$

and for the other two we write herein $m\omega$ and $m\omega^2$ respectively in place of m. Attending only to the first curve, we have

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

$$X^3 + Y^3 + Z^3 + 6mXYZ = 0,$$

as corresponding curves, the corresponding points being connected by the relation

$$\omega x + \omega^2 y - 2lz : \omega^2 x + \omega y - 2lz : x + y - 2lz = X^3 : Y^3 : Z^3,$$

or, for convenience, we may write

$$\begin{aligned} \omega x + \omega^2 y - 2lz &= X^3, \quad \text{giving} & 3x &= \omega^2 X^3 + \omega Y^3 + Z^3, \\ \omega^2 x + \omega y - 2lz &= Y^3, & 3y &= \omega X^3 + \omega^2 Y^3 + Z^3, \\ x + y - 2lz &= Z^3, & -6lz &= X^3 + Y^3 + Z^3. \end{aligned}$$

[p. 524] This is a $(1, 3)$ correspondence; viz. to a given point on the curve (m), there corresponds one point on (l); but to a given point on (l), three points on (m). As to the first case, this is obvious. As to the second case, if the point (x, y, z) is given, then the corresponding point (X, Y, Z) on the other curve will lie on one of the three lines

$$Y^3(\omega x + \omega^2 y - 2lz) - X^3(\omega^2 x + \omega y - 2lz) = 0;$$

each of these intersects the curve (m) in three points: but of the points in the same line it is only one which is a corresponding point of (x, y, z), and the number of the corresponding points is consequently the same as the number of lines, viz. it is $= 3$.

We infer that the above equations lead to a cubic transformation of the quasi-elliptic integral

$$\int du \div \sqrt{\{(lu - \tfrac{1}{2})(2u^3 + 3lu + \tfrac{1}{2})\}},$$

into one of the like form

$$\int dv \div \sqrt{\{(mv - \tfrac{1}{2})(2v^3 + 3mv + \tfrac{1}{2})\}};$$

and this is now to be verified.

We have, as before, the line $z = u(x + y)$ meeting the curve (l) in the points

$$(u^3 + 1)(x^2 + y^2) + 2xy(u^3 + 3lu - \tfrac{1}{2}) = 0;$$

and if similarly through an inflexion of the curve (m) we take the line $Z = v(X + Y)$, this meets the curve in the points

$$(v^3 + 1)(X^2 + Y^2) + 2XY(v^3 + 3mv - \tfrac{1}{2}) = 0.$$

Then if (x, y, z), (X, Y, Z) are taken to be the corresponding points as above, we can obtain v as a function of u. We, in fact, have

$$\begin{aligned} -2lu = \frac{-2lz}{x + y} = \frac{X^3 + Y^3 + Z^3}{-X^3 - Y^3 + 2Z^3} &= \frac{X^3 + Y^3 + v^3(X + Y)^3}{-(X^3 + Y^3) + 2v^3(X + Y)^3} \\ &= \frac{X^2 - XY + Y^2 + v^3(X + Y)^2}{-X^2 + XY - Y^2 + 2v^3(X + Y)^2} \\ &= \frac{(v^3 + 1)(X^2 + Y^2) + (2v^3 - 1)XY}{(2v^3 - 1)(X^2 + Y^2) + (4v^3 + 1)XY}, \end{aligned}$$

or, since we have

$$(v^3 + 1)(X^2 + Y^2) + 2XY(v^3 + 3mv - \tfrac{1}{2}) = 0,$$

that is,

$$X^2 + Y^2 : XY = -2v^3 - 6mv + 1 : v^3 + 1,$$

the equation becomes

$$-2lu = \frac{-6mv(v^3+1)}{(2v^3-1)(-2v^3-6mv+1)+(4v^3+1)(v^3+1)}$$

$$= \frac{-6mv(v^3+1)}{-3v(4mv^3-3v^2-2m)},$$

[p. 525] or say,

$$-lu = m(v^3+1) \div (\square), \quad \text{where the denominator } = 4mv^3 - 3v^2 - 2m.$$

This may also be written

$$-(lu - \tfrac{1}{2}) = 3v^2(mv - \tfrac{1}{2}) \div \square.$$

Proceeding to calculate $2u^3 + 3lu + \frac{1}{2}$, omitting the denominator $(4mv^3 - 3v^2 - 2m)^3$, this is

$$\frac{-2}{l^3}(v^3+1)^3 - 3m(v^3+1)(4mv^3-3v^2-2m)^2 + \tfrac{1}{2}(4mv^3-3v^2-2m)^3;$$

or, observing that

$$m^3 = \frac{-l^3}{1+8l^3},$$

that is,

$$\frac{1}{l^3} = \frac{-1-8m^3}{m^3} \quad \text{or} \quad \frac{m^3}{l^3} = -1 - 8m^3,$$

the numerator is

$$= 2(1+8m^3)(v^3+1)^3 - 3m(v^3+1)(4mv^3-3v^2-2m)^2 + \tfrac{1}{2}(4mv^3-3v^2-2m)^3,$$

which is found to be identically

$$\tfrac{1}{2}(2v^3 + 3mv + \tfrac{1}{2})(v^3 + 6mv - 2)^2,$$

viz. we have

$$2u^3 + 3lu + \tfrac{1}{2} = (2v^3 + 3mv + \tfrac{1}{2})(v^3 + 6mv - 2)^2 \div (4mv^3 - 3v^2 - 2m)^3.$$

and hence

$$(lu-\tfrac{1}{2})(2u^3+3lu+\tfrac{1}{2}) = -3(mv-\tfrac{1}{2})(2v^3+3mv+\tfrac{1}{2})(v^3+6mv-2)^2v^2 \div (4mv^3-3v^2-2m)^4.$$

Moreover, we find

$$ldu = 3mdv \cdot v(v^3 + 6mv - 2) \div (4mv^3 - 3v^2 - 2m)^2,$$

and we thence have

$$\frac{ldu}{\sqrt{\{(lu-\frac{1}{2})(2u^3+3lu+\frac{1}{2})\}}} = \sqrt{(-3)}\cdot\frac{mdv}{\sqrt{\{(mv-\frac{1}{2})(2v^3+3mv+\frac{1}{2})\}}},$$

viz. this differential equation corresponds to the integral equation

$$-lu = m(v^3+1)\div(4mv^3-3v^2-2m),$$

where $8l^3m^3+l^3+m^3=0$, which corresponds to the modular equation.

It may be remarked that, if v is the same function of u', l, in that u is of v, m, l; viz. if

$$-mv = l(u'^3+1)\div(4lu'^3-3u'^2-2m'),$$

then

$$\frac{mdv}{\sqrt{\{(mv-\frac{1}{2})(2v^3+3mv+\frac{1}{2})\}}} = \sqrt{(-3)}\cdot\frac{-ldu'}{\sqrt{\{(lu'-\frac{1}{2})(2u'^3+3lu'+\frac{1}{2})\}}},$$

and consequently

$$\frac{du}{\sqrt{\{(lu-\frac{1}{2})(2u^3+3lu+\frac{1}{2})\}}} = \frac{-3du'}{\sqrt{\{(lu'-\frac{1}{2})(2u'^3+3lu'+\frac{1}{2})\}}},$$

which accords with the general theory of the cubic transformation.

[p. 526] We may inquire into the relation between the absolute invariants of the two curves. Taking the absolute invariant to be

$$\Omega = \frac{64S^3-T^2}{64S^3},$$

where S and T bear the usual significations, we have for the one curve

$$\Omega = \frac{(1+8l^3)^3}{64l^3(1-l^3)^3},$$

and for the other curve

$$\Omega' = \frac{(1+8m^3)^3}{64m^3(1-m^3)^3},$$

and, as above, $8l^3m^3+l^3+m^3=0$: writing herein

$$l^3 = \frac{-\alpha'}{8\alpha'},\quad m^3 = \frac{-\beta'}{8\beta'},$$

the relation between α', β' is simply $\alpha'+\beta'=1$; and the values of Ω, Ω' are found to be

$$\Omega=\frac{64\alpha'(1-\alpha')^3}{(1+8\alpha')^3},\quad \Omega'=\frac{64\beta'(1-\beta')^3}{(1+8\beta')^3},$$

viz. the required relation is given by the elimination of α', β' from these three equations. Or, what is the same thing, writing $\alpha'=\frac{1}{2}+\theta$, and therefore $\beta'=\frac{1}{2}-\theta$, we have

$$(5+8\theta)^3\Omega=4(1+2\theta)(1-2\theta)^3,$$

$$(5-8\theta)^3\Omega'=4(1+2\theta)^3(1-2\theta),$$

and the elimination of θ from these equations gives the required relation between Ω and Ω'.

It of course follows that, if we have a cubic transformation

$$\frac{dx}{\sqrt{\{(a,b,c,d,e\mid x,1)^4\}}}=\frac{Cdx}{\sqrt{\{(a',b',c',d',e'\mid x,1)^4\}}},$$

then the absolute invariants Ω, Ω' of the two quartic functions are connected by the above relation. I have obtained this result, by reducing the radicals to the standard forms

$$\sqrt{(1-x^2\cdot 1-k^2x^2)},\ \sqrt{(1-x^2\cdot 1-\lambda^2x^2)},$$

from the known modular equation as represented by the equations

$$k^2=\frac{\alpha^3(2+\alpha)}{1+2\alpha},\quad \lambda^2=\frac{\alpha(2+\alpha)^3}{(1+2\alpha)^3};$$

viz. the values of the absolute invariants

$$\left(=1-\frac{27}{2J^2},\ 1-\frac{27J'^2}{2J'^2}\right)$$

are

$$\frac{(k^4+14k^2+1)^3}{},\quad \frac{(\lambda^4+14\lambda^2+1)^3}{},$$

but the method of effecting this is by no means obvious.

617. On the Scalene Transformation of a Plane Curve

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 321–328.]

[p. 527] The transformation by reciprocal radius vectors can be effected mechanically by Sylvester's Peaucellier-cell. But, employing a more general cell (considered incidentally by him) which may be called the scalene-cell, we have the scalene transformation in question[1]; viz. if, in two curves, r, r' are radius vectors belonging to the same angle (or say opposite angles) θ, then the relation between r, r' is

$$rr'(r+r') + (m^2 - l^2)r' + (m^2 - n^2)r = 0;$$

or, as this may also be written,

$$r^2 + \left(r' - \frac{l^2 - m^2}{r'}\right) r + m^2 - n^2 = 0.$$

The transformation is, it will be seen, an interesting one for its own sake, independently of the remarkably simple mechanical construction, viz. the scalene cell is simply a system of 3 pairs of equal rods PA, QA; PB, QB; PC, QC (fig. 1, p. 528), jointed together at and capable of rotating about the points P, Q, A, B, C; the three lengths PA, PB, PC (say these are $= l$, m, n) are all of them unequal: in the case of any two of them equal, we have Peaucellier or isosceles cell. The effect of the arrangement is that the points A, B, C are retained in a right line, the distances BA, $= r'$, and BC, $= r$, being connected by the above-mentioned equation; so that taking B as a fixed point, if the point A describe any given curve, the point C will describe the corresponding or transformed curve.

In the case where the given curve is a right line or a circle, we may through B draw at right angles to the curve the axis $x'Bx$: viz. in the case of the circle, **[p. 528]** the axis $x'Bx$ passes through its centre; and we measure the angle θ from this line, viz. we write $\angle xBC = \angle x'BA = \theta$.

[Fig. 1: The scalene cell with hinge points P, Q, A, B, C and rods of lengths l, m, n.]

Suppose, first, that the locus of A is a right line, or a circle passing through B. Its equation is $r' = \dfrac{c}{\cos\theta}$ or $= c\cos\theta$; and we accordingly have for the transformed curve

$$r^2 + \left(\frac{c}{\cos\theta} - \frac{l^2 - m^2}{c\cos\theta\,/\,1}\right) r + m^2 - n^2 = 0,$$

[1] The transformation itself, and doubtless many of the results obtained by means of it, are familiar to Prof. Sylvester; and I abandon all claim to priority.

or else
$$r^2 + \left(c\cos\theta - \frac{l^2 - m^2}{c\cos\theta}\right) r + m^2 - n^2 = 0 :$$
viz. multiplying in each case by $r\cos\theta$, and then writing $r\cos\theta = x$, $r^2 = x^2 + y^2$, the equations become
$$x(x^2 + y^2) + c(x^2 + y^2) - \frac{l^2 - m^2}{c}(x^2 + y^2) + (m^2 - n^2)x = 0,$$
and
$$x(x^2 + y^2) + cx^2 - \frac{l^2 - m^2}{c}(x^2 + y^2) + (m^2 - n^2)x = 0;$$
viz. in each case the curve is a circular cubic passing through the origin B and having an asymptote parallel to the axis of y. The curve is nodal, if $m = n$, viz. in this case the origin is a node: or if $c = \sqrt{(l^2 - n^2)} + \sqrt{(m^2 - n^2)}$.

Suppose next that the locus of A is a circle, centre at a distance $= \gamma$ along Bx' and radius $= h$: we have
$$r'^2 - 2\gamma r' \cos\theta + \gamma^2 - h^2 = 0,$$
viz. if
$$\gamma^2 - h^2 = -(l^2 - m^2),$$
or, what is the same thing,
$$h^2 + m^2 = \gamma^2 + l^2,$$
then we have
$$r' - \frac{l^2 - m^2}{r'} = 2\gamma\cos\theta,$$
[Continued in next instalment, pp. 528–534.]

Arthur Cayley

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617. On the Scalene Transformation of a Plane Curve (continued)

[p. 529] and the transformed curve is

$$r^2 + 2\gamma r\cos\theta + m^2 - n^2 = 0,$$

or, as this may be written,

$$r^2 + 2\gamma r\cos\theta + \gamma^2 - f^2 = 0,$$

where $\gamma^2 - f^2 = m^2 - n^2$, that is, $f^2 + m^2 = \gamma^2 + n^2$; viz. this is a concentric circle radius f.

The theorem may be presented as follows. Consider two concentric circles, centre O and radii h, f respectively; take an arbitrary point B, distance $OB = \gamma$; and taking m arbitrary, determine l, n by the equations

$$l^2 = m^2 + h^2 - \gamma^2, \quad n^2 = m^2 + f^2 - \gamma^2;$$

then drawing through B an arbitrary line to meet the circles in A, C respectively; also describing a circle, centre B and radius $= m$; and through A drawing a line ABC perpendicular to BA to meet the last-mentioned circle in two points P, Q: for these points, the distances from the points A, B, C are $= l$, m, n respectively.

To verify this, take O as the origin, OB for the axis of x, θ the inclination of ABC to this axis, $BA = r'$, $BC = r$; the coordinates of C, B, A are

$$\gamma + r\cos\theta, \quad r\sin\theta,$$

$$\gamma, \quad 0,$$

$$\gamma + r'\cos\theta, \quad -r'\sin\theta,$$

whence, taking (x, y) for the coordinates of P (or Q), the equations to be verified are

$$(x - \gamma - r\cos\theta)^2 + (y - r\sin\theta)^2 = n^2,$$

$$(x - \gamma)^2 + y^2 = m^2,$$

$$(x - \gamma + r'\cos\theta)^2 + (y + r'\sin\theta)^2 = l^2.$$

By means of the second equation, the other two become

$$-2(x-\gamma)r\cos\theta - 2yr\sin\theta + r^2 = n^2 - m^2,$$

$$2(x-\gamma)r'\cos\theta + 2yr'\sin\theta + r'^2 = l^2 - m^2;$$

or, substituting for $n^2 - m^2$, $l^2 - m^2$ the values $f^2 - \gamma^2$ and $h^2 - \gamma^2$, the equations are

$$-2xr\cos\theta - 2yr\sin\theta + 2\gamma r\cos\theta + r^2 - f^2 + \gamma^2 = 0,$$

$$2xr'\cos\theta + 2yr'\sin\theta - 2\gamma r'\cos\theta + r'^2 - h^2 + \gamma^2 = 0,$$

viz. in virtue of the equations of the two circles, these reduce themselves each of them to

$$x\cos\theta + y\sin\theta = 0,$$

which equation, together with the second equation

$$(x - \gamma)^2 + y^2 = m^2,$$

determine (x, y) as above.

[p. 530] Reverting to the case where the locus of A is the circle

$$r'^2 - 2\gamma r'\cos\theta + \gamma^2 - h^2 = 0,$$

this gives

$$r' = \gamma\cos\theta + \sqrt{(h^2 - \gamma^2\sin^2\theta)},$$

$$\frac{1}{r'} = \frac{\gamma\cos\theta - \sqrt{(h^2 - \gamma^2\sin^2\theta)}}{\gamma^2 - h^2},$$

so that for the transformed curve we have

$$r^2 + \gamma\cos\theta\left\{(1-\lambda)r + (1+\lambda)\frac{\gamma^2 - h^2}{r}\right\} + (1-\lambda)\sqrt{(h^2 - \gamma^2\sin^2\theta)} + m^2 - n^2 = 0.$$

Putting for shortness $\dfrac{l^2 - m^2}{\gamma^2 - h^2} = \lambda$, and for r, $r\cos\theta$, $r\sin\theta$, writing $\sqrt{(x^2+y^2)}$, x, y respectively, this is

$$x^2 + y^2 + (1-\lambda)\gamma x + (1+\lambda)\sqrt{[h^2(x^2+y^2) - \gamma^2 y^2]} + m^2 - n^2 = 0,$$

or, what is the same thing,

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

= a bicircular quartic. In the case $\lambda = -1$, it reduces itself to the circle

$$x^2 + y^2 + 2\gamma x + m^2 - n^2 = 0$$

twice, which is the case considered above; and in the case $\lambda = 1$, or $l^2 - m^2 = h^2 - \gamma^2$, the equation is

$$(x^2 + y^2 + m^2 - n^2)^2 = 4\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

so that the curve is symmetrical in regard to each axis. In the case $\gamma = 0$, the locus is a pair of concentric circles, centre B.

The equation

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2 y^2\},$$

which contains the four constants λ, γ, h and $m^2 - n^2$, may be written in the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

(where the constants A, B, a, e are also arbitrary). This is, in fact, the equation of the general symmetrical bicircular quartic, referred to a properly-selected point on the axis as origin, viz. the origin is the centre of any one of the three involutions formed by the vertices (or points on the axis); say it is any one of the three involution-centres of the curve.

To show this, assume

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = x^4 - px^3 + qx^2 - rx + s;$$

[p. 531] then, taking B arbitrary, the equation of the symmetrical bicircular quartic having for vertices the points $x = \alpha$, $x = \beta$, $x = \gamma$, $x = \delta$, is

$$(x^2 + y^2 - \tfrac{1}{2}px + B)^2 = (2B + \tfrac{1}{4}p^2 - q)x^2 + (b + 2\theta A)x + (-s + B^2),$$

in fact, this is the form of the general equation, and writing therein $y = 0$, it becomes $x^4 - px^3 + qx^2 - rx + s = 0$, that is, $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = 0$. Hence, writing for convenience

$$A = -\tfrac{1}{2}p,$$

$$a = 2B + \tfrac{1}{4}p^2 - q,$$

$$b = r - pB,$$

$$c = -s + B^2,$$

the equation is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + bx + c.$$

This may be written

$$(x^2 + y^2 + Ax + B + \theta)^2 = (a + 2\theta)x^2 + 2\theta y^2 + (b + 2\theta A)x + c + 2B\theta + \theta^2,$$

viz. assuming $\theta = -\dfrac{b}{2A}$ in order on the right-hand side to destroy the term in x, the equation is

$$\left(x^2 + y^2 + Ax + B - \frac{b}{2A}\right)^2 = \left(a - \frac{b}{A}\right)x^2 + y^2 \cdot \frac{-b}{A} + \frac{1}{4A^2}(b^2 - 4ABb + 4A^2c),$$

which is of the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2 + f;$$

and if $f = 0$, that is, if $b^2 - 4ABb + 4A^2c = 0$, then it is of the required form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2.$$

We have

$$\begin{aligned} b^2 - 4ABb + 4A^2c &= (r - pB)^2 + 2pB(r - pB) + p^2(-s + B^2) \\ &= r^2 - p^2 s, \end{aligned}$$

or the required condition is $r^2 - p^2s = 0$. But we have

$$r^2 - p^2s = (\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta),$$

as is easily verified by writing

$$p = \delta + p_{\prime}, \quad q = \delta p_{\prime} + q_{\prime}, \quad r = \delta q_{\prime} + r_{\prime}, \quad s = \delta r_{\prime},$$

where $p_{\prime}$, $q_{\prime}$, $r_{\prime}$ stand for

$$\alpha + \beta + \gamma, \quad \beta\gamma + \gamma\alpha + \alpha\beta, \quad \alpha\beta\gamma,$$

[p. 532] respectively. The required condition thus is

$$(\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta) = 0,$$

viz. the origin (that is, the fixed point B of the cell) must be at one of the three involution-centres.

Comparing the equation

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2\{h^2(x^2 + y^2) - \gamma^2 y^2\}$$

with the equation

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

we have

$$\begin{aligned} A &= (1 - \lambda)\gamma, \\ B &= m^2 - n^2, \\ a &= (1 + \lambda)^2 h^2, \\ e &= (1 + \lambda)^2(h^2 - \gamma^2), \end{aligned}$$

and thence $a - e = (1 + \lambda)^2\gamma^2$. Consequently $\dfrac{a - e}{A^2} = \left(\dfrac{1 + \lambda}{1 - \lambda}\right)^2$, which gives λ; and then

$$h^2 = \frac{a}{(1 + \lambda)^2}, \quad \gamma^2 = \frac{a - e}{(1 + \lambda)^2}, \quad m^2 - n^2 = B;$$

viz. we thus have the values of λ, h, γ and m^2-n^2 for the description of a given curve $(x^2+y^2+Ax+B)^2 = ax^2+ey^2$. In order that the description may be possible, a and $a-e$ must be each of them positive.

For the Cartesian a is $= e$, whence $1+\lambda = 0$, and the equation becomes

$$(x^2+y^2+2\gamma x+m^2-n^2)^2 = 0,$$

which is a twice repeated circle; hence the Cartesian cannot be constructed by means of a cell as above.

To obtain a construction of the Cartesian, it may be remarked that, if a symmetrical bicircular quartic be inverted in regard to an axial focus, viz. if the focus be taken as the centre of inversion, we obtain a Cartesian. The axial foci of the curve

$$(x^2+y^2+Ax+B)^2 = ax^2+ey^2$$

are points on the axis, the abscissa $x=\theta$ being determined by the equation

$$e(\theta^2+A\theta+B)^2 - a(\theta^2-B)^2 - ae\theta^2 = 0.$$

The equation referred to a focus as origin is therefore

$$\{x^2+y^2+(A+2\theta)x+\theta^2+A\theta+B\}^2 = ax^2+ey^2+2a\theta x+a\theta^2;$$

then inverting, viz. for x, y writing $\dfrac{k^2x}{r^2}$, $\dfrac{k^2y}{r^2}$ (k arbitrary), we have, as may be verified the equation of a Cartesian.

[p. 533] The inversion can be performed mechanically by an ordinary Peaucellier-cell; the complete apparatus for the construction of a Cartesian is therefore as in fig. 2, viz. we have a cell BAC as before, B a fixed point, locus of A a circle (for convenience of drawing, the arrangement has been made BAC instead of ABC), and we connect with C a Peaucellier-cell $CA'B'$, arms n, n', m', the fixed point B' being on the axis, which is the line joining B with the centre of the circle described by A. This being so, then A describing a circle, C will describe a symmetrical bicircular quartic, and A' will describe the inverse of this, being in general a like curve; but if the position of B' be properly determined, viz. if B' be at a focus of the first-mentioned quartic.

Fig. 2.

then A' will describe a Cartesian. A further investigation would be necessary in order to determine how to adapt the apparatus to the description of a given Cartesian.

A more convenient mechanical description of a Cartesian is, however, that given in the paper which follows the present one [618].

The equation

$$\{x^2+y^2+(1-\lambda)\gamma x+m^2-n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2)-\gamma^2y^2\}$$

may also be written

$$\{x^2+y^2+(1-\lambda)\gamma x-\tfrac{1}{2}(1+\lambda)^2(h^2-\gamma^2)+m^2-n^2\}^2$$
$$= (1+\lambda)^2\{\gamma^2x^2-(1-\lambda)(h^2-\gamma^2)\gamma x-\tfrac{1}{4}(1+\lambda)^2(h^2-\gamma^2)^2-(m^2-n^2)(h^2-\gamma^2)\},$$

viz. the equation is now brought into the form

$$(x^2+y^2+Ax+B)^2 = ax^2+bx+c.$$

Expressing the coefficients A, B, a, b, c in terms of λ, γ, h, m^2-n^2, it appears by what precedes, that we should have identically $b^2-4ABb+4A^2c=0$, viz. this is the equation which expresses that the origin is an involution-centre.

If, instead of the original cell, we consider a new cell obtained by substituting for the arms PB, BQ, the arms pb, bq, jointed on to the points p, q on the arms CP, CQ respectively, and instead of B, making

b the fixed point; then writing $Cp = kn$, $pb = km$, so that the parameters of the cell are l, m, n, k, and taking $Ch = s$, $hA = s'$,

[p. 534] we have $s = kr$, $s + s' = r + r'$, that is, $r = \frac{s}{k}$, $r' = \frac{k-1}{k}s + s'$. Substituting in the equation between r, r', written for greater convenience in the form

$$(r + r')(rr' + m^2 - l^2) + (l^2 - n^2)r' = 0,$$

the relation between s, s' is found to be

$$(s + s')\left(\frac{k-1}{k^2}s^2 + \frac{ss'}{k} + m^2 - l^2\right) + (l^2 - n^2)\left(\frac{k-1}{k}s + s'\right) = 0.$$

On account of the term in s^3, this equation in its general form does not, it would appear, give rise to transformations of much elegance. If, however, $l = n$, then the relation becomes

$$(k - 1)s^2 + kss' = k^2(m^2 - l^2);$$

and in particular, if $k = 2$, then

$$s^2 + 2ss' = 4(l^2 - m^2), \quad \text{or say} \quad (s + s')^2 - s'^2 = 4(l^2 - m^2),$$

viz. taking A instead of b as the fixed point, the relation between the radii AC, Ah is $\rho^2 - \rho'^2 = 4(l^2 - m^2)$; the cell is in this case Sylvester's "quadratic-binomial extractor."

618. On the Mechanical Description of a Cartesian

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 328–330.]

[p. 535] Suppose that in two different curves the radius vectors r, r', which belong to the same angle θ, are connected by the equation

$$r^2 + \left(Mr' + N + \frac{P}{r'}\right) r + B = 0;$$

then, taking one of the curves to be the circle

$$Mr' + \frac{P}{r'} = A\cos\theta,$$

the other curve is

$$r^2 + (A\cos\theta + N)r + B = 0,$$

viz. this is a Cartesian. It perhaps would not be difficult to contrive a mechanical arrangement to connect the radius vectors in accordance with the foregoing equation; but the required result may be obtained equally well by means of a particular case of the relation in question; viz. taking this to be

$$r^2 + (-r' + N)r + B = 0,$$

then, taking the one curve to be the circle $r' = A\cos\theta$, the other curve is the Cartesian,

$$r^2 + (-A\cos\theta + N)r + B = 0, \quad \text{that is,} \quad r^2 + (A\cos\theta + N)r + B = 0.$$

The relation between the radius vectors may in this case be written

$$r' = N + r + \frac{B}{r},$$

which can be constructed mechanically by a simple addition to the Peaucellier-cell, viz. if we joint on to C (fig. 1, p. 536) a rod CD having a slot, working on a pin at A, so that the rod is thereby kept always in the line BAC, then, making B the

[p. 536] fixed point and taking $BA = r$, we have $AC = \dfrac{B}{r}$, whence $BC = r + \dfrac{B}{r}$, or D being a point at the distance $CD = a$, from the point C, and denoting BD by r', we have

$$r' = r + \frac{B}{r} + a,$$

which is an equation of the required form; whence, if the point A describe a circle passing through B, then the point D will describe a Cartesian.

Fig. 1.

The equation of the Cartesian is $r^2 + (A\cos\theta + N)r + B = 0$, viz. this is

$$x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)},$$

or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ay^2,$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point (x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2s^2 = m^2 + 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - a)^2 + y^2$, the function on the left-hand is of the form $(1+l^2)(x^2+y^2+Ax+B)$, whence, assuming $\sqrt{(1 + l^2)} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the same equation $(x^2+y^2+Ax+B)^2 = a(x^2+y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a, such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2);$$

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

(Addition, Feb. 1875.) It is obviously the same thing, but I find that it is mechanically more convenient to derive the Cartesian from the Limaçon $r' = N - A\cos\theta$, by the transformation $r' = r + \dfrac{B}{r}$: I have on this principle constructed an apparatus whereby the Cartesian is described on a rotating board by a pencil moving in a fixed line.

619. On an Algebraical Operation

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 369–375.]

[p. 537] I consider

$$\Omega F(a, x),$$

an operation Ω performed upon $F(a, x)$ a rational function of (a, x); viz. F being first expanded or regarded as expanded in ascending powers of a, the coefficients of the several powers are then to be expanded or regarded as expanded in ascending powers of x, and the operation consists in the rejection of all negative powers of x.

In the cases intended to be considered, F contains only positive powers of a: but this restriction is not necessary to the theory.

The investigation has reference to the functions $A(x)$ of my "Ninth Memoir on Quantics," *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]; for instance, we there have as regards the covariants of a quadric

$$A(x) = \frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}},$$

and consequently, in the present notation,

$$A(x) = \Omega \, \frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}};$$

by a process of development and summation, the value of this expression was found to be

$$\frac{1}{1 - ax^2 \,.\, 1 - a^2},$$

[p. 538] and in the other more complicated cases the value of $A(x)$ was found only by trial and verification. What I purpose now to show is that the operation Ω can be performed without any development in an infinite series; or say that it depends on finite algebraical operations only.

It is clear that if $F(a, x)$, considered to be developed as above contains only positive powers of x, then

$$\Omega F(a, x) = F(a, x);$$

And if it contains only negative powers of x, then $\Omega F(a, x) = 0$.

Consider now $\Omega \, \dfrac{\phi(x)}{x - a}$, where $\phi(x)$ is a function containing only positive powers of x; we have

$$\frac{\phi(x)}{x - a} = \frac{\phi(x) - \phi(a)}{x - a} + \frac{\phi(a)}{x - a},$$

and thence

$$\Omega \, \frac{\phi(x)}{x - a} = \Omega \, \frac{\phi(x) - \phi(a)}{x - a} + \Omega \, \frac{\phi(a)}{x - a} = \frac{\phi(x) - \phi(a)}{x - a},$$

since $\dfrac{\phi(x) - \phi(a)}{x - a}$ is a rational and integral function of x, which when developed contains only positive powers of x, and $\dfrac{\phi(a)}{x - a}$ when developed contains only negative powers of x.

Consider next $\Omega \, \dfrac{\phi(x)}{x^2 - a}$, where $\phi(x)$ is a rational and integral function of x; writing this $= f(x^2) + xg(x^2)$, we have

$$\Omega \, \frac{\phi(x)}{x^2 - a} = \Omega \, \frac{f(x^2)}{x^2 - a} + \Omega \, \frac{xg(x^2)}{x^2 - a} = \frac{f(x^2) - f(a)}{x^2 - a} + \Omega \, \frac{xg(x^2)}{x^2 - a}.$$

As regards the last term, notice that

$$\frac{xg(x^2)}{x^2 - a} = \frac{x[g(x^2) - g(a)]}{x^2 - a} + \frac{xg(a)}{x^2 - a},$$

in which $\dfrac{x[g(x^2) - g(a)]}{x^2 - a}$ is a rational and integral function of (a, x), and therefore when developed contains only positive powers of x, while $\dfrac{xg(a)}{x^2 - a}$ when developed contains only negative powers of x.

We thus have

$$\Omega \, \frac{\phi(x)}{x^2 - a} = \frac{f(x^2) + xg(x^2) - f(a) - xg(a)}{x^2 - a} = \frac{\phi(x) - f(a) - xg(a)}{x^2 - a}.$$

[p. 539] Similarly, if $\phi(x) = f(x^3) + xg(x^3) + x^2h(x^3)$, then

$$\Omega \, \frac{\phi(x)}{x^3 - a} = \frac{\phi(x) - f(a) - xg(a) - x^2h(a)}{x^3 - a},$$

and so on.

Consider now the above-mentioned function

$$A(x) = \Omega \, \frac{1 - x^{-2}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}}.$$

Writing

$$\frac{1-x^{-2}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}=\frac{P}{1-ax^2}+\frac{Q}{1-a}+\frac{R}{1-ax^{-2}},$$

we have

$$P=\left(\frac{1-x^{-2}}{1-a\,.\,1-ax^{-2}}\right)_{a=x^{-2}}=\frac{1}{1-x^{-4}}=\frac{-x^4}{1-x^4},$$

$$Q=\left(\frac{1-x^{-2}}{1-ax^2\,.\,1-ax^{-2}}\right)_{a=1}=\frac{1}{1-x^2},$$

$$R=\left(\frac{1-x^{-2}}{1-ax^2\,.\,1-a}\right)_{a=x^2}=\frac{1-x^{-2}}{1-x^4\,.\,1-x^2}=\frac{-1}{x^2(1-x^4)};$$

that is,

$$\frac{1-x^{-2}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}=\frac{-x^4}{1-x^4}\cdot\frac{1}{1-ax^2}+\frac{1}{1-x^2}\cdot\frac{1}{1-a}-\frac{1}{1-x^4}\cdot\frac{1}{x^2-a},$$

and thence

$$\Omega\,\frac{1-x^{-2}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}=\frac{-x^4}{1-x^4}\cdot\frac{1}{1-ax^2}+\frac{1}{1-x^2}\cdot\frac{1}{1-a}+\Omega\,\frac{-1}{1-x^4}\cdot\frac{1}{x^2-a}.$$

Here, as regards the last term,

$$\phi(x^2)=\frac{-1}{1-x^4}=f(x^2),\quad g(x^2)=0,$$

$$f(a)=\frac{-1}{1-a^2},$$

and

$$\frac{\phi(x^2)-f(a)}{x^2-a}=\frac{1}{x^2-a}\left(\frac{-1}{1-x^4}+\frac{1}{1-a^2}\right)=-\frac{x^2+a}{1-x^4\,.\,1-a^2},$$

and we have

$$A(x)=\frac{-x^4}{1-x^4}\cdot\frac{1}{1-ax^2}+\frac{1}{1-x^2}\cdot\frac{1}{1-a}-\frac{x^2+a}{1-x^4\,.\,1-a^2}.$$

The second term is $=\dfrac{(1+x^2)(1+a)}{(1-x^4)(1-a^2)}$; combining this with the third term, the two together are $=\dfrac{1+ax^2}{1-x^4\,.\,1-a^2}$.

[p. 540] Hence the value is

$$=\frac{1}{1-x^4}\left(\frac{-x^4}{1-ax^2}+\frac{1+ax^2}{1-a^2}\right),$$

which is

$$=\frac{1}{1-ax^2\,.\,1-a^2},$$

being, in fact, the expression for this function when decomposed into partial fractions of the denominators $1-ax^2$ and $1-a^2$ respectively. Hence finally

$$A(x)=\Omega\,\frac{1-x^{-2}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}=\frac{1}{1-ax^2\,.\,1-a^2},$$

as it should be.

For the cubic function, we have

$$A(x)=\Omega\,\frac{1-x^{-2}}{1-ax^3\,.\,1-ax\,.\,1-ax^{-1}\,.\,1-ax^{-3}};$$

the function operated upon, when decomposed into partial fractions, is

$$=\frac{x^{10}}{1-x^4\,.\,1-x^6}\cdot\frac{1}{1-ax^3}-\frac{x^4}{1-x^2\,.\,1-x^4}\cdot\frac{1}{1-ax}$$

$$+\frac{x}{1-x^2\,.\,1-x^4}\cdot\frac{1}{x-a}+\frac{-x}{1-x^4\,.\,1-x^6}\cdot\frac{1}{x^3-a}.$$

Hence we require

$$\Omega\,\frac{x}{1-x^2\,.\,1-x^4}\cdot\frac{1}{x-a}+\Omega\,\frac{-x}{1-x^4\,.\,1-x^6}\cdot\frac{1}{x^3-a}.$$

The first of these is

$$=\frac{1}{x-a}\left(\frac{x}{1-x^2\,.\,1-x^4}-\frac{a}{1-a^2\,.\,1-a^4}\right),$$

which is

$$\frac{1}{1-x^2\,.\,1-x^4\,.\,1-a^2\,.\,1-a^4}\left\{\begin{array}{l}1\\ +x(a+a^3-a^5)\\ +x^2(a^2-a^4)\\ +x^3(a-a^3)\\ -x^4a^2\\ -x^5a\end{array}\right\}.$$

As regards the second, the function operated on may be expressed in the form

$$\frac{-x^9-x-x^5}{1-x^4\,.\,1-x^6}\cdot\frac{1}{x^3-a},$$

Arthur Cayley

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623. On Three-Bar Motion (continued)

[Continued from earlier pages of the same paper.]

[p. 576] Single-infinite series of ∞ and circumscribed triangles, so that, drawing from a point A of the circle tangents to the parabola again meeting the circle in the points B and C respectively, BC will be a tangent to the parabola; this is in fact a known thing, but, repeating the same kind of investigation as before, we can, with the affinity points K on the circle, the parabola described to touch two of the sides of the triangle will touch the third side.

Taking, then, a circle radius $\frac{1}{2}k$, and upon it the three points A, B, C determined by the angles 2α, 2β, 2γ respectively (viz. the coordinates of A are $x = \frac{1}{2}k\cos 2\alpha$, $\frac{1}{2}k\sin 2\alpha$, &c.), and a point K determined by the angle 2ω (suppose for a moment $\omega = 0$), then the lines from K, viz. the lines having for K the equations &c. at the foot of A, and the lines KB with the fixed parabola, are connected by simple relations, viz. writing Q for the gain $\frac{1}{4}k^2\sin 2\gamma$, we obtain

$$P - x = -\tfrac{1}{4}k^2\left(\tfrac{x}{y} - \tan\gamma\right)\left(\tfrac{x}{y} - \tan\alpha\right)\tan\beta,$$

which serves to find the breadths a, b, c of a given trajectory from the bifocus $PQRS\ldots$, and, if necessary, by inverting each or either as alternate ones, make the left-hand column begin with the sign $+$.

[p. 577] 26. The complete role now is for a given trajectory from the bifocus to $PQRS\ldots$, and, if necessary, by inverting each or either ones, make the left-hand column begin with the sign $+$. To find which we begin with a then if these Ω count the intersections with the alternate ones, with the left-hand column to begin with the sign $+$.

26. The complete role now $a+b+c$, and the equation of the conic surface gives $\frac{1}{2}P_1$. Operating with P_x as a tangent, then the calculation is in fact equal to the breadth in a.

27. I give some simple examples.

	0	2	4
$P = a-1$	$-$	$+$	$-$
$Q = a-3$	$+$	$+$	$+$
$R = a-1$	$-$	$-$	$-$
0	P	2	Q

	0	2	4
$P = a-3$	$-$	$-$	$+$
$Q = a-1$	$+$	$+$	$+$
$R = a+1$	$+$	$+$	$+$
0	P	2	Q

In the left-hand column taking the intervals to be considered $0-2$, $0-4$, $2-4$, the binbecome modified as shown are

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$-$	$-$	$-$

viz.

Interval $0-2$, for P gain $+1$, P free; for Q gain -0, Introduction is P;

$-0-4$, $\quad -1$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots PQ$;

$-2-4$, $\quad -0$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots Q$.

And similarly in the right-hand example we have

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$+$	$+$	$+$

[p. 578] viz. these conditions determine the unknown quantities p, R. It is at once seen that we have

$$2\theta - \beta - \gamma = \tfrac{1}{2}\pi - (a - \alpha), \quad \text{that is,} \quad 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

and then

$$p = \Omega \sin(a - \alpha) \div \cos(\theta - \alpha) \div \sin(a - \gamma);$$

from symmetry, we see that the parabola touches also the side AB.

Suppose, next, F, G are points in the circle determined by the angles $2f$, $2g$; retaining α and θ to denote their values,

$$p = \Omega \sin(\alpha - \beta) \sin(\alpha - \gamma), \text{ and } 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

the condition, in order that FG may be a tangent, is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(2\theta - f - g);$$

viz. determining δ by the equation

$$a + \beta + \gamma = f + g + \delta,$$

this is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(a\delta - \delta),$$

or, what is the same thing,

$$\sin(\alpha - a) \sin(\alpha - \beta) \sin(\alpha - \gamma) = \sin(\alpha - f) \sin(\alpha - g) \cos(\delta - \delta);$$

viz. this equation, considering therein δ as standing for $a + \beta + \gamma - f - g$, is the relation which must subsist between f and g, in order that the line FG may be a tangent to the parabola. And then, δ being determined as above, and the point H on the circle being determined by the angle 2δ, it is clear that the lines GH, HF will also be tangents to the parabola, viz. FGH will be an inscribed triangle circumscribed to the parabola; we have a single infinity of such triangles, but it is in fact, the same single infinity in form of these, f, g, satisfy only the relation $a + \beta + \gamma = f + g + \delta$, where δ has the same value but α change. The proposition is obtained as the consequence of an easily established proposition that to determine the determinations of f and α.

[p. 579] that is,

$$(b_1' - a_1')c_2a' - (c_1' - a_1')b_2a' + ac_1(c - b_1) = 0,$$

or, what is the same thing,

$$(b_1' - a_1')a^2 - (c_1' - a_1')b + ac_1(c - b_1) = 0.$$

Suppose, as in the figure, that P is between B and A; then, if $AF = p$, we have $a = ba + cp$, and the equation becomes

$$(b_1' - a_1')(ca + cp)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if a, a' be the dimensions AG, BG, CG, then $ba' = b'a + ca'$, that is, $cn = ba - na$, and the corresponding equation is

$$(b_1' - a_1')(ca' - na)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

We have here

$$BG\colon CF = BP \cdot CO = BC \cdot FO,$$

also,

$$pa + p'a' = AF + AF \cdot AC = FO \cdot AO = FO \cdot OH + FO \cdot a',$$

and

$$pp + p'a' = AP \cdot CF = AG \cdot CG = FO \cdot HH = FO \cdot a,$$

as may be shown without difficulty, p', a', b'' being the distances AB, BH, CH. Hence the equations become

$$c(b_1' - a_1') = \nu a';$$

$$b(c_1' - a_1') = aaa' = 0,$$

showing that, the fact being so in the figure, $b_1' = a_1'$ and $c_1' = a_1'$, are each of them positive, viz. that, in the geometry by the triangle $GCAU$, the radial bars a_0, a_1 are shorter than the sides b, c, respectively. Substituting for a_2, a_3, the values b, b_1, a_2 a c, also, instead of b_1, b_2, b_3, introducing the quantities b_1, a_2, b_3, where

$$b_1, b_2 = b + b_4,$$

$$b_3 = b + b_5,$$

$$b_4 = b + b_6,$$

$$b_5 = b + b_7,$$

these equations become

$$c(b_1' - b_2'c)ca + cba'B = a'a',$$

$$b(b_3' - b_4'c)ca + ba'Ca = a'a',$$

or, as these may be written, partaking the relations $B = aa + b_1 a' \sin BAC$,

$$B'\&'(b_2 - b_4'b)c\,a + ab'\,BCH;$$

$$B'\&'(b_3 - b_4'b)c\,a + ba'c\,AH.$$

[p. 580] All the quantities here so far being represented as positive, and the formula are applicable to the particular figures, but, to present them in a form applicable to any order of the nodes and bar, we have only to write the equations in the form

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cb - b),$$

$$B(\lambda_3 - \lambda_5) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

$$B(\lambda_4 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

and these may be replaced by the equation

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + a),$$

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + b),$$

$$B(\lambda_3 - \lambda_4) = b'ca(a - x) + (b' - x)(s + c),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$BB(\lambda_2' - \lambda_4) = BC \cdot FA \cdot GA \cdot HA,$$

$$BB(\lambda_2 - \lambda_4) = CA \cdot FB \cdot GB \cdot HB,$$

$$BB(\lambda_3' - \lambda_4) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they must for $\lambda \sin(\beta - a)$, $b' \sin(f - a)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$BC(\gamma') - \lambda_4) = FA \cdot GA \cdot HA,$$

$$CA(c_1 - c') = FB \cdot GB \cdot HB,$$

$$AB(b_1 - b_3') = FC \cdot GC \cdot HC,$$

where the signs are so just mentioned. We may say that $\frac{1}{2}(c_1 - c_1')$ is the modulus for the focus A, and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624. On the Bicursal Sextic

[From the Proceedings of the London Mathematical Society, vol. VII. *(1875–1876), pp. 166–172. Read March 10, 1876.]*

In the paper "On the binominal description of certain conic curves," *Proceedings of the London Mathematical Society*, vol. IV. (1873), pp. 105–111, [504], I obtained the binominal sextic as a rational transformation of a bicursal quartic. The theory was in effect as follows: taking Ω, P, Q, R, each of them a function of λ, μ of the form $(*)(\lambda,1)^3(\mu,1)^3$, and considering (λ,μ) as connected by the equation $\Omega=0$ (viz. λ, μ being coordinates, this represents a bicursal quartic), then if $\sigma =$ assume $\sim x:y:z=P:Q:R$, this is rationally connected with the bicursal quartic, viz. the points of the latter curve have a $(1,1)$ correspondence; whence the locus in question, say that curve $U=0$, is binominal. The figure is obtained as the reverse of the curve of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega=0, \quad aP+\beta Q+\gamma R=0,$$

viz. each of these being a quartic curve having the same nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4\cdot 4-4\cdot 3, =4$; and thus the curve $U=0$, $R=0$ have therefore the same 4 common intersections, that is, the curve $U=0$, $R=0$ have therefore the same 4 common intersections, then these are also 4 common intersections of the two corresponding $\lambda\mu$-curves $\Omega=0$, $aP+\beta Q+\gamma R=0$, and consequently the curve $U=0$ is binominal.

The theory assumes a different and more simple form if the second functions Ω, P, Q, R are suppose that these have no common point; the curve $U=0$ is binominal.

[p. 582] rational transformation of the cubic $\Omega=0$, is still a bicursal curve; but the order is given as the number of the variable intersections of the cubics

$$\Omega=0, \quad aP+\beta Q+\gamma R=0,$$

viz. this is $=3\cdot 3-2, =7$. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $=7-k$; and this is therefore the order of the curve $U=0$. In particular, if $k=1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega=0$, $x:y:z=P:Q:R$, where $\Omega=0$, $P=0$, $Q=0$, $R=0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any two curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$, $R=0$. (Observe that we distinguish thus far between the variable and the common intersections of the curves $\Omega=0$, then any linear function of P, Q, R, would have the same number of common intersections.) For, supposing the two curves to be $aP+\beta Q+\gamma R=0$ and $a'P+\beta'Q+\gamma'R=0$ and the cubic any other curve whatever

$$aP+\beta Q+\gamma R=\theta(a'P+\beta'Q+\gamma'R)$$

has the same number of variable intersections with $aP+\beta Q+\gamma R=0$, the points common to these two curves correspond on the cubic the points A_1, A_2, A_3.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any three curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$. For, supposing the three curves to be

$$aP+\beta Q+\gamma R=0, \quad a'P+\beta'Q+\gamma'R=0, \quad a''P+\beta''Q+\gamma''R=0,$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $ax+\beta y+\gamma z=0$, $a'x+\beta'y+\gamma'z=0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_2, viz. to obtain the cubic $aP+\beta Q+\gamma R-\theta(a'P+\beta'Q+\gamma'R)=0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_3, viz. to obtain the cubic to pass through one of these points A_1, A_2, A_3; the third common intersection of $aP+\beta Q+\gamma R$ and $a'P+\beta'Q+\gamma'R$ must then be on the curve; and so on, similarly. And so on, similarly. And so on, similarly. And so on, similarly. And, as before, the three points correspond to a triple point on the bicursal sextic.

[p. 583] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before; the new triple point will correspond to a similar set of three common intersections of the cubics having the same three common points. The theory may be developed if there are six variables instead of three common intersections of the cubics having three of the variable intersections of $\Omega=0$, $aP+\beta Q+\gamma R=0$ count for a triple point; and the same set of three common intersections of $aP+\beta Q+\gamma R=0$, will count for a triple point of the second triple point. As before, we have a number of triple points such as $\Omega=0$, $a''P+\beta''Q+\gamma''R=0$, which thus form (in the same hexagonal locus) the cuspidal sextic. By varying the parameters we obtain the curve $\Omega=0$, and the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience λ, ρ in place of a, μ as made by writing Ω, P, Q, R, each of them in the form $\Omega=aP+\beta Q+\gamma R$, where the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience $\frac{P}{R}$, $\frac{Q}{R}$ in place of x, y, so as to make Ω, P, Q, R each of them cubic function of (λ,ρ); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion: viz. the equations $\Omega=0$, $x:y:z=P:Q:R$ are respectively taken to be

$$\Omega=(a\lambda+\lambda^2\rho)(c+\rho)(a'+b\rho)(c'+a^2\rho^2)=0,$$

$$x:y:z=\lambda\rho:a'+b\rho:c'+a^2\rho^2,$$

respectively as to the form of $\Omega=0$, see the lines down in the figure.

The base curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have thus the three common intersections $(\rho=0)$, $(\rho=0)$, $(\rho=0)$, with the three points represented in the figure as the points of B, D, H in the corresponding to the common intersection $(\rho=0)$.

[p. 584] viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and three intersections by the line $\Omega=0$, viz. those are the points

$$(\lambda,\rho,\mu)=(0,0,1),\quad (0,p,-f),\quad (h,0,-f),\quad (C,F,G).$$

The equation $Q+\theta R=0$ is $a'+b\rho+\theta(c+\rho\mu)=0$; viz. this is, for $\theta=0$, an equation of the second order in λ and ρ in the curve through the three points A, B, H, viz. those are the points

$$(\lambda,\rho,\mu)=(0,0,1),\quad (0,p,-f),\quad (h,0,-f),\quad (C,F,G).$$

To find the tangents at the triple point J, corresponding to the conditions $(\lambda,\rho,\mu)=(0,0,1)$, $(0,p,-f)$, $(h,0,-f)$, we have, for the values

$$\theta=\frac{c}{f},\quad \theta=\frac{c+p}{f}\cdot\frac{c+p}{f-1},\quad \text{or}\quad \theta=\frac{c}{f},\quad \theta=\frac{c+p}{f-1}.$$

Hence, considering for C as origin, the equation is

$$1 + 2a + \theta y = 0; \quad \theta(z + y) + abz(z + y) + \theta\theta(xx + y) = 0;$$

or, the equation $Q + \theta R = 0$ is similarly rendered; and so, similarly, K, M at the points L, M, we have f, a, as before constraint of f, h. As tangent at the conic point.

In fact (and this is well known to be the case for a singular case of a bicursal sextic with two triple points, but in the present case they are conics, C, F, G, are independent of θ, viz. they are the points

$$(\lambda, \rho, \mu) = (0, 0, 1), \quad (0, p, -f), \quad (h, 0, -f), \quad (C, F, G).$$

There are equivalent in virtue of the equation $Q + \theta R = 0$, that

$$(\lambda, \mu) = (0, 0), \quad (\lambda h, -f, 0), \quad \text{(those are also the points)} \quad C, F, G, \text{etc.}$$

To deduce these expressions, writing

$$M = c + 2a(\nu + p),$$

we have $K = (a + p) + c(a + b\mu)$. Hence

$$J = K + Y = (a - p)V\lambda,$$

also, from the original form,

$$J' = b + h\nu p + \lambda(C\eta + 2c) - h\mu(C\eta b + Y),$$

The final values are

$$\Omega = -K^2 + KX + 4a\rho(M + a\mu)R + a(bRY - bzp + Y),$$

$$P = (a - \nu)\mu^2 + (hR + a\mu)\Pi - 4a\nu(b - Z)Y,$$

$$Q = a + KaR + Pq\lambda - a\nu Y + ba\nu c\nu),$$

$$R = b(2a + 4ac\nu c + bR)$$

[p. 585] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before, but here three other common points and the figure of the bicursal sextic. There may also be other singular cubics, see the figure of the three-cubic sextic, see the figure $\Omega = 0$.

If denote like functions $\cos\theta + i\sin\theta$, $\cos\phi + i\sin\phi$ of the angles θ, ϕ, which λ, μ are subscribed to make in connecting set of angle of the bicursal sextic of the binormal cubic, Ω, C_2, R, see, taking X, Y as rectangular rectangular coordinates, x, y, and ν here their first letter for $\cos\theta + i\sin\theta$, $\cos\phi - i\sin\phi$,

$$X = a, \quad \cos\theta + b_2 \sin\alpha,$$

$$X + iY = b\cos C + (\cos\phi)C_2,$$

$$X - iY = b\cos C - (\cos\phi)C_2,$$

viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b\cos C,\ b\sin C)$, the first and third of these on the curve at the point $X = b\cos C$, $b\sin C$ (with from the same point of b, forming, with B and C, a triad of foci.

Verification of Ω.—The equation is

$$\Omega = (a\lambda + \lambda^2\rho)(c + \rho)(a' + b\rho)(c' + a^2\rho^2) = 0,$$

where
$$\Omega = (a\lambda + \lambda^2\rho)(c+\rho)(a' + b\rho)(c' + a^2\rho^2) = 0,$$
or, putting for shortness
$$\lambda = a + 4a + b\mu \quad \mu' \quad \nu' = (X - iY)(b\cos C + i\sin C) + \cos C - i\sin\theta - \nu' + \nu''(a' + \mu\nu),$$
$$M = b\nu + (X - iY)V\lambda + b\nu + ac\nu - a\nu + p\nu - \nu + \nu'\nu' + ac\nu = 0,$$
or, the trial may be made, a, ν being constants, the values $c\nu$ and the dual function K, X, Y.

[p. 586] Writing for homogeneity $\frac{x}{z}$, $\frac{y}{z}$ in place of λ, μ, and $\frac{x}{w}$, $\frac{y}{w}$ in place of x, y, the equations become
$$(ax^3 + a^4\rho^2) + a\rho(b + b_4b\rho)(b + b_4\rho)(c^2 + b_4\rho^2) = 0,$$
and
$$x : y : z : w = \lambda(\lambda + b\rho\mu) : \left(\frac{b}{c} + acb\mu\right) c' : c\rho\nu.$$

Comparing with the foregoing equations
$$x\lambda\rho + f(\lambda^2 + a)\nu = g(c^2 + a)\mu + ac\nu + bu(\lambda^2 + \nu)$$
and
$$x : y : z = \lambda\nu : (a' + b\rho) : (c' + a^2\rho^2),$$
the equations agree together, and we have
$$\rho = a\mu(c' + a)(\mu')z',$$
$$\mu = ecc\nu,$$
$$\Omega = cc\nu(\nu + 1)\nu(bu),$$
$$\beta = cc\nu(\nu + 1)\nu(bu);$$
the integrals at the triple points then are
$$\nu = 0, \qquad \rho = 0,$$
$$a\mu, c\mu(\nu, c) = 0, \qquad a\mu - ac\mu, c = 0,$$
$$\alpha = c\mu(c\mu),$$
$$b\mu = 0, \qquad b\mu = (\nu, c) = 0,$$
viz. writing the rectangular coordinates, and for y substituting the value $\cos C + i\sin C$,
$$X + iY = 0, \qquad X - iY = 0,$$
$$X + iY = b\cos C + (\cos\phi)C_2, \qquad X - iY = b\cos C - (\cos\phi)C_2,$$
$$X + iY = \nu,$$
viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b\cos C, b\sin C)$, and the third pair of three of the points of the figure, viz. the focuses with A, B and C, a triad of foci.

[p. 587] (omitted as overlap)

[p. 588] (continuation)

[p. 589] (continuation)

625. On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines

[From the Proceedings of the London Mathematical Society, vol. VIII. (1876–1877), pp. 53–57. Read December 14, 1876.]

In a congruency or doubly infinite system of right lines, the direction-cosines a, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $a\,dx + \beta\,dy + \gamma\,dz$ must be an exact differential: when this is so, writing $V \int(a\,dx + \beta\,dy + \gamma\,dz) = 0$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface cuts, its differential equation is $a\,dx+\beta\,dy+\gamma\,dz = 0$, and this equation must therefore be integrable by a factor. Now the functions a, β, γ are such that $a^2 + \beta^2 + \gamma^2 = 1$, and they hereby satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. posting from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x+ap$, $y+\beta p$, $z+\gamma p$ (p infinitesimal), the line through this point is parallel to the original line; and consequently the direction-cosines of this line are unchanged by the variations $\Delta x = ap$, $\Delta y = \beta p$, $\Delta z = \gamma p$, respectively. We thus obtain the equations

$$a\frac{da}{dx} + \beta\frac{da}{dy} + \gamma\frac{da}{dz} = 0,$$

$$a\frac{d\beta}{dx} + \beta\frac{d\beta}{dy} + \gamma\frac{d\beta}{dz} = 0,$$

$$a\frac{d\gamma}{dx} + \beta\frac{d\gamma}{dy} + \gamma\frac{d\gamma}{dz} = 0.$$

[p. 591] so that the required equation is a, B, γ will be satisfied if only

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we find

$$\frac{da}{dx} = \frac{dC}{dx}\frac{1}{a+\lambda}, \qquad \frac{d\beta}{dy} = \frac{dC}{dy}\frac{1}{a+\mu}, \qquad \frac{d\gamma}{dz} = \frac{dC}{dz}\frac{1}{a+\nu},$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we have

$$\frac{da}{dx} = \frac{dC}{dx}\cdot a\frac{a+\lambda}{a+\mu} + \frac{d}{dx}\left(\frac{a}{a+\lambda}\right)C = \frac{a^2}{a+\lambda}\frac{dx}{dx} + C^3\frac{dx}{dx}\left(\frac{a}{a+\lambda}\right),$$

and similarly for $\frac{d\beta}{dy}$ and $\frac{d\gamma}{dz}$ such that

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0, \qquad \text{viz.} \quad \frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and thence without difficulty

$$\left(\frac{a^2}{a+\lambda}\frac{da}{dx} + \frac{a^2}{a+\mu}\frac{d\mu}{dx}\right) - \left(\frac{1}{1+d\beta}\frac{d\beta}{dx} + \frac{d\nu}{dx}\right) - \omega = \left(a\frac{dx}{dx} + b\frac{dy}{dy} + c\frac{dz}{dz}\right),$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dy}+\frac{d\mu}{dy}\right)+\frac{1}{2(a+\lambda)}=-m\,a\ (\quad)+(\quad)\,,$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dz}+\frac{a^2}{a+\mu}\right)\frac{dz}{dz}=-\frac{1}{2}\quad(-)+(-).$$

626. On the General Differential Equation $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, where X, Y are the same Quartic Functions of x, y respectively

[From the Proceedings of the London Mathematical Society, vol. VIII. (1876–1877), pp. 184–199. Read February 8, 1877.]

[p. 592] WHERE $\Theta = a + b\theta + c\theta^2 + d\theta^3 + e\theta^4$, the general quartic function of θ, and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0.$$

We have

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ w^4, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

a particular integral of the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

viz. the two constants z, w enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

[p. 593] Take $w = x$; the equation becomes

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{W} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$x(P + \beta\theta + \gamma\theta^2) = (a\theta + \beta) - (a\theta + \beta)\theta(\theta - z)(\theta - z),$$

or, what is the same thing,

$$x(2x + \beta) - z = -(2(a\theta - \theta)(\theta + \theta) - z),$$

$$+2\beta y = -a + (3z - d) - (a + xy + z),$$

where β, γ are indeterminate coefficients which are to be eliminated. Write

$$x^2 = \frac{\sqrt{X}}{xz} = P, \quad y^2 = \frac{\sqrt{Y}}{yz} = Q;$$

then we have $\beta + \gamma = P + Q$, $\beta y + \gamma + Q = z$; giving

$$B + \gamma = 1 + Q - P, \quad B\gamma + Q = z - z.$$

Substituting these values in the first of the preceding three equations, we have

$$x\frac{2(Py - Qx)(x - y) + (Q - P)y}{(x - y)^2} - x = \frac{2 + (Q - P)y - Q}{(x - y)}\,(ay - z + s)\,(x + y + z),$$

that is,

$$\frac{2(Qy-Px)}{x-y}=\frac{2(2+(Q-P)y-Q)}{x-y},\quad (x+y+z)\cdot\frac{2}{x-y}\cdot\frac{1}{x}\cdot\frac{1}{y};$$

or, reducing by

$$Qy-Py-x=a\frac{x-y\sqrt{Y}}{x-y\sqrt{X}}.$$

[p. 594] times, the rational part is

$$M'=c+d(x+y)+e(x^2+xy+y^2),$$

and the two terms involving the operator L are as before stated that, in the particular case where $e=0$, the first equation becomes

$$M'=e+d(x+y)+e(x^2+xy+y^2),$$

and the two results for this case may be perfectly stated $C-e+0dt$.

But it is required to identify the two solutions in the general case where a is not $=0$, I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing

$$\Omega=ax^2+2xe(x+y)e^{-y}+4z(xx^2+2z^2x+4zy)(xe-ry)ee+2xye(x+y)e^{-y}+\frac{1}{2}y(xe+xy+xy+e)(C-exy),$$

then the equation between the variables x, y, C corresponds to the differential equation

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=\frac{dC}{\sqrt{\Omega}}=0,$$

a result which will be useful for effecting the identification. The Abelian solution may be written

$$P\left[\frac{1(xy(\sqrt{X}+y\sqrt{Y})}{(x-y)^2}-x^2\right]=\frac{1}{x-y}\sqrt{\Omega}+4ex(x+y)e^{-y}-4ex(c+ey)=2EX',$$

and substituting for M its value, and multiplying by $(x-y)$, the equation becomes

$$2\sqrt{x}(x-y)(z+X+y\sqrt{Y})(P+y\sqrt{P+Q}+(Q'(x-y)\sqrt{Y}))$$
$$=2(c+d+(x+y)+e(x^2+xy+y^2)e+xze^{-y}2(e\sqrt{X}-x\sqrt{Y});$$

and substituting therein for M its value, and multiplying by $(x-y)$, the equation becomes

$$2x((x+y)(\sqrt{X}-y\sqrt{Y})+2x(x+y)((c+ey)-(x+y)\sqrt{Y}-x\sqrt{Y}))$$
$$=2(c+d(x+y)+e(x^2+xy+y^2))+xz-2(c+ey-y)e^{-y}2(x\sqrt{Y}-x\sqrt{Y});$$

On the left-hand side, the rational part is

$$2X+Y+e(-y)(x^4+2xy+y)+4e^{-y}(c-e^{-y}2(x+y)e^{-y})y+2(c+d)(x-y)(x+y),$$

which, substituting therein for X, Y their values, becomes

$$=2a+b(x+y)+c(x^2+4xy+y^2)+2dxy(x+y)+2ex^2y^2((x+y),$$

and the irrational part is at once found to be

$$=2(a+(a-y)\sqrt{Y}-y(a+(a-y))\sqrt{X}).$$

[p. 595] Multiplying the numerator and the denominator by

$$d(x-y)+3e(x^2-y^2)+2e\sqrt{x}((y-1)(X-y^4Y)),$$

the denominator becomes

$$=(x-y)^2\left[d+3e(x+y)+4e^{-y}(\sqrt{x}-1)\sqrt{X}\right],$$

which, introducing therein the C of Euler's equation, is

$$=(x-y)^2(C-6acC).$$

We have therefore

$$x(x-y)-6ac=(x(x-y)C+b(x-y)+2ay(P+2dx(yP-xQ)+2xx^2(\sqrt{Y}P+y\sqrt{X}Q),$$

$$+2xy(xx-y)\sqrt{X}-x\sqrt{Y}\cdot Vx((y-1)(X-y\sqrt{Y})((x-y)(x\cdot s-2))-2(x-y)\sqrt{Y\sqrt{Y}}.$$

Using b, to denote the same value as before, the function on the right-hand is, in fact,

$$=(x-y)^2\left[Bx-ad+de\sqrt{x}\sqrt{Y}\right],$$

and, this being so, the required relation between x, C is

$$x(\beta x-4ac)=(Bx-ad+de\sqrt{x}\sqrt{Y}),$$

$$((x-y)\sqrt{Y-y\sqrt{X}})((x-y)\sqrt{X}\cdot s-2)).$$

To prove this, we have first, from the algebric

$$\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2=C+d(x+y)+e\sqrt{x}\sqrt{X},$$

to express Φ as a function of C as a variable, gives

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dC}{\sqrt{\Omega}}=0,$$

and we have therefore

$$-\sqrt{\Omega}=\sqrt{x}\frac{dC}{dx}\quad(=-\sqrt{Y}\frac{dC}{dy}),$$

viz. $a\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before

$$M=\frac{C-\sqrt{Y}}{x-y}.$$

we have

$$C=M^2-d(x+y)+e(x+y),$$

and thence

$$\Omega=M^2-x(x+y)+e(x+y)^2.$$

We have

$$\frac{dM}{dx}=\frac{1}{x-y}\left(\frac{X'}{2\sqrt{X}}-\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right),$$

[p. 596] and hence

$$\sqrt{\Omega} = \sqrt{X} - \sqrt{Y}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right],$$

which thus is $(x-y)$, the numerator becomes

$$M' = c + d(x+y) + e(x^2+xy+y^2),$$

say that the particular case as may putting $C = c + exy$.

But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing

$$\Omega = \frac{(C-c)^2(C-4ac)}{C(\theta-y)} - 4a\left[b - c(x+y) + d(xx+y^2) - 2e(xx+xy+y^2)\right](x-y)^2\sqrt{X}\sqrt{Y};$$

and substituting therein for M its value, and multiplying by $(x-y)$, this is

$$\Omega = c(P-p)[2P - 4ac + 2(x-y)^2\sqrt{X}\sqrt{Y}].$$

To prove this, we have first, from the algebric

$$\sqrt{\Omega} = c(P-p)\sqrt{\Omega(P-p)\sqrt{X}\sqrt{Y}},$$

to express Φ as a function of C as a variable. This equation, regarding C as a variable, gives

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\Omega}} = 0,$$

and we have therefore

$$-\sqrt{\Omega} = \sqrt{x}\frac{dC}{dx} \quad (= -\sqrt{Y}\frac{dC}{dy}),$$

viz. $\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before,

$$M = \frac{C-\sqrt{Y}}{x-y},$$

we have

$$C = M^2 - d(x+y) + e(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\frac{dM}{dx} - d - 2e(x+y),$$

that is,

$$\sqrt{\Omega} = \sqrt{x}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right] \quad \text{suppose;}$$

or, writing for shortness X', M' to denote the derived functions $\frac{dX}{dx}$, $\frac{dM}{dx}$, (Y' is afterwards written to denote $\frac{dY}{dy}$), but as the final formula contains only X', $\frac{dX}{dx}$, and Y', $\frac{dY}{dy}$. (This does not assume any defect of symmetry), we find

$$\Omega = N\left[BC - X(X - 2(x+y) + \sqrt{X}\sqrt{Y}\right]$$

$$= D^2\sqrt{X^2} + 2(xx-y)ex^2\sqrt{X^2} + 2(c-0)\sqrt{Y\sqrt{XY}} - X'\sqrt{Y}],$$

[p. 597] viz. this is

$$= [2a + b(x+y) + c\sqrt{x}(y-z) + e(x+y)(\sqrt{Y} - x\sqrt{X}))]$$
$$\times[d(x-y) + 2e(xx + 2xy + yx)],$$
$$+\frac{1}{2}\sqrt{X}(d(x-y) - 4e(xx + 2xy + y^2) - 2exex(x - y + (z+y)e^{-y}((x-y)x + y\sqrt{Y})),$$

which is, in fact, equal to the expression on the left-hand side.

To complete, then, it is required to express $\sqrt{X}$ as a function of x, y symbolic that

$$x(x-y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x+y)x \cdot 4 = (x-y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x-y)(\sqrt{X} + \sqrt{Y}),$$

which may be done by putting therein

$$2P = -X^2X + 2xexy(X-y) + 4ex(\sqrt{x}y - \sqrt{y}x),$$

which when expanded into X as a function of x, y, on X as a function of x, y, of which there are forms in x, y, on $X = 0$, in fact, equal to the corresponding expression on the left-hand side.

So the left-hand side, considering therein down as they immediately present themselves; but, of course, the relation X is $X(x+y)-$ to come $a+(x+y)-$, when these are written in the form

$$-b + dxy = (a+b),$$

which is right.

To X verification of Ω.—The quantity is

$$J\Omega(X-y)X + 8x(c-y)\sqrt{C}(C - 4ac) + 4a(x-y)\sqrt{X}\sqrt{Y}$$
$$= 2L(P - xP\sqrt{X}) - 4a(P\sqrt{X}\sqrt{Y} - 2yP + \sqrt{Y}(c - 4ac))((x-y)\sqrt{Y} - y\sqrt{X})$$
$$+2\sqrt{X}((x-y)\sqrt{Y} - x\sqrt{Y})((x-y)\sqrt{X}(c-1)) = 0;$$

which, throwing out the factor $x - y$, becomes

$$0 = 2\sqrt{X}(c-y) + 2cxy(x-y) + 2cxxy - 2(c+y)e\sqrt{X}\sqrt{Y} - 4\sqrt{X} - 2(x+y) - 2cxy - 2cxy - 2(x+y)y^2,$$

which, throwing out the factor $x - y$, is

$$0 = 2N\left(BC\sqrt{X} - 2(x+y)yP + 2cxy\sqrt{X}\right) + 2(c - 4ac)\sqrt{Y}((x-y)\sqrt{Y} - y\sqrt{X}),$$
$$+2\sqrt{X}((x-y)\sqrt{X} - x\sqrt{Y})((x-y)\sqrt{X}(c-1)).$$

[p. 598] and similarly herein for X, Y their values, and arranging the terms, we find

$$\Omega = a\sqrt{X}\sqrt{Y}\sqrt{X}((x-y)\sqrt{Y} - y\sqrt{X}),$$

where

$$\Omega = a[2X + (a-y)X]$$
$$-2(x-y)yX$$
$$-(xy + (a+y)y)$$
$$-(x-y)(ay-y)$$
$$+By(2X + (x-y)X)$$
$$+2Y(x-y)X,$$

$$\Omega = 4xy(a-y)X$$
$$-2(a-y)y(2X+(x-y)X)$$
$$-4XY$$
$$+2by(x-y)X$$
$$+4cx(x-y)X$$
$$+4ex(x-y)(y-2)X,$$
$$\Omega = -4xy(a-y)X$$
$$+2cx(2X+(a-y)X)$$
$$+4xY$$
$$-2cx(x-y)X$$
$$-4x(x-y)X$$
$$+4x(x-y)(y-2)X,$$

where the same identifications hold down with the formulae above stated, refusing, except differently and averaging, we have here so much.

To reduce these expressions, writing $M = 2x(x+y)$, hence we have

$$J' = 2x + 4xy + e^{-y}((y+y)y + 2xy + Y), \quad 2xy(c+y)(y+Y+Y),$$

which divides by $(x-y)$.

Hence the equation now is

$$(x-y)^2\nabla = (a+b+e)(\sqrt{Y}-y)\cdot x\cdot \nabla((ax-y)y\sqrt{Y}),$$
$$+y(2xx+y) - x(b+e)(a+4ay-a)(c\sqrt{X}-1) + 2x((x-y)x - y\sqrt{Y}-x);$$
$$+ay\nabla(b+e)((x-y)x - y\sqrt{Y} - 4y\sqrt{X})\cdot 2(x-y).$$

The function on the right-hand is, in fact,

$$= 2(L\nabla + (x+y))\cdot ax\cdot \nabla\cdot \sqrt{Y} - y\sqrt{X},$$

and the irrational part is at once found to be

$$= 2(a+(a-y)\sqrt{X} - y(a+(a-y))\sqrt{X}).$$

[p. 599] *Verification of* Ω.—The equation is

$$-X' + 4\nabla Y + (x-y)(X' + (\sqrt{X}-4)y\sqrt{X})M + xyy,$$
$$= 2L(M'\cdot\sqrt{X}-y) + 4xx\sqrt{X^2}M + 2N\sqrt{X^2}M;$$

or, putting therein

$$\frac{X'}{(x-y)} = X' + 8X' - 6\sqrt{X}((\sqrt{X}-y)P)M^2 = 2PxM.$$

this is

$$(x-y)^2 = X' + 4XY$$
$$+2X'\cdot X - 8a(x-y)xP$$

$$+(a-y)(2xx+y)-4(\sqrt{X}\cdot 2(y-y))x\sqrt{X}M,$$
$$-2(\sqrt{X}-y)((x-y)X-xy^2\sqrt{X})-4(c-xy)(c-xy^2)\cdot 2\sqrt{X}\sqrt{Y};$$
$$+(x-y)(2X+4y(c\sqrt{X}+y)P+2(cxy\sqrt{Y}\cdot 2X'+2y\sqrt{X})X^2),$$
$$+(x-y)\cdot\sqrt{X}(c+y)X\cdot xz\cdot 2(X'\cdot x+xy\sqrt{X})-2y\sqrt{X}.$$

We have $YE'F=x(X+1)+de\sqrt{x}(X+Y-4yx+2by+ex^2)$ and thence

$$-X'\cdot 2X+(a-y)y(X'\cdot 4\cdot c\cdot 2\cdot cX)x\sqrt{X}+2\sqrt{X}(c\cdot c\sqrt{X}\sqrt{Y}-((x-y)-x)y\sqrt{X}),$$

or, as this may be written,

$$\Phi=-X'+2\sqrt{X}P.$$

[p. 600] We find, moreover,

$$\Phi'=X'-4(c+y)x+(x-y)((\sqrt{X}-4)y\sqrt{X})Xy\sqrt{X},$$

which divides by $(x-y)$, viz. this is

$$=y(c+(a-y)y)+4y(a+(a-y))+2X+2y(x-y)yX,$$

which divides by $(x-y)$, viz. this is

$$=y(c+a(x+y))\sqrt{Y}-y(b+2ay+2c)\cdot 2y-y\sqrt{X}.$$

Hence above the values reducing, and throwing out the factor X, the equation becomes

$$-b+d\,ay=(a+b),$$

which is right.

To the first line, in the equation immediately preceding, we have

$$J'=-2(x-y)x\nabla x\cdot c=-2(a-y)\sqrt{X}-xPx\nabla cy\sqrt{X};$$

which is

$$=-2X+(a-y)Py\cdot xc+(b+e)((x-y)X-y\sqrt{X}P)x-x;$$

hence, throwing out the factor $(x-y)$,

$$\Phi=aPx\nabla c-x((x-y)X-y\sqrt{X}P)-(a+y)Py\cdot 2xc-2xPx\nabla,$$

which is to come $(x-y)P\sqrt{X}P+2P\sqrt{X}$,

$$0=2XY+2(a+b)x+(x-y)P\nabla+2X\nabla,$$

which, throwing the factor $(x-y)$, is

$$0=-X'+2Xy\nabla+(x-y)\cdot c\cdot cxy\sqrt{X},$$

which, throwing the factor $(x-y)$, becomes

$$0=-X+2\sqrt{X}P,$$

which is the required relation.

[Continuation of paper 623, "On Three-Bar Motion".]

It is to be remarked that the foregoing equation in κ determines a single position of the point K; viz. it determines $\tan\kappa$, and therefore $\sin 2\kappa$ and $\cos 2\kappa$, linearly. The equation is, in fact, a cubic equation in $\tan\kappa$, satisfied identically by $\tan\kappa = i$ and $\tan\kappa = -i$, and therefore reducible to a linear equation.

Write for a moment $\tan\kappa = \omega$, and

$$(\tan\kappa - \tan\alpha)(\tan\kappa - \tan\beta)(\tan\kappa - \tan\gamma) = \omega^3 - p\omega^2 + q\omega - r,$$
$$(\tan\kappa - \tan f)(\tan\kappa - \tan g)(\tan\kappa - \tan h) = \omega^3 - p'\omega^2 + q'\omega - r';$$

also

$$M = \cos f\cos g\cos h \;\div\; \cos\alpha\cos\beta\cos\gamma.$$

Then we have

$$\omega^3 - p\omega^2 + q\omega - r = M(\omega^3 - p'\omega^2 + q'\omega - r'),$$

where

$$p = r + M(r' - p), \qquad q = 1 + M(q' - 1).$$

Substituting these values, the equation becomes

$$\omega^3 - r\omega^2 + \omega - r = M(\omega^3 - r'\omega^2 + \omega - r'),$$

viz. dividing by $\omega^2 + 1$, this is $\omega - r = M(\omega - r')$; or substituting for r, r', M their values,

$$(\cos\alpha\cos\beta\cos\gamma - \cos f\cos g\cos h)\tan\kappa = (\sin\alpha\sin\beta\sin\gamma - \sin f\sin g\sin h),$$

which is the value of $\tan\kappa$, and then

$$\sin 2\kappa = \frac{2\tan\kappa}{1+\tan^2\kappa}, \qquad \cos 2\kappa = \frac{1-\tan^2\kappa}{1+\tan^2\kappa}.$$

It may be further noticed that, if the parabola intersect the circle in a point L, and the tangent at L to the parabola again meet the circle in M, then, if $2l$, $2m$ are the angles for the points L, M, we have l, m, m for values of f, g, h, whence l, m are determined by the equations

$$l + 2m = \alpha + \beta + \gamma, \qquad \sin(\kappa - l)\sin^2(\kappa - m) = \sin(\kappa - \alpha)\sin(\kappa - \beta)\sin(\kappa - \gamma);$$

but as the circle intersects the parabola not only in two real points, but in two other imaginary points, there is no simple formula for the determination of l and m.

To determine the linkage when the nodes are given, suppose that, in the generation by O, the vertex of the triangle OB_1C_1, we have at the node F: then, if τ, σ are the distances of C, B from the node in question, we have, as in the memoir,

$$(b_1^2 + \tau^2 - a_2^2)c_1\sigma = (c_1^2 + \sigma^2 - a_3^2)b_1\tau,$$

that is,

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b_1\tau + \sigma\tau(c_1\tau - b_1\sigma) = 0,$$

or, what is the same thing,

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b\tau + a\tau(c\tau - b\sigma) = 0.$$

Suppose, as in the figure, that F is between B and A; then, if $AF = p$, we have $c\tau = b\sigma + ap$, and the equation becomes

$$(b_1^2 - a_2^2)c\sigma - (c_1^2 - a_3^2)b\tau + ap\sigma\tau = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if p', σ', τ' be the distances AG, BG, CG, then $b\sigma' = c\tau' + ap'$, that is, $c\tau' - b\sigma' = -ap'$, and the corresponding equation is

$$(b_1^2 - a_2^2)c\sigma' - (c_1^2 - a_3^2)b\tau' - ap'\sigma'\tau' = 0.$$

We hence find

$$(b_1^2 - a_2^2)c(\sigma\tau' - \sigma'\tau) + a\sigma\sigma'(p\tau + p'\tau') = 0,$$
$$(c_1^2 - a_3^2)b(\sigma\tau' - \sigma'\tau) + a\tau\tau'(p\sigma + p'\sigma') = 0.$$

But we have

$$BG \cdot CF = BF \cdot CG + BC \cdot FG,$$

that is,

$$\sigma'\tau = \sigma\tau' + a \cdot FG, \qquad \text{or } \sigma\tau' - \sigma'\tau = -a \cdot FG;$$

also,

$$p\sigma + p'\sigma' = AF \cdot BF + AG \cdot BG = FG \cdot BH = FG \cdot \tau'',$$

and

$$p\tau + p'\tau' = AF \cdot CF + AG \cdot CG = FG \cdot CH = FG \cdot \sigma'',$$

as may be shown without difficulty, p'', σ'', τ'' being the distances AH, BH, CH. Hence the equations become

$$c(b_1^2 - a_2^2) - \sigma\sigma'\sigma'' = 0,$$
$$b(c_1^2 - a_3^2) - \tau\tau'\tau'' = 0,$$

showing that, the foci being as in the figure, $b_1^2 - a_2^2$ and $c_1^2 - a_3^2$ are each of them positive; viz. that, in the generation by the triangle OC_1B_1, the radial bars a_2, a_3 are shorter than the sides b_1, c_1 respectively. Substituting for b_1, &c. the values $k_1 \sin B$, &c.; also, instead of k_1, k_2, k_3, introducing the quantities λ_1, λ_2, λ_3, where

$$k_1, k_2, k_3 = \lambda_1 \sin A,\ \lambda_2 \sin B,\ \lambda_3 \sin C,$$

these equations become

$$c(\lambda_1^2 - \lambda_2^2) \sin^2 A \sin^2 B = \sigma\sigma'\sigma'',$$
$$b(\lambda_1^2 - \lambda_3^2) \sin^2 A \sin^2 C = \tau\tau'\tau'';$$

or, as these may be written, putting for shortness $M = \sin A \sin B \sin C$,

$$M^2k^2(\lambda_1^2 - \lambda_2^2) = c\,\sigma\sigma'\sigma'',$$
$$M^2k^2(\lambda_1^2 - \lambda_3^2) = b\,\tau\tau'\tau''.$$

All the quantities have so far been regarded as positive, and the formulae are applicable to the particular figure; but, to present them in a form applicable to any order of the nodes and foci, we have only to write the equations in the forms

$$M^2(\lambda_1^2 - \lambda_2^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$
$$M^2(\lambda_1^2 - \lambda_3^2) = k^2 \sin(\alpha - \gamma) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta).$$

And these may be replaced by the system

$$M^2(\lambda_2^2 - \lambda_3^2) = k^2 \sin(\beta - \gamma) \sin(f - \alpha) \sin(g - \alpha) \sin(h - \alpha),$$
$$M^2(\lambda_3^2 - \lambda_1^2) = k^2 \sin(\gamma - \alpha) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta),$$
$$M^2(\lambda_1^2 - \lambda_2^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$M^2k^2(\lambda_2^2 - \lambda_3^2) = BC \cdot FA \cdot GA \cdot HA,$$
$$M^2k^2(\lambda_3^2 - \lambda_1^2) = CA \cdot FB \cdot GB \cdot HB,$$
$$M^2k^2(\lambda_1^2 - \lambda_2^2) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they stand for $k\sin(\beta-\gamma)$, $k\sin(f-\alpha)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$BC\,(c_1^2-b_1^2)=FA\cdot GA\cdot HA,$$
$$CA\,(a_3^2-c_1^2)=FB\cdot GB\cdot HB,$$
$$AB\,(b_1^2-a_2^2)=FC\cdot GC\cdot HC,$$

where the signs are as just mentioned. We may say that $\pm(a_2^2-b_1^2)$ is the modulus for the focus A; and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624.

On the Bicursal Sextic.

[From the Proceedings of the London Mathematical Society, vol. vii. (1875–1876), pp. 166–172. Read March 10, 1876.]

In the paper "On the mechanical description of certain sextic curves," *Proceedings of the London Mathematical Society*, vol. iv. (1872), pp. 105–111, [504], I obtained the bicursal sextic as a rational transformation of a binodal quartic. The theory was in effect as follows: taking P, Q, R, each of them a function of λ, μ of the form $(*\quad)(\lambda,1)^2(1,\mu)^2$, and considering (λ,μ) as connected by the equation $\Omega=0$, (viz. λ, μ being coordinates, this represents a binodal quartic), then, if we assume $x:y:z=P:Q:R$, the locus of the point (x,y,z) is a curve rationally connected with the binodal quartic, viz. the points of the two curves have with each other a $(1,1)$ correspondence; whence the locus in question, say the curve $U=0$, is bicursal. The degree is obtained as the number of the intersections of the curve by an arbitrary line, or, what is the same thing, the number of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega=0,\qquad \alpha P+\beta Q+\gamma R=0,$$

viz. each of these being a quartic curve having the same two nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4\cdot4-2\cdot4=8$, and thus the curve $U=0$ is in general of the order 8. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the nodes) k common intersections, then these are also fixed intersections of the two curves $\Omega=0$, $\alpha P+\beta Q+\gamma R=0$, and the number of variable intersections is reduced to $8-k$: we have thus $8-k$ as the order of the curve $U=0$. In particular, if $k=2$, then the curve is a bicursal sextic.

The theory assumes a different and more simple form if, in the several functions Ω, P, Q, R, we suppose that the terms in λ^2, μ^2 are wanting. The curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ are here cubics having two common points; the curve $U=0$, qua rational transformation of the cubic $\Omega=0$, is still a bicursal curve; but its order is given as the number of the variable intersections of the cubics

$$\Omega=0,\qquad \alpha P+\beta Q+\gamma R=0,$$

viz. this is $=3\cdot3-2=7$. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $=7-k$: and this is therefore the order of the curve $U=0$. In particular, if $k=1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega=0$, $x:y:z=P:Q:R$, where $\Omega=0$, $P=0$, $Q=0$, $R=0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $\alpha P+\beta Q+\gamma R=0$, there are any two curves which have 3 common intersections with the curve $\Omega=0$. (Observe that we throughout disregard the 3 common points of the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$, and attend only to the 6 variable points of intersection of the curves $\Omega = 0$ and $\alpha P + \beta Q + \gamma R = 0$,—the meaning is, that there are two curves of the series such that, attending only to the 6 variable intersections of each of them with the curve $\Omega = 0$, there are three common intersections.) For, supposing the two curves to be $\alpha P + \beta Q + \gamma R = 0$ and $\alpha' P + \beta' Q + \gamma' R = 0$, then any curve whatever

$$\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$$

has the same three intersections with the curve $\Omega = 0$, say these are the points A_1, A_2, A_3, the coordinates of which are independent of θ. Hence the line

$$(\alpha x + \beta y + \gamma z) + \theta(\alpha' x + \beta' y + \gamma' z) = 0$$

intersects the curve $U = 0$ in six points, three of which, as corresponding to the points A_1, A_2, A_3, are independent of θ, viz. they are the same three points for any line whatever of the series; and this means that the curve $U = 0$ has at the point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0)$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $\alpha x + \beta y + \gamma z + \theta(\alpha' x + \beta' y + \gamma' z) = 0$, rationally determine θ so that one of the three variable points of intersection shall correspond to A_1, A_2, or A_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points A_1, A_2, A_3. The three lines thus determined are the three tangents to the curve at the triple point: and the three branches may be considered as corresponding to the three points A_1, A_2, A_3, respectively.

There is no loss of generality in assuming that the triple point is the point $(x = 0,\ z = 0)$; the condition then simply is that the curves $P = 0$, $R = 0$ shall have three common intersections with the curve $\Omega = 0$; and the tangents at the triple point are $x + \theta z = 0$, θ being so determined that one of the three variable points of intersection shall correspond to one of the three points A_1, A_2, A_3: in particular, if this is the case for the line $x = 0$, then this line will be one of the tangents at the triple point.

The bicursal sextic may have a second triple point, viz. three other nodes may unite together into a triple point. The theory is precisely the same: we must have two other curves $\alpha P + \beta Q + \gamma R = 0$, $\alpha' P + \beta' Q + \gamma' R = 0$, having with the curve $\Omega = 0$ three common intersections B_1, B_2, B_3: there is then a second triple point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0)$$

and, to find the tangents at this point, we must determine θ so that one of the variable points of intersection of the line

$$\alpha x + \beta y + \gamma z + \theta(\alpha' x + \beta' y + \gamma' z) = 0$$

with the sextic shall correspond with B_1, B_2, or B_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points B_1, B_2, B_3. In particular, if, as before, the curves $P = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, and if, moreover, the curves $Q = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, then the bicursal sextic will have the two triple points $(x = 0,\ z = 0)$ and $(y = 0,\ z = 0)$; and it may further happen that the line $x = 0$ is a tangent at the first triple point, and the line $y = 0$ a tangent at the second triple point. The sextic may in like manner have a third triple point, but this is a special case which I do not at present consider.

I write for greater convenience $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, so as to make Ω, P, Q, R each of them a homogeneous cubic function of (λ, μ, ν); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion; viz. the equations $\Omega = 0$, $x : y : z = P : Q : R$ are respectively taken to be

$$\nu(h\nu\lambda + f\lambda^2) + \mu(d\lambda\mu + e\nu\lambda + g\lambda^2) + \mu^2(f\nu + h\lambda) = 0,$$

$$x : y : z = \lambda\mu(a\lambda + b\mu) : \nu^2(c\lambda + d\mu) : \lambda\mu\nu.$$

The four curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have thus the three common intersections

$$(\mu = 0,\ \nu = 0), \quad (\nu = 0,\ \lambda = 0), \quad (\lambda = 0,\ \mu = 0),$$

represented in the figure by the points A, B, C; the curve drawn in the figure is the curve $\Omega = 0$, and the points F, G, H are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

[Figure: cubic curve $\Omega = 0$ with marked points A, B, C at the vertices of the coordinate triangle, and points F, G, H where the cubic meets the sides BC, CA, AB respectively.]

The equation $P + \theta R = 0$ is here $\lambda\mu(a\lambda + b\mu + \theta\nu) = 0$, which intersects $\Omega = 0$ in the points C, A, G, C, B, F, and the three intersections by the line $a\lambda + b\mu + \theta\nu = 0$; viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and the three intersections by the line. Hence, of the six intersections, we have C, F, G independent of θ, or we have $(x = 0,\ z = 0)$ a triple point, say I, corresponding to the three points C, F, G, viz. these are the points

$$(\lambda, \mu, \nu) = (0, 0, 1),\ (0, g, -f),\ (h, 0, -f), \qquad (C, F, G).$$

The equation $Q + \theta R = 0$ is $\nu(c\lambda\nu + d\mu\nu + \theta\lambda\mu) = 0$; viz. the line $\nu = 0$ meets $\Omega = 0$ in the three points A, B, H, and the conic $c\lambda\nu + d\mu\nu + \theta\lambda\mu = 0$ meets $\Omega = 0$ in the points A, B, C, and three other points: hence, rejecting the points A, B, C, the six points of intersection are the points A, B, H, and the three variable points of intersection by the conic; or we have $(y = 0,\ z = 0)$ a triple point, say J, corresponding to the three points A, B, H, viz. these are the points

$$(\lambda, \mu, \nu) = (1, 0, 0),\ (0, 1, 0),\ (h, -g, 0), \qquad (A, B, H).$$

To find the tangents at the triple point I, these are $x + \theta z = 0$, where θ is to be successively determined by the conditions that the line $a\lambda + b\mu + \theta\nu = 0$ shall pass through the points C, F, G [1]; viz. we thus have

$$\begin{array}{lll} \theta = 0, & x = 0, & \text{tangent corresponding to the point } C, (0, 0, 1), \\ \theta = \dfrac{a}{f}, & fx + bgz = 0, & \text{tangent corresponding to the point } F, (0, g, -f), \\ \theta = \dfrac{a}{f}, & fx + ahz = 0, & \text{tangent corresponding to the point } G, (h, 0, -f). \end{array}$$

And similarly, at the triple point J, the tangents are $y + \theta z = 0$, where θ is to be successively determined by the conditions that the conic $c\nu\lambda + d\nu\mu + \theta\lambda\mu = 0$ shall pass through the point H, and shall touch the cubic at the points A, B; viz. we thus have

$$\begin{array}{lll} \theta = 0, & y = 0, & \text{tangent corresponding to the point } H, (h, -g, 0), \\ \theta = \dfrac{c}{f}, & fy + cgz = 0, & \text{tangent corresponding to the point } A, (1, 0, 0), \\ \theta = \dfrac{d}{f}, & fy + dhz = 0, & \text{tangent corresponding to the point } B, (0, 1, 0). \end{array}$$

The two last values of θ are obtained by the consideration that the equations of the tangents to $\Omega = 0$ at the points A, B respectively, are $g\mu + f\nu = 0$, $h\lambda + f\nu = 0$, where λ, μ, ν are current coordinates of a point on the tangent: it may be added that the equation of the tangent at the point C is $h\lambda + g\mu = 0$.

The three-bar curve may be represented by means of a system of equations of the last-mentioned form, viz. $x : y : z = \lambda\mu(a\lambda + b\mu) : \nu^2(c\lambda + d\mu) : \lambda\mu\nu$, where λ, μ, ν are connected as above; or, taking X, Y as ordinary rectangular coordinates, x, y, and z are here the circular coordinates $\dfrac{x}{z} = X + iY$, $\dfrac{y}{z} = X - iY$, and $z = 1$; and the parameters $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ denote like functions $\cos\theta + i\sin\theta$, $\cos\phi + i\sin\phi$ of angles θ, ϕ which are the inclinations of two bars to a fixed line. Using, for convenience, Figure 2 of my paper on Three-bar Motion, (p. 553 of this volume), the curve is considered as the locus of the vertex O of the triangle OC_1B_1, connected by the bars C_1C and B_1B with the fixed points B and C respectively; and we have $CC_1 = a_2$, $OC_1 = b_1$, $OB_1 = c_1$, $C_1B_1 = a_1$, $B_1B = a_3$. Also, to avoid confusion with the foregoing notation of the

[1] Observe the somewhat altered form of the condition: θ is to be determined so that the cubic $\lambda\mu(a\lambda + b\mu + \theta\nu) = 0$ shall touch the cubic $\Omega = 0$ at one of the points C, F, G: but, as the first-mentioned cubic breaks up, and the component curve $a\lambda + b\mu + \theta\nu = 0$ does not pass through any one of these points, this can only mean that θ shall be so determined as that the line shall pass through one of these points, viz. that there shall be at the point, not a proper contact, but a double intersection, arising from a node of the cubic $\lambda\mu(a\lambda + b\mu + \theta\nu) = 0$. And the like case happens for the other triple point; viz. there the cubic $\nu(c\nu\lambda + d\nu\mu + \theta\lambda\mu) = 0$ is to touch the cubic $\Omega = 0$ at one of the points A, B, H; the component conic $c\nu\lambda + d\nu\mu + \theta\lambda\mu = 0$ passes through the points A and B but not through H; hence the conditions for θ are, that the conic shall touch the cubic at A or B, or that it shall pass through H.

present paper, instead of calling it a, I take $BC = a_0$: the angle OC_1B_1 is $= C_1$, and $\cos C_1 + i\sin C_1$ is taken $= \gamma_1$.

Hence, taking the origin at C, the axis of X coinciding with CB and that of Y being at right angles to it: taking also θ, ϕ, ψ for the inclinations of CC_1, C_1B_1, and B_1B to CB, we have

$$a_2\cos\theta + a_1\cos\phi - a_0 = -a_3\cos\psi,$$
$$a_2\sin\theta + a_1\sin\phi = a_3\sin\psi;$$

viz. writing $\cos\theta + i\sin\theta = \lambda$, $\cos\phi + i\sin\phi = \mu$, these give

$$a_2\lambda + a_1\mu - a_0 = -a_3(\cos\psi - i\sin\psi),$$
$$a_2\frac{1}{\lambda} + a_1\frac{1}{\mu} - a_0 = -a_3(\cos\psi + i\sin\psi),$$

that is,

$$(a_2\lambda + a_1\mu - a_0)\left(a_2\tfrac{1}{\lambda} + a_1\tfrac{1}{\mu} - a_0\right) - a_3^2 = 0;$$

viz.

$$(a_2^2 + a_1^2 + a_0^2 - a_3^2) + a_1a_2\left(\tfrac{\lambda}{\mu} + \tfrac{\mu}{\lambda}\right) - a_0a_2\left(\lambda + \tfrac{1}{\lambda}\right) - a_0a_1\left(\mu + \tfrac{1}{\mu}\right) = 0,$$

for the relation between the parameters λ, μ. And then

$$X = a_2\cos\theta + b_1\cos(\phi + C_1),$$
$$Y = a_2\sin\theta + b_1\sin(\phi + C_1);$$

viz. if $x, y = X + iY,\ X - iY$, then

$$x = a_2\lambda + b_1\gamma_1\mu, \qquad y = a_2\tfrac{1}{\lambda} + b_1\frac{1}{\gamma_1\mu},$$

which equations determine the coordinates (x, y) in terms of the parameters λ, μ connected by the foregoing relation.

Writing for homogeneity $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, and $\dfrac{x}{z}$, $\dfrac{y}{z}$ in place of x, y, the equations become

$$(a_2^2 + a_1^2 + a_0^2 - a_3^2)\lambda\mu\nu + a_1a_2(\lambda^2 + \mu^2)\nu - a_0a_1\mu(\nu^2 + \lambda^2) - a_0a_2\lambda(\mu^2 + \nu^2) = 0,$$

and

$$x : y : z = (a_2\lambda + b_1\gamma_1\mu)\lambda\mu : \left(\tfrac{a_2}{\gamma_1}\lambda + a_1\mu\right)\nu^2 : \lambda\mu\nu.$$

Comparing with the foregoing equations

$$e\lambda\mu\nu + f(\lambda^2 + \mu^2)\nu + g(\nu^2 + \lambda^2)\mu + h(\mu^2 + \nu^2)\lambda = 0,$$

and

$$x : y : z = (a\lambda + b\mu)\lambda\mu : (c\lambda + d\mu)\nu^2 : \lambda\mu\nu,$$

the equations agree together, and we have

$$\begin{aligned}
e &= a_2^2 + a_1^2 + a_0^2 - a_3^2,\\
f &= +a_1a_2,\\
g &= -a_0a_2,\\
h &= -a_0a_1,\\
a &= a_2,\\
b &= b_1\gamma_1,\\
c &= \frac{a_2}{\gamma_1},\\
d &= a_1.
\end{aligned}$$

The tangents at the triple points thus are

$$x = 0,$$
$$a_1 x - a_0 b_1 \gamma_1 z = 0, \qquad a_1 y - \frac{a_0 a_1}{\gamma_1} z = 0,$$
$$x - a_0 z = 0, \qquad y - a_0 z = 0;$$

viz. restoring the rectangular coordinates, and for γ substituting the value $\cos C + i \sin C$, for a_0 writing a, and taking $b = \dfrac{b_1 \gamma_1}{\gamma}$, we have

$$\begin{aligned} X + iY &= 0, & X - iY &= 0, \\ X + iY &= b(\cos C + i \sin C), & X - iY &= b(\cos C - i \sin C), \\ X + iY &= a, & X - iY &= a; \end{aligned}$$

viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b \cos C, b \sin C)$, the third two in the point $(a, 0)$: the first and third of these are the points B and C, the second of them is the point A of the figure: viz. the formulae give the point A, forming, with B and C, a triad of foci.

625.

On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines.

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In a congruency or doubly infinite system of right lines, the direction-cosines α, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $\alpha\, dx + \beta\, dy + \gamma\, dz$ must be an exact differential: when this is so, writing $V = \int (\alpha\, dx + \beta\, dy + \gamma\, dz)$, we have $V = c$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface exists, its differential equation is $\alpha\, dx + \beta\, dy + \gamma\, dz = 0$, and this equation must therefore be integrable by a factor. Now the functions α, β, γ are such that $\alpha^2 + \beta^2 + \gamma^2 = 1$, and they besides satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. passing from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x + \rho\alpha$, $y + \rho\beta$, $z + \rho\gamma$ (ρ infinitesimal), the line belonging to this point is the original line; and consequently α, β, γ, considered as functions of x, y, z, must remain unaltered when these variables are changed into $x + \rho\alpha$, $y + \rho\beta$, $z + \rho\gamma$, respectively. We thus obtain the equations

$$\alpha \frac{d\alpha}{dx} + \beta \frac{d\alpha}{dy} + \gamma \frac{d\alpha}{dz} = 0,$$
$$\alpha \frac{d\beta}{dx} + \beta \frac{d\beta}{dy} + \gamma \frac{d\beta}{dz} = 0,$$
$$\alpha \frac{d\gamma}{dx} + \beta \frac{d\gamma}{dy} + \gamma \frac{d\gamma}{dz} = 0.$$

Combining herewith the equations obtained by differentiation of $\alpha^2 + \beta^2 + \gamma^2 = 1$, viz.

$$\alpha \frac{d\alpha}{dx} + \beta \frac{d\beta}{dx} + \gamma \frac{d\gamma}{dx} = 0,$$
$$\alpha \frac{d\alpha}{dy} + \beta \frac{d\beta}{dy} + \gamma \frac{d\gamma}{dy} = 0,$$
$$\alpha \frac{d\alpha}{dz} + \beta \frac{d\beta}{dz} + \gamma \frac{d\gamma}{dz} = 0,$$

and subtracting the corresponding equations, we obtain three equations which may be written

$$\beta\left(\frac{d\beta}{dz}-\frac{d\gamma}{dy}\right)=\gamma\left(\frac{d\gamma}{dx}-\frac{d\alpha}{dz}\right)=\alpha\left(\frac{d\alpha}{dy}-\frac{d\beta}{dx}\right),$$

or, what is the same thing,

$$\frac{d\beta}{dz}-\frac{d\gamma}{dy}=k\alpha,\quad \frac{d\gamma}{dx}-\frac{d\alpha}{dz}=k\beta,\quad \frac{d\alpha}{dy}-\frac{d\beta}{dx}=k\gamma,$$

and, multiplying by α, β, γ, and adding,

$$k=\alpha\left(\frac{d\beta}{dz}-\frac{d\gamma}{dy}\right)+\beta\left(\frac{d\gamma}{dx}-\frac{d\alpha}{dz}\right)+\gamma\left(\frac{d\alpha}{dy}-\frac{d\beta}{dx}\right).$$

We thus see that, if the function on the right-hand vanish, then $k=0$, and consequently also

$$\frac{d\beta}{dz}-\frac{d\gamma}{dy}=0,\quad \frac{d\gamma}{dx}-\frac{d\alpha}{dz}=0,\quad \frac{d\alpha}{dy}-\frac{d\beta}{dx}=0,$$

viz. if the equation $\alpha\,dx+\beta\,dy+\gamma\,dz=0$ be integrable, then $\alpha\,dx+\beta\,dy+\gamma\,dz$ is an exact differential; which is the theorem in question.

But it is interesting to obtain the first mentioned set of differential equations from the analytical equations of a congruency, viz. these are $x=mz+p$, $y=nz+q$, where m, n, p, q are functions of two arbitrary parameters, or, what is the same thing, p, q are given functions of m, n; and therefore, from the three equations, m, n are given functions of x, y, z. And it is also interesting to express in terms of these quantities m, n, considered as functions of x, y, z, the condition for the existence of the set of surfaces.

We have

$$\frac{\alpha}{m}=\frac{\beta}{n}=\frac{\gamma}{1},$$

and thence without difficulty

$$\alpha,\,\beta,\,\gamma=\frac{m}{R},\,\frac{n}{R},\,\frac{1}{R},\qquad\text{where } R=\sqrt{1+m^2+n^2};$$

$$\left(\alpha\frac{d}{dx}+\beta\frac{d}{dy}+\gamma\frac{d}{dz}\right)\frac{1}{R}=-\frac{1}{R^3}\left[m\left(m\,\frac{dm}{dx}+n\,\frac{dm}{dy}+\frac{dm}{dz}\right)+n\left(m\,\frac{dn}{dx}+n\,\frac{dn}{dy}+\frac{dn}{dz}\right)\right],$$

so that the required equations in α, β, γ will be satisfied if only

$$m\,\frac{dm}{dx}+n\,\frac{dm}{dy}+\frac{dm}{dz}=0,$$

$$m\,\frac{dn}{dx}+n\,\frac{dn}{dy}+\frac{dn}{dz}=0,$$

and it is to be shown that these equations hold good.

Writing for shortness $dp=A\,dm+B\,dn$, $dq=C\,dm+D\,dn$, the equations of the line give

$$\begin{aligned}
0&=z\,\frac{dm}{dx}+A\,\frac{dm}{dx}+B\,\frac{dn}{dx}, & 0&=z\,\frac{dn}{dx}+C\,\frac{dm}{dx}+D\,\frac{dn}{dx},\\
0&=z\,\frac{dm}{dy}+A\,\frac{dm}{dy}+B\,\frac{dn}{dy}, & 0&=z\,\frac{dn}{dy}+C\,\frac{dm}{dy}+D\,\frac{dn}{dy},\\
-m&=z\,\frac{dm}{dz}+A\,\frac{dm}{dz}+B\,\frac{dn}{dz}, & -n&=z\,\frac{dn}{dz}+C\,\frac{dm}{dz}+D\,\frac{dn}{dz};
\end{aligned}$$

or, writing

$$\lambda,\mu,\nu=\frac{dm}{dy}\frac{dn}{dz}-\frac{dm}{dz}\frac{dn}{dy},\ \frac{dm}{dz}\frac{dn}{dx}-\frac{dm}{dx}\frac{dn}{dz},\ \frac{dm}{dx}\frac{dn}{dy}-\frac{dm}{dy}\frac{dn}{dx},$$

so that identically

$$\lambda\,\frac{dm}{dx}+\mu\,\frac{dm}{dy}+\nu\,\frac{dm}{dz}=0,$$

$$\lambda \frac{dn}{dx} + \mu \frac{dn}{dy} + \nu \frac{dn}{dz} = 0,$$

then in each set, multiplying by λ, μ, ν and adding, so as to eliminate A, B, C, D, we find

$$\lambda - m\nu = 0, \qquad \mu - n\nu = 0.$$

Substituting these values of λ, μ in the last preceding equations, ν divides out, and we have the two equations in question.

The foregoing equations give further

$$A, B, C, D = -z + \frac{1}{\nu}\frac{dn}{dy}, \quad -\frac{1}{\nu}\frac{dm}{dy}, \quad -\frac{1}{\nu}\frac{dm}{dx}, \quad -z + \frac{1}{\nu}\frac{dm}{dx}.$$

Taking for α, β, γ the before-mentioned values, we find

$$\begin{aligned} \frac{d\alpha}{dy} - \frac{d\beta}{dx} &= \frac{1}{R}\left(\frac{dm}{dy} - \frac{dn}{dx}\right) - \frac{m}{R^3}\,r\left(\frac{dm}{dy} - \frac{dn}{dy}\right) - \frac{n}{R^3}\,r'\left(\frac{dm}{dx} - \frac{dn}{dx}\right) \\ &= \frac{1}{R^3}\left\{(\,\cdot\,)\,\frac{dm}{dy} + (\,\cdot\,)\frac{dn}{dx} + \left(\frac{dm}{dx} - \frac{dn}{dy}\right)\right\}; \end{aligned}$$

and similarly, but using the equations

$$m\,\frac{dm}{dx} + n\,\frac{dm}{dy} + \frac{dm}{dz} = 0, \qquad m\,\frac{dn}{dx} + n\,\frac{dn}{dy} + \frac{dn}{dz} = 0$$

to eliminate the coefficients $\frac{dm}{dz}$, $\frac{dn}{dz}$ which in the first instance present themselves, we find

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy} = \frac{n}{R^3}(\,\cdot\,), \qquad \frac{d\gamma}{dx} - \frac{d\alpha}{dz} = -\frac{1}{R^3}(\,\cdot\,);$$

whence, multiplying by γ, α, β, and adding,

$$\begin{aligned} &\alpha\left(\frac{d\beta}{dz} - \frac{d\gamma}{dy}\right) + \beta\left(\frac{d\gamma}{dx} - \frac{d\alpha}{dz}\right) + \gamma\left(\frac{d\alpha}{dy} - \frac{d\beta}{dx}\right) \\ &= \frac{1}{1+m^2+n^2}\left\{(\,\cdot\,)\frac{dm}{dy} + (\,\cdot\,)\frac{dn}{dx} + \left(\frac{dm}{dx} - \frac{dn}{dy}\right)\right\}; \end{aligned}$$

or we have

$$(1+n^2)\frac{dm}{dy} - (1+m^2)\frac{dn}{dx} + mn\left(\frac{dm}{dx} - \frac{dn}{dy}\right) = 0$$

as the condition for the existence of the set of surfaces.

It is clear that the condition is satisfied when the lines are the normals of a given surface: seeking the surfaces which cut the lines at right angles, we obtain the parallel surfaces; and we are led to the theorem that any parallel surface is the locus of the extremity of a line of constant length measured off from each point of the surface along the normal—or, what is equivalent thereto, the parallel surface is the envelope of a sphere of constant radius having its centre on the surface. I will verify the theorem for the case of the ellipsoid. Taking X, Y, Z as the coordinates of a point on the ellipsoid $\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1$, and x, y, z as current coordinates, the equations of the normal are

$$\frac{a^2}{X}(x - X) = \frac{b^2}{Y}(y - Y) = \frac{c^2}{Z}(z - Z), \quad (= \lambda, \text{ suppose}).$$

We have therefore

$$X, Y, Z = \frac{a^2 x}{a^2 + \lambda}, \ \frac{b^2 y}{b^2 + \lambda}, \ \frac{c^2 z}{c^2 + \lambda},$$

and thence

$$\frac{a^2 x^2}{(a^2+\lambda)^2} + \frac{b^2 y^2}{(b^2+\lambda)^2} + \frac{c^2 z^2}{(c^2+\lambda)^2} = 1,$$

an equation which determines λ as a function of x, y, z.

The direction-cosines α, β, γ of the normal are proportional to $\frac{X}{a^2}, \frac{Y}{b^2}, \frac{Z}{c^2}$, that is, to $\frac{x}{a^2+\lambda}$, $\frac{y}{b^2+\lambda}$, $\frac{z}{c^2+\lambda}$, and the equation $\alpha^2+\beta^2+\gamma^2=1$ then determines their absolute magnitudes: the equation $\alpha\,dx+\beta\,dy+\gamma\,dz=dV$ thus is

$$\frac{\dfrac{x\,dx}{a^2+\lambda}+\dfrac{y\,dy}{b^2+\lambda}+\dfrac{z\,dz}{c^2+\lambda}}{\sqrt{\dfrac{x^2}{(a^2+\lambda)^2}+\dfrac{y^2}{(b^2+\lambda)^2}+\dfrac{z^2}{(c^2+\lambda)^2}}}=dV,$$

viz. the left-hand side, considering therein λ as a given function of V, is an exact differential. We verify this by finding the value of V, viz. writing down the two equations

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}+\frac{z^2}{c^2+\lambda}-\frac{V^2}{\lambda}=0,$$

$$\frac{x^2}{(a^2+\lambda)^2}+\frac{y^2}{(b^2+\lambda)^2}+\frac{z^2}{(c^2+\lambda)^2}-\frac{V^2}{\lambda^2}=1,$$

these are equivalent in virtue of the equation that determines λ: and it is to be shown that, regarding V as given by either of them, say by the second equation, we have for dV its foregoing value. In fact, differentiating the second equation, the term in $d\lambda$ disappears by virtue of the first equation, and the result is

$$\frac{x\,dx}{a^2+\lambda}+\frac{y\,dy}{b^2+\lambda}+\frac{z\,dz}{c^2+\lambda}-\frac{V\,dV}{\lambda}=0,$$

in which substituting for $\frac{V}{\lambda}$ its value from the first equation, we have for dV the value in question. Regarding V as a given constant, the two equations give, by elimination of λ, an equation $\phi(x,y,z,V)=0$, which is, in fact, the surface parallel to the ellipsoid and at a constant normal distance $=V$ from it.

626.

ON THE GENERAL DIFFERENTIAL EQUATION $\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0$, WHERE X, Y ARE THE SAME QUARTIC FUNCTIONS OF x, y RESPECTIVELY.

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Write $\Theta=a+b\theta+c\theta^2+d\theta^3+e\theta^4$, the general quartic function of θ; and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0.$$

We have

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ w^2, & w, & 1, & \sqrt{W} \end{vmatrix}=0,$$

a particular integral of

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dz}{\sqrt{Z}}+\frac{dw}{\sqrt{W}}=0;$$

and consequently the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=0,$$

viz. the two constants z, w must enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

Take $w=\infty$; the equation becomes

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{e} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$e(\theta^2+\beta\theta+\gamma)^2-(e\theta^4+d\theta^3+c\theta^2+b\theta+a)=(2\sqrt{e}\,\beta-d)(\theta-x)(\theta-y)(\theta-z),$$

or, what is the same thing,

$$\begin{aligned} e\,(2\gamma+\beta^2)-c &= -(2\beta\sqrt{e}-d)(x+y+z), \\ e\,\cdot 2\beta\gamma-b &= (2\beta\sqrt{e}-d)(yz+zx+xy), \\ e\gamma^2-a &= -(2\beta\sqrt{e}-d)\,xyz, \end{aligned}$$

where β, γ are indeterminate coefficients which are to be eliminated.

Write

$$x^2=\frac{\sqrt{X}}{\sqrt{e}}=P, \qquad y^2=\frac{\sqrt{Y}}{\sqrt{e}}=Q;$$

then we have

$$\beta x+\gamma+P=0, \qquad \beta y+\gamma+Q=0;$$

giving

$$\beta:\gamma:1=Q-P:Py-Qx:x-y.$$

Substituting these values in the first of the preceding three equations, we have

$$e\,\frac{2(Py-Qx)(x-y)+(Q-P)^2}{(x-y)^2}-c=-\left\{\frac{2(Q-P)\sqrt{e}}{x-y}-d\right\}(x+y+z),$$

that is,

$$e\left\{\frac{2(Qy-Px)}{x-y}+\frac{(Q-P)^2}{(x-y)^2}\right\}-c=-\left\{\frac{2(Q-P)\sqrt{e}}{x-y}-d\right\}(x+y+z);$$

or, reducing by

$$Qy-Px=y^2-x^2+\frac{x\sqrt{X}-y\sqrt{Y}}{\sqrt{e}},$$

$$Q-P=\frac{\sqrt{X}-\sqrt{Y}}{\sqrt{e}}, \qquad =\frac{x^2-y^2+(\cdots)}{\sqrt{e}\,(x-y)} \quad \text{if } M=\frac{\sqrt{X}-\sqrt{Y}}{x-y},$$

this is

$$e\left\{\frac{2(x\sqrt{X}-y\sqrt{Y})}{\sqrt{e}\,(x-y)}-\frac{x+y}{e}\right\}^2+2xy+\frac{1}{e}M^2-\frac{2(x+y)}{\sqrt{e}}M\sqrt{X}+\frac{2}{\sqrt{e}}M\sqrt{Y}=c+d(x+y+z)+e(x+y)^2.$$

We have Euler's solution in the far more simple form

$$M^2=C+d(x+y)+e(x+y)^2,$$

where C is the arbitrary constant. It is to be observed that, in the particular case where $e=0$, the first equation becomes

$$M^2=c+d(x+y+z);$$

and the two results for this case agree on putting $C=c+dz$.

But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, in my *Treatise on Elliptic Functions*, Chap. xiv., further developed the theory of Euler's solution, and have shown that, regarding C as variable, and writing

$$\mathfrak{S} = ad^2 + b^2e - 2bcd + C[-4ae + bd + (C-c)^2],$$

then the given equation between the variables x, y, C corresponds to the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\mathfrak{S}}} = 0,$$

a result which will be useful for effecting the identification. The Abelian solution may be written

$$e\frac{2(x\sqrt{X} - y\sqrt{Y})}{x-y} - e\left\{2xy - x^2 - y^2 + \frac{1}{e}M^2 - 2(x+y)\frac{1}{\sqrt{e}}M\right\}$$
$$- c - d(x+y) = z\big[d + 2e(x+y) - 2M\sqrt{e}\big]\,;$$

and substituting for M its value, and multiplying by $(x-y)^2$, the equation becomes

$$\begin{aligned}2\sqrt{e}\,(x-y)(x\sqrt{X} - y\sqrt{Y}) - e(x^2+y^2)(x-y)^2 + (\sqrt{X} - \sqrt{Y})^2 \\ - 2(x^2-y^2)(\sqrt{X} - \sqrt{Y})\sqrt{e} - c(x-y)^2 - d(x+y)(x-y)^2 \\ = z(x-y)\Big[d(x-y) + 2e(x^2-y^2) - 2(\sqrt{X} - \sqrt{Y})\sqrt{e}\Big].\end{aligned}$$

On the left-hand side, the rational part is

$$X + Y + c(-x^2 + 2xy - y^2) + d(-x^3 + x^2y + xy^2 - y^3) + e(-x^4 + 2x^3y - 2x^2y^2 + 2xy^3 - y^4)$$

which, substituting therein for X, Y their values, becomes

$$= 2a + b(x+y) + c\cdot 2xy + dxy(x+y) + e\cdot 2xy(x^2 - xy + y^2);$$

and the irrational part is at once found to be

$$= 2\sqrt{e}\,(x-y)(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{XY}.$$

The equation thus is

$$2a + b(x+y) + c\cdot 2xy + dxy(x+y) + e\cdot 2xy(x^2 - xy + y^2)$$
$$+ 2\sqrt{e}\,(x-y)(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{XY},$$

$$z = \frac{\text{(above expression)}}{(x-y)\Big[d(x-y) + 2e(x^2-y^2) - 2(\sqrt{X} - \sqrt{Y})\sqrt{e}\Big]},$$

which equation is thus a form of the general integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, and the particular integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0.$$

Multiplying the numerator and the denominator by

$$d(x-y) + 2e(x^2-y^2) + 2(\sqrt{X} - \sqrt{Y})\sqrt{e},$$

the denominator becomes

$$= (x-y)^2\left[\{d + 2e(x+y)\}^2 - 4e\left(\frac{\sqrt{X} - \sqrt{Y}}{x-y}\right)^2\right]$$

which, introducing herein the C of Euler's equation, is

$$= (x-y)^2(d^2-4eC).$$

We have therefore

$$\begin{aligned} z(x-y)^2(d^2-4eC) = \big\{&2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\\ &+2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})-2\sqrt{XY}\big\}\\ &\times\left\{d(x-y)+2e(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\right\}. \end{aligned}$$

Using $\mathfrak{S}$ to denote the same value as before, the function on the right-hand is, in fact,

$$= (x-y)^2\left\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\right\},$$

and, this being so, the required relation between z, C is

$$z(d^2-4eC) = \left\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\right\}.$$

To prove this, we have first, from the equation

$$M^2 = C+d(x+y)+e(x+y)^2,$$

to express C as a function of x, y. This equation, regarding therein C as a variable, gives

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dC}{\sqrt{\mathfrak{S}}}=0,$$

and we have therefore

$$-\sqrt{\mathfrak{S}} = \sqrt{X}\,\frac{dC}{dx} = \sqrt{Y}\,\frac{dC}{dy};$$

viz. $\sqrt{X}\,\dfrac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before

$$M = \frac{\sqrt{X}-\sqrt{Y}}{x-y},$$

we have

$$C = M^2-d(x+y)-e(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\,\frac{dM}{dx}-d-2e(x+y).$$

We have

$$\frac{dM}{dx} = \frac{1}{x-y}\,\frac{X'}{2\sqrt{X}}-\frac{\sqrt{X}-\sqrt{Y}}{(x-y)^2},$$

and hence

$$\begin{aligned} \sqrt{\mathfrak{S}}\,(x-y)^2 &= \sqrt{X}\,(x-y)^2\left\{2M\,\frac{X'}{2\sqrt{X}}(x-y)-d-2e(x+y)\right\}\\ &= -(x-y)X'(\sqrt{X}-\sqrt{Y})+2(X+Y-2\sqrt{XY})\sqrt{X}\\ &\quad+(d+2e(x+y))(x-y)^2\sqrt{X}\\ &= \left[(x-y)X'+2X+2Y+(d+2e(x+y))(x-y)^2\right]\sqrt{X}\\ &\quad+\left[(x-y)X'-4X\right]\sqrt{Y}. \end{aligned}$$

We obtain at once the coefficient of $\sqrt{Y}$, and with little more difficulty that of $\sqrt{X}$; and the result is

$$\begin{aligned} \sqrt{\mathfrak{S}}\,(x-y)^2 = &-\left[4a+3bx+2cx^2+dx^3+y(b+2cx+3dx^2+4ex^3)\right]\sqrt{Y}\\ &+\left[4a+3by+2cy^2+dy^3+x(b+2cy+3dy^2+4ey^3)\right]\sqrt{X}. \end{aligned}$$

We have also

$$\begin{aligned}C(x-y)^2 &= (\sqrt{X}-\sqrt{Y})^2 - d(x+y)(x-y)^2 - e(x+y)^2(x-y)^2\\ &= X+Y-d(x^3-x^2y-xy^2+y^3)-e(x^4-2x^2y^2+y^4)-2\sqrt{XY}\\ &= 2a+b(x+y)+c(x^2+y^2)+dxy(x+y)+2ex^2y^2-2\sqrt{XY},\end{aligned}$$

or, say

$$\begin{aligned}C(x-y)^2 = 2a(x-y)+b(x^2-y^2)+c(x^3-x^2y+xy^2-y^3)\\ +dxy(x^2-y^2)+2ex^2y^2(x-y)-2(x-y)\sqrt{XY}.\end{aligned}$$

We can hence form the expression of

$$(x-y)^2\Big\{2be-cd+dC+2\sqrt{e}\sqrt{\mathfrak{S}}\Big\},$$

viz. this is

$$\begin{aligned}&= (2be-cd)(x-y)^2+2ad(x-y)+bd(x^2-y^2)+cd(x^3-x^2y+xy^2-y^3)\\ &\quad + d^2xy(x^2-y^2)+2de\,x^2y^2(x-y)-2d(x-y)\sqrt{XY}\\ &\quad + 2\sqrt{e}\{[-(4a+3bx+2cx^2+dx^3)-y(b+2cx+3dx^2+4ex^3)]\sqrt{Y}\\ &\qquad + [(4a+3by+2cy^2+dy^3)+x(b+2cy+3dy^2+4ey^3)]\sqrt{X}\},\end{aligned}$$

and this should be

$$\begin{aligned}&= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\\ &\qquad + 2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})-2\sqrt{XY}\big\}\\ &\quad \times\big\{d(x-y)+2e(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\big\}.\end{aligned}$$

The function on the right-hand is, in fact,

$$\begin{aligned}&= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)-2\sqrt{XY}\big\}\\ &\quad \times\big[d(x-y)+2e(x^2-y^2)\big]+4e(x-y)(\sqrt{X}-\sqrt{Y})(x\sqrt{Y}-y\sqrt{X})\\ &\quad + 2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)-2\sqrt{XY}\}\\ &\quad + 2\sqrt{e}\,(x-y)(x\sqrt{Y}-y\sqrt{X})\big[d(x-y)+2e(x^2-y^2)\big],\end{aligned}$$

viz. this is

$$\begin{aligned}&= \big\{2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\big\}\\ &\quad \times\big\{d(x-y)+2e(x^2-y^2)\big\}+4e(x-y)(-xY-yX)\\ &\quad -2\sqrt{XY}\big[d(x-y)+2e(x^2-y^2)\big]+4e(x-y)(x+y)\sqrt{XY}\\ &\quad +2\sqrt{e}\,\Big(\sqrt{X}\big[2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\\ &\qquad\qquad +2Y-(x-y)y\big(d(x-y)+2e(x^2-y^2)\big)\big]\\ &\qquad -\sqrt{Y}\big[2a+b(x+y)+c\cdot 2xy+dxy(x+y)+e\cdot 2xy(x^2-xy+y^2)\\ &\qquad\qquad +2X-(x-y)x\big(d(x-y)+2e(x^2-y^2)\big)\big]\Big),\end{aligned}$$

which is, in fact, equal to the expression on the left-hand side.

To complete the theory, we require to express $\sqrt{Z}$ as a function of x, y. It would be impracticable to effect this by direct substitution of the foregoing value of z; but, observing that the value in question is a solution of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}+\dfrac{dz}{\sqrt{Z}}=0$, or, what is the same thing, that $\dfrac{1}{\sqrt{X}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dx}=0$, $\dfrac{1}{\sqrt{Y}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dy}=0$, we may from either of these equations, considering therein z as a given function of x, y, calculate $\sqrt{Z}$.

Writing for shortness

$$z = \frac{J - 2\sqrt{e}\,y(x-y)\sqrt{X} + 2\sqrt{e}\,x(x-y)\sqrt{Y} - 2\sqrt{XY}}{R - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}},$$

where

$$R = (x-y)^2[d + 2e(x+y)],$$

$$J = 2a + b(x+y) + 2cxy + dxy(x+y) + 2exy(x^2 - xy + y^2);$$

or, if for a moment $z = \dfrac{N}{D}$, then

$$\frac{dz}{dx} = \frac{1}{D^2}\left(D\,\frac{dN}{dx} - N\,\frac{dD}{dx}\right) = -\frac{\sqrt{Z}}{\sqrt{X}},$$

that is,

$$\sqrt{Z} = \frac{\sqrt{X}}{D^2}\left(N\,\frac{dD}{dx} - D\,\frac{dN}{dx}\right) = \frac{\Omega}{D^2} \text{ suppose;}$$

or, writing for shortness X', R', J' to denote the derived functions $\dfrac{dX}{dx}$, $\dfrac{dR}{dx}$, $\dfrac{dJ}{dx}$ (Y' is afterwards written to denote $\dfrac{dY}{dy}$, but as the final formulae contain only $X' = \dfrac{dX}{dx}$, and $Y' = \dfrac{dY}{dy}$, this does not occasion any defect of symmetry), we find

$$\begin{aligned}\Omega = {} & N[R'\sqrt{X} - 2\sqrt{e}\,X - \sqrt{e}\,(x-y)X' + 2\sqrt{e}\,\sqrt{XY}] \\ & - D[J'\sqrt{X} - 2\sqrt{e}\,yX - \sqrt{e}\,(x-y)yX' + 2\sqrt{e}\,(2x-y)\sqrt{XY} - X'\sqrt{Y}];\end{aligned}$$

and substituting herein for N, D their values, and arranging the terms, we find

$$\Omega = \mathfrak{A} + \mathfrak{B}\sqrt{X} + \mathfrak{C}\sqrt{Y} + \mathfrak{D}\sqrt{XY},$$

where

$$\begin{aligned}\mathfrak{A} = {} & -J\,[2X + (x-y)X'] \\ & - 2(x-y)yR'X \\ & \\ & + Ry\{2X + (x-y)X'\} \\ & + 2(x-y)XJ' \\ & + 2(x-y)X'Y,\end{aligned} \qquad \begin{aligned}\mathfrak{B} = {} & JR' \\ & + 2e(x-y)y\{2X + (x-y)X'\} \\ & + 4ex(x-y)Y \\ & - RJ' \\ & - 2e(x-y)y[2X + (x-y)X'] \\ & - 4e(x-y)(2x-y)Y,\end{aligned}$$

$$\begin{aligned}\mathfrak{C} = {} & -4ey(x-y)X \\ & - 2e(x-y)x[2X + (x-y)X'] \\ & - 2R'X \\ & + RX' \\ & + 2e(x-y)y\{2X + (x-y)X'\} \\ & + 4e(x-y)(2x-y)X,\end{aligned} \qquad \begin{aligned}\mathfrak{D} = {} & 2J \\ & + 2(x-y)xR' \\ & + 2[2X + (x-y)X'] \\ & - 2(2x-y)R \\ & - 2(x-y)X' \\ & - 2(x-y)J,\end{aligned}$$

where the terms have been written down as they immediately present themselves; but, collecting and arranging, we have

$$\begin{aligned}\mathfrak{A} &= 2X(-J + Ry - 2Y) + (x-y)[2XJ' + 2X'Y - X'J - 2yR'X + yRX'], \\ \mathfrak{B} &= JR' - J'R - 4e(x-y)^2Y, \\ \mathfrak{C} &= -4XR' + X'R + 4e(x-y)^2X - 2e(x-y)^3X', \\ \mathfrak{D} &= 2J + 4X - 2Rx + 2(x-y)(xR' - R - J').\end{aligned}$$

To reduce these expressions, writing

$$M = d + 2e(x+y),$$

$$\Lambda = c + d(x+y) + e(x^2+y^2),$$

we have $R = (x-y)^2 M$, and therefore $R' = 2(x-y)M + 2e(x-y)^2$; also

$$J = X + Y - (x-y)^2\Lambda;$$

also, from the original form,

$$J' = b + 2cy + d(2xy + y^2) + e(6x^2y - 4xy^2 + 2y^3).$$

The final values are

$$\begin{aligned}
\mathfrak{A} &= -X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b + dxy)M + xyM^2\},\\
\mathfrak{B} &= (x-y)M[4Y + (x-y)Y'] + 2e(x-y)^3Y',\\
\mathfrak{C} &= -(x-y)M[4X - (x-y)X'] - 2e(x-y)^3X',\\
\mathfrak{D} &= 4(X+Y) + 4e(x-y)^4,
\end{aligned}$$

which, once obtained, may be verified without difficulty.

Verification of $\mathfrak{A}$.—The equation is

$$-X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b + dxy)M + xyM^2\}$$

$$= 2X(-J + Ry - 2Y) + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\};$$

or, putting for shortness

$$\Lambda^2 + (-b + dxy)M + xyM^2 = \nabla,$$

this is

$$\begin{aligned}
(x-y)^4\nabla &= X^2 + 6XY + Y^2\\
&\quad + 2X\{-X - 3Y + (x-y)^2\Lambda + (x-y)^3yM\}\\
&\quad + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\}\\
&= -X^2 + Y^2 + 2(x-y)^2X\Lambda + 2(x-y)^3yXM\\
&\quad + (x-y)\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\};
\end{aligned}$$

we have $-X^2 + Y^2 = -(X-Y)(X+Y)$, where $X - Y$ divides by $x - y$, $= (x-y)\Pi$ suppose; hence, throwing out the factor $x - y$, the equation becomes

$$\begin{aligned}
(x-y)^3\nabla &= -\Pi(X+Y) + 2(x-y)X\Lambda + 2(x-y)yXM\\
&\quad + 2XJ' + 2X'Y - X'\{X + Y - (x-y)^2\Lambda\}\\
&\quad - 2yX\left[2(x-y)M + 2(x-y)^2e\right] + (x-y)^2yMX'\\
&= -\Pi(X+Y) + 2XJ' - X'(X-Y)\\
&\quad + 2(x-y)X\Lambda - 2(x-y)yXM\\
&\quad + (x-y)^2X'\Lambda - 4(x-y)^2eyX + (x-y)^2yMX'.
\end{aligned}$$

We have $2XJ' = J'(X+Y) + J'(X-Y)$, and hence the first line is

$$= (-\Pi + J')(X+Y) + J'(X-Y);$$

$-\Pi + J'$, as will be shown, divides by $x - y$, or say it is $= (x-y)\Phi$, and, as before, $X - Y$ is $= (x-y)\Pi$; hence, throwing out the factor $x - y$, the equation becomes

$$(x-y)^2\nabla = \Phi(X+Y) + \Pi(J' - X') + 2X\Lambda - 2yXM + (x-y)[X'\Lambda - 4eyX + yMX'].$$

We have

$$\Pi = b + c(x+y) + d(x^2+xy+y^2) + e(x^3+x^2y+xy^2+y^3),$$

and thence

$$-\Pi + J' = c(-x+y) + d(-x^2+y^2) + e(-x^3+3x^2y-3xy^2+y^3);$$

or, dividing this by $(x-y)$, we find

$$\Phi = -c - dx - e(x^2-xy+y^2),$$

or, as this may be written,

$$\Phi = -\Lambda + dy + 2exy.$$

We find, moreover,

$$J' - X' = 2c(-x+y) + d(-3x^2+2xy+y^2) + e(-4x^3+6x^2y-4xy^2+2y^3),$$

which divides by $(x-y)$, the quotient being

$$-2c - d(3x+y) - e(4x^2-2xy+2y^2),$$

viz. this is

$$= -2\Lambda - (x-y)(d+2ex).$$

Hence the equation now is

$$\begin{aligned}(x-y)^2\nabla = {} & (X+Y)[-\Lambda + dy + 2exy] + 2X\Lambda - 2yXM \\ & + (x-y)\Pi\{-2\Lambda - (x-y)(d+2ex)\} \\ & + (x-y)\{X'\Lambda - 4eyX + yMX'\}.\end{aligned}$$

The first line is

$$(X+Y)[-\Lambda + yM + 2(x-y)ye] + 2X\Lambda - 2yXM,$$

which is

$$= (\Lambda - yM)(X-Y) + 2(x-y)ey(X+Y);$$

hence, throwing out the factor $x-y$, the equation becomes

$$\begin{aligned}(x-y)\nabla &= (\Lambda - yM)\Pi + 2ey(X+Y) - 2\Lambda\Pi + X'\Lambda - 4eyX + yMX' - (x-y)\Pi(d+2ex) \\ &= (\Lambda + yM)(-\Pi + X') - 2ey(X-Y) - (x-y)\Pi(d+2ex).\end{aligned}$$

We have

$$-\Pi + X' = c(x-y) + d(2x^2-xy-y^2) + e(3x^3-x^2y-xy^2-y^3),$$

which is $= (x-y)(\Lambda + xM)$: also $(X-Y) = (x-y)\Pi$, as before; whence, throwing out the factor $x-y$, the equation is

$$\nabla = (\Lambda + xM)(\Lambda + yM) - 2ey\Pi - (d+2ex)\Pi,$$

that is,

$$\nabla = (\Lambda + xM)(\Lambda + yM) - M\Pi,$$

viz. substituting for ∇ its value, reducing, and throwing out the factor M, the equation becomes

$$-b + dxy = (x+y)\Lambda - \Pi,$$

which is right.

Verification of $\mathfrak{B}$.—The equation is

$$J[2(x-y)M + 2e(x-y)^2] - J'(x-y)^2M - 4(x-y)^2Y$$

$$= 4(x-y)MY + (x-y)^2MY' + 2e(x-y)^3Y',$$

which, throwing out the factor $x-y$, is

$$0 = 2M(-J+2Y) + (x-y)M(J'+Y') + 2e(x-y)(-J+2Y) + 2e(x-y)^2Y'.$$

Here $-J+2Y$, $= -(X-Y)+(x-y)^2\Lambda$, is divisible by $(x-y)$: hence, throwing out the factor $x-y$, the equation is

$$\begin{aligned} 0 = {} & M\{-2b-2c(x+y)-2d(x^2+xy+y^2)-2e(x^3+x^2y+xy^2+y^3)\} \\ & + M(J'+Y') + 2M(x-y)\Lambda + 2e(-J+2Y) + 2e(x-y)Y'. \end{aligned}$$

In the first and second terms, the factor which multiplies M is

$$c(-2x+2y) + d(-2x^2+2y^2) + e(-2x^3+2x^2y-6xy^2+6y^3),$$

which is divisible by $x-y$; also $-J+2Y$, $= -(X-Y)+(x-y)^2\Lambda$, is divisible by $(x-y)$: hence, throwing this factor out, the equation is

$$\begin{aligned} 0 = {} & M\{-2c+d(-2x-2y)+e(-2x^2+2xy-4y^2)\} + 2M\Lambda \\ & + 2e\{-b-c(x+y)-d(x^2+xy+y^2)-e(x^3+x^2y+xy^2+y^3)\} \\ & + 2e(x-y)\Lambda + 2eY'. \end{aligned}$$

Here in the first line the coefficient of M is $= e(2xy-2y^2)$: hence, throwing out the constant factor $2e$, the equation is

$$0 = -b-c(x+y)-d(x^2+xy+y^2)-e(x^3+x^2y+xy^2+y^3)+Y'+(x-y)yM+(x-y)\Lambda.$$

The first five terms are

$$= c(-x+y) + d(-x^2-xy+2y^2) + e(-x^3-x^2y-xy^2+3y^3),$$

which is divisible by $x-y$; throwing out this factor, the equation is

$$0 = -c - d(x+2y) - e(x^2+2xy+3y^2) + \Lambda + yM,$$

which is right.

Verification of $\mathfrak{C}$.—We have

$$\begin{aligned} & -2X[2(x-y)M+2e(x-y)^2] + (x-y)^2X'M + 4e(x-y)^2X - 2e(x-y)^3X' \\ & = -(x-y)M[4X-(x-y)X'] - 2e(x-y)^3X' \end{aligned}$$

which is, in fact, an identity.

Verification of $\mathfrak{D}$.—The equation may be written

$$\begin{aligned} 4X+4Y+4e(x-y)^4 = {} & 2X+2Y-2(x-y)^2\Lambda \\ & + 4X - 2x(x-y)^2M \\ & + 2(x-y)\{2(x-y)xM + 2ex(x-y)^2 - M(x-y)^2 - J'\}, \end{aligned}$$

viz. this is

$$= 2X-2Y-4e(x-y)^4-2(x-y)^2\Lambda+2x(x-y)^2M+4ex(x-y)^3-2M(x-y)^3-2(x-y)J'.$$

The first term $2(X-Y)$ is divisible by $2(x-y)$; throwing this factor out, the equation becomes

$$\begin{aligned} & = b + c(x+y) + d(x^2+xy+y^2) + e(x^3+x^2y+xy^2+y^3) - J' \\ & \quad - 2e(x-y)^3 - (x-y)\Lambda + x(x-y)M + 2ex(x-y)^2 - M(x-y)^2. \end{aligned}$$

Substituting for J' its value, the first line becomes

$$c(x-y) + d(x^3-xy^2) + e(x^4-5x^2y+5xy^2-y^4),$$

which is divisible by $(x-y)$; hence, throwing out this factor, the equation is

$$= c + dx + e(x^2 - 4xy + y^2) - \Lambda + xM - 2e(x-y)^2 + 2ex(x-y) - M(x-y),$$

where the sum of all the terms but the last is $= d(x-y) + e(2x^2 - 2xy)$: hence, again throwing out the factor $x-y$, the equation becomes

$$= d + 2ex - 2e(x-y) + 2ex - M,$$

which is right.

Recapitulating, we have for the general integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, or the particular integral of $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} + \dfrac{dz}{\sqrt{Z}} = 0$,

$$z = \frac{J - 2\sqrt{e}\,y(x-y)\sqrt{X} + 2\sqrt{e}\,x(x-y)\sqrt{Y} - 2\sqrt{XY}}{(x-y)^2M - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}},$$

the corresponding value of $\sqrt{Z}$ being

$$\begin{aligned}\sqrt{Z} = \Big\{\sqrt{e}\big[-X^2 - 6XY - Y^2 + (x-y)^4\{\Lambda^2 + (-b + dxy)M + xyM^2\}\big]&\\ + \big[\{4Y + (x-y)Y'\}M + 2e(x-y)^2Y'\big](x-y)\sqrt{X}&\\ - \big[\{4X - (x-y)X'\}M + 2e(x-y)^2X'\big](x-y)\sqrt{Y}&\\ + \big[4(X+Y) + 4e(x-y)^4\big]\sqrt{XY}\Big\}&\\ \Big/\big\{(x-y)^2M - 2\sqrt{e}\,(x-y)\sqrt{X} + 2\sqrt{e}\,(x-y)\sqrt{Y}\big\}^2&,\end{aligned}$$

where, as before,

$$M = d + 2e(x+y),$$
$$\Lambda = c + d(x+y) + e(x^2+y^2),$$
$$J = 2a + b(x+y) + 2cxy + dxy(x+y) + 2exy(x^2 - xy + y^2);$$

also X is the general quartic function $a + bx + cx^2 + dx^3 + ex^4$, and Y, Z are the same functions of y, z respectively.

In connexion with what precedes, I give some investigations relating to the more simple form $\Theta = a + c\theta^2 + e\theta^4$, or, as it will be convenient to write it, $\Theta = 1 - l\theta^2 + \theta^4$.

We have

$$\begin{vmatrix} x, & \sqrt{X} \\ y, & \sqrt{Y} \end{vmatrix} = 0 \quad \text{a particular integral} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0;$$

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \end{vmatrix} = 0 \quad \begin{matrix}\text{the general integral} \\ \text{a particular integral}\end{matrix} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0;$$

$$\begin{vmatrix} x^3, & x, & x^2\sqrt{X}, & \sqrt{X} \\ y^3, & y, & y^2\sqrt{Y}, & \sqrt{Y} \\ z^3, & z, & z^2\sqrt{Z}, & \sqrt{Z} \end{vmatrix} = 0 \quad \begin{matrix}\text{the general integral} \\ \text{a particular integral}\end{matrix} \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0;$$

$$\begin{vmatrix} x^3, & x, & x^2\sqrt{X}, & \sqrt{X} \\ y^3, & y, & y^2\sqrt{Y}, & \sqrt{Y} \\ z^3, & z, & z^2\sqrt{Z}, & \sqrt{Z} \\ w^3, & w, & w^2\sqrt{W}, & \sqrt{W} \end{vmatrix} = 0 \quad \text{of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} = 0,$$

and so on; viz. in taking

$$\begin{vmatrix} x^2, & x, & \sqrt{X} \\ y^2, & y, & \sqrt{Y} \\ z^2, & z, & \sqrt{Z} \end{vmatrix} = 0 \quad \text{as the general integral of} \quad \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

we consider z as the constant of integration: and so in other cases.

It is to be remarked that it is an essentially different problem to verify a particular integral and to verify a general integral, and that the former is the more difficult one. In fact, if $U = 0$ is a particular integral of the differential equation $M\,dx + N\,dy = 0$, then we must have $N\,\dfrac{dU}{dx} - M\,\dfrac{dU}{dy} = 0$, not identically but in virtue of the relation $U = 0$, or we have to consider whether two given relations between x and y are in fact one and the same relation. In the case of a general solution, this is theoretically reducible to the form $c = V$, c being the constant of integration, and we have then the equation $N\,\dfrac{dV}{dx} - M\,\dfrac{dV}{dy} = 0$, satisfied identically, or, what is the same thing, U a solution of this partial differential equation.

Hence it is theoretically easier to verify that

$$\begin{vmatrix} x^2, & x, & \sqrt{X} \\ y^2, & y, & \sqrt{Y} \\ z^2, & z, & \sqrt{Z} \end{vmatrix} = 0$$

is a general solution, than to verify that

$$\begin{vmatrix} x, & \sqrt{X} \\ y, & \sqrt{Y} \end{vmatrix} = 0$$